

Capital Without Usury

*Jewish, Christian, and Islamic Finance
from Scripture to Non-Extractive Capital*

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Preface — Why Usury Still Matters

Somewhere in the world today — at a kitchen table, in a bank's glass office, on a phone screen held in one hand on a bus — a person is agreeing to pay for money. The paper in front of her, or the scroll of it, runs to dozens of pages, and she has been taught by her whole civilization to look for one thing in it above all: the rate. If the number is low, the deal is good; if it is high, the deal is bad; if it is very high, someone may eventually use the word predatory, which is very nearly the last moral word modern finance keeps in stock for a loan. Every other question the document could raise — what happens to her in a bad year, who decided that her need should carry a price, why the lender's money is entitled to grow while her wages are not — has no box on the form. She signs where the flag says sign. So do we all.

It was not always like this, and that is the first thing this book has to say. For the greater part of recorded time, in most of the civilizations that kept records, the deal she just signed would have been examined before it was priced, and examined by professionals. Three great religious traditions, Judaism, Christianity, and Islam, built entire departments of law around transactions like hers: courts that could order the increase given back, scholars who classified such contracts clause by clause, preachers who thundered about them by name, and institutions, some still running, designed to make them unnecessary. The machinery was not ornamental. Confessors' manuals carried chapters on lending the way tax codes carry schedules; charitable pawn banks were chartered because sermons demanded them; partnership deeds were drafted, in more than one legal language, so that the money's conduct could be audited against scripture. Two further traditions, in India and across the Buddhist world, built their own distinct machinery beside the Abrahamic three. The subject of all that law had a name. In English it is usury, and if the word now sounds antique, like a disease no one catches anymore, that is not because the thing it named went away. The thing it named became normal. What went away was the surrounding architecture of questions — a moral architecture of finance, drawn with a jurist's precision and maintained for millennia, which the modern world dismantled without ever quite refuting. This book is about that architecture: what it was, why five traditions thought it worth building, how it was lost, and what it would mean — concretely, contractually, arithmetically — to build it again.

The Wrong Starting Point

Begin with the word itself, because the modern meaning is a trap. Ask almost anyone today what usury is and you will hear something like: charging too much interest. The definition feels self-evident, and it silently decides everything downstream of it. If usury is an excessive rate, then the moral question about lending is a pricing question (where to draw the line), and the answer will always be a number, argued about the way speed limits are argued about, set by statute, revised by lobbying, different in every jurisdiction and every decade. Some rate is fair; some rate is too much; the argument is over which. Modern consumer law has become expert at this, and within its limits the expertise is real: disclosure rules can force a lender to compress his whole product into one comparable number, and comparison is a genuine protection. It is also, on the older view, the smallest of the protections — a civilization checking the price tag on a transaction whose nature it

has stopped examining.

The traditions this book reads did not start with the price. Their question was not how much a loan may charge but what kind of thing a loan is, and behind that a question so basic it tends to embarrass the people who ask it: what is money for? A rate, after all, is just a price, and a price is the answer to a question you have already finished asking — price of what, for whose use, surrendered by whom, under what constraint. The old law of usury lived entirely in those prior questions. It asked what was really being sold when a lender charged for a loan: the money's use, the time, the risk, the borrower's lack of alternatives. And it was prepared to rule, and for a very long time did rule, that some of those things must not be sold at any price, because they were never the seller's to begin with.

That is why the interest rate is the wrong place to start, and why this book will spend so little time on the question that dominates modern discussion of credit. A rate measures the size of a claim; it says nothing about the claim's nature. Two loans at the same twelve percent can be different moral objects entirely — one a priced service between parties who could each have walked away, the other a slow transfer of a family's future to the one party who could afford to wait. The number on the page is identical; everything that matters is elsewhere. It is in who carries the loss if the venture fails, in what the lender truly put at hazard, in what the borrower could and could not refuse, in what the money was doing to the two of them across the time it was owed. The traditions kept their eyes on the elsewhere. Usury, as they used the word, named a relationship before it named a number — a particular way one person's money can come to sit on another person's life — and a relationship cannot be audited with a percentage.

The borrower is the reason the stakes are what they are, and it is worth being plain about her early, because she will be present on nearly every page. Across most of the written record, lending's typical object was survival — grain until harvest, rent until wages, medicine until strength — sought by someone whose alternatives had already run out. A contract signed on those terms is voluntary only in the thinnest sense of the word. The traditions understood this without embarrassment, wrote their lending law as if the desperate borrower were the normal case, and treated the moment of her need as something close to sacred ground: the one moment when another person's money should arrive as help or not at all. Whatever else separates Torah, Gospel, and Qur'an — and plenty does — they agree with one voice here. Need is not a market.

One Suspicion, Three Languages

The traditions arrive at lending from different histories, in different languages, with different pictures of God. Yet each of them, independently, developed a version of the same suspicion: that there is something wrong — not imprudent, wrong — with money that multiplies simply because it is owed. The Hebrew Bible forbids the Israelite to take increase from a poor brother's loan and builds release and restoration into the calendar itself. The Christian scriptures push mercy past the boundaries of kin and prudence, toward lending that expects nothing back, and the Church that read them spent well over a millennium refining a law of just exchange whose center held that a loan, as such, earns nothing. The Qur'an declares trade lawful and *riba* forbidden, warning those who persist of war from God — language the scripture does not use lightly — and fourteen centuries of jurisprudence have been spent drawing the boundary that verse announced. Three bodies of scripture, three legal civilizations, one conviction reached three times: money lent must come back the same sum it went out; gain must be earned by something real. It must come

from somewhere.

Each tradition then built law, and the law is where this book lives, because law is where conviction is forced to become specific. It is one thing to preach against exploiting the poor; every civilization has managed that. It is another to decide whether the lender may keep a grateful borrower's small present, or lodge in the borrower's guesthouse, or buy his wheat at a price fixed before the harvest is in; whether a partnership whose money can lose nothing is a partnership at all, or something else in a partnership's paperwork; whether rent, which everyone permitted, differs in principle from interest, which almost no one did — and if so, where the difference bites. The rabbis of the Talmud, the canonists and scholastic theologians of Latin Christendom, and the jurists of the fiqh schools each spent centuries on precisely such cases, and their three case-literatures, enormous, technical, and for centuries mutually unread, kept arriving at one another's answers. Reading them together, mechanism by mechanism, you watch three legal systems that could not consult each other converge on the same handful of distinctions, the way three bridge-builders on three rivers converge on the arch. The recurring shape is telling you something about the load.

That convergence is the deepest reason Judaism, Christianity, and Islam must be read together rather than in three polite, separate volumes. Read alone, each tradition's usury law looks like an idiosyncrasy — a covenantal quirk, a scholastic obsession, a scriptural severity — and reading alone is how the old polemics got their footing, each tradition free to imagine that its neighbor's practice was the real scandal. Read together, the three look like independent witnesses to one hazard, and their testimony can be cross-examined the way witnesses' testimony always should be. Their agreements identify what the hazard is. Their disagreements, just as instructive, mark the places where a genuine design choice exists and honest law can go more than one way. And the reader gains a second sight no single tradition can supply: whatever blind spot each system carried, the other two tend to see it, name it, and solve it, which will matter enormously in this book's final chapters, when the three toolkits are laid out on one bench.

Two other traditions join the book at its turning points, as guests rather than exhibits, because each answers the same question from ground the Abrahamic three do not occupy. Hindu Dharmaśāstra never prohibited interest; it regulated increase: rates scaled to the venture's peril, ceilings past which no debt may grow. It stands, therefore, as the great structural alternative: the measure, where the others built a wall. Buddhism enters from the opposite side, beginning not with commerce but with renunciation, and its monastic history poses the question every financial institution in this book must eventually face: what becomes of an ideal once it acquires a treasury? Neither guest is decoration. Each holds a piece of the design problem that the Abrahamic traditions handle differently or not at all, and the book pauses twice, once for each, to learn it.

The Braid, Not the Survey

A book about three traditions usually means three books stapled together: the Jewish material, then the Christian, then the Islamic, each in its own chronology, meeting only in a comparative afterword. That design was rejected here on the day this book was conceived, for a simple reason: it keeps the traditions in the very silos this subject needs them out of. The questions that matter — what makes a loan a loan, who may profit from danger, when a partnership is honest, what may be charged for lateness — cut across all three literatures, and any one tradition's answer sharpens the moment it is set beside the other two.

So this book is organized by financial mechanism, not by religion. Each chapter takes up an instrument, and the traditions meet inside it. When the subject is the naked loan, you will read the rabbis on *ribbit*, the scholastics on the *mutuum*, and the jurists on *qard* in the same sitting; when the subject is partnership, the *iska*, the *societas*, and the *mudarabah* share one chapter because they are, structurally, one idea answering to three names. The braid is the book.

The braid imposes a discipline that a survey never has to face, and the reader is entitled to hear it stated as a promise. Each tradition is presented from within — as its own authorities frame it, in its own categories, with its living disputes marked as disputes — before it is compared with anything. No tradition is the trunk from which the others branch; no tradition plays the foil; no tradition is here to serve as the cautionary tale in someone else's sermon. Halakha and fiqh are functioning legal systems with contemporary practitioners — there are rabbinic courts ruling on investment contracts and shariah boards reviewing them this week — and they are written about here in the present tense they deserve. Where a tradition quarrels with itself, and all of them do, vigorously and on the record, the quarrel is reported rather than adjudicated. A reader determined to discover which of these traditions the book privately considers correct should prepare for disappointment.

Read this way, the traditions disclose something that reading them as mere prohibition always hides: they were protecting something, and they knew what it was. A prohibition, examined closely, is a drawing in negative space; trace its edges and you find the outline of what it was built around. The lending law of all three traditions traces the same shape. A person, embedded in a community, moving through time, who must be able to survive a bad year without becoming someone's revenue. An exchange whose terms both sides can actually see. A venture whose gains and losses land on the people who truly shared it. Wealth that moves toward need without being owed anything back. The prohibitions were the fence; that fourfold figure — the person, the visible bargain, the shared fate, the unpriced mercy — was the garden. From prohibition to design is this book's whole trajectory in a phrase: its first movements recover, from the fence-building, what the traditions were protecting; its last movements take the protected thing seriously as a specification and ask what financial architecture would protect it now.

Old Doctrine, Modern Money

It may seem a long way from Deuteronomy to a venture term sheet, and one ambition of this book is to convince you that it is a shorter walk than it looks. The instruments answer to new names (liquidation preference, cumulative dividend, guaranteed return, first-loss tranche), but the moves are old, and the traditions' lawyers, who spent centuries auditing contracts for increase traveling under an assumed name, would find today's documents strenuous but not strange. A preferred share that must be paid its fixed percentage before any partner sees a coin; a certificate that promises repayment at face value no matter what befalls the asset behind it; a penalty schedule that out-earns the loan it claims to discipline: the medieval casuists had files on all of these. They simply filed them under the older, franker headings. Much of this book's second half consists of walking modern financial furniture past three old legal systems and writing down what they notice. Nor is the exercise a museum tour, because nearly everyone alive is inside at least one of these contracts — a mortgage, a pension, a policy, a loan — and usually on the borrowing side of the paper.

Their most unsettling notice concerns not any single instrument but the direction of drift. Every tradition in this book has watched its own safeguards bend back toward the thing

they forbade — this is one of the book's central findings, and it is as much confession as accusation: the permitted workaround that becomes the product, the exception that grows until it is the rule, the old words kept on as labels for instruments that rebuild, payment by payment, what the words once forbade. The book gives the modern chapters of that story their full weight — the long quiet that settled over the usury question in Christian Europe, the workaround become industry standard in more than one tradition, the new secular moralities of finance that inherit the old ambitions and not the old tests. It tells those stories from inside each tradition, in the voices of each tradition's own critics, who are invariably the sharpest available. And the point of the drift story is not that the traditions failed. It is that they kept noticing — that each maintained, sometimes for centuries at a stretch, an internal audit honest enough to tell its ideals from its instruments. That capacity for self-suspicion turns out to be as valuable an inheritance as any doctrine.

Beneath all of it, waiting for the mathematics that could finally state it, lies the question this book regards as the traditions' deepest and least understood — the missing mathematical question of finance. Who survives the path over time? Modern finance is built, to a degree that surprises people meeting the fact for the first time, on averages: the expected return, the mean outcome, the performance of the portfolio as a whole. But nobody lives in an average. A family, a founder, a fund, a village moves through its years in order, each loss resetting the next year's starting point, and somewhere below them all there is a floor — call it ruin — where the story simply stops. An arrangement can be attractive on average and lethal along the path; some of the most respectable structures in finance are exactly that, and the obligation that falls due regardless of the harvest is the oldest of them. The traditions could not write the equations. But look at what they built — the release valves, the ceilings, the harvest-sharing, the pools — and you keep finding machinery whose effect is to stop one bad year from becoming an ending. Where the modern mind optimized the average, the traditions guarded the path. A late chapter will do the arithmetic properly and show that this is not a metaphor: solidarity has a mathematics, the mathematics comes down on the traditions' side, and a young science — ergodicity economics — has lately begun writing proofs underneath their oldest instincts. Usury law, read at this depth, was survival engineering, built long before the tools existed to check its work.

How This Book Moves

The book moves in six long strides — Scripture, law, institutions, evasion, mathematics, design. It may help to walk the route once before setting out.

It opens by clearing the ground. Part I frames the question the whole tradition asks — what is money allowed to do? — locates lending where the traditions located it, between neighbors rather than between abstractions, and assembles the small cast of contract forms, from loan to gift to pool, whose boundaries the rest of the book will patrol. Part II goes to the sources of the fire: the Hebrew Bible's covenant economy, the New Testament's radicalization of mercy, the Qur'an's war against *riba*, each read on its own terms and in its own order, for what it actually says. Part III turns scripture into legal grammar: what makes a loan a loan, why the traditions chased hidden benefit into the smallest corners of social life, how price fraud and overreaching were policed, and what danger genuinely licenses. There the first guest interrupts, and Dharmaśāstra's regulated increase asks whether a wall was ever necessary at all.

Parts IV and V are the constructive heart of the old world. Partnership — *societas*, *iska*, *mudarabah*, the merchant's *commenda* — was the traditions' great escape from usury, the

form in which capital could earn by honestly sharing a venture's fate, and the book lingers there on questions that will sound startlingly contemporary: how capital and labor divide a venture's gains, who absorbs its losses, why profit may come only from profit. Then come the mercy institutions — the *gemach*, the *qard hasan*, the Monti di Pietà, the great engines of alms — which together answer a question modern finance has stopped asking: what credit becomes when it is designed for the borrower's survival rather than the lender's return. The second guest arrives at that part's close, and Buddhist renunciation puts its unsettling question to every institution in the book.

Part VI descends into the engine room: the centuries-long technical debates — actual loss, forgone profit, illiquidity, penalty — in which some of the finest legal minds of Christendom argued over which compensations a lender may take without selling what was never his, while their Jewish and Muslim counterparts ran the same inquiries in their own idioms; the part closes with deception, uncertainty, and the anatomy of consent. If you want to know where this book's analytical machinery was forged, it was there. Part VII tours the instruments those debates produced and constrained — sale-based finance, leases, certificates, insurance and its mutual-aid ancestors, endowments and asset locks — and pauses, toward its close, for the chapter on ergodicity and the mathematics of mutual aid, where the book's quantitative thread is finally tied. Part VIII is the tragedy, told without nostalgia: interest secularized, the question unlearned, and each tradition's modern reckoning — the *heter iska* at industrial scale, Islamic finance's benchmark problem, Catholic social teaching's uneasy silence, and the market's replacement moralities of ESG and impact.

Then the book stops looking backward. Part IX rebuilds the tradition as arithmetic: a ledger that makes every euro of return say what it is for, risk priced without being worshiped, waterfalls and preferences read as the geometry they are, invisible labor made visible, true surplus told apart from the merely available, and the whole assembled into a non-extractive return formula — the traditions' tests restated as something a spreadsheet can check. Part X builds with it: a fund, a contract library, an analyzer, an accounting, and, because every tradition in this book testifies that sincerity alone keeps no institution honest, a closing chapter on the temptations of religious finance and the governance that guards against drift. A conclusion gathers what the traditions teach. Six appendices (glossary, timeline, contract typology, metrics, worked waterfalls, and a reading guide) turn the book from an argument into a reference.

Different readers will want different doors. The practitioner impatient for the design work can go straight to the last two parts and follow the citations backward; the reader who came for history can settle into the first eight and treat the arithmetic as a promised destination; the skeptic who suspects that religious finance is a contradiction in terms is asked only to stay through the partnership and mercy chapters before deciding. The book was built, though, to be walked in order. Its later chapters lean on its earlier distinctions the way upper floors lean on the ones below, and the reader who arrives at the modern material with the old grammar in hand will hear things in a term sheet that cannot otherwise be heard.

A word, finally, on how it is written. This is a work of history and analysis, not of devotion, and it assumes no faith in the reader — only a willingness to entertain the possibility that five civilizations, over spans measured in millennia, kept concluding there was something here worth a jurist's whole career. The technical terms of four legal languages appear because precision requires them; each is explained on arrival and cataloged in the glossary at the back. Scripture and law are quoted sparingly, in named standard renderings, and the book prefers wherever possible to show how a text was read and used by the people

whose law it became. And one rule of respect holds throughout: every tradition speaks here through its own authorities, including its own critics, and never through its rivals' caricatures.

Why, then, does usury still matter? Not because the word deserves reviving for its own sake, and not because the old world was gentle — it was not, and nothing here will pretend it was. Usury still matters because the thing the traditions built their law around has not changed. Need still borrows; money still multiplies while it is owed; and between those two facts, now as ever, ordinary lives are won and lost. Every contract that moves capital settles the old questions somewhere in its clauses — puts a price on someone's need or declines to, hands the bad year to somebody, lets the money share the venture's weather or exempts it — whether or not anyone involved can hear it happening. The traditions heard. They asked their questions out loud, for millennia, in public, with penalties, and they left behind not a nostalgia but a toolkit: definitions, tests, instruments, institutions, and a standard of proof for every coin that claims to have been earned. The wager of this book is that the toolkit still works — that money's longest conversation is not an heirloom but an unfinished project, waiting for builders. The person on the bus, signing where the flag says sign, is owed more questions than the rate. What follows is the long history of the people who asked them, and the beginning of asking them again.

Part I — The First Question: What Is Money Allowed to Do?

Chapter 1 — Sterile Gain and Living Exchange

In the year we call 594 BC, an Athenian merchant and poet named Solon was made sole archon, with extraordinary powers, by a city sliding toward civil war. The trouble was debt. Attica's small farmers had borrowed from their richer neighbors against the only security they had, which was themselves. Stone markers called *horoi* stood planted in fields all over the countryside, each one announcing that the land under it was encumbered, and the men who worked those fields — the *hektemoroi*, "sixth-parters," whose name says they owed a sixth of their harvest, though scholars still argue over the arithmetic — had discovered what every indebted farmer in history discovers eventually: a bad year does not cancel a fixed obligation. It compounds it. When the crop failed, the farmer borrowed against next year. When next year failed too, the creditor took what the law allowed, and the law allowed the man. Athenians were being seized for debt, sold as slaves, some of them abroad. This was not an invading army doing it; it was neighbors, holding paper — or rather holding stones — against neighbors.

Solon's remedy, enacted in his single extraordinary year, was called the *seisachtheia*, the "shaking-off of burdens." He canceled the debts, pulled up the stones, banned all future lending on the security of the person, and set about recovering Athenians who had been sold overseas. We know all this partly because the constitutional history that circulated under Aristotle's name — the *Athenaion Politeia*, written by Aristotle or his school — quotes the poem Solon wrote afterward in his own defense, and it is one of the most remarkable documents in the history of finance: a politician's memoir, in verse, about a debt jubilee. Let the black Earth be my witness, he says — the mighty mother of the Olympian gods, from whom I pulled up the *horoi* planted everywhere; she was enslaved before, and now she is free. He brought home men who had been sold abroad, "one justly, another not," men who had wandered so long they no longer spoke Attic Greek; and those enslaved on their own soil, trembling before their masters, he set free.

Here is the detail worth holding onto. There were no coins in this story. The first true coins had been struck only a generation or so earlier and a sea away, in Lydia — bean-shaped lumps of electrum, a natural gold-silver alloy washed down the river Pactolus at Sardis; Herodotus says the Lydians were the first people we know of to strike gold and silver coin — and Athens would not mint its own until around 560 BC, decades after Solon laid down his office. The debts that nearly wrecked Athens were counted in grain, oil, and weighed silver. Interest-bearing loans, for their part, had been running in Mesopotamia for the better part of two thousand years before anyone owned a drachma; the loan is older than the coin. Money's most dangerous power — the power to bind a future person to a past promise, and to grow while it waits — needed no mint, only a harvest, a ledger, and a bad year.

So the oldest debt crisis on record in the Western tradition arrives before money, in our narrow metallic sense, even exists, and the first great financial reformer answers it not with a rate cap but with a boundary: there are things a loan will no longer be allowed to reach. Solon did not regulate the price of credit; he redefined its jurisdiction. That move — asking not *how much* money may charge but *what money is allowed to do* — is the founding question of this book, and it is the question Judaism, Christianity, and Islam would each, in their own vocabulary and for their own reasons, spend the better part of three thousand years answering.

The Fourth Function

Every economics textbook, more or less since textbooks began, tells you that money does three jobs. It is a *measure* — a common yardstick that lets you compare a goat, a haircut, and an hour of legal advice on a single scale. It is a *medium* — the thing you accept not because you want it but because everyone else will accept it in turn, sparing the shoemaker the chore of finding a hungry farmer who happens to need shoes. And it is a *store* — a way of parking today's surplus for tomorrow's need, wealth in a form that does not rot, rust, or need feeding.

The analysis is ancient, and its most influential early statement belongs to Aristotle. In the fifth book of the *Nicomachean Ethics* he works out how a city of specialists — builder, shoemaker, doctor — manages to trade unlike things fairly, and his answer is that money makes them commensurable: everything can be measured against one thing, and that one thing, he notes, exists by agreement rather than by nature. The Greek word for currency, *nomisma*, comes from *nomos*, law or convention, and Aristotle leans on the pun with evident satisfaction: money is called what it is called because it is not natural but conventional, ours to alter or demonetize as we please. A page later he adds the store-of-value point — money is a kind of pledge for the future, a guarantee that if we do not need a thing now, we can have it when we do. Measure, medium, store: the whole textbook triad, in place before Euclid.

But the textbook list has a fourth item, usually left off because it makes the others look less innocent. Money is also a *claim*. Every unit of it is a kind of standing invitation the rest of society has not yet honored: hold a coin and you hold everyone else's obligation, diffuse and anonymous, to hand over real goods on demand. In that diffuse form the claim is harmless enough — nobody in particular owes you the loaf your coin will buy. The moral temperature changes when the claim acquires a name and an address. Lend the coin, and the abstraction condenses into a relationship: now a particular person owes you, and the ledger that records it is no longer a note about things but a note about *him*. Measure, medium, and store are jobs money does with the world. Claim is the job money does with people, and it is the one function of the four that can wear a face.

The ancients, who lived closer to their creditors than most of us do, saw this with painful clarity. The book of Proverbs — here and throughout, biblical quotations follow the New Revised Standard Version — needs eight words for the sociology of the whole thing: “the borrower is the slave of the lender.” In Solon's Attica that was no figure of speech, and the *horoi* in the fields were its paperwork. A measure measures whether you love it or not, a medium circulates indifferently, a store simply sits; but a claim *ranks* two people, one above the other, for as long as it stands. Of the four, the claim is the only function with a politics.

Notice, too, an oddity in the store-of-value function that the ancients found genuinely strange and we have stopped noticing. Real wealth decays. Grain must be eaten within a few years, cloth wears, a herd must be tended and fed through every winter it survives, and even a house demands its endless small tribute of repair. To hold wealth in the world is to fight rot, and mostly lose slowly. Money alone is exempt. The silver in the strongbox neither eats nor sickens, and — once lent — it does something no barn full of anything has ever done: it grows by itself, on schedule, in the dark. A shepherd's flock might double in three years of good luck and hard labor. A moneylender's silver at the customary rates would double in five with neither. Every farmer who ever mortgaged a field understood the asymmetry in his bones: he had borrowed something immortal and was repaying it with the sweat of something mortal. Interest is the only harvest that ripens in every

weather.

That asymmetry is where the unease begins — the feeling, older than any of the three traditions this book braids together, that when money starts behaving like livestock, something in the order of things has quietly gone wrong.

The Offspring of Metal

You can read the unease in the words themselves, because the ancient world, with a candor later ages abandoned, named interest after baby animals.

In Sumer, where the earliest interest-bearing loans on record were written into clay some four and a half thousand years ago, the word for interest was *mash* — the same word that means “goat kid.” In Greek it was *tokos*, “offspring,” the ordinary word for what is brought forth at a birth. In Latin it was *fenus*, which carries the *fe-* root of fecundity and fetus, the syllable of breeding. The economic historian Michael Hudson, who traced this barnyard vocabulary across three millennia in a study aptly subtitled “*Máš, tokos and fænus* as metaphors for interest accruals,” points out what the metaphor originally made plain: a herd genuinely does throw off kids and calves in season, and a lender’s claim was imagined to breed the same way — a newborn increment arriving with each new moon. The image was not chosen by critics of lending. It was the lenders’ own accounting language, cheerful as a stock ledger. Only later did philosophers turn the metaphor around and ask the obvious question sleeping inside it: a she-goat is alive; in what sense, exactly, does a shekel give birth?

The practice the words described was already old, regular, and legally domesticated long before anyone thought to philosophize about it. In Old Babylonia the customary rates held steady for centuries — one-fifth a year on silver, one-third on barley — and the Laws of Hammurabi, around 1750 BC, simply wrote the custom into stone as a ceiling. Mesopotamian kings coped with the wreckage the same way Solon later would, by periodic clean-slate decrees that canceled consumer debts and sent the pledged family members home. The pattern repeats so regularly across the ancient Near East that it amounts to a rhythm: credit expands, the countryside sinks, the ruler wipes the slate, and the cycle breathes in again. Nobody in Babylon argued that interest was unnatural. It was as natural as the flood, and you managed it the same way, by occasionally opening the gates.

The Greeks were the first to move the question from management to metaphysics, and the man who fixed its terms for the next two thousand years was a doctor’s son from the north, born at Stagira in 384 BC, who spent twenty years in Plato’s Academy, tutored the boy who became Alexander the Great, and settled at last in Athens to run his own school, the Lyceum, from about 335 BC. Aristotle came to money the way he came to everything — constitutions, tragedies, the parts of animals — by asking what it is *for*, and his answer sits in the first book of the *Politics*, in a stretch of argument (1256b to 1258b, in the scholars’ numbering) that may be the most consequential few pages ever written about finance.

Aristotle distinguishes two arts that look alike and are not. The first he treats as part of *oikonomia* — household management, the ancestor of our word “economics” — and its business is provisioning: getting the goods a household or a city actually needs to live well. Acquisition of this kind has a built-in limit, because needs have a limit; the pantry can only hold so much, and a family can only eat so much. The second art he calls *chrematistike* — wealth-getting as an occupation in itself — and in its suspect form its business is not provisioning but accumulation: buying in order to sell, holding in order to gain, treating

the household's tool as the household's purpose. This second art, he observes, has no natural stopping point at all. There is a quantity of grain that means "enough grain." There is no quantity of money that means "enough money," because money, being merely the measure of everything else, sets no measure for itself. A man provisioning a household knows when he is done. A man accumulating money is never done, and Aristotle's diagnosis is that such a man has mistaken the instrument of living for living itself — has turned all his arts, medicine and soldiering and estate management alike, into ways of making coin, as though that were the end and everything else existed for its sake.

Then, near the stretch's close (1258b), comes the verdict that echoed for millennia. Trade for gain already sits low in Aristotle's ranking, an exchange in which men profit from one another rather than from nature. But one practice sits at the very bottom. Usury — the petty weighing-out of small coin at interest — is hated most reasonably of all, he says, because here the gain comes from money itself rather than from the thing money was invented to serve. Money came into being for exchange; interest makes the coin itself the parent of more coin. And here he reaches for the old barnyard word and turns it into an argument: this is why interest is called *tokos*, offspring, for the young resemble their parents, and interest is money born of money. Of all the ways of getting wealth, he concludes, this one is the most contrary to nature. The pun had been sitting in the language for centuries, as innocent as a farmer's ledger. Aristotle picked it up and made it a charge sheet: the lender is running a stud farm for an animal that is not alive.

It is worth being precise about what the argument does and does not claim, because the whole coming tradition depends on the distinction. Aristotle is not saying that surplus is wicked. The farmer's field yields more than was sown; the potter's clay leaves the wheel worth more than it arrived; even the merchant, moving grain from where it is cheap to where it is dear, can be doing the city a service. All of that is increase with a source: nature grows, labor transforms, exchange serves need. His objection lands on one precise spot — increase *without* a source, gain that purports to spring from the sterile instrument itself, from bare time and bare possession. The distinction between the household's economy and the money-getter's art is really a distinction between wealth that feeds life and wealth that merely feeds. And though Aristotle wrote as a pagan philosopher with no covenant, no revelation, and no particular tenderness for the poor, he had built the exact conceptual shelf on which three revealed religions would later find their own convictions already waiting, labeled in Greek.

His Roman readers kept the shelf stocked. Cato the Elder, that flint of republican rectitude, opens his farming manual by noting that the ancestors' laws condemned the thief to repay double, and the usurer fourfold — from which, he adds dryly, you may judge how much worse a citizen they thought the moneylender. Cicero preserves the same Cato in conversation, being asked what was the most profitable part of estate management and answering, with the patience of a man ranking the obvious, good grazing; then adequate grazing; then poor grazing; then plowing. And when the questioner, working down the list, finally asked "What about money-lending?", Cato answered: "What about murder?" The quip is less about lending than about category — the suggestion that somewhere between the pasture and the loan book, a man stops farming and starts feeding on his neighbors.

Not a Matter of Percentages

To modern ears, "usury" means charging too much — an interest rate with the decency dial turned past ten. That meaning is real, but it is a late arrival, and if you carry it backward

into the old texts you will misread every one of them. For roughly two thousand years, in the Latin West and its Greek and Semitic neighborhoods alike, usury did not name an amount. It named an act.

The Latin tells you plainly. *Usura* is just “use” with a suffix: a payment taken for the use of money. The canon lawyers who systematized the Church’s teaching — Gratian’s great twelfth-century compilation is the landmark — defined the sin with arithmetic bluntness: to lend and to expect back anything beyond the principal, anything at all, is usury. Not anything *excessive*; anything. One denarius over, on a loan of a hundred, fell under the same name as fifty. The rabbis drew the line at the same place for loans inside the covenant community, and the Qur’an, in the verse this book will circle back to many times, tells believers who repent of *riba* that they are entitled to their principal — no more. Three legal cultures, one geometry: the loan repays its own amount exactly, and the first coin past that boundary changes the transaction’s species.

Modern usage inverted this. When the English Parliament, in 1545, passed “An Acte agaynst Usurye” that in fact *legalized* interest up to ten percent, the word began its long slide toward meaning only the forbidden remainder — the excess above whatever a legislature currently blesses. That slide matters to this book because it silently replaced a question of kind with a question of degree. Under the old definition, the moralist asks: what sort of transaction is this, and may it yield gain at all? Under the new one, the regulator asks: is the price fair? Both are real questions. But they are different questions, and the traditions this book reads were asking the first one. When a medieval rabbi, a scholastic theologian, and a Basra jurist condemned usury, none of them was running a price check. They were running a classification: this gain has a source, that gain has none; this contract shares the world’s risk, that contract has arranged to be immune to it.

Which brings us to the pair of terms in this chapter’s title. *Sterile gain* is the increase Aristotle diagnosed: a return that claims to grow out of money’s bare existence — out of possession plus a calendar — while the lender contributes no labor, adds no thing, and stands exposed to none of the venture’s weather. *Living exchange* is everything on the other side of the line: the sale that moves a real good to the person who needs it, the hire that pays for a tool’s genuine use, the partnership whose return is carved from a harvest that actually came in, the gift that asks nothing back. The traditions never doubted that money could lawfully pass through all of those and come home larger. The farmer sells at profit; the shipowner charges freight; the partner shares gain; scripture and canon and fiqh bless every one of them, in terms we will spend chapters unpacking. What they refused to bless was the increment with no parent but the coin itself — the gain that would have accrued identically whether the borrower’s venture flowered or drowned, because it fed not on the venture but on the obligation.

Put that way, the old doctrine stops looking like squeamishness about numbers and starts looking like an audit rule. Ask of any increment: where was this born? If the answer runs through soil, sweat, skill, cargo, or shared hazard, the traditions could find a home for it. If the answer is “the loan itself” — if the money claims to have bred — every one of them reached, independently, for the same verdict Aristotle reached, and several of them reached it without ever having read him. Sterility, in this vocabulary, is not an insult to money but a job description. A measure is *supposed* to be barren, the way a yardstick is supposed to stay thirty-six inches long; a ruler that grew while you measured with it would not be a marvel but a fraud. The scandal of usury, for every tradition in this book, was precisely that it asked the yardstick to calve.

Tool, Master, and the Space Between

Beneath the classification runs the first moral distinction of the whole subject, and it can be said in a sentence: money is a tool, and the trouble begins when the tool starts giving the orders. Aristotle's word for the disorder was *chrematistics* — the condition of a man, or a city, in which the means of exchange has quietly promoted itself to the purpose of existence, so that fields are farmed, ships sailed, and skills practiced no longer for what they yield in life but for what they yield in coin. The tool is not wicked; a hammer is not wicked. The question is who is holding whom. And the reason the usurer became the emblem of the inversion — rather than the miser, who merely hoards, or the glutton, who merely spends — is that lending at interest is the one arrangement in which the tool appears to *work by itself*. Every other way of turning money into more money at least routes the increase through the world: a cargo, a workshop, a field. The interest-bearing loan routes it through a person, which makes the borrower the plant and the machinery of the lender's increase.

Watch what the different motions of money do to the space between two people, because everything the traditions later built assumes you can see it. A sale, honestly struck, is the most self-liquidating of human transactions: goods pass one way, price passes the other, and the parties walk off free of each other, equals at the start and equals at the end. A gift binds — anthropologists have filled libraries with the webs of obligation and return that gifts spin — but it binds *laterally*, in a fabric of reciprocity where this year's giver expects to be some year's receiver. The loan is the odd one. It stretches a transaction across time, which no sale does, and for the whole of that stretch it holds the two parties in ranked positions, which no honest sale does either: one waits, the other owes. Proverbs' eight words sit at the bottom of this like a survey stake. Add interest, and the rank acquires a motor — the longer the inferior position lasts, the more it costs, so that time itself, which falls on both parties alike as rain does, becomes revenue for one of them and erosion for the other.

Whether that arrangement is tolerable depends almost entirely on who is borrowing, and why — and here the register must be plain, because for most of recorded history the typical borrower was not a merchant financing cargo but a family financing survival. A subsistence loan is negotiated between a man who can wait and a man who cannot, and every term in it prices that difference. Dependency, in such lending, is the collateral. The farmer who pledges the field eats this winter by mortgaging every winter after, and when the debt finally takes the field, or the daughter, the ledger records a voluntary contract, freely signed. The ancient world knew exactly what these contracts were worth as descriptions of freedom. That is why its oldest financial documents are shadowed by its oldest financial institutions — the Mesopotamian clean slate, the Solonic shaking-off — and why the Hebrew word for interest is not a number but an injury.

Neshekh, the Torah's primary term, means "bite." Its companion terms *tarbit* and *marbit* mean "increase," and the rabbis' later word *ribbit* — cousin, across the Semitic languages, of the Arabic *riba*, "increase" — keeps the same root. The vocabulary is the doctrine in miniature: what the lender calls growth, the borrower wears as a wound. And it matters to the shape of this whole book that Israel's law was asking the relational question centuries before Aristotle asked the philosophical one. "If you lend money to my people, to the poor among you," says Exodus, "you shall not deal with them as a creditor; you shall not exact interest from them." What the verse forbids is not a rate but a stance — the creditor's stance toward a brother — and Leviticus gives the positive command in five words: "let them live with you." Lending inside the covenant is a form of upholding a neighbor whose ruin would be the community's own. Chapter 4 will walk this legislation properly. The

point here is its axis: the Torah never asks what money is — it asks what the loan is doing to the two Israelites at either end of it.

Christianity inherited that axis and pushed it past the boundary of kin. In Luke's Gospel the instruction is to "lend, expecting nothing in return" — a sentence that treats the loan not as commerce to be regulated but as mercy to be perfected, with the enemy included in its reach. The early Church legislated accordingly where it could; the first ecumenical council, at Nicaea in 325, used its seventeenth canon to strip lending at interest from the clergy. For a thousand years the Christian case against usury ran on Scripture and the Fathers. Then, around 1260, the Flemish Dominican William of Moerbeke produced the first Latin translation of Aristotle's *Politics* — a book the Latin West had known only by rumor — and the theologians discovered, with the delight of people finding their family portrait in a stranger's attic, that the pagan had reasoned his way to their conclusion. Thomas Aquinas welded the two arguments together within a generation: usury is unjust, he argued in the *Summa* (II-II, q.78, a.1), because money is consumed in its use like wine, so that charging separately for the money and for its use is selling the same thing twice — selling, in his phrase, what does not exist. Aristotle's sterile coin had become a premise in a theology of justice. But the graft took because the stock was already living; the Church did not learn its unease from the Greek; it learned Greek for its unease.

Islam ran the same convergence in reverse, which is what makes the comparison so instructive. The *Politics*, almost alone among Aristotle's major works, was never translated into Arabic — the Muslim philosophers who handed Latin Europe so much of its Aristotle simply never had this text — so the Qur'an's war on *riba*, and the whole edifice of *fiqh* built on it, owes the philosopher precisely nothing. Yet the Qur'an's own framing turns on the same two categories this chapter has been drawing. Those who devour *riba*, it warns, will claim that trade and usury are alike; the verse — 2:275, here and throughout in M. A. S. Abdel Haleem's rendering — replies that "God has allowed trade and forbidden usury." In a single breath of scripture, living exchange is blessed and sterile gain refused. And when Muslim thinkers later supplied the philosophical commentary, they supplied it from their own resources: al-Ghazali, writing around the year 1100 in his *Ihya Ulum al-Din*, argued that the dinar and dirham were created as means and measures — stones with no use in themselves, made so that everything else could circulate and be valued — and that *riba* is precisely the perversion by which the measure is made an object of trade in its own right, deflected from the purpose for which God fashioned it. It is very nearly Aristotle's argument, arrived at without Aristotle, and it rests on the premise that money's purpose was assigned not by convention but by its Creator.

Three traditions, then, met one question in three vocabularies. The Torah asks it as covenant — whether my brother is still standing upright beside me when the contract closes, or has quietly acquired a master. The Gospel asks it as mercy, auditing what the loan gives against what it takes; the Qur'an, as justice in exchange, sorting the increase that is trade from the increase that devours. None of the three needed the Lyceum to reach its answer, and none of the three treats the question as economic in our narrow sense. What is being audited, in every case, is a relationship — the space between two people with money moving across it — and the shared conviction, older than all three scriptures in its Mesopotamian bones, is that certain motions of money poison that space no matter how politely the paperwork reads.

The Birth of Moral Geometry

Once you ask the question that way, a program follows, and the traditions — being legal civilizations, with courts to run and contracts to enforce — followed it. If some motions of money are life-giving and others predatory, and if the interest-bearing loan cannot be told from the honest sale by the size of the numbers on it, then the moral work has to be done by *form*: by sorting every transfer of wealth into its species and giving each species its own law. What may a loan do that a sale may not? And why may a house earn rent, and a partner take a share, when a lender's hundred florins may earn nothing at all? The answers cannot be found by staring at percentages. They can only be found by drawing the figures precisely and testing what each one does to the people inside it — a discipline it is fair to call moral geometry.

Eight figures will do most of the work of this book. The *loan*, where wealth crosses with an obligation to return it. The *sale*, where goods and price cross once and the parties part. The *rent*, where a thing's use is let while the thing itself comes home. The *partnership*, where money and effort marry and share one fate. The *gift*, where wealth crosses and nothing legal comes back. The *tribute*, where wealth moves because power demands it. The *tax*, where it moves because a community requires it of its members. And the *pool*, where many pay in so that whoever among them is struck may draw out — the figure, as later chapters will argue, in which the traditions' deepest answer to ruin was hiding all along. Every financial instrument ever devised, from a Babylonian barley advance to the capital stack of a venture fund, is one of these figures or a splice of several; and nearly every evasion in the long history of usury is a splice dressed as a single figure, which is why the traditions' lawyers became the world's great connoisseurs of disguise.

Drawing those eight figures carefully — definitions, boundaries, and the tests that tell one from another — is the business of chapter 3, and the chapters after it will spend the rest of the book watching the three traditions build, police, and repair them. But the founding insight of the whole geometry belongs here, at the door, and it is Solon's insight before it is anyone's theology. The Athenians did not solve their debt crisis by finding the right interest rate, and the traditions of Jerusalem, Rome, and Mecca did not frame their law of money by asking what capital could bear. All of them began from the other end, with the person standing inside the contract, and ruled on what must never be done to him — and then, having planted that boundary stone, set about designing every form of exchange, hire, partnership, gift, and common pool that could flourish on the lawful side of it. The prohibition was never the whole of the architecture. It was the surveyor's mark from which a civilization of permitted forms was measured out, and the rest of this book is a walk through the buildings.

Chapter 2 — The Poor, the Brother, the Neighbor, and the Stranger

Long before money was coined, it was lent. The clay tablets of ancient Mesopotamia — surviving by the tens of thousands, baked hard and still legible after four millennia — include contracts by the basketful, and a great many of them are loans: a farmer takes seed barley in the spring and owes barley back at harvest, with its increase; a household takes silver and owes silver the same way. The rates were as settled as anything in that world. Hammurabi's laws, in the eighteenth century BC, simply took them for granted — a third again for barley, a fifth for silver — and even the vocabulary had the calm of long habit. The Sumerian word for interest, *mash*, is the word for a goat kid: increase named for the way a flock increases, a full millennium and more before the Greeks made the same move with *tokos*, "offspring," the pun Aristotle turned against the whole business in the argument chapter 1 has already visited. Interest, in short, is among the oldest institutions in recorded civilization, older than the alphabet, older than coined money by well over a thousand years, and for most of that history nobody blushed about it.

But there is a second genre of document in those same archives, and it tells you what the first one cost. When the barley failed, the loan did not. The debtor pledged what he had — first the field, then the house, then, without embarrassment anywhere in the record, the persons of his household — and a bad year could turn a farmer into a tenant and his children into someone else's servants. So, from time to time, a Babylonian king would proclaim what the Akkadian texts call *misharum*, "justice" or "equity": debts wiped, pledged persons sent home, the slate cleaned. The best preserved of these edicts, from King Ammisaduqa of Babylon, survives nearly whole; Assur had its own word for the release, *andurarum*, and scholars hear that word again behind the Hebrew *deror*, the liberty that Leviticus 25 proclaims in the Jubilee. The Greeks knew both the crisis and the remedy: Solon's *seisachtheia*, the "shaking off of burdens," canceled Athens's debts in 594 BC, freed the citizens enslaved for debt, and banned the practice thereafter. Rome, by the traditional date 326 BC, abolished its own institution of debt bondage, the *nexum*, through the *lex Poetelia Papiria* — after, so Livy tells it, the mistreatment of a bonded young debtor scandalized the city. Everywhere the ancient record shows the same pairing: interest as routine tool, and debt as a slow machine for converting a neighbor's bad season into his servitude — a machine so reliable that whole legal orders had to be interrupted by royal decree just to keep societies from consuming their own members.

The three traditions of this book grew up in the long shadow of that world, and each of them refused a piece of it. None of them refused lending; each, in fact, commands it, with an insistence that surprises modern readers. What they refused was the machine. And the manner of the refusal is the important thing. A royal clean slate treats the debt crisis as weather — it gathers, the king intervenes, it gathers again, and between edicts the machine runs freely. The traditions set out to change the climate instead: to build the restraint into standing law, daily habit, and the community's picture of itself, so that no king's mood was required. Their opening question about a loan was accordingly not the modern one of what money may earn but the older one of what this borrower is to the lender. A brother, said the Torah. A neighbor, said the Gospel, and then redefined the word. A brother in faith and a soul in hardship, said the Qur'an. Everything the coming chapters will build — the contract taxonomies, the partnership law, the charitable engineering — rests on those

answers. This chapter is about the answers themselves: where lending sat on the moral map before anyone drew up the rules.

The Borrower at the Door

Picture the borrower these laws have in view, because he is not the borrower of a modern textbook. He is a smallholder in a village of perhaps a hundred souls. His wheat came up thin, or the tax collector came through, or a child fell sick in the wrong month. He does not walk into an institution and fill out an application; there is no institution. He walks to the house of a man he has known all his life — a kinsman, or a neighbor with two good harvests stored — and he asks. Everyone in the village will know by evening that he asked. The man he asks will sit beside him at the gate for the next forty years. Whatever passes between them is not a bargain between strangers, priced by a market; it is a change in the standing of two households that must go on living side by side. In societies of this kind — and ancient Israel, the Galilee of Jesus, and seventh-century Arabia were all societies of this kind — a loan is an event inside a web of kinship, and everyone concerned can feel the web tighten.

That is why the ancient moral literature of debt reads as kinship literature. The Hebrew Bible's lending laws barely mention the market; they say brother. Where English translations write "kin" or "community member" or "another Israelite," the Hebrew underneath is usually one small word, *achikha*, your brother — the vocabulary of family, not of commerce. And the laws are addressed in the second person singular, to a lender who can see the borrower's face: if you lend to him, if you take his cloak, if your eye is mean toward him. Nothing in that idiom is accidental. Debt in a village is a new relation laid over an old one, and what the law cares about is what the new relation does to the old.

The stakes ran in one direction only. The borrower in view borrows to eat, to sow, to pay the tax — not to expand. If the loan fails, the sequence was as fixed in Israel's memory as in Babylon's archives: the pledge is taken, the field passes, the household follows. The book of Kings preserves the cry of a prophet's widow to Elisha: "Your servant my husband is dead; and you know that your servant feared the LORD, but a creditor has come to take my two children as slaves" (2 Kings 4:1). Centuries later, in the middle of the fifth century BC, Nehemiah heard the same cry at scale from the families of Judah — some mortgaging fields, vineyards, and houses to buy grain in a famine, some borrowing against their land for the king's tax, some already handing sons and daughters into servitude — and the creditors were not foreign conquerors but their own kin. Nehemiah's anger, and the public restitution he extracted, fields and orchards and houses and the interest itself given back, treated the crisis as exactly what this chapter's traditions always understood it to be: a family devouring itself.

Modern categories mislead here in a second way, and it is worth disposing of the confusion early. The ancient world did have commercial credit; Mesopotamian merchant houses ran long-distance trade on it, and Part IV of this book will watch all three traditions build superb instruments for exactly that purpose. The scriptural fire against interest was never primarily about that world. It was about this one: the neighbor whose need has delivered him into your hands. Keep the trader seeking capital and the family seeking bread distinct, because every tradition in this book eventually learned to keep them distinct, and most of the worst readings of the usury laws come from mixing them back together. The texts we are about to read legislate for the second case, the borrower at the door. A rate is a number; a brother is a claim.

The Poor of Your City

The Torah's great interest texts all locate lending the same way before they regulate it. The earliest, in the Covenant Code of Exodus, deserves to be heard exactly, in the New Revised Standard Version's rendering: "If you lend money to my people, to the poor among you, you shall not deal with them as a creditor; you shall not exact interest from them" (Exodus 22:25). Notice what is and is not forbidden. Lending is assumed — commanded, elsewhere. What is forbidden is a posture, dealing with the poor "as a creditor," and a practice, the exacting of interest, for which biblical Hebrew's sharpest word is *neshekh*, the bite. The law then descends immediately into a poor man's evening: if his cloak is taken in pawn, "you shall restore it before the sun goes down; for it may be your neighbor's only clothing to use as cover; in what else shall that person sleep?" (22:26–27). The lender's claim is real, and the law does not cancel it; the law makes it bend, every single sunset, before the body of the borrower. And the reason given has nothing to do with economics: if that borrower cries out, "I will listen, for I am compassionate." The creditor is not the senior party in the room. God has taken sides in advance.

Leviticus widens the frame from a night to a life: "If any of your kin fall into difficulty and become dependent on you, you shall support them; they shall live with you as though resident aliens. Do not take interest in advance or otherwise make a profit from them, but fear your God; let them live with you" (Leviticus 25:35–36) — the Hebrew pairs *neshekh*, the bite, with *tarbit*, the increase, and bans both. Here the interest ban sits inside a command to support. The falling brother is to be caught, held upright, kept beside you, and the refusal of increase is simply what catching him means; you cannot brace a man with one hand and bill him for the bracing with the other. Deuteronomy adds the rhythm of time — the seventh-year release, the warning against the hardened heart and the closed hand as the release year approaches — and everywhere the same address: your brother. Chapter 4 will walk this legal architecture properly, pledges and millstones and Jubilee and all. What matters here is the location. Lending to the poor is not filed under commerce in the Torah. It is filed under rescue, in the same drawer as the gleanings left standing in the field and the tithe set aside for the widow, and the tradition's name for that drawer became *tzedakah* — a noun built on the root for justice. Care for the poor brother is not generosity beyond the law's requirement; it is the requirement.

The rabbis inherited that location and deepened it. Their term of art for forbidden increase is *ribbit*, and the tractate that governs it, Bava Metzia, keeps breaking out of case law into moral astonishment. Rabbi Yosei, in a baraita preserved at folio 71a, marvels at the blindness of lenders at interest: a man called wicked to his face will fight the insult all his days — yet these bring witnesses and a scribe, pen and ink, and solemnly sign a document attesting, in the baraita's own words, that so-and-so has denied the God of Israel. On that page, charging a brother for his need is not sharp practice; it is apostasy with paperwork. A few lines on, Rabbi Shimon ben Elazar reads Psalm 15 — the psalm that lists refusing interest among the marks of the person fit to dwell on God's holy hill — as a working prediction: the fortunes of such lenders collapse and do not rise. And when the sages asked who should be lent to first, they answered with a map of nearness drawn from the wording of Exodus itself (the verse they cite as 22:24; English Bibles number it 22:25): "my people" before others, the poor before the rich, your own poor before the poor of your city, the poor of your city before the poor of another town. Obligation, in this tradition, has an address. Mercy is not a mist hanging evenly over humanity; it runs through channels — household, street, town — so that every poor family is some particular community's first responsibility, rather than everyone's-in-general and therefore no one's.

The Neighbor Who Proved Neighbor

Jesus taught inside this Torah world, in an occupied agrarian province where the borrower at the door was an everyday figure, and his teaching on lending begins from Israel's premises and then leans on them, hard. The Torah had commanded the open hand toward the poor brother and forbidden profit from his need. Jesus, in the Sermon on the Plain, removes the last vestige of return: "If you lend to those from whom you hope to receive, what credit is that to you? Even sinners lend to sinners, to receive as much again. But love your enemies, do good, and lend, expecting nothing in return. Your reward will be great, and you will be children of the Most High; for he is kind to the ungrateful and the wicked" (Luke 6:34–35). Matthew's version of the saying is blunter still: "Give to everyone who begs from you, and do not refuse anyone who wants to borrow from you" (Matthew 5:42). There is no regulation in any of this — no rate cap, no pledge law — only a family resemblance held up as the standard: lend the way your Father gives, to the ungrateful and the wicked, without a meter running, because that is what the children of this particular Father do. The Torah had disciplined the loan to protect the brother. Jesus turns the loan into an imitation of God.

Then there is the question of who counts, and Luke stages it as a legal debate, which is what it was. A lawyer, quoting Leviticus — love your neighbor as yourself — asks the natural lawyer's question about the statute's scope: "And who is my neighbor?" (Luke 10:29). It was a fair question, and an old one. Leviticus 19:18 sits in a chapter addressed to Israel about Israel, and its own thirty-fourth verse had already stretched the command outward — "you shall love the alien as yourself" — so the argument about the boundary was Israel's own argument, running long before the lawyer stood up. Jesus answers with the story every reader knows and none quite domesticates: a man beaten half-dead on the Jericho road; a priest and a Levite passing by on the other side; rescue arriving in the person of a Samaritan, for Luke's audience the sectarian enemy from over the hill. Then the question comes back reversed: "Which of these three, do you think, was a neighbor to the man who fell into the hands of the robbers?" The lawyer's category has been turned inside out. Neighbor has stopped being a status the wounded man must qualify for and become a practice the passerby either performs or refuses. "The one who showed him mercy," the lawyer answers, and hears: "Go and do likewise" (10:36–37).

The church took the reversal literally, and taking it literally changed the economics. If neighbor has no outer edge, the Torah's brother-rule acquires no outer edge either: every borrower is the brother, and what Israel had guaranteed inside the covenant — need met without a meter — Christian teaching would in time claim for anyone in need at all. That is the seed of a remarkable and demanding doctrine. The church's usury ban, when it came, would bind toward every human being, and the Fathers who built it reached for precisely these verses. Chapter 5 will weigh the whole New Testament witness, including the harder texts, and later chapters will watch councils and canonists turn the posture into law — and watch what it cost to mean it, because the cost is the honest part of the story. A boundaryless mercy is a glorious wager. Everything depends on whether the mercy can be sustained everywhere once it is owed everywhere, and a good deal of the history ahead of us, from the Fathers' sermons to the Monti di Pietà, is Christendom straining to make good on the Samaritan.

Devouring the Wealth of Others

Seventh-century Mecca had no king and no bureaucracy; it had clans, a sanctuary, customary honor, and trade. A merchant town in a subsistence landscape is a place of hard asymmetries — caravan fortunes on one street, families a single bad season from destitution on the next — and between them stood no institution but kin solidarity and the lender. The practice the Qur'an confronted there had a name, *riba*, from a root meaning to swell or increase, and the commentators describe its signature move: when a debt fell due and the debtor could not pay, he was offered the choice of paying now or watching the sum grow. Deferral meant doubling. A season's shortfall could compound into permanent clientage — the old machine of Babylon and Athens, running on desert time. The Qur'an's earliest word on the subject, from the Meccan years, already sets two economies against each other; here it is in M. A. S. Abdel Haleem's rendering: "Whatever you lend out in usury to gain value through other people's wealth will not increase in God's eyes, but whatever you give in charity, in your desire for God's approval, will earn multiple rewards" (30:39). In Medina the register sharpens to command — "You who believe, do not consume usurious interest, doubled and redoubled" (3:130), the old practice named to its face — and finally to the fiercest sentence any of the three scriptures aims at this subject: those who persist in *riba* after the warning are to expect "war from God and His Messenger" (2:279). The long passage that carries the threat (2:275–279) also carries the argument. The taker of usury will rise "like someone tormented by Satan's touch"; to the protest that "Trade and usury are the same," the answer comes back flat: "God has allowed trade and forbidden usury."

Hold on to the Qur'an's characteristic verb, because it is doing this chapter's work. Wrongful gain, in the Qur'an's idiom, is eaten. "Do not consume your property wrongfully," runs the verse on corrupt dealing, "intending sinfully and knowingly to consume parts of other people's property" (2:188); and a passage addressing an earlier covenant community — one that assumes the Torah's own ban — condemns "taking usury when they had been forbidden to do so" and "wrongfully devouring other people's property" (4:161). The verb is *akala*, to eat, to devour. Gain from a neighbor's need is not pictured as a fee, a price, or even a theft of objects; it is pictured as a mouth closing on another person's substance. The image belongs to the same moral universe as Nehemiah's assembly: the strong consuming the weak through claims rather than swords. And, exactly as in the Torah, the alternative on offer is not a world without lending but lending relocated. The believer's wealth, the Qur'an insists, already carries the poor inside it — a recognized right, a *haqq*, belonging to the beggar and the destitute (70:24–25) — discharged through *zakat*, the purifying share, through *sadaqah*, the voluntary gift, and through the loan the tradition calls beautiful, the *qard hasan*, whose story belongs to chapter 16. As for the debtor already caught, the instruction is direct: "If the debtor is in difficulty, then delay things until matters become easier for him; still, if you were to write it off as an act of charity, that would be better for you, if only you knew" (2:280). Even the unwinding of old usury is drawn with a draftsman's balance: "You shall have your capital... without suffering loss or causing others to suffer loss" (2:279) — the closing Arabic is a matched pair the tradition treats as a maxim, wrong not and be not wronged. The principal is honest. The increase was the predation.

The community that received these verses heard them as family law, because a family is what the new community was. "The believers are brothers" (49:10): the *ummah* was kinship rebuilt on faith, wide enough to hold the Meccan migrant and the Medinan farmer in one household, and the *riba* ban stood among its founding solidarities — inside this brotherhood, need shall not be farmed. Chapter 6 will read the whole Qur'anic case and the machinery that grew from it. For now, mark the location, because it is the same loca-

tion: like the Torah's, the Qur'an's lending ethic lives where kinship, worship, and survival meet, and it arrived, like Israel's, as the charter of a community surrounded by the older machine and determined not to reproduce it.

The Boundary and the Shield

Every ethic of brotherhood must eventually answer for the man who is not my brother. The three traditions answered differently, and the differences — each with its own cost and its own genius — frame much of the history this book has still to tell.

Deuteronomy answers with a boundary, and the verse must be read exactly, because few verses in Scripture have been read worse: “You shall not charge interest on loans to another Israelite, interest on money, interest on provisions, interest on anything that is lent. On loans to a foreigner you may charge interest, but on loans to another Israelite you may not charge interest, so that the LORD your God may bless you in all your undertakings in the land that you are about to enter and possess” (Deuteronomy 23:19–20; the Hebrew, which numbers these verses 20 and 21, says throughout simply your brother). Within the covenant the ban is total — money, food, anything lent. Toward the *nokhri*, the foreigner, ordinary commercial terms are permitted. Read from within the tradition, this is not a license issued against outsiders; it is the visible edge of a family's internal welfare law. The interest ban belongs to the same cluster as the gleanings, the tithe, and the release year — solidarities a kinship community guarantees its members — and the *nokhri* of this verse is the outside party to whom those solidarities were never owed in either direction: the trader passing through, who lends at interest himself and expects to borrow the same way. The vulnerable outsider was another case entirely. The *ger*, the resident alien, is not the *nokhri*: him the same Torah orders Israel to love — “you shall love the alien as yourself, for you were aliens in the land of Egypt” (Leviticus 19:34) — and him Leviticus 25:35 folds into the very support command we read above, “as though resident aliens... let them live with you.” The boundary marked where the family's guarantees ended, not a wall against the weak.

The rabbis, custodians of the verse, handled the outward permission with visible unease. The Talmud's discussion at Bava Metzia 70b–71a turns the clause over like a suspicious coin — what exactly does *la-nokhri tashikh* permit? — and then hedges whatever it permits: lending at interest to the outsider should be limited to what one needs for a livelihood, the sages ruled, lest the lender learn from his ways; and the precedence rules cut against the profit in any case, for faced with a gentile borrower at interest and a Jewish borrower for free, the free loan to the brother comes first. Whatever Deuteronomy's boundary allowed, the tradition declined to let it become a vocation. Nor was the boundary ever a boundary of mercy: the same Talmud rules that the poor of the nations are sustained together with the poor of Israel, their sick visited and their dead buried, for the sake of the ways of peace (Gittin 61a). One dark sentence must be entered here and then left for its proper place: centuries later, Christian polemic would seize on Deuteronomy's two clauses — a ban Christendom applied to itself, a permission whose exercise it pressed upon Jews and then despised them for — and build from them the slander of the “Jewish usurer,” an afterlife that belongs to its makers and that this book confronts in chapter 17 and Part VIII. The verse in its own house is covenant economics: unlimited duty inward, honest commerce outward.

Islam met the same boundary question and removed the boundary in the other direction: the *riba* ban follows the act, not the relationship. In the law as the great majority of jurists

settled it, a Muslim may not take *riba* from anyone — believer or unbeliever, neighbor or stranger — because the prohibition attaches to the transaction itself rather than to the borrower's membership. The schools did argue an edge case, as legal schools do. A view transmitted in the Hanafi school, resting on a reported hadith that there is no *riba* between a Muslim and a *harbi* — a hostile foreigner — in the abode of war, would suspend the rule in enemy territory; Abu Yusuf within the Hanafi school, and the other schools entire, rejected both the report and the ruling. The shape of the disagreement is the instructive thing: the mainstream position needed no boundary at all. The Qur'an's own catalog of duty had already run past every fence — “to neighbours near and far, to travellers in need” (4:36) — and the law followed it. Where Deuteronomy drew the family's edge, and Christianity dissolved the edge by universalizing the family, Islam wrote the rule into the instrument: some ways of gaining are simply not open to a believer, whoever stands on the other side of the contract.

The Christian dissolution of the edge is where the modern story hides, and one scholar mapped it definitively enough that his subtitle became shorthand. Benjamin Nelson's 1949 study *The Idea of Usury* traced the Deuteronomic pair through Western history under the banner *From Tribal Brotherhood to Universal Otherhood*. The church universalized the brother, so the ban went universal. The Reformation and its heirs then dismantled the ban — Part VIII tells that story — and the universalism quietly inverted. When everyone had been a brother, no one could be charged; when the charging returned, everyone was equally chargeable. The great equality of the modern credit market, Nelson's subtitle implies, is the equality of strangers. His arc runs far ahead of us, but its lesson belongs to this chapter: the scope of obligation and the treatment of the borrower rise and fall together. Deuteronomy's intensity was purchased by its address; Christendom wagered that the intensity could survive without the address; modernity kept the universal address book and canceled the intensity. Each of the three traditions, watching the others, would have a question to ask here, and none of them gets to ask it smugly.

Which returns us, last of all, to why the ban mattered so much to all three — the deepest marker on this chapter's map. Go back to the village of the opening pages. In a subsistence world, ruin is not a tail risk; it is a season that eventually arrives at every door. No single household can stand against it alone — that is what ruin means — and the ancient world offered three answers: the patron, who converts your bad year into your dependency; the clean-slate king, who may or may not come in time; and the community that catches its own. The traditions chose the third answer and built it into law and reflex: the commanded loan and the open hand of Deuteronomy 15; *tzedakah*, and the free-loan practice of *gemilut chasadim* that would one day grow into the *gemach* societies of chapter 15, among the mutual-aid institutions of chapters 27 and 28; the church's common purse and diaconal funds, ancestors of the confraternities and the Monti; *zakat* as the poor's standing right, with the *qard hasan* beside it. Each is a way of pooling a community's strength around whichever member the bad season has found. Seen from this angle, the interest ban is no economic eccentricity bolted onto a religion. It is the design rule that keeps the shield a shield. Aid that charges for itself converts the community's rescue apparatus back into the machine it was built against, and the neighbor's bad year becomes, once again, somebody's yield. Forbid the increase, and help stays help: the strong household's surplus flows to the point of need and comes back, when it can, no larger; the weak household's crisis remains a crisis instead of compounding into a lien on its future. Chapters 27 and 28 will take this architecture seriously on its own terms — the pools, the mutual-aid institutions, and finally the mathematics of why pooling defeats ruin in a way no fairly priced loan can. Here it is enough to state the conviction all three traditions shared, each in its own dialect of

kinship: the community could not keep ruin from coming; it could refuse to let ruin find anyone alone.

That refusal is the ground floor. Above it, the traditions now had to become precise — because “help, don’t harvest” is a location, not yet a law, and the moment a community begins guaranteeing loans without increase, someone asks what counts as a loan, what counts as increase, and whether a partnership, a rent, or a sale is the same thing wearing different clothes. Sorting that out is the business of the next chapter, and the three traditions, sorting independently, arrived at drawers so alike that the rest of this book can usually visit all three with a single label in hand.

Chapter 3 — Loan, Sale, Partnership, Gift, Pool

Consider a single hundred-euro note and the week it is about to have. On Monday it pays a plumber, and the moment the receipt prints, that relationship is over. On Tuesday it goes to a landlord, buying another month under a roof the landlord still owns and must keep from leaking. On Wednesday a neighbor borrows it until payday. On Thursday it is tucked into a card for a niece finishing school, with no expectation except that she not spend all of it sensibly. And on Friday its twin joins dozens of others in the funeral fund of a workers' association, against the day a member's family cannot afford a burial. It is the same note all week — same portrait, same serial number, same faint smell of other people's pockets. Nothing printed on it changes from one hand to the next. Yet in five days it has been five different moral objects: a price, a rent, a debt, a blessing, and a shield.

Every tradition this book reads discovered, early and independently, that the moral analysis of money begins with names rather than amounts. Ask "how much?" first and you will never get anywhere: the same hundred can be mercy in one relation and predation in another, the same rate of return an honest harvest or a quiet strangulation, depending entirely on the structure that produced it. The first question is always "what is this money doing?" — and to answer it you need a vocabulary.

The scriptures treat naming as very nearly the first human job. In Genesis, God forms the animals and brings "them to the man to see what he would call them; and whatever the man called each living creature, that was its name" (Genesis 2:19, in the New Revised Standard Version). The Qur'an gives Adam the same primacy in almost the same breath: "He taught Adam all the names [of things]," says the second sura in M. A. S. Abdel Haleem's rendering, and it is precisely this knowledge of names that ranks Adam above the angels. The lawyers of all three traditions absorbed the working assumption: the world becomes governable when things are called what they are. A creature misnamed cannot be handled; a contract misnamed cannot be judged.

The legal civilizations that inherited the usury question all began with classification. When the emperor Justinian issued his Institutes in 533 — a first-year textbook carrying the full force of law, addressed to "the youth desirous of the laws" — its treatment of obligations opens by sorting them: which contracts arise from the handing over of a thing, which from words, which from mere consent. Six centuries later, confessors' manuals like Raymond of Peñafort's trained priests to ask a merchant not "did you profit?" but "by what contract?" A rabbi consulted about a suspicious payment asks first whether the arrangement was a loan, a sale, a rental, or a deposit, because the whole law of *ribbit* turns on which it was. The qadi does the same for *riba*. Classification is the first act of moral finance, for the same reason diagnosis is the first act of medicine: treatment applied before naming is guesswork, however energetic.

This chapter, the last of Part I, supplies the vocabulary — the family of forms the rest of the book will handle daily. Seven honest members make up the family: the loan, the sale, the lease, the partnership, the gift, the endowment, and the solidarity pool. An eighth arrival, the hybrid, is less a form than a symptom. And one rule tells them apart when the paperwork lies — the rule for the whole book.

What Was Mine Becomes Yours

Strip the loan to its skeleton and four words remain: principal, obligation, time, and return.

Someone hands over money — or grain, or oil, or anything used by being used up — on the understanding that its equivalent will come back. The Latin word is *mutuum*, and the Roman law schools taught their first-years a derivation no modern philologist would sign but that has never been improved on as moral analysis: the thing is called *mutuum*, says Justinian's textbook, because it is given by me to you in such a way that what was mine becomes yours — *ex meo tuum*. Ownership passes at the handover. The coins are the borrower's now, to spend as his own; what he owes is not those coins but their like — in the tradition's compact term, the *tantundem*, the same in quantity and kind. The rabbis reached the same point from their own direction. The Talmud's working rule, stated at Kiddushin 47a, is that "a loan is given to be spent" — *milveh le-hotza'ah nitnah*. The Hebrew word for the loan is *milveh*, and the rule means the lender cannot pretend the money is still his out working for him; it is gone into the borrower's life, and what exists in its place is something else entirely: a personal claim. Islam's word is *qard*, said to come from a root meaning "to cut" — the lender cuts away a piece of his wealth, and it is his no longer. Three languages, one anatomy, and this book will spell the instrument *mutuum*, *milveh*, *qard* whenever it needs to remind you that all three traditions are holding the same object.

Watch what the four skeleton words do. Principal: the amount that must return, no more and — the traditions all add — no less. Obligation: the claim runs against the borrower personally, not against any project the money funded. He owes whether his shop thrives or burns; the debt is deaf to his circumstances, which is its convenience and its cruelty at once. Time: the claim matures on a date, and between handover and maturity time becomes the loan's medium — the dimension the debt lives in and the thing lenders have most wanted to charge for. Return: whatever is stipulated above the principal. And on that last word all three traditions posted sentries, because a loan whose return exceeds its principal has begun charging for something — the borrower's need, the passage of time, the use of money already given away — that the lender, on each tradition's analysis, does not own.

The reasons behind those sentries fill Part III — the anatomy and the loan's careful distinction from investment are chapter 7's business, and chapter 8 shows how far the guards patrol, into gifts, favors, even flattering words. Here the loan needs only to stand in line with its siblings, and what marks it out there is fixity. Sale prices can be haggled and rents lag the market; partnerships fail; gifts wound; pools run dry. The loan is the only contract in this chapter that cannot be disappointed. Its payoff is written before the world has said anything, and it collects regardless of what the world eventually says. Whether that fixity is a mercy or a menace depends on everything else in the relationship — which is exactly why the traditions refused to let the loan pretend to be anything other than what it is.

Price and Rent

The sale is the cleanest of the forms, and the traditions' benchmark for honest gain. A thing changes owners in one direction, a price in the other, and when the exchange completes, the parties owe each other nothing but courtesy. No calendar hangs over them; no claim survives the handshake. Buying cheap and selling dear is how merchants eat, and no tradition in this book condemned the margin as such. What they policed was equivalence — whether the exchange, at the moment it happened, was a fair trade or a disguised taking.

The Qur'an fixes the whole distinction in a single scene. The takers of *riba*, it says, will rise on the Day of Resurrection staggering as if touched by Satan — "That is because they say, 'Trade and usury are the same,' but God has allowed trade and forbidden usury" (Qur'an 2:275, again in Abdel Haleem's rendering). Notice what the verse assumes: the usurers' best defense was a classification argument — increase is increase, the margin on a bale of cloth and the increment on a loan the same gain. The revelation answers not that gain is evil but that the two gains differ in kind, one arising from an exchange of real things, the other from a naked claim on a brother. Everything in this chapter is, in a sense, commentary on that answer, and sura 2 is scripture refusing in so many words to let the taxonomy be flattened.

Blessing the sale obliged the traditions to say what a fair one was, and they answered with a shared realism: just equivalence is a band, not a point. The rabbis drew the band with numbers. The Mishnah — Bava Metzia 4:3 — sets the trigger for *ona'ah*, overreaching, at one-sixth of the price: four silver pieces on a sela of twenty-four. Within a sixth of the going rate, the market is doing what markets do; past it, the wronged party can reclaim the excess or undo the deal, provided he acts within the time it takes to show his purchase to a merchant or a knowledgeable relative. The same mishnah records Rabbi Tarfon, teaching in Lod, generously widening the margin to a third — the merchants of Lod rejoiced — until he added that his rule gave the buyer the whole day to return the goods, whereupon the merchants asked him to please leave them the old rule, and got it. Traders, then as now, preferred a stricter rule briefly enforced to a looser one hanging over the whole day. In Thomas Aquinas's *Summa*, the question on cheating in buying and selling stands immediately before the question on usury — the two sins of exchange, filed side by side — and Christianity ran its inquiry under the name of the just price. Islam added its own refinement, *gharar*: the sold thing must exist, be known, be deliverable; a sale of fog — the unripe harvest, the diver's next haul — is void for excessive uncertainty, because a price cannot be equivalent to a mystery. Chapter 9 walks this ground properly, and chapter 23 carries *gharar* and *ona'ah* into the fine print of modern term sheets.

One seam deserves flagging now. Let the price wait — sell today, collect next spring — and time has crept into the sale; a deferred price above the cash price does quietly what interest does openly, and every tradition noticed. When chapter 24 examines *murabaha*, the cost-plus credit sale on which most modern Islamic finance actually runs, this seam is the entire subject.

The lease is the sale's modest cousin: it sells not the thing but the use of the thing, in slices of time. And it stands on a distinction homely enough to be missed and strong enough to carry the whole usury edifice: some things survive their use, and some things are used by being used up. A house, a field, an ox, a ship — use them for a season and there they still are, a little older, returnable to their owner. Bread is otherwise. To use bread is to eat it; you cannot rent a loaf, because nothing would come back, and a "rental" of bread is just a sale of bread wearing the wrong name — or, if its equivalent is owed back later, a loan of it. Wine, grain, oil, and coin all sit on bread's side of the line. Roman law, with its instinct for shelving, sorted the neighbors. Justinian's four real contracts — the *mutuum* we have met, the *commodatum* (your friend lends you his horse, gratis, and expects that horse back), the *depositum* (custody without use), the *pignus* (the pledge) — are the contracts formed by handing the thing itself over; hiring things and labor for payment lived in the family called *locatio*, home of rent and wages alike. The traditions found both untroubling: payment for the use of a durable thing, like payment for a day's work, buys something real that the seller genuinely gives up.

Look at why the rent is honest, because the logic matters downstream. The landlord who lets a house parts with its use but keeps its ownership — and ownership is not an ornament. The roof that fails, the beam that rots, the fire, the flood: the thing perishes to its owner, as the lawyers said, and the owner must maintain what the tenant merely enjoys. Rent is the price of a use the lessor keeps providing from an asset he keeps risking. Halakha permits renting out a courtyard for payment without a whisper of ribbit anxiety, for exactly this reason, while flatly refusing the “rental” of coins — money has no use separable from spending it, so rent on money is a fiction with arithmetic. The fiqh’s *ijara* is the same instrument given the same discipline: jurists define it as a sale of usufruct — a known use for a known price — and load the ownership burdens, maintenance and structural risk, onto the lessor, so that his income remains the fruit of something he still owns; Christendom’s usufruct and its untroubled centuries of land rent rest on the same floor. And chapter 25 will show what happens when the floor is sanded away — when a “lease” quietly transfers every burden of ownership to the user and keeps only the income, at which point it has become a secured loan in a landlord’s coat.

Side by side, the three exchange forms make a tidy little solar system. The sale transfers everything at once, for a price. The lease transfers use only, for a rent. The loan transfers ownership of a consumable now, against its equivalent later — and because what passes is consumed and what returns is merely equal, there is nothing left over for a price or a rent to honestly attach to. That is the whole geometry of the usury question, drawn in three lines.

The Venture in Common

The partnership begins where a lender’s nerve gives out. Instead of handing wealth across a counter and taking back a fixed claim, the partner commits wealth — or labor, or skill, or all three — into a shared venture, and agrees to be paid out of what happens rather than out of what was promised. Ownership is the hinge, as usual, but it swings the other way: the partner’s capital is never transferred to anyone. It remains his, working in the venture, exposed in the venture, and that continued exposure is precisely what entitles him to a share of the venture’s surplus. The three commitments arrive as a set — the venture shared, the risk shared, the surplus shared — and the traditions treated attempts to keep the third while shedding the second as void, or sinful, or both.

Two consequences give the form its character. First, a partner’s return is a residue, computed after the fact from a real outcome, which means it can be zero or worse; the law of partnership is therefore mostly a law of honest accounting — what counts as profit, who is paid before the split, how a loss is proven — and chapters 12 through 14 will find all three traditions writing that law in parallel, down to the shared insistence that the working partner’s labor is a real contribution that must be really paid. Second, losses do not wander. Money-loss falls on the money’s owner; the laborer loses his sweat and his hoped-for share; and no clause can quietly walk the investor’s loss across the table onto his partner’s side without dissolving the partnership into something else.

Every tradition blessed this form in its own instrument, and the instruments need only be announced here, like guests at a door: Rome and the Church had the *societas*; the rabbi engineered the *iska*; the jurists of Islam codified the *mudarabah* and its shared-capital sibling the *musharakah*; the merchants of the Mediterranean ran them all together into the *commenda* that financed the medieval boom. Part IV tells their stories, opening with a claim worth previewing in one line: partnership is the door all three traditions deliberately left

open in their wall against usury — the sanctioned way for capital to grow — and they policed it so carefully not because they distrusted it but because everything forbidden would try to enter dressed as it. Keep its silhouette in mind, because every costume in Part VIII is sewn to resemble it.

Money That Does Not Ask to Come Back

Every form so far creates a claim. The gift is the form that refuses to — a transfer with no counter-claim, no calendar, no residue except whatever gratitude does on its own. Waving it past as sentiment would be wrong: the traditions built as much law and infrastructure around non-reciprocal support as around any contract in this chapter, treating grace and mercy not as finance's opposite but as one of its layers — the layer that handles what no exchange can.

The vocabulary tells you how deliberately. The Hebrew word for almsgiving, *tzedakah*, is built on the root for justice: what flows to the poor is owed, a rectification, and calling it generosity flatters the giver too much — a point chapters 4 and 18 will develop. Christianity radicalized the gift until it colonized even the loan's vocabulary: "lend, expecting nothing in return," says Jesus in Luke's Gospel (6:35, NRSV) — an instruction that keeps the loan's outward form and guts its essence, since a loan stripped of enforceable expectation is a gift wearing a promissory note. Islam institutionalized the layer twice over: *zakat*, the obligatory annual levy on accumulated wealth whose very name carries the sense of purification — what remains is cleansed by what is given — and *sadaqah*, the voluntary giving stacked above it. And all three traditions, having built the gift, promptly built gift-adjacent lending: the Jewish free-loan tradition of *gemilut chasadim* that became the institutional *gemach*, and Islam's *qard hasan*, the "beautiful loan" made for God's sake and repaid without increase. Those institutions get Part V to themselves — chapters 15 and 16 — alongside chapter 18's almsgiving systems.

One rabbinic text belongs here rather than there, because it is a text about the taxonomy itself. When Maimonides codified the laws of giving, he ranked them — eight rungs — and the top rung is not the largest gift. The highest degree, above which there is none, he writes, is to strengthen the hand of a fellow who has faltered by giving him a gift, or a loan, or by forming a partnership with him, or by finding him work — until he no longer needs to ask (Mishneh Torah, Gifts to the Poor 10:7). Read that list again: gift, loan, partnership, employment. It is this chapter's inventory, redeployed as a ladder. The forms are not interchangeable, Maimonides is saying; they do different things to the person on the receiving end, and the best use of wealth is whichever form restores a man to standing rather than fastening him to your kindness. A taxonomy that can carry that judgment is moral instrumentation, not clerical bookkeeping. The gift has its own failure mode, too — a gift preceding or following a loan can be interest by other means, gratitude hardening into consideration — and chapter 8 patrols that border.

Stretch a gift across time until it outlives its giver and you have the endowment: property dedicated out of private consumption altogether, so that only its fruit ever circulates again. The charter story in Islam is worth telling whole, because the entire legal form hangs from one sentence of advice. Umar ibn al-Khattab had acquired land at Khaybar — the finest property, he said, that had ever come to him — and asked the Prophet what to do with it. The answer, as al-Bukhari's collection preserves it: if you wish, tie up its principal, and let its fruit go to charity. So Umar gave the land on terms that it never be sold, never be given away, never be inherited — its yield to feed the poor and the kinsman, free the

slave, host the traveler, its trustee free to eat from it reasonably and no more. That is the *waqf*: the asset locked out of the market in perpetuity, the income alone flowing. Judaism's cousin is *hekdes*, the dedication that once meant Temple property and came to mean the communal charitable trust; Christianity built the same shape in church lands and pious foundations, wrapped in canon law's deep instinct that certain property must be inalienable. The endowment is the taxonomy's strangest member — a form whose whole point is that the capital can never again be anyone's to extract — and chapter 29 will follow it from Khaybar and the *hekdes* to the modern asset lock and foundation-owned firm.

And then there is the form this book refuses to treat as a footnote: the solidarity pool.

On the ninth of June in the year 136 — the inscription dates itself by the consuls — the worshipers of Diana and Antinous, in the little Latium town of Lanuvium, carved their by-laws in marble in a temple porch. Entry to the association cost one hundred sesterces and an amphora of good wine; dues ran five asses a month, about the price of a few loaves. When a paid-up member died, three hundred sesterces came out of the common chest for his funeral, fifty of it distributed at the pyre. The rules are meticulous in the way only small money demands. Fall six months behind on dues, and your claim lapses even if your will provides for it. Die more than twenty miles from town, and three members must travel out to bury you, with travel money and quadruple penalties for fiddling the accounts. And if a member was a slave, and his master refused to release the body, the club buried an effigy — a funeral of the image, the by-laws call it, so that even a man whose corpse was withheld from his friends did not go unmourned. The membership was freedmen, slaves, the working poor of a provincial town: people for whom one death could mean a nameless pit, and who bought their way out of that fate by the month. The Latin has a euphemism of great tenderness for the event being provided against — the by-laws speak of the member to whom “the human thing” happens, *si ei humanitus acciderit*.

That is the solidarity pool, whole, in one inscription: many small, certain payments against one large, uncertain blow; shared protection against individual ruin. Its logic reverses the rest of the chapter: the exchange forms move value between parties each hoping to end up better off; the pool moves a catastrophe off one member and spreads it thin across all, so that nobody's single worst day is final. A pool is not a partnership — it shares the downside of its members' lives, never the upside of a venture; nobody at Lanuvium collected a dividend when the year's deaths came in light. Nor is it quite a sale, and the difference — protection pooled among members versus protection sold at a margin by someone who profits when claims stay low — is exactly the question chapter 27 asks of insurance and *takaful*, the *fiqh*'s instrument of mutual guarantee, born of Islam's broad duty of mutual assistance.

The rabbis, characteristically, had already legislated for it. A baraita at Bava Kamma 116b rules that the donkey drivers of a caravan may stipulate that if a member's donkey is lost — not through his negligence — the company will stand him another; the sailors may make the same compact, boat for boat. The exclusions are drawn with an actuary's eye, fifteen centuries before anyone was called an actuary. Lost the animal through carelessness, no donkey. Took your boat where boats do not go in that season — the Gemara specifies rope-lengths from shore, one in Nisan, two in Tishrei — no boat. And a driver who says “just give me the donkey's price in coin, and I'll keep guarding the caravan” is refused, because the sages understood incentives: a man made whole in donkeys walks on with the caravan, while a man made whole in silver has one foot pointed toward home. The same folio shares out a caravan's ransom by the money each traveler carries and a jettisoned cargo by weight — proportion tracking exposure, custom governing all of it. Christendom's version was

the confraternity and the guild box, dues against sickness, dowry, and burial, and their industrial-age descendants the friendly societies; the Jewish community ran the pattern through communal funds and guarantor networks. Chapter 2 ended with the community as a shield against isolated ruin; the pool is that shield with by-laws and a treasury.

Why a book about usury elevates the pool to a first-class form is a question chapter 28 answers with mathematics — why averages that look fine across a group can still grind an individual's path to zero, and how pooling changes the survival arithmetic of every path. Here it is enough to fix the vocabulary: ruin is the outcome from which there is no next season, and the pool is the contract-form the traditions built specifically against it. Hold the loan and the pool together one moment longer. Both move money toward a person in trouble. The loan does it while leaving the trouble entirely his, and adds a claim; the pool moves the trouble itself. The distance between those two designs is, in miniature, the distance this whole book intends to measure.

When the Name Lies

Seven names, so far, honestly worn. Now for the eighth, dishonestly built: the hybrid, which is what results when someone assembles two or more licit contracts so that the assembly reproduces the payoff of a forbidden one. Every joint is legal; the skeleton is a loan.

The classic specimen is the fiqh's own textbook case, the *'ina** sale. *Yusuf sells Musa a bolt of cloth for a hundred and twenty dirhams, payable in a year. Musa, in the same sitting, sells the same bolt back to Yusuf for a hundred in cash. The cloth ends the afternoon exactly where it began; what has actually moved is a hundred dirhams into Musa's hands today against a hundred and twenty owed next year. Two impeccable sales; one twenty-percent loan; no riba anywhere on the paper. The jurists' response is instructive twice over: the majority of the schools condemned the device as riba in costume, and the disagreement itself was a disagreement about this chapter's final rule — one Shafi'i line of analysis holding that a judge weighs only the contracts he can see, both valid sales, intentions being left to God; the other schools insisting that two sales engineered to work as one loan must be judged as what they do. The books mark the dispute as an honest ikhtilaf, and it has a precise Jewish twin. The rabbis, having defined the Torah's own prohibition as ribbit ketzutzah* — fixed interest, stipulated at the loan's birth — then fenced it with a category they called *avak ribbit*, "the dust of interest": arrangements that are not biblically forbidden lending but that shimmer with its residue, a deferred price padded above the cash price being the standard example. The dust category is rabbinic legislation, and the Talmud marks the boundary in a lawyer's way, by remedy — a court will claw back fixed interest, but the dust it leaves where it settled (Bava Metzia 61b). Christendom's casuists spent their energies on the same borderland — the redeemable sale, the rent purchased for cash, and above all the triple contract, three separately defensible agreements stacking into one guaranteed return — quarrels that belong to Part VIII. It says everything about the problem's permanence that when the Ottomans codified Hanafi law as the Majalla, its third article — third! — is a rule against being fooled by paperwork, and the drafters' chosen illustration is a hybrid: a sale subject to a right of redemption has the force of a pledge. The empire's civil code opens, in effect, by warning its judges that some sales are loans.*

The hybrid is where evasion begins — begins, because Part VIII watches it flower: the *heter iska* by which Jewish commerce restructured lending as partnership, a mechanism chapter 31 will weigh with real respect and real worry; the *sukuk* engineered until they behave like

the bonds they exist to replace, and the murabaha towers of chapter 32; the lease-to-own that amortizes like a mortgage; and modern finance's own menagerie — preferred stock with guaranteed exits, instruments whose term sheets read like partnership and pay like debt — awaiting the audit of Parts VII and IX. What every hybrid has in common, across centuries of paperwork, is the wager underneath: that the law will read the labels and not the payoff.

Against that wager, every tradition in this book eventually articulated the same defense, and it is the rule for the whole book: legal form must be tested against economic substance.

The test works by interrogation, and the questions are the taxonomy's own. Ask of any instrument where ownership actually sits once the signatures dry, and who is standing under the loss — whose money is exposed to the venture's failure, whose claim survives it untouched. Ask whether the return was written before the world spoke or discovered after it; what, exactly, the charge is a price of — a thing sold, a use provided, a risk genuinely borne, labor genuinely done; and what happens to each party on the venture's worst day. Answer those, and the instrument sorts itself into this chapter's vocabulary no matter what its cover page says. A "partnership" whose investor cannot lose is a loan. A "sale" whose object boomerangs by prearrangement is a loan. A "lease" whose tenant bears every burden of ownership is a loan. A "donation" that unlocks priority access, meanwhile, is a price. The names are hypotheses; the payoff is the evidence.

The traditions said it each in their own register. The fiqh compressed it into a maxim the Majalla codifies as article 3: in contracts, effect is given to intention and meaning, not to words and phrases. The rabbis rarely aphorized the principle; they enacted it, category by category — a legal literature that invents "the dust of interest" is a legal literature that has decided, permanently, to chase substance past the edge of form. The scholastics asked, of every clever instrument the fairs and counting-houses of Europe could generate, whether the contract was named truthfully — whether the thing calling itself a *societas* or a sale was, in the sight of God and the ledger, a *mutuum* with a wig. Even the secular law of our era, with no theology in it, rediscovered the test the moment its revenue depended on it: in 1935 the United States Supreme Court looked at Mrs. Evelyn Gregory's three-day-old corporation — legally flawless, created and dissolved in under a week to dress a taxable dividend as a tax-free reorganization — and disregarded it as "a mere device which put on the form of a corporate reorganization as a disguise for concealing its real character" (*Gregory v. Helvering*, 293 U.S. 465). Substance over form is now taught to every tax lawyer alive, most of whom have no idea they are reciting the oldest rule in moral finance.

For this book, the rule is not a conclusion but a method. Part III applies it to the loan and its disguised benefits; Part IV to partnerships that stop sharing; Part VI to every claimed justification for charging beyond principal; Part VII to the instruments — sale-based, lease-based, certificate, insurance — that live along the borders drawn here. Part VIII is the rule's tragedy, the modern era's long return of form over substance, and Part IX is its arithmetic: chapter 35 will rebuild the test as mathematics, holding legal form against economic payoff and requiring every euro of return to name the real thing it is a return for. When a later chapter calls an instrument honest or extractive, it will be this chapter's test doing the weighing.

Adam's naming, in both scriptures, was true naming — the creature answered to what it was called. Money never stopped requiring the same discipline. Every contract in this chapter is a name for what capital is doing in a human relationship: lent against tomorrow's equivalent, sold for a price, leased by the season, ventured in common, given without

claim, dedicated beyond recall, pooled against ruin. Whatever the paperwork says, one of those things — or the forbidden one — is actually happening underneath. The rest of this book is an audit of names.

Part II — Scriptural Fire: The Prohibition Before the Systems

Chapter 4 — The Hebrew Bible and the Economy of Covenant

The farms the Torah legislates for hang from hillsides. Ancient Israel's heartland was not river country like Egypt or Babylon, where a river delivered the water on schedule, but the central highlands of Canaan: a spine of terraced slopes where a household worked a few acres of barley, wheat, vines, and olives around a stone house and a plastered cistern. Deuteronomy itself dwells on the difference, contrasting the land of Egypt, watered by foot from its unfailing river, with a land that drinks only what heaven sends — "the rain for your land in its season, the early rain and the later rain" (Deuteronomy 11:14, quoted here and throughout in the New Revised Standard Version). The early rain softens the baked ground for plowing, in the weeks we would call October and November; the later rain, in spring, fills the ripening heads of grain. Between them lies the whole year's wager. Plow and sow when the first rains come; pray through the winter; cut barley in the spring and wheat a few weeks after; carry the sheaves to the threshing floor; then the vines, the figs, and last the olives, beaten from the trees as the year closes. Do everything right, and the household eats, seeds next year's field, pays its dues, and lays a little by. And every so often — no farmer needed telling how often — the rain comes thin, or late, or not at all.

The bad year is where biblical economics begins, because the bad year is where a neighbor becomes a debtor. Its anatomy was short and well known. A household that harvests too little must still eat through the winter and must still sow in the fall, and the seed it sows is the grain it did not eat; when there is not enough for both, someone in the village with full storage jars supplies the difference, to be repaid from the next harvest. In the wider region that transaction had a customary price — in Babylonia, as chapter 2 saw, a third again on barley was simply the going rate, written into Hammurabi's laws as casually as a weights-and-measures table. And the wider region also knew, and documented without embarrassment, where the price led when the next harvest failed too: the pledge taken, the field encumbered, the children bonded into a creditor's household. Israel's own memory carried the same sequence — the widow's cry to Elisha that a creditor had come for her two children stands in the book of Kings, where chapter 2 of this book already heard it — and Israel's law carries, as we will see, an entire clause-by-clause acquaintance with it.

What Israel refused was not the seed loan. A highland village without lending would simply have watched its weakest households die at the first drought, and the Torah, far from banning the loan, commands it with an insistence we will meet in its full strangeness below. What Israel refused was the price, and the cascade the price set in motion. The refusal is presented not as economic policy argued from consequences but as covenant — the terms of Israel's tenancy under a God who had brought the nation out of a slave economy and did not intend to watch it rebuild one from the inside. That framing is this chapter's subject. Part I of this book mapped where lending sat on the three traditions' shared moral map and sorted the contract vocabulary; Part II now reads each scripture at depth, in its own house and on its own terms — this chapter the Hebrew Bible, the next the New Testament, the one after the Qur'an. The Torah's house is the oldest of the three, and nearly every room the later traditions built — the ban and its boundary, the protected floor beneath the debtor, the scheduled mercy, the war on permanent ruin — appears here first, in a legal code with one eye always on the sky.

The Bite and the Increase

The Torah prohibits interest three times, in its three great legal collections, and the three texts read less like a statute repeated than like one conviction growing rings.

The oldest formulation, in the Covenant Code of Exodus, chapter 22 has already quoted; it belongs at the head of this chapter too, because everything else in the Torah's credit law is commentary on it. "If you lend money to my people, to the poor among you, you shall not deal with them as a creditor; you shall not exact interest from them" (Exodus 22:25). Two prohibitions, and neither is quite what a modern reader expects. The second bans the charge, and biblical Hebrew names the charge with the word this book met in chapter 1: *neshekh*, from the verb "to bite." Whatever etymological story lies furthest back, the working sense is the borrower's: interest as the wound in the repayment, the piece bitten out of a household each time it settles a debt it took on to survive. The first prohibition is stranger and deeper. Before the verse regulates any amount, it forbids a posture — do not be to him *as a creditor*: do not stand over this man the way the trade stands over its debtors, with the pledge list and the due date and the calm assumption that his need is your opportunity. The verse concedes that the money is owed but refuses to let the owing define the relationship. And the passage immediately shows what that means in practice, descending — as this law always descends — from principle to a poor man's evening: if his cloak is taken in pawn, "you shall restore it before the sun goes down; for it may be your neighbor's only clothing to use as cover; in what else shall that person sleep?" (22:26–27). The creditor holds the pledge by day and surrenders it every sunset, indefinitely, because the law has ranked a shivering body above a security interest. And the sanction attached is not a fine. If that borrower cries out, God says, "I will listen, for I am compassionate." The court of appeal for the poor man is the same God who heard Israel cry in Egypt, and the lender is on notice about how such appeals have gone before.

Leviticus, in the Holiness Code, widens the single night into a household's whole trajectory, and it is worth hearing the full text this time, because Part I quoted only its heart. "If any of your kin fall into difficulty and become dependent on you, you shall support them; they shall live with you as though resident aliens. Do not take interest in advance or otherwise make a profit from them, but fear your God; let them live with you. You shall not lend them your money at interest taken in advance, or provide them food at a profit" (Leviticus 25:35–37). The interest ban here sits inside a support command, and the grammar of the passage makes the support primary: the falling brother is to be caught and held upright — "you shall support them" — and the refusals that follow spell out what catching him must not become. The Hebrew doubles its vocabulary to close the gaps: *neshekh*, the bite, and *tarbit*, the increase — the second word built on the same root of swelling and growth that Arabic would later sound in *riba* — and the ban covers the money loan and the food loan alike, because in a subsistence village grain is lent as often as silver and hunger compounds faster than either. Note, too, the standard the brother is held to: he shall live with you "as though resident aliens," like the *ger*, the protected sojourner — the Torah's category for the person who has no land of his own and must therefore be guaranteed a standing the soil no longer gives him. A kinsman ruined is to be held the way the covenant holds its landless: close, upright, and uncharged. Then comes the reason, and it is the whole theology of this chapter in one verse: "I am the LORD your God, who brought you out of the land of Egypt, to give you the land of Canaan, to be your God" (25:38). Israel lends this way because Israel was rescued; the restraint is part of what the rescue now requires.

Deuteronomy states the law a third time, in the version that would echo loudest down the centuries, and chapter 22 has already walked its famous boundary — the brother and the

foreigner, the covenantal reading, and the long polemical afterlife that belongs to that discussion and to chapter 17 — so here we can attend to the law’s own internal architecture. “You shall not charge interest on loans to another Israelite, interest on money, interest on provisions, interest on anything that is lent. On loans to a foreigner you may charge interest, but on loans to another Israelite you may not charge interest, so that the LORD your God may bless you in all your undertakings in the land that you are about to enter and possess” (Deuteronomy 23:19–20; the Hebrew, which numbers the verses 20 and 21, says throughout simply *your brother*, as the NRSV’s own footnote records). Three features carry the weight. First, the scope: money, provisions, “anything that is lent” — the ban is total as to kind, closing in advance the oldest evasion, which is to shift the increase into some good the statute forgot to name. Second, the address: the law is not written about creditors in the abstract but to a lender facing his brother, the same second-person singular that runs through all three codes, so that no Israelite could hear the verse as regulating somebody else. Third, the strangest feature, the compensation clause: *so that the LORD your God may bless you*. The lender is not told that forgoing interest costs nothing. He is told where the return will come from instead. The increase the loan will not yield, the covenant promises “in all your undertakings” — a transfer of the lender’s yield from the borrower’s hide to God’s providence, which is either the least enforceable clause in the ancient Near East or the most, depending entirely on what one believes about the counterparty. Everything Deuteronomy does with debt runs on that wager, and we are about to watch it double the stakes.

What May Not Be Taken

Interest is only one way a loan can eat a household, and the Torah’s draftsmen knew the others. A loan at zero interest can still strip a debtor bare at the point of security — the pledge demanded up front, the goods seized at default — and so around the ban the law builds a second wall, a list of things a creditor’s hand may not touch. The passages that follow are among the most concrete in Scripture, and they should be read slowly, because each names an object, and each object is a life.

“No one shall take a mill or an upper millstone in pledge, for that would be taking a life in pledge” (Deuteronomy 24:6). The hand mill was two stones, and a household ground each day’s flour between them; hold either stone, and the family’s bread stops until the debt is cleared. The verse does not argue. It states an equation — the tool that makes tomorrow’s bread *is* the life — and by stating it, defines a category the rest of the section fills in: subsistence goods, the small machinery of staying alive, are not collateral. A claim secured on them is not security; it is a hostage-taking.

The pledge law then reaches the debtor’s door, and stops there. “When you make your neighbor a loan of any kind, you shall not go into the house to take the pledge. You shall wait outside, while the person to whom you are making the loan brings the pledge out to you” (24:10–11). The creditor’s claim is valid; the creditor’s entry is not. Inside the house the debtor is still master of his own poverty — he chooses what to bring out, and his wife and children are not made spectators to a search. It is the same instinct the cloak law showed in Exodus, now drawn as a floor plan: there is a line at the threshold that a debt, however genuine, does not cross. Deuteronomy then repeats the sundown rule in its own voice — “If the person is poor, you shall not sleep in the garment given you as the pledge. You shall give the pledge back by sunset, so that your neighbor may sleep in the cloak and bless you; and it will be to your credit before the LORD your God” (24:12–13) — and the bookkeeping image in that last clause deserves its moment. The lender surrenders

his security every evening and is told the surrender is *credited* — righteousness accruing to his account before God, in the very ledger idiom his forbearance has cost him. For the widow the compromise disappears altogether: “you shall not take a widow’s garment in pledge” (24:17) — not until sundown, not with her consent, not at all; where the borrower stands furthest from power the law simply removes the instrument. And in the adjoining verses the same clock that governs the pledge governs the paycheck: the poor laborer’s wages must be paid before sunset, “because they are poor and their livelihood depends on them” (24:14–15), with the same enforcement attached — they might cry to the LORD against you, and the LORD, as Exodus already warned, listens.

Set side by side, these rules describe a perimeter drawn around the debtor’s body and its needs: the bread apparatus, the night covering, the threshold, the day’s wage. Within the perimeter, the creditor’s rights go dark. The debt survives — nothing here cancels an obligation — but the law has decided, in advance and in detail, which assets a bad year is allowed to reach, and the list of what it may not reach is a list of the things that keep a poor household from tipping over the edge it is already standing on. The Mesopotamian archives showed us what unrestrained security practice looks like: field, house, persons. The Torah walks the same inventory in the opposite direction, unpledging item by item whatever life itself requires.

The Seventh Year

The oldest sabbatical law in the Torah is agricultural. “For six years you shall sow your land and gather in its yield; but the seventh year you shall let it rest and lie fallow, so that the poor of your people may eat; and what they leave the wild animals may eat. You shall do the same with your vineyard, and with your olive orchard” (Exodus 23:10–11). Farmers everywhere rest fields; only here is the rest assigned a beneficiary. For one year in seven the fences go down, in effect, and whatever the untended land volunteers belongs first to those who own no land at all — the field’s produce redistributed not by harvest but by access. It is a small law with a large premise: the owner’s claim on his own soil has a rhythm, and in the rhythm’s seventh beat the poor hold the senior right.

Deuteronomy takes that seventh beat and strikes it on the ledger. “Every seventh year you shall grant a remission of debts” (Deuteronomy 15:1) — the release the tradition calls *shmita*. The mechanics follow at once: in that year every creditor “shall remit the claim that is held against a neighbor,” and shall not exact it, “because the LORD’s remission has been proclaimed” (15:2). The passive is exact. The creditor does not grant the release so much as join one already declared over his head; the debt is not his to keep pressing, because the claim, like the land under the fallow law, has an owner senior to him. (Toward the outside trader the rule once again tracks the covenant’s edge — “Of a foreigner you may exact it” (15:3) — the same boundary Deuteronomy 23 drew, and drawn here with the same commercial logic: the release is a solidarity of the household of Israel, and the passing merchant is in the household’s economy but not its family.) Israel’s ledger was commanded to forget, on a schedule, what every other ledger in the ancient world was built to remember forever. The clean-slate edicts of Babylon, which chapter 2 visited, did something similar at a king’s discretion, when crisis or coronation moved him; Deuteronomy removes the discretion and fixes the date, so that release stops being an emergency measure and becomes a property of time itself. Debt in Israel comes with an expiration date stamped at issue. No claim can compound past the seventh year; no bad season can be farmed into a permanent revenue stream; the machine this book’s second chapter described — the one that turns a neighbor’s crop failure into his servitude — is interrupted

by statute before it completes a single decade of running.

The law then does something almost unnervingly clear-eyed: it predicts its own evasion. A lender who knows the remission is coming will simply stop lending as the year approaches — why issue a claim that is about to dissolve? — and Deuteronomy sees him coming: “Be careful that you do not entertain a mean thought, thinking, ‘The seventh year, the year of remission, is near,’ and therefore view your needy neighbor with hostility and give nothing; your neighbor might cry to the LORD against you, and you would incur guilt” (15:9). The command that follows is the Torah’s lending ethic at full strength: “do not be hard-hearted or tight-fisted toward your needy neighbor. You should rather open your hand, willingly lending enough to meet the need, whatever it may be” (15:7–8); “Give liberally and be ungrudging when you do so” (15:10). Read as finance, this is a command to extend credit into a scheduled writedown — to lend, at zero interest, money the calendar may be about to erase. Read as covenant, it is Deuteronomy 23’s wager doubled: the lender is asked to trust the blessing further than any spreadsheet will follow it. The same chapter holds the two sentences that mark how honestly the law knew its terrain: “There will, however, be no one in need among you” if the commandment is kept entire (15:4), and, seven verses later, “Since there will never cease to be some in need on the earth, I therefore command you, ‘Open your hand to the poor and needy neighbor in your land’ ” (15:11). Promise and prognosis, both canonical: the law aims at a community without poverty and legislates for the one that will actually exist.

The prediction of verse 9, it must be said, eventually came true at scale. In the last decades before the Common Era, Hillel the Elder found lenders refusing the poor as the seventh year approached — precisely the transgression Deuteronomy had named — and enacted the instrument the Mishnah records in tractate Shevi’it (10:3): the *prozbul*, a declaration lodging the creditor’s claims with the court, which — so the tradition explains the device — the remission, addressed to the individual creditor, did not reach. A loan under *prozbul* survives the seventh year; the Mishnah itself explains the enactment by quoting the mean-thought verse, presenting Hillel’s device as a rescue of the law’s purpose — keep credit flowing to the poor — at the price of its letter. Whether such engineering rescues a law or retires it is one of the permanent questions of religious finance, and this book will ask it in earnest when the modern descendants of Hillel’s move, above all the *heter iska*, come under examination in Part VIII. Here it is enough to notice what the episode proves about the original: the remission was real enough to bend the credit market, real enough to need engineering around. Nobody builds a bypass for a road that carries no traffic.

The Land Is Mine

Seven sabbatical cycles build to the Torah’s most sweeping economic institution. “You shall count off seven weeks of years, seven times seven years,” Leviticus instructs, and then, on the Day of Atonement of the forty-ninth year, sound the ram’s horn everywhere at once — the most consequential trumpet blast in the history of property law: “you shall hallow the fiftieth year and you shall proclaim liberty throughout the land to all its inhabitants. It shall be a jubilee for you: you shall return, every one of you, to your property and every one of you to your family” (Leviticus 25:8–10). The Hebrew name of the year, *yovel*, is usually traced to that horn; the word for the liberty it proclaims is *deror* — the term in which scholars hear the old Akkadian *andurarum*, the clean-slate proclamation of Mesopotamian kings, taken out of the king’s mouth and fixed, like the *shmita*, into the calendar. (American readers have met the verse without knowing its address: it is the line cast on the Liberty Bell.) In the fiftieth year, land lies fallow as in the seventh; and every

piece of ancestral ground that hardship had forced out of a family's hands goes home. "In this year of jubilee you shall return, every one of you, to your property" (25:13).

The mechanism underneath is stated with a bluntness theology rarely manages: "The land shall not be sold in perpetuity, for the land is mine; with me you are but aliens and tenants" (25:23). Israel does not own the promised land; Israel holds it — every household a *ger* and tenant before the true landlord, each on its allotted ground. From that premise the whole institution follows as lease law. A family in a bad year does not sell its field; it sells the field's remaining harvests until jubilee, which is why the price must be prorated to the years left on the clock, and why the Torah — doing present-value arithmetic long before anyone had a name for it — provides that a seller who recovers his means may buy the land back early by refunding the unused years: "the years since its sale shall be computed and the difference shall be refunded to the person to whom it was sold, and the property shall be returned" (25:27). Failing kin with means to redeem it — the law drafts the family, nearest first, as the default rescue — the land simply waits: "in the jubilee it shall be released, and the property shall be returned" (25:28). What can never happen, whatever the paperwork says, is the thing Leviticus names in its opening clause: sale in perpetuity. Permanent alienation of a household's ground is not forbidden to the parties so much as void against the landlord, like a tenant purporting to sell the building.

The same architecture then reaches the darkest scenario the highlands knew, and the text goes where the creditors' archives of every neighboring civilization went — to the sale of persons — in order to countermand it. "If any who are dependent on you become so impoverished that they sell themselves to you, you shall not make them serve as slaves. They shall remain with you as hired or bound laborers," and only until the jubilee: "Then they and their children with them shall be free from your authority; they shall go back to their own family and return to their ancestral property. For they are my servants, whom I brought out of the land of Egypt; they shall not be sold as slaves are sold. You shall not rule over them with harshness, but shall fear your God" (25:39–43). The reasoning is the land's reasoning applied to bodies: Israelites are already owned, by the God who bought them out of Egypt, and a prior title voids the later sale. Destitution, under this law, can rent out a household's labor for a term, but the person stays beyond purchase. The law does not pretend the desperate sale will never happen — it happened in every subsistence economy on earth — but it converts the transaction, by force of statute, into bounded wage service with a guaranteed exit for the worker and his children together. What the jubilee abolishes is not misfortune but misfortune's compounding: the bad year may cost a family its harvests, its comfort, even its liberty for a term — and then the horn sounds, and the fiftieth year hands the next generation back its land and its labor, and the ruin stops inheriting.

Whether the fiftieth year ever operated at full scale is a question the sources leave open — the tradition itself records the jubilee's lapse, and historians debate how much of Leviticus 25 describes practice and how much prescribes vision. The prophets suggest the machine it was aimed against ran hard regardless: Amos, in the eighth century, cries against those who "trample on the needy," who shrink the measure and inflate the price and rig the scales, "buying the poor for silver and the needy for a pair of sandals" (Amos 8:4–6) — grain fraud and debt bondage in one breath, the exact cascade the jubilee was built to unwind. And the countervision the prophets reach for is the jubilee's own: every household on its own ground, "under their own vines and under their own fig trees, and no one shall make them afraid" (Micah 4:4). Between those two verses — the poor sold for sandals, the farmer secure under his fig tree — runs the entire argument of the Hebrew Bible's

economics. A community answering to the God of the exodus is one in which the second picture is law and the first is sacrilege; and the jubilee, practiced or not, stood in Israel's canon as the standing declaration that no market outcome, however legally arrived at, may permanently overwrite a family's place in the land.

Justice With a Ledger

It remains to name the drawer in which the Torah files all of this, because the filing is itself the finding. The word the tradition settled on for material support of the poor — the commanded loan, the gleanings, the tithe for the widow and orphan, the open hand — is *tzedakah*, and as chapter 3 noted in passing, the tradition hears in that noun its root *tzedek*: justice, rightness, the quality Deuteronomy elsewhere commands Israel to pursue and nothing but. The vocabulary is a verdict. What the poor receive under these laws is not, in the Torah's own grammar, a kindness that reflects well on the giver; it is a debt the giver owed all along, collected. Later languages would file the same acts under other headings — the Latin *caritas*, from love, gives Christian Europe its “charity,” and chapter 5 will watch what that framing beautifully demands and quietly risks — but the Hebrew filing has a hard clarity worth stating on its own terms: a poor man asking bread from a full storehouse is not begging against the owner's rights but presenting a claim the covenant has already validated.

The premise underneath is Leviticus 25:23. If the land is God's and every Israelite a tenant, then the tenancy has terms, and the terms include the poor. Read from inside that sentence, the gleanings at the field's edge are not the landowner's generosity; they are his lease terms. The corner of the field left standing, the forgotten sheaf left lying, the fallow year's open access, the third-year tithe, the sunset return of the pledge, the seventh-year remission, the interest never charged — all of it, in the Torah's accounting, is simply what occupancy of God's land costs, the poor's share written into the master lease before any human owner signed. This is why the Torah can do what pure benevolence never can: legislate the matter, audit it, attach dates and quantities and court-enforceable procedures. Compassion is commanded in these texts, vividly — the hard heart and the closed hand are sins of feeling before they are sins of accounting — but compassion is not left to carry the system. The system is carried by obligation, scheduled and enumerated, precisely so that the poor household's survival never depends on whether its richer neighbor happens to be feeling generous in a drought. Israel's God, the texts keep saying, listens to the cry of the poor; Israel's law aims to make the cry rarely necessary.

Everything this book will later call Jewish anti-usury finance grows from the ground this chapter has walked, and the growing is a story other chapters own; here at the close it is enough to point at the shoots. The commanded free loan of Deuteronomy 15 becomes, in rabbinic hands and then in every Jewish community on earth, the institution of *gemilut chasadim* — deeds of loving-kindness — and its financial arm the *gemach*, the free-loan fund whose modern career chapter 15 follows. The three interest texts become *ribbit* law, a jurisprudence of extraordinary reach and subtlety — the rabbis will chase the forbidden increase into gifts, favors, and even words — whose architecture chapters 7 and 8 lay out. The instinct against exploiting a counterparty's blindness or need becomes *ona'ah*, the overreaching doctrine with its one-sixth threshold, waiting in chapter 9. And the deepest pattern of all — the community organizing itself so that no single household stands alone under a failing sky, the pattern the fallow year and the remission and the jubilee all express in calendar form — becomes the communal risk-sharing whose institutions and whose startling mathematics chapters 27 and 28 take up. Signposts only; each will be met

in its own country.

What Part II must carry forward is the shape of the whole. The Hebrew Bible does not give later tradition a theory of money — Aristotle will supply that, to be grafted on centuries afterward. It gives something prior: a picture of the economy as covenant, in which the lender and the borrower stand before the same God on the same rescued ground, and in which, therefore, a neighbor's need lays a claim on the lender and never pays him a yield. Around that conviction it builds the four defenses this chapter has toured — the banned increase, the protected floor, the scheduled release, the unsellable future — and the two scriptures that follow will each inherit the conviction and rebuild the defenses in their own idiom: a Galilean teacher pressing the open hand past every boundary Israel had drawn around it, and a prophet in Mecca declaring God's war on the swelling of wealth through a neighbor's need. Neither of the next two chapters leaves this one behind. The fire in Part II's title was lit here, on a terraced hillside, in the space between a full storehouse and an empty one, by a law that looked at the oldest transaction in civilization and asked, before anything else, what it was doing to the man at the door.

Chapter 5 — The New Testament and the Radicalization of Mercy

The most familiar prayer in the world contains, at its center, a sentence about debt. “And forgive us our debts, as we also have forgiven our debtors,” runs Matthew’s version (Matthew 6:12) — biblical quotations here and throughout follow the New Revised Standard Version — and the wording is not a translator’s flourish. Matthew’s noun is *opheilema*, the ordinary Greek business word for a sum owed; the “debtors” of the second clause are *opheiletai*, the people on the wrong side of your ledger. Luke’s version of the prayer, taught on another occasion and shorter by two petitions, shifts the first term and keeps the second: “And forgive us our sins, for we ourselves forgive everyone indebted to us” (Luke 11:4). Sins forgiven, debtors released, in one breath, as though the two were a single transaction described from two sides. Scholars have long observed that in the Aramaic Jesus spoke, they very nearly were: one word, *hoba*, meant both the debt you owed and the wrong you had done, so that Matthew and Luke may simply be translating the same petition into the two halves of its own meaning.

It would be a mistake to file this under metaphor, as though “debt” were merely a vivid way of saying “guilt” for people who had never seen a balance sheet. The people who first prayed this prayer had seen little else. First-century Galilee and Judea ran on the machinery chapter 2 walked through: smallholders a bad harvest from dependency, land passing to creditors, tenancy where ownership had been, the tax on top of the rent on top of the loan. How combustible the paperwork had become was demonstrated within a generation of the crucifixion. In the summer of 66, when Jerusalem rose against Rome, one of the insurgents’ first acts — Josephus records it with a historian’s cold eye at *Jewish War* 2.427 — was to carry fire to the public archives and burn the money-lenders’ bonds, precisely to make the recovery of debts impossible and to bring the indebted poor over to the revolt. In that world a prayer about the remission of debts was about as figurative as a prayer about bread, which is to say: it stood one line away from one.

So the Lord’s Prayer asks God for the thing a Galilean debtor most needed and least expected, and then does something no clean-slate edict ever did. It makes the petitioner sign both sides. “As we also have forgiven our debtors”: the request for release is yoked to a report of release already granted, and Matthew’s Jesus, as though anticipating that worshipers would mumble that clause, appends a rider in plain prose: “For if you forgive others their trespasses, your heavenly Father will also forgive you; but if you do not forgive others, neither will your Father forgive your trespasses” (Matthew 6:14–15). Babylon’s kings had canceled debts from above, occasionally, when the countryside was near collapse. This prayer conscripts every person who says it into being a small, daily clean-slate decree — and it polices the conscription. No one is permitted to pray it as a debtor while collecting as a creditor.

That yoking is the engine of this whole chapter, and it is worth saying at the outset what kind of chapter this is. The New Testament contains no lending code. There is no rate cap in the Gospels, no pledge law, no contract taxonomy — nothing resembling the legal architecture chapter 4 found in the Torah or the one chapter 6 will find growing from the Qur’an. What the New Testament does instead is take the Torah’s mercies and remove their ceilings, one by one: the interest ban radicalized into lending without return; the release

years radicalized into forgiveness as a way of life; restitution radicalized into a tax collector giving away half his goods; the community's duty to its poor radicalized into a community with no poor in it. Nothing in what follows steps outside Israel's orbit. Every text below intensifies something Israel's law already commanded — which is exactly what made the result so hard to operate, and why this chapter ends with a problem where the others end with a system.

Even Sinners Lend to Sinners

Begin where chapter 2 left off, on the plain in Luke where Jesus preaches to a crowd that knew the borrower's side of the ledger. The Torah had commanded the open hand and forbidden the bite of interest; the sermon quotes neither law and outbids both. "If you lend to those from whom you hope to receive, what credit is that to you? Even sinners lend to sinners, to receive as much again. But love your enemies, do good, and lend, expecting nothing in return. Your reward will be great, and you will be children of the Most High; for he is kind to the ungrateful and the wicked" (Luke 6:34–35). Matthew's parallel is briefer and aimed at the other party's need rather than the lender's motive: "Give to everyone who begs from you, and do not refuse anyone who wants to borrow from you" (Matthew 5:42).

Look first at what the saying concedes. Reciprocal lending works. Sinners run it profitably among themselves, Jesus observes, and he does not dispute the observation; mutual credit between people who expect repayment is the oldest credit union in the world, and it clears. What he denies is that there is any *credit* in it — and Luke's Greek makes the denial a small pun that English cannot keep. The word rendered "credit" is *charis*: grace. What grace is there, the sentence asks, in a symmetry? Reciprocity is bookkeeping between equals; each party's kindness is priced and covered by the other's. Grace begins where the expectation of return stops, and the sermon locates God's own economy exactly there: he is kind to the ungrateful, rains on the unjust, gives where nothing can come back. To lend "expecting nothing in return" is therefore not presented as a heroic add-on to ordinary ethics. It is presented as the family trait — what the children of *this* Father look like when they hold money and a neighbor lacks it.

Measure the distance traveled from the law to this. Deuteronomy had already commanded the Israelite to lend to the poor brother and forbidden any increase; the rabbis, as chapter 2 saw, ranked the interest-free loan among the highest duties. Jesus stands on that ground and removes the last instrument of self-protection: the expectation itself. The Torah's lender gives up his yield and keeps his claim to the principal; the sermon's lender is told to stop calibrating the loan to its recovery at all. Whether "expecting nothing in return" means surrendering even the principal — whether the loan is to become, at the limit, a gift — readers have argued from the Fathers onward, and the church's lawyers would eventually spend centuries sorting counsel from command. But the direction of the saying is not in doubt. Chapter 1 identified the claim as the one function of money that wears a face and ranks two people for as long as it stands. This teaching dissolves the rank. A lender who expects nothing holds no one beneath him; the loan has been converted from a claim into a kindness, and the borrower at the door — enemy included, for the sentence begins "love your enemies" — walks away owing gratitude and nothing else. Chapter 2 traced how the Samaritan story unfenced the *who* of the command. This is the corresponding surgery on the *what*: mercy without a boundary, now also without a meter.

An Economy of Mercy

The Gospels' word for what happens next is a fiscal word. When Luke's Jesus opens his ministry in the Nazareth synagogue, he reads from the scroll of Isaiah a commission "to proclaim release to the captives" and "to let the oppressed go free," ending in "the year of the Lord's favor" (Luke 4:18–19) — and the noun for release there, *aphesis*, is a term with a paper trail. It is the word the Greek translation of the Torah had used for the seventh-year remission of debts in Deuteronomy 15 and, again and again through Leviticus 25, for the Jubilee liberty itself; and it is, at the same time, the New Testament's ordinary noun for the forgiveness of sins. One word covers the canceled bond, the freed bondsman, and the absolved sinner. English separates "remission" from "forgiveness" and loses the seam; the Gospel's announcement runs along it. Jesus's first sermon, in Luke's telling, proclaims something like a Jubilee — hard to hear otherwise, with Isaiah's release-language in his mouth — and the forgiveness of sins that the church would come to treat as its central business is named, grammatically, as the same act: a release, an *aphesis*, a debt struck from the book.

Luke shows the logic at dinner-party scale first. Challenged by a host who doubts that a sinful woman should be touching him, Jesus answers with a two-line parable: "A certain creditor had two debtors; one owed five hundred denarii, and the other fifty. When they could not pay, he canceled the debts for both of them. Now which of them will love him more?" Simon answers, "I suppose the one for whom he canceled the greater debt," and is told, "You have judged rightly" (Luke 7:41–43). Notice what the parable takes for granted: that cancellation creates something. Not a void — a bond. The creditor who tears up the note does not end the relationship; he transforms it, and the currency it runs on afterward is love in proportion to what was released. Forgiveness, in this picture, is not the absence of a transaction but a transaction of a higher order.

Then Matthew stages the full macroeconomics, in the parable that stands as the Lord's Prayer's shadow side. Peter asks how often he must forgive — "As many as seven times?" — and is told, "Not seven times, but, I tell you, seventy-seven times" (Matthew 18:21–22), and then comes the story. A king settles accounts with his slaves; one is brought in owing ten thousand talents. The sum is deliberately unpayable — a *talanton* was the largest unit of account in the Greek east, and the NRSV's own note reckons a single talent at more than fifteen years of a laborer's wages, which prices this debt at something over a hundred and fifty thousand years of work. Then comes the sentence: "as he could not pay, his lord ordered him to be sold, together with his wife and children and all his possessions, and payment to be made" (18:25). That sentence was not invented for effect. It is the machine every ancient reader knew — the widow of 2 Kings crying out as the creditor comes for her children — here operating at royal scale on a man whose debt no sale could ever cover. The slave begs for time, absurdly ("I will pay you everything"). And the lord does the thing the whole ancient Near East reserved for kings in their mercy: "out of pity for him, the lord of that slave released him and forgave him the debt" (18:27).

Then the freed man goes out, finds a fellow slave who owes him a hundred denarii — a hundred days' wages, real money in a village, nothing against what he has just been spared — and the narrative slows down to watch: "seizing him by the throat, he said, 'Pay what you owe'" (18:28). The plea he hears is his own plea, word for word. He refuses it, and jails the man. When the lord learns of it, the release is revoked in language that gives this chapter its center of gravity: "Should you not have had mercy on your fellow slave, as I had mercy on you?" (18:33) — and the story ends without softening, the man handed over to the torturers, and Jesus pressing the moral onto every listener: "So my heavenly

Father will also do to every one of you, if you do not forgive your brother or sister from your heart" (18:35).

The parable is the prayer, dramatized — the man who asked as a debtor and collected as a creditor — and it fixes the sense in which the Gospels have an economics at all. Mercy, in this teaching, is not issued as a private settlement between the soul and God. It is issued as circulating currency. The forgiven debt is capital placed in the debtor's hands with an implicit mandate: it must move; it exists to be re-spent on the next debtor down the line; and hoarded mercy — received upward, refused downward — is voided retroactively, as though it had never been issued. This is why the tradition's own name for the theme, the economy of mercy, is more than a preacher's ornament. The parable describes a genuine circulation with a genuine failure mode, and its rule inverts the miser's: in this economy the one way to keep forgiveness is to spend it.

From Jericho to the Temple

Luke's Gospel carries its lending theology up the Jericho road, and the last week of Jesus's life gives this book two scenes that every later Christian argument about money would cite. They belong together, and Luke in fact tells them nearly together: a private repair in Jericho, a public boundary in Jerusalem, fifteen miles and a few days apart.

Jericho first. The city sat on a customs road, and at the top of its revenue system perched Zacchaeus — "a chief tax collector," Luke specifies, "and... rich" (Luke 19:2). The tax farming of the Roman east worked by auction: the right to collect a district's tolls was sold, and the collector's income was the spread between what he remitted and what he could extract, a design that made overcharging not a temptation of the job but approximately the job itself. Which is why the crowd's verdict, when Jesus invites himself to the man's house — "Zacchaeus, hurry and come down; for I must stay at your house today" (19:5) — is unanimous and unkind: "He has gone to be the guest of one who is a sinner" (19:7). The scene has its comedy, a rich official up a sycamore tree like a boy; what it builds to is not comic at all. Zacchaeus stands and makes a declaration with numbers in it: "Look, half of my possessions, Lord, I will give to the poor; and if I have defrauded anyone of anything, I will pay back four times as much" (19:8).

The four is not a round number pulled from the air; it is a statute. The Torah's restitution law required a thief who had slaughtered or sold a stolen sheep to "pay... four sheep for a sheep" (Exodus 22:1), and when King David, trapped by Nathan's parable, condemned the rich man who took the poor man's one lamb, he sentenced by the same tariff: "he shall restore the lamb fourfold, because he did this thing, and because he had no pity" (2 Samuel 12:5–6). A Roman ear could hear the figure too: Cato, as chapter 1 noted, boasted that the ancestors' laws had condemned the thief in double and the moneylender in fourfold. By that arithmetic Zacchaeus reads his own occupation exactly. He does not offer charity in the amount of his comfort; he sentences himself under the thief's statute, at the sheep-stealer's rate, to whomever he has bled — and on top of the fourfold, half of everything to the poor. (Whether his Greek verbs announce a resolution or describe a practice already begun, commentators divide; the accounting stands either way.) Jesus's response treats the declaration not as the price of admission but as the evidence of arrival: "Today salvation has come to this house, because he too is a son of Abraham. For the Son of Man came to seek out and to save the lost" (19:9–10). The sentence restores in both directions at once — the defrauded get their money back with damages, and the defrauder gets his kinship back, *son of Abraham*, the brotherhood his trade had cost him. Here is the Gospel's teaching on

unjust gain in one scene: such gain is not laundered by generosity in general, nor dissolved by regret; it is *repaired*, at the wronged party's address, with the law's own multiplier, and the repair is what salvation looks like when it walks into a counting-house.

Then Jerusalem, and the scene that has been argued over ever since. In the temple's outer court, in the crush of Passover pilgrimage, Jesus "drove out all who were selling and buying in the temple, and he overturned the tables of the money changers and the seats of those who sold doves" (Matthew 21:12). To read the scene rightly, start with what the tables were for, because it was not fraud by design. The Torah set a half-shekel due on every adult male for the sanctuary's upkeep, and pilgrims arrived carrying every coinage of the eastern Mediterranean; the dues were payable in acceptable silver — in practice, the standard account runs, the high-grade shekels of Tyre — so somebody had to change money, openly, at published rates. The Mishnah, legislating for this very machinery, gives the timetable: from the fifteenth of Adar the changers' tables were set up in the provinces, and from the twenty-fifth they sat in the Temple itself (Shekalim 1:3), and it even fixes their surcharge, the *kalbon*, at a small silver coin the Sages haggled down by half (Shekalim 1:6–7). The dove sellers likewise served the law's own mercy: doves were the offering the Torah allowed the poor, who could not afford a lamb. The men at the tables were providing a lawful, necessary, regulated pilgrimage service, and the Greek of the Gospels names them by their furniture — *trapezitai*, table-men, from *trapeza*, the table that was antiquity's bank counter.

The protest, then, was not a raid on crooked bookkeeping, and the Gospels do not frame it as one; they frame it — through the scriptures Jesus quotes — as a prophet's claim about what the house is *for*. "It is written, 'My house shall be called a house of prayer'; but you are making it a den of robbers" (Matthew 21:13). Both halves are Israel's own prophets. The first is Isaiah's vision of the temple opened outward — "for my house shall be called a house of prayer for all peoples" (Isaiah 56:7); Mark, quoting one clause more, keeps that ending, "for all the nations" (Mark 11:17), and adds that Jesus "would not allow anyone to carry anything through the temple" (11:16), shutting down the court's use as a commercial shortcut. The second is Jeremiah, thundering at the first temple's door: "Has this house, which is called by my name, become a den of robbers in your sight? You know, I too am watching, says the Lord" (Jeremiah 7:11) — and Jeremiah's image deserves its precision. A robbers' den is not where robbery happens; it is where robbers retire to feel safe after working elsewhere. The charge is not that the changers' spread was extortionate but that the whole apparatus of sacred commerce had become a refuge — piety as the safe-house of an economy whose injustices were committed outside, then absolved within. John's Gospel, placing its version at the ministry's opening rather than its close, sharpens the point to a single boundary-word: coins poured out, tables overturned, "Stop making my Father's house a marketplace!" (John 2:16).

Read from within Israel, as it should be, this is a Jewish prophet enacting a sign in his own sanctuary, with Israel's own scriptures on his lips, in the direct line of Jeremiah at the same gates — a protest about the *location* of commerce, not the existence of commerce, insisting that the one court where God met all peoples, the poor and the foreigner included, must not run first as a market with prayer squeezed in at the edges. One dark afterlife has to be flagged and refused in a sentence: later Christian centuries would twist this family scene into an indictment of Jews as such — the money changers redrawn as "the Jewish usurer" writ large — a slander that belongs to its makers, not to the text, and that this book confronts where it did its damage, in chapter 17 and Part VIII. In its own house the scene pairs with Jericho as the two gestures of one teaching. Zacchaeus repairs gain that was

unjust; the temple action bounds gain that was lawful. Money extracted wrongly must go back fourfold, and money earned rightly must know which spaces and purposes stand above it. Together the scenes give the Gospel's stance on commerce its two gestures — repair of unjust gain, a boundary around lawful gain — a week apart on the same road.

The Talents and the Bankers

And yet the New Testament's most famous money story seems, at first hearing, to run the other way — to scold a man for *failing* to seek interest. The parable of the talents has been quoted at this book's argument for centuries, so it must be read here slowly and fairly.

Matthew places it on the Mount of Olives, in Jesus's last discourse before the arrest, between the bridesmaids who miss the wedding and the sheep and goats of the last judgment — a run of stories about what the household of an absent master should be doing when the master seems delayed. "For it is as if a man, going on a journey, summoned his slaves and entrusted his property to them; to one he gave five talents, to another two, to another one, to each according to his ability. Then he went away" (Matthew 25:14–15). The sums, again, are enormous — years of wages in a single *talanton*, on the NRSV's own reckoning — and the first two slaves trade with the money and double it, and are received identically: "Well done, good and trustworthy slave... enter into the joy of your master" (25:21, 23). The third, who received one talent, "dug a hole in the ground and hid his master's money" (25:18), and his accounting, when it comes, is a small masterpiece of self-exculpation: "Master, I knew that you were a harsh man, reaping where you did not sow, and gathering where you did not scatter seed; so I was afraid, and I went and hid your talent in the ground. Here you have what is yours" (25:24–25). The master turns the speech back on its maker — "You wicked and lazy slave! You knew, did you, that I reap where I did not sow, and gather where I did not scatter?" — and then delivers the line that echoes through the usury debates: "Then you ought to have invested my money with the bankers, and on my return I would have received what was my own with interest" (25:26–27). Luke tells a variant with a nobleman, ten slaves, and ten minas — pocket change by comparison, three months' wages each, per the NRSV's note — with the same hinge: "Why then did you not put my money into the bank? Then when I returned, I could have collected it with interest?" (Luke 19:23).

State the opposing reading honestly, because it is not stupid. Here is Jesus, in a parable where the master transparently figures God, treating deposit at interest as the *floor* of faithfulness — the least a timid man could have done and been spared. The Greek says it without blushing: the money should have gone to the *trapezitai*, the table-men, and come back *syn toko*, with *tokos* — with offspring, the exact noun Aristotle had held up (chapter 1) as the name of money's unnatural breeding. If interest were sin root and branch, the argument runs, the parable could hardly use it as the benchmark of minimal diligence; and when Christians in early modernity began dismantling the usury ban, this verse was reliably produced as Scripture's own smile upon the bank (a story that belongs to chapter 30). Three things, said in order, answer it.

First, the parable's fictional employers are not moral exemplars, and the Gospels' parables never treat them as such. Luke's Jesus builds teaching around a dishonest manager whose shrewdness is commended, and an unjust judge who grants a widow's case to stop her pestering; nobody concludes that fraud or judicial contempt are recommended. The master of the talents is described, inside the story, as "a harsh man, reaping where you did not sow" — a profile of an extractor, harvesting other men's fields, that the Torah-formed

listener was not being asked to admire — and the master himself repeats the description only conditionally, as a blade turned against the speaker: *you knew, did you?* If that is truly what you believed of me, your own logic convicts you — even a coward who thought his master a predator had a lazier option than a hole in the ground. The bankers' line is an argument *a fortiori* inside the slave's own picture of his master, not the storyteller's disclosure of God's investment policy. Building a doctrine of interest on it is like building a doctrine of burglary on the Lord who comes as a thief in the night.

Second, look at what the story actually condemns, because the condemnation, not the banking, is its point. The third slave's sin is fear-driven burial of an entrusted gift. He is the only character who treats the capital as a static object to be preserved against an audit rather than a trust to be ventured; his own verdict on himself is "I was afraid"; and the parable's context — between waiting bridesmaids and a judgment scene in which the saved turn out to be those who fed, clothed, and visited — makes its subject unmistakably the use of what God entrusts during the master's apparent absence. The early church read it exactly so. Chrysostom, preaching through Matthew at the end of the fourth century, took the talents to be "each person's ability — whether protection, or money, or teaching, or anything of the kind" (Homily 78 on Matthew), and read the whole block of parables as instruction in mercy and almsgiving; the Latin homilists handled the coins the same way, as gifts of grace to be spent on neighbors, with the "interest" God seeks being the good the gift does when it circulates. That reading left its fossil in our own speech: English gets the word "talent" — an ability, a gift — from this parable and nowhere else. A civilization does not extract *that* meaning from a story it understands to be about deposit accounts.

Third, weigh the historical company the parable kept. The same church that canonized Matthew banned usury to its clergy at Nicaea within three centuries (chapter 1 met canon 17) and to everyone within a few more, and for a thousand years the theologians who knew this text by heart — who preached it annually — saw no tension whatever between the master's *tokos* and the usury prohibition, because they did not read verse 27 as commercial instruction any more than they read the ten bridesmaids as advice on lamp oil. The interest-apologetic reading is a latecomer, arriving in force only when the ban was already being dismantled on other grounds and a proof-text was wanted. None of this makes the parable an anti-commercial tract, and honesty requires the concession that cuts the other way: the two faithful slaves *trade*, profitably, and are praised for it — venture, increase, and enterprise sit comfortably inside the story's picture of faithfulness, which is why this book's Part IV will find Christian merchants sailing under partnership contracts with clear consciences. What the parable cannot be made to do is bless the one thing its tradition forbade: gain without venture, the guaranteed increment, the hole in the ground's respectable cousin. Its target is the servant who, given a living gift, chose sterility out of fear. On that reading — the church's own, for its first millennium and beyond — the parable does not contradict this book's argument. It states it: capital is entrusted, not owned; it is judged by what it brings forth; and the only strategy it refuses to forgive is burial.

One Body, Common Goods

What did the first Christians actually do with their money? Luke, again, is the one who tells us, in two summary passages in Acts that later ages would read with longing, embarrassment, or both. Of the church in the weeks after Pentecost: "All who believed were together and had all things in common; they would sell their possessions and goods and distribute the proceeds to all, as any had need" (Acts 2:44–45). And of the same community a chapter later: "Now the whole group of those who believed were of one heart

and soul, and no one claimed private ownership of any possessions, but everything they owned was held in common.... There was not a needy person among them, for as many as owned lands or houses sold them and brought the proceeds of what was sold. They laid it at the apostles' feet, and it was distributed to each as any had need" (4:32–35).

Read the mechanism before the mystique. This is not a legislated abolition of property; Luke's own narrative says so a few verses on, when Peter confronts Ananias not for keeping money but for lying about it: "While it remained unsold, did it not remain your own? And after it was sold, were not the proceeds at your disposal?" (5:4). Ownership persisted; what changed was its posture. Sales were triggered by need — "as any had need," twice — and the proceeds pooled at the community's center for distribution, so that the measurable outcome Luke reports is not equality of holdings but the abolition of destitution: *not a needy person among them*, Deuteronomy's old promise for the release-year community, quietly claimed as fulfilled. Judaism knew neighboring experiments — the Essenes held goods in common by rule, as Josephus describes — but the Jerusalem church's version was voluntary, urban, porous, and improvised, less a constitution than a reflex: people who had been taught that mercy must circulate applying the lesson to real estate.

As a property regime it did not last; the Jerusalem church itself was poor within a generation, and the New Testament's next economic document is Paul, moving through his Greek congregations raising relief money for it. His covering argument, in 2 Corinthians, quietly generalizes Acts into a principle that this book will meet again and again: "it is a question of a fair balance between your present abundance and their need, so that their abundance may be for your need, in order that there may be a fair balance. As it is written, 'The one who had much did not have too much, and the one who had little did not have too little'" (2 Corinthians 8:13–15). The quotation is the manna story — wealth that could not be hoarded, gathered unequally yet somehow sufficing equally — redeployed as a rule for money moving between communities: surplus flows toward need, in the expectation that the direction of flow will someday reverse. That is not almsgiving, exactly, and not a loan at all. It is the reciprocity of a pool, described eight centuries before anyone priced one.

Paul had already built the theory, in the image that seeded everything this book will say about solidarity. The congregation, he tells Corinth, is a *body*: "God has so arranged the body... that there may be no dissension within the body, but the members may have the same care for one another. If one member suffers, all suffer together with it; if one member is honored, all rejoice together with it" (1 Corinthians 12:24–26). And to Galatia, in five words that Part VII will spend a chapter unpacking: "Bear one another's burdens, and in this way you will fulfill the law of Christ" (Galatians 6:2) — a command the same page balances, three verses later, with "all must carry their own loads" (6:5), the tradition's own guard against the pool becoming anyone's excuse. A body is the exact opposite of a market in one specific respect: nothing in it is left to bear its fate alone. The hand's injury is not the hand's problem; suffering is not permitted to remain where it lands, but is spread, instantly and involuntarily, across the whole. Chapter 2 called the community a shield against isolated ruin; Paul gives the shield an anatomy. This body-logic — many members, one fate, burdens shifted from the struck to the standing — is the seed of every mutual-aid institution this book will visit, from the burial societies and confraternities to *takaful* and the *gemach*, and eventually of the mathematics of chapter 28, which will show in numbers what Paul asserted in anatomy: that a community which pools its blows can survive what no member could survive alone. Later tradition never quite forgot Acts either; the monastic rule with its common purse, and after it the mendicant experiment,

are the apostolic memory institutionalized, and Part V will watch that memory build credit institutions for the poor.

Which brings this chapter to its unresolved ending — unresolved by design, because the New Testament leaves it so. Tally the estate it bequeaths. A prayer that binds every petitioner to run his ledger as God runs God's. A command to lend expecting nothing, to a neighborhood with no outer edge. A memory of goods held in common, a body in which no member suffers alone, a tax collector's fourfold repair, a temple cleared of tables — and no code. No permitted rate, because no rate is contemplated; no contract law; no answer to the merchant's morning question of what he may charge and stipulate and still be faithful. The omission is intelligible: Israel's law was given to a nation with land, courts, and harvests to govern, and the Qur'an would within a decade of its first revelations be regulating an actual city's contracts; the Gospel, by contrast, was preached by and to people with no economy to run — a teacher who owned nothing, communities that were minorities inside an empire whose markets they could not have reformed if they had tried. Counsel of perfection was the form the situation allowed. But situations change, and this one changed totally: within three centuries the empire itself came to the front, and the people who prayed "forgive us our debts" found themselves operating the debts — landowners, magistrates, eventually the only bankers of a broken world. The Christian problem, which the rest of this book's Christian strand is one long attempt to solve, was born exactly there: how to live a mercy calibrated for the kingdom of God inside a commerce calibrated for scarcity — how to keep books, charge prices, secure loans, and fund ventures with Zacchaeus's arithmetic and the unforgiving servant's fate in view. The first tools were blunt: Nicaea's canon barring the clergy from interest, the Fathers' sermons hammering the lender of last resort. The finer instruments — the just price (chapter 9), the anatomy of the loan itself (chapter 7), the extrinsic titles by which a lender might honestly be made whole (Part VI), the Monti di Pietà lending to the poor at cost (chapter 17) — took a millennium of casuistry that the middle parts of this book will walk through at working pace. All of it, every distinction and decretal and charitable bank, descends from the collision this chapter has described: a tradition commanded to forgive debts, learning, slowly and never quite finally, what it may still ask of a debtor. The prayer, meanwhile, went on being said daily, by merchants among others, one sentence about debt at the center of it — the standing audit before which all the ledgers of Christendom would have to be opened.

Chapter 6 — The Qur'an and the War Against Riba

In the year after Mecca surrendered to the Prophet, so the classical commentators tell the story, two clans fell to quarreling over old money. Banu Amr ibn Umayr, of the tribe of Thaqif in the mountain town of Ta'if, had lent at interest in the days before Islam to Banu al-Mughira, one of the great clans of Mecca. Now everyone concerned was Muslim, the debts had come due, and the lenders wanted what the old contracts promised — principal and increase. The debtors would pay the principal and not one coin more: we do not pay *riba* in Islam. Between the two parties stood the young man the Prophet had left as his deputy over Mecca, Attab ibn Usayd, and Attab did the sensible thing. He wrote to Medina and asked.

What came back, the commentators say, was not a ruling but revelation — two verses of the Qur'an's longest sura, given here in M. A. S. Abdel Haleem's rendering, which this book uses for the Qur'an throughout: "You who believe, beware of God: give up any outstanding dues from usury, if you are true believers" (2:278). And then, for anyone minded to stand on his contracts: "If you do not, then be warned of war from God and His Messenger. You shall have your capital if you repent, and without suffering loss or causing others to suffer loss" (2:279). The lenders of Thaqif took the settlement. The principal came home, the increase died on the page, and nobody elected to try the alternative.

It is worth standing still in front of that second verse. Scriptures forbid a great many things, and all three traditions of this book forbid interest with the severity chapter 2 displayed. But the commentators noticed long ago that this verse has no twin anywhere in the Qur'an: no other offense — not even the ones its theology ranks more gravely — is answered with a formal declaration of war from God and His Messenger against the offender. Murder is not. False witness is not. *Riba* is. The chapter that follows is an attempt to understand why — what this practice was doing, in the Qur'an's moral universe, to draw scripture's only declared war.

The Qur'an is the third scripture of this Part and the youngest of the three. The Hebrew Bible legislated for a covenanted farming people, binding lender to brother (chapter 4); the New Testament radicalized the Torah's mercy until lending expected nothing at all (chapter 5); the Qur'an arrived six centuries later inside a commercial civilization of caravans, contracts, and credit, and it speaks of finance with the most complete voice of the three. It names the sin and prices the exit. It shields the failing debtor. It institutes a poor-fund with officers and categories. It blesses honest trade in the same breath as prayer. And around the burned ground of the forbidden loan it leaves standing every consensual structure of commerce, which its jurists would then spend a millennium elaborating. Chapter 2 climbed this range's peaks. What follows walks it whole: the moral universe the prohibition defends, the trade it exempts, the mercies it commands, and the door it leaves open — the door marked contract — through which everything else in Islamic finance eventually walks.

The Grammar of False Increase

Begin with the word. *Riba* is Arabic for increase — growth, swelling — and the root runs through the Qur'an in its plainest organic sense, as when the dead land, watered, "stirs

and swells and produces every kind of joyous growth" (22:5). The Semitic cousins are all nearby: the Hebrew *tarbit*, "increase," and the rabbis' *ribbit* grow from the same root-idea, as chapters 1 and 2 noticed. The vocabulary itself, in other words, makes a claim on interest's behalf: increase presents itself as growth — natural, seasonal, as inevitable as the swelling of rained-on ground. Nearly everything the Qur'an says about *riba* is a denial of that appearance.

The increase in question was not a modern percentage point. As chapter 2 reported from the commentators, the signature practice of Arabian *riba* was deferral-and-doubling: the debt falls due, the debtor cannot pay, and the creditor extends the term while the balance swells — "doubled and redoubled," as the Qur'an itself describes it (3:130). A herdsman or small trader who missed one season could watch the shortfall compound past any hope of repayment, until his goods, then his land, then his labor were annexed to another man's ledger. The commentators call this *riba al-jahiliyya*, the usury of the Age of Ignorance, and when the Qur'an says *riba*, this practice is in its sights.

The tradition reads its revelation on the subject as a story told in stages, and the staging matters, because it is how the tradition understands its own pedagogy. The first word came in Mecca, before the community had any power to legislate, and it forbade nothing. It re-priced: "Whatever you lend out in usury to gain value through other people's wealth will not increase in God's eyes, but whatever you give in charity, in your desire for God's approval, will earn multiple rewards" (30:39). No command yet — an accounting correction. Increase taken through other people's wealth is recorded, in the books that matter, as no increase at all, while the gift that looks like depletion earns "multiple rewards." Second in the tradition's ordering comes a passage recalling an earlier covenant community, which had been forbidden usury and taken it anyway — "taking usury when they had been forbidden to do so" and "wrongfully devouring other people's property" (4:161). For its first hearers the point was genealogical: *riba* is not a new offense but an old one, banned across the whole prophetic line, and the Qur'an treats the Torah's prohibition as standing law it inherits. Third, the direct command of 3:130 halts the named abuse — "do not consume usurious interest, doubled and redoubled." And last, by the tradition's account among the final passages the Prophet ever received, comes the full case of 2:275–281, from which the war verse went out to Ta'if. The commentators like to observe that the community was weaned from *riba* the way it was weaned from wine, by revelations that tightened in stages; whether the historical sequence ran precisely so is a question for the tradition's own sciences of dating and occasion, and this book takes the story as the tradition tells it — as pedagogy more than archaeology.

That final passage deserves a close reading, because nearly every argument Islamic finance will ever have is compressed inside it. It opens with an image: "those who take usury will rise up on the Day of Resurrection like someone tormented by Satan's touch" — the usurer staggering into eternity like a man possessed. It states the confusion it means to kill: "That is because they say, 'Trade and usury are the same,' but God has allowed trade and forbidden usury." We will return to that sentence; the whole next section lives inside it. It offers terms: "Whoever, on receiving God's warning, stops taking usury may keep his past gains — God will be his judge — but whoever goes back to usury will be an inhabitant of the Fire, there to remain" (2:275). It then restates Mecca's old accounting correction as a law of nature: "God blights usury, but blesses charitable deeds with multiple increase" (2:276). The verb is agricultural — blight, the disease of crops. The increase that looked so inevitable on the contract rots in the field, while the gift, which looked like loss, compounds. Then come the ultimatum and the settlement terms we have already watched arrive at Ta'if; then

mercy for the debtor, which is this chapter's next station but one; and then a closing verse that a report preserved in Ibn Majah's collection — the tradition grades it with its usual candor about such things — has Umar remembering as among the very last revealed, the Prophet dying before he had fully expounded the matter for his community: "Beware of a Day when you will be returned to God: every soul will be paid in full for what it has earned, and no one will be wronged" (2:281).

Notice the moral vocabulary that runs through all four movements, because it never varies. Riba is filed with eating. The earlier community stood accused of "wrongfully devouring other people's property"; the Qur'an's general ban on corrupt gain forbids consuming wealth "wrongfully," "sinfully and knowingly" (2:188, 4:161) — gain from a neighbor's need pictured, as chapter 2 traced, as a mouth closing on his substance. And riba is filed with injustice: the war verse settles the unwinding with a clause the tradition reads as a maxim, wrong not and be not wronged — the capital is honest; the increase was the wrong. What the Qur'an conspicuously never supplies is a theory. There is no discourse here on the sterility of money, no analysis of time and its price; those arguments belong to the jurists, and chapter 7 will sit in their classrooms. In the Qur'an's own register the matter is simpler and more absolute. This gain devours a brother. God refuses to bless it. It ends now.

And it did end — publicly, from the summit of the tradition's memory. At the farewell pilgrimage in the spring of 632, on the plain of Arafat, the Prophet proclaimed the old order's practices trampled under his feet — so Muslim's collection preserves the sermon (Muslim 1218) — its blood-feuds canceled and its riba abolished, beginning at home: the first riba I abolish is our own, the riba owed to Abbas ibn Abd al-Muttalib. Abbas was his uncle, and a lender; whatever else may be said of the war against riba, its first confiscation fell on the victor's family. The same tradition remembers the Prophet cursing the whole apparatus of an interest-bearing loan — the taker, the payer, the scribe who writes it, the two witnesses who attest it; "they are all equal," Jabir reports him saying (Muslim 1598). Readers of chapter 2 will recognize the instinct, for the rabbis of Bava Metzia likewise dragged scribe and witnesses into the dock. A wrong this quiet, both traditions saw, is committed by everyone who lends it their hands.

Trade and Its Counterfeits

"Trade and usury are the same." Of all the objections scripture records its opponents making, this may be the most durable; it is the argument of every age of finance, ours included. My forgone money is a service like the merchant's goods; its price is interest; where exactly is the difference? The Qur'an's reply, we saw, is a line drawn on authority: "God has allowed trade and forbidden usury." But the reply is not a merchant-hater's, and the Qur'an is the last place to look for contempt of commerce. It was revealed in a trading town, to a man who — tradition remembers, and chapter 11 has already told — once carried a widow's goods to Syria for a share of the profit. Its judgment scene is an audit in which "every soul will be paid in full for what it has earned." Its Friday rhythm sends worshipers from the mosque straight back to the market: "when the prayer has ended, disperse in the land and seek out God's bounty" (62:10). Even its gentlest excusal from the night vigil, quoted already in chapter 11, is for those "travelling through the land seeking God's bounty." Commerce, in this book, is not a tolerated vice; it is part of the bounty.

The line between trade and riba therefore runs not between piety and the market but through the middle of the market itself, and the Qur'an leaves its strongest clue to the

line's logic at 4:29: "You who believe, do not wrongfully consume each other's wealth but trade by mutual consent." Two conditions gleam in that sentence. Something real must pass between the parties — trade is exchange, wealth for wealth, want for want — and both must walk in free. A sale enriches by exchange: each side leaves holding the thing it valued more, which is why both can bless the deal. The usurious loan enriches by the calendar: one side's return is fixed in advance and grows out of the other side's inability to close the account, under a consent that was given, if it was ever real, at the exact moment the borrower had nowhere else to turn. The jurists would refine this intuition for a millennium — what exactly a loan is, chapter 7; when capital may earn by sharing a venture's risk instead of charging for its time, chapter 11 and its whole Part — but every refinement is already folded into that one Medinan sentence.

The Qur'an and its Prophet named two further ways of being paid without exchanging, and the three prohibitions travel together through all of later Islamic law. The second is *maysir* — gambling — and it is Qur'anic by name. Mecca's gamblers played a game the commentators describe in detail: marked arrows shuffled and drawn for shares of a slaughtered camel, fortunes riding on the draw. The Qur'an's first word on it is strikingly measured, taking wine and gambling together: "There is great sin in both, and some benefit for people: the sin is greater than the benefit" (2:219). Its last word is not measured at all. Intoxicants, gambling, and divining arrows are "repugnant acts — Satan's doing — shun them so that you may prosper" (5:90); "With intoxicants and gambling, Satan seeks only to incite enmity and hatred among you, and to stop you remembering God and prayer. Will you not give them up?" (5:91). Notice the ground of the objection. It is not probability theory; it is fellowship. The gambler's gain is precisely his companion's loss — wealth migrating on a draw, nothing made, nothing exchanged — and a brotherhood cannot survive a habit whose every win is a brother's injury. Enmity and hatred are not named as gambling's side effects but as its produce, the thing the table reliably manufactures.

The third distortion the Qur'an does not name — the word *gharar* never occurs in it — but the Prophet's practice supplied the rule, and the jurists gave it a career as large as *riba*'s. Muslim's collection preserves the report (Muslim 1513): the Messenger of God forbade the pebble-toss sale, an old bazaar formula in which the deal closed on whatever the thrown pebble struck, and he forbade the sale of *gharar* — the sale of radical uncertainty. The illustrations the tradition's teachers reach for make a little gallery of fog: the fish still in the sea, the milk still in the udder, the calf still unborn, next year's fruit before the tree has set it. In each, someone is paid now for something neither party can see, weigh, or be sure of delivering. A contract, this tradition insists, is a shared act of knowledge — object, price, and terms plain to both sides — and selling fog is not trade but a wager wearing trade's clothes. The jurists accordingly filed *gharar* between *riba* and *maysir* in their catalogs of corrupted exchange, and centuries later the same doctrine would make them squint hard at the insurance policy — is a premium a purchase of indemnity, or of fog? — the question that seeds chapter 27.

Set the three side by side — this is the chapter's one deliberate symmetry — and the shape of the Qur'anic market emerges. *Riba* is gain fixed against a neighbor's need; *maysir* is gain drawn from pure chance; *gharar* is gain taken through the other party's blindness. Each collects from misfortune — need, luck, ignorance — rather than from exchange, and each is pruned away so that what remains is the market of 4:29: real things passing by real consent, gain arriving as the byproduct of a bargain both sides can bless. That is the market the rest of Islamic finance is one long attempt to build in law.

Until Matters Become Easier

Return now to the summit passage, to the verse that stands between the ultimatum and the Day. The war verse has just disarmed the creditor of his increase. The very next sentence hands him a duty regarding his principal: “If the debtor is in difficulty, then delay things until matters become easier for him; still, if you were to write it off as an act of charity, that would be better for you, if only you knew” (2:280).

The verse’s grammar repays attention. The delay is no suggestion; the law schools read it as the insolvent debtor’s entitlement and built from it a jurisprudence of hardship, under which the debtor genuinely unable to pay is to be given time — the verse’s own phrase, until matters become easier — rather than hounded for what he does not have. The second clause goes further while pretending merely to advise. Writing the debt off is called charity — remission as *sadaqah* is the verse’s own equation — and it is “better for you,” with the quiet addendum that the creditor does not yet understand where his own interest lies. Chapter 2 set this verse beside Exodus’s cloak law and the Gospel’s lending that expects nothing, as one of the three traditions’ portraits of the vulnerable borrower. In its home context it does something more specific: it converts the creditor’s forbearance from a commercial concession into a religious act, performed toward God and priced by Him.

The tradition took the conversion literally. Bukhari’s collection (2078) preserves the Prophet’s story of a merchant of no other recorded distinction, who used to lend and would instruct his agents, when they came upon a debtor in hardship, to pass over the debt, in the hope that God might pass over his sins. And God, the Prophet said, forgave him. The story’s economy is exact — patience extended on earth, collected in heaven — and it is the lived form of 2:280: in this tradition the creditor’s gentleness is not written off as bad debt management but held up as the conduct God rewards.

Nor is the strained debtor left to depend on his creditor’s piety alone. When the community’s alms system arrives in the next section, one of its eight chartered classes of recipient will be “those in debt” — the public fund itself standing behind the insolvent, paying down strangers’ debts as a matter of budget rather than favor. The debtor in difficulty has, in this scripture, two shields: the mercy his creditor is commanded to show, and the fund the community is commanded to fill. It is to that fund we now turn.

The Purifying Share and the Beautiful Loan

Everything so far has been prohibition and restraint. But the Qur’an’s economy has a productive side, and its engine is *zakat*. Zakat stands among the five pillars of Islam, beside the profession of faith, the daily prayer, the Ramadan fast, and the pilgrimage — which is to say, it is not the community’s philanthropy but its worship. Abdel Haleem translates it “the prescribed alms,” and the adjective is exact: zakat is owed, not offered — a fixed share of accumulated wealth, classically one-fortieth of monetary wealth each year, with schedules of its own for herds and harvests, collected on a calendar from everyone whose holdings cross the threshold. Chapter 2 noted the premise underneath: the poor hold a recognized right, a *haqq*, inside the believer’s wealth (70:24–25). Zakat is that right made administrable.

The name itself is a claim. The tradition’s lexicographers hear two ideas in the root z-k-w — purity and growth — and both are Qur’anic. “In order to cleanse and purify them,” the Prophet is told, “accept a gift out of their property” (9:103). Wealth, in this picture, comes into a believer’s hands morally unfinished: the poor’s share sits inside it, and while that

share is withheld the whole holding is tainted by it. Paying the share does not diminish the wealth but cleanses it — and, by the arithmetic of 2:276, in which God “blesses charitable deeds with multiple increase,” the cleansing is also the only increase that survives audit. The same root that names the sin thus names the remedy. Riba is the swelling God blights; zakat is the pruning under which wealth is said to genuinely grow — and the tradition planted that shared root at the center of its economy quite deliberately.

And zakat is an apparatus, not a mood. The verse that fixes its uses reads like a budget: “Alms are meant only for the poor, the needy, those who administer them, those whose hearts need winning over, to free slaves and help those in debt, for God’s cause, and for travellers in need. This is ordained by God” (9:60). Eight classes — and the machinery notices itself, for “those who administer them” are salaried from the fund they collect, which tells you this is a fisc with officers and books, not a collection plate. The list intervenes at every point where the old machine of debt once caught people: the poor and needy first; captives bought free; “those in debt,” the class the last section promised, strangers’ debts paid down from the public purse; the stranded traveler far from kin. Riba compounded claims against the strained; zakat, year after year on a fixed calendar, walks a fortieth of the community’s accumulated wealth back down the hill.

Above the owed floor rises the voluntary layer, *sadaqah* — a word the tradition connects to the root of truthfulness, giving as the proof that faith means it. The Qur’an’s definition of goodness includes those who “give away some of their wealth, however much they cherish it, to their relatives, to orphans, the needy, travellers and beggars, and to liberate those in bondage” (2:177), and cherish is the operative word: where zakat is measured on the wealth, sadaqah is measured on the grip it breaks. Its growth imagery is the most exuberant in the book — those who spend in God’s cause are “like grains of corn that produce seven ears, each bearing a hundred grains” (2:261), a seven-hundredfold yield, the honest version of the multiplication riba only pretended to. And because the receiver is a person and not a cause, the manner of giving is policed as strictly as the amount is left free. A gift followed by “reminders and hurtful words” is canceled (2:264); better than open charity is giving “to the needy in private” (2:271). The dignity of the man at the door — the figure chapter 2 set at the center of all three traditions — governs even the etiquette.

Which brings us to the strangest and loveliest instrument in the Qur’an’s economy. Again and again, the scripture that has just abolished interest asks its believers for a loan, and names its own borrower: “Who will give God a good loan, which He will increase for him many times over? It is God who withholds and God who gives abundantly, and it is to Him that you will return” (2:245). “Who will make God a good loan? He will double it for him and reward him generously” (57:11). The promise recurs at 64:17 — “if you make a generous loan to God He will multiply it for you and forgive you” — and in the verse chapter 11 quoted, the good loan stands in a single breath with the pillars themselves: “keep up the prayer, pay the prescribed alms, and make God a good loan” (73:20). This is *qard hasan*, the beautiful loan: in practice a loan to a human being in need, with the ledger entry reassigned. Lend to your neighbor without increase, and heaven records the credit. It remains a loan — the principal is owed back, which is precisely what distinguishes it from sadaqah, and chapter 16 will show how seriously the free-loan institutions take repayment. What it may never do is grow on the lender’s side of the table. The jurists condensed the rule into a maxim — every loan that draws a benefit is riba — a rule of thumb rather than a certified hadith, but one that carries the Qur’an’s whole logic in a sentence. The beautiful loan is barren on earth and fertile in heaven; its interest accrues in the one ledger where the Qur’an is content to let interest compound. The institutions this verse-family grew —

free-loan funds, the *qard hasan* windows of Islamic microfinance — belong to chapter 16, which sets the beautiful loan beside its Jewish cousin the *gemach* and the Catholic works of mercy.

One Body, Many Contracts

The mercies of this chapter presume a community built to carry them. “The believers are brothers” (49:10), and chapter 2 watched that sentence rebuild kinship on faith. The Prophet’s own images for the *ummah* are physiological: in a report both Bukhari and Muslim carry (Bukhari 6011), the believers in their mutual mercy and affection are one body — when one limb suffers, the whole body keeps the night watch with it in sleeplessness and fever. The image is precise: fever is a whole body spending itself on one member’s injury. And Bukhari preserves a working example with evident fondness (2486): the Ash’ari people, the Prophet said, when their provisions ran short on campaign or at home, would gather everything they had into one cloth and divide it equally among themselves — and he claimed them: they are of me, and I am of them. Bukhari’s editors filed that report in the Book of Partnership, which is where it belongs; the pooled cloth is a financial instrument. The old Arabian institution of the *aqila* points the same way — the law retained the custom by which a man’s kinsmen shared the blood-money he owed, liability itself carried as a pool. These are seeds. What grew from them, thirteen centuries on, when the jurists looked at commercial insurance and asked their *gharar* question of it, was *takaful*: mutual guarantee, the risk pool built as solidarity rather than sold as product. Chapter 27 builds it, and chapter 28 does its mathematics — the arithmetic by which pooled ruin is survivable and solitary ruin is not. Here it is enough to mark the seed-bed: a community commanded to be one body will eventually need instruments for carrying one another’s weight.

Now step back from the whole chapter and notice what the Qur’an did not do. For all its fire, it legislates commerce astonishingly little: no partnership statute, no schedule of instruments, no price code — chapter 11 noticed that the contract Islamic finance is proudest of, the *mudarabah*, has no founding verse at all. What the Qur’an does legislate, at the length of the longest verse in the entire book, is procedure. “You who believe, when you contract a debt for a stated term, put it down in writing: have a scribe write it down justly between you” (2:282). The verse runs on for a page: scribes who must not refuse to write, debtors who dictate and guardians who dictate for those who cannot, witnesses summoned and forbidden to withhold, debts recorded with their terms whether “small or large” — and then a protection, “let no harm be done to either scribe or witness.” Set that clause beside the curse of Muslim 1598 and the tradition’s whole theory of documents stands in two sentences: the scribe of a *riba* bond is cursed together with his principals, while the scribe of an honest debt writes under the law’s protection, the difference lying not in the penmanship but in what the paper does to the person it names. Add the command that opens the Qur’an’s fifth sura — “You who believe, fulfil your obligations” (5:1) — and you have its commercial constitution entire: contracts are sacred, contracts must be written, and one contract is banned.

The Sunna then widened the banned zone beyond anything a Meccan creditor would have recognized. Muslim’s collection (1587) carries the report that would keep jurists employed for a thousand years: gold for gold, silver for silver, wheat for wheat, barley for barley, dates for dates, salt for salt — like for like, equal for equal, hand to hand; and if the kinds differ, sell as you please, provided it is hand to hand. From those six commodities the jurists mapped *riba* beyond the loan into the very mechanics of exchange, and gave the branches names: *riba al-fadl*, surplus taken in a like-for-like swap, and *riba al-nasi’a*, in-

crease charged for delay. How far the six generalize — to money as such, to everything measured and weighed, to the paper and pixels that now do money's work — became one of jurisprudence's great standing arguments, school against school, and the tradition never pretended otherwise. Umar's report stands in the collections as its own admission that the Prophet departed before the border was fully surveyed, leaving the survey to the community; the scholars' word for the resulting disagreement-within-the-house is *ikhtilaf*. Whether a modern bank's interest falls inside the word *riba* is a live argument of exactly this lineage, and chapter 32 will watch the tradition conduct it in our own century. This chapter marks only where the arguing began: at a border the Prophet left his community to draw.

And that is the answer to this chapter's last question — why Islamic finance begins with prohibition but develops through contract. The Qur'an cleared the ground and left the building to the law. Its prohibition burned one contract out of the economy, the loan that breeds; everything else it either blessed outright — trade by consent, the written and witnessed debt — or planted as seed: the purifying share, the beautiful loan, the pooled cloth. Into the cleared space came the jurists with their craft: the *qard* defined against its neighbors (chapter 7), capital taught to earn by sharing risk rather than charging for time (the *mudarabah* and *musharakah* of chapter 11 and Part IV), sale and lease and certificate engineered into *murabaha*, *ijara*, and *sukuk* (Part VII), the pools formalized into *takaful* (chapter 27) — every instrument tested, generation after generation, against the rule chapter 3 announced for this whole book, that legal form must answer to economic substance. Part II closes here, with the three scriptures agreed on the ground floor they built in three different voices: the Torah binding lender to brother, the Gospel unbinding mercy from return, the Qur'an declaring war on the increase and dictating the peace. The war against *riba* ended the way wars end — in treaties. The treaties are the contracts, and the rest of this book is largely spent reading their clauses.

Part III — The Legal Grammar of Anti-Usury Finance

Chapter 7 — When a Loan Is Really a Loan

Two farmers each take delivery of a neighbor's grain in autumn. The first takes it as a deposit: the neighbor's sacks sit in his barn, still the neighbor's, awaiting spring. The second takes it as a loan: he pours the grain into his own bin, sows some, eats some, and owes back as much next harvest. In February the river rises and both barns drown. Now ask each man what he owes, and you have this chapter.

The depositary, if he kept the barn as a careful man keeps a barn, owes nothing. The grain that drowned was never his; it was the owner's grain in his keeping, and the owner's loss. The borrower owes every last measure. It does not matter that his fields flooded too, that the loan itself lies under the same water, that no grain of the original delivery still exists anywhere. What drowned in his bin was his own grain — it became his the moment he poured it in — and what he owes is not grain at all but a quantity, an abstraction the flood cannot reach. The depositary keeps your grain; the borrower keeps only your arithmetic.

Every legal system this book visits knows that difference, and each of the three traditions built its usury law on top of it. The rabbis called the loan *milveh*, the Roman and canon lawyers called it *mutuum*, the jurists of Islam called it *qard*, and the three words name one contract with one anatomy: ownership passes, a quantity is owed, time runs between. Part I introduced the loan as one form among eight; Part II watched three scriptures converge on it with commands and warnings; and chapter 11 has already shown the escape route the traditions left open beside it. This chapter closes the circle by asking the technical question underneath all of that: what exactly *is* a loan — and why did its precise legal anatomy, rather than any general suspicion of money-making, convince three traditions that this one contract can never honestly yield increase?

The Anatomy of a Loan

Begin with what can be lent at all. A plow, a house, a mule can be handed over and handed back: the thing returns, the same thing, scratched perhaps but identical in law, and while it is away the owner remains the owner. Roman law kept a whole stable of contracts for such returnable things — chapter 3 met them — and none of them is our subject. The *mutuum* governs the other kind of goods: the ones the Institutes describe as dealt with by weight, number, and measure — wine, oil, grain, coined money (Inst. 3.14). What such goods have in common is that their use *is* their consumption. You cannot use wine and still have it; grain is either seed or bread or someone else's property, but never "borrowed" in the plow's sense; and money — this is the point on which everything turns — money is spent or it is idle. To lend a consumable is necessarily to authorize its destruction, which is to say: to give it away. The borrower does not receive the use of the lender's thing. He receives the thing, whole, ownership and all, because no thinner slice of it can be transferred.

The rabbis fixed the same anatomy in four Aramaic words. A loan, says the Talmud, is given to be spent (Kiddushin 47a) — *milveh l'hotzaah nitna* — and the consequences cascade exactly as in the Roman analysis. The coins the borrower takes become his coins; he may spend them on flour or foolishness; what the lender holds from the moment of delivery is not a residual interest in any object but a personal claim against a man, for a quantity. Chapter 1 called the claim the fourth function of money and the only one that wears a face.

The loan is that function distilled: after the handshake there is no *thing* between the parties at all — only a number, a date, and two people ranked by it.

So the loan's anatomy has four parts, and it is worth laying them on the table as cleanly as the medieval lawyers did, because every argument for the rest of this book pokes at one of them. First, *transfer of ownership*: what was mine becomes yours — the pun on *meum* and *tuum* that chapter 3 reported the medieval jurists hearing inside the word *mutuum* itself. Second, the *principal*: the quantity delivered, which fixes the debt — and fixes it in both directions, a symmetry easy to miss. The same traditions that strip the lender of every increase defend his principal absolutely: the Qur'an's settlement terms at the war verse run "you shall have your capital... without suffering loss or causing others to suffer loss" (2:279), and neither the halakha nor the canonists ever confused the usury ban with debt forgiveness. The wall is not a mercy that costs the lender his money; it is a definition that limits him to it. Third, *time*: a loan exists only as duration; a *mutuum* repaid in the instant of its making is a handshake, not a contract. The term may be fixed or open — the rabbis' default for a loan with no stated term was thirty days (Makkot 3b) — but duration there must be; the gap between delivery and return is the contract's whole body. Fourth, the *obligation*: to return not the thing (it is gone, lawfully gone) but its *tantundem* — as much again, of the same kind and quality, the flood and the harvest and the borrower's fortunes notwithstanding. Notice what these four parts add up to. The lender has exited the story. His goods are gone into another man's ownership; his risk, in the lawyers' sense, is gone with them — come fire or plague, the quantity is owed; all he retains is the claim, ticking quietly toward its date. The loan is the only contract in chapter 3's whole taxonomy in which one party's role is complete on the first day, while the other party carries every hazard the world can supply. That asymmetry is not an abuse of the form. It *is* the form — and everything the traditions permit and forbid about it follows from taking the form seriously.

Rome itself, no enemy of the moneylender, testified to this anatomy in its own strange way. The *mutuum* belonged to the contracts formed *re*, by the physical delivery of the thing, and Roman law held — as a matter of construction, not morals — that the *mutuum* could oblige repayment of the principal and nothing else. A Roman who wanted interest did not write it into the loan; he could not. He took a second contract, a *stipulatio*, a formal verbal promise bolted on alongside the *mutuum*, and collected his interest — his *usurae*, the lawyers' word for sums paid on borrowed money — under that. The pagan empire, which permitted interest freely, still found that the loan contract itself had no room for it: the yield had to ride outside the loan, on a promise of its own. When the theologians later declared the *mutuum* barren by nature, they were not inventing a pious fiction; they were reading the anatomy Rome's own lawyers had drawn, and refusing the bolt-on.

The rabbis read the same anatomy into their law of guardians. The Mishnah (Bava Metzia 7:8) ranks the keepers of other people's property on a ladder of liability: the unpaid guardian answers only for negligence, the paid guardian for more, and at the ladder's top stands the borrower, liable for everything up to sheer accident — because, the rabbis reasoned, all the benefit is his. The borrower of money stands above even that rung: ownership itself has passed to him, so there is nothing left to "guard" at all, only a debt to pay. Liability tracks ownership, and ownership tracks the anatomy. Every legal mind that examined this contract closely — Roman, rabbinic, scholastic, Islamic — came away with the same drawing.

The deposit stands on one side of the loan — the *pikadon* of chapter 11's *iska*, ownership retained, risk with the owner. The partnership stands on the other — money delivered

but not alienated, trading at the investor's peril, which is why its gain is honest. The loan sits between them, and the whole moral geometry of this Part is a matter of respecting the boundaries: each contract has its own anatomy, each anatomy its own honest price, and the oldest trick in finance — chapter 3 called it the hybrid, and Part VIII will watch it industrialized — is to charge one contract's price on another contract's anatomy.

Three Names for One Wall

Set the three traditions' loan-rules side by side and the family resemblance is as close as chapter 11 found among their partnerships — closer, since here even the reasoning matches.

The halakha reasons directly from the ownership transfer. Since the borrowed coins are the borrower's, any increase he pays flows toward a man who no longer owns anything in the transaction — payment for goods already fully conveyed, or else payment for the one thing that did pass between them: the waiting itself. The Talmud's blunt shorthand for the forbidden thing is exactly that — *agar natar*, payment for waiting: any reward for waiting is forbidden, runs the general rule of Bava Metzia 63b — and the Torah's verses (chapter 4 walked them) supply the vocabulary of the *neshekh*, the bite, taken from a brother's flesh while the lender sits still. The rabbis even asked why Scripture should need two words — the bite and the increase, *neshekh* and *tarbit* — for what analysis shows to be a single sin, and Rava gave a characteristic answer (Bava Metzia 60b–61a): so that the lender at interest transgresses twice. One act, two violations, no daylight between them. The prohibition, chapter 2 noted, is total in a way little else in the Torah's civil law is total: it binds lender, borrower, scribe, and witnesses alike, and it admits no floor. The rabbis then spent tractate-loads of ingenuity policing the wall's base — the *iska* of chapter 11 stands on one carefully argued side of it, and the next chapter will watch them chase interest into gifts, favors, and even words — but the wall itself never moved: where the contract is a *milveh*, increase is *ribbit*, and *ribbit* is forbidden.

The Latin tradition arrived at the same wall through the same door, with Aristotle carrying the lamp. Money, the Philosopher had said (chapter 1), is barren; the scholastics sharpened the observation into contract law by way of the *mutuum*'s anatomy. Aquinas's version (ST II-II, q.78, a.1) became the classic, and it is really two arguments folded together. The first is the consumables argument: there are things whose use is their consumption, and for such things use and substance cannot be sold separately — wine's use *is* wine — so a lender who charges for the money and again for the use of the money “sells the same thing twice, or sells what does not exist.” A man may rent out his house, because the house survives its use and remains his; he cannot “rent out” his wine, only sell it; and money, on this analysis, is wine, not house. The second argument reaches back through Aristotle to money's own job description: money was invented as a measure and medium — its proper use is to be spent, alienated in exchange — so a charge for “using” money prices the very act of parting with it, which the sale of the principal has already covered. Either way the charge is not extortion but a kind of metaphysical fraud: billing for a phantom slice that the anatomy of consumables does not contain. On that reasoning the Church legislated early and long — the clergy barred from interest at Nicaea in 325 (chapter 1), the ban universalized in the councils that followed — and the whole magnificent scaffolding of medieval finance, the just price and the partnerships and eventually the extrinsic titles, was erected around a *mutuum* that remained, in itself, barren by definition.

The *fiqh* states the rule with the least apparatus and the most severity. The jurists define

the *qard* as exactly this transfer — ownership in fungible (*mithli*) wealth passing to the borrower against the binding duty to return its like, quantity for quantity, kind for kind — and any stipulated benefit flowing back to the lender voids the arrangement's innocence. The working rule of the madrasas, met in chapter 6, condenses it: every loan that draws a benefit is *riba* — a juristic maxim rather than a certified hadith, but one that carries the Qur'anic logic in seven words. And Islam adds the inversion that gives the rule its distinctive color: the same contract, stripped of every increase, is not merely tolerated but beautiful — the *qard hasan* of chapter 6, the loan whose interest accrues in heaven's ledger only. There is no better proof that the tradition's quarrel was never with lending. The contract is blessed; the increase alone is the enemy.

One refinement shows how precisely the *fiqh* located the poison. The tradition remembers the Prophet himself repaying a loan with an older and better camel than the one he had borrowed — the best among you, he said, are the best in repaying (Bukhari 2305–2306) — and the jurists drew the line accordingly: the *stipulation* is *riba*, not the gratitude. A borrower who freely returns more than he took, unpromised and unexpected, has given a gift; a lender who writes the extra into the contract has taken *riba*, though the sums be identical. The sin lives in the claim, not the coin. The *halakha*, characteristically, went a scruple further and barred even the thank-you — the gifts before and after a loan, the extra courtesies, the flow of unstipulated benefit toward the lender — but that hunt through the loan's shadows is the next chapter's business.

Three legal languages, one wall, and one load-bearing insight beneath it. None of the three traditions bans profit. None bans commerce; none, as chapter 11 showed at length, bans capital's gain as such. What all three ban is gain from *this* contract — because this contract, examined honestly, contains nothing for the gain to pay for. The goods are the borrower's. The risk is the borrower's. The venture, if there is one, is the borrower's. A price needs something priced, and the *mutuum*, once its anatomy is on the table, offers the lender nothing to sell but his neighbor's obligation — or the calendar.

Why a Loan Is Not an Investment

The modern mind files lending under investment, one asset class among many, and the filing is precisely the confusion the traditions' lawyers spent centuries keeping straight. Chapter 11 drew the line from the partnership's side; one paragraph draws it here from the loan's, because the two contracts, which can look identical at the moment money changes hands, are perfect opposites in anatomy.

The investor's money remains his own. It sails on the venture — at his peril, said Aquinas (ST II-II, q.78, a.2, ad 5); half loan, half deposit, said the *Nehardeans*, and the deposit half at the investor's risk; loss follows the capital, said the jurists — and its return, if any, is a harvest from his own exposed field. The lender's money becomes the borrower's. From delivery onward the world's weather falls on the borrower alone: the flooded barn, the drowned cargo, the failed shop are his losses entirely, while the claim against him rides above the water, dry and undiminished. An investment's return is the world's report on what actually happened. A loan's return is a sentence written before anything happened at all — which is exactly why the traditions could bless the one and wall the other. A return that tracks reality is earned inside it, while a return fixed in advance must be extracted from whoever has to live with what actually happened. Chapter 11 ran the two payoffs side by side on the same voyage and the numbers said the rest: the loan treats the lender best precisely when the venture fares worst. Nothing in the loan's anatomy connects the

lender's gain to anything that occurs in the world — and the traditions concluded, with a lawyer's literalism, that it therefore pays for nothing in the world.

The anatomy also explains a fact most depositors would rather not examine: the money in a modern bank account is not, legally, in the bank at all. Hand a warehouse your furniture and it must return your furniture; hand a bank your money and it owes you only your arithmetic — the balance is a debt, the “deposit” a loan to the bank, which is why the bank may lend your money out (it is not your money; it became the bank's on arrival) and why it pays *you* interest rather than charging storage. The old lawyers saw this coming. Deposit a sealed purse and the coins stay yours; deposit loose coin with a money changer, the Mishnah already ruled (Bava Metzia 3:11), and he may put it to work — at which point he holds it as a borrower holds, liable come flood or fire. Fungibility plus permission to use equals a loan, whatever the sign over the door says, and the entire modern debate about banking, interest, and whose risk a failed bank has been running is chapter 3's form-versus-substance rule applied to the shopfront word “deposit.”

The Percentage Fallacy

Here the modern reader raises the obvious objection, and it deserves an honest answer. Surely, the objection runs, the sin in all this is a matter of degree. Five percent between merchants is commerce; fifty percent against a widow is predation; the civilized task is to find the line and cap it. Antiquity itself mostly agreed: Hammurabi's tariff fixed barley loans at a third and silver at a fifth (chapter 2), Rome capped its rates, and when England re-legalized interest in 1545 it did so with a ceiling of ten in the hundred (chapter 1). Regulation by percentage is the oldest and most natural instrument in the drawer.

The three traditions knew that instrument perfectly well, and what is striking is that all three declined it. Their claim is different in kind, not degree: the defect lies in the genus of the charge, not its size. The halakha is the starkest case, because it *does* use thresholds elsewhere — chapter 3 met *ona'ah*, the price-fraud rule with its one-sixth tolerance, a law that expressly forgives small deviations. Ribbit gets no such tolerance. There is no *de minimis* interest in the Torah's law; the forbidden category begins at the first grain above the principal, and the next chapter will find the rabbis extending it past coins altogether, into favors and greetings. The scholastic analysis is rate-blind by construction — Aquinas's double-sale argument works identically at one percent and at forty, since a phantom is a phantom at any price. And when early modern Christians finally pressed the question in its sharpest form — is not a *moderate* charge, to a *rich* borrower, for *productive* purposes, a different thing from the gouging the Fathers cursed? — the papacy answered in 1745 with the encyclical *Vix pervenit*, the tradition's last full-dress statement of its own doctrine: usury has its proper place and origin in the loan contract itself, which by its nature obliges the return of only as much as was received; and the gain being moderate excuses nothing, and the borrower being rich excuses nothing, and the money being put to fruitful use excuses nothing, because none of these facts adds anything sellable to the *mutuum's* anatomy. A percentage is an answer to a question the traditions refused to ask. Their question was never “how much may the loan yield?” but “what in the loan could yield it?” — and to that question a lower rate is no answer at all, merely a smaller specimen of the same nothing.

One clause of *Vix pervenit* points forward, and it should be marked now because Part VI lives inside it. Having sealed the contract's interior, the encyclical allows that reasons may arise *alongside* a loan — reasons “not at all intrinsic to the contract” — that justify recovering

something beyond the principal. A lender who suffers demonstrable loss by lending, or forgoes a real and otherwise certain gain, is not extracting a phantom; he is being made whole for something true. The medieval lawyers called such reasons extrinsic titles, and the long, careful, hazardous work of defining them — actual loss, ceased gain, the risk and the delay — is the story of Part VI. The wall, note, is what makes the titles honest: only because the loan itself yields nothing can a real cost beside the loan be identified, audited, and compensated without the whole distinction collapsing. Kill the wall and the titles become a tariff; keep the wall and they remain what they claim to be — damages, not yield.

Selling What Is Not Yours to Sell

Strip away everything the interest charge cannot be — payment for the goods (they are the borrower's), payment for their use (inseparable from the goods), payment for the lender's risk (he has parted with it), payment for the venture's success (he owns no share of it) — and two candidates remain standing. The tradition examined both, and its two answers are this chapter's destination.

The first candidate is time. The loan's one undeniable content is duration: the lender waits, and the interest meter runs on the calendar. Very well, said the scholastics' opponents in every century: the charge prices the waiting. The thirteenth century's answer came from William of Auxerre, master at Paris, in his *Summa aurea*, and it remains the most beautiful sentence the usury debate ever produced: the usurer sells time — and time is the common possession of all creatures. The sun gives its light, the earth its fruit, and every creature its duration, freely, because none of them made it; a man who meters out the hours as his private inventory has annexed the one estate that belongs to everyone, and even if all men kept silent before the usurers, William says, the very stones would cry out. Innocent IV took the argument up, and the medieval consensus settled on its plain meaning: no one holds title to the calendar. Whatever a lender may someday be compensated for — his losses, his forgone harvests, his demonstrable damages — the sheer passage of time between delivery and repayment is not his property, and a charge calibrated to nothing but that passage is a rent collected on creation itself.

It is worth saying plainly that modern economics answered William rather than refuting him. When Eugen von Böhm-Bawerk built interest its respectable modern foundation at the end of the nineteenth century, the foundation he chose was time itself — present goods outrank future goods of like kind and quantity, runs his *agio* theory, and interest is the premium that ranking commands. The scholastic and the economist agree completely, in other words, about what the naked interest charge prices; they disagree about whether it may be sold. That collision — between time as tradable preference and time as common estate — is not a quaint medieval leftover but a live question this book will press twice more: once when the *lucrum cessans* debates ask whether forgone time-bound opportunities can be real losses (chapter 20), and once when Part IX asks what a defensible price of patience would look like inside a non-extractive fund. William's stones are not yet done crying out.

The second candidate is quieter and uglier, and the traditions saw it with perfect clarity: what the interest charge finally prices, when everything else is stripped away, is the borrower's need. The man at the door — chapter 2 put him at the center of all three traditions, and he must stand here once more. He does not come to the lender the way a buyer comes to a market, weighing offers, free to walk away. He comes because the harvest failed or the ship is late or the child is sick; his consent, as the Qur'an's jurists and the rabbis and the

canonists all observed in their own idioms, is the consent of a man with no alternatives, given at the exact moment his bargaining power is zero. Interest converts that moment into revenue. It is a price set on the distance between the two parties' positions — on the emptiness of one man's hand as such — and this, the traditions concluded, is the heart of the matter and the reason the wall admits no percentage: the borrower's need is a fact about him, not an asset of the lender's. It cannot be owned, so it cannot be sold; it can only be exploited, and a small exploitation differs from a large one as a small theft differs from a large. The Torah says it as kinship — let your brother live beside you. The Gospel says it as mercy that expects nothing. The Qur'an says it as the war verse and the waiting commanded until matters become easier. Underneath all three runs the same finding of fact: in the loan, uniquely among the contracts of chapter 3, one party's weakness is the other party's entire pricing power, and a civilization that lets the meter run on weakness will manufacture more of it.

This is why the loan is the first moral stress test of finance, and why every tradition in this book begins its financial ethics here rather than with the grander instruments. The loan is the simplest contract there is — one delivery, one obligation, no moving parts — and simplicity is exactly what makes it diagnostic: there is nowhere for an unjust euro to hide.

And the test travels, because the anatomy travels. Take the oldest disguise in the book, met in chapter 3 among the hybrids: I "sell" you my cow today for eighty and agree to buy her back next year for a hundred. No loan appears anywhere on the paper — two sales, two prices, all formally in order — yet the cow never moves, and the transaction's skeleton, held to the light, is eighty now against a hundred in a year: a loan at twenty-five percent wearing a bill of sale. Every tradition learned to run the same diagnostic. Strip the names off the paper; follow the money and the risk; ask who holds what, who owes what, and what the world can do to change either answer. If the world can do nothing — if one party's return is fixed, his principal effectively guaranteed, his gain indifferent to everything but the date — then whatever the document is called, its anatomy is a *mutuum* with increase, and the wall applies. Whatever a legal system decides about *this* contract, it has decided about the relationship between money and need, and it will spend the rest of its history either defending that decision or engineering around it. The three traditions decided, independently and in nearly identical words, that the naked loan may carry no price. Everything that follows in this book — the hunt for disguised benefits in the next chapter, the just price in the one after, the risk debates, the extrinsic titles, the partnerships already met, the great modern evasions of Part VIII — is commentary on that decision: the tradition testing, contract by contract and century by century, whether some charge is a real cost honestly recovered or the old phantom in a new coat. The test never changes, because the anatomy never changes. When a loan is really a loan, its price is the principal, whole, and nothing else; and the first question to ask of any instrument, however elaborate, is the question of this chapter: is it, under the costume, a loan?

Chapter 8 — The Forbidden Benefit

A man owes another a hundred dinars. The two have never been close — neighbors of the nodding sort — and this morning, crossing the market, the borrower catches sight of his lender by the date stall and feels a warmth in his chest that was not there before the loan. Before he has examined the feeling he has hailed the man, first, and brightly. A greeting: the smallest coin in circulation. And the rabbis of the Talmud, weighing this least of all transactions, ruled that he may not pay it.

The ruling survives at Bava Metzia 75b, in a baraita in the name of Rabbi Shimon ben Yochai, and it arrives carrying its own derivation. From where do we know that a borrower who was not previously accustomed to greet his lender first may not begin now? From the Torah's own wording. Deuteronomy bans "interest on money, interest on provisions, interest on anything that is lent" — chapter 2 read the verse whole — and the Hebrew behind "anything" is *neshekh kol davar*, the bite of any *davar*. A *davar* is a thing. It is also, in Hebrew, a word. Even speech, the baraita concludes, can be paid as interest; even speech is forbidden.

It is worth standing still in front of how strange, and how serious, this is. Chapter 7 walked the loan's anatomy and found the three traditions unanimous: the contract that transfers ownership of money may carry no price, because it contains nothing further to sell. But a price does not have to arrive as money. Anything a borrower sends toward his lender because of the loan — information, labor, lodging, hospitality, public honor, a place at the front of a queue, a warmth not previously on offer — is value moving in the direction the prohibition guards, and value moving because of a loan is interest, whatever it happens to be wearing. All three traditions saw this, and saw it early. Walling up the front door of the *mutuum*, the *milveh*, the *qard* was the easy part; the benefit that is not money comes in through the windows, and this chapter is about the window locks. The halakha hunted interest into words themselves. The canonists pursued the lender's silent hope. The jurists of Islam quarantined the late fee, the one charge every credit system swears it cannot live without. And behind all three campaigns stand the two questions this law is finally about — when gratitude becomes a price, and when a relationship becomes a lever — together with the institutions each tradition built to keep its own kindness from hardening into consideration.

Interest in Words

The halakha's campaign in the loan's shadows is the most systematic of the three, and its field manual is the fifth chapter of Mishnah Bava Metzia, whose closing paragraphs read like a customs inspection of everything that is not money.

Take favors. Two smallholders trade labor — help me today, I will help you tomorrow — and the Mishnah permits it, on a condition modern readers blink at: like for like. Weed with me and I will weed with you; till with me and I will till with you; but not weed with me and I will till with you, because the tasks differ, one may prove the harder, and the difference between the two days' sweat would be extra labor returned for labor lent — a loan of work repaid with increase (Mishnah Bava Metzia 5:10). Even the calendar is audited. All the dry days of summer count as one, all the rainy days as one; but a man

may not plow beside his neighbor in the dry season against a promise of plowing in the rainy one, when the same furrow costs twice the effort. The Mishnah is pricing weather, in other words, and it prices weather because sweat is currency and currency is regulated.

Take gifts. Rabban Gamliel taught that interest can be paid before a loan exists and after it has ceased to exist, and the Mishnah preserves both scripts. A man has set his eye on borrowing, and sends his hoped-for lender a present with the message: this is so that you will lend to me. That is *ribbit mukdemet*, advance interest. A man has borrowed, repaid to the last coin, and then sends a present with the message: this is for your money, which sat idle with me. That is *ribbit me'ucheret*, interest paid late. Notice what these two rulings do to the loan's timeline. In the first case there is no loan yet; in the second there is no loan anymore; at neither moment could any court find an obligation outstanding between the parties. The Mishnah forbids both anyway, because the benefit is traveling on the loan's account, and the prohibition follows the account, not the calendar.

Then Rabbi Shimon adds the category this chapter takes its Hebrew from: *yesh ribbit de-varim* — there is interest that consists of words. His example is a masterpiece of compression. The borrower may not say to his lender: "Know that so-and-so has come from such-and-such a place." That is the entire case, and on its face it is a scrap of small talk. The commentators unpack it in two directions, and the codes preserve both. Read one way, the borrower is slipping his lender commercial intelligence — a caravan is in, a buyer has arrived, prices are about to move — the kind of tip a merchant would happily pay for, delivered free to this man for this reason. Read the other way, which is Maimonides' way, the sentence runs from lender to borrower as an instruction: watch for so-and-so's arrival, and receive him, feed him, lodge him as befits him — the lender drawing hospitality out of the borrower the way one draws on an account (Mishneh Torah, Creditor and Debtor 5:13). Either direction, the analysis is the same. Market information has a price; a night's lodging and a good dinner have a price; and whatever has a price, flowing lender-ward because of a loan, is *ribbit*.

Maimonides, codifying a millennium later, extended the inventory with a lawyer's thoroughness. The borrower not previously in the habit may not greet his lender first — that much is the baraita — and all the more may he not praise him in public, nor present himself early at the lender's door; nor — the detail that stops every student — may he teach the lender Scripture or Talmud while the loan stands, if he did not teach him before (Creditor and Debtor 5:12). Torah itself, the most precious thing the tradition knows how to give, becomes forbidden freight on the wire between borrower and lender. And the lender's side of the wire is patrolled with equal suspicion: one who lends may not live in his borrower's courtyard rent-free, nor rent from him at a discount, "because this is interest" (Mishnah Bava Metzia 5:2) — the roof over a creditor's head being merely money in another shape. The same mishnah walks on into the adjacent seam, permitting a landlord to charge more for rent paid month by month than for rent paid in advance while forbidding a seller the equivalent premium on a deferred field — the padded credit price that chapter 3 met as the standard specimen of *avak ribbit*, the dust of interest.

Dust is the tradition's own metaphor, and it maps this chapter's terrain exactly. Chapter 3 introduced the grading: *ribbit ketutzah*, fixed interest stipulated at the loan's birth, is the Torah's own prohibition, and a court will claw it back; the rabbinic dust — arrangements that shimmer with interest's residue — is forbidden but stays where it settled, beyond judicial recovery (Bava Metzia 61b). Verbal interest stands at the far end of the same ladder: forbidden with entire seriousness, and beyond any conceivable remedy, because no court can order a greeting returned. Enforcement thins as the benefit dematerializes — coin,

then dust, then words — but the prohibition never thins. That is the point on which the whole structure balances. The rabbis were not confused about the difference between a hundred dinars and a hello; they graded the remedies with perfect precision. What they refused to grade was the principle. *Agar natar*, any reward for waiting, is forbidden (Bava Metzia 63b) — in whatever currency the reward arrives.

The chapter of the Mishnah closes by naming everyone the law holds responsible — lender, borrower, guarantor, witnesses, and, add the sages, the scribe — a totality chapter 7 noted, whose Islamic twin chapter 6 displayed in the curse on the *riba* loan's whole cast, "they are all equal" (Muslim 1598). The Gemara's fine print even allocates the counts: the lender violates every verse on the list, the borrower three, the guarantor and witnesses one apiece. A prohibition that can be violated by a greeting is a prohibition about the relationship itself, and everyone who helps build the relationship stands inside it. The same folio preserves what the lending classes of the era made of all this. Rabbi Shimon reports their table talk: had Moses our teacher known there was profit in the matter, they liked to say, he would not have written the verse (Bava Metzia 75b). The Talmud records the joke the way a prosecutor records a confession — the lenders themselves testifying that the only defect they could find in the law was the money it cost them.

The Hope That Makes the Usurer

Latin Christendom reached the same territory through a different door. Where the rabbis catalogued the benefits one by one, the canonists went after the root of all of them at once: the lender's intention.

Thomas Aquinas states the doctrinal floor with his usual economy in the Summa's question on usury. Anything whose value can be measured by money counts as money — the Philosopher's rule — and therefore to take anything so measurable in return for a loan, "by tacit or express agreement," is usury outright (ST II-II, q.78, a.2). The objections he entertains show how concrete the traffic had become. One objection proposes that remuneration comes, as a gloss on Isaiah has it, not only in things but in words and in service — so surely a lender may accept a little praise, a little help. Aquinas answers that this is exactly why he may not: a man's labor and a man's words are both for hire in every market square — "those who offer for hire the labor which they exercise by work or by tongue" — and exacting either for a loan is exacting a price, the scholastic twin, down to its examples, of ribbit devarim. Another objection asks whether the lender may at least bind the borrower to return the favor with a loan of his own someday; no again, because the obligation to lend back has cash value, and writing it into the deal prices the loan — the Mishnah's weeding rule in Latin dress. What remains permitted is exactly what cannot be priced: the gratuity freely given, where nothing was exacted or due even tacitly, and — Aquinas's lovely concession — "benevolence, and love for the lender," which a lender may seek without sin precisely because no market anywhere quotes them.

But the deep Catholic text of this chapter is a short papal letter of the mid-1180s. A priest had written to Rome with a confessor's problem, and Pope Urban III answered in the decretal the collections call *Consuluit*, from its opening word: your devotion has consulted us. The question ran like this. A man lends money with no interest clause, no agreement of any kind, every stipulation conspicuously absent — but he would not otherwise have lent at all; he lends with the settled purpose of receiving back more than he gave. In the judgment of souls, is such a man to be judged a usurer? A second case rode along with the first: the merchant who sells his wares far above the cash price because payment is deferred.

Urban's answer reached past the whole apparatus of contract law to a single verse. What must be held in these cases, he wrote, is plainly known from the Gospel of Luke, where it is said: *Date mutuum, nihil inde sperantes* — lend, hoping for nothing thereby, the command whose full force chapter 5 measured. Such men, for the intention of gain they harbor, are to be judged to act wrongly, and in the judgment of souls they are to be effectively induced to restore what they took (X 5.19.10).

The canonists drew out what the letter implied and gave it a name that has startled law students ever since: *usura mentalis*, mental usury. Hope makes the usurer, ran their tag. A loan clean on its face, clean in its documents, clean before any earthly tribunal, is usurious in the forum that matters most if the lender's purpose was gain — because the hoped-for profit, being the reason the loan exists, is its price in everything but writing. Hope is the one contract clause that needs no ink. And the doctrine of the two forums did for the Church what the remedy ladder did for the rabbis. The external court judges what can be proven — instruments, payments, agreements; it cannot subpoena a state of mind, and the canonists never pretended otherwise. The court of souls admits evidence no summons can reach. A generation before Urban, Alexander III had answered the archbishop of Genoa about merchants who bought pepper and cinnamon worth five pounds and promised six by public instrument at a term: such a contract, the pope conceded, cannot be called usury by its form, yet the sellers sin unless the future price is genuinely in doubt, and the citizens of Genoa would consult their salvation well if they gave the practice up, “since the thoughts of men cannot be hidden from almighty God” (X 5.19.6). The form was satisfied and heaven was not. When the canonists later built their standing category for arrangements engineered around the ban — contracts *in fraudem usurarum*, in fraud of the usury prohibition — they were generalizing exactly this instinct, which chapter 3 announced as the rule for the whole book: the label is a hypothesis, and the substance testifies.

One border must be marked before leaving the Latin room, because the tradition marked it emphatically. The lender's hope of gain is poison; the lender's repair of real loss is not. A lender demonstrably damaged by lending, or injured by a borrower's culpable delay, claims restoration rather than yield, and the long, contested doctrine of such claims — the extrinsic titles — is Part VI's story. The two categories are this chapter's distinction in different clothes: compensation answers something that happened in the world, while the forbidden benefit answers nothing but the loan's existence. The canonists policed the border with the fraud doctrine precisely because they knew that every disguised yield in Christendom would sooner or later apply for papers as compensation.

The Penalty That Pays the Poor

Islam's front in this war runs through a problem every credit system knows: the borrower who is late.

The classical position is stark, and chapter 6 showed why. The signature usury of pre-Islamic Arabia was precisely a late fee — pay now, or pay more later — the debt “doubled and redoubled” as deadlines rolled past, until the shortfall annexed the debtor's goods and then the debtor. Increase charged for delay is the branch the jurists named *riba al-nasi'a*, and it is no marginal case of the prohibition; it is the original one. Yet the tradition is under no illusion that borrowers are all innocents. The Prophet's own word, in a report Bukhari files under the transfer of debts, is blunt: procrastination in repayment by a man of means is injustice — *matl al-ghani zulm* (Bukhari 2287). And the Qur'an's mercy is calibrated rather than indiscriminate. The debtor in genuine difficulty is entitled to time, “until matters

become easier for him" (2:280, in Abdel Haleem's rendering, as chapter 6 read it) — an entitlement of the insolvent, not a charter for the merely reluctant.

So the *fiqh* inherits a genuine bind. The solvent procrastinator commits a named wrong; the one deterrent every banker reaches for — make delay cost money — is the very engine the Qur'an burned; and letting delay cost nothing invites exactly the injustice Bukhari 2287 condemns. How deep the tradition's scruple runs shows at the retail scale of the problem. A report in Ibn Majah's collection — the hadith scholars grade it weak, and say so, but the jurists quote it for its picture — has the Prophet telling lenders to accept from a borrower neither a gift nor a lift on his riding-beast, "unless they used to treat each other in that manner beforehand" (Ibn Majah 2432). It is the rabbis' greeting rule, discovered independently, down to the same baseline: what the relationship already contained is innocent; what the loan added is suspect. A tradition that scruples over the borrower's mule was never going to wave the late fee through. The madrasas' working maxim, met in chapter 6, covers the whole territory in seven words — every loan that draws a benefit is *riba* — and a penalty clause is nothing if not a benefit drawn.

The solution modern Islamic finance converged on is quietly ingenious. The financing contract — typically a *murabaha* receivable, the credit sale chapter 24 will anatomize — includes an undertaking: if the customer procrastinates, he pays a stipulated amount, and the amount goes to charity. Not to the bank. The standards issued by the Accounting and Auditing Organization for Islamic Financial Institutions, the Bahrain-based body whose rulebook anchors much of the industry's practice, frame the conditions: the clause bites only on the procrastinator, never on the debtor in genuine hardship — the mercy of 2:280 written into the fine print; the sums flow to charitable causes under the supervision of the institution's Shariah board; and they never enter the institution's income, because a benefit drawn from a debt is *riba* however it is booked. The same plumbing extends wherever a forbidden gain condenses inside the machine: the charity account, standing drain of Islamic banking, carries income the institution may not keep out of the building and into the hands of the poor. (Some jurisdictions run a parallel channel called *ta'widh*, compensation for documented actual cost — a solution nearer the canonists' titles, and contested for exactly the reasons Part VI will rehearse.)

Look closely at what the diversion accomplishes, because summaries of Islamic finance routinely miss it. The borrower still faces a deterrent; delay still costs him real money; the clause does the disciplinary work a conventional late fee does. But the lender collects nothing, so the clause cannot become a business. An institution whose arrears feed the poor rather than its own revenue line has no reason to write deadlines designed to be missed, no quiet incentive to prefer its stumbling customers to its punctual ones, no temptation to let the penalty schedule do the earning that the financing margin cannot. Chapter 7 ended with the traditions refusing to price the borrower's need. Here is one of them refusing, in modern contractual machinery, to price the borrower's failure — while conceding that the failure has a price, and routing that price to the one party guaranteed to have done nothing to engineer it.

When Gratitude Becomes Price

In 1925 the French sociologist Marcel Mauss published *The Gift*, the essay that founded the modern study of giving, and its finding, compressed, is that there is no such thing as a free one. Across the societies Mauss surveyed, a gift arrives carrying three obligations — to give, to receive, and to reciprocate — and a present unreciprocated is not a kindness but

a standing debt, felt by everyone concerned, capable of ranking men as surely as a ledger. The lawyers of this book's three traditions had reached the same conclusion a millennium and a half earlier, from the inside. They knew that generosity binds, that the bond can be collected on, and that a creditor is the one man in the village whose gifts are never only gifts. Everything in this chapter so far — the audited greeting, the taxed hope, the diverted penalty — is the legal residue of that knowledge.

Which forces the question the traditions answered with startling unanimity: if gifts bind, when exactly does gratitude — the borrower's honest, human gratitude for rescue — become the forbidden benefit? Lay the three legal systems side by side and their tests align. Gratitude becomes price when it is stipulated. The Prophet repaid a borrowed camel with a better one and said that the best among you are the best in repaying (Bukhari 2305–2306, as chapter 7 told it); the jurists located the sin in the claim, not the camel — the same betterment, written into the contract, is *riba*. It becomes price when it is expected as due. Urban III's lender stipulated nothing; his hope alone was the price; and Aquinas's rule that a gratuity is innocent only where nothing was owed even by tacit agreement draws the identical line, a tacit agreement being an expectation both parties have learned to bill. And it becomes price when it exceeds the relationship's own baseline. The rabbis' borrower may greet a lender he always greeted; Ibn Majah's lender may ride the borrower's beast if riding it was their custom before the loan; what neither may do is let the loan mint new warmth. Spontaneous, unowed, inside the old custom: gift. Solicited, expected, loan-born: interest. Aquinas even supplied the metaphysics, distinguishing the debt of justice, fixed and exactable, from the debt of friendship, whose nature depends more on the feeling than on the size of the favor, and which dies the instant it is made compulsory (q.78, a.2, ad 2). A lender may be loved but never owed love, because gratitude with a schedule has changed species.

The second question is darker, and it will be handled plainly. A debt is not only a claim on money; it is a position in another person's life. The Hebrew Bible states the sociology in six words chapter 1 quoted: "the borrower is the slave of the lender" (Proverbs 22:7). While the debt stands, everything the borrower has — his time, his voice, his standing in the town, in the worst centuries his family's labor and his own body, as chapters 2 and 4 showed — sits inside the creditor's field of gravity. The rabbis legislated against that gravity with a rule almost without parallel in the history of creditor law. Rav Dimi taught that a creditor who knows his debtor cannot pay may not so much as pass in front of him: the verse "do not be to him as a creditor" forbids the walk past the man's stall, the appearance in his lane, because the creditor's mere person, present and silent, is a demand for payment, and a demand pressed on a man with nothing is torment. Rabbi Ami and Rabbi Asi said that one who does so is as though he sentenced his neighbor to the two great ordeals at once, fire and water (Bava Metzia 75b, citing the Psalm). Nothing is being charged in this case; no benefit changes hands; the law here is protecting the borrower's face, the last asset a ruined man owns. And it is recognizing something every debt collector since has understood professionally: a relationship with a debt inside it transmits pressure through channels no auditor will ever see — deference performed, dissent swallowed, service volunteered, the borrower's whole social surface quietly re-angled toward the man who holds his note. Pre-Islamic Arabia ran the dynamic to its terminus, and the Qur'an's word for the whole complex was *zulm*, wrong. When the three traditions stripped the loan of every yield — monetary, material, verbal, mental — this is what they were finally disarming: the conversion of rescue into rank. A loan that can pay its lender nothing, in any currency, cannot make its lender a patron. The ban on the forbidden benefit is, at bottom, a ban on clientage, built so that rescue cannot recruit.

Kindness Without Consideration

The traditions did not leave lending to the kindness of individuals. Each built institutions — and the institutions inherited the problem at scale.

The Jewish free-loan fund, the *gemach* — the name compresses *gemilut chasadim*, deeds of loving-kindness — pools a community's money and lends it at zero, against the wedding, the medical bill, the bridge between jobs; chapter 15 gives it its due. Islam's *qard hasan* funds and microfinance windows institutionalize the beautiful loan of chapter 6 the same way, and chapter 16 sets them beside the *gemach* and the Catholic works of mercy, with Christendom's great anti-usury pawn banks, the Monti di Pietà, following in chapter 17. This chapter's remaining business with all of them is a single design problem, which is its own subject in institutional dress. An individual lender's forbidden benefit arrives as a greeting or a goose. An institution's arrives as a donation.

Free-loan funds run on donated and deposited capital, and donors are human. The man who has given a fund ten thousand expects — not contractually, nothing so crude; expects the way Urban III's lender expected — that when his nephew needs a bridge loan, the queue will discover a certain flexibility. Let the expectation harden into practice and the fund has invented a new instrument: the donation that buys priority access to credit. Chapter 3's diagnostic list already filed the specimen — a donation that unlocks priority access is a price — and its pedigree runs straight back to this chapter's Mishnah, because money sent ahead of a hoped-for loan, "so that you will lend to me," is Rabban Gamliel's advance interest, and interposing a charitable balance sheet between the sender and the loan launders the flow without changing it. The dyad the prohibition polices — benefit moving toward the source of credit because it is the source of credit — has simply been rebuilt with a fund in the middle. Both traditions' free-loan literatures worry the question in its living forms, donor priority and depositor privileges and service charges among them; that casuistry belongs to chapters 15 and 16.

What belongs here is the grammar of the answer — what an institution looks like when it cannot be bought this way. The materials of this chapter assemble it almost by themselves. Separate the doors: the giver's entrance and the borrower's entrance open onto different rooms, and no corridor connects a gift to a place in line. Publish the queue: loans go by stated criteria — need, order of application, the lot if it comes to that — so that priority is never the fund's to sell, which is the institutional form of Aquinas's rule that what may not be exacted may not be tacitly owed either. Let the fund alone be the lender, so that the borrower's gratitude, which is real and right, lands on an entity that cannot enjoy a dinner invitation, cannot be flattered, and has no nephews. Guard the rolls: the borrower's name is the fund's secret, for the reason Rav Dimi gave — a community that knows who owes cannot help collecting in deference — and confidentiality is how an institution declines to become a creditor-patron on its members' behalf; Maimonides' ladder, chapter 3 noted, ranks the anonymous rungs above the known ones for kindred reasons. And where the machine must throw off a benefit somewhere — penalties, idle balances, windfalls — divert it outward on the Islamic charity-account pattern, so that what cannot be prevented is at least made to exit the relationship.

Readers building modern instruments should sit with this grammar longer than its homely examples suggest, because the forbidden benefit is alive in the term-sheet world and almost never audited there. Capital that forswears extraction in cash can still collect in kind: information rights beyond any governance need, first call on the next round, deal flow steered by a grateful founder, the board seat that becomes a megaphone, the public glow

of concessionary money conspicuously performed. These are the greetings and the geese of contemporary finance — benefits with money's measure, flowing to capital because it is capital — and a fund that claims the non-extractive name owes its stakeholders a ribbit-devarim audit alongside its financial one: name every non-cash channel, price each honestly, then close it or divert it. Part IX takes up that audit with arithmetic in chapter 35. The old lawyers' standard is already sufficient to state it: a benefit does not stop being a price because nobody wrote it down.

Maimonides, with whom this Part keeps ending, fixed the design goal in a single rung of his ladder: the highest charity, chapter 3 quoted him, is the gift, loan, partnership, or job that sets a man on his feet until he no longer needs to ask. Read the clause once more against this chapter. The measure of the best help is that it terminates — no residue, no standing warmth owed at the market stall, no client walking behind a patron; the receiver stands up owing exactly what he borrowed and not one greeting more. That is what all the window locks protect. The traditions barred the thank-you not because they despised gratitude but because they wanted the poor man's gratitude to remain his own: spendable nowhere, owed to no one, evidence of nothing but a rescue that asked nothing back.

There remains the daylight side of the market. The loan is now sealed at both ends — principal in, principal out, nothing riding along in either direction, not even a hello. But most of economic life is not lending at all; it is buying and selling, where gain is honest and every tradition in this book says so. The next chapter walks out of the loan's shadows into the bazaar, where the question is no longer whether a benefit may exist but how large it may honestly be — the just price, the overreached buyer, the seller who mistakes a neighbor's ignorance for a margin. That is the law of cheating, and it is chapter 9's. This chapter's work was stranger: the traditions here were policing not the wicked but the grateful, not the sharp deal but the warm one, and what three legal civilizations independently concluded is that a kindness can be a claim in costume. So they wrote, each in its own language, the same unlikely safeguard — protection from the urge to say thank you with interest.

Chapter 9 — Price, Fraud, and Overreaching

Late in the third century, a petition reached the chancery of the Roman emperors from a man about whom history preserves exactly one fact: his father had sold a piece of land, and the family had come to regret the price. His name was Aurelius Lupus. Whether the father was alive or dead, whether the land was vineyard or scrub, what debt or drought or bad judgment had set the figure — none of it survives. What survives is the answer, issued in the names of Diocletian and Maximian and later carried into Justinian's Codex at 4.44.2, and it opens with a word the stern lawyers of Rome did not often reach for. If you or your father sold for a lesser price a thing worth a greater one, the rescript runs, it is humane — *humanum est* — that you recover the land, refunding the price under a judge's authority; or, if the buyer prefers, that you receive what is lacking of the just price. Then the threshold, dropped like an afterthought and destined to run through seventeen centuries of law: the price is deemed too low if less than half the true price was paid.

Nothing in Roman law had talked this way. The classical jurists were bracingly unsentimental about bargaining — in the matter of price, they held, the parties are naturally permitted to get the better of one another, a doctrine sitting undisturbed in the Digest (18.1.38; 19.2.22.3) — and another rescript in the very same title of the Codex restates the orthodoxy at loving length: the buyer comes wishing to buy cheap, the seller to sell dear, and after much wrangling, the one creeping down and the other inching up, they consent to a figure that good faith then protects. Only at the end does that text, too, append the startling exception — unless less than half the just price was paid — in what scholars since 1706 have suspected is the hand of Justinian's editors rather than Diocletian's. Whoever wrote the rule, the Middle Ages received it as authority, named the seller's injury *laesio enormis* — enormous harm — and quietly unshackled it from the rescript's facts, extending it past land to nearly everything salable and past sellers to buyers, until a completed, consented, fully documented sale could be reopened by a court on the strength of arithmetic alone.

Stand back from the rescript and notice what kind of legal object it is, because this chapter is about its family. Chapter 7 walked the wall the three traditions built around the loan: a contract in which any increase at all is forbidden, because the anatomy offers nothing for increase to pay for. The sale is different country. Here gain is honest — “God has allowed trade and forbidden usury,” says the Qur'an's dividing sentence (2:275, quoted here as throughout in M. A. S. Abdel Haleem's rendering), and the rabbis and the schoolmen agreed down to their ledgers — and so the question changes from whether to how much. Every tradition in this book blessed the merchant's margin and then, having blessed it, spent enormous care on three ways the margin goes bad: the lie, where one party prices what the other cannot check; the fog, where neither party can see the thing being priced; and the squeeze, where one party prices the other's desperation. Just price, *ona'ah*, *gharar* — a Latin ideal, a Hebrew tort, an Arabic hazard — and underneath them one conviction that the loan chapters have already made familiar: an exchange is just only between parties who both know what is happening and are both free enough for their consent to mean something.

Equality of Thing and Thing

Thomas Aquinas took up cheating in the Summa's question 77 — immediately before usury, a piece of shelving chapter 3 already admired, since the two questions are the two sins of exchange filed side by side. His opening article asks the whole subject in one line: whether it is lawful to sell a thing for more than its worth.

The answer begins from what a sale is for. Buying and selling, he says, were established for the common advantage of both parties — each needs what the other has — and an institution built for common advantage should not burden one side more than the other. Hence the rule he states with a bookkeeper's calm: the contract should observe "equality of thing and thing," the price matching the worth of what it buys, money having been invented precisely to be that measure. To sell a thing for more than it is worth, or to buy it for less, is on his analysis not sharp practice at the edge of virtue but a plain injustice at its center — taking from a neighbor, through the price, value the exchange never gave him.

Two clarifications immediately soften the geometry without weakening it, and both matter for everything downstream. First, the equality is not arithmetic. The just price, Aquinas writes, "is not fixed with mathematical precision, but depends on a kind of estimate," so that a modest overage or underage does not destroy the equality of justice. Just equivalence is a band, not a point — chapter 3 said it of all three traditions, and here is the Latin instance in its own words. Within the band, haggling is what markets sound like; sin begins where the deviation becomes considerable and known. Second, the moral law and the courtroom run at different tolerances. Human law, made for a people most of whom are not saints, cannot punish everything vice invents; it lets small mutual outwitting pass, Aquinas says, and intervenes only when the deception runs past half the just price — and for that boundary he cites, chapter and verse, the imperial rescripts of our opening. The friar in Paris and the man before Diocletian's chancery stand in a single legal lineage, and Aquinas knew it. But the court of conscience keeps stricter books. Divine law, he goes on, leaves no injustice unpunished: whoever received notably more than he gave owes restitution, whether or not any human judge could be brought to care.

The second article turns from price to thing. A sale can cheat through what is sold as easily as through what is charged, and Aquinas sorts the defects into three: substance, quantity, quality. Wine mingled with water and sold as wine — he reaches for Isaiah's image of silver turned to dross; the false measure, the divers weights of Deuteronomy's warning; the sick animal dressed up as sound. Sell with knowledge of any of these and the sale is fraudulent, the sin real, restitution owed. Sell in honest ignorance and the moral fault vanishes but the debt does not: when the defect surfaces, the seller must make the buyer whole. And the duty runs both ways — the expert buyer who spots gold priced as copper and closes quickly on the seller's mistake is a thief with cleaner fingernails, bound to restitution like his mirror image.

The third article is the one modern readers should read twice, because it asks the disclosure question in nearly modern form: is the seller bound to declare the defects of what he sells? Where the defect is hidden and threatens the buyer with loss or danger — the lame horse sold as a fast one, the tottering house, the spoiled food — yes, absolutely: silence there makes the sale fraudulent and the seller liable. Where the defect is plain to see, the calculus changes; a man selling a one-eyed horse need not hire a crier to announce the eye, and may quietly protect his price against a buyer who would haggle the defect twice. What the article is building, clause by clause, is a jurisprudence of what each party is entitled to know before consent counts — and one hard case at its edge, a merchant, a hungry port,

and a fleet of grain ships just over the horizon, this chapter holds back for a few pages, because it belongs to the quarrel about market price.

The fourth article banks the fire it has just lit. Trade itself, Aquinas concedes, aims at gain, and gain as an end has something disquieting about it — the appetite for it “knows no limit.” Yet gain can be pointed at limited, honorable things: the upkeep of a household, the relief of the needy, the provisioning of a country — and then the merchant’s markup is simply the wages of his labor. The article’s verdict compresses into a rule this book will lean on repeatedly: profit is licensed by the service it prices, not by the leverage that extracts it. For this chapter the point is narrower and sharper. A margin is honest exactly insofar as the price stays inside the band of the thing’s worth; the moment the margin is harvested from the buyer’s ignorance, or from fog, or from his need, it has left the just-price analysis and entered the neighboring question — which is why the *Summa* files usury next door. One clause of that next question already patrols the border between them: to charge a buyer more because he will pay later is, Aquinas rules, usury in a sale’s clothing — “this waiting for the payment of the price has the character of a loan” (q.78, a.2, ad 7) — a seam chapter 24 will find running through the cost-plus instruments of modern Islamic finance.

What Aquinas states as moral theology, the rabbis had long since built as working law, with numbers, remedies, and a clock.

The Measure of the Sixth

The Torah’s sale law begins with a verse addressed, characteristically, to both sides of the counter: “When you make a sale to your neighbor or buy from your neighbor, you shall not cheat one another” (Leviticus 25:14, in the New Revised Standard Version). The Talmud, at Bava Metzia 58b, reads that verse as the charter of monetary *ona’ah* — overreaching, price-wronging — and finds in its near twin three verses later (25:17) a second command aimed at wrongs done with words. From the first command the Mishnah engineered the most precise consumer-protection statute of antiquity.

Chapter 3 met its headline number: *ona’ah* is four silver pieces out of the twenty-four in a sela — one-sixth of the purchase. The benchmark against which the sixth is measured is the going market rate, which is worth underlining: the Mishnah does not ask what the seller paid, or what the thing cost to make, or what a philosopher might think it worth in the sight of heaven; it asks what this thing sells for in this market this week, and starts the meter at deviation from that. The remedies then come tiered, and the Talmud preserves the final calibration in Rava’s ruling at Bava Metzia 50b. Under a sixth, the sale stands and nothing is returned — the law waives what markets ordinarily produce. Above a sixth the sale can be undone — and not only by the victim: at a disparity that size, the Gemara notes, either party may renege. At exactly a sixth, the sale stands but the excess goes back — price repair without contract demolition. The sixth is not a price ceiling; it is the point past which the law stops believing that a market happened.

Every remedy runs on the same clock, and the clock may be the statute’s most humane invention: the wronged buyer can act only “until he can show it to a merchant or his relative.” Not an hour, not a day — the time it takes to get a second opinion. The law assumes that a buyer cannot always price what he is holding, entitles him to borrow a better-informed pair of eyes, and closes the window once he has had his chance, so that sellers are not hostage forever to yesterday’s customers. (Rav Nachman adds, at the same folio, that the window as stated binds the buyer; a wronged seller may go back on the sale at any time.) Rabbi Tarfon’s misadventure in Lod — chapter 3 told it — turns out, in the Gemara’s retelling at 50b,

to hinge on exactly these mechanics: the merchants cheered his widened one-third margin because it waived the sixth itself, then discovered that his generosity came bundled with an all-day return window, did the arithmetic on which rule cost them more, and asked for the Sages back. There is no sturdier testimony to how precisely this law bit than merchants optimizing against its fine print.

Two boundary features complete the machine. The rule is bilateral — “both the buyer and the seller are subject to *ona’ah*,” says the Mishnah (4:4), guarding the amateur seller against the expert buyer as vigilantly as the reverse. And it has carve-outs (4:9): slaves, documents, land, and consecrated property stand outside the rule. The land exclusion earns a raised eyebrow here, because it makes *ona’ah* and *laesio enormis* near-perfect mirror images: the Roman rule protected only the seller of land, the rabbinic rule protected everything except land, and the two systems even share the remedy pair — undo the sale, or top up the price. Legal historians have argued for a century about whether the rules are kin: an early Hebrew-law scholar saw the rabbis behind the rescript; Daniel Sperber read both as parallel responses to the third century’s broken prices; Raymond Westbrook traced the Roman rule past both, to the ancient Near East’s old right to redeem family land sold under debt. The honest verdict is a strong family resemblance with no documented genealogy — two legal civilizations, sharing an eastern Mediterranean and a moral instinct, converging on the thought that some completed bargains should not stay completed.

The same folio that grounds the sale law extends it somewhere a Roman jurist would never have followed. “Just as there is *ona’ah* in buying and selling,” says the Mishnah at 58b, “so there is *ona’ah* in words”: one may not ask a shopkeeper the price of an item one has no intention of buying — raising hopes for the pleasure of dashing them is a market wrong even though no coin moves — nor remind a penitent of his old deeds, nor a convert’s child of his ancestors’. That doctrine, *ona’at devarim*, is the sale-law cousin of the verbal and social benefits chapter 8 patrols on the loan side, and it belongs mostly there; it appears here as a boundary stone, marking how far the rabbis were prepared to chase the idea that transactions can wound. One last contrast completes the picture: the sixth’s tolerance is unique to the sale. Ribbit, chapter 7 found, has no *de minimis* — the loan’s wall starts at the first grain — because a sale has a market to deviate from, and a loan, in the traditions’ analysis, has nothing to price at all.

The Law of the Visible

Islam’s law of exchange starts from the verse chapter 6 called the market’s charter: “You who believe, do not wrongfully consume each other’s wealth but trade by mutual consent” (Qur’an 4:29). Chapter 6 read the verse’s two conditions — something real must change hands, and both parties must enter freely — and left them standing as the market’s constitution. The jurists heard in the second condition an epistemic demand, because consent is only as good as what the consenting mind can see — a principle their tradition received from the Prophet’s market practice in a remarkably concrete form.

The master text, met in chapter 6, is Muslim 1513: the Messenger of God forbade the pebble-toss sale, and he forbade the sale of *gharar* — of radical uncertainty, of fog. Where chapter 6 hung the classical gallery of examples, this chapter can watch the doctrine work on exchange itself, because the collectors did. Bukhari’s chapter on the subject files *gharar* together with a *jahiliyya* specimen so extravagant it needs no commentary: *habal al-habala*, the sale of the offspring of the offspring of a she-camel — payment today for the unborn calf of an unborn calf (Bukhari 2143). Fog squared, and old Arabia traded it. Nearer the

everyday sits the orchard rule: the Prophet forbade selling fruit “until its benefit is evident” — until the crop has set and shown itself sound — and the report adds, with the tradition’s typical evenhandedness, that he forbade it to seller and buyer alike (Bukhari 2194). The buyer is being protected from paying for blossom-scented hope; the seller from the quarrel that arrives with the blight, when the parties discover they never agreed on what, exactly, had been sold. The jurists generalized the pattern into the requirement chapter 3 stated in a clause: the object of sale must exist, be known, and be deliverable. Known — *ma’lum* — is the load-bearing word. A contract, in this tradition, is two informed consents meeting; whatever neither party can see, weigh, or warrant is not yet the stuff of contract.

Around that core the Prophet’s market rules build what can only be called consent machinery. The parties to a sale retain an option — *khiyar* — to cancel so long as they have not yet parted, and the report that carries the rule carries its theology too: if the two “spoke the truth and made things clear,” Hakim ibn Hizam heard the Prophet say, their sale is blessed; if they lied and concealed, the blessing of their sale is wiped out (Bukhari 2079). There stands Aquinas’s third article, centuries early and in a merchant’s Arabic: disclosure is not a courtesy appended to exchange but the thing that makes exchange clean. The fiqh’s standing defect rule, the option called *khiyar al-’ayb*, runs the same way — a concealed fault lets the buyer return or keep the goods once the fog lifts — and two further prohibitions police the integrity of the price signal itself. *Najash*, the planted bidder who runs up the price with no intention of buying, is forbidden outright (Bukhari 2142): a fake buyer manufactures a fake market, and every subsequent “market price” is testimony from a bribed witness. And the caravans were not to be intercepted. Merchants riding toward a town were met, in the old practice, by traders who bought their goods on the road — cheaply, since the sellers had no idea what the town was paying. The Prophet forbade the interception, and gave the rule its remedy: whoever buys from the intercepted caravan, “when the owner comes to the market, he has the option” (Muslim 1519). Read that remedy slowly, because there are law schools that have never put the point so well. The bargain on the road was formally consensual — offer, acceptance, delivery, price. What it lacked was knowledge: the seller had not yet seen the market. So the law leaves the bargain standing, walks the seller to town, lets him look at the going rate — and only then asks whether he consents. The signature on the road, in this tradition, ripens into consent only at the market, where the seller can finally see what everyone else can see.

Three Prices

Set the three rooms side by side and a common architecture emerges, which this book’s later chapters will lean on. All three traditions treat the ordinary market rate as the just price’s working measure; all three refuse to treat it as its own justification; and all three reserve their real severity for a third kind of price — the one extracted where the market’s conditions have failed the weaker party.

Take the measures first. The rabbis, we saw, literally define overreaching as deviation from the going rate. The scholastics, who inherited Aquinas’s “kind of estimate,” ripened it into a doctrine with a name: the just price is the *communis aestimatio*, the common estimation — what the thing fetches commonly, openly, here and now. The civilians had a maxim for the outer claim, rooted in the Digest itself: *res tantum valet quantum vendi potest*, a thing is worth as much as it can be sold for; the theologians accepted it on conditions that amount to this whole chapter in miniature. Alexander of Hales already glosses the just estimate as the price at which a thing “is sold commonly in that city or place”; Albert the Great, as the marketplace’s estimation “at that time”; Bernardino of Siena — a Francis-

can preacher not otherwise soft on merchants — as the valuation “made collectively by the community of citizens.” Cajetan, commenting on Aquinas, states the mature doctrine with lawyerly compression: the just price is what can be obtained from buyers, “assuming common knowledge and in the absence of all fraud and coercion.” At Salamanca, Francisco de Vitoria lectured his way through question 77 and drew the operational rules: where sellers are many and dealing is clean, the common price governs and a seller may take it without scruple; where sellers are few, or collude — monopoly, he held, is fraud owing restitution — the “market price” has stopped being evidence of anything. And Tomás de Mercado, writing for the merchants of Seville’s Indies trade, defined the just price as what is paid publicly in the square, “this week and this time,” with neither force nor deception — while admitting, with a working friar’s honesty, that it shifts like the wind. His colleague Covarrubias supplied the theory in one image: wheat is dearer in the Indies than in Spain though the wheat is identical, because value lives in men’s estimation of it, not in the grain.

This is why Aquinas could resolve his hardest disclosure case the way he did. The merchant who reaches a famine port with grain, knowing a fleet of sellers is a day behind him, may sell at today’s high price without announcing tomorrow’s glut; today’s price is the genuine common estimate of the goods as things now stand, and prophecy, Aquinas concludes, is not a debt of justice — though telling, or discounting, would be “exceedingly virtuous.” The philosophers had argued the case since Cicero, and the tradition’s answer stakes out its market-price doctrine exactly: the common estimation binds and protects both ways, and a merchant is not obliged to bargain against his own honest information. Islam, from its own direction, arrived at a matching restraint on the other actor in the drama — the state. When prices spiked in Medina and the people asked the Prophet to fix them, he declined: “It is God who fixes prices,” Anas reports him saying, “who withholds, who gives lavishly and provides; I hope to meet God with no one among you holding a claim against me for a wrong in blood or property” (Abu Dawud 3451). The same tradition that outlaws the planted bidder and the intercepted caravan refuses to legislate the number itself. Read together, the two rulings are one doctrine: police the process, respect the verdict. A price formed by many hands dealing openly deserves deference precisely because nobody made it; a price formed by fog, fraud, collusion, or a cornered counterparty deserves none, because somebody did.

Which brings this chapter to the third price, and to the register this book reserves for it.

The gravest failure of a market’s conditions is not a lie about the goods. It is need on one side of the counter. Aquinas faces the case in the first article, and his analysis deserves to be walked through slowly, because it draws the finest line in the whole doctrine. Sometimes, he says, the buyer has great need of a thing and the seller would be genuinely hurt by parting with it — think of the plow a craftsman still needs, or the grain a household was holding for the spring. There the just price may rise above the thing’s open-market worth, because it is honestly pricing two things: the object, and the real loss its owner suffers in letting it go. A seller may charge for his own sacrifice. What he may not do is charge for the buyer’s need — where parting costs the seller nothing, the buyer’s desperation entitles him to not one coin, “because the advantage accruing to the buyer is not due to the seller.” And then the sentence that carries the whole tradition: “no man should sell what is not his.” The buyer’s need belongs to the buyer. It is a fact about his hunger, his sick child, his burning field; the seller did not create it, does not own it, and cannot convey it — and a price calibrated to it is therefore payment for nothing the seller ever gave. Chapter 7 heard the identical finding on the loan side, and Aquinas himself joins the two questions with a quiet observation in the usury article: the borrower who pays usury “does not give it

voluntarily simply, but under a certain necessity.” Necessity is where consent goes thin, and where consent goes thin, the strong party’s price stops being evidence of anything except the strong party’s strength. Vitoria, centuries later, gave the thinness a name the schools kept — consent under necessity is “mixed voluntary,” half chosen, half compelled, like the sailor on an old page of Aristotle’s *Ethics* who heaves his cargo into the storm; and Vitoria drew the consequence for goods of basic need in scarce supply: there, the seller’s freedom to take what the desperate will pay simply ends. The famine hoarder, selling bread at hunger’s own price, is every moralist’s standing example, and none of the three traditions has a kind word for him.

The Talmud preserves, at the same folio as the verbal-wrong law, a voice that felt the problem’s other edge. Rabbi Yehuda ben Beteira taught that one who sells a horse, a sword, or a helmet in wartime is exempt from the overreaching rules, “because they hold life” — a man fleeing a battle values the horse beyond any market, pays accordingly, and cannot be said to have been deceived about what it was worth to him. The baraita records the view; the Mishnah’s own list of exemptions, note, resisted every attempt to widen it, the Sages answering another sage’s proposed additions — Torah scrolls, animals, pearls — with “only these.” The tension between those two instincts is real and the tradition let it stand: the desperate buyer’s payment does tell the truth about his valuation, and it is precisely the truth no seller is entitled to harvest. Aquinas, asked the same question, answers without the tension — the advantage is the buyer’s, not the seller’s to sell — and on this the three traditions’ mature doctrine speaks as one. Whatever their differences of idiom, the three doctrines converge here, at the man who must buy: his necessity is a claim on the community’s mercy, chapter 2 found, and it is never, in any of these rooms, a legitimate object of sale.

The Ancestor of the Term Sheet

It remains to say what this chapter’s doctrines become when the market stall grows into a capital stack — compactly, because two later chapters do the technical work.

Modern finance rarely cheats on the tomatoes. Its exchanges fail the old tests in subtler ways, and each of the three failure modes has kept its ancient shape while changing clothes. The lie survives as misrepresentation law knows it — and as the softer arts of the selectively assembled data room. The squeeze survives wherever one party arrives at the table unable to walk away: the founder with six weeks of payroll, the small suppliers of a dominant platform, the borrower in a season the lenders own. And the fog — *gharar*’s territory — has become the most sophisticated of the three, because modern instruments can manufacture it wholesale. A contract eighty pages long, cross-referencing four side letters, is perfectly disclosed and largely unseen; complexity does at scale what the pebble-toss did in the bazaar, which is to obtain a real signature on an outcome the signer has never actually beheld. The traditions’ verdict on such consent was written above: it is the signature on the road, given before the seller has seen the market.

Nowhere does the fog gather more thickly than in the waterfall — the clauses that fix the order in which a company’s eventual value flows out to its claimants. A founder who owns thirty percent of her company’s shares may own thirty percent of almost none of its plausible futures: liquidation preferences, participation rights, and their kin can arrange that in a sale at anything below the promised heights, the preferred money exits whole, and enriched, before the common stock — the founders, the employees — collects at all. Every clause was in the documents. What was never in the documents is the shape of the

payoff — the thing a Genoese notary would have read aloud on the quay — and chapter 37 will show, with arithmetic, how thoroughly a stack of individually reasonable clauses can reproduce the payoff of instruments this book's traditions forbade by name. The doctrinal work belongs to chapter 23, where *gharar*, *ona'ah*, and defective consent are brought formally into the present and the term sheet is read as what it is, a moral document. Here it is enough to fix the principle those chapters will apply: price justice presumes informed, uncoerced consent, and an instrument complex enough to defeat one party's comprehension has recreated, by engineering, the blindness the fog rules exist to forbid.

Seen from this chapter, then, the modern apparatus of transparency stops looking like red tape and starts looking like ancestry. The prospectus that must disclose what the seller knows; the cooling-off period that lets a signature ripen into consent; the plain-language rules; the rescission rights; the anti-fraud statutes that reopen completed bargains — every one of them is a working translation of something this chapter has already read. Aquinas's third article is a disclosure regime; the Mishnah's merchant-or-relative window, a statutory review period complete with sunset. The option granted to the intercepted caravan works as rescission for information asymmetry thirteen centuries before anyone owned the phrase, and the rescript to Lupus reads today like price-based consumer protection with a bright-line trigger. The traditions disagreed about much, and about this they did not: a bargain deserves enforcement in exact proportion as both parties could see it whole. That conviction — not any rate, threshold, or formula — is what the sale law of three faiths bequeathed to the law of contract, and chapter 35 will insist on its corollary when the arithmetic begins: a return that cannot say plainly what it is a price of has failed the oldest test in the book. The rule underneath has not changed since the chancery answered Lupus: no one may keep, as profit, what the other side could not see.

Chapter 10 — Risk, Loss, and the Moral Price of Uncertainty

On the fifth of September, 1234, the papacy published the most consequential law book of the medieval Church: the *Liber Extra*, five books of papal rulings gathered and arranged at Gregory IX's command by Raymond of Peñafort, the Catalan Dominican whose confessor's manual chapter 3 met. Deep in the fifth book, under the title *De usuris*, sits a short chapter addressed simply "to Brother R." — usually identified as Raymond himself, receiving an answer inside his own compilation. It begins with the word that became its name: *Naviganti*. To one sailing.

The instrument on the table was the sea loan, and by 1234 it was old commerce. A financier advances money to a merchant bound for the ports or the fairs, on terms the Mediterranean had used for centuries: if the ship is lost, the debt dies with it, and the lender loses everything; if the venture arrives, he takes back his principal with a stiff premium besides. That lender has not hidden from the world: his money stands exposed to storm, piracy, and war, and if the worst happens, he bears it. Elsewhere in their law the churchmen had made exactly such exposure the mark of honest gain — the investor whose capital trades at his own peril, chapter 7 found, may earn from his own property what a lender can never charge on somebody else's. If the word "risk" could ever unlock the wall around the loan, it should have unlocked it here.

Gregory's answer runs, in a close rendering: one who lends a fixed sum of money to a man sailing or going to the fairs, and — because he takes the peril upon himself — is to receive something beyond the principal, is to be counted a usurer.

The premium fell. Then, in its next sentences, the decretal acquitted two other risk-takers. The man who pays ten solidi today against so many measures of grain, wine, or oil next season, where it is genuinely doubtful whether the goods will be worth more or less when they come due, "is not on this account to be reputed a usurer." And the merchant who sells cloth, grain, wine, or oil on credit, above today's price, is excused "by reason of this doubt" as well, provided he was not otherwise going to sell at the time of the contract. One chapter of law had sorted three risk-takers, condemned one, and cleared two — and the condemned man is the one whose exposure looks most heroic.

Modern finance would flunk this exam, because modern finance has one answer to the question of why capital earns, and it arrives before the question is finished: compensation for risk. The lender charges because he risks; the investor's return rewards uncertainty borne; the premium on anything is the price of the danger in it. The word does all the work, and in most rooms it is never asked to show its papers. The traditions this book reads took risk with total seriousness — their entire licensing system for capital stands on it — and for exactly that reason they refused to let the word operate as an incantation. *Naviganti* belongs to the grammar this Part has been assembling: what a loan is (chapter 7), what may not ride on one (chapter 8), what a price must answer to (chapter 9), and now the closing question — what risk actually licenses, and what it never can. The answer can be stated in advance, because the whole chapter is an unpacking of it. Gain may be earned from a peril genuinely kept; nothing may be charged for a peril given away. The traditions never paid for the word; they paid for the position.

The Peril of the Principal

The canonists' definition of usury was short enough to chant: whatever is exacted beyond the *sors* is usury. *Sors* is the principal, and the word deserves a moment: Latin used it first for the lot a man draws — the fall of chance itself — and the language of business settled on it as the name for capital lent. So the scholastic term of art this section walks, *periculum sortis*, the peril of the principal, carries a reminder in its own grammar: money committed to the future has been handed to chance. Whose chance is the entire question.

Chapter 7 laid out the anatomy this doctrine stands on. In a *mutuum*, ownership passes; and peril rides with ownership — *res perit domino*, the thing perishes to its owner — so the flood that takes the borrower's barn takes the borrower's grain, however recently it was the lender's. The borrower stands at the top of the Mishnah's ladder of keepers, liable even for sheer accident (Bava Metziah 7:8): what he holds is his, and what he owes is arithmetic no fire can reach. The partnership inverts the drawing. He who entrusts money to a merchant by way of partnership, Aquinas rules in the usury question (ST II-II, q.78, a.2, ad 5), has not transferred ownership at all; the money trades at his peril, and he may therefore take a share of the gain as from his own property — *tamquam de re sua*. Roman law had already refused the costume version: a "partnership" in which one party has arranged to be incapable of losing is void, the Digest's lion's partnership (17.2.29.2), whose story belongs to Part IV. Assembled, these pieces make *periculum sortis* the hinge of the whole classification. Ask where the principal can die, and you have identified the contract — and judged its yield. In a partnership the *sors* can sink, burn, or rot in a warehouse, and its death is the investor's loss; that standing possibility is the ground of his share in the gain. In a loan the *sors* cannot die at all, because it has stopped being a thing and become a number. The lender holds no stake in any venture; he holds a claim against a person, and claims do not drown.

But — the schoolmen pressed, being schoolmen — the lender does fear something. The borrower may fail. The claim itself may prove worthless on the appointed day. Is that fear not also a peril of the principal? Here the doctrine drew its finest line, and the modern debate still runs along it. The peril that licenses gain is the owner's peril: exposure to what the world does to your own property while it remains yours. The lender's danger is different in kind — a danger not of the thing but of the claim, the possibility that the personal obligation he holds proves uncollectible. The danger is real; but it is a danger *of* the borrower's failure — his insolvency, his ruin, the emptiness of his hands at the appointed day. To charge for the peril of the claim is to price the borrower's own fragility back to him, and the charge lands heaviest exactly where the fragility is greatest: the man most likely to fail pays the highest premium for the privilege of being doubted. Chapter 7 found the borrower's need at the bottom of the naked interest charge; the "risk premium" is the same finding in actuarial dress, and chapter 9's rule reaches it — nothing may be sold that the seller does not own, and the borrower's precariousness is a fact about the borrower, never an asset of the lender's.

The handbooks of the canonists and confessors eventually listed danger to the principal among the candidate "extrinsic titles" — the reasons alongside a loan (chapter 7 marked the door) that might justify recovering something beyond it — filed with actual loss suffered, gain demonstrably interrupted, and the agreed penalty for default: *damnum emergens*, *lucrum cessans*, *poena conventionalis*, *periculum sortis*. Of the four, the risk title proved the most treacherous, and for a stateable reason: the other titles compensate events — a documented loss, a broken date — while an uncapped risk title prices a permanent condition of lending as such, and so quietly reopens everything the definition had closed. The

long quarrel over when, if ever, danger to the principal can be separately and honestly compensated belongs to Part VI. What belongs here is the baseline the quarrel presupposed, and Naviganti is its plainest statement.

For the sea lender's peril was real — nobody denied it — and it was partial. He stood under exactly one hazard, the catastrophe: storm, wreck, capture. Against every ordinary fortune of the venture he was armored. If prices broke, if the cargo sold at a loss, if the voyage limped home having eaten its own profit, his premium was owed in full, out of the merchant's other goods if need be. His gain was written before departure — fixed in size, indifferent to outcome, contingent only on there being a debtor left to pay it. That is a creditor's payoff with a disaster clause attached, and the Church declined to let the clause consecrate the payoff. The rubric that medieval editors set over the chapter says it with lawyerly finality: he who receives from the debtor anything beyond the principal is a usurer, *even if he takes the peril upon himself*. The sentence has troubled readers for eight centuries, because it seems at first to punish the very exposure the doctrine elsewhere demands; but the refusal is the doctrine. Risk enters the moral ledger through ownership or not at all. The peril that pays is the peril of a thing that remains yours, borne whole, with gain and loss alike flowing from what actually happens to it; a hazard clause bolted to a fixed claim, however sincere the hazard, is not a stake, and partial exposure does not purchase a partner's rights.

Two footnotes to the decretal complete the grammar. First: what the sea lender genuinely supplied — protection against catastrophe — was a real service with a real price, and European commerce would eventually learn to sell it honestly, as its own contract, priced as protection and nothing else, with no loan hiding inside; that story, and its Islamic counterpart in *takaful*, belongs to chapter 27. Second: the decretal's two acquittals read the same principle from the other side. The forward buyer of grain and the credit seller of cloth were cleared "by reason of the doubt" — genuine two-sided uncertainty ran through their contracts, and each party stood to win or lose as the world turned. Where chapter 9's *gharar* rules void the sale of fog — uncertainty about what is being exchanged — Naviganti honors uncertainty of outcome honestly shared. Risk as such neither sanctifies nor condemns; the question is always what work it is doing in the relationship — fog manufactured to blind a counterparty, a clause bolted on to dress a claim, or a genuine openness to loss, kept by the party who claims the gain.

Near to Gain, Far from Loss

The rabbis ran the same analysis, and then did something none of the other traditions quite did: they graded the postures.

The instrument to hold in mind is the *iska*, Babylonia's investment contract, whose skeleton the Nehardean sages fixed as half loan, half deposit (Bava Metzia 104b). Part IV gives the instrument its full portrait; what matters here is the engine under it. The loan half becomes the working merchant's property and his peril, and every coin of its profit is his. The deposit half remains the investor's — a *pikadon*, held in the merchant's hands but perishing, if it perishes, to its owner — and precisely because that half can be lost, its gain may be kept. The investor's return is licensed by his exposure, sized to his exposure, and forfeit with his exposure. Let him extract a promise that his principal comes back whole whatever the market does, and the deposit dissolves into loan — at which point every grain of his "profit" is *ribbit*, with chapter 7's wall standing in front of it. The halakha runs the license and the peril on a single switch.

Then the grading. A baraita preserved at Bava Metzia 70a surveys the terms an investor might write and sorts them, not into permitted and forbidden, but into characters. One who places his money *near to profit and far from loss* — *karov l'sachar v'rachok l'hefsed* — is a wicked man. Near to loss and far from profit: a pious man. Near to both, or far from both, sharing the venture's fortunes as they come — that, says the baraita, is the quality of everyone: the ordinary honest stance. It is worth pausing on what kind of text this is. The rabbinic literature does not usually do character sketches of term sheets. The tradition here looks straight past an instrument's name to its payoff — to the corner of the possible outcomes where the drafting has seated its author — and judges the shape itself. A man whose contract clusters him beside the gains and walls him off from the losses has, whatever the paperwork says, arranged himself a lender's safety under an investor's yield, and the baraita calls him what the tradition calls the usurer. The pious man has overshot in the other direction, accepting terms tilted against himself to stay unimpeachably clear of the line; the tradition admires him and does not require him. Between them stands the measure of everyone: real exposure, both directions open. Long before anyone drew a payoff diagram, the halakha was reading them — and grading the reader by where his contract seats him.

Then the exception that proves the design. One class of investor is expressly permitted the wicked man's terms: orphans. The court that guards a dead man's minor children may place their inheritance close to profit and far from loss — the permission stands on the same folio — because an arrangement at that careful distance from true ribbit falls only under the rabbis' own fence, the "dust of interest" chapter 3 met, and the Sages waived their fence for wards who cannot fend for themselves. Even for orphans, the biblical wall itself never moved. When one sage reported in Shmuel's name that orphans' money may simply be lent at interest, Rav Nachman's retort has lost none of its temperature: because they are orphans, shall we feed them what is forbidden? Orphans who eat what is not theirs will follow their parent. The folio closes with the machinery of the protected position: Rav Ashi has the court itself place the orphans' money, and only with a vetted man — quiet, uncontested estates, trustworthy, obedient to the law without need of the Sages' ban. Read whole, the design is exact: the shelter of a loss-remote position is real wealth, the tradition knew its worth to the coin, and it rationed that shelter to the one investor who cannot stand in the open. For the grown and the able, a seat near the gains could be had only with a seat near the losses.

One more ruling from the same page, small and nearly perfect. The household of Mar Ukva — minor orphans again — owned a copper kettle, kept and rented out for them at the house of Mar Shmuel, who would weigh it before each rental and weigh it on return, charging both a rental fee and the price of the metal the kettle had lost. For anyone else, the Gemara notes, the general rule forbids the combination. Take rent, and the wear is yours; the copper burns on the owner's account, which is what makes him owner enough to charge rent at all. Be paid for the wear, and the kettle was effectively a loan in the renter's hands — and "rent" on a loan is interest. (The kettle itself, Rav Nachman explains, was innocent even for adults: metal measurably burned away and paid for by weight is copper sold, not a premium, the vessel's remaining fortunes staying on its owners.) Rent or wear, never both — this entire chapter in a kitchen implement: income from property is honest while the property's decay is yours; shift the decay to the user and keep charging, and the charge changes species.

The jurists of Islam wrote the same rules from their own sources and compressed them into the tradition's most quotable maxims. In the *mudarabah* — capital from the *rabb al-mal*, the

master of the capital; skill, travel, and labor from the *mudarib* — monetary loss falls on the capital and only on the capital; the working partner loses his months and his effort, which the law counts as loss enough. No clause may make the *mudarib* guarantee the money back: a guarantee converts the venture into a loan and the “profit share” into *riba*, exactly as the guaranteed *pikadon* collapses the *iska*. In the *musharakah*, where both sides bring capital, profit may follow the agreement — the schools argued the permissible spreads, as schools do — but loss follows capital, strictly and pro rata, and the books carry the rule in a maxim attributed to the caliph Ali himself: profit as the partners stipulate; loss according to the capital. The modern standard-setters have re-inked the line without moving it: today’s Shari’ah standards still bar the *mudarib* from guaranteeing the capital, and still require partnership losses to track capital shares whatever the profit ratio says. Behind the contract rules stand two proverbs of the law. *Al-kharaj bi-l-daman* — the yield goes with the liability — is preserved as a hadith: profit follows responsibility, Aisha reports the Prophet ruling, and the collection files the sentence under the case of a purchased slave who worked for his buyer and was then returned for a defect (Sunan Abi Dawud 3508, graded hasan). The buyer kept those weeks’ earnings, the jurists explain, because the slave, had he perished in the same weeks, would have perished to him: whoever stood to bear the loss had title to the fruit. Its twin, *al-ghurm bi-l-ghunm* — loss goes with gain — circulates in the schools as a juristic maxim rather than a text of revelation, and when the Ottomans codified Hanafi commercial law in the Majalla, both went in among the opening principles: the benefit of a thing is the return for the liability borne for it (article 85); disadvantage accompanies enjoyment, and burden runs in proportion to benefit (articles 87 and 88).

Three legal languages, and by now the book’s most familiar music. The *halakha* runs license and peril on one switch, and grades the postures in between. The *fiqh* titles the fruit to whoever would have borne the loss. The canonists ask where the sors can die. And all three void, in nearly identical words, one specific instrument: the guarantee that leaves the name “partner” on a man the contract has made incapable of losing. Which means all three had already answered, centuries in advance, the argument modern finance regards as its trump.

The Actuary’s Defense

That argument deserves to be stated at full strength, because it is serious, and because the traditions’ answer is more interesting than a refusal. It runs: lending is exposure. Some borrowers fail. A lender who may never receive more than his principal must, across a career of such failures, end poorer than he began; interest is therefore not gain at all but restitution in advance — the pooled price of the failures, spread across the book. A banker states the position in two words, risk compensation, and a credit manual states it in a formula: the fair charge on a loan begins from its expected loss, the chance of default multiplied by what a default would cost, and a portfolio priced at expected loss earns the lender, on average, nothing at all from risk. On this account the interest line in the ledger is an actuarial fact, as morally blank as a rainfall table. Note what the argument concedes at the door: it accepts the traditions’ premise that a charge must answer to something real, and offers the mathematics of expectation as the something.

The mathematics itself is not the issue — the traditions are sometimes accused of innumeracy, and Rome could already average a lifetime. When the jurist Aemilius Macer needed to value bequests of yearly maintenance — support left to freedmen and dependents, worth everything or little depending on how long the pensioner lived — he recorded the schedule in use and credited it to Ulpian: for a beneficiary up to age twenty, count thirty years’

purchase; from twenty to twenty-five, twenty-eight; from twenty-five to thirty, twenty-five; and so on down a sliding scale of shortening expectations (Digest 35.2.68). The purpose was estate arithmetic — valuations feeding the computation of the heir's protected quarter — and the table is, in effect, a life-expectancy curve: proof that pricing uncertain futures by their averages was a working tool of the same legal civilization whose loan law this Part reads. The traditions' quarrel with the actuary was never his multiplication.

The quarrel is with what the multiplication is asked to launder, and it comes in two findings. The first is about the guarantee. The traditions did pay for risk — handsomely, lawfully, in every instrument the last section walked — but they paid for it out of outcome: the investor's compensation for uncertainty was a share in whatever actually happened, collected only if something collectible happened. A guaranteed return cannot be a price of risk, because the guarantee is the subtraction of the risk it claims to price. Watch when the "risk premium" on a loan actually collects. In every season the borrower prospers — every season the feared thing fails to happen — the premium arrives in full, payment for a danger that visited no one. And in the season the feared thing does happen, the premium does not pay out the way an insurer's reserve pays out; it joins the principal in the claim that pursues the failing borrower through whatever he has left. A charge that collects when the harvest comes in and keeps collecting when it fails is less a price of peril than a toll no peril can interrupt. Interest-as-risk-compensation casts each borrower as the underwriter of his own possible ruin — he pays the premium every good year, and if ruin comes, he also absorbs the loss; whatever that arrangement is, it is not compensation.

The second finding takes the argument at its mathematical best: the perfectly fair book, priced at exact expected loss, not a coin of margin anywhere. Across such a book, the interest charges accomplish precisely this: the many borrowers who repay in full fund the losses inflicted by the few who cannot. Painted in one color, that is chapter 3's solidarity pool — members covering one another's catastrophes — except that every flow runs backward. In the burial club at Lanuvium, the common chest existed for the member the disaster struck. In the actuarially fair loan book, the member the disaster strikes is the one participant the pooled premiums do nothing for: his neighbors' payments protect the lender's average, while his own debt survives his ruin and keeps collecting. The arithmetic balances, but in the wrong ledger: actuarial fairness is a property of the lender's book, and no borrower lives in the book; each lives on a single path through time, a fact whose full weight this chapter's last section measures. On that path, the "fair" premium bought its payer nothing but the right to be doubted at a price. And this is the generous version, where the premium really is set by a shy actuary at expected loss. Chapter 9 already reported what prices become wherever consent runs thin. The expected-loss formula is the risk charge's floor and its alibi; the borrower's need remains its ceiling; and the distance between the two is collected from exactly the people least able to audit it.

The Conservation of Risk

Nothing in the traditions requires a lender to be careless — the usury doctrine is sometimes caricatured as a demand that creditors go naked into the world, and it never was. The traditions protected the principal with every honest tool they had. Roman law kept the *pignus*, the pledge, among its basic contracts (chapter 3 shelved it); the Torah's loan law presupposes pledges and spends its severity on their mercies — the millstone that may not be taken, the cloak that must come home by nightfall — rules chapter 4 walked; and the Qur'an, immediately after its great instruction on writing debts down, provides for the traveler beyond the reach of scribes: "If you are on a journey, and cannot find a

scribe, something should be handed over as security" (2:283, in M. A. S. Abdel Haleem's rendering, as throughout this book). The Prophet himself died, Aisha reports, with his iron armor pledged to a Jewish lender for thirty measures of barley (Bukhari 2916) — a plain working pawn securing a plain working debt, and no jurist ever heard *riba* in it. Security for the sors is the loan's own logic. The wall, chapter 7 insisted, defends the principal in both directions.

The moral turn comes one step later, and the traditions took it with their eyes open: every device that protects a claim works by moving a peril, and a peril moved is not a peril borne. Risk, in any venture, obeys a conservation law. The world fixes the size of the possible loss — the harvest fails or it does not, the shop burns or it does not — and no drafting has ever destroyed an ounce of it. What contracts do is assign seats. Collateral moves the loss into the borrower's pledged property: if the debt fails, the lender is made whole out of a thing the borrower stops owning. The guarantor takes the seat beside the borrower, his estate underwriting a venture he does not run. Seniority — the clean modern geometry of the queue — moves the loss backward, onto whoever stands behind you: the junior lender, the unsecured supplier, the employees owed their wages, the shareholders; "senior" means, precisely, that someone else's claim burns before yours. Each device is, in its place, legitimate, and some are mercies. None of them shrinks the loss by a coin. The modern participle "de-risked" describes a delivery, not a disappearance.

And therefore the kettle rule scales. An income justified by peril dies with the peril — the owner may charge rent while the copper burns on his account, and not after. A claim that has armored itself, collateral posted, guarantees signed, seniority stacked, may be prudent, honest, even charitable; what it may not do is keep charging for the danger its own paperwork certifies it no longer holds. Protection or premium: a lender may take either, and taking both is charging twice for one uncertainty — once to remove it and once for bearing it — the double sale chapter 7 met at the bottom of the naked loan, resurfaced in a risk manager's vest.

The modern capital stack runs the old devices in series and gives the result a paradoxical name: risk capital. An instrument can be equity in the pitch meeting and a debt in the waterfall — preferred shares that exit whole, and enriched, before the common; participation that collects twice; ratchets that reprice yesterday's shares when tomorrow disappoints; redemption rights that turn a "share" back into a claim on a date. Every clause is disclosed, so chapter 9's fog rules sleep; but the stack's payoff, held to the light, is the sea lender's — armored against every ordinary fortune of the venture, exposed only to outright catastrophe, gain written in advance. Chapter 37, "Waterfalls, Preferences, and Synthetic Usury," runs the arithmetic on how completely a tower of individually reasonable protections can rebuild the payoff these traditions forbade by name.

What this section still owes is the other half of the conservation law: the seats the loss is moved to, because someone sits in each of them.

The founder signs the personal guarantee — standard furniture in small-business lending — and the company's "risk capital" is thereby secured on her house. Her unpaid years, the salary she did not draw, the pension she did not fund, are capital in everything but the ledger; chapter 38, "Founder Labor, Worker Claims, and Invisible Contribution," takes up what that contribution is owed. The workers hold no instrument at all and absorb the variance first: when protected capital must be made whole, payroll is the adjusting variable, and people who were offered none of the upside discover they had been carrying downside all along. The mission absorbs it next — the enterprise steered, quarter by quarter,

toward whatever makes the protected exit likelier, until the purpose the capital was raised to serve has been spent as a buffer for it. And past the company's edge the loss lands on people who never saw the documents: the town that keeps the emptied plant, the suppliers paid last or never, and — where the protected claims grow large enough to frighten governments — the public, whose rescue of the creditors is the conservation law operating at national scale. Part VIII will name names. The grammar is already complete: the risk was never destroyed, only moved to seats that could not refuse it, while the fee for bearing it was collected somewhere else.

One Path Through Time

There remains the deepest version of the actuary's comfort, and it needs this Part's last word, because the traditions answer it with their whole design rather than with any single rule. It says: grant everything above, and still, across the whole economy and the long run, priced risk comes out fair — losses real but scattered, premiums pooled, the system's expected outcome sound. The modern name for what is wrong with that comfort turns on a distinction this book will lean on to its final chapters: the difference between an ensemble average and a time average.

An ensemble average is computed across many parallel cases at one moment — a thousand borrowers, a thousand harvests, a thousand identical wagers side by side, summed and divided. It is the natural mathematics of anyone holding a book of other people's ventures, and for him it is truthful: he really does live spread across the ensemble, collecting its average. A time average is a different object: what one person's single sequence actually accumulates, season upon season, each result compounding on the last. The two numbers can part company entirely — when they do, the process is non-ergodic — because a sequence, unlike an ensemble, can be ended. Losses on a single path do not scatter; they arrive in order, and some of them are doors that lock. The farmer who loses two harvests back to back does not experience the decade's average harvest. The family that sells its land, the debtor bound into service — chapter 2 stood them at the front of this book — have reached the state the mathematics calls absorbing: the place from which no later average can retrieve you. The chance that a given path ends there also has a name — ruin probability — and on it stands a finding the book's later mathematics will prove: a system of wagers can be favorable in ensemble average and still carry, for those who must live it in sequence, a ruin probability that climbs toward certainty with the years. The average is a fact about the portfolio; the person on the path needs a different number — whether his sequence survives.

Set the fixed claim inside that frame, and its work shows plainly. Interest, collateral, guarantee, seniority: each transfers variance away from the bookholder's ensemble, where it would have been diluted a thousandfold, onto single human paths, where it compounds and can absorb. The lender's book grows smoother and the borrower's life spikier; the instrument does nothing else. This is why the traditions' strangest-looking laws read, from here, like engineering. The pledge that must come home at nightfall, the release of the seventh year, the rule that waiting may earn nothing, the insistence that capital's gain ride capital's own peril — the vocabulary is ours and not theirs, but every one of those rules lowers somebody's ruin probability at the expense of somebody's yield. And the instrument the traditions built squarely for the problem — chapter 3 gave it a place of its own in the taxonomy — is the pool: many small certain payments set against the single ending blow; an anti-ruin reserve, a solidarity reserve, standing between each member's path and the floor. The pool is the one financial structure whose entire purpose is the time average

of its weakest member. That is why a book about usury keeps returning to it, and why chapter 28 — “Ergodicity, Solidarity, and the Mathematics of Mutual Aid” — gives its mathematics a chapter of its own.

The rule, then — the one this chapter was built to state, and the closing stone of Part III’s grammar. No one may charge for a risk already shifted away. In full: a return claimed as the price of risk is honest exactly insofar as the claimant still holds that risk when the return is collected, and the test runs on the payoff, never the name. Name the peril precisely — of what, to whom, in which seasons. Find where it sits once every device in the documents has finished operating — the pledge, the guarantee, the queue, the founder’s signature, the payroll, the town. Then watch whether the return arrives out of the outcome or regardless of it. If the peril named turns out to live somewhere else — in the guarantor’s estate, the workers’ jobs, the junior claims, the public’s rescue — then the charge for it prices a phantom, chapter 7’s phantom, dressed in the one modern word no one thinks to question — a word that, in this grammar, buys nothing without the position.

Chapter 7 defined the loan; chapter 8 chased its disguised benefits; chapter 9 disciplined the price; this chapter has disciplined the alibi. One visit remains before the partnerships: the grammar assembled here belongs to three traditions, and Comparative Interlude I, “Dharma, Risk, and the Regulation of Increase,” asks how far beyond them it reaches. Then Part IV opens the gate in the wall, and the instruments come through it — *societas*, *iska*, *mu-darabah*, *commenda*, the contracts of peril genuinely kept. They are the constructive answer to this chapter’s rule, built by lawyers who had learned its lesson from the inside: the only honest seller of risk-bearing is the one still bearing it, and no one may price a peril he has already handed away.

Comparative Interlude I — Dharma, Risk, and the Regulation of Increase

In the third book of the Arthaśāstra — ancient India’s treatise on statecraft, ascribed by tradition to Kautilya, the Brahmin chancellor said to have set the first Maurya emperor on his throne around 320 BC, and read by modern scholars as a composite that reached its present form in the first centuries of our era — there is a chapter on the recovery of debts, and near its head stand four sentences that amount to a complete philosophy of credit. Here they are in R. Shamasastri’s translation: “An interest of a pana and a quarter per month per cent is just. Five panas per month per cent is commercial interest. Ten panas per month per cent prevails among forests. Twenty panas per month per cent prevails among sea traders.”

Read the tariff slowly, because most of this interlude is folded inside it. Fifteen percent a year is *just* — the Sanskrit word is *dharmya*, the rate that accords with *dharma*, with right order. Sixty percent a year is commercial — *vyavaharika*, the rate of business. One hundred and twenty percent is the lawful price of money that must cross wild country to come home, and two hundred and forty percent is the lawful price of money that goes to sea — *samudra*, the sea-traders’ rate. Then come the teeth. Lenders who exceed the tariff are to be “punished with the first amercement,” and — a detail I know no parallel to in the Western codes — so are the people who stood by and listened: the “hearers of such transaction” are fined half as much apiece. A few lines later the chapter states its stakes as a principle of government: the dealings of creditors and debtors are matters on which “the welfare of the kingdom depends,” to be kept under standing scrutiny.

A reader who has come through the last three chapters will feel the ground shift underfoot. Chapter 7 walked the wall: three traditions that examined the loan’s anatomy and concluded that no increase at all could honestly ride on it. Chapter 8 watched the wall patrolled outward into gifts, favors, and flattering words; chapter 9 found the same civilizations refusing to let even an honest sale harvest a buyer’s blindness or need. Against that landscape the Arthaśāstra’s tariff looks, on first reading, like everything Part III has stood against — interest not merely tolerated but graded, published, and enforced. The first reading is wrong, and the way it is wrong is the reason this interlude exists.

Hindu Dharmaśāstra belongs in a history of usury because it asked the very question the book’s three traditions asked — what may increase lawfully, from whom, by how much, and at whose cost — and gave the other structurally serious answer. Where Jerusalem, Rome, and Mecca built a wall, the lawbooks of India built a measure: increase lawful within bounds, graded by circumstance, stopped by ceilings, and policed by the king. If the history of usury is told only by the prohibiting traditions, moral seriousness about money quietly becomes a synonym for prohibition, and every regulated credit system since looks like a lapse from grace. The Sanskrit lawbooks are the standing proof of a second architecture — older than most of the prohibitions, elaborated with equal rigor, and haunted by the same fear of the devouring lender. This interlude reads that architecture from inside, in its own categories, before asking what the rest of the book may carry away from it.

The Lawful Increase

The literature is a shelf rather than a book, and the shelf spans something like a thousand years. Its oldest stratum is the Dharmasutras — terse manuals of right conduct from the last centuries before our era, of which Gautama's, Baudhayana's, and Vasishtha's will concern us here — followed by the great verse codes: the Manusmṛiti, the "Laws of Manu," composed by the second or third century of our era on the scholars' best reckoning, and the Yajñavalkya Smṛiti, later and tighter, whose medieval commentary, the Mitakshara, remained working authority in Indian courtrooms into modern times. The Arthaśāstra stands a little apart, a manual of *artha* — wealth, power, statecraft — rather than of dharma proper; but on debt the two genres speak one language, down to the numbers.

All of it rests on something older still. The Sanskrit word for lending at interest, *kusida*, is already in the Vedic ritual texts, and the usurer — *kusidin* — turns up in the Shatapatha Brahmana, centuries before any surviving law code; historians of the literature trace the vocabulary of debt back to the Rgveda itself. Deeper down lies a fact about the religious imagination: in the ritual books' famous accounting, a twice-born man arrives in the world already owing — study to the seers, sacrifice to the gods, a son to the ancestors — so that existence itself opens in arrears. The word for debt, *rina*, covers both registers; what you owe the moneylender and what you owe the cosmos share a noun. A civilization that pictures life as born indebted was never going to file the repayment of loans under mere commerce, and it did not: in the lawbooks, paying one's debts is dharma, and the duty, as we shall see, outlives the debtor.

Increase itself received the same name the Semitic languages chose. Yajñavalkya's term for interest is *vṛiddhi*, "growth" — a third language family reaching independently for the metaphor that chapter 1 traced through the Sumerian goat-kid and the Greek *tokos*, increase named for the way living things increase. And lending at that increase was, within bounds, simply lawful. Manu lists lending at interest among the seven lawful modes of acquiring property, in the same breath as inheritance, purchase, and the wages of work (10.115), and again among the ten livelihoods open to anyone in times of distress (10.116). Vasishtha assigns money-lending to the Vaishya — the farmer-merchant order — as part of his ordained calling, alongside agriculture, trade, and cattle (2.19). Even the number is stated with remarkable unanimity across the shelf. Gautama fixes the legal interest for money at five mashas a month on twenty karshapanas (12.29) — the commentators do the arithmetic: one and a quarter percent a month, fifteen percent a year. Vasishtha announces the same rate in his own name: "Hear the interest for a money-lender declared by the words of Vasishtha" (2.51; the English here, as for Manu and the Dharmasutras throughout this interlude, is Georg Böhler's). And Manu adopts it citation and all, allowing the lender "the eightieth part of a hundred" monthly, the increase "allowed by Vasishtha" (8.140). One lawbook citing another by name, across what may be half a millennium: the tradition understood itself as a single long conversation, and on this number it never wavered. Fifteen percent a year was, for context, gentler than the twenty percent that silver loans carried as standard in Hammurabi's Babylon (chapter 2). The *dharmya* rate was not an indulgence; it was the ancient world's ordinary credit, trimmed.

Yet the same shelf despises the usurer, in terms as hot as anything in the sibling traditions. The lawbooks' word is *vardhushika* — built, like *vṛiddhi*, on growth — and it does not mean everyone who takes interest; it means the man who lives by increase beyond measure. Vasishtha forbids a Brahmin or a Kshatriya to lend at interest in the usurer's fashion (2.40); quotes a verse extending the name to the commodity profiteer, who buys cheap, sells dear, and is "called a usurer and blamed among those who recite the Veda" (2.41);

and then quotes the tradition's most extraordinary sentence on the subject, in which the crime of usury is weighed, in a divine balance, against the gravest crime this literature knows. Brahman the Creator, the verse says, "weighed in the scales the crime of killing a learned Brahmana against the crime of usury; the slayer of the Brahmana remained at the top, the usurer sank downwards" (2.42). The heavier pan sinks. In a Brahmanical lawbook nothing outweighs the killing of a learned Brahmin — and the usurer's pan went down. Manu keeps the same figure at arm's length in the small print of ritual life: the *vardhushika* stands among the Brahmins who may not be invited to the ancestral rite (3.153); food given to a usurer, says a nearby verse, finds no place in the world of the gods (3.180); usury sits in the catalog of secondary sins (11.62). And the upper orders' access to the trade is rationed by station: a Brahmin or Kshatriya must not lend at interest at all, Manu rules — except in distress, for sacred purposes, and then only "to a very sinful man at a small interest" (10.117); Vasishtha concurs, confining the permission to the utterly wicked and letting it take double on gold and treble on grain and no more (2.43–44). The law of extremity opens the door a crack, names the borrower it may be practiced on, and posts a limit even there.

Hold the two halves together, because together they are the doctrine. Lending at increase is lawful — listed beside inheritance and honest work, assigned to a *varna* as a calling, priced at the eightieth part. And the usurer is damned — outweighing the Brahmin-slayer in the scales, barred from the rites, his food unfit for the gods. The halves do not collide; they define each other. What the tradition blesses is increase within the measure; what it curses is increase beyond it; and the boundary between the two is not a sentiment but a number, published in the code. Manu's verse on over-rate lending supplies the grammar of the whole system: interest stipulated "beyond the legal rate" is unrecoverable in court, "a usurious way" of lending that the law refuses to touch (8.152). Within the rate, lawful increase; past it, usury. English law would not stumble into that distinction, as chapter 1 told it, until the statute of 1545 legalized ten in the hundred and left "usury" as the name for the excess. The Sanskrit lawbooks had been running the same grammar for well over a thousand years by then, inside a sacred law, with heaven's own scales behind it. A ceiling is a prohibition that has learned to count.

The Grammar of Lawful Debt

Whatever separates the Indian lawbooks from the traditions of Part III, they drew the loan's anatomy the same way, and a reader of chapter 7 will recognize every bone. Recovery of debt heads Manu's list of the eighteen titles of law under which all litigation falls — "the first is the non-payment of debts" (8.4) — before deposits and pledges, sales, partnerships, boundaries, assault. Civil law begins at the loan. The claim is for a principal; it matures across time; it binds the borrower personally — and not only the borrower. Yajnavalkya makes the duty concrete: the father dead, or long abroad, or broken by incurable illness, his debt falls to the son, and failing the son, to the grandson (2.50–51); Kautilya's debt chapter says it with the old word attached — the sons of a dead debtor pay the principal with its *kusida*. The lawyers of British India would later call this the son's pious obligation and trace it straight to these verses. But the lawgivers filtered what may descend, and the filter tells you what they thought debt was for: sons are not bound by a father's debts of liquor, lust, or gambling, nor by his unpaid fines and tolls, nor by idle promises (Yajnavalkya 2.47); Gautama's list of debts that die with the debtor likewise shields the heirs from his vices (12.41). An obligation that reaches across generations reaches, on these terms, only if it was honest to begin with.

Security is drawn with equal care, and in the two registers chapter 3 taught. The pledge — *adhi* — comes in a custody form and a use form, and the codes legislate the difference exactly where the Roman lawyers drew it for the *pignus*: by asking who eats the pledge's fruit while the debt runs. A pledge given for keeping may not be touched; the creditor who uses it by force gives up his whole interest and makes good the wear (Manu 8.144), and even unauthorized use costs him half (8.150). Where the pledge is fruitful by design — a field that crops, a cow in milk, a thing whose keeping is itself an income — the loan bears no interest at all (Manu 8.143; Gautama 12.32). The reasoning runs on the oldest rule in this book: a charge must name the real thing it prices. The usufruct pledge already pays the lender in fruit; a meter running beside it would charge twice for one waiting. Yajñavalkya sorts the whole field accordingly — the custody pledge, the enjoyment pledge that can never be forfeited (2.58–59) — and surrounds it with debtor protections of surprising delicacy: a pledge is never lost by mere lapse of time (Manu 8.145); a creditor who refuses a tendered repayment watches the interest stop, the debtor depositing the sum with a middleman and walking away from the meter (Gautama 12.33; Yajñavalkya 2.44); the redeemed pledge must come back, or the creditor stands as a thief. Kautilya adds a roll of persons against whom interest may not accumulate at all — the performer of long sacrifices, the sick, minors, students still in their teachers' houses, the destitute — and cuts off debts neglected for ten years. There are even three named kinds of surety — for appearance, for honesty, for payment — with liability parceled among them and their heirs (Yajñavalkya 2.53–54).

None of this makes the codes a debtors' charter. The same books bless the creditor's self-help with a directness that startles: a debtor who runs to the king complaining that his creditor is collecting the debt on his own authority is fined a quarter of the sum for his trouble and made to pay the rest (Manu 8.176); Yajñavalkya confirms the rule and adds the crown's percentage of whatever the court recovers (2.40–42). The Dharmaśāstra is a working creditor's law, written by men who assumed lending would go on and meant for it to go on in order.

Set the apparatus beside Part III and the shape of the difference comes clean. The instrument is the one chapter 7 anatomized — principal, time, personal obligation, security — and half the guardrails would look familiar to any rabbi or canonist in this book. The difference sits at a single joint. Where the Talmud's governing rule, met in chapter 7, reads *any reward for waiting is forbidden*, Manu's reads: the reward for waiting is the eightieth part, month by month. The two sentences disagree about everything except the thing they agree on most deeply — that the reward for waiting is precisely where a law of money must plant itself, name the object, and govern it. One civilization governed it to zero. The other governed it to a number. Neither was willing to leave it to the parties.

Four Rates for Four Roads

Now the tariff proper, because its grading is the most instructive thing on the shelf.

Kautilya's four rates are not four intensities of appetite; they are four theaters of loss. The dharmya rate — fifteen percent a year — prices money at rest in settled country, lent and repaid under the same roofline of law. The commercial rate, at five in the hundred monthly, prices money that has gone to work in trade, where returns run higher and failures run with them. The forest rate, ten monthly, prices money traveling the wild tracts — *kantara*, the caravan country of tolls, tigers, and men with no tolls to pay — and the sea rate, twenty in the hundred each month, prices money afloat. Yajñavalkya carries the top tiers into the

dharma literature proper: those who go through wild country pay ten in the hundred, he rules, and seafarers twenty (2.38, in my close rendering), adding — a liberality worth remembering later — that parties of any class may instead pay the increase they have set for themselves. The axis of the ladder is exposure. Not the borrower's hunger, and not, on these rungs, his station: the statistical weather over the venture the money rides.

The tradition says so itself, which is what keeps this section from being a modern reader's flattering retrofit. When Vijnaneshvara glossed Yajnavalkya's verse in the Mitakshara around the year 1100, he explained the premium in so many words: these borrowers take the money into places where there is fear of destruction of life and wealth alike, he writes, and the lender may take the high rate because even the principal itself stands in doubt (my close rendering). There is the modern theory of the risk premium — expected loss priced into the promised return — stated as settled law in a commentary that Indian courts were still treating as authority in the twentieth century. Modern finance distinguishes the rate that pays for time from the spread that pays for default. The Dharmaśāstra wrote the spread into statute, venture by venture, and capped each tier: the sea rate runs sixteen times the just rate, and the sixteenth is the lid. Chapter 10 asks, of this book's three traditions, when a return is genuinely the price of risk and when "risk" is a word doing costume work; the Sanskrit codes give the bluntest answer on record — publish the risk grades, attach them to roads rather than to persons, and fine everyone in the room when the grade is exceeded.

Two honesty clauses belong here. First, the codes do not say how the sea loan behaved when the ship was lost — whether the claim died with the cargo, as it did in the Greek sea-loan the canonists would one day argue over, or survived against the borrower and his sons like any other debt. The verses price the fear; they do not narrate the wreck. Second, the premium compensates the lender's average, across many caravans and many sailings; it does nothing for the particular borrower whose venture failed and whose obligation, in this tradition as in the others, descends. A risk-graded rate protects the lending; it does not, by itself, protect the ruined. That distinction — between compensating an average and sparing a person — is exactly the gap into which the sibling traditions poured the pool and the partnership (chapters 3 and 11), and Part IX will need both ideas.

Kautilya's chapter even holds the neighboring instrument in its hand. Interest on trading stock, it rules, shall be one-half of the profit, settled as each year expires — a profit-share standing calmly in the middle of the debt statute. Where the return is graded by the venture's hazard, the loan has already begun edging toward the partnership the other traditions preferred; the Indian codes simply kept both on one shelf and priced the difference.

One more verse belongs to the sea. In Baudhayana's Dharmasutra, the list of deeds that cost a twice-born man his caste opens with a specific item: "making voyages by sea" stands first among them, ahead of stealing a Brahmin's property and false witness about land, and the prescribed penance runs three years (2.1.2.1–2, 10). The same ocean that carries the statute's highest lawful rate is, on the same shelf, the place where a Brahmin's world ends. The rate map and the purity map agree about where order gives out; they price the crossing differently — one in interest, one in penance.

Around the whole ladder stands the state. Over-rate lenders are fined; the bystanders who heard the bargain are fined half; the over-rate stipulation is void in court (Manu 8.152); and the field of creditor and debtor is declared a matter on which "the welfare of the kingdom depends." Rates in this system are not discovered but declared. What that declaration can teach a modern instrument-builder, the last section will collect — after the ladder's darkest

rung is faced.

The Price of a Person

So far the ladder has graded ventures. One rung of it grades human beings, and nothing in this interlude matters more than stating that rung exactly.

Manu, having given the pledge-secured rate, continues: “Just two in the hundred, three, four, and five, he may take as monthly interest according to the order of the castes” (8.142). Yajnavalkya compresses the same rule into a line — an eightieth part monthly where there is a pledge; otherwise two, three, four, and five in the hundred, by the order of the varnas (2.37, my rendering) — and Vasishtha carries it as received wisdom, quoting the two-three-four-five verse “as has been declared in the Smṛiti” (2.48). The rule is pan-Smṛiti; no single text owns it. Spelled out, it says: where a loan is unsecured, a Brahmin borrower pays two percent a month, a Kshatriya three, a Vaishya four, a Shudra five. Twenty-four percent a year at the top of the social order; sixty percent a year at the bottom. Same coin, same lender, same absence of collateral. The borrower’s birth is the rate card.

The commentators transmit the rule without discomfort. Medhatithi, Manu’s great early commentator, glosses it flatly: from the four castes, the Brahmana and the rest, the lender takes the four rates respectively. The Mitakshara specifies the domain — these are the rates where no pledge is given — and the specification sharpens rather than softens the thing: the moment the law’s eye leaves the collateral and falls on the person, what it prices is the person’s rank. A modern reader can assemble an actuarial defense, and some modern readers have — unsecured lending prices the borrower’s means; the lower varnas were, by the same law’s design, the poorer; the ladder merely shadows the distribution of wealth. The texts offer no such argument. Their stated axis is birth: *varnanam anupurovashah*, in the order of the classes. Where the lawbooks explain other rates, they explain them by hazard — the Mitakshara’s fear for life and wealth. Here they explain nothing, because inside the world of the codes there was nothing to explain: the varnas were the order of things, and the price of money bent to that order as it bent to the monsoon.

And the rate card is only the gentlest of what rank does to debt in these books. When an insolvent debtor cannot pay, Yajnavalkya rules that a debtor of low caste shall be made to work the debt off; an impoverished Brahmin is instead to pay by degrees, as means arise, his body left out of the ledger (2.43). Manu grades the same rule by relative rank: the debtor makes good the debt even by his labor, if he is of the creditor’s caste or a lower one; a debtor of higher caste than his creditor pays gradually (8.177). The same literature knows *kayika*, corporal interest, increase paid in body: Manu’s list of forbidden forms includes it (8.153), Gautama’s list of recognized forms names it (12.34–35), and the commentators gloss it variously as interest connected with work or as the milk and muscle of a pledged animal — the tradition argued with itself here, as legal traditions do. But wherever increase may lawfully be paid in labor, and wherever insolvency lawfully converts the low-born debtor’s body into the repayment schedule, the border between debtor and bondsman thins, and in these codes it thins by rank. Chapter 2 stood in this territory with the widow of 2 Kings and the families of Nehemiah, and the register there is the register here: this is the machinery by which a bad season becomes servitude, and here the machinery is graded, bearing hardest on those standing lowest.

Now the comparison this book cannot avoid, and must make with care. Chapter 2 read Deuteronomy’s double rule — no interest on loans to “another Israelite”; “On loans to a foreigner you may charge interest” — and found the visible edge of a family’s internal

welfare law: zero inside the covenant, ordinary reciprocal commerce outside it, the vulnerable resident alien folded within the protections. Set Manu 8.142 beside that verse and the structural kinship is real. Both laws refuse to price a loan as if the parties were anonymous; both write the relationship between borrower and lender into the lawful rate; in both, the identity of the person across the table changes what money may do. And that is exactly why the two must not be flattened into each other. Deuteronomy's line is a boundary between inside and outside — binary, reciprocal, the foreign merchant charging interest in his turn. The varna rule is a gradient running through the inside of one society — fixed at birth, admitting no crossing, and pointed in a single direction, the rate rising as standing falls, whoever holds the money. The one differential lifts the meter off the brother; the other sets the meter's speed by the borrower's birth, slowest for the best-placed, fastest for the least. A boundary of mutual aid and a tariff of rank can resemble each other structurally and remain, morally, different objects; this book ranks no tradition against another, and it also declines the false comfort of pretending the difference away. Unequal moral pricing — persons sorted by birth into different prices for the same coin — is the danger this rung carries, and the tradition's own texts supply the evidence on both sides of it.

Both sides, because the lawgivers plainly felt pressure at this joint. With a pledge, the ladder vanishes: every borrower, of whatever varna, pays the flat eightieth part. By consent, says Yajñavalkya, parties of every class may set their own rate (2.38). And the same shelf that graded persons produced the weighing verse, and barred the usurer from the rites, and rationed the whole trade for its own elite. The pressure never became repeal. The ladder stood as long as the codes stood, and its lesson stands accordingly: hierarchy, handed a rate card, will collect its own maintenance. Any credit system that prices the borrower rather than the venture rebuilds the rung with new labels — wherever the poorest pay the highest rates because they are the poorest, because their lack of collateral and standing and alternatives is itself the thing being priced, the price of money is maintaining the hierarchy it claims merely to measure. Chapter 7 gave the analysis in advance: the borrower's need is a fact about him, not an asset of the lender's. The varna rates are that analysis inverted into statute — lowness itself made pricable — and Part IX's audit will need them as the canonical example of what a rate may never be indexed to.

The Lawful Measure

What, then, does the book carry out of this room?

Begin where the tradition itself would begin: dharma is measure before it is regulation. The word descends from a root meaning to hold, to uphold — dharma is what keeps each thing in its place and each act at its right size — and the lawbooks apply it to interest without strain: there is a right size for increase, as there is for a fine, a tax, a gift. The Arthashastra's name for the fifteen percent rate is dharmya, according-to-right; not "customary," not "whatever the market found." In chapter 9, the book's three traditions, asked what a just price was, pointed to the clean market's common estimate and then policed the conditions that keep it honest. The Sanskrit codes trust the market less and the statute more: the just rate is not discovered in the bazaar but declared in the book, tiered, capped, and enforced against lender, borrower, and bystander alike. Different machinery; same refusal underneath, and it is the refusal this whole Part has been assembling: what the strong can get is not the measure of what they may take. Market absolutism — the doctrine that the agreed number is just because it was agreed — finds no more shelter in Manu than in the Mishnah or the Summa. Someone must be able to say *enough* and mean it enforceably. In these codes the someone is dharma, with the king for its bailiff.

Inside that premise, the retainable inventory is concrete, and it answers to four words. Ceilings: every tier of the tariff is a maximum past which the courts go deaf — the over-rate stipulation is void, the over-rate lender fined, the listeners fined with him. Bands: the system runs on a small published set of lawful rates rather than a single number or an open field — the pledge-secured eightieth, the bounded unsecured rates, the venture tiers — supple enough to price real differences, stiff enough to refuse invented ones. Context: the meter answers to events — accrual stops against the sick, the student, the minor, the destitute; stops when repayment is tendered and refused; is forfeited when the creditor helps himself to the pledge; and compounding as such sits on Manu's forbidden list (8.153), increase upon increase being increase unmoored from anything. Proportionality: the total is capped as well as the rate. Interest on money, Manu rules, "shall never exceed the double" of the principal — run the loan as long as you like, the meter dies when the accumulated increase equals what was lent (8.151); Gautama lets a long-outstanding principal double and then stop (12.31); Yajñavalkya tiers the stops by commodity, two-fold for gold, three for grain, four for cloth, eight for liquids (2.39). Indian lawyers came to call the money rule *damdupat*, and it is no museum piece: courts in the Bombay tradition apply it to this day — at no moment may the interest claimed exceed the principal outstanding. A debt under *damdupat* can be heavy; it cannot be bottomless. Time alone can never carry it past twice itself, which means the one thing compounding cannot do, inside this system, is convert delay into doom. In the vocabulary this book will use when the mathematics arrives, a total cap is a ruin limiter: a rate ceiling slows the borrower's worst case, and a doubling ceiling bounds it.

Which brings the interlude to its modern assignment. When Part IX — "Moral Mathematics: Making the Traditions Computable" — sets out to encode what the traditions know, and chapter 43 sketches its "Moral Geometry Contract Analyzer," the Dharmaśāstra's bequest arrives as a set of dials that none of the book's other traditions states so explicitly. A non-extractive simulator that took this interlude seriously would encode: a ceiling on every permitted return, with over-ceiling terms void rather than merely embarrassing; rate bands tiered by the venture's documented hazard, with the borrower's identity inadmissible as a pricing input; a hard total cap, *damdupat*-fashion, so that no chain of rollovers and accruals can carry any claim past a fixed multiple of what was actually advanced; and automatic accrual stops keyed to hardship and to tender. Price the road, never the man. The four rates for four roads are the retainable core, and the four rates for four varnas are the standing warning — the same architecture of differentiation swung from ventures onto persons, and holding, for as long as the texts governed, the hierarchy that wrote it.

This interlude closes Part III, and it earns its place at the close by widening the Part's claim. Chapter 7 found three traditions refusing the naked loan any price at all; chapter 8 chased the refusal into gifts and words; chapter 9 held even honest sale prices to what both parties could see; chapter 10 weighs risk, the last and best of the reasons a fixed return gives for itself. That grammar is a wall grammar: nothing rides on the *mutuum*, and whatever claims to must name a title outside it. The Dharmaśāstra, asked the same question across the same centuries, answered in a measure grammar: increase may ride, within a published band, under a total cap, at rates answering to the money's real exposure, with the king watching. Wall and measure disagree profoundly, and no chapter of this book will referee them. But stand where they overlap and notice what neither will tolerate: increase answerable to nothing — the rate with no ceiling, the claim with no stop, the price that reflects only what the desperate will sign. On that refusal the two architectures are one tradition. Dharma, in the lawbooks' own sense, is what holds things in their places; in the matter of money it held a line, and the line's inheritance — ceilings, bands, context, proportion — passes to Part

IX with its warning label still attached. Part IV now returns to the door the book's three traditions kept open beside their wall: the partnership, where capital earns its increase the older way, by sharing the fate of the venture it feeds.

Part IV — Partnership as the Great Escape from Usury

Chapter 11 — Societas, Iska, Mudarabah, Commenda

On a summer day in 1156, in a portico by the harbor of Genoa, two men stood before a notary named Giovanni — a public scribe, a *scriba*, whose surviving register happens to be the oldest notarial cartulary in the world — and did a thing merchants around the Mediterranean had been doing for centuries. One of them had money. The other had a ship's berth, a strong back, and an appetite for the sea. The money changed hands; the notary's pen scratched out the terms; and when the traveler came home from Alexandria or Bougie — if he came home — the investor would take back his capital and three-quarters of whatever profit God and the market had provided. The traveler kept the rest. If the venture drowned, the investor's money drowned with it, and the traveler had spent a season of his life for nothing. Giovanni Scriba's cartulary, kept from 1154 to 1164, is stuffed with these agreements, hundreds of them, in a Latin shorthand so compressed it reads like a man writing during an earthquake.

Fourteen hundred miles to the southeast, in the old quarter of Cairo called Fustat, Jewish merchants were storing their worn-out documents in a synagogue attic — the famous Geniza — and among the letters and marriage contracts historians would later find there sits the same handshake, over and over, in Hebrew and Judeo-Arabic. Some of those partnerships followed the form the rabbis of Babylonia had engineered centuries before. Others followed the form of the surrounding Islamic courts, and the Geniza merchants, wonderfully, had a name for the difference: they called the latter *qirad al-goyim* — “the qirad of the gentiles” — as opposed to partnership done according to the *iska*. A Fustat trader shipping flax to Sicily could tell you, precisely, which tradition's rules his money was traveling under.

And centuries before either register was written, the caravans of Mecca had run on the same device, capital from one hand and labor from another, so routinely that when Islam arrived and burned guaranteed increase out of its economy, nobody thought to forbid this — the Prophet himself, tradition remembers, had once carried a wealthy widow's goods to Syria on exactly such terms, and married her afterward.

Three traditions. Three languages, three law codes, three sets of scholars who mostly could not read each other. And one idea, arrived at with the convergent inevitability of birds and bats both discovering wings: if money may not breed as a loan, let it work as a partner. Part III watched Judaism, Christianity, and Islam wall off the loan — the *mutuum*, the *milveh*, the *qard* — as a contract in which any increase is forbidden. This chapter is about the gate all three built in that wall, and left open on purpose. The rule of the gate was everywhere the same: come in as a participant, not a creditor. Own a share of the venture rather than a claim against the person, and let the money sail with the ship instead of standing on the dock demanding payment whether or not the ship returns.

The traditions did not regard this as a loophole. That is the point most modern readers miss, trained as we are to suspect that every exception is a lawyer's tunnel under the law. Partnership was not the tunnel; it was the road the law was built to protect. The prohibition and the permission were two halves of one moral geometry — no gain without exposure; gain from exposure is honest — and to see how deliberately each tradition drew that line, it helps to watch each of them draw it. The three chapters that follow will walk the partnership's anatomy — labor's claim, loss's allocation, profit's definition — but first

the instruments themselves deserve their portraits.

The Money That Stays Yours

Start with Rome, because Christian Europe did. Roman law knew the *societas* as one of its consensual contracts — formed by simple agreement, no ritual required, just consent between partners to pursue gain in common. Roman partnerships ranged from the total — the *societas omnium bonorum*, partners pooling everything they owned down to the last spoon — to the surgical: one venture, one voyage, one consignment of grain, dissolved the day the accounts settled. Commerce ran on the surgical kind. The seventeenth book of the Digest, title two, *pro socio*, is the classical law of partnership, and buried in it at 17.2.29.2 is one of the great sentences of legal history. The jurist Ulpian, writing his commentary on Sabinus, reports an older opinion: Aristo relates that Cassius — a jurist grand enough to have been consul in the year 30 — held that a partnership cannot be formed on terms where one party takes only the profit and the other bears only the loss. Such an arrangement, Cassius liked to say, was a *societas leonina*: a lion's partnership, after the fable in which the lion goes hunting with his fellow animals and then claims every portion of the kill for himself.

Void, said Cassius. Not sinful — Roman jurists did not deal in sin — simply not a partnership at all. The name did the arguing for him. A partnership in which one side cannot lose is a predation wearing the costume of an agreement, and Rome refused to enforce the costume. The rule cut in one direction only: a silent partner who risked his capital while another did the work was no lion — the law's objection was never to unequal contributions, only to a party who had arranged to be incapable of losing. Nineteen centuries before anyone wrote an equation about risk and return, the principle was sitting in the Digest with a fable attached: whoever claims the gain must stand in the way of the loss.

Christian moral theology took that Roman chassis and drove it straight through the usury debates. The difficulty, recall, was the *mutuum*: the loan of money or any consumable, in which — as the medieval wordplay had it — what was mine becomes yours. Ownership of the coins passes to the borrower. And if the money is his, then charging him for the use of it is selling him something that already belongs to him, which is why the scholastics classed usury with theft rather than with sharp bargaining. But the same logic that condemned the loan acquitted the partnership, and Thomas Aquinas states the acquittal with the tidiness of a man closing a ledger. In the Summa's question on usury — ST II-II, q.78, a.2, ad 5 — he distinguishes: the lender transfers ownership and so may claim nothing beyond the principal; but he who entrusts his money to a merchant or craftsman by way of a partnership has not transferred ownership at all. The money remains his. The merchant trades with it, the craftsman works with it, at the owner's risk — and therefore, Aquinas concludes, the investor may licitly seek a share of the resulting profit — *tamquam de re sua*, as from a thing that is his own.

It is worth pausing on how much work that little clause — *at the owner's risk* — is doing. The coins in the merchant's sea chest look identical whether they arrived as a loan or as a *societas* share. The voyage is the same, the cargo is the same, the profit may be the same to the penny. What differs is who is standing under the outcome. The lender has stepped out from under it: storm or no storm, his claim is fixed, and that fixity is exactly what the tradition refused to let anyone sell. The partner has stayed under it. His return is a harvest from his own field, not a toll on someone else's need, and if the field floods he eats nothing. Same coins, opposite verdicts, and the entire difference is the address of the risk.

Later scholastic centuries would stress-test this distinction with the enthusiasm of engineers cracking beams — could you insure the capital, sell your uncertain profit for a certain one, stack three licit contracts into one guaranteed return? — and Part VIII of this book will watch the *contractus trinus* try precisely that. But the baseline held for the whole medieval conversation. Investment was never the sin. The sin was wanting investment's reward with the loan's immunity — wanting, in Cassius's terms, to hunt with the animals and dine like the lion.

Half Loan, Half Deposit

The rabbis of Babylonia got there earlier, and by a characteristically different route: not a treatise but an instrument, not a distinction defended but a contract engineered.

The problem facing the Talmudic sages was practical. Torah law, as we saw in chapter 4, forbids *ribbit* between Jews with a severity that spares no one — lender, borrower, witnesses, even the scribe who writes the bond. But Babylonian Jewry lived by commerce, commerce runs on outside capital, and a community that enforces its credit ban too crudely simply starves its own traders. What the rabbis produced, and what the tractate Bava Metzia debates at length, is the arrangement called *iska*: an investor places capital with an active merchant, the merchant trades with it, and the returns are shared. The Nehardean sages — Nehardea being one of the great Babylonian academy towns — gave the device its lasting legal skeleton, preserved at Bava Metzia 104b: “this *iska* is half loan and half deposit.” *Palga milveh, palga pikadon*.

The formula sounds like an accounting technicality and is actually a small masterpiece of moral engineering. Take the investor's hundred coins and cleave them, conceptually, down the middle. Fifty are a loan to the managing merchant: they are his now, his to risk, and — because they are a loan between Jews — utterly barren; he owes back exactly fifty, and every grain of profit those fifty earn belongs to him alone. The other fifty remain a deposit: the investor's property, held in the merchant's hands, traded at the investor's risk. Their profit is the investor's; their loss is the investor's. The merchant walks the market carrying both halves in one purse — half of it as borrower, half as trustee — and the investor ends up exposed on exactly half his capital, which exposure is what entitles him to a return on it. No part of his gain is *ribbit*, because no part of it comes from the loan half. It comes from his own fifty coins, working in another man's hands, at his own peril. The Talmud even records the rabbis' satisfaction that the split was an enactment “agreeable to both parties”: the merchant gets working capital he could not otherwise touch, the investor gets a lawful yield, and the Torah's wall stands unbreached.

The structure had consequences its architects debated with gusto. If half the merchandise is really the manager's own borrowed property, mused the Nehardeans, then presumably he may sell his half and drink beer with the proceeds — a piece of logic with which Rava, to the relief of investors everywhere, refused to go along: the money was given for business, and business it must do. The exchange is preserved at the same folio, and it is hard not to love a legal literature that stress-tests its financial instruments against the specific scenario of the fund manager heading to the tavern.

One more rabbinic safeguard deserves its place here, because it shows how fine the sages' nose for disguised interest was. The merchant, notice, spends his days working the investor's half as well as his own. If he does that management for free, what exactly is his free labor? It is a benefit flowing to the man who lent him the other fifty — which makes it, the rabbis saw, interest paid in sweat rather than coin. So the Mishnah (Bava Metzia

5:4) flatly forbids setting up a shopkeeper for half the profits unless the investor pays him a wage for his trouble; the Talmud, at 68b, sizes the wage “as an idle laborer” — roughly, what a man would accept to leave his own work and mind yours. Later authorities let the payment shrink to a token, but the principle never shrank: the working partner’s toil is a real contribution, and it must be really paid for. Here, in embryo, is the idea the next chapter will meet in all three traditions under the name *sechar tircha* and its cousins — labor is never free, least of all when pricing it at zero would smuggle interest in through the back door.

The same watchfulness guarded the deposit half itself. Let the investor extract a promise that his fifty coins would come back whole regardless of the market, and the deposit stopped being a deposit: a guaranteed principal is a loan, whatever the document calls it, and any yield on it ribbit. The *iska* worked only so long as the investor’s half genuinely traded at the investor’s peril. Later practice knew a second way to compensate the manager — tilt the profit split in his favor rather than pay a wage — but no drafting could tilt away the investor’s exposure without collapsing the whole structure into forbidden lending. In time the arrangement acquired a standard document, the *shtar iska*, whose descendants we will meet doing heavy modern lifting in chapter 31.

Capital and Caravan

Islam inherited the oldest version of the device and gave it the most explicit theology of labor. Mecca, before the revelation, was a merchant republic: a town of brokers and caravan-masters where fortunes went out on camelback twice a year, and where a standing arrangement matched idle capital with able young men. Tradition remembers that Muhammad himself, in his twenties and known around town as *al-Amin*, “the trustworthy,” carried the merchandise of the widow Khadija to Syria on such terms — her capital, his labor, the profit shared — and that she was sufficiently impressed by the returns, and by the man, to propose marriage. When the Qur’an later declared war on *riba*, it never said a word against this ancient arrangement. The jurists grounded its licitness in consensus and in the Prophet’s own tacit approval: the community had practiced it under his eyes, and he had let it stand. It is a nice irony of comparative law that the contract Islamic finance is proudest of has no founding verse at all — it entered the law the way most good institutions do, by working.

What the jurists did supply was vocabulary and rules. The scholars of Iraq called the contract *mudarabah*, a name they heard inside the Qur’an’s own idiom for commerce — those whom sura 73, verse 20, describes (in M. A. S. Abdel Haleem’s rendering) as “travelling through the land seeking God’s bounty.” To journey striking out across the earth, *daraba fi’l-ard*, was to trade; the *mudarabah* was the contract that let a man do it on someone else’s money. (The same verse, in the same breath, urges the believer to “make God a good loan” — and there, packed into one sentence of scripture, is nearly this whole book: commerce blessed, lending turned toward mercy. We will unpack the “good loan” in chapter 16.) The scholars of the Hijaz, meanwhile, called the same contract *qirad*, from a root said to mean “to cut” — the investor cuts off a portion of his wealth and hands it to the agent. Two names, one architecture, and the architecture is severe. The *rabb al-mal* — the master of the capital — provides money and only money. The *mudarib* provides skill, effort, and time. Profit is divided by a ratio fixed in advance — a half, a third, two-fifths — and the ratio must bite on actual profit: stipulate the investor a guaranteed sum, any guaranteed sum, and you have not made a hard-nosed *mudarabah*, you have voided it, because a guaranteed sum is a loan’s payoff wearing a partnership’s clothes. Loss falls where own-

ership sits: monetary loss eats the capital, and the capital alone. The *mudarib* loses his months, his labor, and his hoped-for share, which the jurists regarded as loss enough. He is a trustee of what he carries, an *amin*, answerable for negligence and for breach of his instructions but not for the sea, the price of pepper, or the will of God. The books filled in the working details with their usual appetite: the investor could scope the mandate to a route, a commodity, or a season, or grant an unrestricted commission; the agent's travel and lodging came out of the venture's purse, but he could not draw a salary alongside his share — pay a man twice, once for his labor and once from the profit, and the profit share stops measuring anything. The *mudarib*'s real collateral was his name. In a trading world built on repeat engagements, an agent who came home with suspicious losses had taken his last commission.

Where both sides bring capital, the contract shades into *musharakah* — the sharing partnership, from the *fiqh*'s broader law of *shirka*, whose varieties the jurists catalogued with taxonomic relish: partnerships of capital and of labor, of credit and of standing, universal mergers of whole estates and narrow ventures scoped to a single bale of cloth — and here the jurists left us a maxim their books attribute to the caliph Ali himself: profit is as the partners agree; loss follows the capital. The second half was unanimous across the legal schools, and it is the load-bearing half. Partners may negotiate profit shares that honor unequal work, unequal skill, unequal risk appetite — on how far those shares could stray from the capital shares, the schools argued, as schools do. But loss was not negotiable. Loss obeys ownership arithmetic: a man who put in a third of the money bears a third of the shortfall, and no contract clause can move his loss onto his partner's side of the table. Partners may bargain over the fruits, but not their way out from under the tree.

And the contract scaled. With the investor's leave a *mudarib* could place part of the capital onward with a second agent, partnership nesting inside partnership the way a caravan's goods split at a junction city; capital placed in Fustat might travel through Aden to the Malabar pepper ports and come home, years later, as an accounting sworn before witnesses. The Geniza preserves the correspondence of exactly such ventures — Goitein spent his last decades editing what scholars call the India traders' archive — and what the letters show is a commercial world of astonishing reach running almost entirely on relationship: no banks, no state enforcement worth the name across most of the route, just family, reputation, and the shared grammar of *qirad* and *iska* binding a merchant in Egypt to partners he might see twice in a decade. The partnership contract was not merely how the pious avoided usury; it was the load-bearing institution of Mediterranean and Indian Ocean commerce for half a millennium.

Set the three systems side by side and the family resemblance is unmistakable. The Roman-Christian *societas* says: the money remains the investor's, so it earns at his risk. The *iska* says: half remains the investor's, so that half earns at his risk. The *mudarabah* says: the capital remains the *rabb al-mal*'s — the very title says so — and so its loss is his and its profit divisible. Three legal grammars, one deep structure: ownership retained, risk retained, gain therefore honest. None of the three traditions would let the investor be paid for time. All three insisted he be paid, if at all, from outcome.

The Sea's Common Grammar

Which brings us back to Genoa, and to the contract the Latin West called the *commenda* — because the Mediterranean is where these three grammars stopped being parallel and started being neighbors.

The commenda was the *societas* built for salt water. In its one-sided form, an investor — the *stans*, the one who stays — entrusts capital to a *tractator*, the one who goes. The tractator sails to Alexandria, Tunis, Constantinople; buys, sells, dodges pirates and customs men; and comes home to an accounting. The standard division gave the stans his capital back plus three-quarters of the profit; the tractator, who had contributed nothing but himself, took a quarter. In the two-sided form — Genoa's notaries called it *societas maris*, the partnership of the sea — the traveler put in a third of the capital against the investor's two-thirds and took home a full half of the profit: his quarter for the labor, as it were, plus his proportional share for his money. In Giovanni Scriba's register it is this bilateral form that dominates the early decades, the young men of Genoa evidently preferring to sail with some skin of their own in the game. Nor were the investors all merchant princes: the registers show widows placing dowry money, artisans venturing savings, clergy quietly present — the commenda was how a town of modest means lent itself to the sea, as a partner rather than a creditor. Loss, in either form, followed the rule every tradition in this chapter has already taught us: money-loss to the money, labor-loss to the laborer. If the ship went down, the stans lost his capital and the tractator lost his voyage, and neither owed the other anything.

Notice what the commenda is not. It is not a loan to a shipmaster, and by the thirteenth century that distinction had papal teeth. The older instrument, the sea loan — money advanced for a voyage, repayable with a hefty premium only if the ship arrived, so that the lender's whole claim sailed with the hull — had flourished on the theory that the lender's assumption of shipwreck risk justified his premium. In 1234, in the decretal *Naviganti* gathered into Gregory IX's *Liber Extra*, the papacy shut the theory down: one who lends to a man sailing or going to the fairs, taking the risk upon himself in exchange for something beyond the principal, "is to be counted a usurer." Canonists have puzzled over the decretal's logic ever since — it seems, at first read, to condemn exactly the risk-bearing the schoolmen elsewhere required — a puzzle that belongs to the *periculum sortis* debates we met in chapter 10. What matters here is the practical effect: the Church had blessed the partnership form and cursed the loan form, and Mediterranean capital flowed obediently toward the blessing. The commenda became the standard vehicle of the medieval boom not despite the usury prohibition but substantially because of it. When the moralists closed the toll road, the money took the ship.

And whose invention was the ship? Here sits one of the most charming unresolved questions in economic history. The commenda surfaces in the Italian ports in the tenth and eleventh centuries looking remarkably like the qirad the Islamic world had been operating for three hundred years — same one-voyage horizon, same capital-and-labor split, same loss rule — and in 1962 the historian Abraham Udovitch put the question directly in the title of a famous article: "At the Origins of the Western Commenda: Islam, Israel, Byzantium?" His suspect was the qirad, carried north by exactly the merchants we met in the Geniza — men who traded across the religious frontier as a matter of routine and kept the paperwork to prove it. Byzantium had its own candidate contract, the *chreokoinonia*; the rabbis, as we have seen, had the *iska*; and the honest verdict of later scholarship is that the parentage cannot be settled, because the Mediterranean's merchants borrowed whatever worked and rarely footnoted their sources. Which is, for this book's purposes, the finding. The lawyers of three faiths, each reasoning from their own scriptures toward the same two commitments — no gain without risk, no charge for time alone — produced instruments so alike that a millennium later, professional historians cannot always tell whose contract they are reading. Fustat's traders could operate under any of them and did. The sea had a common grammar, and its first rule was that capital travels as a partner or not at all.

The Price of Participation

It remains to say plainly what the partnership form buys, morally speaking, and what it costs — because the cost explains the next seven centuries.

What it buys is truth in the payoff. A loan's return is written before the venture begins; a partner's is discovered after it ends. The lender's claim is a fact about the calendar — so many months, so much increase — and the calendar, as the scholastics never tired of observing, belongs to no one. The partner's claim is a fact about the world: it asks what actually happened when human effort met uncertainty, and it pays out of that answer or not at all. Interest, in the traditions' unanimous judgment, prices the borrower's need and calls it the price of time. Profit-share prices the venture's outcome, which is the only thing any investor honestly owns a piece of. That alignment — return tracking reality — is why all three traditions, so fierce against the barren loan, threw the gate open to the fruitful partnership without a flicker of embarrassment.

Run the two contracts side by side on the same venture and the geometry shows plainly. A hundred coins go to sea. As a loan at a fifth, the investor is owed a hundred and twenty whatever happens: if the voyage comes home worth two hundred, he takes his hundred twenty and the merchant keeps eighty; if it comes home worth ninety, he takes all ninety and a claim on the merchant's house for the balance. As a commenda on the standard split, the same two hundred pays him a hundred seventy-five — his capital plus three-quarters of the hundred in profit — and the same ninety pays him ninety, and closes the file. The loan treats the lender best precisely when the venture fares worst, which is to say precisely when mercy would run the other way. The partnership's arithmetic bends with reality, and that bending is a large part of what the traditions meant by justice.

What it costs is certainty, and certainty is a drug with a market. The partnership form forces the capitalist to live inside the venture's actual story, and the venture's actual story is known, in the first instance, to exactly one person: the man with the books. So the whole apparatus of participation runs on verification, and each tradition built the machinery its own way — the rabbis with graded oaths and witness requirements on claims of loss, the fiqh with the mudarib's trustee liability for negligence and overreach, the Genoese with the notarized accounting that awaited every tractator on the quay. And before a profit can be shared it must be defined, which is its own quarrel: what expenses come off the top, whose wage is paid before the split, what a prudent venture holds back against the next voyage. The traditions all discovered, each in its own vocabulary, that profit is not a number lying in the till but a judgment about what the venture owes before its owners eat — a judgment chapter 14 takes up under the banner that profit must come from profit. Monitoring is partnership's tax, and moral hazard its weather: the manager gambling boldly with money mostly not his own, the investor squinting at a loss report he cannot check from shore. None of this machinery ever worked perfectly, all of it cost money, and every florin of friction whispered the same temptation into the investor's ear: wouldn't you rather just be owed the money?

There is the door through which the old sin walks back in. Not brazenly — brazen was forbidden — but by increments of guarantee, each dressed in the vocabulary of partnership. Fix the "profit share" as a sum certain. Insure the capital out of the venture's own pocket. Set the evidentiary bar for proving a loss so high that no loss is ever proven. Every tradition in this chapter will spend centuries fighting exactly these engineerings of its own escape route: the rabbis policing an iska that threatened to harden into paper (chapter 31 tells that story), the fiqh confronting instruments that replicate a bond's payoff in a partnership's

costume (chapter 32), the scholastics disputing the triple contract that assembled a guaranteed return from three separately licit pieces (Part VIII again). The pattern is so regular it deserves stating as a law: whatever gate a tradition opens in its wall against usury, capital will spend the following centuries trying to widen into a road for guaranteed return. Partnership solved the problem of honest gain, and in the same stroke created the problem of dishonest partnership.

Modern readers keep rediscovering the shape without the scruple. The limited partnership at the heart of today's venture capital — passive investors, an active managing partner, profits split by pre-agreed ratio, losses falling on the capital — is the commenda's recognizable descendant, and scholars have traced the lineage. What the modern version quietly reintroduced is everything this chapter's lawyers labored to keep out: preferred returns, liquidation preferences, guaranteed fees drawn win or lose — the lion's clauses, notarized. Part IX will put those instruments under the traditions' lens one by one.

But that is the coming story, and it should not sour this chapter's finding, which the three traditions would each recognize in the others' languages. The *societas*, the *iska*, the *mu-darabah*, the commenda: four names for the one bargain every moral economy of capital keeps offering — profit, gladly, in exchange for genuine exposure. The license to gain was purchased with the willingness to lose. Much of what follows in this book is the story of people trying to get a refund.

Chapter 12 — Capital and Labor in the Same Venture

Somewhere in Roman Palestine, around the turn of the third century, a man with money made a man without money an offer that sounds, across eighteen hundred years, positively generous. Mind my shop. Sell my oil and my wine, keep the books, sweep the floor, and every coin of profit we split down the middle. Half for capital, half for work. A modern reader hunting for the villain in that sentence comes back empty-handed; the Mishnah heard it and reached for the word that this book has been circling since chapter 2. One may not set a shopkeeper up on half profits, rules Bava Metzia 5:4, unless the investor also pays him for his trouble — because without that payment, the arrangement is interest.

Interest paid to whom, for what? Follow the rabbis' arithmetic, which chapter 11 has already half-assembled. Under the *iska's* logic, an investment placed with a working merchant is half loan and half deposit: half the stock becomes the shopkeeper's own, half remains the investor's, traded at the investor's risk. The shopkeeper's daylight hours, though, are not so divisible. He minds the whole shop — his half and the investor's half — and if nothing pays him for tending the investor's half, then his sweat on that half is a free benefit flowing to a man who has also made him a loan. Chapter 8 taught the category: a benefit that travels alongside a loan is interest, whatever it is made of. It can be made of money, of favors, of first refusal on a vineyard. Here it is made of labor. Unpaid work does not disappear; it becomes interest.

That small ruling is the seed of this chapter, and this chapter is the hinge of Part IV. Chapter 11 introduced the partnership instruments as the traditions' great lawful gate through the usury wall: the *societas*, the *iska*, the *mudarabah*, the *commenda* — capital joining a venture as participant rather than creditor. But a venture is made of two scarce things, and only one of them is money. The other is a stretch of a human life: months, skill, attention, standing, shoe leather. Get the price of money wrong and you have usury, which Part III anatomized. Get the price of work wrong and you have something the traditions found equally intolerable and modern finance finds barely visible at all — the partner whose contribution is real, whose loss is real, and whose name appears nowhere in the arithmetic. The three traditions, each in its own legal dialect, refused to let the working partner be priced at zero. Chapter 13 will ask who bears a venture's losses and chapter 14 what may honestly be called its profit; this chapter asks the prior question — what a venture is made of, and what each ingredient may claim. How the traditions answered, and what their answers add up to, is worth the telling, because the answer is about to go missing for several centuries, and Part IX will have to go looking for it.

What Each Brings

Begin with the obvious contribution, since the traditions never denied it: money is a real input. The whole quarrel of this book is with money that pretends to breed by itself; money that feeds an actual venture — buys the stock, fits the ship, pays the rent while the kiln is built — is doing honest work, and every tradition said so by the very act of blessing the partnership. But notice the terms of the blessing, because they are stricter than they look. In the *societas*, Aquinas ruled, the investor earns because ownership never left him: the money trades at his risk, and its gain is his gain *tamquam de re sua*, as from a thing of his own. In the *iska*, the investor's half earns precisely because it remains his deposit,

in peril. In the *mudarabah*, the capital belongs to the *rabb al-mal* from the contract's first hour to its last, and so does every monetary loss. Capital's claim on a venture's surplus, in other words, was never a payment for being money. It was a payment for being *ventured* — for standing exposed inside the story. The moment capital arranges to be safe, its claim dies: that was chapter 10's rule, and Part IV simply restates it in the affirmative. Capital contributes by being consumable, spendable, and losable, and it earns exactly as far as it stays that way. Its claims even come first in the dissolution arithmetic — the deposit half restored, the *rabb al-mal*'s coins counted back, the stans repaid before any split — and nothing in this chapter unseats that priority; what this chapter presses is what stands beside money on the schedule of contributions, not whether money belongs there.

Labor's standing took longer to say out loud, and it is a Roman jurist who says it most plainly. The Digest's title on partnership preserves, at 17.2.29, a little staircase of rulings from Ulpian that the medieval lawyers would walk up and down for centuries. First step: if partners never specified their shares, the shares are equal. Second step: may one partner take two shares or three, and another only one? Yes, says the text — *placet valere* — provided he brought something more to the partnership, "whether of money, of labor, or of any other thing": *vel pecuniae vel operae vel cuiuscumque alterius rei causa*. There it is, in a list of three words, the whole doctrine: *opera* stands next to *pecunia* as contribution of the same rank, convertible into profit share at whatever rate the partners' honest judgment sets. And then the third step, which still surprises law students: Cassius held — with Sabinus supplying the condition — that a partnership may even be formed in which one partner bears no share of the money-loss at all, yet shares the gain, *if his labor is worth as much as the loss*. The text gives the reason in a clause that deserves rescue from the footnotes: "for often a partner's industry contributes more to the partnership than money" — *plerumque enim tanta est industria socii, ut plus societati conferat quam pecunia* — and it gives examples: if he alone sails, if he alone travels. This is the same Cassius whom chapter 11 met naming the lion's partnership, the *societas leonina*, void because one party had arranged to take gain without exposure. The two rulings sit in adjacent sentences of the same fragment, and together they draw the line with Roman economy: a man who risks nothing and takes profit is a lion; a man who risks his season, his industry, his one irreplaceable voyage — that man has stake enough, and may be spared the money-loss because he is already standing under a loss of his own. The laborer was never the lion. The laborer was the other animal, the one who actually hunted.

What counts as labor's stake, once you start itemizing it honestly, turns out to be a longer list than a timesheet. There is skill — the *industria* of Ulpian's clause, the difference between a man who can sell flax in Sicily and a man who can lose money in any port in the Mediterranean. There is time, though the traditions handled time with tongs: waiting, by itself, earns nothing — the rabbis' *agar natar*, the reward for waiting, is chapter 7's forbidden core, and none of the three systems ever paid the clock as such. What they paid was time *filled*: months of attention spent inside the venture, which is not waiting but working. The distinction sounds thin until you watch it operate. Two men each part with a hundred coins for a year. The first spends the year at home while his coins sit in a stranger's shop, and the year, as a year, entitles him to nothing — that was the whole scandal of Part III. The second spends the year walking beside his coins, buying and haggling and keeping the books by lamplight, and the traditions will pay him for every month of it — not because the calendar turned but because he filled it. The coins and the calendar are identical; what differs is a person's presence inside the venture, and presence is what the instruments of chapter 11 were engineered to price. There is reputation, and it is astonishing how much of the old commercial law is really a law of names: chapter 11 noted that the *mudarib*'s true

collateral was his standing in a world of repeat engagements, and the Geniza merchants ran ventures from Egypt to the pepper coast on little else. A man who lends his good name to a venture has put something real and ruinable into it — names, unlike coins, cannot be re-minted. And there is the contribution the modern economist reaches for first and the traditions touched last and most gingerly: opportunity foregone, the profit the working partner might have made elsewhere with the same months. The scholastics would eventually build an entire disputed doctrine — *lucrum cessans* — on that thought, and this book quarantines it where it belongs, in chapter 20, among the technical debates. For now it is enough to mark the asymmetry: the traditions were quick to see labor spent and slow to credit labor hypothetically redirected, and the caution was deliberate. Spent months can be witnessed. Imaginary alternative months can be claimed by anyone, about anything, in whatever amount flatters the claimant.

So the schedule of honest contributions stands at something like this: money ventured; work performed; skill applied; standing staked; and, under heavy guard, alternatives genuinely surrendered. Every instrument in chapter 11 is a machine for pairing the first item with the middle three. The question that separates the traditions from the term sheets of our own day is what happens to the items that are hard to count.

The Wage for Trouble

The rabbis' answer runs through the shopkeeper we have already met. What the Mishnah demands for him, and what the Talmud at Bava Metzia 68b proceeds to size, is a wage: the investor who sets a man trading on half profits must pay him for the toil he spends on the investor's half. The Talmud's measure is a small masterpiece of practical psychology — pay him “as an idle laborer,” which the tradition unpacked as roughly what a man of his trade would accept to down his own tools and mind someone else's. Not a full market salary for the work, notice; the shopkeeper is, after all, also working his own half and keeping its whole profit. The wage prices the marginal imposition — the trouble — and the halakhists' shorthand for the category became *sechar tircha*, the wage for trouble, the payment that keeps the working partner's sweat from silently subsidizing his financier.

Two refinements show how carefully the principle was held. First, the wage's size collapsed over the centuries into a formality: later authorities let the parties stipulate a token — the codified rule in the Shulchan Arukh's laws of interest, at siman 177, is that even a single dinar suffices. That looks at first like the rule decaying, and is nearly the opposite. The dinar is doing declaratory work: it forces the contract to *say* that the manager's labor is a compensable contribution and not a courtesy, and once said, the rest of his compensation can ride honestly in a tilted profit split — the *iska*'s other standard device, met in chapter 11, giving the working partner a fatter share of gains than of losses precisely because his work is worth something. What the halakha would not tolerate was zero: the pretense that management of another man's money is free. Second, the codifiers' gloss draws the boundary from the other side: where both parties put in capital and one happens to manage, the wage requirement relaxes — a true two-sided partnership is not a creditor-debtor pair in disguise, and the manager's own capital, working under his own hands, already pays him. The rule was never a labor tax on partnership as such. It was targeted, with the rabbis' usual marksmanship, at exactly the configuration where unpaid labor could function as interest: one man's money, another man's sweat, and a loan half sitting in the middle of the relationship. Price the sweat, even at a dinar, and the categories come clean: what the worker does for the deposit half is service, sold; what he does for his own loan half is his own business; and nothing is left traveling, unnamed and unpaid, alongside a loan.

It is worth pausing to admire what the rabbis have done here, because it is the mirror image of everything Part III condemned. Part III's villain was the charge without a contribution — the lender billing for risk he had shifted away, for time he did not own, for need he did not create. The sechar tircha rule attacks the opposite fraud: the contribution without a charge. Both rules serve one ledger-keeping instinct: every real input priced, no phantom input paid. A tradition that spent chapter 8 hunting benefits that flow *to* the lender unpriced was never going to miss the benefit that flows *from* the worker unpriced — the same crime with the arrow reversed. Chapter 8 watched the rabbis count a borrower's unaccustomed greeting as interest; a season of unaccustomed shopkeeping stood no chance of slipping past them.

The Mudarib's Portion

Islamic law reached the same destination by the boldest route of the three: it made the profit share itself the wage.

Recall the mudarabah's severe geometry from chapter 11. The *rabb al-mal* brings capital and only capital; the *mudarib* brings effort and only effort; profit divides by a pre-agreed ratio; monetary loss falls on the capital alone, the working partner losing his months and his hoped-for share. Inside that geometry, the ratio is not an incentive bonus sitting on top of the mudarib's compensation; it is his compensation, entire. His labor has no other price: no salary, no retainer, no management fee accruing quietly whatever the venture does. The classical jurists were explicit about the exclusion, and their reasoning cuts to the structure's heart: a fixed draw for the mudarib would be a claim against the venture that pays regardless of outcome — and a payment regardless of outcome is precisely the payoff shape the whole contract exists to banish. Pay the worker like a creditor and you have smuggled the creditor back in through the servants' entrance; worse, in a losing year his fixed wage would eat into the rabb al-mal's capital, making the laborer a lien on the very coins whose loss is supposed to be the capitalist's own clean burden. So the rule held: the mudarib eats from the profit or he does not eat from the venture at all. Modern codifications state it with regulatory bluntness — Malaysia's central-bank standard on the contract, to take one living example, provides that the mudarib “is only entitled to profit” for the works integral to the venture “and shall not earn any additional fee for such works” — while preserving the classical safety valves: the venture's own direct expenses, the mudarib's road and lodging among them, come out of the mudarabah's purse rather than the worker's, and the financier remains free to hire the same man, by a separate contract at a separate fee, for work genuinely outside the venture's scope. The line the jurists are holding in all of this is the line between reimbursing a venture's costs and guaranteeing a worker's gain.

The tradition liked to teach the contract's texture through a story its compilations preserve about Abbas ibn Abd al-Muttalib, the Prophet's uncle, a man who would hand over mudarabah capital only on conditions: do not carry my money by sea, do not take it down a valley, do not buy livestock with it — and if you do, you stand guarantor for the loss. The report has the stipulations reaching the Prophet, who allowed them; modern codifications still cite it as the contract's pedigree. Whatever the chain's strength — the compilers transmit it at report level — the jurists drew from it the design point that matters here: the rabb al-mal may fence the mandate, by route, commodity, or season, because the money is his to fence; what he may not fence is the outcome, because the outcome was never anyone's to sell.

The severity, though, ran both ways — and this is the part modern summaries most often miss. If the *mudarib*'s ratio is his wage, then the *mudarib*'s labor is never, under any failure of the arrangement, simply forfeit. Let the capitalist promise funds and never deliver them after the agent has begun the work: the jurists' remedy — echoed in the same modern standard — is that the *rabb al-mal* must cover the expenditures incurred and pay the man "fair and reasonable wages," what the classical books call the wage of his like, *ujrat al-mithl*. The same remedy backstops a *mudarabah* that turns out to be void for a defect in its terms: the invalid contract converts, retroactively, into an employment at a fair wage, because whatever the paperwork's failings, the labor actually happened. Sit with that for a moment. The tradition's default is that work performed creates a claim that survives the death of the very contract that priced it — the ratio while the partnership stands, the fair wage when it falls, and at no point zero. A legal system reveals its priorities in its wreckage rules, and when a *mudarabah* collapses, the *fiqh*'s first question is the one this whole chapter has been circling: has the man who worked been paid?

One more feature completes the picture. The profit ratio, being labor's price, was genuinely negotiable — a half, a third, two-fifths, whatever the parties' judgment of the work's worth and the market for agents would bear, with the schools' usual disputes at the edges. What it could never be was a number pinned to the capital: the classical rule, again preserved in the modern codifications, voids a "share" expressed as a guaranteed sum or as a percentage of the capital amount, because either device unhooks the worker's pay from the venture's truth. The ratio prices the pairing of one man's money with another man's months, and it must bite on the profit that pairing actually produced. Chapter 14 will make that principle — returns defined only on real outcomes — the spine of its argument; here it is enough to see whose interests the rule was written in. A ratio on actual profit is the only wage that cannot be faked in either direction: the capitalist cannot promise it and then starve it, and the worker cannot collect it out of a venture that produced nothing.

Payment for Real Work

Latin Christendom's version of the principle arrives from an unexpected quarter: not from its partnership law, which it largely took over from Rome with the Digest's staircase intact, but from its agonized defense of the merchant himself.

The schoolmen inherited from the ancient world, and from a strand of patristic suspicion, the sense that commerce as such — buying to sell dearer — hovered near sin. Aquinas's rehabilitation of the trader, in the question of the *Summa* immediately before his question on usury, turns on a single move: he looks at the merchant's margin and asks what it is *for*. Gain sought as an end in itself is the appetite that "knows no limit," and it earns from Thomas the coldest of acquittals — not sinful in itself, merely base when it serves nothing. But gain directed to keeping a household, to relieving the needy, to stocking one's country with what it lacks — that gain, he writes, is sought "not as an end, but as the wage of his labor": *non quasi finem, sed quasi stipendium laboris*. The margin, morally read, is a paycheck. Trade is work — the sourcing, hauling, storing, risking, judging that chapter 9 watched the just-price doctrine struggle to see inside the price — and the man who does that work may eat from it, in measure, like any other worker. The phrase became a load-bearing category of the whole scholastic economics: *stipendium laboris*, payment for real work, the title under which human effort collects inside transactions that superficially look like money moving against money.

Set the phrase beside the partnership doctrine and the Christian schema snaps into the

same two-column shape the other traditions drew. Capital's title is ownership-at-risk: the investing partner earns *tamquam de re sua*, because his money stands exposed inside the venture. Labor's title is the stipendium: the working partner, the merchant, the agent earns because his effort is a real cost and a real contribution. Two titles, two different justifications, and the entire scholastic case against usury can be restated as the insistence that these are the *only* two — that every florin of return must trace to ventured property or expended work, and that the lender of a *mutuum*, having transferred ownership and performed no service, holds neither. The loss-side doors — the damages and foregone gains of the extrinsic titles, Part VI's subject — open off the same corridor, but those are compensations for what lending cost the lender, never payments for what he contributed; the schema does not confuse making a lender whole with making him a wage-earner. When Benedict XIV's *Vix pervenit* reaffirmed the old doctrine in 1745, chapter 7 noted, it carefully left the door open to titles arising outside the loan itself; the wage of labor had walked through that door centuries earlier and no serious voice ever tried to shut it on honest work. Even the institutions this book will meet in Part V ran on the principle: when the Monti di Pietà charged the poor a small fee on pledge loans, the defense that carried the day — chapter 17 tells the story — was precisely that clerks must be paid and buildings kept, cost and labor recovered, profit never taken. The tradition that would not let money earn as money insisted, with the same breath, that people must earn as people. Its horror of the one was the exact shadow of its respect for the other: usury takes without contributing, and there is no surer sign of the taking than that somewhere nearby, someone's work is going unpaid.

The Invisible Partner

Now walk the three doctrines forward eight centuries and stand them in front of a startup's capitalization table.

The founder of the modern venture is the most heavily contributing partner any of this chapter's lawyers ever imagined. She brings the skill and the *industria*; she brings the months — commonly years — of labor; she stakes her professional name on the venture's fate as thoroughly as any Geniza agent staked his on a cargo; she has usually surrendered, concretely and provably, a salary elsewhere, the very *lucrum cessans* the scholastics fenced with conditions, here made verifiable by the job offer she declined. There is a small historical irony buried in that fact: the scholastics fenced *lucrum cessans* with conditions precisely because foregone profit was easy to claim and hard to witness, and the modern founder is the rare claimant who can document her foregone alternative to the dollar — she is also the claimant for whom the modern instruments make no line at all. Frequently she also lends the venture money in the most honest of forms: the below-market salary, the payroll she skips in a bad quarter — capital contributions in every substantive sense, extracted in sweat rather than wired in. The folk term for all of this is “sweat equity,” and the folk term is a confession: everyone can see the equity is bought with sweat; the books just have no line for the purchase.

Because here is what the capitalization table records: money. It records it beautifully — to the cent, by date, with the price per share carried to four decimals. Capital contributions enter the venture's permanent memory the moment they arrive. Labor's contributions arrive differently: they trickle in nightly, they never pass through a bank account, and the accounting system books them not as contribution but as cost — a salary expense, kept artfully low — or does not book them at all. The founder's stake, the common stock, is what the arithmetic leaves over after every priced claim is served, and chapter 10 already

toured the devices that push it to the queue's end: preferences, seniorities, guarantees, each one shifting risk onto exactly the participants whose contribution the ledger never learned to see. What the ledger cannot see, the ledger will not pay. The result is a structure the traditions had a name for. Recall what the *societas leonina* actually was: not, in practice, a contract announcing that one party takes all the gain — no one signs that — but any arrangement whose *payoff*, once every clause has operated, leaves one party's contribution compensated and the other's absorbed. A venture in which capital's input is priced to four decimals and labor's input is priced at whatever residue the waterfall happens to leave has not escaped the lion so much as modernized his bookkeeping. The difference from the fable is that no clause announces the division; it emerges, politely, from the arithmetic.

And the founder, note, is the lucky one: she at least appears on the capitalization table, however junior her stock. Below her stands a stratum the documents see even less. Employees who took thin salaries against options; contractors paid late through a tight quarter; the spouse whose second income floated the household while the venture consumed the first — co-investors all, by any substantive test this chapter's lawyers would recognize, and none of them anywhere in the arithmetic. Chapter 38 gives them their own accounting, under worker claims and invisible contribution; chapter 10 already exposed the mechanism that makes their seat the worst in the house: shifted risk rolls downhill toward the participants least able to spread it, and unpriced contribution rolls with it.

The traditions' three answers, set side by side, read like a single design brief written in three languages. From the rabbis: no venture may treat the working partner's labor as free, and the pricing must be *explicit* — a stated wage, even a token, so that the contract itself confesses labor's claim, with the working partner's tilted share doing the substantive paying. From the fiqh: the worker's share is a first-class claim defined on real outcomes, immune to being fixed away, and it survives even the contract's own death — the venture that consumed a man's months owes him the wage of his like though every other clause should fail. From the schoolmen: work is a title to return as fundamental as ownership itself, and any structure whose returns cannot all be traced to ventured property or expended labor is collecting for a phantom. Put affirmatively, as the design brief this book will hand to Part IX: a venture's surplus belongs to its contributions, and the schedule of contributions is longer than the wire transfers. Money ventured, yes — first out of the arithmetic, as every tradition agreed, standing at its own risk. But also months spent, skill applied, names staked, salaries foregone, risk absorbed by the people least able to spread it. Ulpian's three-word list — *pecuniae, operae, cuiuscumque alterius rei* — is a contribution schedule; the traditions' whole labor doctrine is the insistence that the schedule be filled in honestly and paid in the order mercy and truth require. What that means in the vocabulary of modern instruments — how a term sheet would actually price founder labor, worker claims, the contributions no wire ever carried — is the assignment of chapter 38, which revisits this chapter's three terms by name, and of the surplus arithmetic of chapters 39 and 40. The old lawyers, who priced a shopkeeper's trouble at a declaratory dinar and made a dead partnership pay a living wage, would not have found the assignment strange. They would have found it overdue — eight centuries of compound arrears on the one debt their ledgers never let go invisible.

Chapter 13 — Loss Allocation and the Truth of the Contract

Return to the portico by the Genoese harbor, and let the year turn. In chapter 11 two men stood before the notary Giovanni Scriba and signed a commenda: one man's money, the other man's voyage, the profit to be divided on the standard split. Now it is the following spring, say, and the traveler has come back up from the quay with his sea chest, his account book, and nothing else. The venture is gone. Prices broke in Alexandria; the cargo sold for less than it cost to load it; what the pirates and the port fees left, the homeward freight consumed. Before witnesses, as the contract requires, he renders his accounting — every purchase, every sale, every charge — and the investor, who is out three years of savings, asks his questions and hears the answers. And then the two men do the thing this whole Part has been building toward, the thing that made the instrument worth a chapter of anybody's book. They part.

No debt follows the traveler up the hill. No lien settles on his house, no claim passes to his heirs, no ledger anywhere records that he owes the investor so much as a bent denarius. The money is simply gone, the way a harvest is gone after hail, and the two of them walk away from the wreck unequally bruised but equally free. One street over, that same spring, some shopkeeper who financed his stock with a plain loan is learning the other arithmetic: his venture failed too, and his creditor is owed one hundred twenty, today, out of whatever he still owns. The venture's death did not touch the debt. Chapter 10 watched that claim ride out every storm that killed the things it paid for; a loan, whatever else may be said of it, does not know how to lose.

Who loses, then, if the venture fails? The partnership instruments answer with a map — this loss to this man, that loss to that one, drawn before the venture begins — and the map is the subject of this chapter. Prosperity flatters every contract; failure cross-examines it. When the returns are good, loan and partnership look like cousins: money went out, more money came back, everyone is smiling. The difference between them is written entirely in the bad year, in who eats what when there is less at the end than there was at the beginning. The traditions understood this, which is why their partnership law spends so little ink on dividing gains and so much on allocating losses: the gain side takes care of itself, and the loss side is where the contract either tells the truth or is caught lying. Three questions organize everything they built. Where does honest loss fall — loss that arrived by weather, war, or the ordinary treachery of markets? When does blame move it — what must a manager have done before his partner's loss becomes his debt? And what happens to the whole structure when someone tries to draft his way out of his seat? The answers, across three legal civilizations, agree so closely that by the chapter's end we will be entitled to call the agreement what it is: a geometry.

The Ledger of Failure

Chapter 12 itemized what each partner brings to a venture — money, months, skill, standing. Failure reads the same schedule backward. When a venture dies an honest death, every real contributor loses the very thing he contributed, in its own coin and nobody else's.

Capital loses money, and only money. That limit deserves more admiration than it usually gets. The investor's downside in every instrument chapter 11 walked — *societas, iska, mudarabah, commenda* — is bounded at the sum he ventured: the venture can consume his stake entirely, and then it must stop, because nothing in the contract converts his loss into anyone's debt or reaches past the ventured purse into the rest of his estate. The fiqh states the boundary as cleanly as anyone: monetary loss falls on the capital, up to the capital, and the *rabb al-mal* has no complaint beyond it. His hundred could become zero; it cannot become minus twenty.

Labor loses the months. The traveler's season, the *mudarib*'s year of roads and warehouses, the working partner's daylight — spent in full, recoverable never. The books of the fiqh put it with a plainness that has worn well: when a *mudarabah* loses, the financier loses his money and the agent's labor has gone in vain. That loss is as real as the contribution was, and the traditions' refusal to pretend otherwise is half the moral architecture of this Part. The Roman lawyers, chapter 12 found, built the other half of the same thought into the Digest's partnership title: a partner may be spared any share of the money-loss where his labor is worth as much as the loss — *si tanti sit opera, quanti damnum est* — because, the jurists reasoned, a man's industry is itself a stake, and often a greater one than coins. Read from the failure side, the ruling says something exact: the laboring partner was never exempt from loss at all. His loss was simply denominated in a different currency, and he pays it first, automatically, the moment the venture fails — before the investor has even opened the account book. And there is a third loss the ledgers never captured, which chapter 11 met as collateral: the manager's name. A venture that dies under a man's hands leaves a residue of doubt on him, deserved or not, in exactly the world of repeat engagements where his name was his fortune.

Notice what the traditions did not do with these losses: they did not try to equalize them. No rule anywhere requires the investor to reimburse the traveler's wasted season; no rule bills the traveler for a tenth of the sunken capital. The two losses are incommensurable — coins against months — and the instruments let each fall where it grew. What looks at first like rough justice is, on a second look, the only honest accounting available. Each partner ventured what he ventured; failure takes precisely that. The alternative designs all smuggle in a guarantee somewhere — of the investor's money by the worker, or of the worker's wage by the investor — and the traditions rejected both directions by name, the first as the very definition of disguised lending, the second, more interestingly, with a precision the fiqh spelled out and chapter 12 already met. When a *mudarabah* collapses for a defect in its terms, or a financier promises capital and never delivers it after the work has begun, the law pays the worker — expenses, plus the wage of his like, *ujrat al-mithl* — because the arrangement itself failed, and the failure of an arrangement is somebody's doing. But when the arrangement holds and the market simply answers no — the goods bought soundly and sold at a loss, the season honestly bad — the worker's remedy is nothing, because a ratio of a profit that never existed is nothing. The line could not be drawn finer: the law compensates labor for a broken contract, never for an unlucky one, because a venture's ordinary death is nobody's doing.

Nobody's doing: that is the premise the whole allocation rests on, and it is also the premise that wrongdoing destroys. The rest of this chapter is about what happens at that boundary — and each tradition, it turns out, patrolled it with the same three-part grammar: honest loss lies where the map says; caused loss moves to the causer; and the map itself may never be redrawn by guarantee. We take the traditions in turn, starting with the one whose instrument confesses its own hybrid nature right in its name.

Two Purses in One Coat

The *iska*, the rabbis' investment contract, is half loan and half deposit — *palga milveh*, *palga pikadon*, in the Nehardean formula chapter 11 unpacked from Bava Metziah 104b — which means it is also two loss regimes stitched into one coat, and the stitching is where its honesty lives.

The loan half loses like a loan. Those fifty coins became the manager's property on the day the *iska* was signed, and property perishes to its owner: if the stock they bought burns, sinks, or sells at half price, the loss is the manager's alone, and he owes the investor fifty at settlement in every possible world. On this half he stands where chapter 7 put every borrower — at the top of the Mishnah's ladder of keepers, liable even for sheer accident, because what he holds is his and what he owes is arithmetic. The Mishnah's ladder, met at Bava Metziah 7:8, grades the four keepers of other people's property by the benefit they draw: the unpaid guardian, who swears about what happened and goes home; the paid guardian and the renter, who answer for theft and loss but not for catastrophe; and the borrower, all benefit and all liability, who pays for everything up to and including the wolf that could not have been foreseen. The *iska*'s loan half simply is that fourth rung, and the rabbis never pretended otherwise.

The deposit half loses like a deposit. Those fifty remain the investor's property, trading in the manager's hands at the investor's peril, and when the market takes them the loss is his — that peril being, as chapter 10 established, the entire license for his share of the gain. The Mishneh Torah, codifying the instrument in its partnership laws, states the two regimes with a bluntness that settles every boundary question: the administrator is responsible for the loan half "even if it is destroyed by forces beyond his control," while the other half "is considered an entrusted object" at the investor's risk — "if the half that is considered an entrusted article is stolen or lost, the administrator is not liable to pay," which is exactly why that half's profit belongs to the investor (Sheluchin veShutafin 6:2, in Eliyahu Touger's rendering). Notice how far the owner's peril runs: theft and loss on the entrusted half stay on the investor's account, not the manager's. The manager is a keeper of that half, never its underwriter — and what no keeper escapes, on any rung of the Mishnah's ladder, is negligence. What negligence means for money, the Talmud made concrete enough to touch. Money, rules Shmuel at Bava Metziah 42a, has only one proper safekeeping: burial. The same page tries the case of a man who deposited coins with a fellow who tucked them into a willow hut, where they were promptly stolen — a keeping Rav Yosef dissected with a lawyer's relish: against thieves, he ruled, the hut was effective safeguarding; against fire it was negligence; and a keeping that begins in negligence stays liable however the blow happens to land — the rule the Gemara, after recording a dissenting version, fixed as the law. On the same folio, a bailee who tells the owner he cannot remember where he put the money collects Rava's shortest ruling in the tractate: every "I do not know" is itself negligence — go pay. And when professional treasure-diggers grew common enough that buried coins stopped being safe, the law moved the standard — money now wanted hiding in the roof beams; then, when dismantlers learned to search beams, between the bricks of a wall; then, when tappers learned to sound walls, in the handbreadth by the floor or the ceiling, where tapping tells nothing — because negligence is measured against the current state of the burglar's art, not against the customs of one's grandfather. The Shulchan Arukh's laws of interest, chapter 12 found at siman 177, keep the whole machinery in working order.

One more rule from the same chapter of the code completes the loss map, and it shows how honestly the machinery was built. Where no separate wage is paid and the parties stipulate

nothing, Maimonides rules, the default division is asymmetric: the manager takes two thirds of any profit and bears only one third of any loss, the gap between the fractions being his wage for handling the entrusted half (6:3). Read that against chapter 10's baraita and the design leaps out. Near to profit and far from loss — the posture the tradition brands wicked in an investor — is the law's own default for the working partner, and it is not a lapse but a payment. The asymmetry that would be usury when capital buys it with drafting is justice when labor earns it with months; loss allocation, the rabbis are saying, follows contribution, and the manager's months are a contribution the settlement cannot give back.

Stand back from the details and look at what the instrument is doing. One venture, one purse, one man walking the market — and two complete, opposite answers to the question who loses, running side by side in the same coat. Market ruin on the loan half: the manager. Market ruin on the deposit half: the investor. Negligent ruin anywhere: the negligent. The *iska* never resolves the tension between loan and partnership; it declares it, half and half, and dares either party to claim confusion. A modern instrument that behaved this way would state on its face which fifty of your dollars were guaranteed and which fifty were genuinely at sea — and it is exactly the absence of such candor, chapters 37 and 38 will argue, that lets modern capital collect a partner's premium on a creditor's terms. The rabbis' contract could not lie about its downside, because its downside was its definition.

The Trustee's Hands

The *fiqh* sorted every holder of another person's property into one of two postures, and the pair does more work in Islamic commercial law than any other distinction in this chapter. Property can sit in a *yad al-daman*, a guarantor's hand — the borrower's hand, the usurper's — where the holder owes the thing back whatever befalls it. Or it can sit in a *yad al-amana*, a trust-hand: the depositary's, the agent's, the partner's, the *mudarib*'s — where the holder answers for his conduct but not for the world's, and the thing, if it perishes innocently, perishes to its owner.

Everything the tradition says about loss in the *mudarabah* unfolds from filing the *mudarib* under the second hand. He is an *amin*, a trustee. The capital in his saddlebags is not his and never becomes his; its market fortunes belong to the *rabb al-mal* as completely as its profits would have; and no clause may move him to the other hand — the classical rule, still printed in the modern standards chapter 10 cited, that the *mudarib* shall not guarantee the capital, and that a *mudarabah* stipulating such a guarantee has stopped being a *mudarabah*. The maxim the books attribute to the caliph Ali carried the loss half of the doctrine across the schools without dissent: loss follows the capital, whatever the profit agreement says. The severity runs both ways, as this book keeps finding. The financier who wants his money warranted has an instrument for that — it is called a loan, and it earns nothing; the financier who wants earnings has the *mudarabah*, and it warrants nothing. What he cannot have, in any school of the law, is the hybrid: the trustee's yield from a guarantor's hand.

But the trust-hand is a posture of conduct, and conduct can forfeit it. The jurists' liability switch has three settings, and the modern codifications state them in a formula so compact it reads like a fuse rating: the *mudarib* bears no loss absent his *ta'addi*, his *taqsir*, or his breach of the specified terms — transgression, negligence, or violation of the mandate. Transgression is doing what the contract never authorized; negligence is failing the care the undertaking demands; breach is crossing a fence the financier drew. Chapter 12 met

the fences in the story of Abbas ibn Abd al-Muttalib, who released capital only on conditions — no sea carriage, no valleys, no livestock — and the fiqh's rule for the far side of the fence completes that story: step outside the mandate, and everything that happens out there happens to you. Inside the fence, outcomes belong to God and the market and therefore to the capital; outside it, or beneath the standard of care, or past the edge of honesty, they belong to the man who put them there. When the switch flips, the trustee becomes a debtor — not because he guaranteed anything, but because the loss stopped being weather and became conduct. The modern standards run the same current, in words worth quoting because a chapter about drafting should show the drafting: a manager of investment capital "is a guarantor of the capital, at its nominal value, in case of his negligent acts or omissions or his non-compliance with the investor's conditions" — whether he manages as *mudarib*, partner, or agent (AAOIFI's 2008 sukuk resolution, restating its standards).

Here is the fiqh's version of the line every tradition in this chapter draws, and it is worth stating at its full width because it answers one of the ToC's plainest questions — what separates ordinary business failure from culpable loss. Ordinary failure is the contract operating. The *mudarabah* prices uncertainty by seating the parties inside it; a season that answers no is the instrument telling the truth, and the loss it delivers is as lawful as the profit it might have delivered. Culpable loss is the contract violated: the venture's outcome manufactured by a party rather than found by the world. The first kind of loss stays on the map; the second moves, always, to its maker. And past negligence stands fraud, for which the tradition reached not for a partnership rule but for scripture. "You who believe, do not wrongfully consume each other's wealth but trade by mutual consent," runs the Qur'an's commercial cornerstone (4:29, in M. A. S. Abdel Haleem's rendering, as throughout this book) — a verse chapter 9 set at the foundation of the fraud rules, and chapter 23 will meet again at consent's edge. A manager who reports a loss that never happened, or hides a profit that did, has passed beyond mismanagement into consuming his partner's wealth by falsehood, and the whole apparatus of unlawful devouring — the Qur'an's own word for it — closes over the act.

One evidentiary note completes the portrait, and it says something about the tradition's temperament. The trustee's hand carries a presumption: the *mudarib*'s account of an innocent loss stands unless the financier can prove the switch flipped — fault must be alleged and proved rather than presumed from an empty purse. A legal system that wanted to flatter capital would have run the presumption the other way, and made every failed agent a defendant. The fiqh trusted first, structurally, and spent its vigilance where chapter 11 located it: in the world of repeat engagements and carried names, where a suspicious loss cost an agent his livelihood more surely than any court could. How the traditions policed loss claims when documents replaced reputation — the oaths, the witnesses, the settlement formulas — belongs to the next chapter, where the *heter iska*'s machinery gets its own accounting.

Weather and Wrongdoing

Rome, as usual, had laid the grammar out first, and Latin Christendom, as usual, inherited the grammar and supplied it a conscience.

The *societas* allocated loss by agreement against a default of equality: partners share as they stipulate, the Digest's partnership title says at 17.2.29, and absent stipulation they share equally — with the one absolute boundary chapter 11 made famous, that no partner may be seated where he cannot lose at all, on pain of the arrangement being a lion's partnership

and void. Within the map, the Roman question was the same one the fiqh asked: when does a partner's loss become his partner's fault? The answer sits farther down the same title, at 17.2.72, in the tri-grade grammar Roman lawyers applied to almost everything. For *dolus* — fraud, the designed injury — a partner answers always. For mere accident, *casus*, he answers never; accident lies where ownership put it. The middle grade is the interesting one. A partner answers his partner for *culpa*, for negligence — but the Digest declines to hold him to the meticulousness of the ideal custodian, the *exactissima diligentia* the law demanded elsewhere. The standard is *diligentia quam in suis*: a partner owes the common venture the care he habitually gives his own affairs, no more — and the fragment appends the tartest sentence in this chapter's sources by way of justification. Whoever takes on a careless partner, it observes, has only himself to blame — *de se queri debet*. You chose him; his ordinary sloppiness was part of what you chose; the law will protect you from his betrayals but not from his character.

It is a strange little standard at first meeting — liability graded to the defendant's own habits, so that the naturally careless man is judged by a gentler yardstick than the naturally careful one — and then its wisdom opens. Partnership is not a service purchased from a stranger; it is a joining of fates, entered with eyes open, and the law declines to let one partner demand, after the fact, a partner different from the one he shook hands with. What the standard preserves is precisely the weather. A man trading with his customary competence who guesses wrong about the price of pepper has not wronged his partner; he has been the partner he always was, in a world that answers to no one. *Culpa* begins where a man falls below his own norm — where the loss traces not to the world's uncertainty, nor to his settled nature, but to a lapse: care withheld from the common purse that he would have spent on his own. The Roman line and the fiqh's line land within a hair of each other, drawn from opposite ends: *ta'addi* and *taqsir* measure the agent against the undertaking's objective demands, *diligentia quam in suis* measures him against his own demonstrated self, and both are patrolling the same border between the loss that happens to a venture and the loss that is done to it.

The schoolmen took over the whole chassis — Aquinas's acquittal of the investing partner, chapter 11 found, turns on the money trading at the owner's risk, its gain licit *tamquam de re sua* — and what Christian moral theology added was less a new rule than a new insistence on why the rule mattered. Gain is honest, in the scholastic account, exactly where loss was possible and genuinely shareable: the merchant's profit is the *stipendium* of his labor and the fruit of his ventured property, and his losses are part of the same calling — the misjudgments of an honest trader are his own to eat, covered by no one, exactly as his skill is his own to sell. What the schoolmen would not allow was the sorting of loss by anything other than truth. A partner's fraud, in the confessors' analysis, belongs to theft and restitution rather than to usury — the sin reclassified by causation, because a man who manufactures his partner's loss has taken property, whatever instrument the two of them signed. And a financier who arranges immunity from loss while collecting a partner's gain commits the opposite falsehood: his contract asserts an exposure that does not exist. Naviganti, chapter 10's decretal, condemned the sea lender on precisely that reading — his premium was fixed, his exposure partial, his claim alive through every ordinary failure of the venture — and the same pontificate that blessed the commenda cursed him. Rome's civil lawyers, the rabbis of Babylonia, the jurists of Baghdad and Cordoba, and the canonists of Bologna disagreed about nearly everything else in this book. On the proposition that loss must lie where the contract's truth puts it — on ownership for weather, on the wrongdoer for wrong — they speak, without having conferred, in unison.

The Geometry of Downside

Which brings us to the guarantee, the one device every tradition in this chapter prohibited by name, and the reason this chapter carries the title it does.

A guarantee, chapter 10 taught, destroys no risk; it readdresses the parcel. Within a partnership, that readdressing has a specific and fatal meaning. The instruments of Part IV all run on one engine — gain licensed by exposure — and a clause that removes a partner's exposure while preserving his gain does not merely tilt the deal; it stops the engine and leaves the body rolling on momentum and paperwork. The traditions' verdicts are interchangeable to a degree that still startles. Guarantee the *iska's* deposit half home whole, and the deposit dissolves into loan and the investor's yield into *ribbit* — the boundary chapter 11 marked as the structure's one unforgivable clause. Extract a guarantee of capital from the *mudarib*, and the *mudarabah* is void, the "profit share" *riba* in a partner's costume — the rule the standard-setters still print. The rabbis' character sketch at Bava Metzia 70a, which chapter 10 read, gives the arrangement its lasting portrait: the man seated near the gains and far from the losses is the *baraita's* wicked man, whatever his documents say. And when Latin Christendom's financiers finally engineered the thing in its perfected form — the *contractus trinus*, three separately licit contracts stacked into one guaranteed return, partnership plus capital insurance plus the sale of uncertain profit for a certain one — the papacy answered on October 25, 1586, with a bull whose first word is "detestable." Sixtus V's *Detestabilis avaritia* condemned every partnership pact providing that "the very principal, or capital be always safe" in the managing partner's hands, or that a fixed sum be paid "yearly, or monthly" as gain; declared such conventions "illicit and usurious," raising no obligation civil or natural; and then, in the same breath, restated this chapter's whole doctrine as the honest alternative — partnerships of money and labor formed "honestly, sincerely and with good faith," the contractors to cast up "the fruits, costs and losses, and let them share and divide them fairly and justly." That the fight did not end there — that Lessius and the century after him would defend the triple contract until the doctrine itself gave ground — is Part VIII's story, and chapter 11 already posted its signpost. What matters here is the shape of the objection, which never changed and which aimed past the greed at the falsehood: the triple contract labeled as partnership a position from which loss had been drafted away.

Why is the guarantee corrupting, rather than merely generous? Because the loss it removes from one seat does not leave the theater. Show me where the losses may not go, and I will tell you what the contract is. Route the venture's downside away from capital and it must land on the working partner — the arrangement every tradition condemned first and most fiercely, the worker underwriting the financier — or on some stranger to the venture, or on the truth itself: the loss unreported, renamed, rolled forward, buried in an account nobody may audit, which is the corruption chapter 14 will find waiting inside the *heter iska's* settlement clauses and chapter 32 inside instruments that replicate a bond in a partnership's clothing. The modern standard-setters patrol that corner with drafting of their own, and their texts repay reading. The guarantees standard forbids stipulating any guarantee in a fiduciary contract — for partnership managers it is "not permitted... to guarantee the capital, or to promise a guaranteed profit," nor may such contracts be "marketed or operated as a guaranteed investment" (AAOIFI Shari'ah Standard 5, art. 2/2/1) — because, the same standard explains, a guarantee from the man managing the investment "turns the transaction into an interest-based loan." What survives is exactly the residue the old lawyers would have watched: a guarantee volunteered outside the contract's walls, by a third party or even by the manager himself acting, the standard says, "in a different ca-

capacity" (art. 2/2/2), and the disclosed reserve account — built from the venture's own earnings against its own bad seasons, which is no guarantee at all but the pool of chapter 3 wearing modern dress. The seam strains where it always strained: a promise that may not be sold inside the contract stands just beside it. Chapter 26 meets the arrangement doing heavy lifting under modern *sukuk*, and the tradition's own critics, chapter 32 will show, have said the sharpest things about it. The diagnostic, in every century, is the one this chapter has been assembling: find the seats the drafting has made incapable of loss, and you have found the parties who are no longer partners, whatever the cover page announces.

That diagnostic deserves its full name, because Parts IX and X of this book intend to use it as an engineer uses a level. Call it the moral geometry of downside. Every venture, chapter 10's conservation law says, carries a fixed load of possible loss, and every contract distributes that load across the people in the room — visibly or invisibly, honestly or not. The upside arrangements tell you what the parties hoped; the downside map tells you what they are. Read a modern financing the way the rabbis read the baraita's postures — by where each party sits when the venture fails — and the familiar instruments confess with remarkable speed: the liquidation waterfall is nothing but a loss map with the losses at the bottom, drafted distance from ruin, priced and sold by the tier; chapter 37 runs that arithmetic in the traditions' terms. The seats nearest the loss, chapter 38 will show, are held disproportionately by the people who never saw the map — workers, founders, the small and the late — and chapter 10's sober accounting of shifted risk is really this chapter's doctrine read at industrial scale. And beneath the whole geometry sits the fact the traditions grasped in their bones and the mathematics of chapter 28 will state exactly: a loss allocated to a seat is lived there as a sequence rather than an average, and some seats can be ruined — which is why the seating of the downside deserves to be read as a deal's substance and not its fine print.

The old lawyers would want one more thing said, because everything in this chapter can sound like a machinery of blame, and it was built as a machinery of peace. An honest venture, honestly failed, ends. That is the gift the instruments of chapter 11 gave the people who used them: the investor's loss stopped at his stake, the traveler's at his season, and nobody left the accounting owing anybody a life of installments. The traditions built contracts that knew how to die — cleanly, all losses suffered, all losses final — because they had seen the alternative, the claim that failure cannot kill, walking through every farm and workshop of the ancient world with a ledger in its hand. Loss allocation is the truth of the contract, and the truth had a mercy in it. But an ended venture leaves one question standing, and it is the last one Part IV must answer. The losses have found their seats; the survivors are looking at what remains in the till; and before anyone touches it, somebody must say what part of it is actually profit — what the venture must first pay its costs, its people, its reserves, and its future before anyone may call the remainder his. That is chapter 14's question — why profit must come from profit — and the traditions' answers to it turn out to guard the same wall this chapter has been walking, from the inside.

Chapter 14 — Profit Must Come from Profit

The caravan is home. Say it came up from the Yemen coast with the winter convoy, and that the *mudarib* who took it out three years ago is sitting tonight in a Fustat courtyard with his investor, a lamp, a scale, and the sacks and bales of everything the venture now is: pepper sold and silver taken, flax traded for indigo, a bad debt in Aden written on a scrap of paper, a donkey nobody has valued yet. Tomorrow, before witnesses, the accounting. And at the center of the accounting sits a question this Part has been circling for three chapters and must now answer head-on, because every rule the traditions built comes to a point exactly here, at the table where the money is divided. The investor's capital comes off the top first — that much every school agreed. The ratio bites what remains. But what remains? Which of these coins are the venture given back, and which are the venture's *increase*? Until that question is answered, the ratio has nothing to bite; and how a legal system answers it — what it lets the parties call profit, and when — turns out to be the deepest test of whether its partnerships are honest.

Chapter 12 priced the labor. Chapter 13 seated the losses. This chapter completes the arch: it asks what may honestly be called profit, and it defends the answer the three traditions gave in unison — profit is what a venture actually produced, counted after everything the venture owes, and nothing else may wear the name. The negative form of the rule sounds almost too simple: no one may claim profit before real profit exists. Yet each tradition spent centuries fortifying that simplicity against a single, endlessly renewed assault — the attempt to write the profit down in advance, as a number, in the contract itself. So the chapter starts with the assault.

The Number Written in Advance

Recall what makes a partnership lawful in every legal system this book reads: the investor's return rides the outcome. Now watch what a fixed return does to that architecture — not a tilted ratio, which chapter 12 defended, but a stipulated *number*: pay me a hundred and twelve next Michaelmas; pay me two dirhams a month per hundred; guarantee me, as the bull *Detestabilis avaritia* condemned in 1586, “yearly, or monthly... a certain sum” for the life of the company.

The corruption is not that the number is greedy. It may be modest — modesty was never the point, and when Benedict XIV's *Vix pervenit* met the argument that a *moderate* charge on a loan should pass, it rejected the adjective outright: the doctrine was never about the size of the gain but about its source. The corruption is that the number is *written*. A stipulated return is knowable before the venture does anything, which means it is not a share of what the venture does; it is a claim that merely uses the venture as its address. The moment the investor holds it, three things die quietly inside the partnership. His exposure dies, because a fixed claim is precisely what chapter 13 watched the guarantee create — a seat from which loss has been drafted away, the near-to-profit-far-from-loss posture the baraita calls wicked. His interest in the venture's truth dies, because a man owed a hundred and twelve in every state of the world has no further question to ask about the world; the books may say anything. And the working partner's claim dies last and worst, because a fixed charge, unlike a ratio, does not shrink when the harvest does — it eats out of the venture's substance in the bad year, out of wages and stock and seed corn, exactly the way chapter

10 watched the sea lender's premium survive every ordinary failure of the voyage. Rome's lawyers had refused the naked version — the lion who takes gain without loss — and the traditions refused the actuarial version dressed as a partner's share. The stipulated number is the loan's payoff smuggled into the partnership's clothing, and all three systems, meeting it, said the same word: void.

Run the two payoffs through one bad year and the incidence shows plainly. A venture of a hundred comes home worth ninety-eight. Under a ratio, the investor's share of profit is a share of nothing: the two-coin loss falls where chapter 13 seated it, and the books close. Under a stipulated twelve, the venture owes a hundred and twelve out of ninety-eight, and the twelve is paid — if it is paid — out of capital, out of wages withheld, out of stock sold at the wrong moment; out of the venture's body, since its increase does not exist. The fixed number does not merely misdescribe the bad year. It bills it, and a charge that collects out of substance in precisely the seasons there is no increase is this book's oldest acquaintance in its newest coat.

Their alternative is the instrument this Part has been assembling piece by piece: the *ratio*. A half, a third, two-fifths of profit — the commenda's three-quarters to the *stans*, the Talmud's tilted splits, the *mudarabah*'s freely bargained fractions. It is worth pausing on what an odd object a ratio is, because familiarity has made it invisible. A ratio is a number that cannot be computed. Signed on the dock, it says nothing at all about what the investor will receive; it is an algebra with one variable missing, and the missing variable is the future. Only when the venture comes home — when the pepper is sold, the debts are collected or mourned, the costs are paid — does the ratio acquire a value, and the value it acquires is manufactured out of the venture's actual story and nothing else. A ratio asks the world what happened; a rate tells the calendar what is owed. That single design difference carries the whole moral load. The rate-holder is a creditor whatever he is called, because his claim matures by the mere passage of dates — the *agar natar*, the wage of waiting, that chapter 7 found at the forbidden core. The ratio-holder cannot collect a coin until there is truth on the table: an accounting, a valuation, a real remainder. The traditions did not merely prefer the ratio; they built their entire licensing of investment on the fact that a ratio is, structurally, a question — and a question, unlike a demand, keeps everyone honest who must answer it.

The Ratio's Discipline

The fiqh states the prohibition with the most doctrinal machinery, and its machinery repays a close look because every gear exists to protect the same thing: the connection between the word "profit" and an actual remainder.

Begin with the rule chapter 11 met at the contract's founding: the *mudarabah*'s profit clause must be a fraction of profit — of *profit*, not of anything else. Stipulate the *rabb al-mal* a fixed sum and the contract is void; stipulate him a percentage of the *capital* — ten on every hundred, however the venture fares — and it is equally void, because a percentage of capital is just a fixed sum wearing arithmetic. The modern codifications keep both bans in force: no guaranteed profit, no pre-determined amount to either party, the working partner's ratio his only compensation. The schools quarreled, as chapter 11 noted, over how far profit shares could stray from capital shares in the *musharakah* — the Shafi'i and Maliki masters required profit to track capital exactly, invest two-fifths and take two-fifths, any contrary clause voiding the contract, while the Hanafi and Hanbali schools let agreed ratios diverge from capital shares, since one partner's work, skill, or standing may be worth

paying for; chapter 12's whole argument lives inside that second position. But notice what even the strict schools were policing: proportion between partners — fairness in the division of a real remainder. About the form of the claim there was no quarrel anywhere: profit shares bite realized profit; loss follows capital, pro rata, in every school; and nothing upstream of realization may be promised, priced, or paid as if it were profit.

Then the discipline that gives the rule teeth: you may not call it realized until the capital is whole. The jurists' working principle — there is no profit until the capital is made good — runs through the classical books as consensus, and it does an enormous amount of quiet work. Interim gains are not yet anyone's; a good first year followed by a bad second has produced, in the law's eyes, whatever the whole story produced, and early distributions are advances against a truth not yet known. The modern standards preserve the principle with accountants' precision, and the drafting is worth quoting because it shows a medieval doctrine alive inside modern paragraph numbering. The *mudarib's* entitlement to his share arises when profit appears, says the AAOIFI *mudarabah* standard, but "this entitlement is not absolute, as it is subject to the retention of interim profits for the protection of the capital"; it hardens into a right only at distribution, "when actual or constructive valuations take place" — the final division made on what the assets actually sold for, or, where the venture continues, on a sober fair-value reckoning. And the same article reaches the subtlest corner of the counting-room with a rule the scholastics would have stood and applauded: receivables — the venture's uncollected promises — enter the reckoning at their cash-equivalent value, "after the deduction of a provision for doubtful debts," and in valuing them "neither time value (interest rate) nor discount... shall be taken into consideration." Sit with that for a moment. Inside the arithmetic that decides what may be called profit, the tradition polices its own founding prohibition: even here, especially here, time is not money. A paper claim on next year is worth its honest collectibility, not a penny more for the waiting, because the moment waiting acquires a price inside the profit computation, the whole quarantine has failed from within.

The standards even domesticated the impatience. Profits may be distributed early, "on account" — the phrase is the drafters' own — provided the distribution is revised when the real valuation arrives: early money is provisional money, an advance against a truth not yet known, callable back if the truth disappoints. What may never be arranged is the reverse. The 2008 resolution forbids a manager to undertake to cover shortfalls with loans in the same breath that it permits the disclosed reserve chapter 13 met — because smoothing paid out of the venture's own retained truth is honest, and smoothing paid out of borrowed pretense is the old sin with better cash management.

The *halakha* ran the same discipline through its own instrument. The *iska's* settlement, chapter 13 showed from the Rambam's codification, unwinds the halves in order — the loan half restored in every world, the deposit half restored from what the venture holds, and only then a division of increase, computed on what actually happened, with the manager's asymmetric fractions doing their labor-pricing work. And Rome, with characteristic economy, had put the whole principle in the design of the *societas* itself: the partnership's gain was whatever the common venture held above its contributions when the accounts were struck — Cassius's condition, chapter 12 found, priced even a partner's *labor* against the *loss*, which is to say the Roman lawyers were netting contributions and outcomes against each other from the very beginning. Three legal languages again, and one grammar: profit is a conclusion. It sits at the end of an audited story — capital restored, costs paid, claims proven — and every attempt to move it forward in time, toward the signing table and away from the truth, is the same attempt this book has been tracking

since the wall went up: to be paid as an owner of outcomes while standing somewhere outcomes cannot reach.

The Oath and the Fee

And yet the most instructive document in this chapter is the one where the discipline bends — because watching exactly where and how a tradition lets its own machinery flex shows what the machinery was for.

Chapter 11 introduced the *shtar iska*'s descendant, the *heter iska*, and chapter 31 will follow its enormous modern career, in which the instrument stands behind the interest-bearing operations of banks serving observant communities. Here is the joint on which everything turns. An *iska* investor's return, we have seen, is licensed by the deposit half's genuine exposure, and chapter 13 left the verification question standing: the venture's truth lives with the manager, so how does the investor ever collect without simply taking the manager's word? The classical documents answered with evidence law. The condition the tradition attributes to the Maharam of Lublin, and which the later standard forms adopted, runs in two halves: a claim of *loss* is believed only on the testimony of acceptable witnesses; a claim of *low profit* — less than the figure the parties named — is believed only under a biblical oath. Both halves are exquisitely calibrated to the theology of testimony: losses are public events, the kind witnesses can know; profits are ledger events, the kind only an oath can reach. And the documents added a third clause, the one that made the instrument's fortune: the manager may *decline the oath* and pay the named figure instead — the specified return, which the modern forms allow to be a fixed sum, a percentage, even a number keyed to prevailing interest rates — and upon paying it he owes no verification at all.

Run the machine forward and watch what it produces. An observant merchant takes financing under a *heter iska*. The venture does what ventures do. At term, he can open his books to witnesses, swear a biblical oath about his profits — an act the tradition regards with a seriousness modern readers must work to recover; the documents could even stipulate the oath be taken publicly, in the synagogue — or he can pay the named figure and be done. Almost everyone pays. The oath is heavy, the books are private, the figure was set with exactly this outcome in mind; paying-in-lieu is not an abuse of the instrument but, as the halakhic literature itself says plainly, a basic premise of how it functions. The result, visible from any distance: a stream of fixed periodic payments, sized like interest, timed like interest, keyed — in the modern forms — to the same benchmarks as interest, flowing under a document whose every clause is partnership.

Scale then finished what convenience began. The modern instrument's furthest form is the *heter iska klali* — the general *iska*, one master document executed by an institution and covering, prospectively, every depositor and borrower who ever walks in. The compendia record the halakhists' own unease in nearly comic register: some proposed hanging the document on the wall of the establishment, others objected that a sign nobody reads binds nobody; the banks of Israel fold the *heter iska* into their standard terms, and the posekim reason, with impeccable contemporary realism, that a customer is bound by an institution's known conditions whether or not he has read them. The debate is its own portrait of the instrument's late career: a partnership form so successful at wearing a loan's face that the last live question is whether anyone on either side of the counter remembers the partnership is there.

The tradition knows. That is what makes this section instructive rather than scandalous,

and the internal debate is precise about where the live nerve sits. May a manager who *knows* there was no profit pay the fee anyway, buying his way past an oath about a fact he could truthfully swear to? A minority of authorities balks exactly there — at the point where the payment stops summarizing an uncertain truth and starts replacing a known one. The majority answers with the instrument’s own logic: the fee was bargained as the price of *not verifying*, the investor genuinely cannot know the books, the manager is buying peace and privacy, and both parties walked in with open eyes. Meanwhile the same literature patrols the boundary that would turn candor into farce: a document naming as valid witnesses only men in a far city who cannot possibly know the venture is invalid — verification that cannot occur is not verification; an *iska* written over investments that are inherently unverifiable, hidden holdings no witness could ever attest, cannot carry a witnesses-clause at all. The line the *poskim* are holding, through every refinement, is the line this whole chapter defends: the structure must keep a real, live channel to the venture’s actual outcome. Narrow the channel, price the passage, let commerce buy its convenience — but the moment the channel is drawn on paper in a way that can never carry truth at all, the instrument stops being an *iska* with a toll and becomes a loan with a costume. Whether the modern instrument, at scale, still keeps the channel open — whether anyone, anywhere in a heter *iska* banking system, is really being asked about outcomes — is the hard question of chapter 31, and the tradition’s own *posekim* are its sharpest interrogators. What Part IV takes from the document is its unembarrassed honesty about the tension: the heter *iska* is a machine for living beside the profit rule under commercial pressure, and its drafters left the seams showing — the witnesses, the oath, the fee — precisely because a seam you can see is a seam a court can police.

Proportion and the Actual

Latin Christendom’s contribution to this chapter is a pair of measuring instruments: one for the *share*, one for the *substance*.

The share first. Scholastic partnership doctrine inherited Rome’s rule that shares follow agreement against a default of equality, valid wherever a partner “contributed something more — of money, of labor, or of any other thing” — and moralized it into a doctrine of proportion: gain is honest when it answers to contribution, each partner’s fraction a judgment about what each brought. Chapter 12 walked the components — ventured capital earning *tamquam de re sua*, work earning its *stipendium laboris* — and the just-price machinery of chapter 9 supplied the background conviction that exchanges answer to something real. What the schoolmen added at this chapter’s doorstep is the other instrument: a test for whether there is anything real to share. Their word for it was *actual*. Gain had to trace to an actual venture, actual labor, actual cost, actual peril — and the most quotable statement of the principle came, fittingly, from the same council session that approved the era’s great experiment in non-profit lending. In 1515, blessing the Monti di Pietà — whose small charge was allowed as “a moderate sum for their expenses and by way of compensation,” intended “exclusively to defray the expenses of those employed,” and “provided that no profit is made therefrom,” a story chapter 17 tells whole — the Fifth Lateran Council paused to define the enemy, and its definition is the entire chapter in one sentence: “For, that is the real meaning of usury: when, from its use, a thing which produces nothing is applied to the acquiring of gain and profit without any work, any expense or any risk.”

Read the definition backward and it becomes a construction manual. Honest gain requires a thing that produces — a venture, a field, a workshop, something with actual fruit in it — and it requires the claimant’s presence inside the producing: his work, his expense,

his risk, in some real combination. Usury is the null case on both axes at once: barren source, absent claimant. That is why the tradition could bless the Monti's charge in the same breath — costs actually incurred, recovered without a grain of profit, fail both halves of the definition's accusation — and it is why the scholastics' term *lucrum cessans*, profit merely hoped for elsewhere, spent centuries under suspicion the moment claimants tried to collect it as if hope were harvest; Part VI re-opens that file.

The same instinct for the actual disciplined the doctrine's own exceptions. When *Vix per-venit* restated the tradition in 1745, chapter 7 noted, it left carefully standing the titles that can arise alongside a loan — but every title on that list is a claim about something actual: a loss suffered and shown, a gain concretely interrupted, a default that in fact occurred. The door the encyclical held open was a door for facts, never for numbers written in advance. And where contribution itself needed pricing, the tradition let real markets do the measuring. The commenda's standard splits — three-quarters against a quarter in the one-sided form, half against half where the traveler also ventured a third of the coin — are chapter 11's textbook figures, and they are proportionality quoted at the going rate: the Mediterranean's own long-run judgment of what a season of a man's life was worth beside a season of a man's money. The Catholic instruments and the fiqh's converge from opposite workshops onto the same specification: the one polices the claim's proportion to real contribution, the other polices the profit's derivation from real remainder, and both refuse the word "profit" to any number that was written before the world answered.

The Till, Truly Counted

Which leaves the counting itself — and here the old doctrine turns its face directly toward the modern reader, because the assault on the profit rule did not end when stipulated returns went out of fashion. It moved inside the books.

Consider what must be true before the pile on the table contains any profit at all. The capital must be back — both traditions' first rule, Rome's netting, the standards' retention-for-protection. The costs must be paid: the venture's own keep, the road and the warehouse, the *mudarib*'s lodging that chapter 12 saw charged to the venture's purse rather than the worker's hide. The wages must be paid — not as a courtesy after the split but as a claim before it, the entire lesson of *sechar tircha* and the idle laborer's wage: labor is a cost of producing the surplus, and a "surplus" computed by leaving labor unpaid is not a surplus but a confiscation with good bookkeeping. The venture's known wounds must be provided for — the doubtful debt in Aden written down at what it is honestly worth, the standards' provision "for doubtful debts" doing in modern dress what the sworn accounting did on the quay. And something must be held against tomorrow — the reserve that chapter 13 watched the modern standards bless as the licit alternative to guarantee, the venture's future standing as a quiet claimant at its own settlement table. Only after that procession — capital, costs, wages, provisions, reserves — does a remainder appear that the traditions were willing to call by the name. Profit is a remainder, and a remainder cannot be promised. It can only be *found*, at the end, by counting honestly; which is why chapter 11's formula — a partner's return is discovered, never written — was always an accounting doctrine wearing a moral one, or the reverse; the traditions would not have recognized the distinction.

Honest counting, notice, is not free. Every device in this chapter costs somebody something — the sworn accounting on the quay, the witnesses, the valuations actual and constructive, the provisions argued line by line — and chapter 11 named the temptation that expense

whispers into every investor's ear: wouldn't you rather just be owed the number? The written return is, among its other characters, a refusal to pay the audit's toll. The traditions kept paying it, century after century, because the toll purchases the one thing the fixed claim can never deliver — a payout that is true. Modern practice has even learned to skip the counting altogether: a distribution financed by fresh borrowing declares a surplus no operation ever produced, profit conjured out of an enlarged claim on the future, the stipulated return's logic run backward through the till.

Now name the modern inversion, because the ToC does: profit before wages, reserves, and mission. A venture that declares distributable surplus while its workers are underpaid against any honest measure of their contribution has moved labor's claim below capital's in the very computation chapter 12 spent itself ordering the other way. A venture that distributes while running its reserves to the bone has spent its own future as if the future were income — handed tomorrow's anti-ruin cushion, chapter 10's vocabulary, to today's claimants. And the mission — the purpose the enterprise exists to serve, the care of the town, the craft, the customers, the commons — can be liquidated into "surplus" a quarter at a time without any single act of theft, merely by counting it as nothing; the charters Part X studies exist, in large part, to make that particular zero impossible to book. Each of these produces a number that looks like profit, audits like profit, and distributes like profit, and each is the stipulated return's quieter descendant: a claim paid out of the venture's substance rather than its increase, extraction booking itself as fruitfulness. The old lawyers had a whole taxonomy for gain that traced to no real remainder, and Part IX — where founder labor, worker claims, true surplus, and the non-extractive return formula get their modern accounting, chapters 38 through 40 — is in large part the exercise of running their taxonomy through a contemporary income statement. Chapter 39 will even take this chapter's question for its title: what, truly counted, is surplus?

The traditions' answer, assembled across this whole Part, fits in a breath. Surplus is fruitfulness — the *tokos* question of chapter 1 finally answered in the affirmative register. Aristotle called interest the unnatural offspring of a barren parent; the partnership doctrines are the traditions' portrait of the natural kind: real increase, born of a fruitful venture and present human effort, counted after every honest claim, shared in proportion to contribution, by people who stood inside the story. Where those conditions hold, all three traditions not only permit the gain — they praise it, defend it, and build their commercial civilizations on it. Where any fails, the gain is extraction wearing increase's face, whatever instrument delivers it: a stipulated sum, a guaranteed partnership, a fee that replaces a known truth, a distribution carved out of wages or reserves or mission. So Part IV closes by adding its third stone to the rule of the gate. Chapter 12: every real contribution priced. Chapter 13: every loss in its true seat. And now: no profit claim before real profit exists — the payout waits for the world, counts everything the venture owes, and takes only what is genuinely left. Three rules, one architecture, and a striking omission at its center: nowhere in it is there any payment for money as money, time as time. The traditions did not forget. They are about to show us what they built in that empty space — institutions that lend at the price of costs, or at no price at all, where the return sought is not a remainder but a neighbor kept afloat. Part V is the story of finance whose profit rule is that there should be none: the *gemach*, the beautiful loan, and the Mountains of Piety.

Part V — Cost Recovery, Mercy, and the Institutions of Non-Extractive Credit

Chapter 15 — The Gemach and the Beautiful Loan

On a December evening in 1892, eleven men met in the Wilner Synagogue on Henry Street, on the Lower East Side of Manhattan, and put their savings on the table. The pile came to ninety-five dollars. They were immigrants from the Russian Empire's Jewish towns, and what they founded that night they did not have to invent, because every one of them had grown up inside it: a fund that would lend money to the poor — their neighbors, the peddler short a cart, the seamstress behind on a machine payment, the family between disasters — and would charge, in the founders' own words, no interest or any expense whatsoever. Not low interest. Not cost plus a little. Nothing. The Hebrew Free Loan Society of New York is lending on Henry Street's terms still, a hundred and thirty years later, and by its own accounting has turned that first ninety-five dollars over into hundreds of millions lent — with something like ninety-nine cents of every dollar coming faithfully home.

Every part of that sentence violates what a modern reader knows about credit. Unsecured loans to the working poor are supposed to be the riskiest paper there is; here they repay at rates a prime mortgage book would envy. Lenders are supposed to need yield to cover losses; here there is no yield, and hardly any loss. And charity is supposed to flow downhill and disappear; here it flows out, comes back, and goes out again, indefinitely. This chapter is about the institution that performs these violations on purpose — the Jewish free-loan fund, the *gemach* — and about the very old idea it runs on. Parts III and IV of this book walked the wall against usury and the partnership gate through it. Part V enters the wall's third structure, the one easiest for finance to overlook because it makes no one rich: the credit of mercy, lending stripped of gain entirely. The *gemach* is its Jewish form, and the traditions will braid from here — the *qard hasan*, Islam's "beautiful loan," in the next chapter; the Monti di Pietà after that. But the *gemach* comes first, because its parent category is not, strictly speaking, finance at all.

The Kindness That Repays

The name says so. *Gemach* compresses *gemilut chasadim* — deeds of loving-kindness — a category the rabbis ranked among the pillars of the world and studied with the same precision they gave to contracts. The tractate Sukkah preserves the classic comparison, and it repays slow reading. Acts of kindness, the Sages taught, are superior to charity in three respects. Charity is performed with one's money; kindness with one's person and one's money. Charity goes to the poor; kindness to the poor and the rich. Charity goes to the living; kindness to the living and the dead (Sukkah 49b). On the same page, Rabbi Elazar sharpens the point through a verse from Hosea — sow according to charity, reap according to kindness — and glosses it like a farmer: one who sows may or may not eat, but one who reaps certainly eats; and he adds that charity itself is rewarded from Heaven only according to the kindness inside it — by the care taken, not the sum transferred. Kindness, in this accounting, is not charity's soft cousin. It is the larger and more exacting category, of which giving money away is one instance — and lending money for nothing, strange as the arithmetic sounds at first, is a higher one.

The tradition states the ranking outright, in a line this book has been walking toward since Part I. Greater is the one who lends than the one who gives charity, taught Reish Lakish —

and greater than both, the one who throws money into the common purse, taking a poor man in as his partner (Shabbat 63a). The commentators supply the reason, and it is not economic: the gift shames; the loan spares. A man handed money stands, however gently, revealed as an object of pity. A man lent money remains a man of affairs — someone whose signature is worth something, whose repayment is expected, who will settle his account and owe nothing to anyone. And a man taken into partnership is not even a borrower; he is a colleague. The ladder climbs away from humiliation, rung by rung, and money's terms are what each rung is made of. This is why the same page of Sukkah reads "walk humbly with your God" as the burial of the friendless dead and the dowering of the poor bride — kindnesses done, the Gemara notes, without fanfare. Dignity is the whole design constraint. The Torah had set it before the ladder was built: "If you lend money to my people, to the poor among you, you shall not deal with them as a creditor; you shall not exact interest from them" — chapter 2 read the verse's two prohibitions as one, a barred posture and a barred charge, because a man dealt with "as a creditor" has a patron whether or not he has an interest bill. Maimonides' famous ladder of charity, met in chapter 3, hangs from the same hook: its highest rung is not the largest gift but the gift, loan, partnership, or job that sets a man on his feet "until he no longer needs to ask." The best help ends the asking. A loan — repayable, repaid, finished — is engineered to end it.

Nor was the lending optional. Chapter 4 read the great chapter of Deuteronomy where the duty lives: when a member of your community falls into need, you shall not be hard-hearted or tight-fisted, but open your hand, "willingly lending enough to meet the need, whatever it may be." Lending to the fallen neighbor is a commandment, in the full legal sense, and the Torah armed it in advance against the lender's own arithmetic. The seventh-year release — every sabbatical year canceling debts among Israelites — threatened to do exactly what any credit analyst would predict: dry up loans to the poor as each release approached, the very meanness of heart the text warns against by name. When the prediction came true, Hillel the Elder answered with the *prozbul*, the device chapter 4 told whole, assigning debts to the court so that they survived the year of release. Read from this chapter's angle, the *prozbul* is the free loan's first great institutional patch: a legal system so committed to interest-free credit for the poor that it engineered around its own beloved release rather than let the lending stop. The *gemach* descends from both halves of that commitment — the commanded open hand, and the practical carpentry that keeps the hand full.

The baraita of Sukkah pays out one more design specification: kindness reaches the rich as well as the poor. A *gemach*, unlike an alms fund, does not means-test its front door, and the practice is wiser than it first looks. The comfortable householder in a bad cash-flow week borrows from the same counter, on the same terms, as the family in real trouble — which means the counter carries no stigma, because standing at it announces nothing. Charity marks its recipients by definition; a commons does not mark its users at all. Making the free loan universal is what makes it dignified, and the rabbis had said so in principle many centuries before the mutual societies and credit unions rediscovered it.

So the free loan is not a discount instrument, interest waived as a favor. It is a different instrument, built backward from the borrower's standing rather than forward from the lender's return. Repayment is not the price of the kindness; repayment is part of the kindness — the mechanism by which the borrower exits the transaction owing nothing, shamed by nothing, a full participant who kept his word. That is why the prohibition chapters 7 and 8 anatomized sits so comfortably underneath it. A loan that may carry no increase, no benefit, not so much as an unaccustomed greeting, is a loan that cannot make its lender a

patron — and a fund built of such loans is a machine for rescue without clientage.

Revolving Mercy

Now watch what the institutional form does to the arithmetic of giving, because this is the *gemach*'s quiet genius. A dollar given is a dollar gone — gloriously gone, but gone. A dollar lent at zero comes back, and goes out again, and comes back again. Interest teaches money to breed; mercy teaches it to circulate. The founders' ninety-five dollars did not fund one rescue in 1892 and retire; it has been out working continuously for thirteen decades, the same capital wearing successive emergencies, and the society's hundreds of millions in cumulative lending were revolved out of a base a tiny fraction of that size. The *gemach* is a perpetual-motion machine for chesed in exactly the sense the physicists forbid and the accountants permit: nothing is created, nothing extracted, and the wheel turns for as long as borrowers keep faith — which, at ninety-nine percent, is functionally always. The fund's enemies are not defaults; they are inflation, which quietly shrinks the wheel, and idleness, capital sitting unlent. Its friends are small sums, short terms, and the community's long memory. Run the wheel's arithmetic on a modest scale and the leverage shows. A fund of ten thousand dollars, lending five-hundred-dollar bridges repaid over ten months, puts roughly twenty households a year across their bad week; in a decade it has made two hundred rescues, and the ten thousand dollars are all still there, ready for the third decade and the fourth. No endowment, no yield, no growth — and no depletion. The gift model would have needed two hundred separate gifts; the wheel needed twenty donors, once.

And because the form is so simple — a pool, a ledger, a rule of zero — it proved wildly generalizable. The money *gemach* is only the parent. In the neighborhoods where the institution thrives, above all in Israel and the denser diaspora communities, there are *gemachim* for nearly everything a household might need briefly and cannot cheaply own: wedding dresses and bridal jewelry, lent gown by gown down a street of brides; tables, chairs, and dishes for a celebration; baby carriages, cribs, and car seats; power tools; folding beds for guests; medical equipment — crutches, wheelchairs, nebulizers, hospital cots — kept in spare rooms and lent at any hour; interest-free emergency funds for medical bills, tuition, rent between jobs. Some are institutions with offices and boards; a great many are a family's closet and a telephone number, a private *mitzvah* run for decades with the discipline of a lending library. It is worth pausing on what this ecology is, structurally: a community deciding, object by object, that access matters more than ownership and that need should be met from a common shelf rather than priced at a private counter. The rabbis' category stretched to receive it without strain, because *gemilut chasadim* had never been about money as such. It was about carrying one another — with one's person and one's money, as the *baraita* says — and a wheelchair lent at midnight is the doctrine in its plainest modern dress.

The modern free-loan societies are the ecology's formal tier, and their genealogy runs straight through the immigrant fraternities. The eleven men of Henry Street were copying the *landsmanshaftn* mutual-aid pattern of the towns they had left; sister societies rose in city after city across the Americas, and the model professionalized across the twentieth century into agencies with underwriting desks, guarantor files, and repayment plans — everything a small bank has except the yield. Today more than fifty accredited agencies, federated in an international association, oversee some \$364 million in free-loan assets by the association's own count, from the North American cities to Israel, where the member organization Ogen lends at the national scale and the informal *gemach* ecology beneath

it runs denser still. Their loan books read like biographies of ordinary precariousness: security deposits, dental work, citizenship fees, a used car to reach a job, the gap between a layoff and the next paycheck. The sums are modest; the leverage on a life is not.

The Rules of the Beautiful Loan

A fund that touches lending, however saintly its purpose, stands inside the full force of the law this book spent Part III mapping — and the halakhic literature around the *gemach* is the proof that mercy got no exemptions.

Start where chapter 8 left the problem: benefits. A *gemach* runs on donated and deposited capital, and every coin of it is analyzed. The donor's money is charity and raises chapter 8's question — the donation that hopes to buy a nephew's place in the queue — which the design grammar built there answers and this chapter need not rebuild: separate doors, published criteria, the fund alone as lender, the rolls guarded. The depositor's money is subtler, and the analysis is a small classic of the genre: money placed with a fund that will lend it out is not a deposit at all in the halakha's eyes but a loan to the fund — and a loan, even to a charity, is still a loan, so nothing may flow back to the depositor beyond his principal: no premium, no priority, no quiet benefit in any of chapter 8's currencies. His entire return is the *mitzvah* itself — his idle savings out doing rescue work between his own uses of them — and the literature's willingness to say so plainly is the whole prohibition in miniature: there is no party, however generous, from whom increase on a loan may lawfully travel.

Then the guarantors — and here the tradition shows its least sentimental face. The great modern advocate of the free-loan fund was Rabbi Yisrael Meir Kagan of Radin, the Chofetz Chaim (c. 1838–1933), whose treatise *Ahavat Chesed* — the love of kindness — argued communal lending into a program and whose rule for it has the ring of a man who kept books as well as commandments: a *gemach* must take proper precautions that its money comes back — guarantors, security, documentation, “and the like.” Mercy, in other words, does not underwrite carelessly; it underwrites *better*, because the capital it risks belongs to the next borrower. The standard instrument is the *arev*, the co-signer the halakha has regulated since the Mishnah counted him among a loan's responsible parties: the fund collects from the borrower first and the guarantor only after, and the guarantee itself is halachically clean here for a reason worth savoring — the intricate case law about guarantors entangled in interest, which can treat a co-signer as a lender for prohibition purposes, simply has nothing to bite on when the loan carries no interest at all. A zero loan has clean paperwork all the way down. And the guarantor requirement quietly performs a second service: it recruits the borrower's own community into his rescue, two or three names on a form standing surety not as a humiliation but as a statement — we know him; he is good for it — which is, among other things, how a fund achieves a repayment rate that embarrasses the credit-scoring industry. The wheel turns on trust, but the trust is organized.

What the fund may not do is charge for turning. Late fees on a loan are increase for delay — the oldest *riba* there is, chapter 6 said, and the halakha's *agar natar* refuses it with equal flatness — so the *gemach*'s remedies for the slow payer are patience, the guarantor, and the community's regard, in roughly that order. Administrative reality presses, as it presses on every institution in Part V — clerks, ledgers, storage for the wheelchairs — and how far a fund may recover bare costs without the recovery curdling into yield is a live question in the literatures, handled with the same instruments this Part will watch the Monti's defenders build in chapter 17. The instinct, everywhere, is Henry Street's: “or any

expense whatsoever” is the ideal, donations carry the overhead, and the loan itself stays perfectly, symbolically clean. The literatures keep annotating the edges with their usual appetite. Whether tithe money may seed a loan fund is a standing question, precisely because a loan is not a gift and comes back; how a fund’s paper rides through the sabbatical release in the Land of Israel is another, answered in practice by Hillel’s own instrument, the *prozbul* still doing after two thousand years the job it was built for. None of the annotations loosens the core rule; all of them exist because communities actually run these funds and actually meet the edges. And around the whole operation stands the discipline of discretion. The rolls are guarded for Rav Dimi’s reason — a community that knows who owes cannot help collecting in deference, and the creditor who may not even walk past his insolvent debtor is certainly not entitled to publish him. The nineteenth-century societies took confidentiality as seriously as solvency; the Gemara’s “without fanfare” is their operating manual. A *gemach* that bragged of its borrowers would have destroyed with its left hand what its right was building — the possibility of being helped and remaining, in every public sense, unhelped.

Underneath all the rules runs the fund’s true collateral, which no clause names. Chapter 11 found the Geniza merchants running ventures across two seas on reputation — the trader’s name as his working capital — and observed that the arrangement priced honesty better than any court could. The *gemach* is the same name-economy, opened to people who own nothing else. The applicant with no assets, no credit file, and no salary history has one thing the community’s fund can see and the distant bank cannot: he is known — his family, his history, his word, the two neighbors willing to sign under his name. The fund lends against that knowledge, and the astonishing repayment rates are the knowledge proving out. There is a hard institutional lesson folded in here, worth stating before the *qard hasan* funds and the Monti raise it again: information the size of a community is itself a form of capital, and mercy finance works not because it ignores risk but because it is better informed about the only risk that matters — character — than any scoring model has yet managed to be.

The Bridge at the Locking Door

Why does a small, brief, zero-interest loan work such disproportionate rescue? Chapter 10 built the vocabulary for the answer, and it is worth restating here in plain clothes, because the *gemach* is the vocabulary made brick and ledger.

A life is lived in sequence, one month after another, each standing on the last. Most financial shocks are temporary in themselves — a fever, a layoff, a burst pipe, a season’s bad receipts — but a sequence can be ended by a temporary thing. Miss the rent and lose the apartment; lose the apartment and lose the job that depends on the address; sell the tools to eat and lose the trade that needs the tools. The mathematics chapter 28 will formalize calls these locking doors absorbing states, and the chance of passing one a ruin probability; the ancient world needed no formalism to know that a household can fall through a bad month into a place from which no later good month retrieves it. Stand at that door and look at the two instruments on offer. Credit at interest — the storefront kind that clusters, in every era, exactly where the emergencies live — bridges the month by adding a permanent charge to a temporary problem, so that the borrower crosses the gap carrying a new weight sized precisely to how desperate he was; the very seasons that cannot pay are the seasons that pay most, and one bad month becomes a long correspondence with ruin. The free loan bridges the same gap and adds nothing: the shock passes, the loan is repaid out of recovered ordinary income, and the sequence continues as if the door had never

appeared. It is the cheapest anti-ruin technology ever devised — a few hundred dollars, at the right week, against the collapse of a household — and it is priced at zero not from naivety about money but from clarity about what is being purchased. A community that lends to its own at zero has decided that no neighbor's bad month should have to turn a profit.

The ancient world's alternative at that same door, chapter 2 showed, was the patron — rescue priced in dependency, the bad year converted into a lifetime of owed deference — and the storefront lender waiting at the same door is only the patron with a license and a rate card. Where the mercy funds are absent, the traffic through the locking door does not stop; it reroutes, to whichever lender waits there, on whatever terms desperation signs. That is why the *gemach* belongs to this book's infrastructure chapters rather than its charity ones. It is not an ornament on a community's economy. It is load-bearing: the difference, household by household and year by year, between a town whose members stumble and recover and a town that slowly sorts itself into creditors and the permanently fallen.

This is what the ToC's phrase *solidarity infrastructure* means, and the word infrastructure is doing real work. Chapter 2 watched the traditions choose, against the patron and the clean-slate king, the third answer to subsistence risk — the community that catches its own — and chapter 3 gave the catching its taxonomy: the pool, many small contributions standing between each member and the single ending blow. The *gemach* is that pool in credit form, permanent, replenishing, addressed to the one kind of catastrophe a loan can actually fix: the temporary one. It does not pretend to fix the permanent kind — the chronic shortfall, the disability, the widow with nothing coming in; for those the tradition built the outright-transfer institutions, *tzedakah* as obligation and its Islamic sibling *zakat*, which chapter 18 will set beside it. The two systems are calibrated to the two shapes of trouble: gifts for the falls no bridge can span, bridges for the stumbles that only need a rail. Run them together and a community has built, out of donations and discipline, what a later century would call a social safety system — except that this one's working capital never depletes, because most trouble is temporary, and temporary trouble, caught early and charged nothing, pays itself back.

What the Money Learns

It remains to collect what this chapter teaches the rest of the book — because the ToC's last bullet asks the impertinent question directly: what can venture finance possibly learn from a charity closet full of wedding dresses?

Begin with the fact that the tradition itself refuses to keep the two worlds apart. Reish Lakish's ladder does not stop at the free loan; it climbs past it, to the partner who throws money into the common purse with the man he means to help. The summit of the mercy scale is an instrument from Part IV — the working partnership, capital and labor sharing fate — which means the rabbis saw a single continuum where the modern mind sees a wall between philanthropy and finance. At the bottom, the gift, for the fall no structure can catch. Above it, the bridge loan, for the stumble. At the top, the partnership, for the climb — the instrument that does not merely rescue a man's month but joins his venture, because the deepest dignity is not being spared embarrassment; it is being worth investing in. Mercy finance and venture finance, on this ladder, are not opposites but altitudes of one climb.

So the lessons travel upward. First: circulation is a design choice. The *gemach* proves that capital can be structured to revolve — out, back, out again, serving serially — rather

than to extract and exit; the evergreen and revolving structures Part X will examine, funds built to recycle their returns into the next venture rather than distribute them out the top, are the gemach's shape scaled up, and chapter 41 will draw the line explicitly. Second: discipline is not extraction's property. The free-loan societies underwrite, take guarantors, document, and collect — everything a bank does — and their repayment rates show that rigor and mercy are not trade-offs; what they never do is let the rigor become a revenue line, and the distinction between disciplined capital and extractive capital is exactly the one Part IV spent three chapters drawing. Third: dignity is an engineering specification, not a tone of voice. Confidentiality, standard terms, published criteria, the borrower's exit owing nothing — chapter 8's grammar and this chapter's practice — can be written into any instrument, including a term sheet; clientage is a design failure, and it is optional. And fourth: the pool beats the premium. The gemach community handles the risk of its members' bad months not by pricing each member's fragility back to him but by standing a common reserve between everyone and the door — the anti-ruin logic chapter 10 stated and chapter 28 will prove, already running for centuries on donated capital and a ledger.

There is a last observation, and Part IV's lawyers would have enjoyed it. Run the gemach through the three tests that Part just finished building and it passes without breaking stride: no contribution goes unpriced, because nothing is charged for anything; no loss sits in a false seat, because the donors chose their seat with their eyes open and called it giving; no phantom profit is declared, because none is sought. The tests were engineered to catch extraction dressed as increase, and an instrument with nothing to extract walks through every checkpoint with empty pockets and a clean bill. That emptiness is no absence of finance but finance with the polarity reversed — all of the machinery, none of the take — and holding it beside the term sheet is the most clarifying exercise this book knows.

None of this, the tradition would insist, is utopian bookkeeping. The wheel on Henry Street has turned through depressions, wars, and pandemics; the closets of Jerusalem lend their thousandth wheelchair as reliably as their first. What keeps the machinery honest is the same rule that founded it — nothing may ride back on a loan, not interest, not benefit, not standing, not so much as a greeting — enforced not as austerity but as the price of the thing being built: help that leaves the helped intact. Islam, watching the same human facts through its own revelation, arrived at an instrument so similar that the two traditions' vocabularies translate almost word for word — a loan the Qur'an itself calls beautiful, lent to the neighbor and accounted, astonishingly, as lent to God. Its story is next.

Chapter 16 — Qard Hasan and the Beautiful Loan in Islam

There is a verse of the Qur'an in which God asks to borrow money. Not a tithe, not an offering — a loan, by name, with terms: "Who will give God a good loan, which He will increase for him many times over? It is God who withholds and God who gives abundantly, and it is to Him that you will return" (2:245, in M. A. S. Abdel Haleem's rendering, as throughout this book). When the verse was first heard in Medina — so the classical commentators tell the story — a man of the Ansar named Abu al-Dahdah went to the Prophet to make sure his ears had not betrayed him: does God, who needs nothing, ask us for a loan? He does, came the answer, and what He intends to repay with is Paradise. Abu al-Dahdah did his accounting on the spot. He owned two orchards, he said, and nothing besides; let the better of them stand lent to his Lord — six hundred date palms. The Prophet's reply, as the commentators keep it: whole orchards of laden clusters were already waiting for Abu al-Dahdah on the far side of the books.

The story is the tradition's own; the verse is scripture, and it does not stand alone. Five more times, in suras revealed years apart, the Qur'an returns to this strangest of requests, in which the party asking to borrow is described in the same breath as the source of everything the lender has. The verse itself performs the reversal: it is God who gives abundantly, says the sentence in which God asks to receive. Whatever this is, it is not fundraising; it is a doctrine of where money can be sent, and the tradition that received it heard something exact: a loan, with a real borrower's obligations attached — gratitude and multiplied repayment — offered to anyone willing to put wealth where no earthly return can follow it.

The previous chapter ended at this door. The *gemach*, we saw, is Judaism's machine for lending stripped of gain, and its close observed that Islam, reading the same human facts by its own light, built an instrument so similar that the two traditions' vocabularies nearly translate each other. This chapter is that instrument's own story: the *qard hasan*, the beautiful loan — what the verses say and how the jurists secured them; why a loan that must be repaid ranks, in this tradition's eyes, above a gift that need not be; why the instrument is not an investment and was never meant to be; how it grew institutions, from the rotating pots of Iranian households to the mosque courtyards of Lahore; what a mercy fund may honestly charge, and how a service fee goes bad; and what this layer of capital — the layer that does not grow — contributes to everything above it.

The Borrower Who Owns Everything

Start with the verses, because Islamic law starts there and because their language repays the close reading chapter 6 gave the war verses. The request of 2:245 recurs almost word for word in the sura of Iron: "Who will make God a good loan? He will double it for him and reward him generously" (57:11). Seven verses on, the same sura names the company the loan keeps: "Charitable men and women who make a good loan to God will have it doubled and have a generous reward" (57:18) — givers of charity and makers of the loan, side by side in one sentence, two instruments of one mercy. The sura called al-Taghabun adds what may be the most startling clause in the family: "if you make a generous loan

to God He will multiply it for you and forgive you," it promises, and then describes the borrower — "God is ever thankful and forbearing" (64:17). A scripture in which God is the party giving thanks: the tradition read the clause with lasting wonder. And in the night-vigil sura the loan joins the tradition's innermost circle, in the verse chapter 11 read for the merchants and chapter 6 for the alms: "keep up the prayer, pay the prescribed alms, and make God a good loan" (73:20) — prayer, *zakat*, and lending to God in a single breath, followed by the ledger's rule: "Whatever good you store up for yourselves will be improved and increased for you."

Scholars who have counted observe that these verses — with one more we will come to — are the only places the Qur'an uses the word *qard* at all. The scripture never once applies the word to money lent at the market; the long verse on debts, with its scribes and witnesses, speaks a different vocabulary. In the Qur'an's own usage, the only loan worth the name is the one made to God, and the hadith use the plain word *qard* without the ornament. The ornament is the Qur'an's: *hasan*, from the family of *ihsan*, the beautiful, the excellent — the word the tradition uses for doing a thing as if you could see God watching. The noun underneath is homelier. *Qard* is said to come from a root meaning to cut: the lender cuts a piece from his wealth and hands it away, as chapter 3 heard the jurists say of the *qirad*'s cousin root. Cut a piece of what you own, says the whole phrase, and cut it beautifully.

The verses build an economy of deferred and multiplied return. Chapter 6 watched the Qur'an declare that increase taken from a neighbor's need is no increase at all — that God blights it — while charity, which looks like depletion, compounds. The good-loan verses run the same arithmetic in its generous direction. The multiplication promised is not a metaphor the tradition softened into sentiment; it is stated as flatly as any commercial term — doubled, many times over, rewarded, forgiven — and it is precisely located. None of it accrues on earth. The Qur'an that forbade a believer to collect one dirham of increase from his brother promises him increase without limit from God; the whole moral engineering of the instrument sits in that relocation. The appetite for return is not condemned; it is redirected toward the one creditor — the phrase is exact, for in this transaction the believer is the creditor — who cannot be impoverished by paying and cannot be pressed for payment.

And the tradition never doubted where the coins themselves go. The neighbor takes delivery; heaven takes the debt. Commentators from the earliest generations read the verses through the practice they commanded: to lend to God is to lend to His servants in their need, without increase, without benefit, without the posture chapter 2 heard Exodus forbid — and the loan so made is accounted upward, entered against a name that does not default. One verse makes the mechanism nearly explicit, and it matters most to this book's braid. In the fifth sura the good loan appears in the covenant of an earlier community: "God took a pledge from the Children of Israel," the passage runs, and the pledge's terms include "if you keep up the prayer, pay the prescribed alms, believe in My messengers and support them, and lend God a good loan, I will wipe out your sins" (5:12). The Qur'an, in other words, does not present the beautiful loan as an innovation. It reads the instrument back through the whole prophetic line — the same line along which, chapter 6 found, the *riba* prohibition had always run. Israel too was asked for this loan. Deuteronomy's open hand — chapter 4's text, chapter 15's institution — is in the Qur'an's account the same command in an older dispensation. When the free-loan fund of the synagogue and the free-loan fund of the mosque translate each other almost clause for clause, the Qur'an's response would be that of course they do; the creditor is the same.

So the definition the jurists secured is already visible in the verses. A *qard hasan* is a loan

of money or any fungible — ownership passing, the like amount owed back, the anatomy chapter 7 drew — carrying no stipulated benefit of any kind, made from goodwill toward a borrower who needs it, in the expectation that God repays what the borrower cannot. Benevolent lending, the standard English has it. The law's whole later apparatus is a set of walls around that little word benevolent.

Mercy with a Due Date

The Sunna's editors filed with the same quiet wit chapter 6 found in Bukhari's. In Ibn Majah's collection, the chapter on lending sits inside the Chapters on Charity — the loan shelved among the gifts, which is the doctrine in a single act of librarianship. And the report that anchors the chapter arrives wrapped in a story about its own enforcement, told by a man named Qays ibn Rumi (Ibn Majah 2430; the graders account it sound). Sulaiman ibn Udhunan had lent Alqamah a thousand dirhams until his stipend came; when it came, he collected roughly, and Alqamah paid and went away angry. Months later Alqamah was short again, and came back and asked — and Sulaiman said yes, and counted it an honor, and called to Umm Utbah for a sealed leather bag. Here are your own dirhams, he told the astonished borrower; I never touched one of them. Then what, Alqamah asked, made you treat me so harshly the first time? What I heard from you, said Sulaiman: your own report, from Ibn Mas'ud, that the Prophet said no Muslim lends to another twice but it counts for him as charity given once. The lender had been persuaded by his own borrower's scholarship, and the persuasion took the form of the same coins going out again — a one-man revolving fund, the gemach's arithmetic at the scale of two neighbors and a repented grudge.

A second report in the same chapter climbs higher and carries a health warning the tradition itself affixes. On the night of the ascension, the Prophet is made to say, he saw written over the gate of Paradise that charity is repaid tenfold, but a loan eighteenfold; and when he asked Jibril why a loan should outrank charity, the angel answered: because the beggar may ask while he still has something, but the borrower does not borrow except from need (Ibn Majah 2431). The hadith scholars grade the report weak, and say so, and this book quotes it as the jurists always have — for its picture, not its chain. The picture is the tradition's dignity analysis compressed into one sentence of angelic underwriting. A request for charity may misjudge itself; a request for a loan — which must be repaid, and which the asker therefore makes against his own future labor — is need's honest signature. The accountant blinks at the arithmetic over the gate; the moralist reads it at once. Charity answers a hand held out; a loan answers a head held up. Readers of the last chapter have met this ranking before, in the rabbis' ruling that the lender outranks the almsgiver — chapter 15 climbed that ladder to its summit and it will not be re climbed here. It is worth one sentence of wonder that two traditions, reading two scriptures, ranked mercy's instruments in the same unobvious order for the same stated reason.

Believing in the borrower means expecting him to repay, and here the tradition is as unsentimental as the gemach's rulebook. The qard hasan is a real debt. The borrower owes the principal as absolutely as any commercial debtor — the fiqh never invented a soft loan whose repayment was optional; that instrument is called *sadaqah*, with its own law and its own chapter later in this Part. What distinguishes the loan is that everything beyond the principal runs on a strict one-way valve. From the borrower's side, generosity at repayment is virtue itself: the Prophet, repaying a borrowed camel with a better one, said that the best of you are the best in repaying — chapter 7 told it from Bukhari — and the classical rule permits any excess the borrower freely adds at settlement, provided nothing of the

kind was stipulated, promised, or quietly prearranged. From the lender's side, the same excess demanded — or expected, or engineered by custom — is *riba*, full stop; the jurists' working maxim, every loan that draws a benefit is *riba*, covers the beautiful loan precisely because there is nothing a benefit could be but the corruption of it. (A maxim, chapter 6 was careful to mark, not a certified hadith.) Between the borrower's licensed gratitude and the lender's forbidden expectation runs the line chapter 8 spent its whole length drawing, and the *qard hasan* lives or dies by it. As for delay: the solvent procrastinator commits the injustice Bukhari 2287 names, and may be pressed; the genuinely strained debtor holds 2:280's entitlement — "if the debtor is in difficulty, then delay things until matters become easier for him" — and the merciful write-off that verse commends stands ready as the loan's last conversion, back into the charity it was always accounted as.

Which brings us to a question this chapter owes a blunt answer: why is this instrument not an investment? The answer organizes everything this book has said about Islamic finance into one sentence. The Shariah leaves capital exactly two honest postures toward money placed in another's hands, and they are mirror images. The partner's posture — the *mu-darabah* of chapter 11 — surrenders every guarantee to keep its claim on yield: the capital rides the venture, the return is unknowable, and a guaranteed dirham voids the contract. The lender's posture surrenders every yield to keep its guarantee: the principal is owed absolutely, come flood or famine, and one stipulated dirham of increase voids the loan's innocence. Guarantee or yield; choose. The seat between the two — principal protected and increase collected — is the seat the Qur'an burned, as chapters 6 and 7 told. Seen from that geometry, the *qard hasan* is no investment performing badly; it is the guarantee-side posture performed perfectly, and the jurists filed it accordingly: not among the commutative bargains at all but among the acts of accommodation and kindness — a gratuitous contract, they say, a donation not of the money but of the money's use for a season, with its like coming faithfully home. An investor asks what his money will become. The maker of a *qard hasan* is not asking anything of the money; he is asking something of the month — that it carry his neighbor past whatever stood in the road — and the only balance sheet his return appears on is the one Abu al-Dahdah traded an orchard against.

Brotherhood at Scale

An instrument this personal scales the way the *gemach* scaled: it becomes a fund, and Islam has built the fund in every register from a housewife's notebook to an act of parliament.

The humblest form never needed an institution. Across the Muslim world, households run what the microfinance literature calls a rotating savings-and-credit association: a circle of neighbors, colleagues, or cousins each contributes a fixed sum on a fixed rhythm, and the gathered pot goes to each member in turn, by lot or by need, until the circle closes. Iranians call the arrangement by the loan's own name, *qarz al-hasanah* in the Persian pronunciation; the Gulf knows it as the *jam'iyyah*, Pakistan as the committee. The jurists who examined the form found it clean: no member's benefit is stipulated from any borrower. Each participant who takes the pot repays it forward into the turns of the others; what a lender receives comes not from his debtor but from the circle's continuing life, and the scholars who parsed the contract stack concluded, under every analysis, that nothing rides on the loan that the law forbids. Village Iran, one economist notes, ran the same architecture in milk — each household with a cow pouring its measure into a common pot that each took whole in turn — and the jurists of the day saw nothing to condemn. The pool, chapter 3's fifth name for money, keeps passing its audits.

Above the circles stand the funds, and above the funds, in one country, the law itself. Iran's bazaars and neighborhoods have long grown free-loan funds — the *sandug*, the box — run on donated and deposited money; and when the Islamic Republic rewrote its banking law in 1983, the beautiful loan went into the statute: banks were authorized to take deposits under the *qard hasan* title, the depositor lending the bank his balance at zero, and dedicated *qarz al-hasaneh* banks followed in time, one chartered at national scale in 2007. The everyday form is closer than most depositors think: in Islamic banking everywhere, the ordinary current account is juridically a *qard* to the bank — an interest-free loan from customer to institution, repayable on demand, on which the bank may promise nothing, not interest, not prizes — and the *halakha*, chapter 15 found, analyzed money deposited with a *gemach* the same way. The person who funds a lender is himself a lender, and mercy's plumbing must be *riba*-proof at every joint.

The institution that made the *qard hasan* famous in our century began with a widow and a civil servant in Lahore. In 2001, Dr. Muhammad Amjad Saqib was working in the Punjab Rural Support Programme, a conventional microcredit operation, when — as the organization he founded tells its own story — a widow raising four children came asking for help that no interest-bearing product could honestly give. Saqib and a few friends answered from their own pockets, and the first loans went out at nothing: no interest, no fees, no costs, repay when you can stand. Out of that beginning grew *Akhuwat* — the name is the Arabic for brotherhood — which by 2020 had made, on its own count, four and a half million such loans, one hundred twenty-eight billion rupees, some eight hundred million dollars, to more than three million families (later counts run past six million loans), and which reports, year after year, that better than ninety-nine percent of it (the institution says 99.9) comes home. The reader who met the Hebrew Free Loan Society's figures in chapter 15 will recognize the epistemics — these are the institution's own books, marked as such — and the pattern: two organizations on two continents, lending to the poor at zero, reporting repayment the collateralized industry cannot match. It begins to look like a finding.

The name was chosen with a scene in mind, and the scene is in Bukhari. When the Muslim emigrants reached Medina with nothing, tradition remembers, the Prophet paired each with a local believer in a bond called *mu'akhat*, brotherhood — and Sahih al-Bukhari preserves the pairing of Abd al-Rahman ibn Awf with Sa'd ibn al-Rabi', who told his new brother: I am the richest of the Ansar; take half my wealth (Bukhari 2048). Abd al-Rahman's answer completes the tradition's picture of dignity from the other side: I am not in need of all that, he said; is there a market where trade is done? There was — the market of Qaynuqa — and he went at first light. Both halves are the teaching. The offer had to be whole-hearted and it was; the refusal had to be possible and it was; and within days, Bukhari's narration adds, the man who chose the market over the gift had traded his way from butter and dried yogurt to a marriage portion of gold. An institution named for that exchange is claiming what rescue should feel like — a brother's hand, not a patron's — and where it should point: toward the borrower's own earning power, as fast as possible.

The model runs on five stated principles, and three of them would have made the Chofetz Chaim nod. Loans begin at ten thousand rupees — some thirty-five dollars — and carry no interest and, by *Akhuwat*'s own rule, no service charges or financial costs of any kind; donations carry the overhead. Disbursement happens in religious buildings — mosques and, deliberately, churches — after prayers, before the community, in rooms where no overhead is billed and every witness is free. And borrowers are invited, once on their feet, to become donors themselves — pennies a day, entirely voluntary, the wheel recruiting its

own next push. A researcher from Cambridge who went to look, Kamal Munir, reported the texture: head-office staff sitting on the floor of a single room; employees volunteering a share of their time; borrowers with several debts outstanding who repay Akhuwat first, against every prediction of the textbooks, because this is the lender they cannot bear to lose; women and non-Muslims walking into mosques, and Muslims into churches, because that is where the lending is. The public ceremony is the *gemach*'s dignity problem solved by the opposite engineering. The *gemach* guards its rolls — help so private it never happened. Akhuwat performs the loan in the sanctuary — help so public it carries no shame, reframed from a private embarrassment into an act of communal worship in which the borrower is a participant, not a case. Privacy erases the event; liturgy transfigures it. Both work; which works better is a question about a community, not about mercy.

The backdrop should be stated plainly, because it is Akhuwat's own reason for existing. Conventional microcredit, whatever its founding hopes, lends to the world's poor at rates researchers put between thirty and fifty percent a year and sometimes beyond, justified by the real costs of tiny loans; and its recovery practices, documented in the villages around Lahore, can lean on the one asset the very poor still hold: their standing before their neighbors. Debt collected by shame, from women with nowhere further to borrow, in communities where shame is ruin: this is the machinery chapter 2 met in older costumes, and it is what a mosque courtyard full of zero-interest capital is built against. Most Islamic microfinance, it should be said, runs on priced instruments — above all the markup sale this book reaches in chapter 24 — and the industry argues internally, in the *ikhtilaf* register chapter 32 will observe, over whether gratuitous lending can be more than a boutique. Akhuwat's books are one answer. The *qard hasan* windows that Islamic banks maintain — funded, the Indonesian scholars' council rules, from the bank's own capital, set-aside profits, and entrusted charity, never from customers' investment accounts — are another. The beautiful loan remains what it was in the verses: not the economy's engine but its mercy, a layer deliberately exempted from the market's logic and guarded from its appetites.

The Price of Running Mercy

Guarded, above all, at the seam where mercy meets its own payroll. A fund is not a friend with a sealed bag. It has clerks, ledgers, offices, motorbikes for the loan officers, tea for the committee; someone must pay for it all, or the fund dies of its own administration. Henry Street's founders answered with an ideal — no interest or any expense whatsoever — and let donors carry every cost. The question presses on every free-loan institution in this Part, and twentieth-century Islam answered it in public, at the scale of an intergovernmental bank. When the Islamic Development Bank asked the Islamic world's assembled jurists what it could charge on its loans, the International Islamic Fiqh Academy, meeting at Amman in 1986, gave the answer that has governed since: a fee for loan-related services is permissible "within the limit of the actual expenses"; and any fee beyond actual service-related expenses is prohibited, "because it is considered as *riba*." Actual costs, recoverable; one dirham more, and the old sin has a new name.

The industry's standard-setter turned the ruling into machinery. AAOIFI's Shariah standard on the loan — the nineteenth in the Bahrain body's rulebook chapters 13 and 14 have already consulted — restates the classical core: a stipulated excess for the lender is *riba* whatever its form, quality or quantity, object or benefit, stipulated at signing or extracted at extension, and custom is no defense. On fees it is stricter than most secular regulators would dream of being. An institution may recover the actual cost of servicing a given loan; the calculation must be precise, transaction by transaction; the method must be approved

and supervised by the institution's Shariah board; and — the clause with teeth — indirect costs do not count. Staff salaries, building rent, general expenses: not chargeable. Those belong to the institution's own budget, which is to say to its donors and shareholders, because the alternative books mercy's overhead onto the borrowers least able to carry it and calls the result a fee.

The reason for all this precision is the oldest reason in this book: the fee is where the profit hides. A charge labeled cost recovery that is set as a percentage of the loan — two percent of principal, so much per month outstanding — has confessed in its own arithmetic: real service costs do not double when the loan doubles, and do not accrue with the calendar; a fee that scales with amount and term is a rate, whatever the paperwork calls it, and a rate on a qard is *riba* with a receipt stapled to it. The diagnostic is the same one chapter 8 taught for benefits in kind — ask what the payment varies with; anything that varies with the loan is the loan's price — and the institutional grammar is chapter 8's too: separate doors, donors carrying the machine's costs, windfalls diverted outward. Christianity fought this exact battle over its own mercy banks, and settled it, at a Lateran council, in the same terms — recovery of expenses, yes; profit, no — as chapter 14 noted in passing and chapter 17 will tell properly. Akhuwat's answer is the most radical available and the easiest to audit: charge nothing at all, and publish what the donors paid. Between Henry Street's nothing whatsoever and Amman's actual expenses runs the entire honest range; everything outside it is a margin in a fee's clothes, and the traditions wrote these rules because they knew that disguised profit is the form in which extraction always asks to be let back in.

The Floor Under the Market

Set the two mercy institutions side by side, as chapter 8 promised this chapter would, add the third tradition's, and the family portrait almost composes itself. The *gemach* and the qard fund run the same machine: a pool of donated and deposited capital; loans at absolute zero; guarantors and community knowledge doing the work of collateral; repayment expected, and achieved; the capital revolving, serially, indefinitely — out, back, out — so that a founding gift keeps rescuing for a century. Their vocabularies translate: *gemilut chasadim*, deeds of loving-kindness, is what the *gemach*'s name compresses; *ihsan*, beauty-in-doing, is what the qard's adjective invokes; and both traditions rank the loan above the gift with the same argument about the borrower's face. Their safeguards translate too — the depositor who may take back only his principal, the donor whose gift buys no place in the queue, the fee that may never outgrow its receipts. Where they differ is instructive: the *gemach*'s counter is unmarked and universal, its rolls guarded like state secrets; Akhuwat's courtyard is public and liturgical, the loan handed over before the congregation — the same dignity pursued down two opposite roads, as the last section watched. And the Qur'an, as we saw, would regard the deep agreement as no coincidence at all, having itself recorded the good loan among the terms of Israel's covenant. As for Christendom: its scripture set the same stone — "lend, expecting nothing in return," the Gospel word of chapter 5, which chapter 8 watched a pope apply to a lender's unwritten hopes — and its tradition catalogued mercy into works, fed the hungry, ransomed the captive, buried the dead, and in time built gratuitous credit its own great institutional house in the Italian cities, the *Monti di Pietà*, whose story belongs to chapter 17. Three traditions, commanded toward the same borrower, built the same machine three times.

Name what the machine is, and the chapter's last obligations fall into place. Beneath every economy's growth layer — the partnerships of Part IV, the priced instruments of Part VII, the venture stacks of Part IX — the traditions insisted on a stratum of capital with a different

job description entirely: the non-surplus layer, money organized to hold rather than to grow. Its instruments are the ones this Part collects — the gemach's closets and ledgers, the qard funds and the bank windows, the chest the next chapter opens, the alms chapter 18 weighs — and its defining renunciation is the one this chapter has anatomized: principal guaranteed, yield forsworn, the venture layer's exact inverse. It would be a mistake to read the renunciation as capital failing or resting: the layer works constantly, and pays its returns in a currency the market does not quote. Mercy capital collects its wages in things that do not happen. The tools not sold in a bad month, the child not pulled from school, the shop that did not close, the widow who did not sign whatever the last lender at the door was offering: that is the yield, and a community holding such a layer is richer by every disaster it quietly declined to have.

The logic underneath was built in chapter 10 and needs only one plain paragraph here, with its mathematics still on deposit at chapter 28. A household lives one path through time, month standing on month, and some of the places a path can reach do not give a family back: the formal name is an absorbing state, and the chance of entering one is a ruin probability. Averages across many households — the ensemble a portfolio sees — never register such places; the one path a family actually walks, the time average of chapter 10's vocabulary, registers little else. Most shocks are temporary, but a temporary shock at the wrong moment ends a sequence permanently, and everything afterward, all the good months included, is lost with it. The non-surplus layer is the community's anti-ruin reserve stationed exactly there, at the moment of the shock, lending the one thing that converts a permanent fall back into a temporary problem — principal, for a season, priced at zero, because any price charged at that moment compounds the very desperation it feeds on. This is why the layer must not seek yield: its entire product is the survival of the paths that cross it, and yield extracted at the crossing is subtracted from exactly that.

Which yields, in closing, the design brief this Part has been assembling — because zero-interest capital is not a museum piece, and the traditions' rules read like a specification for anyone building the layer today. Fund it as a gift, run it as a loan: donors capitalize the wheel once, borrowers return what they drew, and the layer, uniquely among financial institutions, never needs to grow to survive, only to not shrink, which makes inflation and idleness its real enemies. Recover costs with receipts, never with percentages: Amman's actual-expenses line is an auditable spec, and any fee that varies with loan size or term has failed it. Engineer the dignity explicitly, by one honest strategy or the other: the guarded roll or the public liturgy, standard terms either way, and repayment expected as a form of respect — expectation being the one compliment no scoring model ever pays. Lend against knowledge instead of collateral: the guarantor, the family appraisal, the community's long memory — the name-economy of chapter 15, running just as surely through Akhuwat's courtyards — information the size of a neighborhood, which prices character better than any bureau. And seat the layer deliberately in the community's capital structure, sized to the shocks that actually arrive, upstream of the growth layers and unembarrassed beside them; chapter 41 will draw the modern fund designs, and the evergreen structures of Part X are this wheel scaled up. None of this is charity's afterthought; it is infrastructure, as load-bearing as anything in Part IV, and the traditions that built it did not think they were being soft; they thought they were being accurate about what a community is for.

The beautiful loan, then, stands delivered: scripture's own instrument, God's declared debt, mercy with a due date and a repayment rate that embarrasses the professionals. Judaism runs its version from unmarked counters; Islam runs its own from courtyards and statutes. Christianity built its mercy lending into a chest with pledges in it, in the hill towns

of Italy, and gave the chest a mountain's name. Chapter 17 opens it.

Chapter 17 — The Monti di Pietà and Catholic Anti-Usury Credit

On the thirteenth of April, 1462, the council of Perugia met to consider a proposal their Lenten preacher had spent the season urging on them, approved it, and voted three thousand florins toward it. The preacher was Michele Carcano, a Franciscan of the strict Observance, down from Milan; the design is credited to two brother friars, Barnaba da Terni and Fortunato Coppoli of Perugia, though the friars argued the paternity for generations and a rival town still claims priority. The thing was simple to describe and, as the next fifty years would prove, ferociously hard to classify: a public fund from which a poor citizen could borrow a small sum against a pledge left at a counter, paying not what his need would bear but a charge reckoned on what it cost to keep the counter open. The Perugians called it a *monte di pietà*. *Monte* was the ordinary Italian word for a fund — Venice had called its public debt one for three centuries — a mountain, a heap, money piled for a purpose; *pietà* named the purpose: pity. Chapter 14 closed Part IV at this institution's gates and called it by its English name, the Mountains of Piety; this chapter is the story it promised.

The Monte is the third of Part V's mercy institutions, and the hard case. The *gemach* of chapter 15 lends at zero, out of donated capital, to a community that knows its borrowers by name; the *qard hasan* of chapter 16 is Islam's answer in the same register. The Monte di Pietà charges. Not much — a few percent, sized to wages and rent and candle wax — but the charge rides on a loan, and Part III established that the loan may carry no increase at all. Chapter 15 left the question standing in exactly these terms: how far may a mercy fund recover its bare costs before the recovery curdles into yield? The Monti answered — at city scale, under two generations of theological fire, with the matter settled at last by an ecumenical council — and their answer, cost recovery as a discipline with instruments and a ceiling, is among the most useful things these centuries left.

The chapter carries a second obligation, stated at the door. The Monti were preached into existence, and the preaching had two edges. The same sermons that raised the capital demanded the removal of the Jewish lenders the new fund was designed to displace — lenders who were in the towns lawfully, by the towns' own invitation — and in one city the Lenten cycle was followed within weeks by a blood libel that destroyed the Jewish community entirely. The charity and the persecution came braided — one movement, one pulpit, often one man — and a history that told only the kind half would falsify both. This chapter tells the two in the proportion the record gives them, and it tells the dark one plainly.

Poverty's Bankers

Begin with the paradox: Europe's first network of charitable banks was built by the religious order most radically sworn against property. The friars minor owned nothing by rule, and it was precisely this order that produced, across the fourteenth and fifteenth centuries, Christendom's most forensic writing on money, price, and merchant capital; renunciation sharpened the analytical eye. The Observance, the reform branch that swept the order in the fifteenth century, added a preaching machine of extraordinary reach — itinerant friars on a Lenten circuit, whole towns in the piazzas — for penitence, peace, and

the reform of the town's economic sins. Usury stood high on that list, and by mid-century the preachers had noticed that denunciation lends no one money.

An institutional answer had been tried: a London fund of 1361, endowed by the city's bishop to lend on pledges free, its costs paid from the capital itself, ate itself by plain arithmetic — every year's expenses carved from the corpus — and closed. The Franciscan design paired charity's capital with a users' contribution to costs, then did what no theorist had managed: assembled the capital, town by town, out of thin Umbrian air. A Monte's founding stock was charity gathered retail — the Lenten sermons themselves, processions, legacies, civic allocations like Perugia's florins, alms boxes by the counter. Padua collected for its Monte every Easter; Gubbio taxed bequests one percent; at Spello the notary drawing a will was required to ask the testator whether he had remembered the Monte. The begging order begged capital, and put the town's conscience on retainer for the upkeep. And a fund so gathered revolved, as chapter 15 watched the *gemach's* capital revolve — the florins that bridged one winter came back and bridged another — except that this wheel charged a few percent for turning, because clerks and vaults are not free, and London's dead bank had shown what happens when a fund pretends they are.

It spread by imitation. Orvieto founded its Monte the year after Perugia; Monterubbiano in the Marches followed in 1465, founded by a Franciscan and a Dominican together — a detail worth keeping for later; Viterbo came in 1469, Siena in 1472, Genoa in 1480, Milan in 1483. By the century's end the institution was general across central and northern Italy, and the papacy had been drawn in early: sixteen papal acts blessed one Monte or another before the matter reached a council, and the settling bull would recite five approving popes by name. Rome itself took until 1539 to open its own.

Bernardino of Feltre

The man who turned the Monte from an Umbrian experiment into a peninsular institution was born Martino Tomitani at Feltre, in the Venetian foothills, in 1439, eldest of nine in a noble house. At seventeen he was a law student at Padua; that Lent James of the Marches preached the city, and by May the law student was a friar, renamed for the order's most famous voice, Bernardino of Siena. Ordained in 1463, he made an improbable heir to the trade — tiny, shy, and afflicted with a stammer — but the stammer lifted, and what emerged was one of the century's most consequential popular preachers: crowds too big for churches, sermons in squares and fields, the era's signature bonfire theater, onto which a town threw its dice and cards and charms. Towns at feud engaged him as a peacemaker.

From 1484, when he established the Monte at Mantua, the institution became his signature. His missions raised funds behind a painted banner of the Man of Sorrows or the Pietà; where no Monte existed he founded one, and where one limped he refounded it — through the Venetian territories, Emilia, Romagna, the Abruzzi. The emblem he spread became the institution's own: a small green mountain of three mounds, topped with a cross or a standard reading *Curam illius habe* — "Take care of him," the Samaritan's words to the innkeeper, two coins pressed into his hand for the trouble — mercy initiating, a paid service carrying it, the coming theology in four Latin words on a banner. Bernardino enforced its practical half inside the movement: against the purists who wanted the lending free, he insisted that a Monte must charge for its keep, because the free funds died, and a dead fund rescues no one. On the point he was right, and a general council would say so.

That is one half of the man. The other half must be stated with the same precision.

The usury Bernardino preached against had a face, and he named it. Following his order's established line, he presented the Monte not merely as relief for the poor but as the instrument that would make the towns' Jewish lenders dispensable — and he preached, openly and repeatedly, that they should then be dispensed with: expelled, city by city, from Italy and the Tyrol.

The lenders he meant were not interlopers; they were there by invitation, under a standard instrument called the *condotta* — the same word the cities used for hiring a mercenary captain: a negotiated charter, running a term of years, licensing a Jewish banking household to lend at capped rates, taxing its profits, spelling out its duties. Florence wrote its charter in 1437 and capped the rate at twenty percent; by mid-century nearly every city in central and northern Italy except Venice and Genoa had at least one licensed Jewish lending house. Shut out across most of Italy from the guilds and from land, Jewish families found in lending one of the few livelihoods left open to them; the towns, for their part, wanted both the credit and the tax. Under the community's own law the trade was lawful: the Torah forbids interest inside the covenant community and permits it toward the outsider — chapter 2 read Deuteronomy 23 whole, from within, with the rabbinic hedges around the permission — so a lender under *condotta* stood square with the law of Moses as his tradition read it and with the law of the town whose seal was on his charter. The rates the preachers denounced were, in the main, rates those same towns had negotiated, licensed, and taxed. What the sermons did was re-describe this arrangement as a crime in progress and its practitioners as a people preying on Christians — the “Jewish usurer” of the polemical afterlife that chapter 2 flagged at its first appearance in this book. In fifteenth-century Italy that figure was the working premise of a campaign with a practical logic: town councils do not tear up profitable contracts over doctrine; they tear them up over fear.

Then Trent. In Lent of 1475 Bernardino preached in the city, whose Jewish community was three households — those of Samuel, a moneylender settled there since 1461; Tobias, a physician; and Engel — living under a residence permission the prince-bishop, Johannes Hinderbach, had renewed six years before. The sermons vilified them. On Holy Thursday, the twenty-third of March, a tanner's son named Simon, two and a half years old, went missing toward evening. The searches over the following days included the Jewish households themselves; they had heard the rumor already moving through the city — that the Jews had taken the child — and they understood what such a rumor could become. On Easter Sunday a cook in Samuel's house, going down to the cellar for water, found the boy's body in the channel that ran beneath the property. Samuel and two companions went to the city's magistrate and reported the discovery themselves. The official summary of the case, drafted for circulation three years later, omitted that fact.

The whole community was arrested, men and women. Under judicial torture, over weeks of sessions, confessions were extracted that matched the ritual-murder story the interrogators required. Fifteen men, Samuel among them, were condemned and burned at the stake. The women were held in prison and released three years later, after papal intervention. The Jewish community of Trent ceased to exist, and Jews did not live in the city again for some three centuries.

Rome's response must be recorded with the same care: it is neither the rescue one hopes for nor the endorsement one fears. Sixtus IV ordered the proceedings suspended that August and sent an investigating commissioner, Battista dei Giudici, bishop of Ventimiglia. Trent obstructed him — kept him first from the prisoners, then from the records — until he withdrew down the valley to Rovereto and summoned the bishop and the magistrate to answer for the trial. Hinderbach sent, instead of himself, a dossier vindicating his conduct, and

circulated it widely. In Rome, where he had powerful friends, a committee of cardinals exonerated him and censured the investigator. The bull that closed the file in June 1478 accepted that the trials had been conducted in legal form, declined to affirm what they claimed to have proven, and reasserted — citing a decree then more than two centuries old — the Church's standing protection of the Jews and the unlawfulness of ritual-murder prosecutions. Giudici wrote an apologia for the Jews and a defense of himself, and disappears from the story.

The dead child did not. Miracles were logged at his tomb — more than a hundred within the year — the story traveled through the new printing presses, pilgrims came, and the emperor-to-be Maximilian promoted the devotion. Simon entered the Roman Martyrology in 1583, and in 1588 Sixtus V gave the local veneration formal recognition. The cult of Simon of Trent stood for nearly five centuries. It was abolished on the twenty-eighth of October 1965 — the same day the Second Vatican Council published *Nostra aetate*, its declaration on the Church's relation to the Jewish people — when the archbishop of Trent suppressed the devotion and ended the annual procession. The murdered community received its acknowledgment, such as it is, in the twentieth century.

Bernardino's part must be stated at exactly the granularity the evidence supports. He did not conduct the trial, sit on the tribunal, or govern the city; he had preached his Lent and moved on. What the record establishes is the sequence — he preached against the Jews of Trent in the weeks before the accusation, and the accusation followed — and the judgment of the standard scholarly reference is blunt: the *Encyclopaedia Judaica*'s entry on him states that his preaching was “mainly responsible” for the blood libel at Trent. His later career kept the shape: the sources track an attempt at Reggio the following year, then Bassano and Mantua; the expulsion of the Jews from Perugia in 1485 and from Gubbio in 1486, at his instigation; Ravenna in 1491, where the expulsion was accompanied by the destruction of the synagogue; Campo San Pietro in 1492; Brescia in 1494, the year of his death. In Florence in 1488 his preaching brought crowds to the edge of sacking the Jewish quarter, and the Medici government expelled him from the city to prevent it; six years later the Medici fell, and under Savonarola's republic Florence founded its Monte and decreed the expulsion of its Jewish bankers. The wave that went out from Trent did its own work besides: Vicenza banned Jewish lending in 1479 and expelled its Jews in 1486.

Bernardino died at Pavia on the twenty-eighth of September 1494 and was venerated immediately. The Church confirmed his cult — beatification, in its categories — under Innocent X in 1654; a cause for his canonization was opened in 1870 and has not concluded. He stands on the Franciscan calendar today as Blessed Bernardino of Feltre. So the Church's own record now holds, side by side and unresolved: a suppressed cult built on a libel against the innocent; protection bulls reissued while the trials they condemned were vindicated on procedure; and a beatus whose sermons stand behind both the mountains of mercy and the campaigns of expulsion. This book does not resolve the record either. It sets it down whole, because the institution the rest of this chapter examines was built inside it — and because pretending the counter can be inherited without the sermon is how a tradition forgets what it owes.

The Counter and the Pledge

Watch the counter at work — the borrower is a composite, but every rule she meets is documented. A woman comes to the Monte with a copper pan. The appraiser values it, and the fund lends her up to two-thirds of that value in small coin, on the spot. The

notary enters the loan in the register — name, object, sum, date, term — and she takes a ticket. If she redeems within the term, she pays the principal plus the charge, four to twelve percent a year depending on the town, and carries her pan home. If she cannot, the pledge goes to public auction, and here the statutes show their purpose: whatever the pan fetches beyond the loan and the accrued charge is returned to her. The Monte was forbidden to profit even from her default — the forfeiture, a commercial lender's dearest moment, was here a settling of accounts that sent the surplus home with the borrower.

Everything about the design says consumption bridge, not commercial credit. The loans were capped small — six florins at Perugia, four at Gubbio — sums that fix a roof or feed a family to the next fair, not stock a warehouse. The staff was a handful — a director, the *depositarius*; the appraiser; the notary; the auction sellers — salaried, or in some statutes paid a share of the surplus. And the charge obeyed a rule that inverts everything a modern reader expects of a lender: profits, if any, were folded back into the capital, and a Monte whose surplus grew was required to cut its rate. The price chased the costs downward. Where endowment covered more of the wage bill the borrower paid less, and the ideal, written into more than one foundation, was a fund endowed to the point that the charge fell to nothing. Lombardy tried starting at nothing — free funds in the London mold — and they ended as London's had, absorbed or converted, which is why Bernardino of Feltre kept insisting on the charge: the fee was not a compromise of the mercy but its life support.

Why pledges? Because the pledge is what makes lending to strangers at cost possible at all. Chapter 15's *gemach* underwrites with knowledge and needs no collateral; a city fund serving thousands cannot know anyone, so the object stands in for the acquaintance. The pledge asks no questions, requires no guarantor, publishes nothing: whoever brings a pan of sufficient copper gets the loan, known or unknown, and walks away owing an object rather than an explanation. The pledge is where subsistence credit has always lived — chapter 2 met it at the root of the Torah's lending law, the cloak that must go home by sundown — and the Monte did not abolish the poor man's pawn; it took the one instrument the poor reliably possessed and rebuilt the counter behind it so that the instrument stopped being a trap. The design's limit is part of its honesty: a pledge loan reaches only those who own something pawnable. Below that floor the Monte could not go, and the tradition knew it — what waits below is the outright gift, whose disciplines are chapter 18's ground.

Set the counter beside chapter 15's locking door and the Monte's place comes clear. The shock that ruins a household is usually temporary — the fever, the dead donkey, the slow month — and what decides whether it locks into permanent ruin is the price of the bridge across it. The lenders at that door priced the bridge at what desperation would pay: twenty percent under a lawful charter, more beyond it. The Monte parked itself at the same door and priced the same bridge at cost — a few percent on six florins, with the surplus of any auction coming back — so that the emergency, though still an emergency, no longer compounded. It is chapter 15's rule — no one's bad month as anyone's profit center — generalized to a city of strangers and billed at cost.

Work, Expense, and Risk

The charge, though, had to survive the theologians, and it very nearly did not. The objection was the tradition's own logic applied without blinking. The wall Part III built says the *mutuum* may carry no increase: whatever rides back on a loan beyond the principal is usury, and the size of the charge was never the test. The Monte's fee rides back on a loan. Worse, said the objectors, look at who pays it: the fund's running costs, which char-

ity ought to bear, are collected exclusively from the poor at the counter — the one class of persons the prohibition exists to shield. A good end cannot launder a forbidden means; call the fee what you like, the Monte sells loans, and Christ commands lending that hopes for nothing back. The case was pressed in disputation and in print — at Faenza in 1494 the Augustinian Bariani published against the funds under the title *De Monte Impietatis*, the mount of impiety, and four years later a young Dominican of formidable precision named Tommaso de Vio, whom the schools would come to know as Cajetan, wrote the most thorough attack of all. The quarrel is remembered as Franciscans against Dominicans, and the memory is not quite fair: a Dominican had co-founded the Monte at Monterubbiano, Dominican jurists wrote opinions for the funds, and the most famous Dominican in Italy — Savonarola — presided over the founding of Florence's. It was a dispute about doctrine, and both sides were defending the same wall.

The defenders answered that the charge never touched the wall. Nothing is sought from the loan as such: the loan itself is gratuitous, and would remain so if a benefactor paid the fund's expenses some other way. What the borrower's fee buys is everything around the loan — the appraisal, the register, the vault and its custody, the auction that liquidates honestly — real services, really costed, which someone must pay for — by an old rule of law, most fairly the one who uses them. The fee, in the vocabulary this book built in Part III, is not the forbidden benefit flowing to a lender because of a loan; it is the *stipendium laboris* of chapter 12, the wage of actual work, plus the recovery of actual expense, priced by a ledger anyone could audit, owed to a fund that keeps no profit. Even the canonists' subtlest test — chapter 8's *Consuluit*, where the mere hope of gain makes the usurer — finds nothing to bite: the Monte hopes for nothing from its lending; its charge is fixed by its costs and falls as endowment rises. A charge pinned to costs inherits a ceiling; a charge pinned to appetite inherits none. That is the institution's moral geometry, and why the fee could be defended by men who would sooner have gone to the stake than defend interest.

The matter reached the Fifth Lateran Council, and on the fourth of May, 1515, in the bull *Inter multiplices*, Leo X settled it — in a document remarkable, among the era's rulings, for the respect it pays the losing side. It states the objectors' case in their own terms, down to their definition of usury — chapter 14 quoted it whole — turning on gain sought “without any work, any expense or any risk”; it states the defenders' case, that “nothing is being sought nor hoped for from the loan as such” and that a fund may take “a moderate and necessary sum” for its upkeep, “provided that no profit is made therefrom”; and it commends both parties, the one for its “zeal for justice,” the other for its “love of piety and truth.” Then it rules, in the standard conciliar English: “We declare and define, with the approval of the sacred council, that the above-mentioned credit organisations, established by states and hitherto approved and confirmed by the authority of the apostolic see, do not introduce any kind of evil or provide any incentive to sin if they receive, in addition to the capital, a moderate sum for their expenses and by way of compensation, provided it is intended exclusively to defray the expenses of those employed and of other things pertaining (as mentioned) to the upkeep of the organisations, and provided that no profit is made therefrom.” Such lending, the bull goes on, “is meritorious and should be praised and approved. It certainly should not be considered as usurious.”

Read the ruling against the definition and the architecture snaps into place: usury is gain without work, expense, or risk; the Monte's charge is work and expense without gain — the definition run in reverse, and the council plainly meant the mirroring. Every clause is load-bearing. *Moderate*: sized, not open-ended. *Exclusively to defray the expenses of those employed*: the wage bill and the upkeep, an auditable list, nothing else. *No profit therefrom*: the fund

itself may not fatten — the clause that makes the charge a recovery rather than a price. Two riders complete the design. The council ruled that it would be “much more perfect and more holy if such credit organisations were completely gratuitous,” and directed that the faithful be prompted with indulgences to endow the funds so the borrowers’ charge could shrink — cost recovery licensed as a floor whose direction of travel is down, toward the gemach’s zero. And for the future, anyone who dared “to preach or argue otherwise by word or in writing” against the approved funds would incur immediate excommunication. One bishop, of Trani, spoke against the ruling on the council floor. Cajetan, by then the Dominican general, was absent from the session, and afterward let the position go.

Notice what the settlement did and did not settle. In blessing the Monti, the council operationalized the three words of its own definition: work, expense, and risk are the things whose presence unmakes usury — the licit titles, as the tradition would call them, under which something beyond the principal may honestly change hands. The Monte’s charge ran on the first two; the third, risk, the counter mostly engineered away, the pledge absorbing it — but the word stands in the definition, and the great technical debates of the next age would be fought over precisely these headings: what counts as compensable loss, as real forgone gain, as genuinely imperiled capital. Those are the doors into Part VI — *damnum emergens* in chapter 19, *lucrum cessans* in chapter 20 — and *Vix pervenit*, chapter 7 noted, would still be holding the same line in 1745: titles extrinsic to the loan, never a price for the loan itself. What the council did not settle — no ruling can, because it is a boundary to be watched rather than a theorem that stays proved — is the drift. Costs pad; wage bills grow; “upkeep” is a word with appetite. The settlement is a discipline rather than a formula: a ledger open to audit, a charge that falls as endowment rises, a fund forbidden to profit, a governance that keeps asking whether this year’s charge answers to this year’s costs. Chapter 15’s question — how far may recovery go before it curdles — got its answer here: exactly as far as the audited ledger goes, and not one soldo further.

The Mountain That Remains

It remains to say what the Monti were not, and what became of them — both halves answer this chapter’s closing obligations.

They were not venture funds, though superficially the Monte looks like the very thing Part IV forbade — a fixed charge riding on a loan — and the difference is total. A partnership’s return is discovered after the venture; a lender’s interest is written in advance and priced to the lender’s advantage; the Monte’s charge is written in advance and priced to the ledger. It seeks no share of anything the borrower does with the money, claims no upside, compounds nothing, and is forbidden by charter to enrich even the fund that collects it. The Monte was designed to make no one rich, including itself. It financed consumption, not enterprise — the bridge across the bad month, not the venture toward the good year; its assets were pans and cloaks, not cargoes; its growth ambition was, by rule, to shrink its own price. In Part V’s taxonomy the Monte fills the tier between Part IV’s partnership capital, which earns by standing inside ventures, and the mercy capital of chapters 15, 16, and 18, which earns nothing at all: cost-price capital, working at scale for strangers, taking home its expenses only. The drift record shows how live the boundary was — loan caps crept from six florins toward a thousand ducats, staff took percentages, funds took deposits and shaded into municipal banking — and Siena’s Monte, chartered in 1472, is the famous case: re-founded in 1624 on the security of the grand duchy’s Maremma pastures, the *paschi* that gave it its name, it is today Banca Monte dei Paschi di Siena, commonly called the oldest bank in the world still doing business. The charity that outlived every empire around it is

now a commercial bank with a stock price — at once proof of the design's durability and a lesson in what happens when cost recovery goes unwatched.

That survival is the first reason the Monti still matter: proof, running five centuries, that lending to the poor at cost is an operable institution, not a sermon's daydream. The second is the lineage. Ypres had a Monte from 1534; France planted them from Avignon in 1577; the Paris fund re-founded under Louis XVI in 1777 still operates at its original address, renamed after 1918 as the city's municipal credit institution, and describes its work today, in its own words, as social and solidarity finance — the fifteenth century's instrument wearing the twenty-first century's name for itself. Madrid's Monte de Piedad, founded in 1702, became the root from which Spain's montes and then its nineteenth-century savings banks grew. Mexico City's Nacional Monte de Piedad, founded in 1775 on a silver magnate's three hundred thousand pesos, is lending against pledges beside the capital's main square right now. In Italy in 1896, five hundred fifty-six Monti were still at work. Wherever the model traveled, it carried the Lateran grammar with it.

And the third reason is that the argument the Monti settled has come back in modern dress. The living debate over lending to the poor — what a mission lender may charge, when "sustainability" has quietly become profitability, whether the poor should carry an institution's overhead or a donor should — is Faenza 1494 and Session 10 rerun, usually without the participants knowing they have predecessors. The tradition's answer bears repeating because it is not a number: a charge pinned to audited costs and nothing else; no profit from the fund's own poor; surplus returned to capital and the rate cut; endowment as the engine that moves the price toward zero; governance that treats the boundary as a watch, not a wall. Every piece of that answer transfers. None of it transfers automatically.

Which brings Part V to the formulation its title promises: cost recovery without capital domination. Set the mercy institutions side by side — the *gemach* lending at zero on a community's knowledge of its own; the *qard hasan*, chapter 16's beautiful loan, at zero within its own tradition's frame; the Monte lending at cost to a city of strangers; behind them, chapter 18's layer, the outright gift for the fall no loan should touch — and the common grammar is plain. In all of them capital is present, organized, audited, disciplined; in none of them does capital rule. The charge, where there is one, answers to the cost ledger and only to it: no yield, no compounding, no claim on the borrower's future beyond the small sum and the settled fee. Capital here is a servant with its wages paid exactly — the wages keep the servant alive, and the rule keeps the servant a servant. That is the principle Parts IX and X will carry into modern fund design, where cost recovery reappears as management fees and stewardship budgets, and the Lateran clauses translate with almost embarrassing directness.

Two memories, then, leave this chapter together, and they must leave it together. At the counter: the pan weighed with both hands, the small loan at the price of its own book-keeping, the auction surplus counted back into a poor woman's palm — a five-century demonstration that credit can be built to serve the poor at cost, and one of the anti-usury tradition's most honorable works. And in the square outside: the sermons that raised the fund by teaching the town to see its licensed Jewish neighbors as predators; the expulsions that followed the missions; Trent, where three households were annihilated for a crime that did not happen, weeks after the Lenten preaching, while the instrument of mercy went on being founded from the same pulpits. The Church that approved the counter also beatified the preacher, and took until 1965 to suppress the cult his era built on the libel. Both memories are true; neither cancels the other; and the institution's descendants inherit them together. What an honest history can do is keep them in one frame: take from the Monti

the cost-recovery discipline, among the best ideas anyone has had about mercy at scale, and take with it the knowledge of what it cost when mercy for one people was preached as a campaign against another — so that whoever rebuilds the counter, and it deserves rebuilding everywhere, knows which part of the inheritance must be refused.

Chapter 18 — Almsgiving, Zakat, Tzedakah, and the Common Good

The Talmud's charity law carries, in a few columns of the tractate Bava Batra, a complete picture of collection day. Two officers walk the town — two, never one, because authority over a community may not be exercised by fewer than two, and because a man alone with a charity box is at the very least alone with the appearance of temptation. They keep within sight of each other on their rounds. If a collector finds a coin in the street, it goes into the charity purse then and there, in public. The coins are counted out one at a time, never two by two, so that no onlooker can imagine a sleight. And when the fund is given out, the two collectors are not enough: distribution takes three, because — this is the Gemara's own reasoning, not a commentator's flourish — deciding who receives from the poor-fund, and how much, is a proceeding in monetary law, and money cases take three judges (Bava Batra 8b).

The same page lays out what the officers are running. A town in this law operates two standing systems. The *tamchui*, the food platter, is collected and distributed daily and feeds any poor person who stands in the town that day, local or stranger. The *kupah*, the coin fund, is gathered and disbursed each Sabbath eve and belongs to the town's own poor — the weekly budget of a community that has made its neighbors' subsistence a civic charge. The townspeople may convert one fund into the other as need shifts. And the men who run the apparatus are simultaneously honored and audited: the Talmud reads Daniel's verse about "they who turn many to righteousness" shining like the stars as a description of charity collectors — the same men it polices like suspects. Dignity for the office; procedure for the man.

None of this is the etiquette of a collection plate; it is the machinery of a court, run on the premise that what moves through it is owed. That premise is this chapter's subject. Part V has so far been walking the credit of mercy — the *gemach* of chapter 15, mercy revolving at zero interest; the *qard hasan*, Islam's beautiful loan, in chapter 16; the Monti di Pietà of chapter 17. Each of those instruments is a bridge, and chapter 15 ended on the limit of bridges: a loan carries a household across temporary trouble and can do nothing for the permanent kind — the widow with nothing coming in, the laborer whose back is gone, the household whose gap repeats every month and always will. For the falls no bridge can span, the traditions built transfer systems: *tzedakah*, *zakat*, almsgiving — money that goes out and is not owed back, there being nothing it could honestly be owed back from. This chapter is about those systems: the doctrine of ownership underneath them, the institutions each tradition built, the ladder that ranks help by what it does to the helped, and the design lesson the whole apparatus teaches — that a community's capital comes in kinds, and the kinds must not be confused.

Tenants and Trustees

Begin with the traditions' answer to a question modern law thinks it has settled: what does it mean to own something?

Israel's answer is written into the land law this book met in chapter 4. The fields of the covenant could be sold, mortgaged, and inherited — the Torah is no stranger to title — but

they could not be alienated forever, and the reason given is a landlord's: "The land shall not be sold in perpetuity, for the land is mine; with me you are but aliens and tenants" (Leviticus 25:23). Every deed in Israel was a lease from the same Landlord, and the Jubilee was the lease's reversion clause. The Psalmist states the title plainly: "The earth is the LORD's and all that is in it, the world, and those who live in it" (Psalm 24:1). Inside that sentence, the farmer's harvest was never wholly his to begin with. The portions the law reserves for the poor — gleanings, forgotten sheaf, the field's corner, the poor-tithe — sit in the grant itself, obligations that came with the ground, and chapter 4 walked those fields. The owner is real; he is simply not senior.

Islam frames the same subordination as trusteeship. "Believe in God and His Messenger," runs the fifty-seventh sura in M. A. S. Abdel Haleem's rendering, "and give out of what He has made pass down to you" (57:7) — and the Arabic participle behind that clause, *mustakhlafina fihi*, is heard by the commentators in its root sense: wealth you hold as trustee, for the true Owner and for those who come after. The Prophet's own image survives in Muslim's collection: the world is sweet and green, and God has installed you in it as His vicegerents, to see how you act (Muslim 2742). Chapter 2 met the legal consequence: inside the believer's wealth sits a recognized right, a *haqq*, belonging to the beggar and the destitute — a claim standing before any asking. A trustee, unlike an owner, can be in arrears to the beneficiaries without having borrowed a thing.

Christianity reasons its way to the same place through the schoolroom. When Thomas Aquinas asks in the *Summa* whether private property is lawful at all, he answers that ownership is licit and even necessary, for three homely reasons: people care better for what is theirs; affairs run in better order when each thing has its keeper; and peace holds better when each is content with his own — anyone who has shared a kitchen can verify the third. But that is ownership as management. As to use, he continues, a man ought to hold external things "not as his own, but as common, so that, to wit, he is ready to communicate them to others in their need" (ST II-II, q.66, a.2). Procurement private, use common: the formula the church would later name the universal destination of goods and carry into its modern catechism — the earth entrusted "to the common stewardship of mankind," every owner "a steward of Providence" (CCC 2402, 2404). The Second Vatican Council gives the doctrine its plainest modern sentence: "God intended the earth with everything contained in it for the use of all human beings and peoples" (Gaudium et Spes 69).

Three vocabularies, then — tenancy, trusteeship, stewardship — and one structure: title is real, and title is junior. None of the three traditions is against property; all three are against its pretension to be absolute, having seen what the pretension licenses. If my surplus is wholly mine, then what I give the poor is largesse, revocable at my mood, and the beggar at my gate is a petitioner. If my surplus carries a lien — the tenant's rent, the trustee's distribution, the steward's account — then the man at the gate holds a claim, and the transfer systems this chapter now walks are less the traditions' generosity than their debt service.

The Debt Called Justice

The Hebrew word says it first. *Tzedakah*, the tradition's term for giving to the poor, is built on *tzedek* — justice, righteousness — a fact chapters 3 and 4 flagged and this chapter must cash out. In biblical Hebrew the word means righteousness at large; in the language of the rabbis it settled onto the poor man's portion specifically, and the settlement is a thesis: what flows to the poor rectifies, and the giver, whatever his feelings on the day, is paying rather than bestowing.

The law treats it accordingly: Maimonides codifies giving as a positive commandment, and refusal as the violation of a negative one — the averted eye of Deuteronomy 15 — before a word is said about generosity of spirit (Gifts to the Poor 7:1–2). And because it is law, it is enforceable. The Talmud’s charity collectors may press the well-off for their assessment — may seize a pledge for it, the Gemara says, even on the eve of the Sabbath — though the same passage turns fierce against collectors who lean on the strained, reading Jeremiah’s threat against Israel’s oppressors as aimed at over-zealous fund-raisers. The page then gives the doctrine a face: the sage Rava compelled Rav Natan bar Ami — the verb is not softened — and took four hundred zuz from him for the fund (Bava Batra 8b). Modern readers wince at the story, and the wince measures the distance between two conceptions. You cannot be compelled to be generous; generosity dies under compulsion. You can certainly be compelled to pay what you owe, and the rabbis had concluded, from the tenancy clause of the last section, that the rich man’s surplus already carried the poor man’s portion inside it. Rava was not extracting a donation; he was executing a judgment.

The same conviction built the apparatus of the opening pages, and the kupah-and-tamchui pair encodes chapter 2’s map of nearness — your own poor before your city’s, your city’s before another’s — as administration: the weekly fund guarantees the neighbors, the daily platter catches whoever hunger has washed into town, and no one is outside both. A community running this system has accepted something remarkable: that some of its members will be net recipients for life, and that this is a budget line rather than a failure.

And the measure of the obligation is taken — this is the tradition’s most striking stroke — on the recipient’s life, not the donor’s. The governing phrase comes from Deuteronomy’s open-hand command — *dei machsoro*, enough for his lack, whatever it may be — and Maimonides presses it to its end: the poor man is owed according to what he lacks, clothed if he lacks clothing, furnished if he lacks household goods, helped to marry if marriage is the lack; and if he was a man of standing who kept a horse and a servant before his ruin, then, resources permitting, a horse and a servant, “for you are commanded to fill his lack, and you are not commanded to make him rich” (Gifts to the Poor 7:3). The horse scandalizes every first-year student of the passage, and it is meant to. The obligation is indexed to the wound, and a man’s wound includes the life he has fallen from. Generosity measures what the giver can spare; justice measures what the poor man lacks — and *tzedakah*, the word insists, is the second kind of thing.

One more verse belongs here. The support command of Leviticus 25:35 — chapter 2 quoted it whole — reads in the NRSV “you shall support them,” but the Hebrew verb underneath, *vehechezakta bo*, is built on the root for strength: you shall strengthen him, take hold of him, brace him. The halakhic midrash reads the tense of the verse as a timing instruction — steady the faltering man while he is still on his feet, for one hand can hold him up now where five cannot raise him later — and Maimonides will take the very name of his ladder’s top rung from this verb. The Torah’s charity, at its most ambitious, is not a payment made after the fall but a hand under the arm before it.

The Fortieth Share

Islam took the same conviction and gave it the most fully specified administrative form of the three: a permanent, calculated, collected wealth transfer, standing among the five pillars of the faith. Chapter 6 read *zakat*’s scriptural charter — the purifying share, owed and not offered, “the prescribed alms” in Abdel Haleem’s exact adjective — and this chapter’s business is the machine the charter specified.

Zakat has a threshold, the *nisab*: classically twenty gold dinars or two hundred silver dirhams (in the jurists' later metric, roughly eighty-five grams of gold or five hundred ninety-five of silver), below which a household owes nothing, because the law refuses to tax subsistence into charity. It has a clock, the *hawl*: the wealth must have rested with its owner a full lunar year, so that the levy falls on settled holdings rather than passing income. It has a rate — one-quarter of one-tenth, 2.5 percent, on monetary wealth — with the older pastoral and agricultural schedules running beside it. And it has a spending side written into scripture itself, in a verse that reads, as chapter 6 observed, like a budget: "Alms are meant only for the poor, the needy, those who administer them, those whose hearts need winning over, to free slaves and help those in debt, for God's cause, and for travellers in need. This is ordained by God; God is all knowing and wise" (9:60). The manuals argue, as manuals do, over the edges — how far "for God's cause" stretches beyond its classical reading, whether a modern asset is more like coin or like cattle, what to make today of the hearts to be won, a category that in the founding generation was frank statecraft, goodwill purchased from chiefs whose tribes would follow. The schools mark these as the disagreements they are. What no school disputes is the list's center of gravity: the poor and the needy first, the slave to be freed, the debtor to be rescued — the fund standing at the exact doors where chapter 2 found the old machine of debt catching people.

The founding instruction survives in Bukhari's collection, in the Prophet's charge to Mu'adh ibn Jabal, sent as his officer to Yemen: teach them the faith; then the prayer; then, if they obey, that God has made obligatory upon them an alms *taken from their rich and returned to their poor* (Bukhari 1496). Bukhari filed the report under a chapter heading that is itself the doctrine — zakat is taken from the rich and rendered to the poor — and the charge closes with two restraints that mark the founding of an actual administration: do not take the best of their possessions, and fear the cry of the wronged, for between it and God there is no veil. A collector who must leave the finest camel in the herd is a collector under law; a fisc warned against the oppressed man's prayer is a fisc that knows exactly how tax-gathering goes wrong. The early community treated the whole system as constitutive — when tribes after the Prophet's death proposed to keep the prayer but withhold the alms, the first caliph went to war over it, the tradition records, refusing to let the pillar be split from its brothers. Later centuries mostly left the levy to conscience, mosque, and trust rather than the treasury, but its legal architecture never loosened: the believer computes and delivers, audited or not.

Set the machine beside the last section's and the family resemblance is exact — a wealth already encumbered, a share owed and calculable, recipients holding right rather than receiving favor — but zakat adds two refinements of its own. The first is in the base: zakat bites accumulation, not income. The fortieth falls on wealth that has sat the year through, so an idle hoard is pruned annually while wealth pushed into trade, herds, or partnership can outgrow its levy. Whatever a believer's intentions, the arithmetic preaches the sermon the Qur'an states outright in the verse chapter 6 set at the center of its economics: "God blights usury, but blesses charitable deeds with multiple increase" (2:276). Riba is increase the verse says God blights; zakat is decrease the verse says God multiplies. The shared root — *z-k-w*, purity and growth at once — does the summarizing. The second refinement is in the payer's soul. "In order to cleanse and purify them," the Prophet is told, "accept a gift out of their property" (9:103): what purifies is the paying, and what is purified is the remainder. Until the fortieth goes out, the poor's share sits inside the holding like an unsettled lien; zakat is the annual clearing of title. A believer who has paid it holds the other thirty-nine fortieths, for the first time that year, cleanly.

Two footnotes complete the picture. Alongside the owed fortieth stands the little *zakat al-fitr* of the fast's end — a measure of staple food owed for every member of a household, children included, before the festival prayer, so that no one meets the feast day hungry — and past both stretches *sadaqah*, the voluntary giving chapter 6 read, which no nisab measures. There remains the modern question of magnitude. The development institutions have tried for years to count global zakat, and the honest summary of their work is the spread itself: estimates of the potential pool run from two hundred billion dollars a year to a trillion, while the formally collected figure — state boards, licensed funds — comes to perhaps ten or fifteen billion. The gap is not all evasion. Most zakat moves the way the tradition has always half-preferred it to move: privately, hand to hand, unrecorded — the Qur'an itself commends giving "to the needy in private" (2:271) — which makes the world's oldest standing wealth transfer also its least countable. The estimates should be held loosely; the institution they fail to measure should not.

The Bread in the Barns

The Christian doctrine arrives not as a statute but as a sermon, and the sermon has a date. In the late 360s famine struck Cappadocia, and in Caesarea a priest named Basil — soon to be the city's bishop — went into the pulpit with the parable of the rich fool, the man in Luke's Gospel who answers abundance by planning bigger barns. Basil preached at the barn-owners of his own congregation in the second person. What the rich man calls his own, the sermon presses, is precisely what he is holding out of other people's survival: the bread going stale in his stores while men starve, the cloak folded in his chest while men shiver, the silver buried while the destitute are ransomed by no one — an inventory of withheld rescue, delivered to hearers who owned the inventories in question. The sentence has never stopped being quoted since, and the quoting is itself a small lesson in how the church holds property. When the canon lawyers gathered the sentence into the Decretum, they filed it under Ambrose's name; Thomas Aquinas quotes it from them the same way — "It is the hungry man's bread that you withhold, the naked man's cloak that you store away, the money that you bury in the earth is the price of the poor man's ransom and freedom" — while the Second Vatican Council's footnote points instead to Basil's homily on the barns; and Ambrose, who had read his Basil, really did preach the sentence's twin in his own attack on the rich, the *De Nabuthae*. The Middle Ages never kept the paternity straight, which is forgivable in a sentence whose entire argument is that some property is common.

Paul VI, quoting that Ambrose treatise in 1967, chose the lines that make the logic explicit: "You are not making a gift of what is yours to the poor man, but you are giving him back what is his. You have been appropriating things that are meant to be for the common use of everyone. The earth belongs to everyone, not to the rich" (*Populorum Progressio* 23). A reader who has just come through this chapter's Hebrew pages will recognize the grammar at once. Restitution, not largesse; the poor man's own, not the rich man's spare — *tzedakah's* claim, in Latin. The traditions did not borrow this from each other; each read the same Landlord's title and drew the conveyancing conclusion.

What the scholastics added was precision about the duty's force, and Aquinas's question on almsgiving is where the precision lives. Almsgiving, he begins, is an act of charity "through the medium of mercy" — the very word says so, *eleemosyna*, from the Greek for mercy, the ancestor of the English "alms" (ST II-II, q.32, a.1). Filing alms under charity might seem to soften the duty — love cannot be legislated — but Aquinas closes that escape in the fifth article. Loving your neighbor in word only is not loving him; succoring his need

is what the love consists of when he is needy; therefore almsgiving out of one's surplus — the *superfluum*, what remains when your own state and dependents are honestly provided for — “is a matter of precept,” commanded, not counseled (q.32, a.5). Perfection may be a counsel; the poor man's rescue is an order. And where the need is extreme, the tradition's oldest and hardest sentence applies, the one Aquinas quotes and Vatican II still quotes from the canon law: feed the man dying of hunger, because if you have not fed him, you have killed him. Withholding, at the limit, is not stinginess; it is the killing the canon names.

Then Aquinas takes the step that startles modern property lawyers, in the question on theft. Since the natural order destines inferior things to human need, “whatever certain people have in superabundance is due, by natural law, to the purpose of succoring the poor”; the division of property is human law, and human law cannot repeal natural law's senior lien. Ordinarily the owner is the lien's own trustee, dispensing his surplus by his judgment. But where necessity is manifest and urgent — the starving man before the locked granary — the title simply gives way: taking what survival requires, “openly or secretly,” is not theft or robbery at all, for in necessity all things are common, and what the desperate man takes becomes, by his need, his own (ST II-II, q.66, a.7). He may take it for his starving neighbor too. The doctrine is fenced with conditions, but its teeth are real, and the modern church has not filed them down: the Catechism still teaches that the universal destination of goods “remains primordial” over every property right (CCC 2403). The formal architecture of that teaching — *Rerum Novarum* to the present, the common good, the whole edifice of Catholic social doctrine — belongs to chapter 33; its credit institutions, the *Monti di Pietà*, to chapter 17. What belongs here is the foundation stone all of that stands on, and the stone is Basil's: the surplus was never absolutely yours, so the bread you release to the hungry was already, in the oldest sense of this chapter's oldest word, theirs.

The Ladder and the Stack

It remains to ask the question the traditions press hardest: given that the transfer is owed, what is the *best* way to make it? Judaism's answer became the most famous ranking in the literature of charity, and it turns out to be, on inspection, a theory of capital.

When Maimonides finished codifying the laws of gifts to the poor, he closed with a ladder: eight levels, each above the next. The top rung this book has climbed twice already — in chapter 3, where its four instruments turned out to be the taxonomy of Part I redeployed, and in chapter 15, beside the Talmudic ladder of Reish Lakish that ranks the lender above the alms-giver and the partner above them both. It bears its exact weight here a third time: the highest degree, above which there is none, is to strengthen the hand of a fellow who has faltered — the verb of Leviticus 25:35, *hamachazik*, the brace under the tottering man — by giving him a gift, or a loan, or by forming a partnership with him, or by finding him work, until he no longer needs to ask (Gifts to the Poor 10:7). Gift, loan, partnership, employment: four contract-forms from chapter 3's family, ranked here not by their yield but by their exit — the rung's test being whether the help ends the asking. Dependence made unnecessary outranks dependence relieved, and the summit of mercy, as chapter 15 found in the rabbis and Part IV in the contract law, is not a bigger gift but a shared venture, the poor man converted back into a colleague.

Then the ladder descends, and the descent is as instructive as the summit. Below the hand-strengtheners stands the giver in double blindness — he does not know who received, the poor man does not know who gave — which Maimonides calls the commandment done purely for its own sake, and illustrates twice: the Temple's Chamber of the Silent, where

the discreet left money and the shamefaced poor of good family drew it out, neither seeing the other; and, close behind it, the town charity fund, the kupah of this chapter's opening pages, provided its administrator is faithful and wise (10:8). Pause on that second example: the anonymous *institution* outranks nearly everything a private giver can do with his own warm hands. Bureaucracy, so often mercy's caricature, is here mercy's technology — two collectors, three judges, guarded rolls — precisely because the machinery deletes the one thing a personal gift cannot delete, the donor's face and the recipient's knowledge of it. Below the double blind come the single blinds, each with its Talmudic exemplar: the giver who knows his recipient while remaining unknown — the sages who went in secret and dropped coins in poor men's doorways; then the recipient who knows his giver while remaining unseen — the sages who walked with coins bundled in a cloth slung behind them, so the poor could come and take without being watched (10:9–10). Then the open gifts, ranked by initiative: into his hand before he asks; into his hand after he asks — the asking itself being a cost the better giver spares him (10:11–12). And at the bottom, two rungs that measure what money alone cannot: giving less than one ought, but with a pleasant face; and giving, however much, with sadness (10:13–14). Even the final rung is still *tzedakah* — the coin arrives, the obligation is technically met — but the ladder has been measuring something else all along: what the transfer does to the receiver's dignity, from the man braced before he falls down to the man handed his own portion in a manner that makes him eat shame with it. The other traditions built the same rungs. The Qur'an commends the private gift over the published one and voids the charity followed by hurtful reminders, as chapter 6 read; the Gospel tells the giver to keep his left hand ignorant of his right; and the deeds of *gemilut chasadim* that chapter 15 read are done, the Gemara says, "without fanfare." Where the traditions legislate manner this insistently, they are telling you what the transfer is *for*: not the mover of the money but the standing of a human being, with the money only the vehicle.

Read as a whole, then, the ladder is a portfolio doctrine, and it forces the distinction the rest of this book will trade on. Call the two kinds mercy capital and investment capital. Investment capital (the *societas*, *iska*, and *mudarabah* of Part IV, the instruments that will fill Parts VII and IX) buys a claim on outcomes; its return is discovered in the venture's story, and its virtue is discipline, the alignment of gain with real exposure. Mercy capital buys outcomes directly: the month bridged, the tools replaced, the dowry provided, the debt lifted, the man set back on his feet. Investment capital is priced by what it returns; mercy capital is priced by what it prevents. Its yield is real — rescue, standing, the ladder's exits — but it books nowhere, accrues to someone else, and cannot be repatriated, which is why no market supplies it and every tradition in this book commands it.

And some of it must not seek return at all — not as a counsel of saintliness but as a finding about arithmetic. A household's life is one path through time, and some doors on that path lock behind you — absorbing states, in the vocabulary chapter 10 built and chapter 28 will prove, the chance of passing one being a ruin probability. Trouble comes in two shapes. The temporary shock (the fever, the layoff, the burst pipe) interrupts an otherwise sufficient income, and a zero-interest bridge, as chapter 15 showed, is its complete cure, repaid out of recovery. But the permanent shortfall — the disability, the widowhood, the wage that will never meet subsistence — has no future surplus for any loan to draw on. Credit aimed at that gap does not fix it; it schedules it, converting a standing insufficiency into a growing claim, and the claim into the locked door. For the permanent shape of trouble the only instrument that works is the transfer, renewed for as long as the gap endures — which is precisely how Deuteronomy talks about it: "Since there will never cease to be some in need on the earth," the command runs, "Open your hand to the poor and needy neighbor

in your land” (15:11). The gift layer of a community’s capital is its anti-ruin reserve, spent at the doors where bridges cannot reach, and demanding a return from that layer is not prudence but a category error, paid for in households.

So the traditions, without ever writing the word, converge on what a modern designer would call a capital stack — layers of money with different jobs, different prices, and different permissible returns. At the base, the transfer layer: *tzedakah*, *zakat*, the alms of precept — return forbidden, because its recipients are exactly the people from whom nothing can honestly come back; sized, as the *kupah* and the fortieth size it, by the census of permanent need, not by what happens to be left over. Above it, the mercy-loan layer — *gemach* and *qard hasan*, principal sacred, increase zero — revolving across temporary trouble: chapter 15’s wheel, chapter 16’s loan. Above that, cost-recovery credit for borrowers the free funds cannot reach: chapter 17’s story. Then investment capital proper, the partnership instruments, taking genuine risk for genuine return among ventures that can bear both. Standing across all of them, the solidarity pool of chapter 3’s taxonomy — Lanuvium’s burial chest, the *takaful* circle, the mutual aid of chapter 27. And beneath the whole structure, where a community can manage it, the endowment: Khaybar’s answer, the principal tied up and only the fruit spent, which is chapter 29’s subject and gets no more than this clause. The layers are not rivals: a community with only a gift layer stays poor; one with only an investment layer eats its weakest members, one locking door at a time. The stack is the traditions’ composite answer to the question Part I opened — what may money do to a neighbor? — and its integrity depends on the audit rule of chapter 3: every layer must be what it says it is. A gift that expects gratitude’s yield is a loan in costume, and chapter 8 showed how carefully the law hunts that costume. A loan to a household below the floor is a transfer wrongly booked, and it will be collected — this is the cruelty — as if it were not. An investment in the desperate is the oldest wrong in this book wearing its newest clothes. When Parts IX and X come to design funds in earnest — the Non-Extractive Return Formula of chapter 40, the fund architecture of chapter 41, with chapter 28’s mathematics underneath — the stack drawn here is the frame they will build in, and the proportions among its layers will turn out to be a community’s true moral accounts, more honest than any mission statement.

The chapter’s opening asked why the rabbis staffed a charity box like a courtroom, and the whole answer is now visible: because the money in the box was owed — by tenants, by trustees, by stewards — to claimants the law could name; because owed money must be collected, allocated, and guarded with judicial care; and because the collection was aimed, rung by rung and category by category, at the one outcome all three traditions agree wealth exists to secure: a community where no member’s ruin becomes another’s revenue and no rescue is priced in the standing of the rescued. That is what the old books mean by the common good — not a sum but a floor, laid gift by gift, fortieth by fortieth, under every neighbor at once. On that floor the traditions were willing to let commerce build as high as honest partnership could carry it. The great technical debates of Part VI — what a lender may recover, what a loss is worth, what time may never be priced at — all assume the floor is in place and being paid for. It was the first thing they built, and they built it, deliberately, out of money that never comes back.

Comparative Interlude II — Buddhist Renunciation and Institutional Finance

On a railing medallion from the great stupa of Bharhut, carved in central India around the second century before our era, a work crew is unloading an oxcart. The ground beneath the workers is wrong: it is drawn as a grid of small squares, edge to edge, as if the earth had been tiled. The squares are coins. A label cut beside the scene explains, in five words of Brahmi, one of the most famous purchases in the history of religion — in the epigraphists' rendering: Anathapindika gives the Jeta Grove, "bought for a layer of crores."

The story behind the label is told in the Vinaya, the Buddhist order's own book of discipline, at Cullavagga VI.4. Anathapindika, a banker of Savatthi and the most open-handed layman the tradition remembers, wanted a park near the city where the Buddha and his monks could spend the rains. The park he judged perfect belonged to Prince Jeta, who did not care to sell, and who named, as one does to end a conversation, an absurd price: cover the ground with gold and it is yours. Anathapindika sent for his carts. The prince protested that he had been joking; the arbitrators he appealed to disagreed — a price named is a price named — and the sale stood. The tradition puts the total at eighteen crores, a number best read the way it was offered, as the sound a fortune makes when it empties. The grove became the Jetavana, the most storied monastery of the early texts.

Hold the image, because this interlude lives inside its contradiction. The community that received that carpet of money was founded on the renunciation of money — not its regulation, not its moral pricing, its renunciation, person by person and coin by coin. A Buddhist monk, then and now, is forbidden by his ordination vows to accept gold and silver at all. The order whose founding real-estate transaction is commemorated in stone as a field of coins is the same order whose members may not touch one. Neither the tradition nor this book reads that as hypocrisy. It is something more interesting: the two halves of a single institutional design, a man without money serving inside a community that must nonetheless be housed, fed, clothed, and endowed. What happens when the most radical personal rejection of money in religious history has to run institutions anyway — that is this interlude's subject, and by the end of it the drama will have run from a grove in Kosala to a treasury in Chang'an so large an emperor confiscated it, and the lessons will be two words this book will carry into its final part.

First, though, the disclaimer that guards everything that follows: Buddhism is not simply another anti-usury legal tradition, and it would falsify the tradition to treat it as one. Interlude I could stand beside Parts I through III as a rival answer to their shared question — where Jerusalem, Rome, and Mecca built a wall against increase, the Sanskrit lawbooks built a measure, a tariff of lawful rates with the king for enforcer. The Pali canon offers neither wall nor tariff. No verse of it forbids lending at interest; no rule of it caps a rate. The five trades the Buddha barred to his lay followers — dealing in weapons, in human beings, in meat, in intoxicants, and in poison (Anguttara Nikaya 5.177) — make a pointed little list precisely for what it omits: the moneylender is not on it. Debt in these texts is not a sin the law must prohibit; it is a suffering the wise avoid — and, as we will see in a moment, the canon describes the sufferings of the borrower with the cold accuracy of someone who had watched a collections process run its course. The tradition's quarrel is not with the loan contract. It is with something underneath all contracts: with grasping itself, the craving

— *tanha* — that the second of its Four Noble Truths names as the engine of suffering. The Abrahamic traditions of this book locate the usury problem between persons: an injustice done by a lender to a borrower. The Buddhist analysis locates the money problem inside a person: a bondage the handler fastens on himself. Everything else in this interlude — the strange double structure of monk and monastery, the endowments, the treasuries, the drift — unfolds from that relocation.

The Monk Without Money

The rules come first, because the rules are the tradition's own statement of how seriously it means this. In the Patimokkha, the code of some two hundred twenty-seven rules every monk lives under, three consecutive rules in the class called *nissaggiya pacittiya* — offenses requiring forfeiture of the object and confession — deal with money. Rule 18: a monk may not accept gold and silver, may not have another accept it for him, may not even consent to its being placed beside him or kept for him. Rule 19: he may not buy and sell with money. Rule 20: he may not engage in barter. Taken together, the three rules wall the renunciant off not only from the coin but from the market's habits of mind — pricing, dickering, calculating an exchange — and the Vinaya's own gloss closed the obvious loophole at once: "gold and silver" extends to whatever is used in business, any medium of exchange whatever. There is no personal-use exception, no *de minimis* threshold, no intention test worth gaming: the money is forfeit and the offense stands regardless of what the monk meant to do with it.

How much weight the tradition put on these rules is best heard in its own voices. In a discourse preserved at Samyutta Nikaya 42.10, the matter comes up as gossip at court in Rajagaha: word in the king's assembly is that the Sakyan-son contemplatives accept money after all. A headman named Maniculaka, sitting in the assembly, defends the monks — money "is not allowable" for them; they "have given up gold & jewelry, have renounced money" — and then, being a careful man, goes to the Buddha to ask whether he has misrepresented the teaching. He has not, the Buddha tells him, and the confirmation rises into one of the sharpest sentences in the canon, here in Thanissaro Bhikkhu's translation: "For anyone for whom money is allowable, the five strings of sensuality are also allowable." Money is not condemned as unjust; it is diagnosed as appetite in bearer form — take the coin and you have taken everything the coin can buy. A monk needing thatch may seek thatch, the discourse continues, needing wood may seek wood, a cart, a workman; "but by no means do I say that money may be consented to or sought for in any way at all." He may seek any particular thing he needs, but never the universal solvent of things. Elsewhere the Buddha lists the four stains that keep contemplatives from shining "like the sun and moon": drink, sex, accepting gold and silver, and wrong livelihood (Anguttara Nikaya 4.50). It is worth pausing on the company money keeps in that list. Nothing there is a crime against a counterparty. All four are corruptions of the renunciant himself.

A century after the founder's death — the texts' own count — the order proved it had meant all of it. At Vesali, the monks of the Vajji country had begun accepting money from the laity, setting out a collecting bowl on the observance day and inviting coins into it. A visiting elder named Yasa refused his share and told the laity to their faces that the practice was unlawful; the local monks turned on him for saying so. The quarrel he ignited became the Second Council: seven hundred elders convened to rule on ten disputed practices, of which the tenth and gravest was the money in the bowl. The council ruled against all ten. The tradition's first great institutional crisis, the one its chroniclers place at the head of the road to its first schism, was, at bottom, an audit finding about cash handling — and the

strict party won it. Whatever else may be said of the Sangha's later career with money, the tradition put its refusal on the record early, unanimously, and at the cost of its own unity.

And yet — this is what keeps the Buddhist chapter of this book from being a simple story of heroic abstinence — the same canon is, for the layperson, calmly and even warmly pro-wealth. The householder's discourses assume he will earn, save, and lend. The Sigalovada Sutta, the Buddha's long advice to a young householder, praises the man who gathers wealth harmlessly "like to a bee that honey gathers" and then hands him a budget in verse, here in Narada Thera's rendering: "One portion for his wants he uses, two portions on his business spends, the fourth for times of need he keeps" (Digha Nikaya 31). Half of the lay budget is working capital; a quarter is reserves. Another discourse, given to Anathapindika himself, catalogs the four blisses available to a man of means, and one of them is *debtlessness* — "owes nothing—large or small—to anyone at all" (Anguttara Nikaya 4.62). Its dark twin (Anguttara Nikaya 6.45) walks the other direction down the same road, rung by rung, in the frame the sutta itself supplies — "for one who partakes of sensuality": poverty is suffering; then "getting into debt is suffering"; then "interest payment is suffering"; then being served notice, then being hounded, and then, when a destitute debtor cannot pay, bondage. The ladder ends where chapter 2 of this book began, in the debtor's unfreedom, and the canon reports it without euphemism as the ordinary machinery of its world. That is the shape of the whole teaching: for the household, prudence — earn righteously, keep reserves, owe nothing; for the renunciant, zero — not a low rate but no rate. The line is not drawn through the transaction, as Deuteronomy drew it, or through the rate, as Manu drew it. It is drawn through the person: on one side of ordination money is a tool to be managed; on the other it is a live wire.

The Steward at the Gate

But the order has to eat. Four requisites — robes, food, lodging, medicine — do not materialize from doctrine, and the almsround, that daily flow of cooked food from lay bowls to monastic ones, is itself an anti-accumulation device: what arrives today is eaten today. The moment the community grew past what a morning's walk could feed, someone had to handle value on its behalf. The Vinaya's answer is a figure this book has met in other robes: the steward.

The device is drawn with the precision of a statute. A donor who wishes to provide for a monk's robe may not hand the monk money — but he may deposit the sum with a lay steward, a *kappiya-karaka*, "maker of what is allowable," and tell the monk so. Thereafter the monk may accept whatever allowable thing the fund produces; the money itself he never touches, commands, or owns. The regulating rule (*nissaggiya pacittiya* 10) even quantifies how hard he may lean on a dawdling steward: six verbal reminders, or up to twelve silent appearances — the monk simply standing there, robed, saying nothing — and a verbal prompt may be traded for two silent ones; past the limit, whatever the fund finally yields must be forfeited. Even the permitted sentences are scripted. Asked who his steward is, a monk may say, "That person is the steward"; he may not say, "Give it to him," for that would be command of money, nor "He will buy it," for that would be trade by proxy. The line between information and instruction is walked syllable by syllable.

It is easy to smile at the arithmetic, and wrong to. What the prompting rule and the scripted answers actually record is a tradition refusing to lie to itself. The drafters of the Vinaya understood perfectly that a renunciation of money that merely relocated money into a servant's hands would be a fiction within a generation — that "the monk has no money"

can be made true in exactly two ways, by the monk having none or by the sentence being redefined. So they built the separation as engineers build it: custody split from benefit, request split from command, with bright-line protocols at every joint and forfeiture as the penalty for leaning through the wall. Chapter 3 gave this book its audit rule — every arrangement must be what it says it is — and chapter 8 watched the rabbis patrol the loan's perimeter against benefits in costume. The kappiya-karaka protocol is the same jurisprudence pointed inward: not a loophole in the renunciation but its load-bearing wall, the honest price of having an ideal survive contact with lunch. What the monk renounces, the monastery must administer.

For the monastery, as an institution, may own what no monk may touch. Gifts in the donation formulas run to “the Sangha of the four quarters, present and to come” — to the order as a corporate body outliving every hand that serves it — and that body may hold parks, buildings, bedding, bronze, land. The Jetavana was exactly such a holding: no monk owned a square foot of it, and the community owned all of it, coins-in-the-ground and all. Within a few centuries of the founding, the two-tier design — propertyless persons inside a propertied perpetuity — was complete across the Buddhist world. The reader of this book has seen the pattern before, in the waqf and the hekdesch and the monastic lands of Christendom that chapter 29 will treat together. What the Buddhist case adds is the sheer voltage of the differential: nowhere else does so absolute a personal zero sit inside so capable a corporate purse. That differential is a design achievement, and it is also — the rest of the interlude must now say it plainly — a standing gradient, down which value flows from the person of the monk toward the property of the order.

The Imperishable Principal

Follow the endowments, because the endowments are where the tradition's financial genius shows first. Sometime around the year 120 of our era, a donor named Ushavadata — son-in-law of the satrap Nahapana, a man whose gifts are itemized across western India — had a benefaction cut into the wall of a cave monastery at Nasik. Three thousand *karshapanas*, the inscription records, given as *akshaya-nivi* — imperishable principal, permanent endowment — and invested, on the donor's instruction, with the weavers' guilds of the town of Govardhana: two thousand with one guild at one percent a month, the interest to clothe the monks; one thousand with another at three-quarters percent, the interest to feed them. The principal, the stone specifies, is never to be repaid — “their interest only to be enjoyed.” The deposit was proclaimed in the guild assembly, the *nigama-sabha*, and then carved above the monks' heads, rates and all, where it has now outlasted the guilds, the satrap, and the coinage by nineteen centuries.

Stand before that inscription with the rest of this book in mind and the strangeness blooms. Here is a monastic community whose members may not touch a coin, living — by design, by published contract — on interest income; and not usurious interest either, but rates sitting squarely inside Interlude I's lawful bands, placed with commercial borrowers through the town's own institutions. The monastery has not lent anything; it could not. It has been made the beneficiary of a perpetuity, with the guilds as the working debtors and the donor as the settlor, and the whole arrangement flows from one insight: alms feed monks for a day, but a principal that never dies feeds them forever.

The northern monastic codes took the further step, and took it in the Buddha's own voice. The Mulasarvastivada Vinaya — the discipline that governed the great monasteries of north India and, later, Tibet — contains rules that startled even its modern editors, and

the scholar who put them before Western readers, Gregory Schopen, gave his study a title the material earns: “Doing Business for the Lord.” When donors endowed perpetuities for the upkeep of buildings, the text reports, the funds at first sat idle or were misplaced, and the donors complained: my gift was to be inexhaustible, and it is being exhausted. The Buddha of this Vinaya responds with an instruction: “for the sake of the community a perpetuity for building purposes is to be lent on interest” — lent, by the monastic corporation, at increase. And the text immediately shows the community learning, by failure, what every lender in this book learned. Lent to the donors themselves, the perpetuities dissolved in disputes; lent to the great, they were not repaid, rank refusing dun; lent to the poor, they could not be repaid. So the rule matured into an underwriting standard: lend against a pledge worth twice the loan, under a written contract, sealed and witnessed. Loan paperwork, in a book of scripture. Schopen notes that the legal vocabulary is the dharmashastra’s own — the monks were borrowing the lawbooks’ instruments as well as their debtors — and a reader of this Part will note something else: the doctrine of the endowment has arrived, by internal logic, at the exact place the Talmud reached when it let the orphans’ money be placed “near to profit and far from loss” (chapter 10), and the place the Monti’s defenders reached when they priced the pawnshop’s costs (chapter 17), and the place chapter 29’s founders will formalize. Money that must serve a mission forever cannot be permitted to die of idleness; and the moment a tradition admits that, it is in finance, whatever its founders renounced.

The Inexhaustible Treasury

China is where the consequences arrive at scale, and the standard economic history of the arrival — Jacques Gernet’s *Buddhism in Chinese Society*, a book built out of monastery account books, pawn tickets, and loan contracts — reads in places like the file of a central bank examiner. In China, on Gernet’s showing, the Buddhist monastery did not merely receive endowments; it became, for several centuries, the most sophisticated financial institution the empire had. The pawnshop — lending against a deposited pledge, the poor man’s collateralized credit — enters Chinese history inside the cloister: the earliest specimen the historians can produce is a man named Chen Pin pawning a bolt of hemp cloth at a monastery treasury at the end of the fifth century, and before the Tang the business was effectively a monastic monopoly. The institution outlived its origins by a millennium and a half; later Chinese usage knew such funds as “long-life treasuries,” and Yang Lien-sheng’s classic study counts the pawnshop as merely the first of four Chinese money-raising institutions — with the mutual financing association, the auction, and the lottery — that either began in or grew up beside the monasteries. Mercy-lending against pledges, the mechanism this book watched Italian friars fight to license in chapter 17, was operating in Buddhist China nine hundred years before Perugia.

The oasis of Dunhuang, whose sealed library cave preserved the paperwork, shows the other face. There the monasteries were the granaries of the countryside, and the surviving contracts record seed loans — wheat, millet, hemp — advanced to farm households at planting and repaid at harvest, seven or eight months later, at rates that ran as high as fifty percent of the principal. The loans were written before witnesses; the monastery kept the ledger. What the ledger meant in a bad year, the reader of chapters 2 and 4 already knows: a household that borrowed its seed at fifty percent and lost the crop stood at the top of the oldest slide in this book, and the institution at the bottom of the slide, holding the paper, was the one whose monks had vowed never to touch the coin. This is said plainly, not prosecutorially: the monasteries also fed the district in famine, buried the poor, and lent to

smooth the very seasons they profited by; the account books record charity and increase side by side, written by the same hands. But the gradient was real, and the tradition's own conscience knew it.

Nothing shows that conscience — and its defeat by scale — like the Inexhaustible Treasury. The Sanjie or Three Levels movement, founded by the monk Xinxing in the sixth century, taught that the age was too degenerate for fine doctrinal distinctions and that the surest practice left was giving — universal, indiscriminate, boundless giving, channeled through an institution built for the purpose at the Huadu monastery in Chang'an: the *wujinzang*, the Inexhaustible Treasury. The name is the Nasik inscription's theology in Chinese dress — the same adjective, the same dream of a principal that never runs out. Donations poured in anonymously, penitents leaving silk, money, and treasure without so much as a name in the register, and the Treasury poured them out again: alms to the poor and sick, loans to the needy — interest-free, the accounts insist — and, the historians add, the repair of monasteries far beyond the capital. It was a solidarity fund of the kind chapter 3 taxonomized and chapter 15's founders would have recognized on sight, running at a scale none of them ever touched. And its very success indicted it. A treasury that inexhaustible was a power — anonymous, doctrinally suspect, and outside the state's sight; the movement was proscribed repeatedly, and in 713 the emperor Xuanzong ended the question by liquidating the Treasury outright and breaking up the school's institutions. The tradition's most radical experiment in mercy finance died of its own balance sheet — not because the mercy failed, but because the accumulation the mercy required had become, in the state's eyes and perhaps in fact, an institution unto itself. The instrument of boundless giving was now financial infrastructure, and the infrastructure, not the giving, is what the emperor saw.

The pattern's afterlife can be traced in a single word. In Japan, the Buddhist vocabulary of inexhaustibility gave its name — *mujin* — to the humble rotating credit ring: neighbors pooling fixed contributions, the pot going to each member in turn, the temple often the organizer and the meeting hall — mutual aid in its simplest geometry, kin to the pools chapter 27 will survey. The word first surfaces in the records in 1255; through the Muromachi centuries the rings ran as interest-free mutual aid organized by temples and shrines; and then the drift resumed its patient work — the rings commercialized, the operators incorporated, a Mujin Business Law arrived in 1915 to regulate the companies, and a mutual-bank statute of 1951 converted the survivors, most by the mid-1950s, into banks proper. A noun that began as monastery treasury theology ended, seven centuries on, as a regulated category of Japanese banking — the whole arc of this interlude compressed into one word.

Holy Suspicion and Administrative Realism

Institutional drift — the slow migration of a founding purpose toward the interests of the institution built to serve it — is not a Buddhist problem. It is the recurring weather of this entire Part, and of every tradition in this book. The Monti di Pieta of chapter 17 drifted from emergency pawn-charity toward deposit banking until one of them, five and a half centuries on, was simply a bank. The heteriska of chapter 14 drifted from a negotiated partnership between named men toward boilerplate on a wall, and chapter 31 will weigh what was lost in the lamination. Islamic finance conducts the same argument about itself today, in its own journals, and chapter 32 will let its critics speak. The waqf, the endowment, the church living — chapter 29 will find administrators' interests grown over every one of them like root systems under a road. Buddhism's place in this catalog is not the worst case; it is the clearest, the nearest thing the history offers to a controlled experiment.

Here is the tradition that renounced money most absolutely at the level of the person — no rate, no handling, no touch — and it produced monastic landlords, empire-scale treasuries, and the pawnshop monopoly of medieval China. If drift were a symptom of doctrinal laxity, the strictest doctrine should have drifted least. It drifted most, because drift is not doctrinal at all. It is structural: a perpetual fund, a steady inflow of gifts, and a class of administrators will accumulate, whatever the founding sermon said, exactly as water will find the low ground. The founders vow poverty; the thing they found endures past them, and an enduring thing fed by endowment accumulates as a matter of course.

What, then, does the guest leave on the table as it departs? Two things, and the heading of this closing section has already named them.

The first is holy suspicion. Buddhism's distinctive contribution to the moral analysis of money is not a rule about transactions but a diagnosis of the handler: money binds; grasping scales; the appetite does not announce itself but compounds in silence. The tradition aimed that diagnosis at the individual heart — and its own history, read honestly, extends it one step further, to the corporate person. Institutions crave too; their craving compounds. An endowment does not lust, but it grows; a treasury does not grasp, but it retains; and a community of the selfless can, as a body, exhibit every acquisitive behavior its members have individually renounced. When E. F. Schumacher went looking for an economics of sufficiency rather than maximization — the 1966 essay that gave the field its name, "Buddhist Economics" — this suspicion of calculation and accumulation was the well he drew from. The design consequence for Part X is direct. A fund built to be non-extractive toward its borrowers must also be built suspicious of itself — audited against its mission and not merely its books, its accumulation capped or sunset or forcibly recycled, its administrators structurally unable to convert the pool's endurance into their own estate. Chapter 41 will draw those instruments; chapter 45, whose subject is the temptations of religious finance in every tradition, is in a sense this section grown to full length. The Buddhist gift is the null hypothesis those chapters must design against: assume drift. No fund is exempt because its founders were sincere. The sincerest founders in this history built the biggest treasuries.

The second is administrative realism, and it is the first lesson's necessary partner, because suspicion without administration curdles into a piety that changes nothing. The Vinaya is the standing proof that the two can be married. The same tradition that diagnosed craving refused to pretend that the requisites would materialize by merit: it counted the promptings, scripted the steward's sentences, and when the perpetuities came, it wrote the pledge ratio and the witnessed contract into the discipline itself. That is not the ideal corrupted. That is the ideal surviving — renunciation kept honest for twenty-five centuries precisely by the exactness of its bookkeeping. The lesson for the fund-design chapters is the same one the steward protocol teaches at the gate: an ideal that refuses to specify its own administration will be administered anyway, by drift; the mission that intends to outlast its founders must be drafted into instruments with at least the care the Vinaya spent on a robe fund — the vow written like a contract, because the institution that inherits a vow will read it like one.

So the guest departs as the first one did, refereeing nothing. Buddhism does not join the argument that runs through the rest of this book — wall against measure, title against evasion, the lawful price of waiting. It stands at the argument's edge and asks the question none of the disputants quite asks: not what may be charged, but what the charging does to the one who charges; not whether the institution's returns are just, but what the institution is becoming while it earns them. Part VI now returns to the disputants. The great technical

debates ahead — beginning with *damnum emergens*, the lender's actual loss — are the wall traditions' own centuries-long audit of every reason a lender ever gave for taking more back than he gave. The interlude's question travels alongside that audit from here to the end of the book, posted over every ledger in it: what the money, so carefully minded, is doing to its minders.

Part VI — The Great Technical Debates

Chapter 19 — Damnum Emergens: Actual Loss

For five Parts this book has watched three traditions build. Parts I through III raised the wall: a loan may bring nothing back beyond itself, because a lender who charges for the use of what he has handed over sells what he no longer owns. Part IV opened the wall's one honest gate — the partnerships in which capital earns by standing genuinely exposed. Part V walked the institutions raised against the wall's inner face, lending at zero or at cost to the very people the wall was built to shelter. What remains is the work a serious legal system performs on its sternest rule once the rule is standing: it holds hearings on the hard cases against it. The five chapters of this Part are those hearings — an audit, the last interlude called it, “of every reason a lender ever gave for taking more back than he gave” — and the remarkable thing about the audit is who conducted it. Not the wall's enemies. The canonists, the rabbis, and the jurists of the *fiqh* ran it themselves, century after century, against their own rule, because they had noticed what every honest auditor eventually notices: a rule that cannot tell the claims against it apart will end by enforcing injustices of its own.

Begin, as the auditors did, with the strongest claim on the docket. A man has lent a hundred florins for a year at nothing — precisely as the law he prays under commands. He is no banker. The money was his margin against his own obligations, and he chose the repayment date because his rent to the wool merchants falls due the week after. The borrower fails. Nothing dramatic: prices broke, a cart was late, the money is not there. And now the lender's own promises begin to cost him. He borrows the hundred from a stranger and pays for the accommodation; or he sells cloth into a soft market at a forced-sale price; or he misses his own date and owes a penalty to a man harder than himself. When the season settles he is poorer for having lent — not poorer the way a man is poorer for a daydream that failed to arrive, but poorer in coin, by an amount his clerk can put a figure on. The question of this chapter is what the wall says to him.

All three traditions answered, in their different grammars: be repaired. And all three said it nervously, with conditions attached like seals down the margin of a charter, because they could see both cliffs at once. Refuse such a lender, and the prohibition becomes a machine for punishing obedience — the man who lent gratuitously, as commanded, is made to fund his borrower's failure out of his own substance, and the commandment quietly teaches prudent men never to lend at all. Admit him carelessly, and every loan in the market will promptly discover a wound, since lenders bleed with wonderful convenience once bleeding pays, and the wall stands hollowed from the inside. Between the cliffs the traditions cut a path, and its file carries five headings that make this chapter's itinerary: damage, the wound itself; expense, the costs it drags behind it; default, the failure that inflicts so many of both; impairment, the wound capital takes inside a venture; and recovery, the arithmetic by which any of it may honestly be made good. The medieval lawyers gave the file the name this chapter wears. Of the four titles chapter 10 met standing beside the loan — the agreed penalty, the interrupted gain, the imperiled principal, and this — *damnum emergens*, the loss arising, was the claim the tradition trusted first, because it is the only one that looks entirely backward: at something that has already happened, to a person it happened to, in an amount the world has already fixed. The audit opens here because if any claim can cross the wall honestly, it is this one. It opens here also because the ways this claim can be faked — and they are many, and modern, and mostly legal —

teach a reader how to test all the others.

To Avoid a Loss

The doctrine's oldest bones are Roman, and they were never about lending. When a Roman's contract was broken, the law asked what the breach had cost him — not as a rate but as a difference: *id quod interest*, "that which is between," the gap separating the position he holds from the position performance owed him. Out of the phrase grew a noun, *interesse*, that would live two lives, a compensation word for a thousand years before it became the modern world's blandest word for the price of money; chapter 22 watches its older self still at work, policing the penalty clause. What matters here is the shape of the Roman question. It does not ask what money earns. It asks what an injury measures. A claim built in that shape has a ceiling fixed by the injury and a direction fixed by the past, and both were about to matter enormously to lawyers Rome never imagined.

Latin Christendom carried the Roman grammar into a world where the *mutuum* had been declared barren — the anatomy Part III walked — and met the collision case at once: the gratuitous lender genuinely damaged by his borrower's failure. The canonists' answer settled earlier and more easily than almost anything else in the usury literature, because it asked nothing exotic of anyone. Everywhere else in life the principle already governed: break your neighbor's fence and you mend it; withhold what is owed and answer for what the withholding wrecks. Extending the same justice to a damaged lender required no theory that money breeds, no price on time, no new fondness for finance — only the admission that making a loan does not strip a man of the protections every other injured party enjoys. So the surplus crossed the wall under strict escort: not as the loan's yield but as an injury's repair, owed because equality between two persons had been disturbed by a wrong, and measured — this is the discipline — by the disturbance, never by the lending. The loan itself stayed barren; the injury, being no part of the loan, did not.

The formulation the whole tradition would cite came from Thomas Aquinas, in the usury question of the *Summa Theologiae*. Article 1 of that question holds the wall itself — taking usury for lent money is unjust, the double sale chapter 7 walked. Article 2 then asks the auditor's question, "Whether it is lawful to ask for any other kind of consideration for money lent?", and its first reply is this chapter's charter. "A lender may without sin enter an agreement with the borrower for compensation for the loss he incurs of something he ought to have," Aquinas writes, "for this is not to sell the use of money but to avoid a loss" (ST II-II, q.78, a.2, ad 1, in the English Dominicans' rendering). Then, in the same breath, the fence: the lender "cannot enter an agreement for compensation, through the fact that he makes no profit out of his money: because he must not sell that which he has not yet and may be prevented in many ways from having."

The architecture deserves a slow look. The permitted payment is defined by what it is not: not a sale. Nothing has been sold, because nothing passed to the borrower beyond what the loan itself moved; the payment un-does rather than purchases, which is why the identical florin can be honest in one ledger and usury in the next — its moral species fixed not by its size but by what it answers to. And between the permission and the fence sits a sentence most summaries skip, worth rescuing for what it shows about the man writing. It may also happen, Aquinas notes, that the borrower avoids a greater loss than the lender incurs, "wherefore the borrower may repay the lender with what he has gained." The reply sees both households at once — a borrower who used the money to dodge a catastrophe of his own, repaying the smaller wound his rescue inflicted out of the larger one it spared

him — and that double vision is the temper of the whole doctrine. This is repair between neighbors, not tribute to capital.

Around the charter the schoolmen built fences, few and strict. The loss must be actual — arisen, nameable, not the standing condition of being without one's money, which is simply what lending is; a lender who calls the bare absence of his coin an injury is pricing time again, and the tradition sent that claim to a different file, where chapter 21 will find it. It must be caused — by the borrower's failure or by the transaction's own external burden, not by the lender's imprudence or the ordinary weather of markets; a borrower answers for the losses his conduct made, not for his creditor's whole commercial life. It must be shown — narrated, dated, totaled; a wound that cannot answer questions is a story. It must be proportioned — repair runs to the injury's edge and stops, and whatever crosses that edge has left the title and needs another name. And it may be counted once: a delay already avenged under a penalty clause may not bleed again as damages — one wound, two names — though the penalty's side of that seam belongs to chapter 22's file. None of these fences is decorative. Remove any one and the title turns, quietly and lawfully, into the very thing it was built against.

What happened next is the part of the story modern finance would recognize as its own biography. Commerce thickened; money stopped sitting still; chains of obligation ran through the fairs so that one missed repayment could topple a row of promises, each fall a documentable cost. The title standardized — the safest claim in the confessors' manuals — and its scope grew with the commercial world's genuine capacity to be wounded. The theologians of Salamanca spent their best casuistry on it: Domingo de Soto holding the cautious line, insisting the word "damage" never float free of proof; Luis de Molina integrating the title with the civil lawyers' machinery, so that a sum agreed in advance against breach is honest exactly insofar as it estimates probable, provable, real loss, and a disguised rate exactly insofar as it does not; Leonardus Lessius, whom the merchants of Antwerp consulted and who is remembered for carrying his questions onto the exchange itself, mapping how loss, foregone gain, and risk overlap in a working merchant's affairs. Then Juan de Lugo, in Rome, gathered the family into a synthesis — his great treatise on justice appeared in 1642 — reading foregone profit and imperiled principal as varieties of one broad category of compensable detriment. It was the most powerful statement the doctrine ever received, and the most dangerous, for a reason the reader can finish before the sentence does: the broader "loss" grows, the less it excludes, and a category that once repaired burned barns becomes available to anything a clever claimant can describe as a wound. When Benedict XIV's *Vix pervenit* restated the center in 1745 — from the loan itself, never; from genuine titles outside it, possibly — the restatement was necessary, chapter 7 saw, precisely because the century's best minds had built this title an empire. The door that document holds open is the door this chapter walked through; the audit's remaining work is making sure of what comes through with it.

A Loss Must Testify

The halakha ran the same audit inside a different courtroom, because its actual-loss problem lived inside its own investment contract. Chapter 13 drew the *iska's* loss map: half the capital a loan, the manager's property and the manager's peril; half a deposit, trading at the investor's risk, that risk being the entire license for the investor's gain; negligence anywhere moving the loss to the negligent. On that map, the rule that the investor bears the deposit half's honest losses is the load-bearing wall — and it creates an exquisite problem of knowledge. The venture's truth lives with the manager. He kept the books, walked the

markets, watched the cargo sell or fail to sell. If his bare word could seat a loss on the investor's capital, what would the investor's exposure be worth? Every mediocre year could be reported as a wounded one, and the very openness to loss that makes the investor's gain lawful would make him his manager's insurance.

So the tradition priced the claim in the currency courts respect: evidence. The machinery, which chapter 14 walked clause by clause through the *heter iska's* documents, runs on a division its drafters took from the theology of testimony itself — a claimed loss is believed only on the word of acceptable witnesses, because losses are public events, the kind the world sees; a claimed shortfall of profit is believed only under oath, because profits are ledger events, known finally to the man who kept the ledger. What matters to this chapter is not the machinery but the principle under it. Loss-language is cheap, and the halakha made it testify. A loss that wants to cross from the manager's mouth to the investor's account must arrive with its papers in order — which is, note, the same demand the scholastics were writing into the confessional in the same centuries, aimed the other way. The Latin doctrine polices the lender who claims a wound against his borrower; the halakha polices the manager who claims one against his financier's capital. Opposite claimants, and one shared instinct: whoever brings the claim, an actual loss is a fact with witnesses, or it is a story.

The Talmud had set the tone long before the standard documents. Rava rules on the manager of an *iska* who suffered a genuine loss, then labored until later gains replaced it, and told the investor nothing (Bava Metzia 105a): at settlement he may not reopen the ledger and say, bear the old loss with me. The Gemara answers him out of his own motives: for this you toiled and made it good — so that men would not call you a loser of ventures. The ruling is small and its edges are sharp. An actual loss is an event in an account, not an anecdote available at settlement; a wound the venture's own later gains have healed has ceased to exist, whatever it cost the manager's pride to heal it; and a loss claim is presented fresh or not at all. Buried in the ruling is an accounting principle this chapter will shortly meet again wearing Arabic: recovery precedes division. Gains fill the hole before anyone may call them gain.

And the halakha's own lender — the creditor this chapter opened with, damaged by his borrower's failure? Here the rabbis were stingier than the schoolmen, and the stinginess was policy. The wage of waiting is forbidden outright — chapter 7's *agar natar* — and how the law disciplines a delaying debtor without ever selling him time, along with which of a creditor's losses from delay it deliberately declined to price, is chapter 22's file. But one paragraph of the codes shows the tradition drawing exactly this chapter's line in its own hand. A litigant who wins his case collects, as a rule, nothing for the cost of the litigating: the codified rule is blunt that the loser is not obliged to make good his opponent's expenses, though the suit dragged the winner to a court in another city (Shulchan Arukh, Choshen Mishpat 14:5). Going to law is the price of living under law; process is nobody's wound. But let the defendant refuse the court — ignore the summons, force the plaintiff to spend money compelling his appearance — and the same paragraph reverses: now he pays all the expenses his defiance caused. Read the two halves together and they are causation doctrine in miniature. The ordinary costs of pursuing a right lie where they fall; the costs manufactured by a man's wrongful conduct follow the conduct home. A defendant defying the court's summons and a borrower defaulting in Molina's Castile were answerable, on this point, to one idea — expense caused by fault is a compensable wound, and expense caused by life is not.

Furnishing the Reason

Islam's actual-loss doctrine begins where its usury law begins, in the settlement terms of the war verse. When the Qur'an commanded believers to abandon their outstanding increase or know themselves at war, it stated the repentant lender's entitlement in a clause chapter 6 read whole: "You shall have your capital if you repent, and without suffering loss or causing others to suffer loss" (2:279, in M. A. S. Abdel Haleem's rendering, as throughout this book). *La tazlimuna wa-la tuzlamun*, runs the Arabic close: you do not wrong, and you are not wronged. The clause is this entire chapter in one sentence. The lender is restored to his capital, all of it — the settlement does not fine him for repenting — and he is stopped at his capital: above it he wrongs, below it he is wronged. Restoration full, enrichment none — the boundary the schoolmen spent their casuistry fencing is drawn here in the founding text, as a condition of peace.

Inside the fiqh's own investment contract the doctrine turns technical, and the technical word is the right one: impairment. Chapter 13 filed the *mudarib* among the trustees — the capital in his saddlebags never his, its market losses falling on the *rabb al-mal* as completely as its profits would, his liability switching on only at fault. The modern standards state the file with an accountant's flatness the classical jurists would have recognized as their own. "A loss is depletion from the value of capital," states Malaysia's central-bank standard on the contract — loss defined as a fact about the capital rather than a feeling about the year — and the trustee is not answerable "for any impairment of asset" unless the loss traces to his "misconduct," his "negligence," or his "breach of specified terms" — *ta'addi*, *taqsir*, *mukhalafah al-shurut*, the fault grammar chapter 13 anatomized, now wearing its Arabic labels: weather to the owner, fault to its author. But the clause this chapter exists to notice is the quieter one beside those: in the case of loss, the standard directs, the *mudarib* shall "furnish the reason for the occurrence of loss." There is the witnesses-rule again, in regulatory prose. A loss is not a number the trustee reports; it is an event he must narrate — what happened, when, from what cause — because the presumption run in the honest trustee's favor is a presumption about honest men, not a dispensation from bookkeeping. The fiqh and the halakha, sharing so much anatomy in these contracts, share this too: the capital's owner bears the true wounds, and truth is established, never merely asserted.

Then the standards do the arithmetic of healing, and it is Rava's arithmetic. The classical rule — the schools state it as consensus, and the industry's standard-setters print it still — is that a running *mudarabah* has no profit until the capital stands whole: losses are made good from the venture's own gains before any gain exists to divide, and what the working partner draws mid-venture he draws provisionally, on account, his unsettled share described in the standards' own idiom as held back as a protection for the capital until the final reckoning, when the venture is wound up in fact or valued as though it were and the account finally speaks. Set that beside the manager of Bava Metziah 105a, who filled the hole and could not afterward dig it up again, and the two traditions are keeping one ledger discipline in two languages. Recovery precedes distribution. A venture's wounds heal out of the venture's own flesh first; "profit" is the name of what remains once the capital has been made whole; and impairment is measured on the whole story — netted, cumulative, final only at the end — never cherry-picked transaction by transaction toward whichever reading flatters the claimant. Chapter 14 built its argument on the profit half of that sentence. This chapter adds the loss half, and notes the symmetry: a tradition that would not let profit be declared before it existed was never going to let loss be charged before it emerged.

One more piece completes the strand. When the modern industry met the lender-side

problem squarely — the financed customer in default, the institution genuinely out of pocket — it reached for this chapter's title and split repair from punishment with a cleanliness the schoolmen would have envied: compensation capped at the institution's actual, shown loss, and everything punitive routed away to charity — machinery that belongs to chapter 22's file and gets its full hearing there, critics included. What the Islamic strand contributes to this file, distinctively, is the refusal at its foundation. At no point, in any of its instruments, is "loss" allowed to become an asset of the lender's. It is a fact about capital, borne by capital's owner, moved only by fault, escorted always by an account of itself, and bounded on both sides by the war verse's terms of peace — no wronging, and no being wronged.

The Average and the Event

Modern credit does something with loss that no scholastic imagined, and the difference is the sharpest analytical edge this chapter can hand a present-day reader. A modern lender prices loss before it happens. Chapter 10 stood in front of that machinery at full strength — expected loss, the chance of default multiplied by its cost, priced into every loan on the book so that the many who repay fund the losses of the few who cannot — and reported what the arrangement does to the people who must live it one path at a time. What belongs to this chapter is the categorical point underneath: an expected loss and an actual loss are not the same kind of object, and the whole title hangs on the difference. An expected loss is a price; an actual loss has an address. The expected loss lives in the lender's models — a property of a portfolio, smooth, attached in advance to everyone, caused by no one, suffered by no one in particular. The actual loss lives in the world: it happened on a date, to a person, from a cause, in an amount. *Damnum emergens* compensates the second and only the second — after the event, from its author, in its measure. The discipline is not squeamishness about mathematics. A charge collected in advance, from every borrower, whether or not the wound ever occurs, is not repair; what it sells is the possibility of loss, and the possibility of loss is simply lending itself, the one thing the wall forbids the lender to sell at all.

None of this outlaws prudence. An institution may hold reserves; one that did not would be gambling with money the community trusted to it, and the funded reserve is among this book's honored instruments. But a reserve is the pool's tool — built from named contributions against the book's bad years, at home in chapter 27's architecture — and the discipline is naming. Book reserve-building as "damages" and a communal buffer has been converted into a per-borrower fine for a wound no borrower caused. The moral ledger begins at the label, and the label must say what the money is for.

The same sorting settles the question this chapter's brief calls administrative loss, because "administration" names two different things a lending institution spends. The first is the ordinary running of the counter — the appraiser, the clerk, the vault, the ledger. That is not a wound at all; it is work and expense, and the tradition titled it honestly as such: the *stipendium laboris* of chapter 12, the recovery the Fifth Lateran Council licensed for the Monti in the settlement chapter 17 told — "a moderate sum for their expenses and by way of compensation," "intended exclusively to defray the expenses of those employed," and, the clause bearing all the weight, "provided that no profit is made therefrom." The mercy institutions of Part V run the same discipline at their own chosen severities: the *gemach* ideal of chapter 15, where donations carry the overhead and the loan comes home clean; the *qard hasan* rule chapter 16 reported, actual servicing cost per transaction and nothing indirect — salaries, rent, the building itself expressly unbillable to the borrower, because

the alternative loads mercy's overhead onto the people least able to carry it. Across all of them the shape is identical: sustainability licensed, priced off an auditable cost ledger, pointed downward by design; profit from the need to borrow, never. The second thing "administration" names is different in kind — the workout, the collection file, the second appraisal, the lawyer: administration that a particular default made necessary. That is *damnum* proper, caused expense chargeable to its causer, on proof, at cost, after the sorting of cannot-pay from will-not-pay that every tradition demands before any remedy fires. Two files, one counter. An institution that keeps them separate can sustain itself without extracting; an institution that merges them has found the oldest door out of the wall — bill the running of the shop to the borrowers as "losses," and the shop will profit by the poor exactly as if no wall stood.

When Repair Becomes Revenue

Every fence this chapter has walked was built because someone had been caught climbing it, and the climbs have not changed in five centuries; only the drafting has improved. The classic moves are four. The wound can be invented — a loss asserted that never occurred, a fee that answers to no event. It can be inflated — real damage priced past itself, the hundred-florin injury healed at three hundred. It can be pre-priced — written into the contract as a standing sum payable however the world turns, at which point it has stopped describing anything and become the price of time under a nobler name. And it can be double-billed — one delay collected as penalty, again as damages, again as fee, four clauses nursing one wound, each collection past repair being precisely a surplus with no title under it. Behind all four stands the oldest test the tradition owns, older than any of the drafting: watch the direction of the lender's hope. Chapter 8 met it in *Consuluit* — the mere hope of gain from a loan is enough to convict — and it reads a damages clause as easily as it read a friendly gift. A lender who dreads his borrowers' wounds holds a compensation clause; a lender who can budget for his wounds has begun to sell them.

Now read a present-day facility agreement with the file open. The default apparatus — the rate step-up, acceleration, cross-default — chapter 22 audits entire; around it stands the quieter loss-recovery furniture, and it sorts with surprising speed. Costs-and-expenses clauses, under which a defaulting borrower pays the documented outside costs of enforcement, are the defiant defendant's rule and Molina's integration in modern boilerplate: honest *damnum*, so long as the invoices are real, itemized, and not run through a collections operation that turns a margin on them. Break-costs clauses — the charge when a borrower repays early or fails to draw, computed as what the lender actually loses unwinding the funding arranged for him — are the doctrine's most exact modern artifact and its neatest boundary marker: computed from the actual cost of the funds, the clause repairs a caused wound; computed from the profit the lender had hoped to earn across the loan's remaining life, the same clause steps, mid-formula, out of this chapter's title and into the next one's — from the loss that emerged to the gain that ceased. Increased-costs clauses, which pass to the borrower the expense of new regulation landing on the lender, stretch the causal fence to its creaking point: the loss is real and documentable, but the borrower did not cause it, and a claim whose causation runs "you were nearby when my world got expensive" has quietly become mutual insurance billed as damages — an arrangement with an honest name and an honest home, neither of which is this file. The reader now owns the method, and it travels: name the event, trace the cause, demand the papers, measure the repair, count once, and watch the hope.

For the builders — and Parts IX and X of this book intend to build — the file compresses to

a working grammar. Trigger on events, never on the calendar alone: no damage without a happening. Document as an institutional habit, not a courtesy to auditors: the invoice is the modern witness. Cap and proportion: repair ends at the wound's edge. Net honestly: recovery already achieved — the mitigation that worked, the resale that covered, the hole the venture's own gains refilled — comes off the claim first, as Rava ruled and the standards still rule. Book loss-recovery in its own line at zero margin, so the ledger itself can testify that the line is repair; and watch the budget for appetite — chapter 22 built that audit for penalty income, and it reads identically here, since a lender whose year balances on projected borrower failures has inverted his purpose whatever each clause recites. Route anything above shown cost away from the house; chapter 22 found that conduit built, at industrial scale. And keep the mercy sort upstream of everything, because a wound-claim pressed home against the destitute repairs nothing and ruins someone, and every tradition in this book wrote that boundary before it wrote this title. Chapter 40 will hold the grammar to arithmetic — its return formula carries actual cost recovery and actual loss recovery as named, separate, bounded lines — and chapter 44 will give the ledger its audit. What they inherit from this chapter is a title already disciplined: backward-looking, event-born, proven, proportioned, counted once.

The audit's first hearing can close with its finding, and the finding vindicates the wall rather than breaching it. The traditions never required the lender to bleed for his obedience; they required him to prove. Damage, expense, default, impairment — each may be repaired, to the coin, from its cause, and such repair is the one payment above principal that leaves the wall standing, because it looks only backward, at what is already true. The next claim on the docket refuses that direction. It is the merchant's claim, and it is not a lie: had I kept my money, I would have earned with it. Aquinas heard it and answered in a single clause — "he must not sell that which he has not yet" — and the tradition spent the following centuries discovering how much commercial truth was pressing against that fence from the other side. That hearing is chapter 20, "Lucrum Cessans: The Profit That Was Not Made," and it opens where this chapter closes: at the property line between what a lender lost and what he merely failed to gain.

Chapter 20 — *Lucrum Cessans*: The Profit That Was Not Made

Two men in the same lane lend the same sum on the same day. The first is a wool merchant with his capital already committed: the fleeces bought, the cart hired, the buyer at the autumn fairs half promised, the whole small machine of a season assembled and pointed down the road. His neighbor comes to the door in genuine need, and the merchant — because the man is his neighbor, and the traditions this book reads had spent centuries telling him so — unloads the cart, sells the fleeces back for what he paid, and lends the money interest-free until winter. The second lender lives across the lane with the same hundred in a strongbox, where it has sat since Easter doing what coins do in boxes. He lends it on the same terms, to the same sort of neighbor, with the same open hand.

At winter both are repaid in full, to the coin. And both, standing in the lane, say the same sentence: *I am the poorer for it. I could have earned more elsewhere.*

One of them is telling the truth about a loss. The other is telling a truth about money — which is a different thing, and the difference is this chapter. The merchant can name his elsewhere: that buyer, those fairs, this wool, a margin his trade earned last year and the year before. The strongbox has no elsewhere; it has every elsewhere, which amounts to none — an infinite address book with no house in it. Out of the space between those two sentences the scholastics built the most contested instrument in their whole toolkit: *lucrum cessans*, the gain that ceases, the profit that was not made — the claim that a lender may sometimes be compensated not for what the loan took from him but for what it prevented him from getting.

Chapter 10 shelved the four extrinsic titles and called the risk title the most treacherous, because an uncapped charge for danger prices a permanent condition of lending. This title earns a different superlative: the most dangerous. Not the most abused — every title was abused — but the only one whose corrupted form eventually stopped needing an advocate. Generalize the merchant's claim to the strongbox, let every coin bill the world for its imaginary career, and you do not merely breach the usury wall; you replace the civilization that built it with one in which the wall looks insane. That civilization is ours. The claim this chapter examines is no longer an exception argued by lenders; it is the axiom taught to every first-year finance student under the name opportunity cost. Chapter 7 promised that the *lucrum cessans* debates would ask whether forgone time-bound opportunities can be real losses. They can. Whether that answer can be kept from devouring everything else these traditions concluded — that is the file, reopened.

The Claim from a World That Did Not Happen

Begin with what makes the claim strange, because its strangeness is load-bearing. Every other compensation the traditions weighed points at something in this world. Actual loss — the file of chapter 19 — has a wound to show: a bill paid, a barn burned, a cost incurred because repayment failed. The penalty of chapter 22 has a broken date. Even the risk title of chapter 10 points at a position actually held. *Lucrum cessans* points at nothing here at all. It asks a court, or a confessor, to compare the world as it stands with a second world that never happened — the one where the loan was refused and the wool went to the fairs

— and to bill the borrower for the difference. The lender's purse is exactly as full as his contract promised. His grievance lives entirely in the subjunctive.

That is why the title is elastic in a way no honest wound can be. A loss already suffered has edges; witnesses can walk around it. A profit that was prevented has whatever size the claimant's imagination allows — the merchant would have cleared ten, or say twenty; the deal would surely have closed. Chapter 12, pricing labor's contributions, met this exact asymmetry and quarantined it here: the traditions paid readily for months actually spent inside a venture and balked at months hypothetically redirected, because a spent season leaves witnesses and a redirected one leaves only the claimant's word. The quarantine was wise. This is the one title where the evidence for the claim is, by construction, a story about a world nobody inhabits.

And yet the claim would not die, because sometimes it is true. The medieval lawyers met it long before anyone had a term for it. William of Auxerre had ruled the calendar off the market — the usurer sells time, which is the common possession of all creatures, the argument chapter 7 walked — and that ruling holds here in a way worth noticing: what the merchant lost was not time, which belonged to him no more than to his neighbor, but a use he had built inside it. The canon law's first accommodations came, tellingly, not in loans but in sales. The same decretal chapter 10 read for the risk title, *Naviganti*, cleared in its quieter clauses the man who pays now for grain deliverable next season, and the seller who charges above today's price for payment at a term, wherever the future value was genuinely in doubt — *verisimiliter dubitatur*, as the Latin verified there has it; and the decretal's cast — grain, wine, oil, a man on the road to the fairs — kept the question commercial rather than speculative. Future value had entered the moral arithmetic of price. If a thing could be worth more now because of what it was concretely about to become, then a man deprived of that becoming had perhaps lost something real.

The first canonist to let the reasoning touch the loan itself was Hostiensis — Henry of Susa, cardinal-bishop of Ostia, remembered as the most brilliant canonist of his century, whose rhymed count of thirteen lawful exceptions the schools memorized. Among the thirteen stood the gain that ceases, and the shape of his opening mattered more than the fact of it. The claim was not for everyone. It belonged to the lender who was *not* in the business of lending — the merchant whose money was ordered toward trade and who lent it away from that order, at cost to himself. The restriction carries the whole logic. A professional lender who pleads lost profit is pleading that this loan cost him the chance to make another loan at interest: the title would justify usury by pointing at usury, a circle the canonists refused to walk. The alternative gain had to be licit, which meant it had to be something other than the thing under judgment. From its first appearance, *lucrum cessans* was admitted only as the exception of a man whose real vocation lay elsewhere — and everything dangerous in its later history is the story of that restriction wearing away.

One more thing belongs in the file before the theologians take it up, and it is a date. The phrase every modern reader reaches for — opportunity cost — entered economics in 1894, as the title of an article in the *Quarterly Journal of Economics*, a few years after Friedrich von Wieser had systematized the doctrine under the name alternative cost. The concept the phrase names is treated today as timeless common sense, the first axiom of choice. It is worth insisting that the axiom is younger than the bicycle, while the question of whether a forgone gain can be a real loss had by then been argued for six and a half centuries. The medievals did not lack the concept. They had it, examined it, and put conditions on it that the modern form quietly dropped. What follows is what the conditions were.

The Slow Yes of the Schoolmen

Aquinas answered first, and his answer — a no with a door in it — governed the debate for as long as the debate stayed honest. The text is a few lines in the usury question, and this book has touched its edges twice; here it must be read whole. A lender, he allows, may without sin agree with the borrower on compensation “for the loss he incurs of something he ought to have, for this is not to sell the use of money but to avoid a loss” — the clean restorative title that chapter 19 files, quoted here as throughout in the English Dominicans’ rendering. But compensation for the profit he would have made with the money is another matter: the lender “must not sell that which he has not yet and may be prevented in many ways from having” (ST II-II, q.78, a.2, ad 1).

Sit with the reason, because it is better than a rule; it is an observation about the future. The merchant’s ten of profit, so vivid in the lane at winter, was never his. It depended on the road staying open, the buyer staying solvent, the price staying up, the wool staying dry — a season’s worth of contingencies, any one of which could have eaten it. To charge the borrower for that ten as though it were a possession is to sell a certainty the seller never owned: the borrower pays in hard coin for a profit that existed only in the weather of one possible year. Worse, the compensated lender ends up better off than the merchant he claims to be — he collects the venture’s imagined fruit without its labor, its costs, or its risk, a harvest with the farming subtracted. Aquinas saw where the logic led: every lender would discover, upon lending, that he had been on the verge of something lucrative. The no was a fence around exactly that meadow.

But Aquinas built the door himself, sixteen questions earlier, and the later tradition walked through it carrying his own words. In the treatise on restitution he asks what is owed when one man injures another not by taking what he has but by “preventing him from obtaining what he was on the way to obtain” — blocking a wage, an inheritance, a sale already approaching. Something is owed, he answers, and then comes the sentence on which the whole later doctrine of foregone profit balances: not the full amount, “because to have a thing virtually is less than to have it actually.” His example is the one this book has been waiting for since chapter 14 asked whether hope could ever be collected as if it were harvest: the sower. A man who has sown his field does not yet have his harvest — *nondum habet messem in actu, sed solum in virtute* — he holds it “not actually but only virtually,” as seed holds wheat; and one who wrongfully destroys the sown field owes for the crop accordingly, at a discount the size of the world’s uncertainty, “according to the condition of persons and things” (ST II-II, q.62, a.4). Read the two texts together and the apparent contradiction becomes an architecture. In the loan contract, profit not made may not be sold — that is q.78, guarding the *mutuum*. In the court of restitution, a future good wrongfully prevented may be compensated — that is q.62 — but only as what it was: a probability, priced as a probability. Hope is compensable at hope’s price, never at harvest’s. The whole conservative discipline this chapter ends with is already in that sentence, and it is Aquinas’s, not ours.

What pushed the tradition through the door was not theory but Provence. Peter John Olivi — born near the coast at Sérignan around 1247, handed to the Franciscans at Béziers at about twelve, censured at Paris in 1283, cleared at Montpellier, dead at Narbonne in March 1298, his tomb demolished twenty years later — spent his last years in a town full of merchants, and did them the unusual honor of watching what their money actually was. In the *Tractatus de contractibus*, the treatise on contracts he wrote there in the mid-1290s, he observed that money firmly committed by its owner’s settled purpose to a probable profit is no longer mere coin: it carries “a certain seminal character of profit, which we commonly

call capital” — *quod communiter capitale vocamus*, the merchants’ own word, taken up by a theologian for the first time. His books were confiscated in his lifetime and condemned after it; Bernardino of Siena kept a manuscript of the treatise all the same, and a century and a half later carried whole passages of it, word for word and without the dangerous name, into the sermons that made him famous — the doctrine traveling under a saint’s voice because it could not travel under its own. Olivi’s point was narrow, and its narrowness was its power. He did not say money breeds; the *mutuum* stayed barren. He said that not all money is situated alike: coin inside a working commercial process — his word seminal, seed again — differs in economic kind from coin in a box, and interrupting the process can cost its owner something as real as trampled grain. The claim of the wool merchant in our lane is Olivian, and the claim of the strongbox is not, and the distinction between them — a productive alternative actually in train, against the general fertility of money — is the hinge on which every later formulation turns.

The fifteenth century said the careful yes. In the commercial cities the case kept arriving in the confessional, and the great pastoral theologians — Antonino, the pastoral theologian of merchant Florence; Bernardino of Siena (1380–1444), Franciscan preacher to the same world — admitted that a lender genuinely torn from a concrete trade could be compensated for it, while spending most of their ink on the abuses: the lost profit invented after the fact, the habitual lender in a merchant’s costume, the gain valued at fantasy prices, the borrower’s need quietly doing the pricing. Cajetan, the great Dominican commentator on Aquinas, supplied the Thomist reconciliation: a future gain sufficiently determinate — a transaction arranged, a practice standing, a market waiting — could be treated as *virtually* possessed, so that its loss fell under q.62’s restitution rather than q.78’s ban; everything then hung, as it still does, on how much work the word virtually is allowed to do. Conrad Summenhart at Tübingen, whose seven-part treatise on contracts was printed at Hagenau in 1500, folded the title into a general anatomy of exchange, asking of every payment what it actually paid for. And Salamanca inherited the settled question and disciplined it. Domingo de Soto, the school’s Dominican conscience, pressed the universalization danger — admit the title to everyone and the prohibition dies in its sleep. Luis de Molina wrote the conditions into their mature form: a real alternative use; an intention actual or at least virtual, standing in the lender’s ordinary practice rather than invented for the occasion; a gain foreseeable with reasonable probability; a causal chain from this loan to that loss; a valuation kept sober. Leonardus Lessius, pressing the logic of money markets to the edge where the bare deprivation of liquidity itself becomes a burden, opened the neighboring file this book keeps closed until chapter 21. And Juan de Lugo, gathering the whole tradition, fixed the three conditions that became the common teaching — the *sententia communis* of the manuals. The loan must truly cause the profit to cease. Nothing may be demanded beyond the value of the profit that ceased. And the compensation must not be taken up-front — because a lost future gain, deducted at signing, has become a guaranteed present one, and a guaranteed present return on a loan is the thing itself. When Benedict XIV restated the usury doctrine in 1745, chapter 7 noted, *Vix pervenit* left standing the titles that arise from reasons “not at all intrinsic to the contract”; the gain that ceases had by then carried its conditions for two centuries, a lawful claim exactly as wide as its proof.

Notice what the slow yes never conceded. At no point in four centuries of development did the tradition hold that money as such earns, that time as such is billable, or that the market’s going rate is anyone’s entitlement. The title that emerged is a tort dressed as a title: a specific injury, done by this loan to that venture, proven by the claimant, discounted for the world’s uncertainty, capped at the injury’s size, collected after the fact. Everything else — every claim in the family of the strongbox — the schoolmen went on calling by the

old name.

A Grievance, Not a Debt

The halakha heard the same claim and gave an answer of a wholly different shape: it looked the loss in the eye, named it, and declined to make it a debt.

The text is short enough to hold in the hand. One who idles his fellow's purse, rules Rebbi Isaac in the Jerusalem Talmud — takes money entrusted for trade, say, and lets it lie — has against him "nothing but a grievance," a *tar'omet*: a grievance the court acknowledges and declines to price. And the same passage draws the line that shows exactly where the wall stands: idle a man's *field* and you owe him, for a field bears fruit by nature; the fallow season is a harvest that was really coming, and its destroyer pays. The purse is different in kind. Coins bear nothing by nature — that has been the traditions' finding since chapter 7 — and so the law, asked to compensate the purse's imaginary year, refuses to presume a counterfactual it spent its whole jurisprudence declining to presume. Chapter 22 met this ruling among the costs of delay; read from this chapter's angle it is something more pointed: the halakha's direct verdict on *lucrum cessans*. The claim is not dismissed as false — the grievance is real enough to be named in law, which is its own kind of honesty. It is dismissed as unpayable, because the only meter that could price it is *agar natar*, payment for waiting, the forbidden core itself in chapter 7's gloss. A tradition that would not let a lender charge for the days was not going to let him charge for what he dreamed the days contained.

The refusal was not innumeracy about commerce, and the proof is what the same law pays without blinking. The working partner's months are compensated scrupulously — the idle-laborer wage of chapter 12, sized at what a man would take to down his own tools — because months spent inside a venture are witnessed fact, not counterfactual. Labor's forgone alternatives get a real price; money's forgone alternatives get a sympathetic shrug. The asymmetry is deliberate and evidentiary: it is the same line Molina would draw with intention and causality, drawn instead with the court's oldest instrument, the question of what can be proved.

And then the tradition built its lawful channel, whose logic is the most elegant answer to *lucrum cessans* any of the three traditions produced. The problem: a man with genuinely productive uses for his money — the Olivian lender, the man with real elsewhere — cannot be paid for forgoing them, because on a loan that payment is *ribbit*. The halakha's solution, met in chapter 11 and priced in chapter 14, was not to relax the rule but to dissolve its premise. Recharacterize the financing as an *iska*, half loan and half deposit, and the money's productive career no longer stops at the borrower's door: the deposit half remains the investor's own capital, trading in the borrower's hands at the investor's peril, and what the investor collects is not compensation for a profit forgone elsewhere but his share of a profit sought *here*, in the venture his money actually entered. The forgone gain is never paid because it is never forgone. Where the schoolmen disciplined the counterfactual, the rabbis abolished it — by making the lender a participant in the factual instead. The *heter iska*'s settlement machinery — the named figure, the heavy oath, the fee that buys the oath's silence, sized in the modern forms even to prevailing rates — is chapter 14's file and chapter 31's reckoning, and this chapter borrows only its skeleton: the named figure is, in substance, a conservative model of expected gain, agreed in advance, collectable only because the capital that earns it genuinely stands in the wind. Whether the modern instrument keeps that condition true at banking scale is precisely the question chapter 31 holds

open. The logic, though, is complete already: the halakha's answer to "I could have earned more elsewhere" is: then go earn it — and if you would rather lend, we will write you into the earning, at your own risk. It is the one answer in this chapter that makes the lament structurally impossible to utter.

Time in the Price, Never in the Loan

The fiqh drew its boundary with a draftsman's economy, and the boundary runs straight through the middle of this chapter's subject: time may take a share of a price; time may take nothing of a loan.

The second half is the older and harder rule, and it is absolute. Money in a *qard* has no rental rate; increase stipulated for the term — at signing or, in the original abomination, at maturity: the jahiliyya creditor's pay-or-increase, which chapter 22 found standing at the doctrine's birthplace — is *riba* without further analysis. No version of "I could have earned more elsewhere" survives contact with this rule, because the rule is a refusal to let money's time be priced at all, and a lost hypothetical return is nothing but money's time wearing a costume. The strongbox lender of our lane has no case in the fiqh, and neither, on a loan, does the merchant.

The first half is the seam, and the jurists never pretended otherwise. A sale may carry a deferred price above the cash price — the *murabaha*'s installment markup, the credit sale the Prophet himself used — and the manuals say why with unembarrassed clarity: the deferment takes a share of the price, as the jurists' own formulation has it. Chapter 24 walked the seam at full instrument length — the ownership interval, the disclosed cost, the price that never grows once set — and watched each tradition patrol its own version of the line: the Mishnah's rent-and-sale pair, the canonists' doubt clause, Aquinas's flat ruling that the naked deferral premium "has the character of a loan." The point this chapter needs from the seam is narrow: the fiqh's acknowledgment that time has value is real, and it is quarantined inside contracts about things, where the premium can answer to ownership, risk, and service. What may never happen is the migration of that value out of the sale and into the debt as such. The entire modern architecture of Islamic finance lives on one side or the other of that line.

Which makes the tradition's modern encounter with *lucrum cessans* the most clarifying in this chapter, because it happened in committee, in writing, within living memory. In 2000 the Islamic Fiqh Academy at Riyadh took up the penalty clause — the contractor who delivers late, the supplier who defaults — and resolved the question with two sentences that map the whole terrain. A penalty clause is permissible "in all financial contracts except when the original commitment is a debt," where it is "usurious in the strict sense." And the loss it may compensate "includes the actual financial loss suffered by the partner, any other material loss and the certainly realisable gain that he misses due to his partner's default or delay" — though never "moral prejudice" (Resolution 109 (3/12)). Read the second sentence twice, because there it is: the "certainly realisable gain that he misses" — *lucrum cessans*, admitted by name in everything but Latin, under the strictest formulation of the probability condition anywhere in this chapter: not probable gain, "certainly realisable" gain. The first sentence holds the boundary: everywhere except the debt. A builder's default may cost you a harvest, and the harvest may be billed; a borrower's default may not, because on the debt the only thing "lost" is money's time, and money's time is not a compensable good. The tradition's own standard-setters then split, as living traditions do, over how far actuality can stretch: Malaysia's regime lets an Islamic bank keep *ta'widh*

— compensation capped at its actual, demonstrated loss — while routing every further *gharamah* to charity; the AAOIFI standards refuse the bank's income even that, allowing only the charity-routed donation clause of chapter 22, with critics on each side of the gap. It is a genuine *ikhhtilaf*, argued in the tradition's own categories, and this book reports rather than referees it. But notice what nobody in the dispute defends: compensating a financier's forgone market return on a debt. On that, the committee room of 2000 and the schools of the classical centuries hold one line.

And yet the industry those jurists supervise runs, in bulk, on a number that comes from exactly the place the doctrine forbids. The markup on the murabaha that carries, by the industry's own much-quoted estimate, up to four-fifths of Islamic finance was priced for decades against LIBOR — the interbank interest rate itself — and, since the benchmark's demise in 2023, against its successors. Chapter 24 gave the defense its full hearing: Usmani's ruling that "merely using the interest rate as a benchmark" does not corrupt a sale that is genuinely a sale, and his own caveat in the next breath, that the institutions "should get rid of this practice as soon as possible." What this chapter adds is the name of the danger. A benchmark is a generic market return; a generic market return is the strongbox's claim, the exact object four centuries of conditions were built to exclude; and an industry built to honor the prohibition, pricing its permitted seam off the forbidden meter, has imported through the back office the one number its doctrine exists to refuse. Whether that is prudence, contradiction, or an honest transition still under way is the quarrel of chapter 32, and it is the tradition's own quarrel, conducted by its own scholars in its own vocabulary. Here it stands as the chapter's sharpest modern exhibit: the benchmark problem is *lucrum cessans* generalized — the market's elsewhere, priced into every contract, unexamined — arriving inside the very legal system best armed to recognize it.

The Horse and the Gate

Everything the three traditions concluded can now be said in two sentences. A productive alternative — particular, licit, intended, interrupted — can ground a real claim, cautiously priced. A generic market return can ground nothing, because it is not a fact about the lender; it is a fact about the market, and the market lost nothing. The whole moral geometry lies between those sentences, and the modern mind, trained on the second sentence's logic as if it were the first's, has mostly lost the ability to feel the difference. So state it operationally. The market return is what anyone could have earned, which is why it is proof that no one in particular lost it. A claim of foregone profit is exactly as strong as its particularity: the moment it can be made by any holder of money about any loan, it has stopped describing an injury and started describing a wish.

This is why "I could have earned more elsewhere" — our second lender's sentence, and every term sheet's — is not enough, and the traditions' conditions, laid side by side, are simply five questions the sentence cannot answer. Earned it where — in a named venture, or in the whole weather of the market? Earned it how — was the alternative licit, or is the claimant billing his neighbor for a forgone extraction, the circularity Hostiensis closed in the thirteenth century? Was it intended — standing in the claimant's actual practice before the loan, or discovered afterward, when grievance went looking for arithmetic? Did this loan cause its loss — or would the gain have arrived anyway, or failed anyway? And what was it worth — the brochure's number, or the sober figure that survives Aquinas's discount, net of the costs, the labor, and the risk the real venture would have charged for it? A claim that passes all five is the wool merchant's, and every tradition in this book would hear it. A claim that fails them is the strongbox's, and the sentence that carries it is not evidence

of a loss; it is a description of money, spoken in the tone of an invoice. The last question belongs to the borrower, and it is the oldest in this book: who is being asked to pay? Where the answer is a person in need, the traditions' verdict does not soften with the rate — the lender's imagined world is being billed to the one party in the room who never lived in it.

Now watch the horse come through the gate, because the image is exact. A Trojan horse is not a battering ram; it enters as a gift, welcomed, because it looks like something the city already honors. *Lucrum cessans* entered the usury doctrine as compensation — restitution's cousin, justice for wool merchants, a tort with conditions. And inside it, folded small, sat the entire future logic of capital: if a forgone gain can be a loss, then money has a normal yield it is always forgoing; if money has a normal yield, every loan deserves it; if every loan deserves it, the yield is not a claim to be proven but a baseline to be assumed — and whoever asks capital to serve without it is imposing a cost, which must be measured, priced, and recovered. Run that chain to its end and the burden of proof has silently changed hands. The schoolmen made the lender prove his elsewhere; the modern order makes the borrower — the enterprise, the mission, the patient project — prove why capital should ever accept less than its elsewhere. Böhm-Bawerk built interest's modern foundation on the premium of present over future goods, answering William of Auxerre rather than refuting him, as chapter 7 showed; Wieser and the marginalists generalized the alternative into the very definition of cost; and by the time the phrase opportunity cost had a journal article to its name, in 1894, the exception had finished becoming the axiom. Nothing in the machinery is arithmetically wrong. The discount rate, the hurdle rate, the cost of capital — each is the strongbox's sentence, compounded and institutionalized: true as mathematics, and morally identical to the claim the traditions spent six centuries refusing, that the world owes money its imaginary career. The Fifth Lateran definition chapter 14 quoted named usury as gain sought "without any work, any expense or any risk"; the generalized opportunity-cost claim is precisely such gain, sought not by argument but by definition — the wall not breached but declared, from inside the new axioms, never to have made sense at all.

The traditions' reply is not to burn the horse. It is the reply they gave at every gate in this book: open it, and price what is actually inside. Forgone gain is sometimes real; therefore model it — conservatively, which this chapter's sources specify with surprising completeness. Compensate after the fact, on proof, never as an upfront deduction — Lugo's third condition, because a lost future paid at signing is a guaranteed present, which is interest. Compensate the particular, never the index: a named alternative, in the claimant's real practice, causally severed by this financing — Molina's conditions, Hostiensis's restriction, the Yerushalmi's evidentiary instinct. Price hope as hope: discount the claimed gain for every contingency the real venture would have faced — the sower holds his harvest only in virtue, and the compensation runs "according to the condition of persons and things," net of costs, labor, and risk, never gross. Cap it at the loss: the title restores; it must never out-earn the venture it mourns. Demand a licit elsewhere: a forgone extraction is not a compensable injury. Refuse the double count: the same deprivation may not be billed once as lost profit, again as delay, again as liquidity — the files of chapters 19, 21, and 22 name distinct wounds or they name nothing. And refuse the universal: a clause that appears in every contract has stopped compensating anybody for anything. A claim built to those rules is rare, small, late, and provable — which is to say, it looks nothing like a return on capital, and everything like what the tradition said it was: damages. Chapter 12 noted the irony that the one modern claimant who can meet this standard to the dollar — the founder with the declined offer letter in her files — is the one for whom the instruments make no line; the arithmetic of honoring her, and of refusing the strongbox politely, be-

longs to Part IX, where chapter 35 turns these conditions into working models and chapter 40 builds the return formula that keeps them honest.

So the answer promised in chapter 7 can be given whole. Can a forgone, time-bound opportunity be a real loss? Yes — when it was a particular man’s particular venture, lawful in itself, already in train, killed by the loan, and mourned at its honest, discounted, after-the-fact price. That loss the traditions saw, named, and — each in its own grammar — paid: the schoolmen by a tort with conditions, the rabbis by writing the lender into the venture, the jurists of Islam by admitting the “certainly realisable” harvest everywhere except the one contract where money pretends to be a field. What none of them would pay is the other claim — the elsewhere of anyone, owed by everyone, forever — and their refusal is the finding this chapter has met at every turn: the profit that was not made is not a possession, and billing the world for it does not become justice by being called compensation. The deprivation of liquidity — the burden of simply being without one’s money, the subtlest of the family’s claims — is the file of chapter 21; the boundary between honest uncertainty and manufactured blindness closes Part VI in chapter 23.

Chapter 21 — *Carentia Pecuniae*: The Deprivation of Liquidity

For twenty chapters, “without money” has meant the borrower. It was the borrower’s empty hand the covenant codes shielded, the borrower’s cloak that had to go home by sunset (chapter 4 carried it there), the borrower who stands in the proverb chapter 1 set at the front door of this whole subject: “the borrower is the slave of the lender” (Proverbs 22:7, NRSV). In the moral grammar the scriptures built, the lack of money is a condition to be relieved; it is the reason lending exists.

This chapter is about the moment the lack of money changed sides. Late in the long Catholic argument over usury, in the generation when moral theology kept office hours within walking distance of the great exchanges, a Latin phrase began doing new work: *carentia pecuniae* — the lack, absence, or deprivation of money. Not the borrower’s lack. The lender’s. For the term of a loan, the argument ran, the lender is himself a man without his money. He cannot spend it, place it, or answer an opportunity with it; he endures, for a season, the peculiar poverty of the fully solvent; and that deprivation — not any loss he can document, not any venture he can name, just the bare state of being without — is a real burden, and a real burden is worth a price.

Part VI has been reading the lender’s honest claims one file at a time: actual loss is chapter 19’s file, the profit that was not made is chapter 20’s. This is the third file, and the strangest, because its claim is the thinnest and the widest at once. Thin, because it asserts no event — nothing burned, nothing missed, nothing broken. Wide, because if it is good once it is good always: every loan, by definition, leaves the lender without his money for a term. A title that rides on the definition of lending is no exception to the usury prohibition but its photographic negative, and the question of this chapter is whether the tradition could admit the claim without unsaying everything it had said since the Psalms. The question was fought out by name, in print, by some of the most commercially literate minds Christian theology ever produced, and this chapter will watch the fight.

One thing should be said plainly before the lawyers take over. Across the table from every lender who was ever “without his money” sits a person without money in the older sense — that is why he came. A doctrine of compensated deprivation would pay the solvent man for his temporary lack out of the pocket of the man whose lack is the emergency. The traditions never forgot which of those two deprivations their scriptures were about, and neither will this chapter.

Ready Money

Begin with what the claim gets right, because it gets something right. Money in hand is more than the goods it could buy; it is the power to act before anyone knows what acting will require. The merchant with cash meets the fire, the bargain, the tax collector, and the funeral with the same instrument, and meets each on the day it arrives. The merchant whose capital is out on loan meets them with apologies. The old classroom examples of wine and wheat never quite capture this, because a cask drunk is simply gone, while money lent is only elsewhere — still yours in law, owed back whole, and useless to you at exactly the moments when money matters most. Sixteenth-century commerce knew

this in its bones. On the exchange at Antwerp and at the fairs of Castile, readiness had a market of its own: the man who could pay today was in a different business from the man who could pay at Easter, and the difference between them was priced daily in the terms of bills and deposits. Modern finance has names for what those merchants saw — liquidity preference, the liquidity premium, the term premium — and a yield curve to draw them with; chapter 7 met the theory's respectable ancestor in Böhm-Bawerk's *agio*, the premium present goods command over future goods of like kind. None of this is fantasy. To be without one's money for a term is a real constraint, and the longer the term, the realer it gets.

The question is what kind of reality it is, and the neighboring files of this Part supply the measuring sticks. A lender who, because his money was out, had to borrow at cost to meet an obligation of his own can show a wound: that is actual loss, chapter 19's business, compensable because demonstrated. A lender whose money was already committed to a particular venture, and who let the venture go in order to lend, can show an interruption: that is chapter 20's business, and the tradition hedged it with demands for concreteness and causation precisely because it reaches into the future. *Carentia pecuniae* claims neither. It names no cost incurred and no venture interrupted; it says only, I was without my money, and being without money is as such a disadvantage. Strip away the Latin and the three claims sit in plain view: this loan cost me something; this loan cost me that; this loan was a loan.

And hold the mirror steady while the distinction settles. The borrower's side of the table knows everything the merchant knows about readiness, without the vocabulary. A household one repair away from ruin is living the economics of liquidity, not reading about them; chapter 15 drew the line between the temporary liquidity shock and the permanent ruin it becomes when no one will lend. The very fact that makes ready money precious to its owner — that life's demands arrive unscheduled — is the fact that brought the borrower to the door. Any doctrine of compensated deprivation must answer for that symmetry. The lender's *carentia* ends at the due date; the borrower's lasted until the loan.

So the question Part VI must put to this title is exact: is the deprivation of liquidity a burden standing outside the loan, which justice may weigh the way it weighs loss and risk — or is it the loan itself, wearing its own definition as a grievance? The tradition did not duck. It put the title on trial.

The Trial of a Title

The shelf of honest claims was already crowded. The classic extrinsic titles had earned their places by pointing at something genuinely outside the *mutuum*: a documented loss; a concrete forgone gain; the genuinely imperiled principal that chapter 10 called *periculum sortis*; the penalty for culpable delay that chapter 22 will put on trial next. Around them stood the working claims institutions live by — *stipendium laboris*, the wage of real administrative labor, and the recovered expenses that chapter 17 watched the Monti defend all the way to the Fifth Lateran Council. *Carentia pecuniae* was proposed as one more shelf-mate, and the proposal did not come from nowhere. It came up from the exchanges, where the lack of money visibly had a price among merchants long before it had a theology.

The first great commercial moralist to rule on it said no. Luis de Molina (1535–1600), the Jesuit whose *De iustitia et iure* brought scholastic doctrine as close to working markets as it had ever stood, rejected the title on grounds that cut to the bone of the contract: the obligation of not reclaiming one's money for the term is not a separable service the lender renders

on top of the loan; it is inherent in the *mutuum* itself. Granting time is what lending is. To charge for the granting is to charge for the loan — and beyond the analysis Molina saw the arithmetic, warning (as the historian Fabio Monsalve summarizes him) that the concession would be “extremely dangerous in opening the doors to usury.” A claim available to every lender on every loan amounts to a general repeal, however modest each invocation looks.

Then came Leonardus Lessius (1554–1623), the Jesuit of Louvain whom the merchants of Antwerp consulted on the morality of their dealings, and who went to the exchange himself to question the most experienced of them. In the usury chapter of his *De iustitia et iure* (1605), Lessius did what no scholastic of his rank had done: he formulated the title and accepted it. When a lender binds himself for a term not to call his money home, that self-binding, he argued, is a real obligation distinct from the loan — in Monsalve’s rendering, “an obligation that can be separated from the loan contract itself and is worth a price or compensation.” And the price was not to be invented by theologians; it was already there, in the common estimation of the merchants, who paid daily to have money ready and charged daily to go without it. “Regarding interest as the market price for money,” writes Wim Decock, the episode’s modern historian, “Lessius took the first step to integrate the doctrine of interest-taking and money-lending within the doctrine of just pricing.” The scholarship’s own name for the episode is the title of Toon Van Houdt’s study of it — “Lack of money” — and its verdict on the stakes has been steady since John T. Noonan’s *Scholastic Analysis of Usury* (1957): Lessius is credited, in Decock’s words, with formulating and accepting the title “in an unprecedented way... thereby undermining traditional scholastic teaching on interest and usury.”

Be fair to the man, because he was no smuggler. Lessius was describing an economy that existed. Antwerp’s money market was real; the burden of parting with liquidity was real; and merchants who would never dream of gouging a poor neighbor were pricing one another’s patience on the exchange every morning. He also saw the difficulty in his own position, and said so: on a bourse as busy and as international as Antwerp’s, it had become impossible to know any given lender’s situation and motives — and the whole diagnostic method of the older doctrine ran through exactly that knowledge. Every earlier title was tested lender by lender, burden by burden: show the loss, name the venture, prove the peril. A title tested only against the market’s going rate cannot be examined case by case; it can only be blessed wholesale. Lessius answered that the just price of money’s lack, like the just price of wheat, could be left to common estimation. His critics answered that wheat is not the thing the usury prohibition exists to protect.

The greatest of those critics arrived a generation later, and he cannot be accused of rigorism. Juan de Lugo (1583–1660), the Madrid-born Jesuit who taught at the Roman College and whose *De iustitia et iure* (1642) so impressed Urban VIII that its author was made a cardinal the following year, was the most permissive systematizer the doctrine of the titles ever had. It was Lugo who gathered the lender’s honest claims into a single architecture: “There are three chief titles, which can cleanse a *mutuum* from the stain of usury: emergent loss, risk, and cessant gain; and these three can be comprehended under one; emergent loss” — so runs Monsalve’s translation. Read the roster twice and notice what is missing. The theologian willing to comprehend forgone profit and imperiled principal as losses of a sort refused to read the bare lack of money as one of them. Admit *carentia pecuniae* broadly, Lugo held, and “no contract could further be considered as usurious.” The title would not have joined the doctrine so much as replaced it. A century later, Benedict XIV’s *Vix pervenit* (1745) restated the boundary the whole quarrel had been testing — gain from the

loan itself remains usury, whatever real titles may stand outside the loan — and chapter 14 has already shown that sentence doing its work.

So: Catholic title, or late-Scholastic pressure valve? On the books, the answer is clean. *Carentia* never joined the four; it stands in the tradition's lists as an adjacent claim — formulated late, accepted by one great commercial moralist, and rejected on either side of him in time by the two others best equipped to love it. In function, the answer is already in the question. It was a pressure valve: the point where the doctrine's old seal against pricing time met the rising pressure of a commercial world that priced time daily — and hissed. The valve mattered; through it the scholastic system vented enough commercial reality to stay credible in Antwerp. But the tradition's own verdict was the sober one. Hold the valve open and the tank drains: everything the prohibition guarded flows out through "the lack of money," renamed. Within a century and a half of Lugo, the states of Europe had begun licensing openly what his generation still debated in Latin — chapter 30 will tell that story — and *carentia pecuniae* settled into the role it has played ever since, the ghost grammar underneath the phrase "time value of money." None of this was the tradition failing; it was the tradition working: its own casuists testing a claim against the rule, in public, with reasons, and mostly holding the line at the exact spot where the line was hardest to hold.

Duration Without a Price

The halakha did not meet this claim late, and did not meet it as an open question. It met it at the beginning, and named it. The Talmud's shorthand for forbidden interest, chapter 7 taught, is *agar natar* — payment for waiting; any reward for waiting is forbidden, runs the rule of Bava Metzia 63b. What the seventeenth century debated as a possible exception, the rabbis had installed as the definition of the offense. A lender's compensation for being without his money is not near the line in Jewish law; it is the line.

The tradition even ran the experiment. A Babylonian sage, Rav Chama, rented out coins the way other men rent tools — one peshita, an eighth of a dinar, per dinar per day — and defended the business with a jurist's question: how is this different from renting out a hoe? Chapter 25 told the answer as part of the law of leases. The hoe comes home the same hoe, its wear known and chargeable to its owner; dinars are given to be spent, and what comes home is other dinars — so the daily fee was never rent for a surviving thing but a price for the term of a loan. The report closes with the Talmud's flat epilogue: all of Rav Chama's money was lost. It is as close as any tradition's texts come to a pure *carentia* charge — no loss shown, no venture named, a posted daily price for the bare absence of money — and the law refused it while it was still one sage's experiment.

Duration itself the halakha handles with structure rather than price, because a loan simply is a duration. An unspecified loan stands for thirty days — chapter 7 carried the default term from Makkot 3b — which is the law building time into the loan's anatomy as part of what the lender gives. What the lender may never do is bill for what he gave. He may not collect the time's worth as damages either — chapter 22 will show that refusal standing firm even where the borrower is at fault — and he may not harvest the waiting in kind: no lodging below the going rate in the borrower's courtyard (the mishnah chapter 25 carried), no favors, no first place in line, not even the flattering words chapter 8 watched the law confiscate as *ribbit devarim*. The wall runs on both sides of the due date, and it runs through every currency a benefit can be paid in.

Where duration lawfully earns, it earns as something other than duration. In the *iska* — half loan, half deposit, the instrument chapter 11 anatomized — the deposit half rides the

venture as capital genuinely at risk, and when the turnover is slow, the investor's eventual share is larger because the venture ran longer and risked longer, not because a clock accrued against a debtor. Time enters as venture time. The working partner's own months are paid as *sechar tircha*, a wage for labor, under chapter 12's rule that the man must never work for nothing; and the *heter iska* (chapter 14; chapter 31 keeps the hard questions) carries the same insistence to industrial scale — the financier recast as investor, the settlement machinery running for exactly as long as the capital is outstanding, so that what looks from outside like interest-for-time is drafted, and must be defended, as profit-share-for-venture. Even the famous asymmetry chapter 24 met — rent may rise for deferred payment, but a sale price may not — draws the same boundary through time itself: a house endures, so its use through time can be honestly sold; coins are consumed, so theirs cannot. That is chapter 25's consumptibility doctrine standing guard at the border of the calendar.

Here, then, is the Jewish answer to the lender's carentia, and it has the economy of a proof. Duration is real, so the law gives it structure: a default term, a venture whose profits may honestly reflect its length, a wage for the months a manager works. Benefit is real, so the law confiscates it wherever it leaks toward the lender. What survives is a rule the seventeenth-century casuists would have recognized as the position Molina defended and Lessius could not hold: duration may deepen a stake; it may not draw a wage.

Two Kinds of Lockup

Islam's first answer to the deprivation of liquidity is to canonize it. The *qard hasan* — the beautiful loan whose anatomy chapter 16 owns — is capital lockup embraced on purpose: the lender cuts a piece from his wealth (the jurists, chapter 16 heard, say the word *qard* comes from cutting) and hands it away for a term, expecting back the like amount and nothing else. The deprivation is not a cost the instrument regrets; it is the substance of the gift, and the tradition's whole reward apparatus — chapter 16 traced it — consists in relocating the return to a ledger no bank keeps. The law then walls the lockup against every earthly leak: the juristic maxim chapter 6 was careful to grade as a maxim rather than a hadith — every loan that draws a benefit is *riba* — and the modern standards chapter 16 reported, which bar stipulated excess in any form, at the contract or at its extension. That last clause deserves a slow reading in this chapter, because the extension of a loan is precisely a purchase of further deprivation. The fiqh will let the lender suffer longer. It will not let him sell the suffering.

The saver's version of the question arrives at the other counter of the same bank, and it is carentia pecuniae in modern dress: I am willing to be without my money for a year — what is that worth? A conventional bank answers with a rate stepped up by tenor, which is the Lessius position operating as furniture. The fiqh's standard instruments answer with the *mudarabah* investment account, and the difference is the whole argument of this Part run through a deposit slip. The depositors' money enters a common pool as investment capital, the bank standing to it as *mudarib*, working partner, the depositors as *rabb al-mal*, owners of the capital — chapter 11's contract, wearing a teller's window; chapter 27 met its cousins in the takaful operator's fee. Duration is then recognized not through a promised rate but through weightages: multipliers assigned to each class of account and published in advance, by which a year's money takes a larger share than a week's money of whatever profit the pool actually earns. One much-studied early case, an Islamic bank in Bangladesh, assigned full weight to its thirty-six-month deposits and scaled the rest below them, and stated the rationale in a sentence that repays attention: "The longer the period of deposit, the greater the risk they bear with regard to fluctuation of the rates of profit." A typical

published schedule — this one posted by a Pakistani Islamic bank in the summer of 2026, declared before the month began — ran from just under 0.5 for seven-day money to 1.2 for the largest year-long deposits, the pool's net income split half-and-half between bank and depositors and then divided among the depositors by weight.

Read the mechanism against the doctrine and every joint lines up. The long depositor is not paid for waiting; he is paid a thicker slice of what happened, because for twelve months his capital stood in the way of whatever happened — including loss. The no-guarantee rule chapters 10 and 13 established holds the floor: the *mudarib* may not guarantee the capital, and in a bad year the pool's loss lands on the depositors "in the ratios of investment," as one bank's published mechanism puts it — the weighted account sharing the bad year as faithfully as the good. Lockup buys weight in real outcomes; it does not buy a rate against the calendar. That is the *fiqh*'s *carentia* ruling, rendered as retail banking. And the tradition polices its own counter: chapter 32 will give the floor to the industry's internal critics, who ask how much of this survives competitive deposit markets — how often the weighted share is smoothed until it tracks a benchmark and the stake behaves like the rate it replaced. The instrument's honesty, as always in this book, is a property of its practice rather than its drafting; chapter 26 watched the same drift overtake certificates that promised their money back at par.

Two kinds of lockup, then, and one refusal. At the mercy layer, deprivation is the gift itself, and pricing it would unmake the instrument. At the investment layer, deprivation buys standing in a venture's real fortunes, loss included. What neither layer will sell — what no layer of this tradition will sell — is the term as such.

Weighing Patience

Now the modern files, and first the honest kernel inside the old title, because there is one. Time really does compound something: exposure. Capital that can be lost is likelier to meet its loss in five years than in one; a longer commitment holds the investor's money in harm's way through more seasons, more markets, more chances for the world to turn over. So a longer term can justify more compensation — but derivatively, through the risk title, and only for capital that is genuinely at risk. This is duration-weighted capital at risk: the months multiply the peril, and the peril, not the months, is what earns. Chapter 10's conservation grammar decides the hard cases: guarantee the principal, collateralize it, make it senior and enforceable regardless of outcome, and duration weights nothing, because zero exposure multiplied by any number of months is zero. The Islamic weightage table prices duration exactly this way — the case study said it out loud: the long depositor bears more fluctuation risk — and the *iska*'s slow venture does the same. When Part IX builds the arithmetic, this is chapter 36's file, capital at risk over time; the doctrine it will formalize is already visible here in a Bangladeshi deposit schedule and a Babylonian partnership.

When is liquidity deprivation real, then — real enough to weigh? The tests assemble themselves from the Part's own files, and none of them is new. It is real when it is demonstrated: the lender who had to borrow at cost to cover his own obligation holds a receipt, and chapter 19's discipline pays receipts. Specificity is the second test — a committed use interrupted, under chapter 20's rules of concreteness and causation, not a mood of missed possibility. Structure is the third: capital bound into a venture's fate for the term, where the deprivation rides as exposure and is paid as participation. And whatever passes must be proportionate and counted once — not collected as a liquidity premium, again as a risk premium, and a third time as a fee, the stacking every tradition in this Part condemns.

Genuine cases exist and deserve saying plainly. A fund that forgoes recallability so that a workshop can carry inventory through the winter has given the workshop something real, and what it gave will show up honestly — as documented cost if covering it cost anything, as deepened exposure in the risk accounting, as a longer wait for its share of surplus. Every one of those is a title the tradition already knows how to weigh. Notice what is not on the list: a charge for the calendar itself.

And when does liquidity deprivation become ordinary interest? It announces itself, once you know the three signatures. The first is universality — the burden claimed is one every lender bears on every loan, the definition of lending resubmitted as a grievance. A burden everyone can claim is not a title; it is a rate. The second is clock-linearity — the charge is computed from elapsed time and nothing else, so much per month outstanding, indifferent to any event that would let a court or a confessor test it. The third is outcome-indifference — the claim collects in the venture's good years and its bad ones alike, senior to wages and reserves, payable by the failing as punctually as by the flourishing. Where all three run together, no removal of demonstrable burdens shrinks the charge, because the charge never rested on them: subtract the shown cost, the named venture, the possible loss, and the "liquidity premium" stands unmoved, which identifies it as payment for the use of money over time — the thing whose definition occupied the first chapters of this book. The steelman deserves its due: some commitment charges pay for something real, a standby facility's reserved capital and standing administration among them. The discipline is not to deny such costs but to name them under their true titles — labor, expense, exposure — where they can be sized, tested, and audited. Whatever residue insists on being priced per month for nothing per month has told you its name.

Which leaves the last design question: modeling patient capital without extracting from need. The grammar is short, because the whole chapter has been writing it. Pay patience in weight, not rate: longer-committed capital takes a larger share of realized surplus — the weightage logic, the *iska* logic — so that patience deepens participation instead of accruing against the borrower's calendar. Let duration size the risk compensation of genuinely exposed capital, chapter 36's arithmetic. Recover the demonstrable costs of illiquidity as costs — documented, proportionate, once. Then make patience survivable instead of expensive: chapter 28 showed that a lender's own path through time is where ruin lives, and a fund with an anti-ruin reserve — a solidarity reserve sized to its worst plausible stretch — can afford to wait, where a fund that waiting can kill will charge for waiting as surely as fear compounds. (That is a time-average problem, in chapter 28's exact sense, and its mathematics stay there.) Finally, the ledger line itself: when Part IX assembles the non-extractive return formula, chapter 40 will carry an entry called liquidity deprivation, sitting among the titles a fund may honestly claim. This chapter's discipline travels with the line: it must be capable of reading zero, it must read zero wherever no test in this chapter puts a number in it, and on the mercy layer of the capital stack — the *gemach* and *qard hasan* floors of chapters 15 and 16 — it must read zero by design, because there the deprivation is the gift. Chapter 41's stack will hold funds to exactly that; chapter 7's old promise, that Part IX would ask what a defensible price of patience looks like, comes due there with this chapter's answer attached: patience has a price only where patience carried a burden something other than a clock can measure.

End where the phrase began, on the other side of the table. *Carentia pecuniae* was a lender's Latin for a season's inconvenience, and the seventeenth century's best minds disagreed honestly about whether the season could be sold. But the people the phrase describes without any Latin at all are the ones this book started with: the household without

money for the repair, the farmer without money for the seed, the debtor of chapter 22's coming pages who has sold what there was to sell. Their deprivation of liquidity is not a premium forgone; it is the emergency the whole architecture of non-extractive credit exists to answer. The scholastics put the solvent man's lack on trial and, at their best, refused to let it become a wage — not because liquidity is imaginary, but because the man across the table had less of it, and a doctrine built to shield his lack could not end by billing him for the lender's. The lender's deprivation ends on the due date, and this chapter has shown where it may honestly be weighed. The borrower's deprivation is why the doctrine exists, and it is never for sale. What happens when the due date passes and the money does not come — the debtor late, the penalty clause armed, discipline and temptation in the same drawer — is the business of chapter 22.

Chapter 22 — Poena Conventionalis: Penalty, Delay, and Default

The oldest late-payment law in this book is not about a loan at all but about a day's wages. "You shall not keep for yourself the wages of a laborer until morning," Leviticus commands (19:13; NRSV, as throughout this book), and Deuteronomy repeats the deadline with the reason attached: pay the hired hand before sunset, "because they are poor and their livelihood depends on them," lest the unpaid worker cry out against you to God — verses chapter 4 set among the Torah's shields for the vulnerable (24:14–15). Islam kept the rule and gave it an image no employer forgets: give the worker his wages, the Prophet said, before his sweat dries (Ibn Majah 2443). Notice who is late in these texts. Not a borrower — an employer. Scripture's paradigm delinquent is not the poor man who cannot pay his creditor but the comfortable man who has not paid his worker, and the first thing the traditions know about lateness is that it is a species of withholding, and that withholding what is owed is something the strong do to the weak at least as often as the reverse.

But a debt binds in both directions, and about the borrower's side of the calendar the traditions are just as blunt. "The wicked borrow, and do not pay back," says the thirty-seventh Psalm. The Talmud counts repaying a creditor as a *mitzvah* — a commandment, on the principle the passage attributes to Rav Pappa (Ketubot 86a) — and the same page tries out, deadpan, the debtor who announces that this particular commandment does not suit him; the Gemara's answer reaches for a *baraita* about a man ordered to build a *sukkah* and declining, which is to say that repayment belongs among the commandments a court enforces, not the ones left to conscience. The Prophet's ruling on the same character has the compression of a proverb: procrastination by the rich man — *matl al-ghani* — is injustice, *zulm* (Bukhari 2287, which chapter 8 met). A companion hadith supplies the judicial teeth: the delay of one who has the means, *layy al-wajid*, makes his honor and his punishment lawful — and the transmission carries its own gloss from Ibn al-Mubarak: his honor, meaning he may be spoken to harshly; his punishment, meaning he may be imprisoned (Abu Dawud 3628). On this much, three legal civilizations speak as one. The solvent debtor who lets the date go by is not managing his cash flow; he is holding what has stopped being his.

Then there is the other late debtor. He has sold what there was to sell, and the money is not there. For him the Qur'an legislates in the sentence that follows its war verse, and the sentence is a command aimed at the creditor: "If the debtor is in difficulty, then delay things until matters become easier for him; still, if you were to write it off as an act of charity, that would be better for you, if only you knew" (2:280, in M. A. S. Abdel Haleem's rendering, as throughout — chapter 6 read the verse whole). The rabbis built the same mercy into the creditor's very manners: a lender who knows his borrower cannot pay may not so much as walk past the man, rules the Talmud, because the creditor's face is itself a demand — do not be to him as a creditor (Bava Metzia 75b, from chapter 8). One calendar, one missed date, two opposite moral facts: the first lateness is a wrong the debtor commits; the second is a hardship he suffers.

That is the moral problem of lateness, and it would be manageable if it involved only those two characters. It involves a third. Whatever machinery the law builds to discipline the willful delayer will also reach the desperate one, because from the creditor's window the

two can look identical. And whatever that machinery pays the creditor becomes, coin for coin, a reason to wish for delay. A due date needs teeth; the whole question is how to give it teeth without giving anyone an appetite. The medieval lawyers packed the question into the term this chapter carries as its title — *poena conventionalis*, the penalty agreed in advance — and their centuries of worrying at it produced the most portable moral instrument in this Part. So we begin with them.

A Penalty, But Not in Fraud

Rome, which charged interest without embarrassment, had already built both halves of the machinery. A promise could be armed in advance: beside the main obligation the parties could stipulate a *poena*, a fixed sum owed on non-performance — the *stipulatio poenae*, all-purpose stiffening for contracts of every description, no more scandalous to a Roman jurist than a door bolt. And delay had a doctrine of its own, *mora*. A debtor did not fall into *mora* merely because the sun had set on the date; the classical rule wanted culpable delay, ordinarily established by a demand, and it entertained excuses — bare lateness, the jurists held on Papinian's authority, does not make *mora* (D. 12.1.5). But once *mora* attached, the world tilted against the delayer: the risk of the thing owed moved onto him, and in the contracts governed by good faith — sale, hire, partnership — interest, *usurae*, ran on unpaid money from default forward, as damages for the withholding; the Digest's title on interest and fruits (22.1) is much occupied with the details. One contract stood outside this weather. The *mutuum* was construed so strictly that not even delay could make it yield, and a lender who wanted anything past his principal needed the separate promise chapter 7 described. Lay Rome's map on the table, then, and read the strange result: money delayed earned interest as damages nearly everywhere in the law of honest dealings — except inside the loan itself.

Latin Christendom inherited the map and added a conviction: the *mutuum* is not merely strict but barren by nature, on the anatomy Part III walked. Which raised the drafting question at once. May a loan carry a clause reading, if you do not repay by Michaelmas, you owe ten more? Formally the ten is no price of the loan. It is conditional; it answers a breach; the borrower holds the key to never owing it. The canonists answered yes — with a proviso they welded into the tag itself: *poena nec in fraudem*, a penalty, but not in fraud of the usury prohibition. The rhymed list by which the schools remembered the lawful ways past the wall — thirteen exceptions by the count the tradition ties to Hostiensis, the age's great canonist — opens on exactly those words. Everything lives in the proviso. A penalty clause was never lawful as a form; it was lawful as a truth, and the tradition spent the better part of four centuries specifying what had to be true.

Four conditions, as the confessors' manuals eventually settled them, and together they make a compact grammar. The penalty must be conditional — attached to a genuine breach, not to the mere passage of the term. It must be avoidable — the date realistic, the borrower in actual possession of a road to punctuality, because a date built to be missed converts the clause from consequence into plan. It must be proportionate — sized to the wrong and its burden, not calibrated to what the money would have earned at interest. And it must be unwanted. That last condition sounds like sentiment and is in fact the load-bearing wall. The Franciscan John Duns Scotus is remembered in this literature for pressing the question that outlives every drafting fashion: when repayment arrives on the day, is the lender glad? A true penalty is a deterrent, and a deterrent succeeds by never firing; a lender who profits when deterrence fails has acquired an investment in its failure, and a contract that pays its author to hope for breach has turned the moral world upside

down — the borrower's stumble has become the lender's payday.

Set beside interest, the honest penalty shows its species difference plainly. Interest is a rate: it binds to time, accrues on every loan in the portfolio, and is expected — it is the plan. A penalty is a verdict: it binds to a fault, fires once, on the failed case only, and — if honest — is hoped never to fire. A fine wants a culprit; a meter wants only a clock. The difficulty, which the canonists saw as early as anyone, is that parchment can dress the one as the other with very little effort. Shorten the date until missing it is a certainty, size the “fine” at what interest would have been, roll the defaulted loan into a fresh loan with a fresh penalty, and the verdict repeats itself into a rate. The test the tradition kept applying — does this clause punish a wrong, or price the term? — cannot be run on the words of the contract, because the words are identical in both cases. It runs on dates, sums, expectations, and the direction of the lender's hope.

The earliest Paris teachers already distrusted the pocket the penalty fell into. Robert of Courçon — an Englishman who taught theology at Paris and died in 1219 preaching the Fifth Crusade at Damietta — is remembered for a radical prophylactic: where a late-payment penalty is admitted at all, let the surplus go to the poor rather than into the creditor's purse. The clause keeps its deterrent face; the creditor loses his stake in its firing; the arithmetic of temptation is simply unplugged. In the thirteenth century the proposal reads like a footnote. File it carefully anyway, because this chapter will meet it again, built at industrial scale, in a legal system the Paris masters never read.

The Fine and the Meter

The halakha's version of the problem can be said in one sentence: the remedy that first suggests itself, a charge for the extra days, is the very thing the law forbids. Chapter 7 found the Talmud's shorthand for forbidden interest — *agar natar*, the wage of waiting (Bava Metzia 63b) — and a charge that grows with the borrower's delay is nothing else. The days after the due date are still days; a fee of so much per late week prices them as surely as any contract rate prices the weeks before; the wall against *ribbit* has no gate on its far side. The rabbis did not lack calendar discipline. A loan with no stated term falls due, by default, in thirty days (Makkot 3b), and repayment, as the opening said, is a commandment the court itself will enforce. What the law would not do, at any severity of delay, was let the lender meter it. A meter is a meter on either side of the due date.

What remains is the fine, and the halakha's handling of it is a study in exactness. A one-time fixed sum, triggered by the missed date and indifferent to everything after it, does not price duration; it punishes an event. The working doctrine carried in the practical *ribbit* literature runs along that line: a late charge phrased as a genuine *kenas* — a fine for the wrong of not paying on time, not a payment for further use of the money — can pass, in the debts where the interest prohibition stands at rabbinic strength: the unpaid rent, the open invoice, the price of goods and services sold on terms. Let the fee recur, though — a fresh charge for every thirty days outstanding — and the mask slips: a fine assessed per unit of time is a purchased extension, whatever the drafting calls it, and the arrangement is back inside *ribbit* and reaching for help. On a true money-loan the scruple tightens further still, and there the practice routes the whole question through the *heter iska*. The line the poskim hold is precisely the canonists' line, drawn with the halakha's own instruments: the event may be punished; the duration may never be sold.

The *heter iska*, chapter 14 showed, is what carries the modern load. Under the *iska* form the financier is not a creditor awaiting his principal but an investor whose capital is still out

working, and the settlement machinery — the oath no one swears, the fee paid in its stead — simply continues for as long as the capital remains outstanding. Delay does not trigger a penalty; it extends a partnership. Whether that answer answers — whether anything real still crosses the channel the drafting keeps open — is the standing question of chapter 31, and it presses hardest exactly here, where the continuing payments most resemble the meter the law set out to forbid.

And beneath the whole apparatus, held apart from it with deliberate care, stands the debtor who cannot pay. The Torah's collection law reaches him before any fine does: the creditor stands outside the house while the pledge is brought out; the poor man's cloak goes back at sundown; the millstone and the widow's garment may not be taken at all — chapter 4 read the list, each object a life. The Talmudic law of execution added a floor the codes preserve under the name *mesadrin*, the court's arranging for the debtor. When the court collects a judgment debt, Maimonides rules, the debtor is told to bring every movable thing he owns, down to the last needle — and then the court hands back: food for thirty days, clothing for twelve months, his bed and bedding, his sandals, his tefillin; if he is a craftsman, two of every tool of his trade, the carpenter keeping his two awls and his two planes (Mishneh Torah, Creditor and Debtor 1:7). The farmer's ox team and the sailor's boat are property, and they are sold; the hands' capacity to work is not. Collection may empty the estate, but it may not unmake the person. Add the manners rule already met — the creditor forbidden even to show himself to the borrower he knows to be empty — and the halakha's package stands complete: the solvent delayer is compelled by the court; the delay itself is never sold to the lender at any price; and the broke man keeps his tools, his clothes, and his month of bread. There is discipline everywhere in this system, and there is no revenue in distress anywhere in it.

The Procrastinator and the Poor Box

The fiqh begins from the two hadith the opening quoted, and its classical toolkit follows from them with a directness modern readers find startling. The procrastinating solvent debtor is an offender — *matl al-ghani* is *zulm* — and the offense lifts his ordinary protections: his honor, so that the complaint against him may be made loudly and in public; his punishment, so that the judge may jail him until he pays. The tradition armed the creditor with the complaint, the courtroom, and in the stubborn case the cell — everything, that is, except a charge. Not a dirham may flow to the creditor because the debt ran late.

The reason sits at the tradition's foundation. Increase upon delay is not a marginal case of *riba*; it is the original one. The *riba* of the age of ignorance, chapters 2 and 6 recounted, was precisely the creditor's question at maturity — will you pay, or will you increase? — the debt rolling forward and swelling with each forbearance until it consumed the debtor whole. A default-interest clause is that question written down in advance and signed. When the Qur'an declared its war, the enemy actually in the field was not some modest origination fee; it was compounding default interest, the meter that starts when the debtor stumbles. The fiqh's refusal here is less an application of the doctrine than the doctrine itself, standing on the ground where it was born.

For the debtor in genuine difficulty, the verse of 2:280 governs, and the tradition institutionalized its mercy rather than leaving it to temperament. The insolvent is entitled to time, not merely eligible for it. Chapter 6 told of the merchant whose standing instruction to his agents — pass over the man in hardship — earned him, the Prophet said, God's own passing-over (Bukhari 2078). And when the community's alms system arrives, debtors

stand among its eight chartered classes of recipient, the public fund paying down strangers' debts as budget rather than favor. Insolvency, in this architecture, is not a pricing problem but a claim on the community.

Then the twentieth century handed the tradition a version of the problem its classics had never faced at such scale. An Islamic bank's book is heavy with fixed-price receivables — above all the *murabaha* sale, in which the institution buys goods and resells them to the customer at a disclosed markup payable over time. The price is fixed at signing; it is the whole point that it never grows. But the arithmetic has a consequence: a debtor who delays a conventional bank meets a meter, while a debtor who delays an Islamic bank meets, on the classical rules, no charge at all. The arithmetic was noticed, as arithmetic about money always is, and institutions serving millions of customers found themselves running, in effect, a standing referendum on *matl al-ghani*. The industry's standard-setters had to answer the old question under new load: how do you discipline the procrastinator without selling him time?

Their answer reinvented Robert of Courçon. The AAOIFI standards, chapter 8 reported, permit the financing contract to carry the debtor's own undertaking: if he procrastinates, he will pay a stated amount — to charity. The clause bites the willful delayer exactly as a late fee would; the money moves, under the Shariah board's supervision, to charitable causes; and none of it may enter the institution's income. Insolvency never triggers it — the undertaking reaches procrastination only, never inability. Malaysia's central bank runs a two-tier refinement of the same anatomy, in force since the start of 2012: *ta'widh*, compensation capped at the institution's actual loss from the default, which may be recognized as income; and *gharamah*, any penalty above that, which may not be — it is channeled to charity under the oversight of the institution's Shariah Committee. Read the two tiers slowly, because they are this entire chapter compressed into regulatory prose: repair may be kept; punishment must be given away.

The tradition's own jurists argue about the residue, as living traditions do. Critics — chapter 32 will give them the floor — ask whether a “donation” that arrives with the predictability and sizing of a late fee is a fee in pious dress; whether *ta'widh*, benchmarked carelessly, reopens the door the schools nailed shut. Defenders answer from the incentive analysis this chapter has been assembling all along: destination is not a detail. An institution that cannot keep penalty money has no reason to want penalties fired at all; its only remaining stake in the clause is deterrence, which is the only stake a just clause ever had. And with that, the design seeded in thirteenth-century Paris — sever the discipline from the revenue — stands built at industrial scale, by a legal system that owed Paris nothing and had never heard the suggestion. A penalty no one may pocket is a penalty no one will farm. Two traditions reached that sentence independently, seven centuries apart, which is about as close to a law of moral engineering as history supplies.

What Delay May Honestly Cost

Between punishment and price sits a third thing: repair. Delay can injure. The creditor who was to be paid on the first of June may have obligations of his own timed to that money, and a debtor whose culpable lateness causes a real, showable loss has done more than annoy. The Catholic tradition met this squarely, and its clearest sentence is in the *Summa*. A lender, Aquinas allows, may agree with the borrower on compensation for a loss he incurs — this, he writes, is “not to sell the use of money but to avoid a loss” (ST II-II, q.78, a.2, ad 1, in the English Dominicans' rendering) — while in the same breath he

bars the lender from selling the profit he merely hoped for elsewhere, “that which he has not yet and may be prevented in many ways from having.” His disputed questions are read as extending the allowance to our case, recognizing the debtor’s wrongful delay as a source of compensable harm (De malo, q.13, a.4, ad 14). Out of this grew the vocabulary of *interesse* — in its oldest sense not a rate but a difference: the gap between where the creditor stands and where timely performance would have put him, owed because of fault, measured by proof. Rome’s *mora* doctrine supplied the working grammar — culpability, demand, excuse — and the boundaries write themselves from everything this Part has established: the loss must be actual, the fault genuine, and the same wound may not be billed twice under two names, once as penalty and again as damages — a double-counting the manuals condemned and modern fee schedules have quietly perfected. What counts as an actual loss, what the profit that was not made may claim, and what the bare deprivation of liquidity is worth are the files of chapters 19, 20, and 21; this chapter borrows only their trigger: fault, shown, after a date that could have been kept.

The halakha, offered the same channel, largely declined it — and the refusal is instructive. The Jerusalem Talmud preserves the rule in Rabbi Isaac’s inference: one who idles his fellow’s purse — takes money to trade with and lets it lie — leaves its owner “nothing but a grievance” against him (Yerushalmi Bava Metzia 5:3); the term is *tar’omet*, a resentment the court will hear about but not remedy. Leave a man’s field fallow, the same passage continues, and you owe him, for a field bears fruit by nature. Money’s increase is exactly what the law declines to presume; a creditor’s coins, delayed, have earned nothing in the law’s eyes, because coins earn nothing — chapter 7’s axiom holding its line even where it costs an honest claimant real compensation. The tradition preferred an uncompensated loss to a compensable fiction, having watched, the rabbis would say, where the fiction leads. Its working relief runs once more through the *iska*: recast the financier as an investor and the idle-money problem dissolves into the partnership’s ordinary accounting. The *fiqh*’s modern *ta’widh* is the same tension resolved one notch the other way — actual cost, and only actual cost, may cross — with the tradition’s own critics standing at that gate and fearing what the rabbis feared: that “actual” is a word benchmarks find ways to inflate.

Hold the three positions up together and the deep rule of this section appears. Repair looks backward at a wound; a price looks forward along a duration. The moment a “compensation” is computed from the calendar alone — so much per month outstanding, whatever happened — it has changed species in the hand that holds it. That is the fulcrum on which the modern question turns, and we can now turn with it.

Remedies That Prefer Repayment

A present-day loan agreement carries a default apparatus the medieval drafters would recognize at a glance, sharpened by better printing. The centerpiece is the default rate: upon an event of default, the interest margin steps up — a point or two above the contract rate is the standard drafting — and in most agreements the elevated interest compounds. Around it cluster the late fee, the acceleration clause that matures the whole debt at once, the cross-default provision by which one stumble trips every agreement the borrower has signed, and the recovery of enforcement costs. Each element has a stated rationale. The whole assembly asks to be read by exactly the test the canonists built, and it is worth watching the modern law read it first.

The common law built a penalty test of its own, and its history is a mirror worth a paragraph. Equity’s rule against penalties, England’s highest court has itself observed, “origi-

nates in the concern of the courts to prevent exploitation in an age when credit was scarce and borrowers were particularly vulnerable” — and the old reports read accordingly: in the 1690s, in cases like *Lady Holles v Wyse* and *Strode v Parker*, uplifts on defaulted debts were struck down as penal. In 1915, in the *Dunlop* case, Lord Dunedin gave the doctrine its long-serving formula — a clause survives as a genuine pre-estimate of loss and falls as a penalty — and among his working tests sits one the scholastics could have drafted: where the breach consists only in not paying a sum of money, a clause demanding a larger sum is presumptively penal. For most of three centuries, in other words, England’s secular courts shared the canonists’ suspicion of the priced default. Then the road forked. In 1996, in *Lordsvale Finance v Bank of Zambia*, a commercial judge upheld the now-universal default uplift — one percent, in that loan — on the reasoning that it is not a deterrent at all: a borrower in default is a worse credit risk, and the higher rate merely prices him, a term “commercially justifiable” so long as its dominant purpose is not to deter breach. The Supreme Court confirmed the framework in 2015 in *Cavendish Square Holding v Makdessi*: the true test of a penalty is now whether the clause imposes a detriment “out of all proportion to any legitimate interest of the innocent party in the enforcement of the primary obligation,” since “the innocent party can have no proper interest in simply punishing the defaulter.”

Stand the two regimes face to face and they are mirror images. The modern rule forbids punishment and blesses pricing: the default rate survives review precisely because it is not a penalty — because it compensates the lender for involuntarily extending credit to a riskier man, which is to say, because it is a price for time and risk. The old rule blessed punishment and forbade pricing: the poena survived scrutiny precisely because it was not compensation for time — because it answered a wrong, and the lender was required not to want it. Each system polices proportion with genuine vigilance; each permits what the other prohibits; and on the specific instrument of default interest their verdicts pass head-on in the corridor — the most defensible charge in the contract, says the one; the oldest form of the forbidden thing, say all three traditions in this book, remembering the jahiliyya creditor at maturity. This book does not arbitrate between legal systems. It only asks the reader to see plainly what each is afraid of. The common law fears the creditor as punisher; the traditions feared the creditor as beneficiary — the man whose income statement quietly acquires a line that does better in bad years.

What, then, does a lender build who accepts the traditions’ constraint — who wants a due date with teeth, and discipline without profit from distress? The materials of this chapter assemble into a working grammar, and none of it is utopian; every element already exists somewhere in current practice. The calendar comes first: a repayment schedule matched to the borrower’s actual rhythm of cash — harvests, invoices, seasons — because a date designed to be missed is not discipline but bait, and that abuse was cataloged eight hundred years ago. Triage comes before remedy: a hardship process that sorts the cannot-pay from the will-not-pay, on the rule of 2:280 and of AAOIFI’s drafters — remedies aim at procrastination, never at inability — with rescheduling on honest terms for the genuinely stuck. Repair runs at cost: documented collection and administration expenses, itemized so that no wound is billed twice, nothing computed from the mere calendar — the ta’widh tier, the Thomist allowance. Discipline, where the relationship wants it, takes the shape the halakha teaches: a fine — fixed, capped, once per event, never compounding, never accruing — and, on the Courçon-AAOIFI design, given away: routed to charity under real oversight, so that the institution holds no stake in its own borrowers’ stumbling. And the books are audited for appetite: if next year’s budget balances only because default charges are projected as revenue, the model has inverted, whatever each clause says. That is Scotus’s question, asked of a portfolio.

Two more elements close the design, and both are old. A road home: grace periods, cure rights, reinstatement — remedies ordered toward restoring the relationship, so that a stumble is an event in the loan's life rather than the end of the borrower's. And a floor under collection: the instinct of the mesadrin rule — the thirty days of food, the tools of the trade, the bed — which survives, recognizably, in every modern exemption statute that keeps a bankrupt's clothes and work tools out of the estate. A default remedy that cannot tell a debtor from his tefillin has stopped collecting a debt and begun collecting a person, and every tradition in this book wrote that boundary in its plainest prose.

Part X will build lending institutions on this grammar. One assumption, though, has run silently beneath the whole chapter: that the borrower understood the clause — that the default apparatus he signed was, to him, information rather than ambush. Where complexity itself defeats comprehension, the law of lateness shades into a different failure altogether, a failure of consent, and that is the business of chapter 23, "Gharar, Ona'ah, and Defective Consent." Here the chapter can rest with its finding. Across three traditions — and one secular cousin that keeps rediscovering their questions — the honest late-payment regime sorts three payments into three separate pockets: repair to the wounded, punishment to the poor box, and to the lender nothing but his own money, home at last. All three traditions were trying to write the same signature page: a contract whose best outcome, for every party at the table, is the boring one — the money back on the day, and the penalty clause dying unread, of old age.

Chapter 23 — Gharar, Ona'ah, and Defective Consent

One day — Muslim's collection preserves the scene — God's Messenger passed a heap of grain offered for sale in the market, pushed his hand down into the pile, and drew his fingers out damp. What is this, owner of the grain? Rain fell on it, Messenger of God. The answer that followed has governed the commercial conscience of a civilization ever since: why did you not put the wet grain on top of the heap, where people could see it? Whoever deceives is not of me (Muslim 102).

The scene rewards a slow look, because every detail in it carries law. The defect was real but shallow — a hand's depth from daylight. The seller told no lie; asked a direct question, he answered it truthfully. What he had done was arrange the pile so that the truth lay just beneath the reach of an ordinary buyer's attention. And the remedy the Prophet prescribed was neither a price control nor a punishment but a rearrangement of information: put what is wrong where eyes will find it. A market runs on consent; consent runs on sight; and a seller who stages his goods so that the other side cannot see what it is buying has struck at the market itself, whatever his mouth was doing at the time.

This chapter is about that staging — ancient and modern. A venture financing today arrives not as a heap of grain but as a stack of paper: the term sheet, then the stock purchase agreement it summons, the investors' rights agreement, the voting agreement, the amended and restated certificate of incorporation, side letters, schedules, exhibits — hundreds of cross-referenced pages, every clause of them, in the lawyers' word, disclosed. Nothing in the stack is hidden the way the wet grain was hidden, and nothing needs to be. A modern instrument can be arranged so that its truth requires an excavation the signer has neither the time, the training, nor the bargaining position to perform — and the arranging can be done clause by lawful clause, by drafters who never once misstate a fact. There is a violence in this that our ordinary vocabulary of market wrongs fails to name, because that vocabulary was built around the lie. No one lied. Everyone signed. And one party walks away holding rights over the other's future that the other never actually beheld. The violence stays hidden because it leaves no witness who can say what, exactly, was done to him — which is the oldest reason for hiding anything.

The old law, it turns out, was not naive about this. It kept two distinct doctrines for the two ways a bargain goes blind — deception, where a person makes the darkness, and uncertainty, where the world does — and it policed them by different methods because they are different wrongs. What the old doctrines never quite anticipated is the instrument complex enough to produce uncertainty's blindness on purpose: fog with an author. Reading that instrument takes both doctrines at full strength, plus the line between them. So this closing chapter of Part VI turns from the lender's claims — actual loss in chapter 19, the profit that was not made in chapter 20, the deprivation of liquidity in chapter 21, penalty and default in chapter 22 — to the integrity of the bargain itself. Chapter 9 walked the three traditions' common sale law: the lie, the fog, and the squeeze. Here each doctrine gets its own room and its finest machinery — the fiqh's law of fog, the halakha's law of overreach, the Latin law of fraud and just price — then the moral difference between uncertainty and deception, and then the present: the term sheet, the waterfall, and the tools that might yet read them for the people who cannot.

The Sale of Fog

The master text came in chapter 6 and worked in chapter 9: the Prophet forbade the pebble-toss sale, and he forbade the sale of *gharar* (Muslim 1513). The classical teachers built the doctrine the way courts build anything, case by case — the fish still in the sea, the unborn calf, next year's fruit — and chapter 6 hung that gallery. What this chapter needs is the engineering underneath: what uncertainty actually is, in this law, and how much of it kills a contract.

Definition first. The modern codification of the doctrine — AAOIFI's Shari'ah Standard 31, issued in September 2007 and devoted entirely to "Controls on Gharar in Financial Transactions" — states it with useful bluntness: *gharar* is "a state of uncertainty that exists when the process of concluding a transaction involves an unknown aspect"; it is the condition of "results that may or may not materialize." The Ottoman codifiers had already distilled the classical sale rules into three short sentences that chapter 3 met in passing: the thing sold must be in existence; the thing sold must be capable of delivery; the thing sold must be known to the purchaser (Majalla, arts. 197, 198, 200). Known, the code goes on, means known "by its state and description which distinguish it from other things" — this quantity of red corn, this field within these boundaries (art. 201). Sell what does not exist, what cannot be delivered, or what neither party can describe, and there is no contract to enforce, because there was never an object for two consents to meet on.

Now the crucial refinement, because stated baldly the rule would abolish commerce. No contract on earth is free of unknown aspects. The standard's own examples of tolerable fog are chosen with a drafter's wink: a man who buys a house has usually not seen its foundation; a man who leases by the month has signed for months that differ in length. So the jurists graded the fog. Minor *gharar*, *yasir*, is "the degree of Gharar that a contract could hardly avoid, and is not sufficient to generate dispute," and it never harms anything. Excessive *gharar*, *fahish*, is uncertainty that has become "a dominating and distinctive aspect of the contract," uncertainty "capable of leading to dispute," and it voids the deal outright; the assessment shifts with place, time, and normal practice — 'urf, custom, the same arbiter chapter 9 watched the schoolmen appoint for the just price. Between the two lies a middle grade — the underground crop, the melon that cannot be known unless broken — which the law, having flagged, lets pass. And for uncertainty to kill a contract at all, the standard requires four things at once: it must sit in an exchange-based contract, not a gift, for one may donate into fog what one may never sell out of it; it must be excessive; it must attach to the contract's primary subject matter and not to some corollary — the unborn calf cannot be sold alone, but it rides along inside its mother without harm, exactly as the milk rides in the udder of a sold sheep; and it must serve no genuine need, *hajah*, that could not be met by any cleaner instrument. None of this is a phobia of the unknown; it is a working commercial law, asking of every deal whether the thing these parties are consenting to is actually visible to them both.

Which is why *gharar* must not be confused with risk, a confusion this book has been dismantling since chapter 10. The traditions license — insist on — uncertainty about how a venture will fare: the investor whose capital trades at his own peril earns honestly from precisely that exposure. The *mudarabah* investor knows exactly what he holds, a defined share of an undefined outcome; the *gharar* buyer holds an undefined thing. Uncertainty about the future is the human condition, and pricing one's own exposure to it is the whole honest side of finance. Uncertainty about the object — what am I buying, what am I owed, what does this clause do — is a drafting choice, and the fiqh's verdict on manufacturing it was never in doubt.

The doctrine's finest hour, though, is its great exception, because the exception shows what the rule was for. *Salam*, the forward purchase, sells what does not exist: the Prophet arrived in Medina to find farmers being paid now for dates deliverable in two and three years, and rather than forbid the practice he specified it. Whoever pays in advance, he said, let it be for a known measure, a known weight, to a known term (Bukhari 2240) — *ma'lum*, three times in one sentence, the same load-bearing word the Majalla would build into article 200. The Ottoman code duly requires the salam contract to state type, variety, quality, quantity, and the time and place of delivery (art. 386). Notice what has happened: the nonexistent may be sold after all, provided description does the work that sight cannot. The ban was never on the future; it was on the unknown, and the unknown can be cured — measured, weighed, dated, specified — until both parties see the same object through the paper. Fog, in other words, is not a fate but a variable under the drafters' control, and this law made its reduction a condition of enforceability.

One late clause completes the portrait. The standard's modern drafters, working through the old prohibition of the pebble-toss — the bazaar sale that closed on whatever the thrown stone happened to strike — paused to note that the sale is equally void when conducted “by using a programmed machine to determine the sold good irrespective of the choices of the seller and the buyer.” A ruling aimed at a jahiliyya market game, reissued in 2007 against the algorithm: the fiqh has never believed that randomness improves with automation.

The Waiver That Knew Its Number

The rabbis, chapter 9 showed, built their sale law with numbers: *ona'ah* — overreaching, price-wronging — begins at one-sixth of the going rate, on the strength of Leviticus's command that neighbors not cheat one another in buying and selling. Under a sixth the market is doing what markets do; at a sixth the excess goes back; past it either party can unwind the deal entirely, and the remedies run on the most humane clock in ancient law: the time it takes to show the purchase “to a merchant or to a relative” (Bava Metzia 50b). That machinery needs no rebuilding here. What this chapter needs is the folio that follows it, because at Bava Metzia 51a and 51b the Talmud takes up precisely the questions the modern term sheet poses: what knowledge is worth, and whether protection can be signed away.

Start with knowledge, because the Gemara does. Why, it asks, does the return window bind the buyer while a wronged seller may go back at any time? Because the buyer is holding the goods: wherever he goes he can show them, and someone who knows the market will tell him whether he erred. The seller holds nothing — he learns what his goods were worth only when merchandise like his happens to pass before him (BM 51a). The remedies are calibrated to each party's access to the truth: the window closes when knowledge could have arrived, and stays open while it could not. On the same page the law prices knowledge from the other side. A man was selling silk strips, announcing six coins though they were worth five, and privately ready to take five and a half; a buyer sized up the situation, reasoned that paying five and a half would read as waiving the difference, and paid the full six intending to sue for the overcharge. Rava sent him away empty-handed, as Rav Chisda would send away a jewelry buyer who tried the identical maneuver at sixty against fifty: the *ona'ah* protections were taught for one who buys from a merchant, not from a householder selling his own effects — a man does not part with the furniture of his life “except for extra money,” as the Gemara has it. Readers of chapter 9 will recognize the doctrine from its Latin room: Aquinas's seller, who may charge above the market for the loss that parting inflicts on him, though never for the buyer's need. Two

traditions, from opposite directions, meet at the same fine line — attachment is a real cost and may be priced; and a buyer who can see the premium and pays it scheming has not been wronged by it.

Then the folio arrives at the waiver, and produces the chapter's most quietly modern page. What of the seller who writes candor into the contract itself: I sell this on condition that you have no ona'ah claim against me? Rav says the claim survives the clause; Shmuel says it does not — and the Talmud's resolution, which the codes adopted, splits the difference exactly where a disclosure lawyer would. A blanket waiver, stipulated in the ordinary way, *stam*, is void, and Rav's stated reason deserves engraving: does the other party know there will be an overcharge, that he should consciously waive it? He signed hoping there was nothing to waive. But a waiver made explicit, *mefadesh*, stands, and the baraita at 51b writes out the script: a seller who says, this item that I am selling you for two hundred, I know to be worth only one hundred, and I sell it on condition that you have no ona'ah claim — against him there is no claim; likewise the buyer who names the mirror-image numbers and buys anyway. Maimonides codified the pair with his usual finish: the blanket clause fails because the waiving party "does not know how much money he is forgoing"; the explicit clause binds because, in Eliyahu Touger's rendering of the rule, "all conditions that are accepted by both parties are binding in cases of financial law" (Mishneh Torah, Sales 13:3–4). The principle could be printed on the cover of every disclosure statute since: protection against overreach is waivable — by a consent that can see what it is waiving. The waiver that stands is the waiver that knew its number. Consent to an unquantified, unexamined disadvantage is, in this law, not consent to anything.

Two edges complete the room. The same chapter of the code blesses the trader who deals "in a faithful manner" — I bought this for such-and-such and take such-and-such as profit — and lifts the ona'ah rules from him entirely (Sales 13:5): candor about cost and markup replaces the market benchmark, a device whose enormous modern career under the name *murabaha* belongs to chapter 24. And the tradition extended overreach past money altogether — *ona'at devarim*, the wrong done with words alone, met in chapter 9 — a boundary stone marking how far the rabbis would chase the intuition that transactions can wound. Between the numbered waiver and the wounding word runs one straight conviction: the law of the market is a law about what the weaker mind in the room was actually able to know.

The Lie with an Author

Rome came at the problem through the character of the wrongdoer, and gave the West its permanent definition of fraud. The praetor's edict promised an action "where things shall be said to have been done by evil deceit," if no other action lay — fraud as the law's last resort, catching what no tidier claim could reach (D. 4.3.1.1). But what was *dolus malus*? The jurist Servius defined it as a machination to deceive, "where one thing is simulated and another done" — fraud as the false front. Labeo, a generation on, objected that the front is optional: a man can be circumvented without any pretense at all, and one can pretend without wronging anyone, as those do who dissemble to protect their own affairs or another's. So Labeo widened the definition to the shape it kept: *dolus malus* is *omnem calliditatem fallaciam machinationem ad circumveniendum fallendum decipiendum alterum adhibitam* — every cunning, every trickery, every machination employed to circumvent, trick, or deceive another. Ulpian, reporting the dispute two and a half centuries later, closed it in four words: Labeo's definition is the true one (D. 4.3.1.2). Sit with what the true definition does. It cuts fraud loose from the false statement. Arrangement, staging, engineered opacity —

cunning employed to circumvent — are fraud under Labeo's net even where every uttered word was accurate. The merchant with the rain-damp grain, who spoke no falsehood in his life and stacked his pile dry-side-up, is caught, six centuries in advance, by a definition written for the Roman forum. And the praetor, the next fragment adds, said *malus* deliberately, because the old lawyers knew a *dolus bonus* — the ordinary shrewdness of dealing, the stratagem against an enemy or a brigand (D. 4.3.1.3). The law of fraud begins by conceding that some cunning is just commerce, and then asks, of the rest, the only question it ever asks: was this cleverness employed against the person across the table?

Christian moral theology took the Roman chassis, as usual, and supplied the tribunal Rome lacked. Chapter 9 walked Aquinas's question 77 clause by clause; here it needs one paragraph, read from this chapter's angle. Exchange exists for the common advantage of both parties, so the contract must observe equality of thing and thing — an equality "not fixed with mathematical precision" but resting on "a kind of estimate," a band whose working measure is the community's common estimation. Cheating enters three ways — through the price, through the thing (the threefold defect of substance, quantity, and quality), and through silence, for the third article makes hidden defects that threaten loss a matter of mandatory disclosure while leaving the one-eyed horse to speak for itself. And the tribunal's jurisdiction runs past the courtroom's: human law, Aquinas says — citing, chapter and verse, the rescript this book met at the head of chapter 9 — leaves small mutual outwitting unpunished and intervenes only past half the just price; the court of conscience holds the gainer to restitution wherever the equality notably broke, whether or not any judge could be brought to care. Commutative justice is the name of that stricter ledger: person to person, thing for thing, the claim true or the claim owed back. On the desperate buyer — the man whose necessity would pay anything — the doctrine's finding was chapter 9's sober center and is only recalled here: the seller may charge for his own loss, never for the buyer's need, for the buyer's need is not his to sell; and the usury question next door heard the same finding in the borrower's voice, who pays "not... voluntarily simply, but under a certain necessity."

Now set the three traditions side by side and ask the question this chapter exists to sharpen: for a completed bargain to be reopened, must there have been a lie? Three legal civilizations, three considered answers. Rome's late law said no: the rescript to Aurelius Lupus reopened a land sale on arithmetic alone — less than half the true price, *humanum est*, undo it or top it up — though the classical jurists had preached free outwitting, and scholars have suspected since 1706 that the humane rule is Justinian's editors writing over Diocletian's name; the Middle Ages, unbothered, named it *laesio enormis* and generalized it. The rabbis said no with a finer gauge: the sixth reopens or repairs the sale with no inquiry into anyone's honesty — *ona'ah* is not a fraud tort but a strict-liability geometry, and a perfectly innocent seller returns the excess like anyone else. The Hanafi codification said yes, and drew its lines with actuarial candor: the Majalla defines *ghabn fahish* — flagrant lesion — as one-twentieth in merchandise, one-tenth in animals, one-fifth in real property (art. 165), and then rules that lesion alone, without deceit, reopens nothing; only where one party has actually misled the other — *taghrir*, "deceit is cheating," art. 164 — does flagrant lesion let the victim cancel (arts. 356–357). Three thresholds, three philosophies of finality; and then, in the middle of the Ottoman article, an exception that shows exactly where its drafters' hearts were: where the property of orphans is sold at flagrant lesion, the sale is invalid with no deceit needed — and the endowment and the public treasury are protected like the orphans. The parties who cannot look after themselves get the strict-liability rule; everyone else is presumed able to haggle. Whether one prefers Rome's half, the rabbis' sixth, or the Hanafis' graduated ladder, the family resemblance is the finding: every one

of these systems fixed a number past which it declined to presume that consent had been real.

Underneath the divergence sits the distinction the ToC of this book asks for by name, and it is worth stating with care: the moral difference between uncertainty and deception. Deception has an author. It is a wrong done by a will to a person — Labeo’s cunning *employed*, the wet grain *placed* — and the law’s response is therefore personal: rescission, restitution, the *actio de dolo*, whoever deceives is not of me. Uncertainty has no author. The udder is opaque to seller and buyer alike; no one made the fog, so there is no one to punish for it — and the law’s response is therefore structural: it condemns the contract, not the man, refusing to let certain bargains exist because their object cannot be seen by either party. The law of fraud is a police force; the law of *gharar* is a building code. The one hunts wrongdoers after the harm; the other forbids the harm’s architecture in advance, indifferent to intent, the way a code forbids the staircase that will eventually break somebody’s neck no matter how honest the carpenter. Ona’ah, characteristically, holds the middle: a strict-liability rule about outcomes, blind to malice, timed to knowledge. Three instruments, because there are three distinct ways for a consent to die — poisoned, starved, or overrun — and a mature legal system, the traditions quietly agree, needs all three. What none of them doubted is the junction case: fog sold by the one party who can see through it. That is deception wearing uncertainty’s coat — the Majalla’s *taghrir* joined to *ghabn*, the dry grain arranged over the wet — and it is the precise address, this chapter will now argue, of a great deal of modern financial drafting.

Consent That Can See

Muslim’s collection preserves a case that shows the whole apparatus bending toward a single human being. A man told the Prophet that he was cheated — regularly, helplessly — in his purchases. The answer was not a lecture on diligence. It was a formula, placed in the man’s mouth: when you conclude a sale, say *la khilaba* — no deception (Bukhari 2117; Muslim 1533a). The jurists read the formula as reserving to him an option to revisit his bargains, and the classical law generalized the device: parties to a sale may stipulate an option, for a fixed period, to cancel or ratify (Majalla, art. 300). Muslim’s text adds one detail, and the transmitters kept it with a tenderness that does them credit: ever after, whenever he traded, the man would say the words — and the text spells his version with a letter gone astray, *la khiyabah* for *la khilaba*. The formula protected him anyway. The law had not asked him to become sharper than he was; it re-engineered his consent — stretched it over a review period — so that his signature ripened only after his understanding had time to catch up. Consent, in this tradition’s hands, stopped being a moment and became a process with a built-in second look.

Around that instinct the *fiqh* assembled what can only be called consent machinery, met in chapter 9 as scattered rules and visible here as a system. The option of misdescription: sell the cow as a milker and let her prove dry, or sell a stone by lamplight as a red ruby that dawn shows to be yellow, and the buyer chooses freely between keeping and canceling (art. 310). The option of inspection: buy a thing unseen and the purchase stays open until sight closes it (art. 320). The option for defect: every unconditional sale silently warrants soundness, a defect being whatever “persons competent to judge” say depreciates the price (arts. 336–338). Each option is a dated, structured right to re-decide — and each dissolves on disclosure: the vendor who declares the defect and finds a willing buyer owes nothing further (art. 341). Even the modern standard’s rule for selling the unseen is at bottom a comprehension test: description suffices only when it covers “all those characteristics

which could affect the price” — description, that is, that puts the buyer where a viewer would stand. (The Hanafi code, honesty requires adding, will also enforce a blanket sold-with-all-faults clause (art. 342) — the one point in this chapter where a tradition takes the other side of the waiver question the halakha answered with its named number; the reader may hold both designs in mind and notice that each polices the abuse the other tolerates.)

The halakha built the same machine with its own parts, and this chapter has already toured them: a return window timed to the arrival of a second opinion; remedies keyed to which party could check the price; a waiver valid only when it names the figure being waived. And Latin Christendom’s version has been visible since chapter 9: the disclosure article; Cajetan’s just price obtained “assuming common knowledge and in the absence of all fraud and coercion”; Vitoria’s verdict that necessity makes consent merely “mixed voluntary,” half chosen, half compelled. Three traditions, one architecture, and the Qur’an’s commercial cornerstone states its constitution in a single verse this book keeps returning to — “You who believe, do not wrongfully consume each other’s wealth but trade by mutual consent” (4:29, in M. A. S. Abdel Haleem’s rendering). Chapter 6 read the verse as the market’s charter and chapter 9 set it under the fraud rules; here, at consent’s edge, its load-bearing clause finally bears its full weight. Mutual consent, in every room this chapter has visited, means more than two signatures. Disclosure is its floor; comprehension is its test; a consent that could not see the object — because fog hid it, because complexity buried it, because desperation shut the time to look — is defective in exactly the way these three laws spent fifteen centuries specifying. The signature proves that consent was given; it has never, in any of these traditions, been allowed to prove that consent saw.

The Term Sheet, Read Aloud

Chapter 9 closed by promising that this chapter would read the term sheet as what it is, a moral document. Read it so now.

A term sheet is a short paper — a few pages, mostly declared nonbinding — that fixes the skeleton of a financing before the long documents grow around it: how much money, at what valuation, with what preference, participation, and protective rights. Its brevity is its charm and its cover. In the taxonomy this book built in chapter 3, a term sheet is a proposal about names — which claims will be called ownership, which will behave like debt, which party’s downside is being quietly relocated — and by the time the definitive documents arrive, its skeleton has become load-bearing and unmovable. It is a moral document in the exact sense this chapter has assembled: it distributes sight and blindness, exposure and immunity, between parties of unequal knowledge, before a single dollar moves. The traditions would not be impressed by its page count. They would ask their three questions. Is the object of consent visible to both parties? Has anyone’s advantage been harvested from the other’s inability to check? And is any cunning here employed to circumvent?

Consider, plainly, the machinery the answers must survey. A liquidation preference entitles the investor, when the company is sold or wound up, to take his money back — once, or some multiple of once — before common shareholders receive anything. If the preferred also participates, it then shares in what remains, collecting twice from the same exit. Preferences stack: each financing round typically sits senior to the last, so that a company which has raised in four rounds carries a tower of prior claims between its employees’ stock and any dollar of proceeds. Anti-dilution provisions reprice earlier preferred shares when a later round comes in lower — under a full ratchet, as if the earlier investors had bought at the new, lower price — with the difference conjured out of the common holders’

ownership. None of these clauses is secret, none is a lie, and severally each has a rationale an honest scholastic would hear out. Assembled, they can produce a payoff structure in which the paper's headline — you own thirty percent of your company — is true in exactly one scenario, the triumphant one, while across the wide middle range of ordinary outcomes the holder of that thirty percent collects little or nothing. The number on the certificate describes a possible world; the waterfall decides the actual one; and the two can be arranged to part company almost entirely. Rome had a name for the arrangement in which one party has contracted never to lose while calling the other a partner, and chapter 11 told its fable; chapter 37 — “Waterfalls, Preferences, and Synthetic Usury” — runs the full arithmetic of how a stack of individually reasonable clauses can rebuild, in equity's costume, the payoffs this book's traditions forbade by name.

Here the register must go plain, because the ToC's bullet says what needs saying: there are waterfalls that founders do not understand, and they are signed anyway. The founder at that table is very often chapter 9's desperate party in modern dress: weeks of payroll left, a team she recruited on promises, savings and reputation and years of work fused into the company — livelihood, not portfolio. The financing is offered as a package, at speed, in a dialect the other side of the table drafts for a living and she may be reading closely for the first time; the recitals will state that she is sophisticated, was represented, had opportunity to inquire. Aquinas heard that kind of consent seven centuries ago and called it what it is — given under a certain necessity; Vitoria would call it mixed voluntary; the Gemara would ask its one devastating question — did she know, that she should waive? A blanket acknowledgment that one understood and accepted an unquantified disadvantage is precisely the *stam* waiver the halakha refuses to enforce, precisely the consent-without-sight the fiqh's building code exists to condemn, and where the imbalance of comprehension is not accidental but priced in — where the fog is maintained because it harvests — Labeo's definition is already satisfied, whatever the recitals say. The wrong here is not that capital protects itself, nor that preferences exist. It is that the instrument's truth has been made structurally unreachable for the party with the most at stake, and that her signature is then treated as proof she reached it.

The traditions do not leave the analysis in despair; they leave it a specification. Consent needs a second pair of eyes — the Mishnah's remedy clock exists so the buyer can carry the thing to a merchant or a relative who knows. Disclosure must reach comprehension — description covering all the characteristics that affect the price, in the standard's words, not a data room the size of a national archive. A waiver binds when it names its number — two hundred against one hundred, said aloud, both directions. And the reading must pierce names to substance — in contracts, says the Majalla's third article, effect is given to intention and meaning and not to words and phrases, which is Gregory v. Helvering's principle from chapter 3 speaking Ottoman Turkish. Every item on that list is buildable, and this book's last Part intends to build them. A contract analyzer — an instrument that takes the stack as input and returns the moral geometry as output: who is paid, in what order, at every exit value; which clauses moved the losses, and onto whom; what the thirty percent is actually worth across the futures the parties can foresee; where the drafting departs from the plain design and by how much — is not a novelty grafted onto this tradition. It is the tradition's own machinery, retooled: the merchant-or-relative window, the named number, the description that stands in for sight, delivered before signature rather than litigated after. Chapter 43 — “A Moral Geometry Contract Analyzer” — undertakes the building; what this chapter contributes is the standard such an instrument must meet and the reason it deserves the old, strong word *justice* rather than the weak modern word *transparency*. Transparency describes the documents; justice describes the parties — and these three tra-

ditions, disagreeing about nearly everything else, agreed for fifteen hundred years that a bargain binds in the measure that both parties could see it whole. The analyzer's whole ambition is to make that old sentence enforceable at modern complexity: to stand beside the person about to sign — founder, borrower, the man in the market with his formula and his hope — and stay there until the moment the tradition was always guarding, the moment the signature becomes true, because this time, before she signed, she saw.

Part VII — Instruments: How Contracts and Institutions Try to Escape Usury

Chapter 24 — Sale-Based Finance: Murabaha, Credit Sale, and Just Price

Among the domestic details that Aisha remembered about her husband is one that a careless reader of Part II would call impossible. The Prophet of Islam, the man through whom heaven had declared its war on *riba*, bought his household's grain on credit. The report sits in Bukhari's chapter on sales without a flicker of embarrassment, under a heading that is itself a small manifesto: *the Prophet purchased foodgrains on credit*. He took delivery of food from a Jew, payment deferred, and handed over his iron armor as the pledge (Bukhari 2068). The tradition preserves the ending too, and it is one of the most quietly affecting facts this book will touch: when he died, the armor was still in pawn, against thirty sa's of barley (Bukhari 2916; chapter 10 met the report among the pledge rules). The most famous enemy usury ever had ran his kitchen on deferred payment, dealt across the confessional line to do it, and died owing an honest debt, secured by an honest pledge, to the man who held his armor and his word.

Nothing in that scene violates anything Part II built, and seeing why is the work of this chapter. The obligation the Prophet carried was not a loan that had bred; it was the price of grain actually delivered, owed to the one who had delivered it. The Qur'an's dividing sentence — "God has allowed trade and forbidden usury" (2:275, quoted as throughout in M. A. S. Abdel Haleem's rendering) — does not merely tolerate the credit sale; the written and witnessed debt that chapter 6 admired presupposes it, scribe, term, and all. And yet everyone who has read chapter 3 can feel the ground shift underfoot. Let the price wait, that chapter warned, and time has crept into the sale; a deferred price above the cash price does quietly what interest does openly. The instant goods move today against money promised for the spring, one party is financing the other, and every question this book has spent six Parts sharpening comes crowding into the market stall. The live question is what the deferred premium prices: genuine uncertainty, since nobody knows what the goods will be worth at the term; or the seller's real service and cost in an accommodation that would not otherwise happen; or nothing but the waiting itself — *agar natar*, payment for waiting, the forbidden thing as chapter 7 defined it — in which case two innocent-looking prices have engineered a loan at interest, and everything Parts I through VI concluded applies in full. Part VII is where the traditions stop debating and start building: five chapters of actual instruments, the machines by which real economies financed themselves for centuries while the wall against usury stood. This first machine is the most used and the most watched — finance dressed as what it must genuinely remain, a sale.

The Space Between Sale and Loan

Begin where the traditions began, with the conviction that sale and loan differ in kind and not in paperwork. Chapter 7 laid the loan on the table: ownership of a consumable passes, a bare quantity is owed back, the lender exits the story on day one and holds nothing but arithmetic and a date. The sale has an entirely different anatomy. Ownership passes here too — that is not the difference — but it passes against a price, and the price answers to a thing in the world. The seller must have something real to convey and warrant; the buyer receives something that can rot, break, flourish, or disappoint; and the price itself, chapter 9 spent its length showing, is policed in all three traditions by the measure of the thing's

worth — the just price's band, *ona'ah's* sixth, *gharar's* demand that the object be known. A sale can go wrong in a dozen honest ways, and each tradition's law patrols all of them: defect, misdescription, false measure, and behind these the lie, the fog, and the squeeze that chapter 9 mapped. The loan can go wrong in only one way, nonpayment, because it consists of nothing but the payment. That difference is the whole moral geography. The merchant's margin is licensed because it prices real things and real service: Aquinas's fourth article called honest gain the wages of the merchant's labor; the rabbis, we are about to see, exempted the candid trader who names his cost; the *fiqh*, whose Prophet had been a merchant before he was a messenger, built its commercial law on the blessed premise of 2:275. A sale's gain is exposed to everything that can happen to goods, prices, and buyers, while the naked loan's increase — chapter 7 closed the question — pays for nothing that exists. On one side of that line the traditions built markets; on the other they built the wall.

The credit sale is the place where the line must be walked rather than drawn, because deferral imports the loan's one undeniable content — duration — into the sale's honest architecture. Sell for cash and the just price does all the work. Sell for autumn payment at a premium over today's price, and the premium prices something; the opening already named the candidates. Three traditions ran that analysis independently, on the same transaction, with tools from three legal languages. Their answers agree so closely that this chapter can tell them as one story with three tellers, and where they disagree at the margin, the disagreements map — with real precision — where the danger actually lives.

The Sale That Confesses Its Costs

The *fiqh's* contribution is a contract that solves the buyer's oldest problem — how do I know this price is fair? — by candor rather than by market reference, and it begins centuries before anyone thought of banking with it. Classical Islamic law knows a family of contracts the jurists call trust sales, *bay' al-amanah*, for situations where the buyer cannot see a going rate and must rely on the seller's honesty about what the goods cost him. Their names declare the mechanics: in *tawliya* the seller passes the goods on at his cost exactly; in *wadi'a*, below cost, a disclosed discount; and in *murabaha*, the family's working member, at cost plus a disclosed profit: I paid a hundred for this cloth, I sell it to you at a hundred and ten, and both numbers are on the table. The Prophet's market, chapter 9 found, was already a jurisprudence of visibility: the option that lasts until the parties part, the planted bidder forbidden, the intercepted caravan handed its option once the seller reaches the town rate; the report that carries the option rule carries the theology too: truth and disclosure bless a sale, concealment blights it (Bukhari 2079). The *murabaha* extends the same instinct into the price itself. Where the ordinary haggling sale — *musawama*, no cost named — leans on the market to keep the price honest, the trust sale leans on disclosure, and the law leans on the discloser with corresponding weight: misstate the cost, and when the *khiyanah*, the breach of trust, comes to light, the classical rules arm the buyer with remedies running from price adjustment to walking away.

The *halakha*, working the same counter, built the same stall. Chapter 23 met it in the Rambam's code and promised it to this chapter: the trader who deals *be-amanah*, on faith — who says, in the code's own formula, "I purchased this article for such and such, and I am making this and this amount of profit" — stands outside the *ona'ah* rules entirely (Sales 13:5). The sixth, the merchant-or-relative window, the whole apparatus of chapter 9 exists to police prices formed in fog; a price whose anatomy is publicly declared needs no such policing, because the buyer knew exactly what he was paying and for what. And the Talmud's fine print on faith-dealing, chapter 23 found at Bava Metziah 51b, reads like

a murabaha manual drafted in Aramaic: the faithful dealer may load into his declared cost the porter, the camel, the lodging — the real expenses of bringing these goods to this counter — but not a wage for his own labor, which must live openly in the named profit and nowhere else. Real costs in the cost; the dealer's own gain in the margin, by name. Two traditions, no documented contact on this point, one architecture: where the market's thermometer fails, candor is the instrument, and the candor must itemize.

What the twentieth century did was take this small, honest, merchant's contract and make it carry a banking industry. When the first commercial Islamic banks opened their doors in the mid-1970s, they faced a structural famine: the partnership instruments this book spent Part IV admiring — *mudarabah*, *musharakah*, capital married to enterprise and paid from real remainders — demand an appetite for enterprise risk that a deposit-taking institution struggles to run at scale. The device the industry's own histories credit to the Jordanian banker Sami Homoud supplied the alternative: murabaha to the purchase orderer. The customer wants a machine, a car, a cargo. The bank — not a dealer in machines, cars, or cargo — buys the asset at the customer's request, takes title, and resells it to the customer at cost plus a disclosed markup, payable in installments. Three parties now, and a promise holding the triangle together: the customer undertakes that once the bank has bought the thing, he will buy it from the bank. The Islamic Fiqh Academy, weighing the arrangement in Kuwait in 1988, blessed it on conditions that compress the whole doctrine: the sale is lawful provided the institution genuinely possesses the goods first, as the law of sale requires, and carries their risk until delivery; a promise binds morally, and may bind legally where the other party has spent money on its faith, with breach answered by actual damages; and the parties' mutual promises are tolerable only so long as somebody retains an option — bind both sides from the outset and the "two sales" have collapsed into one prewritten financing. The modern standards pile the same conditions into paragraph numbering. The institution must own the asset, legally or beneficially, before it sells it, and have taken possession, actual or constructive; it must disclose the cost (Malaysia's central-bank parameter makes nondisclosure fatal, the murabaha "null and void ab initio") and state the markup separately; and once the price is set it is set forever, because any increase for extending the term would be the old meter starting up inside the new machine. The AAOIFI standard adds the discipline chapters 8 and 22 already walked: the late customer meets no growing price, only the charity-routed pressure of the donation clause; the insolvent meets the mercy of 2:280; and the institution's income never includes a coin conjured by anyone's delay.

Notice what all this regulatory architecture is protecting, because it is the honest heart of the instrument. Between the bank's purchase and the customer's, there is an interval, an hour or a week, in which the goods are the bank's: its title, its risk, its problem. If the crane drops the machine on the dock, that is the bank's machine in pieces. If the customer lawfully walks away from a promise the Academy left him room to walk from, the bank owns a car it never wanted. The fiqh's old maxim prices exactly this seat: *al-kharaj bi-l-daman*, the yield follows the liability, a ruling the tradition carries as the Prophet's own (Abu Dawud 3508; chapter 10 met it among the risk rules). The collection files the maxim in a chapter about a purchased slave put to work and then returned for a defect — a human being held as property, recorded here plainly, as chapter 9 recorded the exclusion lists — and the jurists' reading of the case is this section's entire doctrine: the weeks' wages belonged to the buyer because the weeks' risk had been his. Entitlement tracks exposure. The murabaha's markup is earned in the hours the goods are the seller's problem. Every clause of the standards — the ownership, the possession, the itemized cost, the price that never grows — exists to keep those hours real.

The Doubt and the Dust

Latin Christendom met the credit sale early, and met it exactly where Islam's bankers would meet it eight centuries later: at the port, among merchants, with the paperwork in order and the conscience uneasy. In the second half of the twelfth century the archbishop of Genoa put his city's habit to Pope Alexander III. Traders were taking pepper and cinnamon worth no more than five pounds cash and promising six pounds at a fixed term. Was this usury? The reply, preserved in the Decretals as the chapter *In civitate tua*, is one of the most precise instruments in the canonical toolbox, because it answers on two levels at once and declines to confuse them. As to form: a contract of this shape "cannot be reckoned under the name of usury" — *non possit censeri nomine usurarum* — for there is no mutuum here, only a sale at an agreed price. As to substance: the sellers nonetheless incur sin, *nisi dubium sit* — unless there be doubt — "whether the goods would be worth more or less at the time of payment." And then the pastoral close, worth hearing for its tone alone: the citizens of Genoa would consult well for their own salvation if they ceased from such contracts, since the thoughts of men cannot be hidden from almighty God.

Read the doubt clause slowly, because the entire Catholic doctrine of sale finance hangs on it. If pepper deliverable against autumn payment may genuinely be worth seven pounds or four by autumn — prices move, ships sink, gluts happen — then six-for-five is not a disguised interest charge; it is today's honest estimate of an uncertain thing, a price with real ignorance inside it, and both parties stand inside the same ignorance. The premium prices the doubt. But where the goods' future value is not really in doubt — where everyone knows the pepper will fetch five in October, and the extra pound is simply the price of October itself — the sale's form is a costume, and underneath it a man is being paid for time, the one estate that chapter 7 heard William of Auxerre rule off the market altogether: the calendar is nobody's inventory. Half a century later, Naviganti's less famous clauses codified the test with the jurists' extra scruple, chapter 10 showed: the seller charging above the going price for deferred payment is excused *ratione huius dubii*, by reason of this doubt, where the goods' value at the term is genuinely uncertain — and the decretal adds a telling condition, that the seller was not otherwise going to sell, so that the premium compensates a real accommodation rather than harvesting a routine one. Aquinas then stated the negative rule with the bluntness chapter 9 quoted, and it must be requoted here because it is the exact center of this chapter: to charge a buyer more because he will pay later is usury in a sale's clothing — "this waiting for the payment of the price has the character of a loan" (ST II-II, q.78, a.2, ad 7) — and the mirror holds, buying cheap because of prepayment condemned in the same breath, with one homely exception: the seller who freely knocks something off for early money has given a rebate, not priced a financing. The scholastics, in short, refused to let the sale's form settle anything. The just price of chapter 9 supplies the baseline — the cash price, the common estimate — and the deferred premium above it must then point to something real. The only something the credit sale itself contains is genuine, shared uncertainty about value at the term; so doubt became the doctrine's load-bearing wall. A civilization that could not read hearts wrote the rule for the one court that could: the Genoese contract passed the courtroom and failed the confessional, and the tradition held both verdicts at once for five hundred years.

The rabbis drew the same line with a draftsman's economy, in the Mishnah this book has been circling since chapter 3. One may increase the rent, rules Bava Metzia 5:2, but one may not increase the price of a sale — and the Mishnah, as always, teaches by pairing. Rent a man your courtyard and say: pay me now, and it is ten selas for the year; pay month by month, and it is a sela a month, twelve for the year, and it is permitted. Sell a man your

field and say: pay me now, and it is yours for a thousand zuz; pay at threshing time, and it is twelve maneh, twelve hundred, and it is forbidden. The two cases are arithmetically identical, a fifth more for the waiting, and halakhically opposite, and the reason is the anatomy this Part exists to teach. Rent falls due as shelter is enjoyed, month by month; the landlord quoting ten-now is discounting his own future claims, not charging for time, because the monthly rate is the true price of a year of use paid as the use arrives — a logic chapter 25 will build a whole instrument on. The field's price, by contrast, is owed whole when the field changes hands; every zuz of the twelve hundred above the cash thousand is a naked charge for the months between delivery and threshing — *ribbit* in a bill of sale. The category the rabbis assigned it tells its own story: the deferred markup is *avak ribbit*, the dust of interest — not the biblical fire itself, chapter 3 explained, but its residue, forbidden by the rabbis and left where it settles rather than clawed back by courts. And the Talmud's discussion of *tarsha* — the credit sale at a single quiet price, never decomposed — shows the sages walking the cliff edge with open eyes. Rav Nachman permitted the merchant who names one deferred price, higher than today's market would pay but never stated as today's-price-plus-waiting: there, he ruled, nothing has been fixed for the waiting as such. Rav Papa, himself a brewer, ran his beer trade on that theory and defended it in the first person — the beer would not have spoiled, he did not need the money, the accommodation ran from him to the buyer. The Talmud preserves the answer of Rav Sheshet, son of Rav Idi, in the same breath, and it turns the ledger around: why follow yourself, he asked, rather than your customers? A man with coins in hand would buy at today's cheap rate; only the man whose hands are empty agrees to the dearer future one. Seen from the empty side of the counter, the quiet premium was pricing the emptiness — chapter 7's finding about the naked loan, surfacing inside the permitted sale. The lane the halakha finally left open is deliberately narrow and deliberately unmarked: a deferred price survives only while it never confesses to being a price for deferral. The moment the two prices stand side by side in one mouth — a thousand now, twelve hundred at threshing — the dust becomes visible, and the law sweeps it. A tradition does not fence a permission that finely unless it knows precisely which animal keeps trying to get through.

The Costume Test

Set the three rooms side by side, as this book keeps doing, and the shared design is unmistakable. All three traditions bless the sale and its margin. All three permit the sale to carry deferred payment — commerce could not breathe otherwise, and the Prophet's own armor stood pledge for the principle. All three then post a guard at the same gate: the deferred premium must trace to something other than the calendar — to real doubt, said the canonists; to a price that never itemizes the waiting, said the rabbis; to disclosed costs, real ownership, and a markup earned by genuine dealing, said the jurists of Islam. Which means all three need what chapter 7 called the first question to ask of any instrument, and what this Part will apply five times: a test for when the sale is real.

By now the test has no mystery left in it. Ask of any sale-shaped financing: Did the financier ever own the thing — own it in the world, not in a folder? Did its risk actually sit with him for a real interval, so that the crane, the blight, or the market could have made his problem worse while the goods were his? Is the price a price — fixed once, answering to the thing's worth and the deal's real costs — or does it grow with the calendar, compound with delay, reprice at rollover? And could the transaction be disappointed by anything other than nonpayment? Yes down the line is a sale that finances a purchase, the thing every tradition blessed. No down the line is a loan wearing a sale's clothes — and

the traditions' case files on the costume are among the most instructive documents they produced, because each tradition caught its own.

The fiqh's specimen, chapter 3 exhibited: the 'ina double sale, cloth sold dear on credit and bought back cheap for cash in the same sitting, condemned by the majority of the schools precisely because the cloth goes nowhere and the two prices net to money-now against more-money-later. Christendom's twin wore a Spanish name. In the *mohatra* — usually traced, with some justice, to an Arabic word for hazard — the needy party buys velvet dear on credit and sells it back cheap for cash, often to the same merchant, and walks out with today's money owing tomorrow's premium. The casuists who defended it on the elegant ground that each sale, taken alone, was valid were mocked by name in Pascal's Provincial Letters, and in 1679 the Holy Office under Innocent XI condemned the defense in a proposition whose Latin deserves its place in this book's museum of costumes: *Contractus mohatra licitus est* — "the contract of mohatra is lawful, though made with the same person, and with a reselling of the same thing, before agreed on, with a design of gain" — condemned, with its sixty-four companions, as "at least scandalous and pernicious in practice." Rome did not bother to dispute that the paperwork was in order. It condemned the arrangement for what it did.

The modern chapter of the file is being written by the tradition with the largest sale-finance industry to police, and its jurists have shown the old reflexes. Take the murabaha's cousin *tawarruq*, the turning-to-silver: buy a commodity on deferred payment, sell it at once for cash to a third party, and you have financed yourself through two genuine sales — a device the classical schools debated but many tolerated, since unlike the 'ina the goods really move and the parties really differ. Then watch what scale does. In the banked version (organized *tawarruq*, the industry's own label) the institution sells the customer metal warehoused an ocean away, on deferred payment at a markup, and simultaneously, as the customer's agent, sells the same metal back into the market for spot cash; parcels in London Metal Exchange warehouses are the standard prop, bought and resold within minutes by parties who will never see, store, or want a gram of them, and the same plumbing runs much of the industry's interbank liquidity. In 2009 the Islamic Fiqh Academy — the same body that had blessed the murabaha triangle in 1988 — looked at the assembled machine and refused it: organized and reverse *tawarruq* are impermissible, resolution 179 rules, because the simultaneous, prearranged legs amount to "a deception... in order to get the additional quick cash" — paper sales netting to *riba* on the trailing obligation. The resolution reads, across seven centuries and a confessional divide, like the second paragraph of *In civitate tua*: nobody disputes the forms; the assembly is judged by what it does. The metal in the warehouse is the mohatra's velvet with a commodities code, and the tradition's own supreme juridical body said so — which matters for how this book reports the whole subject. The critique of sale-shaped lending is not an outsider's sneer at Islamic finance, any more than Innocent XI was an outsider to Catholic casuistry or the Gemara's editors to Rav Papa's beer; it is the sound every one of these traditions makes when it catches its own instrument standing too close to the wall.

What the Markup Prices

One habit of the modern instrument draws more fire than any other, and it makes the cleanest laboratory for this chapter's last question. Ask what the plus should actually be, and the industry's working answer, for most of its history, was: cost plus roughly what the interest would have been. Institutions financing by murabaha, wrote Taqi Usmani, the Pakistani judge-scholar who did as much as anyone to build and police the industry's stan-

dards, “determine their profit or mark-up on the basis of the current interest rate, mostly using LIBOR” — the London interbank rate that served, until its disgrace and abolition, as the world’s reference price for money. The practice was criticized from inside on the obvious ground, and Usmani’s reply is the classic statement of the instrument’s whole defense: if the murabaha fulfills all its conditions — real asset, real ownership, real sale — then “merely using the interest rate as a benchmark for determining the profit of murabahah does not render the transaction as invalid, haram or prohibited, because the deal itself does not contain interest.” His illustration has become industry furniture: one brother sells liquor, the other sells lawful soft drinks, and if the second prices his lemonade to match the first’s wine, his trade may be imitative, but it does not thereby become the wine trade. Even the defense’s author would not leave it there: it remains true, Usmani wrote in the next breath, that “the Islamic banks and financial institutions should get rid of this practice as soon as possible,” building benchmarks of their own, because a lawful trade that steers by the meter of an unlawful one has chosen a strange star. The halakha, chapter 14 found, arrived at the same uneasy accommodation from its own side: the standard heter iska forms let the fee-in-lieu be keyed to prevailing interest rates — the partnership document borrowing the loan market’s number while refusing its name. And when LIBOR itself finally died (the bank panel ended in June 2023, the synthetic remnant was switched off in September 2024), the world’s contracts migrated to the successor rates, and the sale-based industry migrated with them, repapering without missing a payment date. The thermometer changed; the temperature did not.

The honest answer to what that proves comes in halves. A benchmark, the defenders are right, is only a thermometer. Financing has a going price in every economy; a bank that must cover deposits, costs, and losses cannot price its margins off the weather; and the just-price tradition itself, chapter 9 showed, defines fairness by the common estimate — which, for the forbearance of money, is the interest rate, whether or not anyone says so aloud. The critics are right about the rest. An instrument whose margin tracks the loan market’s price, repricing with its cycles, booked and stress-tested and securitized alongside its products, has told you what market it actually lives in — and by a much-cited estimate, murabaha in its various dresses carries up to four-fifths of the world’s Islamic-finance volume, a figure the tradition’s own scholars quote as complaint far more often than as boast. When a benchmark is borrowed long enough it stops being a thermometer and becomes a thermostat: the number no longer reads the price of money but sets it, delivering the loan market’s return through a sale’s paperwork, at scale. That is the “referendum” chapter 22 watched the late-payment rules run, extended to the whole balance sheet — and it is why the industry’s own consciences, from the Fiqh Academy’s tawarruq ruling to Usmani’s get-rid-of-this-practice, keep insisting that the costume test be applied from inside the house.

So the moral geometry of cost-plus finance, this chapter’s closing assignment, can be stated as an audit any reader can now run. Decompose the markup. Part of it prices the goods themselves — the just-price band of chapter 9, policed by candor where the market is thin: the murabaha’s named cost, the Rambam’s faithful dealer, real expenses honestly loaded. Part of it prices service — procurement, expertise, the accommodation of a buyer who could not pay today: the wages-of-labor title every room recognized, *stipendium laboris*, *sechar tircha*, the murabaha’s named profit. Another part prices the ownership interval — the hours or weeks the goods were the financier’s problem, the crane and the defect and the walked-away promise, *al-kharaj bi-l-daman*’s seat. And some of it may price genuine doubt — In civitate tua’s *dubium*, Naviganti’s excuse — value honestly unknown at the term, both parties inside the same fog. Every one of those components is real, auditable, and blessed, and a sale-based financing whose margin decomposes into them completely

is what the traditions said it was: trade, permitted and honorable. Whatever remains after they are priced — the residue that answers to nothing but the distance between today and the due date — is the old tenant back in the building, on a bill of sale this time, and no quantity of paperwork changes what it pays for. The instrument is not condemned by this audit; it is constituted by it. The murabaha and the licit credit sale live exactly on the wall, closer to the forbidden thing than any structure the traditions allowed before or since, and they stay honest by the disciplines this chapter has walked — ownership that is real, risk that is borne, costs that confess, doubt that is shared, a price that never grows — enforced by regulators in this world and, where *In civitate tua* left the remainder, in the next. Chapter 32 will weigh what becomes of those disciplines under an industry's full commercial pressure, and chapter 35 will turn the audit into arithmetic. The next two chapters stay in the workshop, with the instruments the traditions trusted further because their honesty is structural rather than policed: chapter 25, the lease — finance built on things that survive their use, where payment may honestly follow time because use is delivered through time — and chapter 26, the certificate, where ownership itself is divided into paper and sold to many hands. The sale financed the Prophet's own table; the lease and the certificate financed the temples, roads, and fleets of all three civilizations. The question, as always, is less whether the instruments can be honest than whether the people running them at scale can afford to be — and what has to be true, in law and in design, before the honest version is also the profitable one.

Chapter 25 — Lease-Based Finance: Ijara, Rent, and Usufruct

Two workshops on the same street burn down on the same night, and in the morning two machinists stand in the ash with two pieces of paper. The first man was renting his lathe: a true lease, the machine's owner across town. His workshop is gone and his trade with it, but his paper died in the fire too — rent stops where use stops, and the lathe that no longer exists is its owner's loss, not his. The second man's paper is called a lease as well. It says so at the top. But read down the page: he insures the machine, he maintains the machine, he bears every risk of the machine's destruction, and his monthly payments — calculated, had he looked closely, as principal plus a return on it — continue in full for the remaining four years, fire or no fire, because a clause the trade calls hell-or-high-water says they do. Same street, same fire, same word at the top of the page, and two opposite fates, because the two papers are opposite contracts. A Roman jurist named Celsus wrote the second man's lesson centuries early. Usufruct is a right in a body, he said — *ius in corpore* — and when the body is destroyed, the right necessarily dies with it (D. 7.1.2). A right to use survives exactly as long as there is something to use. The insurers, notice, already know which paper is which: on the first machine the owner across town files the claim, because the loss is his; on the second, the "lessee" files it, with the "lessor" named as beneficiary, and the label on the paper has already confessed.

This chapter is about the instrument the fire cannot embarrass — the lease, the second machine in Part VII's workshop, and in every tradition this book reads the most serenely lawful way that capital has ever earned a return through time. The loan of Part III may charge nothing; the sale of chapter 24 lives on the wall, policed clause by clause. The lease stands in open country, so uncontroversial that the usury literature of three faiths mentions it mostly as the contrast case — the thing interest is *not*. That serenity has exact anatomical reasons, and they are the chapter's first business. Its second business is the modern instrument that borrowed the lease's name and none of its anatomy — the financial lease of the second machinist, in which chapter 3 promised we would watch the floor sanded away — and the tests, ancient and modern, that tell the two apart.

The Thing That Survives Its Use

Recall the wall's foundation, because the lease is built on the same stone, right side up. Chapter 7 found the loan's anatomy in the nature of consumables: wine, grain, and coin are used only by being used up, so their use cannot be sold apart from their substance — a lender who charges for the money and again for its use "sells the same thing twice, or sells what does not exist." That argument has a flip side, and Aquinas states it in the same article, as the tradition's own boundary marker: the letting of a house raises no usury question at all, because its use can be granted month by month while the house itself endures and stays the owner's. There, complete, is the lease's license. A durable thing holds two separable goods — the thing itself, and the flow of service it gives off month by month — and law can assign them to different people. Rome, whose lawyers organized everything, gave each assignment its instrument. Under *locatio conductio*, the broad contract of letting and hiring that chapter 3 met in the taxonomy, an owner lets the use of a thing, or his own services, or a whole work, against the rent the Romans called *merces*. And where the

split needed to survive for a lifetime, Rome carved it as a property right and defined it in a sentence the civil law still recites. *Usus fructus est ius alienis rebus utendi fruendi salva rerum substantia*: usufruct is the right of using and enjoying things that belong to another — *salva rerum substantia*, the substance of the things preserved (D. 7.1.1). Read that ablative clause slowly, because the whole morality of rent lives inside it. The usufructuary may take the fruit; he must keep the tree whole for its owner. Use flows to one party, substance stays with the other, and the price paid for the flowing use is as honest as any price in commerce, because something real is delivered every single day the contract runs — this month's shelter, this season's harvest, this year's turning of the lathe. The old taxonomy of chapter 3 even kept a free cousin beside the priced instrument, and the pair teaches the anatomy better than either alone. In the *commodatum*, the gratuitous loan-for-use, a neighbor borrows the hoe itself and returns that hoe — no rent, a favor between households, ownership never blinking; in the *locatio*, the same hoe goes out against coin. Both are possible for exactly one reason: the hoe outlives its work. Set the *mutuum* beside them and the family portrait is complete — the contract for things that die in the using, where nothing can come back but a quantity, and where the traditions therefore let nothing extra come back at all.

Now watch the anatomy exclude money, in the traditions' own courtroom. If use can be sold where use is separable from substance, a clever mind will ask why coins should be different, and the Talmud preserves the mind that asked. Rav Chama, a Babylonian sage, would rent out dinars, the Gemara reports, at a peshita per dinar per day, and his theory was exactly the one this chapter has been building: in what way, he asked, is this different from renting out a hoe? The Gemara's answer is the entire doctrine of Part III in a farmyard image. The hoe comes back to its owner *as is* — the same hoe, its wear visible and known, its deterioration all the while on the owner's account. The dinars do not come back as is; a borrower does not return the same coins he took, for the coins are given to be spent, and what wears away in the spending can never be weighed. The "rent" on a dinar is therefore not a price for use flowing from the owner's surviving property; it is a charge on a loan, ribbit with a landlord's vocabulary, and the tradition remembers the episode with an ending it plainly regarded as a verdict: all of Rav Chama's money, the same page records, was lost (Bava Metzia 69b). He had tried the oldest maneuver in this book — chapter 7 called it charging one contract's price on another contract's anatomy — and the anatomy will not carry it. A hoe can be rented. A house can be rented. A dinar can only be lent, and the wall of Part III begins where the rentable world ends.

The Rent Book of Three Traditions

Around that shared anatomy each tradition kept its own rent book, and the entries are worth reading side by side — partly for the harmony, and partly because each tradition patrols the lease's border with the loan at a different gate.

The halakha's gate is the lender. Rent as such never troubled the rabbis: the courtyard, the field, the animal, the kettle hire out for payment in every tractate without a whisper of anxiety, and chapter 24 met the Mishnah blessing rent that rises for deferred payment — ten selas for the year paid now, a sela a month paid monthly — precisely because each month's rent falls due with each month's shelter, use paid as use arrives. What the same Mishnah forbids, in its opening line, is the landlord-tenant relation flowing toward a creditor: one who lends his fellow money may not live in his courtyard free of charge, nor rent from him below the going rate, because this is ribbit (Bava Metzia 5:2). The house is rentable; the lender is not the one who may harvest it. Free shelter running to the man who holds your

note is interest wearing a tenant's face — the disguised benefits of chapter 8, patrolling the lease's side of the street. And the rental relation itself, the rabbis kept honest by the ladder chapter 7 climbed: in the Mishnah's ranking of keepers, the borrower stands at the top, liable even for sheer accident, because all the benefit is his — while the paying renter stands a rung below, not liable for the accident that destroys the thing, because the benefit is split: he has the use, but the owner has the rent, and ownership's risks stay where ownership stays. The kettle of Mar Ukva's household, chapter 10 told, was weighed before each hiring and weighed after, the missing copper charged to the renters' use and the fee licensed by the owners' wearing metal. Use priced, substance guarded, risk seated with title — the halakha's lease is Paul's definition running in Aramaic.

The Latin gate is the conscience, and its rent book is the calmest document in this book's Christian archive. Christendom rented land for a thousand years — glebe and vineyard, mill and meadow, the whole feudal economy a lattice of use granted against render — and its moralists, so quick to smell usury in a deferred sale, hardly stirred. The reason is the one Aquinas gave: the house survives its use and remains the lessor's, so the rent prices a real and continuing conveyance, not a phantom. Usufruct itself lived comfortably inside canon law's world; the usufructuary who took the fruits *salva substantia* owed nothing to the usury question at all. The scholastics' worries began only where the lease's clean geometry was borrowed by instruments that did not share it — the *census*, the rent purchased for a lump of cash, where a capital sum went one way and a "rent" flowed back the other, and the schoolmen had to ask whether a man had bought a genuine stream of fruits or merely dressed a loan in an annuity's clothes. Those quarrels, chapter 3 promised, belong to Part VIII, and this chapter needs from them only their premise: nobody in the quarrel doubted the honest lease. The fight was always over who deserved to borrow its reputation.

The fiqh's gate is definition, and its rent book opens, characteristically, with revelation smiling on the whole business. When Musa flees to Midian in the Qur'an's telling, it is a daughter of the household that shelters him who speaks the tradition's charter of hire: "Father, hire him: a strong, trustworthy man is the best to hire" (28:26, in Abdel Haleem's rendering) — and the prophet is duly engaged, for wages, for a term of years. Hiring use and hiring labor sit in the same lawful house, exactly as they did in Rome's *locatio*. On that floor the jurists built the *ijara* with their usual precision: chapter 3 gave the definition — a sale of usufruct, *manfa'a*, a known use for a known price — and the schools' machinery enforced each word. The use must be known and specifiable, or gharar takes the contract; the price must be known, or the same; and — the clause that does this chapter's work — the thing whose use is sold must be the kind of thing that has a use apart from its consumption. Money fails the entrance exam here as everywhere: the fiqh's rental instrument simply cannot be written over dirhams — Rav Chama's hoe question, answered the same way in Arabic. What the three rent books share, beneath the different gates, is the settlement this Part keeps finding: the traditions were never against capital earning through time. They were against capital earning from time alone — and the lease is their standing proof that the distinction is real, an instrument in which every month's payment buys that month's delivered, perishable, unrepeatable use.

One wing of the rent book deserves its own sentence before the chapter turns, because Rome's category and the traditions' care both cover it: the hire of persons. *Locatio* let a man's services as well as his mule, and the scriptures that police the lease of labor are among the fiercest texts this book has quoted — wages owed before sunset because the laborer is poor and his life depends on them, in Deuteronomy's command from chapter 4; "you shall not keep for yourself the wages of a laborer until morning," in the Leviticus

clause chapter 22 carried; the worker paid “before his sweat dries,” in the Prophet’s word from the same chapter. The logic is this chapter’s own, run at human stakes: hire prices delivered labor as rent prices delivered use, day by day as it flows, and holding back the payment after the use is spent converts an honest hire into a forced and secret loan — the laborer financing his employer overnight, at the price of his supper. The traditions guarded the small landlord of one pair of hands more jealously than any owner of courtyards.

The Burdens That License the Rent

If the lease’s income is honest because ownership stays home, then everything turns on what “ownership” is made to mean — and here the traditions speak with one voice, in the lawyers’ oldest maxim about loss. *Res perit domino*, chapter 3 already carried it: the thing perishes to its owner. Ownership is not a name on a registry; it is the seat where the thing’s own fortunes land. The rent is licensed by the roof: whoever owes the repairs may charge for the shelter.

The modern fiqh has written that sentence into regulation with an exactness worth quoting, because the AAOIFI ijara standard reads like the old anatomy annotated by accountants. The lease must be preceded by acquisition — the institution acquires the asset or its usufruct before any lease of it can bite (SS 9, art. 3/1): you cannot sell a flow from a tree you do not yet own, the same gate chapter 24 watched the murabaha standards build. The lessor “must accept responsibility for any defects... which impair the intended use” and “may not exclude his liability” for them (5/1/5) — no contracting out of the ownership seat. He may not shift the major maintenance, the keeping-the-thing-capable-of-service, onto the lessee (5/1/7); the insurance is his to arrange and his to pay (5/1/8). If late payment tempts him toward the meter, the standard bars the door with the discipline chapter 22 walked: no increase in rent may be stipulated for delay (6/3). Even the modern hunger for floating rates got its answer in the lease’s own grammar rather than the loan’s: rent for later periods may float on a benchmark — the thermometer of chapter 24, admitted here too — but only under “a clear formula which is not subject to dispute,” and capped with a ceiling and a floor (5/2/3), so that what varies is still a price being agreed, not a meter running free — the lease may borrow chapter 24’s thermometer precisely because the thing being priced is real and the cap keeps the borrowed number a pricing aid rather than an engine. Note too what the maintenance rule does and does not forbid: the wear of use — oil, filters, the scratches of the lessee’s own hands — may honestly sit with the user, for it is his use that makes them; what may not move is the owner’s kind of upkeep, whatever keeps the thing capable of service at all. The line between a user’s wear and an owner’s substance is the whole lease drawn in a maintenance schedule. And at the bottom of the standard sits Celsus, translated into paragraph numbering: on the total destruction of the leased asset, the ijara terminates (7/1/3). The body gone, the right gone, the rent gone. No clause in the classical instrument can make a payment outlive the thing it pays for.

Sit with how much moral work that battery of burdens does. The lessor’s return arrives through time, month after month, and yet it is not *agar natar*, not a wage of waiting — because at every moment of the waiting he is long a real asset with real enemies: rot, rust, fire, obsolescence, the defect that surfaces in year three, the roof that owes him a winter’s repairs. His income is the fruit of something he still owns, and the owning costs him something every day. Strip those burdens away one by one — make the tenant insure, make the tenant repair, make the tenant carry the destruction risk, promise the owner his stream in every state of the world — and with each clause the owner’s seat empties a

little more, until what remains is a man collecting a schedule: near to profit, far from loss, the baraita's exact posture, achieved not by guarantee this time but by upholstery. The burdens are not friction in the lease; they are the license for its price, and every one of them is auditable — which is why chapter 10's rule scales here without strain: the income justified by the owner's peril lives exactly as long as the peril does, whether the thing at risk is a widow's kettle or a fleet of aircraft.

The Landlord's Coat

Which brings this chapter to the second machinist's paper, and to the promise chapter 3 made when it first showed the honest lease its floor: we would watch what happens when the floor is sanded away — when an instrument “quietly transfers every burden of ownership to the user and keeps only the income, at which point it has become,” in that chapter's words, “a secured loan in a landlord's coat.”

The financial lease did not need a conspiracy to evolve; it needed only the drift this book has watched in every century. A manufacturer's finance arm, a bank's leasing desk, discovers that owning ten thousand machines is a nuisance but financing ten thousand machines under a landlord's title is a business. So the paper keeps the lease's name and sheds its seat, clause by clause. The lessee maintains — not the wear of use, which even the classical law might apportion, but everything, structure and substance, the whole of the owner's ancient duty. The lessee insures, naming the lessor as beneficiary. The lessee bears destruction: the hell-or-high-water clause of the opening, standard furniture in equipment finance, obliges the payments though the machine burn, sink, or seize. The term runs the asset's whole useful life, so nothing of substance ever really returns to the “owner”; at its end a purchase option for a dollar, or a token sum, tidies the title into the hands that held everything else all along. And the payments themselves are built like the thing they are: cost of the machine, plus a return on that cost for the time outstanding — an amortization schedule wearing rent's name. Run chapter 24's costume test and every answer comes back reversed. Did the financier own the thing in the world, or in a folder? Did its risk ever sit with him for one insurable hour? Is the payment a price for delivered use, or a claim that matures by dates? Could the arrangement be disappointed by anything but nonpayment? The fire already answered. A payment that outlives its machine was never rent.

The lease even acquired its own double sale, completing the costume family chapter 24 catalogued. In the sale-and-leaseback, a company sells its own building or fleet for cash and leases it back in the same set of signatures — two impeccable contracts, and the net of them is money now against a schedule of payments later, secured on an asset that never moved and never changed hands in any sense the eye could catch. Run honestly, with the burdens genuinely migrating to the buyer-lessor, the device can be a legitimate way to put sleeping capital to work; run as drafting, with the “seller” still insuring, maintaining, and bearing the fire, it is the 'ina of chapter 3 with a title deed — and the reader who has come this far can now grade any given specimen without help.

What is striking — and this book keeps finding the pattern — is who said so first and most plainly: the systems with something at stake in the word “lease.” The accountants surrendered the label a generation ago. Since the first of January 2019, the main international standard requires the lessee of nearly any substantial lease to book a right-of-use asset and, beside it, a lease liability for the payments owed (IFRS 16) — the drafting profession's own confession that the tenant of such paper is, in substance, a debtor who has financed an acquisition, whatever the document's title page says. Substance over form:

Gregory v. Helvering's rule and the Majalla's third article, chapter 3's twin authorities, arriving at the ledger. And the fiqh, with the largest religious lease-finance industry on earth to police, patrols its own version of the drift in its own voice. Usmani's warning to his industry is the chapter's thesis from inside the tradition: leasing "was not originally a mode of financing," and when institutions adopt it as one, "it is not sufficient... to substitute the name of 'interest' by the name of 'rent'" — "there must be a substantial difference between leasing and an interest-bearing loan." The instrument the industry built at the boundary, the *ijara muntahia bittamleek* — the lease ending in ownership, Islamic finance's lease-to-own — shows the guardrails the tradition insists on when it walks this close to the edge. The rentals may resemble a financing schedule; the customer may fully intend to own the machine at the end. But the standard keeps the anatomy honest at the joints: throughout the term the lessor's ownership burdens stay put — maintenance, insurance, destruction — so the *ijara*'s rent remains licensed rent (the fire in year nine still lands on the lessor, and the rent still dies with the machine, 7/1/3); and the transfer of title at the end may not simply happen inside the lease. It rides on a promise recorded in a document separate from the lease itself (8/1), and when the day comes, "a new contract" must be drawn — a gift or a sale, executed then, at its own moment — "since the ownership to the property is not automatically transferred" (8/3). Two real contracts, in sequence, each with its own anatomy, rather than one financing wearing two masks. Whether the industry's practice honors the seams or merely draws them — the question chapter 24 asked of the benchmark, and chapter 32 will ask of the whole — is a live quarrel among the tradition's own scholars, which is precisely the health this book keeps pointing to. A tradition still arguing about its instruments is a tradition still awake at the gate.

One more figure stands in this section, and the register owes him plainness: the second machinist himself. The financial lease concentrates its comfort upstream and its catastrophe downstream — the small operator who loses shop, machine, and livelihood in one night and still owes four years of payments on the ash has been placed, by drafting, in the seat the traditions spent three thousand years refusing to let the weak occupy: every burden of ownership, no shred of its protection. Whatever else the old rent books were, they were agreed on this: the seat that collects the income keeps the risk, and a document that separates the two has not discovered a new kind of lease but rediscovered a very old kind of loan — secured, this time, more thoroughly than the old law would ever have allowed, for the coat keeps the collateral and the claim together: repossess the machine, sue for the balance, and call the whole arrangement rent.

The Lease That Stays Honest

It remains to say what this chapter's instrument looks like when it is doing what the traditions built it to do — because Part VII's assignment is not autopsy but architecture, and the lease, honestly run, is the most naturally non-extractive financing instrument in the whole toolbox. It requires no partnership appetite, no profit audit, no oath at settlement; it only requires that ownership mean what it says.

The specification assembles itself from everything above, and it reads like a checklist because it is one — the ToC's own design bullet, rendered as clauses. The financier owns the asset in the world: acquired before let, registered, insured at his expense, his problem when it fails (3/1; 5/1/5–5/1/8). The rent prices delivered use: it begins when the asset is usable, pauses when the asset is not, scales — where the parties wish — with hours run or output taken, and may float only on a formula with a ceiling and a floor (5/2/3). The term is shorter than the asset's life, or honestly renewed at renewal's real price. Destruc-

tion terminates (7/1/3); delay meets triage and, if discipline is needed, the charity-routed pressure of chapter 8, never a meter (6/3; the mercy of 2:280 for the genuinely stuck). Any path to ownership at the end is a real second contract at a real moment — promise, then transfer, seams showing (8/1; 8/3). Nothing in that specification is exotic; every clause is in force somewhere today, in standards this chapter has quoted. The instrument scales institutionally, too, and always has: the great endowments this Part will reach in chapter 29 — waqf, hekdeshe, church property, the asset-locked estates of Steward Ownership's ancestors — earned much of their keep across the centuries precisely as standing lessors, patient owners letting real things at honest rents. An economy could finance its workshops, its vehicles, its looms and kilns and tractors this way at any scale it chose — the owner-financier earning a rent licensed by roofs he genuinely must fix, the user paying for use and walking whole out of the fire. Productive assets financed without extraction is not a utopia; it is a rent book, kept the old way.

The lease also hands Part VII its next instrument. An owner with one warehouse can lease it; an owner who needs to raise the warehouse's price from a hundred savers must cut its ownership into paper — certificates that entitle each holder to a slice of the asset and its rents — and the moment ownership becomes paper, every question this chapter settled comes back at scale: whose burdens, whose risk, whose fire? It is no accident that when modern Islamic finance went looking for an instrument to stand where bonds stand, it reached first for the lease — the classic sukuk template is a leased asset cut into certificate form — nor any surprise that the drift this chapter watched in equipment finance reappears there, as chapter 26 will show. The fire test travels; chapter 26 will strike the match under paper. What this chapter adds to the book's argument is a quiet, load-bearing distinction the traditions never let go: time was never the enemy. The lease collects through time, month after month, and every tradition blesses it — because what time delivers under a lease is use, real as bread, from a thing the owner goes on owning, insuring, repairing, and losing. The usurer's sin was never patience; it was selling the calendar while owning nothing that lived inside it.

Chapter 26 — Asset-Backed Certificates: Sukuk, Bonds, and Participation

This Part opened in Genoa, with a pope answering an archbishop about merchants' paper, and it is deliberate architecture that the Part's third instrument brings us back — to the same city, two centuries and more on, standing in the counting hall of one of the most remarkable financial institutions of the late medieval world. In 1407, after more than a century of piecemeal expedients, the Republic of Genoa gathered its public debts into a single house, the Casa di San Giorgio. The republic's creditors, great and small, held *luoghi*: shares of a nominal hundred lire apiece, "without materiality," as the great historian of the Casa puts it, existing as lines in a register. A luogo could be transferred to anyone by simple registration in the official creditors' ledgers; the shares were actively traded, and speculated in, for four centuries. And what the paper held was not a bare promise. Each underlying *compera* was, in the standard definition, the right to collect, for the holders' benefit and for the entire term, a named tax — already levied or created for the purpose — so that the man who bought a luogo bought a seat at a particular fiscal table: a gabelle already levied or created for the purpose, a duty of the port — revenues with names, yields that rose and fell with the trade of the city. Genoa had invented, or perfected, something this book has not yet examined and the modern world cannot live without: the claim cut into equal pieces and sent traveling — capital raised from a crowd, paper in ten thousand hands.

Paper at scale is Part VII's hinge, because everything the last two chapters established was established between people. The murabaha's candor, the lease's burdens — these are disciplines of a relationship, enforceable because somebody is looking somebody in the eye. A certificate abolishes the eye contact. The holder of piece four hundred and seven has never seen the aircraft, the tax farm, the venture; he may hold his piece for an afternoon; he bought it from a stranger and will sell it to a stranger. The question this chapter asks is the whole moral content of securitization, and it can be put in five words: what does the paper hold? Because two lineages of certificate run through history side by side, alike to the eye and opposite in anatomy. One is paper that *owns* — a fractional seat in a real thing and its real fortunes. The other is paper that *is owed* — a fractional loan, the mutuum of chapter 7 cut into coupons and set loose to compound among strangers. Every tradition this book reads blessed the first kind in principle and built versions of it in practice; every tradition's wall stands against the second; and the drift from the first to the second — accomplished quietly, clause by clause, while the name on the paper never changes — is the most consequential costume story in modern religious finance.

Paper That Is Owed

Take the bond first, because its anatomy is already in the book. Strip the trading floor away and a conventional bond is chapter 7's contract in formal dress: a sum delivered, ownership of it passing utterly to the issuer; a promise of the sum's return at a date; and between delivery and return, the coupon: increase, calculated on principal and calendar, precisely the *agar natar* the traditions walled out, now stamped in convenient denominations. Nothing about cutting the loan into a thousand equal pieces changes the analysis; it changes the scale and the sociology. The bond made the creditor's seat fungible: anyone's money

could occupy it, briefly, anonymously, and the seat's terms never renegotiate because no one sits in it long enough to be anyone. Chapter 2 watched the Torah's lender bound to a brother whose face he knew, and watched the modern credit market complete what Nelson called the journey to universal otherhood; the bond is that journey's vehicle. A debt owed to everyone in general is a debt owed to no one in particular, and no one in particular has ever forgiven a coupon.

Two honesties must be entered at once, because the traditions enter them. First: the certificate was never the sin. Writing debt down is not merely tolerated but commanded — the written and witnessed debt of chapter 6 — and the halakha's whole civil law runs on instruments; a paper acknowledging that a hundred is owed, transferring for what buyers will pay, offends nothing in any of the three traditions so long as the hundred is all that ever comes back. The trouble is the coupon, not the certificate. Second: the traditions saw early and clearly that paper is a different kind of thing from the things it points at. The Mishnah's *ona'ah* carve-outs, chapter 9 listed them, exempt documents from the price-fraud rules — because a *shtar* is not a commodity with a market worth but a claim, worth whatever the promise inside it is worth, and the ordinary machinery of overreaching has no purchase on it. The market learned the same lesson from the trading floor. A bond's price moves with nothing but arithmetic — when the going rate of money rises, yesterday's coupons are worth less, and the paper falls without a single fact about any venture changing anywhere — because there is no venture; the certificate's entire interior is a rate and a date, and its market value is the calendar repricing itself. Ownership paper cannot move that way. Its price carries news — a lean harvest, a grounded fleet, a good year at the port — and the difference between paper that trades on news and paper that trades on arithmetic is chapter 7's whole distinction, quoted daily at market close.

And so the traditions' honest lane for debt paper is narrow and real: the certificate may carry principal across hands and across time, and it may not breed. A written qard, a *shtar* for a hundred that a hundred discharges, a bill that moves a debt to a new holder for what buyers freely pay — all of this is commerce in claims, blessed and ancient. The line is crossed at the coupon, and crossed again, more quietly, wherever paper is issued at a discount that is simply the coupon folded into the price. What the wall demands of debt's paper is exactly what it demanded of debt: that the claim come home the size it left. Sitting inside that dry procedural rule is the insight this whole chapter runs on: paper has no value of its own; it is a window, and everything depends on what stands on the other side of the glass — a thing, with fortunes; or only a promise, with a date.

Paper That Owns

The other lineage is older than the bond and, in every tradition, cleaner. If a venture, a field, a building can have one owner, it can have forty; the law needed only to learn to write fractions. The rabbis wrote them in partnership deeds — the *shtar iska* of chapters 11 and 13 is already a certificate, paper whose holder owns half the venture's deposit-half and eats its outcome, and nothing in principle stopped such deeds from finding new hands. The instrument's modern institutional career, chapter 14 showed, runs through the *heter iska klali* — one master partnership document standing behind an entire bank's book, every depositor holding, in form, a sliver of a venture rather than a claim to interest; paper that owns, at national banking scale, with all the questions about whether anyone remembers the ownership that chapter 31 will press. The *fiqh* wrote fractions with its usual definitional care, and its modern standard-setters have compressed the lineage into a definition this chapter can hang everything on. Investment *sukuk*, runs the AAOIFI standard's for-

mula, are “certificates of equal value representing undivided shares in ownership of tangible assets, usufruct and services” — undivided shares in ownership: the holder of a sukuk certificate is not owed the aircraft’s price; he owns, with ten thousand co-owners, an undivided sliver of the aircraft, the way heirs own a field before partition. The classic structure, as chapter 25 promised, is the lease bound into paper: the holders’ undivided aircraft is leased to the airline, the rent flows through to them pro rata, and every doctrine of the last chapter comes with the package — their rent licensed by their ownership burdens, their income dying, in principle, with the asset in the fire. The plumbing is modern — the holders’ undivided title is typically parked with a trustee, the certificates evidencing each stranger’s fraction — but the doctrine riding in the pipes is chapter 25’s, unchanged: distributions are rent, rent is licensed by ownership’s burdens, and if the aircraft is lost the certificates hold ash and an insurance claim, not a right to be made whole by anybody’s promise. Paper multiplies the owners; it must not be allowed to subtract the ownership. The Fiqh Academy’s sukuk resolution states the resulting rule in one clause: the return on all such certificates “should be stemming from income-generating assets” — from the named kitchen, not from the calendar.

And Latin Christendom, though the textbooks rarely put it this way, ran one of history’s great experiments in participation paper. The luoghi of San Giorgio were not bonds in the modern sense, whatever later centuries made of the family they founded. A luogo’s yield was not a stipulated coupon; it was a *provento*, a distribution of what the assigned taxes actually brought in, and the holders stood, as a body, in the fiscal weather of Genoa: rich years and plague years, war and the loss of colonies, all of it passing through to the paper. (Whether a luogo’s yield was fruit or disguised interest was itself argued, on and off, for as long as the Casa stood — the census quarrel of Part VIII in miniature, and proof that the drift this chapter tracks was visible to consciences long before it had a compliance department.) The scholastics’ comfort with such instruments followed the logic this book has traced for three chapters: a man who buys, for a price, the fruits of a named and fruitful thing has bought something real, and bears its fortunes; the quarrels — and they were real quarrels, running for centuries, about when a purchased revenue-stream is a harvest and when it is a disguised loan — belong with the census family and the monti of Part VIII, where the early modern age will fight them at full length. (A disambiguation, before the Part VIII door closes: the papal and civic *monti* of public finance are not the Monti di Pietà of chapter 17 — the same word, a mountain of money, naming a lending charity in one century and a debt consolidation in another.) What matters here is the design point Genoa proves: participation paper is not a modern invention struggling against ancient categories. It is the ancient category — the *societas* of chapter 11, given a register and a crowd — and for centuries it financed ports, fleets, and public life on exactly the terms the traditions could bless: many small owners, real assets, named revenues, outcomes shared.

The three traditions’ paper, set side by side, agrees on the load-bearing point with an exactness that should no longer surprise any reader of this book. A certificate is honest when it conveys a genuine fraction — of a thing, of a venture, of a named stream of fruits — and its holder eats what the fraction earns and loses what the fraction loses. The Talmud’s deed-holder awaits the venture’s actual accounting; the luogo-holder’s *provento* shrinks when the port is quiet; the sukuk-holder’s distribution is rent from an aircraft he co-owns, and the Fiqh Academy requires him to bear his proportional risk as the price of his proportional gain. Participation, in other words, scales. What does not scale — what the crowd, the register, and the secondary market grind away at from the first day — is the appetite for participation’s terms. Ten thousand strangers did not buy the paper to stand in fiscal weather. They bought it to be paid, evenly, quarterly, at par; and the history of the certifi-

cate is the history of drafting that tries to give them exactly that while keeping ownership's name on the instrument. Notice, before the drift is walked, what the three old papers all got right about transfer itself: the ledger. Genoa's line in the creditors' register, the sukuk trustee's record, the deed in the partner's strongbox — in each, the paper travels freely while the obligations of ownership attach, entire, to whoever holds it at the moment the roof leaks. Transferability was never the corruption; the corruption is drafting that lets the benefits travel while the burdens stay behind with someone else.

The Four Things a Certificate Can Hold

Before the drift can be watched honestly, the reader needs the audit, and it has four layers. Ask of any certificate what its holder can actually reach — a question a careful buyer can put to any prospectus, and the first module of the analyzer chapter 43 will build — and every security ever drafted holds one of four things. It can hold the thing itself, true fractional ownership: the asset's title runs to the holders or their trustee, and if every promise around it burned away tonight, the holders would still, at law, own the aircraft. It can hold a usufruct, not the tree but its fruit for a term: chapter 25's right in a body, alive exactly as long as the body — the sukuk standards themselves list usufructs and services beside tangible assets, so that paper may hold a term of a thing's use, or of a service to be rendered, and the holder's income is the fruit of that term and dies at its end. It can hold a revenue assignment (no title, no usufruct, but a named stream): a named tax, a toll, a schedule of lease payments, flowing to the paper as long as the stream flows — the compera's direct descendants live on in every instrument that finances a bridge from its own tolls, and the moral character of such paper turns entirely on whether the stream's failure is the holder's weather or somebody's guaranteed problem. Or it can hold a promise about all of the above, and here is where the reader must slow down, because the fourth layer devours the other three. A guarantee that the holder will receive his capital back at par, whatever becomes of thing, fruit, or stream — a binding undertaking by the issuer or manager to repurchase the assets at the stated price “regardless of their true or market value at maturity,” in Usmani's exact and exasperated description of the market's standard clause — does not add safety to ownership; it replaces it. The holder who cannot lose no longer owns anything; he is owed something, by the very party whose performance was supposed to be the venture's, and the thing, fruit, and stream have quietly become collateral for a loan whose name is still Arabic. A guarantee is how paper resigns from ownership.

The fiqh's standard-setters, to their credit, wrote this analysis before the critics did, and chapter 13's fetched texts carry it verbatim into this one. A sukuk manager — *mudarib*, partner, or agent — guarantees the capital only “in case of his negligent acts or omissions or his non-compliance”; beyond fault, trust capital may not be guaranteed at all, and instruments may not be “marketed or operated as a guaranteed investment.” Combine agency with guarantee and the drafting itself, says the standard, “turns the transaction into an interest-based loan” — *tantamount to riba*. A genuinely independent third party may volunteer a guarantee of capital, in a different capacity and for no fee — the narrow gate, and note how narrow: the moment the guarantor is anyone inside the venture's economics, the paper has flipped layers. The Fiqh Academy's resolution draws the same two lines with a jurist's economy: no guarantee by the manager except for transgression or negligence, and no redemption at nominal value — the exit price must be the market's or an honestly agreed one, never par-by-promise. Redemption at par is the guarantee wearing its maturity-date clothes; the fixed number of chapter 14, written not into a partnership this time but into ten thousand certificates; and the same verdict follows it: void where the

tradition can reach it, riba where it cannot.

The Drift Back to Bonds

Now watch the drift run, because the ToC's word is exact — nobody legislated the certificate back into a bond; it drifted, one accommodating clause at a time, under the pressure the section above named: the crowd wants par and coupons, and the industry wants the crowd. And the crowd deserves its steelman before its correction, because the crowd is not a villain. It is the schoolteacher's retirement account, the small trader's reserve, the modern versions of every widow this book has met holding paper because paper is what a person of modest means can hold; such holders cannot audit a venture and know it, and their hunger for a certain number is the hunger for safety of people who have exactly one number to lose. The traditions never mocked that hunger. They answered it with different institutions — the pools and endowments this Part is walking toward — rather than with ownership's name on a creditor's terms. The distributions came first. A true participation pays what the assets earned, which is lumpy; the market's sukuk paid, in Usmani's words, on rates "not linked to the expected profits from the enterprise, but to the costs of financing or to the prevalent rates of interest" — LIBOR's thermometer again, wired now not to a markup but to an ownership instrument. The excess went next: whatever the assets earned above the benchmark was assigned to the manager as an "incentive," which inverts partnership's whole geometry — the crowd holds ownership's name and collects a creditor's stipend, while the insider collects ownership's upside for managing the costume. Shortfalls were smoothed by the manager's advances, until the 2008 resolution barred the manager from undertaking to lend against them, leaving only the honest, disclosed reserve chapter 13 met. Even the accounting rules inside the standards hold the line chapter 14 admired: receivables enter the profit reckoning at their honest collectible value, with "neither time value (interest rate) nor discount" admitted into the computation — the tradition refusing, down in the bookkeeping, the very quantity the drift smuggles in up on the term sheet. And underneath the cash mechanics ran the deepest clause of all, the lawyers' distinction the industry itself coined: *asset-backed* versus *asset-based*. In a true asset-backed issue, the assets are actually sold — perfected, bankruptcy-remote, reachable — and the holders' fate genuinely rides on them. In the asset-based issue that came to dominate the market, the "sale" is beneficial only: title never perfects, the holders' recourse at failure runs to the originator's balance sheet like any unsecured lender's, and the assets are scenery. Paper that owns, as drafted; paper that is owed, as enforced.

The tradition's own conscience said all of this first and loudest. Usmani's paper — written from inside the industry's own standards body — concluded that "most of the Sukuk that have been issued are identical to conventional bonds with regard to the distribution of profits," and that the market's favorite mechanisms exist precisely so the instruments "take on the same characteristics as conventional, interest-bearing bonds"; his summary rule for repair is five words of pure Part III: "reward always follows after risk." The AAOIFI board answered with the February 2008 resolution chapter 13 fetched — the guarantee bans, the reserve discipline, the no-par-repurchase rule — and the Fiqh Academy's 2009 Sharjah session, the same sitting that refused organized tawarruq in chapter 24, issued the sukuk resolution quoted above. Three internal verdicts in two years, from the tradition's most senior juridical machinery. Whether the market obeyed — and what it means that the certifying and the structuring are so often paid for by the same issuer — is a question chapter 32 will weigh with the whole industry on the scale.

But one specimen belongs here, because it put the drift's every layer before a court, in

public, with dates. In June 2017, Dana Gas — a Sharjah company with some seven hundred million dollars of *mudarabah* sukuk outstanding since 2013 — announced that its own certificates were “not Shari’a compliant and... therefore unlawful under UAE law,” suspended the distributions, and took injunctions in three jurisdictions to hold off its creditors. The market was appalled in both directions at once, and both directions matter for this chapter. The holders discovered what layer their paper really held: when the *mudarabah*’s compliance was impeached, their protection turned out to be the English-law purchase undertaking — the par-repurchase promise — which the English High Court, in November 2017, held “valid and enforceable in accordance with its terms” precisely because it was an ordinary English contract whose obligations did not depend on the Islamic structure at all. The paper was enforced, and what was enforced was not ownership of anything; it was the fourth layer, the promise, standing alone in an English courtroom, severed from the structure it had been drafted to complete. And the issuer’s maneuver taught the opposite lesson to the same audience: a compliance doctrine invoked for the first time at the moment it became convenient to stop paying looked, to analysts and to the tradition’s scholars alike, less like scruple than like a borrower discovering theology at the due date. The affair ended as such affairs end — a June 2018 restructuring, holders exiting at ninety and a half cents or rolling into new three-year paper — but its testimony stands. When the costume was contested, neither side defended the ownership the certificates proclaimed. They fought about the promise, because the promise was always the instrument.

From Backed to Locked

The repair the ToC points to is not better backing. It is a different relationship between the paper and the thing, and the old institutions already modeled it. “Backing” is a snapshot: assets stand behind the paper today, and tomorrow they are swapped, released, repledged, or — the asset-based lesson — revealed never to have left the issuer’s hands at all. What is merely backed can be quietly unbacked; what is locked has taken vows. Genoa’s compere locked: the assigned tax belonged to the holders for the entire term, by the instrument’s own definition, not as collateral they might race other creditors to but as the very thing they had bought. The endowments this Part reaches in chapter 29 — *waqf*, *hekdes*, church property, and their modern descendants in Steward Ownership’s asset locks — are the lock’s full development: property bound to a purpose so that no holder, manager, or successor can strip it. Between the snapshot and the vow runs the whole spectrum of honesty in securitization — the bare representation, the negative pledge that merely promises not to encumber, the perfected security interest, the true sale, and at the far end the purpose-bound corpus that no one may alienate at all — and the traditions’ verdicts, assembled across this chapter, sit at the vowed end. Drafting can locate an instrument anywhere on that spectrum; what it cannot do is make the middle of the spectrum feel like the end to the person holding the paper, which is why the disclosure duties of chapter 23 travel with every certificate this chapter has blessed.

So the specification the ToC asks for — what non-extractive securities would require — can be written as the audit run in reverse, and every line of it is already in force somewhere in the instruments this chapter has quoted. The transfer is true: title perfects, the holders or their trustee reach the thing itself at failure, and the originator’s insolvency is the paper’s weather, not its grave. The return stems from the named fruits — the Academy’s income-generating assets, the compere’s salt tax — and distributions track what the assets earned, smoothed only by the disclosed reserve, never by advances that are loans in waiting. Losses land on the holders *pro rata*, as the price of their gains; the manager is paid as a

worker or a partner — a fee, a ratio — and never stands as guarantor of the capital his own conduct steers, save for his proven fault. There is no par-by-promise: exits, redemptions, and repurchases happen at market or honestly appraised value, so the certificate can be disappointed, which is the entire moral point. The waterfall is disclosed whole, chapter 23's standard, so the crowd knows which layer it is buying. And where the financed thing serves a purpose beyond its yield — the utility, the housing, the working forest — the asset is locked to that purpose, chapter 29's move, so that the paper's travels through ten thousand anonymous hands can never carry the thing itself out the door. And because certificates travel, the specification must travel with them: every resale re-poses the four-layer question to a new stranger, and honest paper is paper that answers it identically in every hand — the tenth holder reaching the same thing, bearing the same weather, reading the same disclosed waterfall as the first. A security that is participation at issuance and something else by the third trade has merely scheduled its costume change. None of this is exotic; all of it is costly, in exactly the way honesty has been costly in every chapter of this Part: it surrenders the sale-ability of certainty. Paper that owns cannot promise par, and the crowd pays less for weather than for promises. The non-extractive security is, in the end, participation priced as participation — the oldest door in the book, cut into equal pieces without cutting away the doorway — and whether a market can be built that prefers it is not a drafting question but the civilizational one this book's last Parts take up in earnest — chapter 40's return formula, chapter 41's fund design, and the sample waterfalls of Appendix E are, among other things, sketches of what such paper would pay and to whom.

One instrument remains in Part VII's workshop before the endowments close it — the one that answers a different fear. Sale, lease, and certificate finance what people can do; none of them touches what people dread. The pool — insurance, takaful, the mutual aid that chapter 2 called the shield and chapter 28 will render as mathematics — is where the traditions took the logic of participation and pointed it at catastrophe itself, and chapter 27 opens that door. The paper of this chapter will be waiting when the book returns to design: a certificate, honestly cut, is how a community owns a big thing together, and the last Parts of this book are, among other things, an argument that the crowd deserves better instruments than the coupon — paper with weather in it, paper that knows what it holds. Genoa's luoghi held the taxes of a real port, and the paper rose and fell with the ships. That, and not the promise of par, is what it has always meant for capital to be in the world rather than over it.

Chapter 27 — Insurance, Takaful, Mutual Aid, and Risk Pools

Modern insurance entered history in a costume, and the costume was a loan. Chapter 10 left Mediterranean commerce holding a problem: the sea loan — capital advanced against a voyage, the debt dying with the ship, a stiff premium if she arrived — stood condemned by name in the *Liber Extra* of 1234. The lender's premium was payment for carrying the sea's peril, and Naviganti refused to let a hazard clause consecrate a creditor's gain. But the peril did not go away because the premium did. Ships still sank, no merchant could stand alone under a cargo's whole fate, and carrying the sea remained real work with a real price. So the notaries of the Italian ports went to work, and the historian who assembled their paper trail, Florence Edler de Roover, found the solution taking shape under exactly the dress this Part has learned to check for. In the earliest contracts the carrier of the risk appears as a borrower: he acknowledges receipt of a loan, money he had never in fact received, repayable if and only if the ship failed to arrive. If she docked, the fiction dissolved and he kept his true premium; if she was lost, he "repaid" what he had never borrowed, and the merchant was made whole. Other drafts ran the same protection through a fictitious sale. The Church's usury doctrine had as yet no shelf for protection sold plain, so protection traveled under names the law already trusted — and the older scholarship de Roover engaged read Naviganti as the push that sent the merchants hunting. By 1329 the documents of Palermo show the premium contract stepping into daylight; by the 1370s the notarial books of Genoa carry premium insurance as ordinary business; the earliest policies that survive at all come from the Italian ports of the 1340s.

The joke history was telling here runs at this Part's expense, instructively. Three chapters have watched suspect things dress as honest ones — loans as sales, loans as leases, bonds as ownership. Here the traffic ran the other way: an honest thing entered the world wearing the suspect thing's clothes, because it was so new it had no name of its own. And the unbundling underneath the disguise was the very analysis Naviganti had demanded: the sea loan had charged for two services — capital for the venture, and a shoulder for the venture's catastrophe — through one premium riding one debt, and the decretal struck the bundle. The market's eventual answer, once the costumes wore out, was to sell the two services separately: the merchant found his capital where capital was lawfully found, and the insurer carried the sea alone, for a premium that paid for the carrying and for nothing else. That is the story chapter 10 promised this chapter would finish. Insurance did not evolve out of gambling, and it did not evolve out of charity; it evolved out of the usury question — the honest remainder left standing when the Church audited the sea loan and struck the part that was a loan.

The instruments of the last three chapters finance what people intend: the purchase, the workshop, the fleet cut into certificates. This chapter's instrument faces the other way, toward what nobody intends — fire, wreck, death, the failed season — and toward the dread of it that shadows every household this book has visited. Chapter 3's taxonomy gave the instrument its own first-class name, the solidarity pool, and posted at its door the question this chapter exists to answer: whether protection is pooled among members or sold at a margin, and what each arrangement does to the people inside it. But an older accusation must be answered first, for it has been leveled at this instrument in every century of its life, with real evidence behind it. The premium is certain; the payout hangs on chance. Is the

whole business a wager?

The Wager and the Fog

State the resemblance at full strength before answering it. A man pays a small sum now; a contingency decides whether he collects a large sum later; the counterparty keeps the small sum either way. Drawn as a payoff diagram, the fire policy and the bet on the fire are the same picture, and for a long time the law of the world's busiest insurance market could not tell them apart either. Eighteenth-century London ran the experiment to its limit. A Londoner with nothing aboard a ship could buy a policy on her safe arrival — the trade wrote such contracts "interest or no interest," meaning the buyer need prove no stake beyond the paper itself — and a Londoner could insure a stranger's life and follow the coffee-house odds on his surviving the season. Nothing in the instrument's form distinguished these wagers from a merchant's honest cover: same clauses, same premium, same contingent payout. Parliament eventually drew the line the form could not hold. The Marine Insurance Act of 1745 voided the ships' wagers — policies written interest or no interest were declared null and void — and in 1774 the Life Assurance Act did the same for lives: experience had shown, its preamble runs, that insuring lives and events "wherein the assured shall have no interest hath introduced a mischievous kind of gaming." The remedy became the common law's permanent doctrine: no policy without an insurable interest — a stake the buyer already holds, such that the insured-against day would genuinely wound him — no recovery beyond that stake's value, and the beneficiaries named on the face of the policy. The wager-buyer and the honest policyholder are told apart, in the end, by a single question: which day is each of them waiting for? A man with cargo in the hold dreads the storm his policy covers. A man holding a naked wager on the same ship has been handed a reason to pray for the storm — and, the eighteenth century learned, sometimes to arrange it. Chapter 22 heard the canonists ask the same question of the penalty clause — the lender must not be made to want the default — and the insurable-interest doctrine rebuilds that finding in an underwriter's office: a contract of protection stays honest only while its holder still needs the protected thing to survive.

Islam had run the analysis on gambling itself eleven centuries earlier, and its terms cut to the same joint. The Qur'an's word is *maysir*, and its original target, the commentators say, was the arrow-game of the jahiliyya — shares in a slaughtered camel riding on marked shafts drawn blind. Its first pass at the subject is surprisingly patient: "There is great sin in both, and some benefit for people," it says of wine and gambling together (2:219, here as throughout in M. A. S. Abdel Haleem's rendering). The patience does not last: gambling stands among the "repugnant acts — Satan's doing — shun them so that you may prosper" (5:90), and the ground given is not probability but fellowship: "With intoxicants and gambling, Satan seeks only to incite enmity and hatred among you" (5:91). Chapter 6 read the doctrine whole: the winner's gain is carved from the loser's substance, with nothing made and nothing traded, and enmity is what the table reliably produces. Against that doctrine the underwriters' statutes read like a Meccan finding in Georgian dress, and the distinction both were groping for can be said plainly. The gambler manufactures his peril; the insured already owns one. The world put the fire in the workshop and the storm across the sea lane; the wagerer's danger exists only because the stakes created it. Insurance and gambling are not neighbors on a spectrum of prudence but opposite answers to chance — one transfers a risk, the other conjures one — and every sound insurance doctrine in the world is a set of fences for keeping the second out of the first's clothing.

But the fiqh had a second question waiting, and it proved the harder one. Chapter 6 planted

it at this chapter's door: is a premium a purchase of indemnity, or of fog? Grant the peril real and the interest genuine, and look at the exchange itself. A known premium goes out against an unknown return — whether any indemnity will ever be paid, when, and how much, none of it knowable at signing. Run chapter 23's building code over that structure and the verdict writes itself: uncertainty seated in the contract's primary subject matter, excessive by any customary measure, inside an exchange-based contract. Sold as a sale, protection is a sale of fog. That analysis stands behind the modern story's opening ruling: meeting in Jeddah in December 1985, the Islamic Fiqh Academy resolved that the commercial insurance contract with its fixed periodical premium "contains major elements of deceit that void the contract and is therefore prohibited by Shariah" — the Academy's own English renders as deceit what its later resolutions name precisely: a *muawada*, a compensation contract, shot through with excessive *gharar*. Two exits, then. The common law kept the sale and policed it: insurable interest, indemnity caps, solvency rules. The fiqh judged the sale itself unsalvageable and moved the whole arrangement to a different shelf of chapter 3's taxonomy, and the building code's own conditions show the door it used: *gharar* voids exchanges, not gifts. A donation may walk into uncertainty that no sale may enter — chapter 23 marked the clause — so let the contributions be donations and the fund a common thing, and no one has sold indemnity to anyone; the uncertainty rides in a gift, where the law lets it ride. The 1985 resolution named the lawful alternative in one phrase: "the contract of cooperative insurance, founded on the basis of charity and cooperation." The shelf's name is the pool, and the traditions had been building on it for two thousand years before the modern argument made them say so exactly.

Three Rooms of Shared Weight

The antique floor under all three rooms is the burial club, and chapter 3 read this book's specimen off a temple porch at Lanuvium: the worshipers of Diana and Antinous, in the year 136, a hundred sesterces and an amphora of good wine to join, five asses a month thereafter, three hundred sesterces from the common chest when a member died, fifty of them distributed at the pyre. The by-laws' fussiness already anticipated everything this chapter will meet. Dues six months in arrears void the claim, a will notwithstanding; a death beyond twenty miles sends three members out with travel money; a slave member whose master withholds the body is buried in effigy. Its members were the town's small people — freedmen, slaves, laborers — and five asses a month stood between them and the pit without a name. Hold two of the club's plainer facts, because the chapter will lean on them. The dues were the same five asses for every member. And the chest paid for funerals and for nothing else: a year in which few members died enriched nobody. The light years were banked against the heavy ones, which is the entire actuarial science of the arrangement, and all it ever needed.

The rabbis, chapter 3 also showed, legislated for the same instrument with the same matter-of-factness, and its case law patrols the pool's hardest border. The donkey drivers of a caravan may stipulate that a member's lost animal will be replaced from the company; the sailors may compact boat for boat; and the exclusions arrive in the same breath — lost through his own negligence, no donkey; sailed where boats do not go in that season, measured in rope-lengths from shore, no boat (Bava Kamma 116b). The pool covers misfortune and refuses to cover carelessness, because a pool that pays for carelessness manufactures the very losses it exists to share — the moral hazard of the 1745 preamble, legislated against in a baraita. Even the payout's form is policed: the driver who asks for the animal's price in coin, promising to keep guarding the caravan, is refused — replacement in kind

keeps him walking with the company; cash points him home. Deeper than the case law lies a piece of constitutional grammar. The halakha's name for standing under another's obligation is *arev* — the guarantor at a loan's side, the neighbor whose signature chapter 15 watched carry a borrower's name — and the tradition took that exact legal figure and stretched it across the whole people: all Israel are guarantors for one another, the Talmud rules (Shevuot 39a). It is a remarkable sentence — a people constituted not by shared sentiment but by mutual co-signature — and the institutions followed the grammar. The *kupah* and *tamchui* of chapter 18 ran subsistence as a civic charge; the guarantor networks of chapter 15 priced character where no collateral existed; and against the fear the Lanuvium members paid dues toward, the communities built the *chevra kadisha*, the holy society. Its work is the guarding, washing, and shrouding of the dead, the burial of the friendless, the company kept with mourners through shiva; membership is an honor; its customary fast day is the seventh of Adar, the day tradition assigns to the death of Moses. The first of the modern societies was organized in 1564 by Eliezer Ashkenazi — in conscious imitation, the historians note, of the Christian guilds around it — and its regulations were codified by Judah Loew, the great rabbi of Prague. Note the premium's currency: the chevra's members do not pay dues; they serve, holding in common a fund of labor and presence rather than asses and sesterces, and answering Lanuvium's fear — that a human being might leave the world unaccompanied — with the same guarantee: no one's last day is his alone.

Christendom built its pools wherever its people already gathered. The medieval confraternities and guild boxes — parish fraternities, trade brotherhoods — kept dues against sickness, dowry, and burial, and walked their dead to the grave with candles and masses: the collegium's furniture, baptized. When the industrial age dissolved that world, the pattern reassembled itself at the scale of the new towns. England's friendly societies, gathering members' contributions into a common sum for emergencies before the seventeenth century was out, became the self-built welfare system of the working class: the Oddfellows, organized in 1810, kept a surgeon at every lodge; the Rechabites pooled the abstainers; the miners of Tredegar built a medical-aid society that doctored a whole town; as late as 1945, Parliament was told the societies still counted some eight and a half million British members. The ritual and the archaic titles are easy to smile at; underneath them sat sickness pay, the doctor, and the funeral, owned by the members who drew on them. And Leo XIII, writing the Church's charter document on capital and labor, canonized exactly this form. His diagnosis in *Rerum Novarum* was that the old guilds had been abolished and nothing put in their place, leaving working men "isolated and helpless" before their employers' hardness; his remedy pages praise the record — "History attests what excellent results were brought about by the artificers' guilds of olden times" — and list, among the age's genuine consolations, the "societies for mutual help" and the benevolent foundations provided for the workman and for his widow and orphans in sudden calamity, sickness, and death (*Rerum Novarum*, 1891; chapter 33 takes up the doctrine entire). The encyclical that founded modern Catholic social teaching blessed the mutual-aid society by name, as the guild's lawful heir.

Islam's room, chapter 6 showed, was furnished at the founding. The law retained the *aqila*, the old custom by which a man's kinsmen shared the blood-money he owed — liability itself pooled, centuries before premiums were; Bukhari preserves the Prophet's delight in the Ash'ari people, who pooled their short provisions in one cloth and divided them equally, and his claiming of them — they are of me, and I am of them — filed by Bukhari's editors, with perfect instinct, in the Book of Partnership (Bukhari 2486); the believers in their mutual mercy are one body, sleepless and fevered together when any limb is hurt (Bukhari 6011); and over all of it stands the Qur'an's command: "help one another to do

what is right and good" (5:2). These are the seeds; the twentieth century formalized them by verdict. The 1985 resolution's third clause called on the Muslim countries to build the co-operative alternative, and an industry answered. *Takaful* — the word is built on the root of *kafala*, guarantee: mutual standing-for — is the pool engineered to the fiqh's specifications. Each participant's contribution enters the fund under a *tabarru'* commitment, a donation rather than a price, the genus-move of the last section performed in the contract's opening clause; the fund belongs to the participants in common; and the operating company stands outside it as hired administration — a *wakil*, an agent managing for a disclosed fee, or a *mudarib* sharing declared profits on the fund's investments, or a hybrid of the two. When the Academy returned to write criteria — Oran in 2012, Riyadh in 2013 — it drew the lines with a draftsman's economy: cooperative insurance is participation in "a non-profit fund"; the manager is an agent contracting on the fund's behalf; contributions and their investment returns are the fund's own; and the arrangement, not being a compensation contract at all, may carry the uncertainty that voids a sale. The industry's standard-setter keeps a dedicated standard on the instrument's plumbing — AAOIFI's Shari'ah Standard 26, on Islamic insurance — and even the wholesale layer is rebuilt on the same principle: reinsurance among such funds runs as *retakaful*, as the 1985 resolution's second clause required.

Three rooms, then, and one architecture. Membership, not customership; the payout an obligation among people who stand guarantor for one another, not a seller's performance owed a claimant; and every room policing the same seams — negligence excluded on the rope-length rules of Bava Kamma, arrears lapsed on the marble at Lanuvium, the manager fenced off the fund at Riyadh. The differences are the humane part: the chevra washing its dead, the lodge sitting with its sick, the takaful fund settling its claims — one instrument, three currencies of the same promise.

The Actuary's Two Fairnesses

In 1762 a group of Londoners founded the Society for Equitable Assurances on Lives and Survivorships on a proposition then novel: that a life policy's price should be computed. The mathematician James Dodson had worked out the method from mortality records — the Northampton figures — so that each age paid in proportion to the dying it actually brought through the door, and the society's chief officer took a job title nobody had held before: actuary — the first use of the term. The house that coined it was a mutual, owned by its policyholders. Fairness-as-computation was born inside fairness-as-membership: the first actuary was hired not to sharpen a seller's margin but to keep a club's promises to its own members. The computation itself was no scandal to anyone in this book; Rome priced lifetimes on Ulpian's sliding table when estates required it, as chapter 10 told, and the traditions' quarrel, chapter 10 found, was never with anyone's arithmetic.

The quarrel is with what the arithmetic is asked to decide, and it turns on two fairnesses the modern argument habitually runs together. Actuarial fairness is a property of prices: each member contributes what his own peril is expected to cost the pool, nobody rides free, and the strong subsidize the weak by not a penny. Moral fairness is a property of the arrangement: the pool exists so that a blow that would end one household lands, survivably, on many; its test is not the ledger's symmetry but the worst day of its weakest member. In a club of rough equals the two fairnesses coincide — at Lanuvium every member's death was equally certain and every member's five asses the same. They come apart the moment peril tracks poverty. Price every risk to its bearer with perfect accuracy and the frail pay most for medicine, the low ground pays most for the flood, the old pay most for the years — each person handed back his own danger, exactly measured, with a service charge

— until at the limit the pool has quietly dissolved into a mirror: perfectly fair prices and nothing protected. Chapter 10 caught the same inversion inside the loan, where the “risk premium” lands heaviest exactly where fragility is greatest; Comparative Interlude I filed the standing warning about rate cards keyed to what a person is rather than to what a venture faces; and the modern pricing engine can walk the same road in fresh vocabulary whenever its factor tables shadow poverty or birth. The arithmetic in such cases is fair, and the arrangement is not — and the traditions, given the choice, price the venture and shelter the person. Their own pools knew both principles and mixed them deliberately: the caravan at Bava Kamma shares ransom by the money each traveler carries and jettison by weight — contribution honestly tracking the goods at stake — while Lanuvium’s flat dues and the chevra’s flat duty declare the members equal before death, whatever they carry. What none of them tolerates is the third thing: the price that expels the very person the pool was raised to shelter. Solidarity, in a pool, has an exact meaning — not an actuarial error but the members’ standing decision that the strong may knowingly overpay so that the weak stay covered: cross-subsidy chosen with open eyes, as policy, because the pool’s product is membership and not price.

And the two fairnesses come with two possible owners — this Part’s oldest question, arriving at the instrument built to escape it. Chapter 3 named the fork when it first showed the pool, and the sentence can now be paid in full: the difference is “protection pooled among members versus protection sold at a margin by someone who profits when claims stay low.” In the pooled form — the club chest, the friendly society’s box, the takaful fund — premiums are the members’ own money held in common; a payout is the fund doing what it was gathered for; the surplus of a light year stays home or goes home. In the proprietary form the premiums are a seller’s revenue, the claims are his costs, and the distance between the two is his margin — a margin that widens with every claim resisted, every exclusion enforced, every fragile customer re-priced out the door. A member’s claim is the pool at work; a customer’s claim is the seller’s loss. Nothing in that geometry requires the seller to behave badly; a great deal in it pays him to. Name the seat with chapter 10’s precision: near the gain of the light years, at a managed distance from the heavy ones — the posture the baraita graded. The traditions’ objection is not that such men are wicked but that the arrangement has quietly turned protection back into a product and the protected back into counterparties — and the instrument this chapter has been assembling was invented, three times over, to be the alternative.

The Pool’s Own Books

A pool is a promise with a ledger, and the ledger keeps three books: the cost of running the thing, the reserve that keeps it standing, and the surplus of the light years. The marble at Lanuvium kept all three — travel allowances and officers’ duties for the first, banked dues for the second, the arrears rule guarding both (a pool must be able to say no to the lapsed, or the faithful underwrite the faithless), and a quadruple fine for fraud, because funds tempt and the by-laws knew it. Run the three books honestly and the pool is sound; corrupt any one of them and the pool becomes, by inches, one of the things this Part has spent four chapters unmasking.

Administration first. Somebody must keep the rolls, price the risks, chase the arrears, settle the claims, and be paid for it — the question chapter 16 met at the mercy funds, back at industrial scale. The takaful architecture answers with the discipline of Amman: the operator’s *wakala* fee is disclosed, agreed in advance, and priced as work — the actual-expenses logic chapter 16 watched the Fiqh Academy write for an interest-free bank’s ser-

vice fees. The test underneath is indifference. An administrator honestly paid earns the same whether the year runs light or heavy; the moment his compensation improves when claims are refused, he has reacquired, inside the mutual's vocabulary, the very short position the form exists to abolish. Hence the Riyadh criteria's seating of the manager as agent and nothing more — and the standing argument among the industry's own jurists over operators who take performance incentives out of underwriting surplus, a quarrel chapter 32 will hear in full. The rule the old pools model is stricter and simpler: pay the steward wages, never a share of the misfortunes his members were spared.

The reserve is the second book, and it is where the pool's true product sits. A fund holding only its expected claims has provided for the average year, and the average year is not what anyone joined for; the pool's whole product is the bad year survived. So honest pools over-hold, and they layer: mutuals reinsure with mutuals, takaful funds cede into retakaful pools. Where a fund runs dry anyway, the industry's standard rescue is worth savoring: the operator advances the deficit as a *qard hasan*, an interest-free loan repaid from future surpluses — chapter 16's beautiful loan, employed as shock absorber of last resort inside a regulated industry. Chapter 10 gave this book the name for all this machinery — a solidarity reserve, an anti-ruin reserve, capital whose job is not yield but the survival of the paths that cross it — and in the pool the name stops being vocabulary and becomes a line item.

The third book is the surplus, and it is where the two lineages of this chapter finally refuse to be confused. A light year leaves money in the chest. In the pooled form that remainder belongs to the members: the Riyadh criteria say it plainly — the insurance surplus “can be retained in the fund or distributed, totally or partially, among policyholders” — and if a cooperative fund is ever wound up, its residue goes to charity or back to the participants, the manager excluded by name from the estate. Lanuvium's chest ran the same book — funerals, travel money, the banked light years, and never a distribution, there being no one the money could go to who was not the pool. In the proprietary form the same remainder bears a different name — underwriting profit, the owner's earnings — and the light year has become yield. Nothing else in the two forms differs as starkly as this single ledger line, and everything follows from it: the owner of the surplus is the owner of the pool, and the pool's owner decides what a bad year means — the event the fund exists for, or the event the business must survive.

Which is the honest frame for the modern controversy: commercial insurance and profit extraction. Say the steelman first, and mean it. The shareholder carrier does real protective work; its capital absorbs the catastrophe year that would sink a small mutual, for a mutual of the poor pools correlated poverty — one flood can take the fund and the members together, and the mathematics of when pools themselves need pooling belongs to the next chapter. Professional reserving, solvency regulation, worldwide reinsurance markets: these are engineering, not villainy, and the traditions' own institutions borrow all of them freely. The critique the traditions press is narrower, structural, and pressed by each from inside its own house. The Fiqh Academy's 1985 verdict and its 2012 restatement are the fiqh's own scholars ruling on the contract's anatomy, not an outsider's sneer at commerce. Leo XIII's remedy for the worker left “isolated and helpless” was not a product to be bought from strangers but an association owned by the men it protected. The mutual movement's sharpest witness is its own history: in the late twentieth century many of the great member-owned carriers demutualized — the members' common funds re-badged as shareholders' equity, the surplus book moved by a stroke of drafting from one lineage to the other. And honesty compels the mirror critique, which the fiqh's own critics supply: a takaful opera-

tor collecting rich wakala fees, plus incentives from surplus, plus investment shares, can rebuild a stock insurer's economics inside a mutual's vocabulary — the costume test of chapter 24 travels here as everywhere in this Part. Meanwhile the strict tradition keeps its own mercy clause, and it belongs in the record beside the verdicts: the need provision chapter 23 met in the gharar standard tolerates commercial insurance where no takaful alternative exists — severe hardship, no permissible substitute. The doctrine's purpose is protection; a purity that leaves the widow uncovered has mistaken the instrument for the point.

The Shield with By-Laws

Chapter 2 closed its map of the ancient world with the community as a shield against isolated ruin, and chapter 3, showing the pool, called it that shield furnished with by-laws and a treasury; this section's title repeats the phrase on purpose, because the survey can now cash it. The pool is infrastructure in the exact sense chapter 15 claimed for the gemach — load-bearing civic plant, not decoration and not charity in a different hat — and a community without one sorts, season by season, into the lucky and the ruined. In the capital stack chapter 18 drew, it takes the seat that chapter reserved: across the layers, catching the loss that arrives uninvited while transfers, mercy loans, and investment capital do their own work. A free loan carries a household over a bad month, chapter 15 showed; only a pool can take a catastrophe that landed on one roof and spread it across a hundred. Why a pool defeats ruin where even a fairly priced loan cannot — why averages comfort the bookkeeper and betray the household, what pooling does to the arithmetic of a single life lived in sequence — is the entire business of the next chapter, "Ergodicity, Solidarity, and the Mathematics of Mutual Aid." This chapter's contribution is the census: the thing has been built, and built, and built again — the burial chest on the temple porch, the caravan compact in the baraita, the chevra with its basins and shrouds, the confraternity's candles, the lodge's surgeon, the takaful fund under its regulator, the temple-organized *mu-jin* rings of Comparative Interlude II, the immigrant landsmanshaftn that taught Henry Street's founders their trade. Much of what the welfare states later nationalized was first built here, by members, at nobody's margin.

So the design grammar, gathered from every room and readable as a specification. The peril must pre-exist and belong to the member — no manufactured wagers; the Georgian statutes and the verses on maysir guard the same gate. Contributions enter as the members' own — dues or donation, tabarru' or the club's five asses — never as a seller's revenue. The fund belongs to those it shelters, and its administrator is a wage-earner inside it, paid for work, indifferent to claims, fenced off the surplus. Exclusions police conduct, never misfortune: negligence out, rope-lengths published, the arrears line drawn plainly and known to all — strict about conduct precisely because it must never be predatory about calamity. Reserves are sized to the bad year, and pools pool upward — retakaful, mutual reinsurance — so that no single fund's weather is final. The surplus goes home to the members or forward to the mission, never out as the light year's yield. And outside capital may serve the pool — reserve, backstop, the operator's beautiful loan — but never own it. Every clause on that list is in force somewhere today, from the Riyadh criteria to the statute book of 1774 to a marble porch in Latium; the specification is not a proposal but an inventory.

A pool holds a treasury, though, and a treasury raises the question every institution in this book eventually meets: how money meant to serve forever is kept from being spent, seized, or drifted away. The pool's chest wants a spine — property locked to mission beyond any

member's or manager's reach — and that instrument, the endowment, closes the Part in chapter 29, with the mathematics of mutual aid standing between, in chapter 28. Close, then, where the whole subject is tenderest, back on the porch at Lanuvium. Chapter 3 lingered over the by-laws' euphemism, and it will bear one more look: the rules provide for the member to whom the human thing happens — *si ei humanitus acciderit*. The human thing happens to everyone; the pool has never been anything but the contract by which people arrange, in advance and together, that when it comes the cost will find many shoulders and the person will find his own kind waiting. Sale, lease, and certificate put price tags on goods, on use, on fractions of ownership, and this book has spent three chapters auditing the tags. The pool prices only membership, and pays out in the one currency the traditions never let the market quote — the difference between facing the worst day alone and not.

Chapter 28 — Ergodicity, Solidarity, and the Mathematics of Mutual Aid

A stranger sits down at your kitchen table, produces a fair coin, and offers you the most generous game you have ever been shown. Each round, your money rides on one toss. Heads, he adds half again to everything you have; tails, he takes forty percent of it. That is the whole game — in the words of the physics paper that made it famous, “toss a coin, and for heads you win 50% of your current wealth, for tails you lose 40%.” Any accountant would bless it on the spot. On every hundred dollars staked, half the time you gain fifty, half the time you lose forty, so the average round hands you five percent — a rate of return most funds would advertise in gold leaf. The gamble comes from Ole Peters’s 2019 paper in *Nature Physics*, “The ergodicity problem in economics”; the gamble is his, and the arithmetic that follows is this book’s own, simple enough to check at the same kitchen table.

Play one round, and the accountant is right. Play on, your whole stake riding each time, and something begins to go wrong that no single round explains. Take one heads and one tails, in either order: your money is multiplied by 1.5 and by 0.6, which is to say by 0.9. Every fair pair of tosses — every stretch of the game in which your luck is exactly average — costs you a tenth of everything. Start with a hundred dollars and flip ten times: the average outcome, the number the accountant computed, has grown to about \$163; the middling player, five heads and five tails, holds \$59. After a hundred flips the average player holds more than \$13,000 — and the middling player holds fifty-two cents. Nor is the middling player unlucky; he is the typical case. Count the paths: after ten tosses, only about thirty-eight in a hundred players are ahead at all; after a hundred tosses, fewer than fourteen in a hundred; after a thousand, about one player in seven thousand. And still, at every one of those horizons, the average across all players is climbing at its serene five percent per round — because that average is being carried by a thinning aristocracy of impossible luck. After a hundred flips, half the entire ensemble average sits on paths that flipped seventy-two heads or better: about six players in a million. Everyone else is subsidizing a statistic.

Where, then, does the famous five percent live? Peters’s answer deserves to be quoted, because it is the hinge of this chapter: the expected value averages over “an ensemble of copies of myself that cannot be accessed.” The five percent is real — for a syndicate that could stake a thousand tables at once, collecting the lucky paths along with the ruined ones. No such syndicate sits on your side of the kitchen table. You will play the rounds in order, each on whatever the last one left, and for you the operative number is not the average of the outcomes but their geometric mean — multiply, then take the root — which for this game is $\sqrt[10]{1.5 \times 0.6} = \sqrt[10]{0.9}$, about 0.949. Your money, on its actual path through time, decays at roughly five percent a round in the very game that grows at five percent a round on average. Both numbers are correct; they answer different questions — one asks what happens to everyone at once, the other what happens to you, in sequence, over a life.

This book has been promising the present chapter since its first Part. Chapter 3 raised the pool to a first-class contract form and deferred the reason to the mathematics here. Chapter 10 built the grammar — two averages, one path, the doors that lock — and left a note payable. Chapters 15 and 16 put their *gemach* and their beautiful loan on the scale and

said the weights would arrive in chapter 28. And chapter 5, watching Paul tell Corinth that the congregation is a body, promised that this chapter would show in numbers what Paul asserted in anatomy. Those debts come due now. The plan is plain: first the disease, stated exactly — why averages flatter and sequences kill; then the cure the traditions built, from a burial club in Latium to fourteen million friendly-society members; then a model village, run twice, to watch the cure work; and last, what it means to build the whole finding into the machinery of a non-extractive fund.

The Average That No One Lives

Begin with chapter 10's two definitions, because everything else stands on them. An ensemble average is computed across many parallel cases at one moment — a thousand borrowers, a thousand harvests, a thousand coin games summed and divided. A time average is what one single sequence actually accumulates, step after step, each result compounding on the last. A process is *ergodic* when the two agree — when the experience of the many, averaged, matches the experience of the one, lived out. The word is borrowed from physics, where it earns its keep: gas molecules in a box genuinely do visit, over time, the states the ensemble occupies at an instant, which is why the physicist may swap one average for the other. Peters's charge against economics is that it inherited the swap without the license — that "expected utility theory and its descendants," in his paper's framing, assume ergodicity indiscriminately, in a subject whose central processes are growth processes, and growth is precisely where the swap fails.

It fails because growth multiplies. Here the mathematics touches the ground of ordinary life, and it is worth standing on it. Some fortunes add: a wage earned this month does not multiply next month's wage. But most of what this book calls capital compounds — the harvest that becomes seed, the shop whose stock buys stock, the herd, the fund, the reputation — and a household's year is built on the year before it, materially, the way this year's flock is built of last year's lambs. For additive processes, the two averages keep company, and expectation is an honest guide. For multiplicative processes they part — such a process is non-ergodic, to give the failure its name — and they part in one direction: the time average runs at or below the ensemble average, and volatility is what pries them apart. The coin game is only the cleanest specimen. Its rounds average five percent up; its paths grind five percent down; the gap between those numbers is the price of living in sequence, paid by everyone who cannot be a syndicate.

Now add the fact the coin game politely omits, the one chapter 10 placed at the bottom of its grammar: real paths have floors. The game lets a player shrink to fifty-two cents and keep flipping. A household cannot. Below some line the tools are sold, the land is mortgaged, the family's labor is pledged — chapter 2 stood at that door — and the sequence does not continue in any form the earlier years would recognize. The mathematics calls such a place an absorbing state, and the definition is mercifully exact: a state that, once entered, is never left; every path that touches it stays. Chapter 15 called them locking doors and promised the formal name; here it is, doing formal work. The chance that a given path reaches one within some horizon is its ruin probability, and with those two definitions the finding chapter 10 stated as a promissory note can be paid in full. A system of ventures can be favorable in ensemble average and still carry, for anyone who must live it in sequence, a ruin probability that climbs toward certainty as the rounds accumulate. The coin game proves it by exhibition: expectation up five percent a round, the typical path down five percent a round, so that against any fixed floor — subsistence, the last tool, the seed corn — the typical path is always traveling toward the door, and the fraction of players still ahead

thins from thirty-eight in a hundred to fourteen to one in seven thousand — favorable in ensemble, ruinous in sequence, which is not a paradox but a division of labor between two arithmetics, one of which was never about you.

State it morally now, because the failure of averages is genuinely moral, not technical. Expected value is the native arithmetic of whoever holds many positions at once — the casino, the diversified book, the insurer, the treasury. For such a holder the ensemble is not a fiction; he lives spread across it, and chapter 10 already conceded him his truth. The failure begins when his arithmetic is used to judge arrangements for people who hold exactly one position: their own life. Chapter 13, reading liquidation waterfalls, sent this chapter a sentence to prove, and it is proven now: a loss allocated to a seat is lived there as a sequence rather than an average, and some seats can be ruined. A deal that is fair in expectation can therefore be lethal in fact, and the lethality is invisible to the very calculation that certified the fairness — the average survives its own catastrophes, because the paths that die simply weigh less and less of it, while the paths that soared weigh more. The old traditions, who never wrote an expectation in their lives, graded contracts by exactly the thing expectation cannot see. The baraita chapter 10 quoted sorts investors by where their terms seat them — near to profit and far from loss, the wicked man — and reserves its one exemption for orphans, the paths least able to absorb a blow. That is time-average reasoning, running in Aramaic, centuries before anyone needed a word for ergodicity. An average survives its losers; a person does not.

One more turn of the screw, and the section has done its work. Nothing above requires malice, and this chapter is not yet accusing anyone; the coin game has no villain — the stranger at the table offered a mathematically generous bet. That is the deepest and least comfortable point. Ruin does not need a usurer; multiplication plus volatility plus a floor will manufacture it out of fair gambles, free of charge. The farmer of chapter 10, who does not experience the decade's average harvest, is not the victim of a swindle; he is the victim of a sequence — two bad years back to back, in a life that is compounded, not averaged. What instruments do — the fixed claim that collects in the bad year, the guarantee, the meter — is aim the grinding at somebody, as Part III showed. But the grinding itself comes standard with time. Which is why the traditions' answer to it could not be merely a prohibition. Refusing to add insult was Part III's law; something still had to carry the injury. The answer had to be an institution.

The Body and the Burial Club

The traditions met ruin not with a theory but with dues. The theory, such as it was, came first and came as anatomy. The congregation, Paul told Corinth, is a body, arranged by God so that "the members may have the same care for one another. If one member suffers, all suffer together with it" (1 Corinthians 12:25–26, NRSV, as chapter 5 carried it); and to Galatia, in the command chapter 5 flagged as the seed of this whole Part: "Bear one another's burdens" — balanced, three verses on, by "all must carry their own loads," the fence the tradition posted against the shared burden becoming anyone's alibi. Chapter 6 found the identical anatomy in the Prophet's words: the believers in their mutual mercy are one body — when one limb suffers, the whole body keeps the night watch with it in sleeplessness and fever (Bukhari 6011). And beside the image, the practice: the Ash'ari people, short of provisions on campaign, gathered everything into one cloth and divided it equally, and the Prophet claimed them — they are of me, and I am of them — a report Bukhari's editors, who knew a financial instrument when they saw one, filed in the Book of Partnership (2486). Even liability had its cloth: the *aqila*, the kin group that shared a

man's blood-money, carried into law. A body, chapter 5 observed, is the one arrangement in which suffering is not permitted to remain where it lands. Read with this chapter's eyes, that is a statement about variance: the body is an ensemble with a circulation, and the whole design problem of mutual aid is to give a community what anatomy gave the body — a way for a blow that lands on one member to be borne by all of them at once.

Now watch the anatomy acquire a treasury, because the mathematics begins with the dues. Recall the little club chapter 3 read from the marble at Lanuvium — the worshipers of Diana and Antinous, freedmen and slaves and the working poor of a Latium town, AD 136. Entry cost a hundred sesterces and an amphora of good wine; dues ran five asses a month; a paid-up member's death drew three hundred sesterces from the common chest, fifty of it spent at the pyre. Run this book's kind of arithmetic on those numbers, at the empire's four asses to the sesterce, and the design leaps off the stone. The funeral benefit equals twelve hundred asses: two hundred forty months of dues — twenty years, to the month. A laborer saving his own five asses monthly against his own funeral is making a bet that he will be given two decades of health and discipline before what the by-laws called the human thing happens to him; most such bets, the last section showed, are lost not by imprudence but by sequence. A membership paying the same five asses together funds every funeral as it comes, from the first month onward. The club could do for each what none could do for himself, and the reason is not generosity but geometry: the club holds all its members' paths at once — the coin-game syndicate, built by the poor, for stakes of dignity, the ensemble average made into a fact by a body large enough to live in it.

From that inscription the roll of the traditions' pools can be read almost in one breath, because the same shape recurs wherever the traditions kept their books. The gemach of chapter 15: eleven men at a Henry Street synagogue in 1892 pooling ninety-five dollars into a fund that has since lent, by its own accounting, more than \$380 million at zero interest, with repayment near ninety-nine percent — a pool whose payout is a bridge rather than a burial. The *qard hasan* of chapter 16, and its modern industrial form at Akhuwat, lending through mosques and churches with repayment the institution reports above ninety-nine percent. The *takaful* of the *fiqh* — mutual guarantee, the participants' own fund standing under the members' losses rather than a shareholder profiting beside them — which chapter 27 builds in full and this chapter borrows for one clause. The confraternities of Christendom, prayer brotherhoods whose statutes ran to sick-beds, dowries, and graves; the craft guilds whose chests did the same for the trade, mutual aid annexed to the work-bench; the monastery treasuries and temple funds Comparative Interlude II watched administer what monks had renounced, down to the Japanese *mujin*, the temple-organized ring of rotating credit. None of these borrowed from the others; each grew from its own scripture's anatomy, which is precisely what makes the convergence evidence.

And when the industrial age arrived, the shape scaled. England's Parliament noticed the societies of workingmen pooling weekly pennies against sickness and burial, and in 1793 gave them their first statute — "the first act for the encouragement and relief of friendly societies," the 1911 *Encyclopaedia Britannica* calls it, Rose's Act, which designated them, in a phrase too good for a statute, "societies of good fellowship." By 1905 the Registrar's rolls that grew from that act counted nearly fourteen million members and fifty million pounds in accumulated funds — "the mutual insurance societies of the poorer classes," the *Britannica* says plainly, by which working families sought "to aid each other in the emergencies arising from sickness and death." Fourteen million memberships, decades before the welfare state: the largest financial institution most of those families would ever touch was a pool. The same article, tracing the pedigree back through the guilds to the ancient world,

offers a sentence this chapter could have written for it: “The origin of the friendly society is, probably in all countries, the burial club.” Lanuvium, in other words, never stopped. The porch inscription and the Victorian actuary’s rate-book are one institution, nearly eighteen centuries apart, solving one problem: the single path cannot carry its own worst day, and a membership can.

Hold the whole roll-call up at once — *gemach*, *qard hasan*, *takaful*, *collegium*, confraternity, guild box, friendly society, *mujin* — and one claim establishes itself: religious solidarity was anti-ruin infrastructure — infrastructure, not mood. Every institution on the list has by-laws, a treasury, defined benefits, defined contributions, membership rolls, and sanctions; every one of them stands between its members’ paths and an absorbing state, which is what chapter 10 meant in calling the pool an anti-ruin reserve; and nearly every one of them is wrapped in worship — dues collected at the monthly meeting, benefits owed to the brother in the pew, the whole apparatus consecrated. The wrapping, the next section will argue, is not decoration. But the load-bearing fact comes first: where the scriptures said body, the communities built chests, and the chests are the only financial technology in this book whose entire purpose — chapter 10’s definition, to be repaid shortly — is the survival of the person the average forgets.

A Village, Run Twice

Now for the numbers this chapter owes chapter 5, and the fairest way to pay is to build the community ourselves and break it twice. What follows is a model village of this book’s own — every rate chosen for transparency, every result computed and checkable, none of it a claim about any historical place. Twenty households. Each scrapes out six units a year to set against disaster, from an income of about a hundred. Each year, each household independently faces a one-in-twenty chance of a blow costing a hundred units — a full year’s income: the ox dies, the boat is lost, the roof burns, the funeral must be paid. We watch for twenty years, and we use the words defined above exactly: a blow that cannot be paid is ruin, and ruin is absorbing.

Run the village first with every household on its own path — each family saving its six units a year in its own jar, the way prudence is usually preached. The arithmetic of futility is visible before any dice roll. Six units a year means a household needs seventeen years of uninterrupted saving before one blow becomes survivable; through the first sixteen years, the jar’s ninety-six units or less simply loses to the hundred-unit blow, every time. Meanwhile the blows will not wait: over twenty years, each household runs about a sixty-four percent chance of taking at least one. Simulate the village twenty thousand times over and the count comes out steady: on average, more than eleven of the twenty households are ruined within the twenty years — about fifty-seven in a hundred, across the whole run of trials. And here is the sentence to read twice, because the whole chapter turns on it: the village as a whole was never poor. Its households set aside a hundred and twenty units a year, every year, against blows that averaged one hundred a year. Summed across the ensemble, provision exceeded misfortune throughout — the accountant’s ledger balances with room to spare, in the very decades that more than half the families fall through the floor: enough on average, fatal in sequence. The coin game has come to the village, and it did not even need a gambler.

Now run the same village again, changing nothing but the address of the money. The same six units per household per year — no one pays a penny more — go into one common chest instead of twenty private jars, and blows are paid from the chest as they come. Everything

the solo run lacked is suddenly present. The chest holds a hundred and twenty units by the end of the first year: from the very first year, the blow that no household could survive alone is survivable — chapter 5's promise, first installment. Simulated the same twenty thousand times, the average count of ruined households falls from more than eleven in twenty to fewer than two — from fifty-seven in a hundred to about nine. Same coins, same blows, same twenty years; six times fewer deaths of households. And notice with proper astonishment what did not change. Pooling moved no ensemble number at all: each household's chance of taking a blow is still sixty-four percent; the expected blows per household are still exactly one; the dues each pays over the twenty years, one hundred twenty, still exceed the hundred of expected loss by the same margin as before. Judged by expectation — judged by the arithmetic that certified the coin game — the pool accomplishes precisely nothing. Every one of its effects lives in the dimension expectation cannot see: the sequence. Where the solo household met a lumpy, rare, fatal loss, the member meets a small, certain, scheduled fee, and the lump falls on the chest; the mechanism converts catastrophe into dues — pennies on a schedule against a catastrophe that keeps none — which is to say it converts a multiplicative shock, the kind that ends compounding paths, into an additive charge, the kind a path shrugs off. There is the mechanism, stated as exactly as this book knows how: mutual aid changes nothing about the average and everything about the time average, because it re-seats loss from paths that can be absorbed onto a pool that cannot — the blow, chapter 10's conservation law reminds us, is never destroyed; it is delivered to a bearer built for it.

Why is the chest a bearer built for it? Because the chest, unlike any member, genuinely holds many fates at once — twenty paths, whose bad years mostly refuse to coincide. Independence is the quiet engine here, and it deserves one plain paragraph. In any given year the village expects one blow; the chance of a doubled year, two blows or more, is about twenty-six percent — roughly one year in four, and there is the source of the pool's residual deaths, to which the next section attends. But watch what mere numbers do: a town of a hundred households, same dues, same hazard, expects five blows a year and suffers a doubled year — ten or more — only about once in thirty-five years. The yearly claim on the chest steadies as membership grows, not because any household's luck improved but because luck, summed, cancels; the law the actuaries call large numbers is nothing but the ensemble average becoming trustworthy for the entity big enough to live in it. Run the town of a hundred through the simulation and per-household ruin falls below two in a hundred. So the finding assembles itself: the ensemble average, a fiction for any single household, becomes a working fact for the chest that holds them all. Mutual aid is the art of building something that *lives* in the ensemble so that no one has to.

But the mathematics, standing alone, is a machine with no fuel, and here the word spiritual earns its place in a chapter of arithmetic. Every number above assumed things numbers cannot enforce: that dues keep arriving in the years no blow has come; that the healthy do not drift away and leave a chest of the fragile; that claims are honest; that the treasurer is. These are the pool's persistence conditions, and they are exactly what the traditions' religious wrapping was engineering. Lanuvium's by-laws lapse the claim of a member six months behind on dues — the chest defends itself against the fair-weather brother. The Talmud's caravan compact, chapter 3 showed, excludes the negligent loss and refuses the member who wants his benefit in cash while the caravan walks on. Marble and baraita between them had met moral hazard and adverse selection, and legislated for both, long before either term existed. The monthly meeting is a premium-collection technology; the roll of names read aloud, a persistency audit; the oath, the feast, the funeral procession in which the whole society walks — each binds the healthy year to the stricken one, which is

the pool's entire actuarial problem. Rose's Act called the societies "good fellowship," and the phrase turns out to be load-bearing: fellowship is what keeps a pool funded between disasters. This is what it means to call pooling a spiritual and mathematical technology, one artifact with two working faces — the mathematics needs the covenant to persist, and the covenant needs the mathematics to be more than a wish.

The Reserve and the Two Kinds of Pool

The pooled village still lost families — about nine in a hundred — and an honest chapter asks where. The simulation answers: at the beginning, and in the pile-ups. A chest that starts empty meets its first doubled year — one year in four, we computed — with a hundred and twenty units against two hundred of claims, and someone's blow goes unpaid. Age cures this; a chest that has run surplus for a decade holds a buffer. But the cure for the young pool was carved into the Lanuvium porch all along, hiding in the entry price: a hundred sesterces to join — eighty months of dues, paid at the door. The founding cohort walks in carrying a reserve. Run the village a third time with exactly that device, each household paying a one-time five units so the chest opens with one full blow already banked, and per-household ruin falls from about nine in a hundred to below six; bank two blows and it falls toward three; give the town of a hundred its one-blow reserve and ruin drops to about one in a hundred. The gradient is the lesson: every unit of idle money standing in the chest buys survival for paths that will never know which year it saved them. This is the instrument chapter 10 named for the whole book — the anti-ruin reserve, the solidarity reserve — and the village gives its sizing rule plainly: a solidarity reserve is sized to the pile-up, not the average year, because the average year, as this chapter has been at pains to show, is the one year that never needs it. Idle is the accountant's word for such money and the wrong one. Its product is continuation — the difference between the two simulation columns — and a community that lets the reserve be milked for yield in the calm years has quietly sold the very thing the chest existed to hold.

The same geometry climbs the scale of institutions, and two modern rungs need naming before the pools divide. First: missions walk paths too. A fund, a school, a congregation, a movement lives one sequence, exactly as a household does, and chapter 10's conservation law already caught the modern habit of spending the mission as the buffer of last resort — the purpose steered, quarter by bad quarter, into whatever keeps protected capital whole. A mission reserve is the same chest built for an institution's purpose: capital stationed between the mission and its worst year, so that one funding drought or one failed venture is a blow absorbed rather than an identity abandoned. Grown permanent — locked past the reach of any board's bad quarter — the mission reserve becomes the endowment, and that is chapter 29's business, one door down. Second: portfolios are villages. A fund holding twenty ventures holds twenty paths, and the pooling mathematics runs without translation: the fund layer, which genuinely lives across the ensemble, is where predictability honestly belongs — the steady distribution cadence, the reserve, the smoothing — precisely so that no single venture's worst year has to be mortgaged to a calendar. The design rule runs: schedule the chest rather than the household — promise cadence at the pool, patience at the path. Portfolio resilience, in this book's terms, is nothing but the village model wearing a term sheet, and chapter 41 ("Designing a Non-Extractive Fund") will hold the industry to it.

Which forces the distinction this Part has been saving, because everything in it converges here: the same actuarial table serves two opposite masters. Chapter 10 found the inversion first, in the actuarially fair loan book — premiums pooled across borrowers, exactly like

dues, except that the pooling protects the lender's average while each borrower keeps his own ruin; Lanuvium's chest run backward, the many paying so that the bookholder's line comes out smooth. Set that beside the burial club and the difference is not degree but direction, and it can be read off any pool with two questions. Whose ruin does the design treat as the failure state — the capital's, or the participant's? And who lives on the smoothed line once the pooling has done its work? In the collegium, the gemach, the friendly society, the takaful fund, the answers run: the member's ruin is the failure state; the member's path is the one smoothed; surplus in a light year thickens the reserve or returns to the members — nobody at Lanuvium, chapter 3 noted, drew a dividend when deaths came in few. In the pool run for investor return, the participants' variance is the raw material and the investor's smoothness the product; the participants remain on their jagged paths, now paying a fee for the privilege, and surplus is the point. Both are pools; both cite the law of large numbers; they are morally opposite instruments, and the difference is invisible on a rate card — you find it by asking, of the ensemble the pool has built, who gets to live there. Chapter 27 walks this line through the modern insurance industry and its takaful counter-model, where it becomes a question of contracts; here it is a question of arithmetic served, and chapter 10's definition can finally be repaid with its proof attached: the solidarity pool is the one financial structure whose entire purpose is the time average of its weakest member. The village numbers are that sentence, worked.

One sober paragraph before the machines, because the two-pools test also catches the failure the traditions were likeliest to commit themselves. A pool defends its members; the boundary of membership is therefore where its mercy stops, and a chest can be honest inside and cruel at the gate — the sick excluded as bad risks, the poorest priced out of the dues, the stranger outside the brotherhood's roll. The friendly societies' rate-books knew the temptation; so did every guild that closed its list. The traditions' own correctives have run through this book — the tzedakah and zakat that are owed, not pooled, chapter 18's floor under everything; the insolvency mercies of chapter 22 that catch the debtor no chest caught — and they matter here because a solidarity architecture is judged, like every instrument in this book, at its worst seat: a community that pools brilliantly for its members and lets the gate do its extraction has moved the usury to the door.

The Simulator's Conscience

One assignment remains, and it sounds technical while being in fact the chapter's landing: implementing ergodicity in a non-extractive capital simulator. Investors have begun to hear the word from their own side of the table. A small literature now presses time-average reasoning on practitioners — Graham Boyd and Jack Reardon's *The Ergodic Investor and Entrepreneur* (2023) is the flag-carrier, arguing that capital formation is multiplicative and order-dependent, and that sound strategy works “with, not trying to prevent and control, unpredictability”; Boyd's venture builder raises portfolio companies on the approach. The argument of this chapter is that the traditions got there first by millennia, and that a fund claiming their lineage should be able to prove it — in the running code of its own models, not in a mission statement.

What would it mean, concretely? A simulator is a machine for living many lives quickly: describe a venture's fortunes — good years and bad, multiplying as fortunes do — then describe a contract laid on it, and let the pair run a few thousand times. The ergodicity is implemented the moment the machine is forbidden to summarize its runs the accountant's way. Not one number, the expected collection, but three, reported side by side: what the capital collects across the ensemble; what happens to the median path — the venture's

time-average growth, with the claim's drag on it stated; and the ruin probability, the fraction of runs that end in the absorbing state, counted as a first-class output and never netted against the winners. The whole moral force of the instrument is in that refusal to net. Two contracts can collect identically in expectation while one of them manufactures ruin and the other manufactures none — the fixed claim that bites hardest in the worst year against the contingent claim paid from actual surplus, Part III's oldest lesson re-run as simulation — and only the third number can tell them apart. Ruin, the doctrine of restitution would add, is not a cost to be offset but a wrong to be undone, and a simulator with a conscience scores it that way.

Then the machine implements this chapter, and the switch is almost embarrassingly simple, because the village model above already is the prototype. Run the same book of ventures solo and pooled — with and without the solidarity architecture: the common chest, the entry reserve, the contingent claim, the scheduled pool above patient paths. Read the three numbers again. If the design is honest, the before-and-after table will look like the village's: collections nearly unmoved, ruin collapsed, median paths alive and compounding. Solidarity becomes something measured rather than admired; a "solidarity reserve" that does not move the ruin column is a marketing line, and the simulator will say so. The specification travels in one sentence: judge every contract on the time axis, at the worst year, with ruin priced beyond offset, and let pooling prove itself in the ruin column or not at all. Building that machine in earnest — the doctrine made computable, the return formula, the fund design, the contract analyzer — is the work of Part IX, "Moral Mathematics: Making the Traditions Computable," from chapter 35 ("From Doctrine to Model") through chapter 40's Non-Extractive Return Formula to chapter 41's fund and chapter 43's analyzer, with worked metrics and waterfalls in appendices D and E. This chapter supplies the part of the machinery that two thousand years of chests already tested.

So the book's accumulated promises can be settled. Chapter 5 promised that this chapter would show in numbers what Paul asserted in anatomy — that a community which pools its blows can survive what no member could survive alone — and the numbers are now on the table: the blow of a hundred against the jar of ninety-six, lost; the same blow against the chest of a hundred and twenty, carried, in the first year and every year after. The members of the burial club at Lanuvium never saw the equations their dues were solving; they knew the funerals happened, every one, and that no member's family stood at a pit alone. Paul never computed a geometric mean; he said body, and meant that suffering would not be permitted to remain where it landed. The friendly societies' fourteen million never heard the word ergodic; they paid their pennies at the monthly meeting of a society of good fellowship, and the pennies did what the mathematics of this chapter says pennies, pooled, must do. The theorem was lived long before it was stated, and stating it subtracts nothing from the living — it explains, rather, why the living worked, and why every tradition this book reads arrived at the same instrument from different scriptures: because ruin over time is a single mathematical fact about compounding lives, and the pool is its single mathematical answer. What remains, before Part VII closes, is the oldest and strangest of the chests — capital dedicated so completely to its purpose that no one may ever extract it again. The endowment is the solidarity reserve grown permanent, and chapter 29 is its story.

Chapter 29 — Endowments and Asset Locks: Waqf, Hekdesh, Church Property, and Steward Ownership

The finest piece of land Umar ibn al-Khattab ever owned worried him. He had acquired it at Khaybar, the great oasis north of Medina; nothing so valuable, he said, had ever come into his hands, and he went to the Prophet and asked what to do. The answer, as Bukhari's collection preserves it, took one sentence: if you wish, tie up the land itself, and give its fruit to charity. Muslim's parallel keeps the instruction as the jurists would carry it for fourteen centuries: "If you like, you may keep the corpus intact and give its produce as Sadaqa." So Umar dedicated the land, on conditions the tradition remembers like a liturgy: the land never to be sold, never to be given away, never to be inherited; its yield to the poor and to kinsfolk, to the freeing of slaves, to God's cause, to the traveler and the guest; and the man who administers it free to eat from it in reasonable measure, and to feed a friend, so long as he hoards nothing for himself (Bukhari 2737; Muslim 1632a).

Chapter 3 met this deed early and called the form it founded the strangest member of the book's taxonomy; this chapter is its home. Every other instrument in this book moves property between owners. Umar's deed moves property out of ownership altogether: afterward the land belonged to no one who could be named — not to Umar, who could not sell it; not to the poor, who held a claim on the fruit but no title to the tree; not to the trustee, who kept nothing beyond his reasonable bread. A deed of sale changes the owner; a deed of dedication abolishes the office. What remains is a purpose with property attached — the drawer where property goes to stop being property — and all three traditions built that drawer, furnished it, litigated it, and trusted it with their mercy's infrastructure.

Part VII ends here by design. Its first five chapters walked instruments that discipline what capital does while it stays owned; chapter 26 closed on its distinction between backed paper and locked paper, and promised the lock its full development where the Part ends. This is the room. It answers the question every instrument of this Part raises: rules can keep an owner honest, but who guards the thing itself after the founder dies, when every rule has an heir ready to argue? The traditions' answer was a deed written against the future — *waqf*, *hekdesh*, the church's inalienable patrimony — and the modern movement called Steward Ownership is, this chapter will argue, the same deed learning to speak company law.

The Deed That Cannot Die

Islam gave the form its most complete legal career. The instrument is the *waqf* — the word is usually glossed as a stopping, a making-stand-still — and the jurists' shorthand tracks the Prophet's sentence exactly: detain the origin, let the fruit go free. The founder is the *waqif*; the deed, a *waqfiyya*; the property detained, the corpus; between corpus and beneficiaries stands the manager the law calls *mutawalli* or *nazir*, whose bread Umar's deed already regulated. The corpus must be a thing that survives its use — chapter 25's test at the door, since a *waqf* of what perishes in the using would be a gift wearing a deed. The dedication, once made, binds: AAOIFI's *waqf* standard states the classical rule in regulatory English — the *waqf* "is binding upon the Waqif from the instance of the creation of Waqf, and he cannot resile from it" (Shari'ah Standard 60, art. 2/3). And the founding

deed's three bars — no sale, no gift, no inheritance — became the institution's signature: property outside the market and the family, answerable only to its purpose.

What that legal move built is easy to underestimate, because the modern eye files it under charity; it was closer to infrastructure. For a millennium and more the endowed deed carried much of the Islamic world's public life: the mosque and its college (al-Azhar in tenth-century Cairo, the Qarawiyyin at Fez a century older, both waqf-fed from their founding, as the histories tell it), the hospitals, fountains, baths, caravanserais, the Ottoman soup kitchens. How much of the economy the deeds finally held is hard to know; the honest report, as with chapter 18's zakat volumes, is the spread itself: a fifth of Egypt's arable land, half of Algeria's, a third of Tunisia's, as much as three-quarters of what became Turkey's, one much-quoted figure gives the empire's twenty thousand waqfs an income equal to a third of the state's revenues. Hold the numbers loosely and the conclusion stands: one of history's great answers to who should hold capital that exists to serve.

Notice how the locked thing pays, for chapter 25 made a promise on this chapter's behalf. A corpus that can never be sold can still be let, and the classic waqf earned exactly as that chapter blessed — shops let to merchants, orchards to farmers, bathhouses and mills to their keepers, the rents spent on the purpose: the standing lessor at institutional scale, a patient owner letting real things at honest rents, its income licensed by roofs it genuinely had to fix. Doctrine kept the lease's priorities: maintenance of the corpus before every other expenditure, the trustee charged with "management, maintenance, development and investment," the endowment commanded to keep its capital alive, not merely intact. Interlude II reached the same conclusion from inside a tradition that had renounced money altogether — an endowed mission cannot let its fund lie idle and survive — and the Islamic deed formalized the instinct as a trust duty. It even built the lock a controlled exception, *istibdal*, substitution: where the corpus fails — the shop burned, the district emptied — the asset may be exchanged for another of equivalent value, under conditions the modern standard states with visible suspicion: genuine benefit, no favoritism, prompt replacement, the competent authority's approval. The lock was never a fetish about an object: it binds value to purpose, and lets the object change only under audit, at full worth, with the purpose holding every card.

The tradition kept two quarrels alive inside the institution, and both are its own conscience at work. The first concerns the family. Classical law recognizes the *waqf ahli*, the family endowment, whose named beneficiaries are the founder's descendants, charity taking the remainder when the line fails. Its defenders point to the founding deeds themselves: Umar's conditions fed kinsfolk. Its critics (jurists, judges, eventually legislators across the Muslim world) observed what it became in practice: a way to hold estates together against the Qur'an's own inheritance shares, to favor sons over daughters by a stroke of the pen, to breed hereditary trustees with every incentive to milk and none to mend. Codes ended the quarrel differently, generation limits here, dissolution there; but mark who made the argument: the sharpest critique of the perpetual deed as a dynastic shelter came from inside the tradition that invented it.

The second quarrel cut into this book's own subject. By the sixteenth century, founders across Anatolia and the Balkans were endowing not land but money — the cash waqf, a capital sum locked as corpus and lent at a return that paid the teacher, the imam, the lamps. A modern historian's study of the affair is titled, with perfect economy, "Usurious Piety"; it describes the instrument as "the establishment of a trust with money the interest from which might pay the salary of a teacher, or preacher, or even unashamedly pass into the pocket of the founder." The empire's jurists went to war over it. Civizade, chief mufti until

his dismissal in 1541, attacked on both classical fronts: money is no durable corpus, and return on lent money is *riba* by every rule of Part III. Ebussuud, the most powerful shaykh al-islam the Ottomans ever had, defended them on custom, public welfare, and the plain fact that whole provinces' pious institutions lived on them. Birgivi, the age's great rigorist, answered in 1572 with a treatise whose title translates as the drawn sword. Imperial authority settled practice in the funds' favor — court registers show whole towns owing them money by the early seventeenth century — but the doctrine never fully surrendered, and the argument reads today like Faenza and the Fifth Lateran run in Ottoman Turkish — a mercy institution's need for income colliding with the wall against increase; chapter 17 watched Christendom hold exactly that debate about the Monti. No tradition plays the innocent of this story, and none the villain; the collision is structural, and every room found it.

The Treasury and the Poorhouse

The Hebrew instrument is older, and its biography stranger. Hekdesh is built on the root k-d-sh, holy: to declare property hekdesht was to move it, in the rabbis' idiom, out of common ownership into the ownership of the Most High, and at first the word meant Temple property alone. The Torah legislates for it as trespass: "When any of you commit a trespass and sin unintentionally in any of the holy things of the LORD, you shall bring, as your guilt offering to the LORD, a ram without blemish from the flock... And you shall make restitution for the holy thing in which you were remiss, and shall add one-fifth to it and give it to the priest" (Leviticus 5:15–16). From that verse the rabbis built the doctrine of *me'ila*, sacrilege: private benefit taken from consecrated property is trespass against Heaven, undone as trespass is undone — restitution, an added fifth, atonement. Property could be placed where misusing it stopped being a tort and became a sin: the same fence Umar's deed built, staked out a millennium earlier.

The dedication law carried an engineering refinement. The rabbis distinguished two grades of sanctity: *kedushat ha-guf*, sanctity of the body — the animal fit for the altar, holy in its substance, irredeemable — and *kedushat damim*, sanctity of value: property consecrated for the treasury, redeemable for money, the sanctity passing from object to coins and traveling on. Where the waqf locks the thing, the hekdesht of value locks the worth and lets the thing go — a consecrated field could be sold, the price, now itself hekdesht, doing the holy work. And among the carve-outs from the *ona'ah* price-fraud rules that chapter 9 read at Bava Metziah 4:9 stands consecrated property — not a license to cheat the treasury but a category statement: the overreaching rules referee bargains between owners, and one party here has no self to overreach.

Then the Temple fell, and the word had to decide what it meant in a world without its object. The halakha answered twice, and both answers instruct. First, the referent migrated: since we no longer have the Temple, ruled the medieval authorities — the ruling comes down in Ri Migash's name — an unqualified consecration now means consecration for the synagogue or for the poor. The instinct of dedication survived its address. But second, and here the tradition chose the opposite engineering from Islam's, the migrated hekdesht was deliberately stripped of the Temple's legal armor: the special rules of Temple property, the codifiers held, do not apply to consecrations for the poor, which follow the same laws as the property of the common man. No *me'ila*, no sacral untouchability — post-Temple charity property is ordinary property with an extraordinary owner, the community, standing in for purposes that cannot hold deeds. Hence the town of Bava Batra 8b, as chapter 18 walked it, may convert the food platter's funds into the coin fund and back as need shifts:

what the community dedicated, the community's court may redirect. Two philosophies of one lock: the first trusts the deed against everyone including the future; the second trusts the living community and arms it with authority over its own dedications. The waqf's rigidity preserved fountains for centuries and also preserved obsolete deeds no one could lawfully fix; the community's flexibility kept charity supple and left it exactly as strong as the community's own standing — and this book declines to referee.

That last clause carries a weight in Jewish history that must be stated plainly. The waqf lived under sovereigns who enforced it; the church's patrimony lived under kings who prayed inside it. The Jewish community's property lived on revocable charters, in towns from which the community could be — and repeatedly was — expelled; expulsion did not distinguish the synagogue's courtyard from the merchant's stock. A people whose tenure was conditional could not bind acres to purposes forever, and its genius ran to what could move and revolve: the *kupah's* coins, the *gemach's* revolving fund of chapter 15. The endowment's work — a permanent floor under communal mercy — was done less by locked land than by disciplined circulation: an asset lock of its own kind, enforced by covenant rather than by title. And the word itself took a sober last turn. In the towns of Eastern Europe, in the last centuries before the destruction, *hekdesch* came to name a building: the communal hospice-poorhouse where the itinerant poor and the broken slept. The scholar who studied it calls it a hospital-cum-poorhouse and, without softening, one of the grimmest institutions in East European Jewish society; by the late nineteenth century its connection to care of the sick was often nominal. The word that began as the treasury of Solomon's successors ended as the name of the shed at the edge of the *shtetl*. That descent is not one tradition's scandal; it is what this Part's institutions look like when a community's own margin collapses. An endowment is a promise about a future that includes poverty, war, and neglect; its history is written by its administration through the bad centuries — which is where this chapter is headed.

The Patrimony of the Poor

Who owns a cathedral? The question sounds scholastic and is in fact Christianity's oldest administrative problem, and the church answered it early, in council. In 451 the Council of Chalcedon, the same council that settled Christology for most of the Christian world, enacted two property canons in one sitting. Canon 26 ordered a universal trusteeship: "every church having a bishop shall have also a steward from among its own clergy, who shall manage the church business under the sanction of his own bishop" — the *oeconomus*, the house-manager, so that the church's goods should not be handled without witness. And canon 24 forged a lock: "Monasteries, which have once been consecrated with the consent of the bishop, shall remain monasteries for ever, and the property belonging to them shall be preserved, and they shall never again become secular dwellings." Together the two canons hold this chapter's whole apparatus in the year 451: the asset lock and the independent trustee, legislated in one afternoon.

What the property was for had been answered earlier. The patristic conviction chapter 18 read in Basil's barns — the surplus already belongs, by God's destination, to the need it can meet — hardened into an administrative maxim: the goods of the church are the patrimony of the poor, *patrimonium pauperum*. The bishop held much and owned nothing; what he ran was the poor's estate. The old Roman rule of the fourfold division made the trusteeship arithmetic: church revenues split into four portions — one to the bishop, charged with hospitality; one to the clergy; one to the poor; one to the fabric and repair of the churches — decreed by Simplicius and Gelasius in the late fifth century and quoted as

settled tradition by Gregory the Great to Augustine in newly missionized England. The poor hold a standing quarter of church revenue, not as alms at the administrator's discretion but as co-beneficiaries written into the division — the universal destination of goods, chapter 18's doctrine, running as bookkeeping. When the modern Catechism teaches that ownership makes its holder "a steward of Providence" and that the universal destination of goods "remains primordial," it repeats what the fourfold division assumed in the fifth century; the endowment is that doctrine's institutional form, and a pious foundation is what the universal destination looks like when somebody notarizes it.

On those foundations medieval Christendom built its own waqf civilization. The monastic estates — granges, glebes, vineyards, mills — were Europe's standing lessors, exactly as chapter 25 said: patient corporate owners letting real assets at rents that funded prayer, learning, hospitality, and relief century after century. The endowed hospital carried the sick and old on rent-rolls; the chantry locked a capital sum to a priest's stipend and, often riding along, a school; the cathedral chapter fed its canons from divided prebends. Canon law wrapped the whole estate in the instinct chapter 3 flagged from the start — presumptive inalienability, transfer only for grave cause, permissions designed to make dispersal slow — and the modern Code still carries the architecture. The church holds temporal goods for named purposes: divine worship, the decent support of ministers, "works of the sacred apostolate and of charity, especially toward the needy" (c. 1254 §2). Administrators must manage "with the diligence of a good householder" (c. 1284). Alienating the stable patrimony requires the competent authority's permission (c. 1291) — *istibdal's* cousin. The accepted wills of the faithful who give for pious causes "are to be fulfilled most diligently" (c. 1300), and the pious foundation remains a defined category (c. 1303): property accepted with a perpetual or long-term obligation attached — the waqfiyya's Latin twin, alive in current law.

The lock made enemies, and the most instructive early enemy was the state — not yet as confiscator but as actuary. Feudal kingship lived on the accidents of mortality — reliefs when a landholder died, wardships for his heir, escheat when a line failed — and land conveyed to a religious corporation stopped dying: the church never came of age, never married, never left an heir; its hand was a dead hand, *mortmain*, and every acre under it fell out of the feudal revenue system forever. The English crown moved from complaint to statute: Magna Carta's 1217 reissue barred the collusive gift-and-lease-back to religious houses, and in 1279 Edward I's Statute of Mortmain forbade conveying land to the church at all without royal license, on pain of forfeiture. The statute concedes even as it restricts: perpetuity had grown attractive enough to need pricing. The dead hand was the endowment form working exactly as designed, shifting costs onto a fiscal system built around death — a quarrel between the lock and the future's representative that has never really closed. The church kept critics on nearer ground, and their standing complaint was not the lock but its capture — the benefice, the endowed "living," held for its income by men who never saw the parish.

The Trustee and the Dead Hand

Set the three rooms side by side and one figure stands in all of them. The waqf has its mutawalli — the standard prefers nazir — appointed first by the founder, charged with management, maintenance, development, and investment, entitled to remuneration, and held to what the standard calls a fiduciary capacity: liable not for outcomes but for "misconduct, negligence, or breach of Waqif's terms and conditions" (SS 60, art. 3/7/5). The hekdeshe has its trustee under the community's court — itself, the rabbinical judgments

say, the higher guardian of every dedication — and its *gabbaim*, whose procedures chapter 18 walked: collection in pairs, distribution by three, coins counted one by one. The church has its *oeconomus* under the bishop's sanction, and canon 1284's good householder. And Umar's deed, before any legal science existed, already held the whole office in one clause: eat from it in reasonable measure, hoard nothing. Compensation licensed, enrichment barred, the line patrolled by judgment. That clause is the seed of everything the modern lawyer calls fiduciary duty — care, loyalty, the trustee no richer from the trust than his agreed keep — and it sits in the charter deed itself because the founders understood from the first hour that the danger would live not with thieves but with staff.

The traditions' own histories supply the case file, and Interlude II told this book in advance what it would show: every tradition's endowed institutions found, sooner or later, with their administrators' interests grown over them. So they were. The hereditary *mutawalli* dynasties of the later *waqf* centuries treated trusteeships as family livings; the reform literature of the nineteenth and twentieth centuries — Muslim officials and jurists, long before any outside critic — catalogued decayed deeds, vanished registers, the corpus leased to the trustee's cousins at nominal rents. The benefice bred its absentees and pluralists in every reforming century, as the church's own preachers said — an endowment converts purpose into revenue, and revenue attracts men who prefer it without the purpose. The *hekdesh*'s guardians got the Talmud's double treatment chapter 18 recorded, honored as the stars of Daniel's verse and audited like suspects, because the sages assumed drift rather than hoped against it; and the *halakha*'s rule on accounting catches the balance the problem demands. Strictly, the law compels no reckoning from collectors of established integrity — the proof-text is the Temple-repair verse in Kings, where the money went to the foremen without accounting, “for they dealt honestly” — yet the codes' glossators add in the same breath that a public fund's trustees should render accounts anyway, periodically and discreetly, to stay above suspicion. Trust extended, verification institutionalized, dignity preserved: three requirements pulling against one another, held in one procedure. Every fund designer since has been rediscovering the triangle.

The instruments the traditions built against drift are this chapter's most exportable cargo. Dual control and public counting at the point of collection. A trustee distinct from the beneficiary, under an authority distinct from both — *qadi*, court, bishop — with the power to remove. Compensation fixed to the office, never to the corpus's appetite: the reasonable bread, the Monti's salaried *depositarius*, the fee chapter 16 watched the jurists cap at documented cost. Investment pitched conservative, the endowed corpus taking the seat the Talmud licensed for the orphans' money — near to profit and far from loss, chapter 10's phrase — because capital that must last forever has no business courting ruin for yield. The controlled exits — *istibdal*, redemption, the community's conversion power, the common law's *cy-pres*, by which a court steers an obsolete charitable purpose to the nearest living one rather than watch a deed die of literalism. Over it all, the founding document, written against the founder's own successors. On one thing all the traditions agreed: perpetuity is cheap to declare and expensive to administer, and the administration is the institution.

All of which disciplines the endowment's first enemy, the one inside the walls. The second enemy does not negotiate. A lock binds everyone who accepts the legal order that made it; it cannot bind the power that rewrites the order, and the modern era's opening act, in country after country, was to rewrite. England, 1536 to 1541: the monasteries dissolved by statute — some eight hundred twenty-five houses — their lands vested in the crown and sold, in what a standard account calls the largest legally enforced transfer of property in English history since the Norman Conquest. France, the second of November

1789: the National Assembly placed the goods of the church “at the disposal of the nation”; the sales as *biens nationaux* began within the year, and in February 1790 the monastic orders themselves were dissolved. Joseph II had already emptied the contemplative houses of his empire in the 1780s; Spain and Portugal followed in the 1830s. The waqf met the same century more slowly, to the same effect: colonial administrations confiscated outright — France took Algeria’s *habous* for the public domain in 1844 — while reforming and republican states centralized, folding the deeds into ministries whose spending on the purposes shrank decade by decade; the new Turkish republic gathered the empire’s endowments into a state directorate, and Egypt dissolved the family waqfs by decree in 1952. These facts are set down plainly, because the people under them lost schools, hospitals, and bread; the lesson is structural rather than partisan: the deed that cannot die can be killed. Every asset lock is a promise made inside a legal order and inherits that order’s mortality. The old drafters knew it: the waqfiyya calls down God’s judgment on violators, the church wielded excommunication, the hekdeshe lived under the covenant’s own sanctions. The modern drafter has no such artillery, and the design conclusion travels to Part X: a lock survives on legitimacy — on a society that wants the locked purpose served — and legitimacy is earned by governance, transparency, and visible service, or not at all.

Capital That Cannot Be Sold Back

The modern translation began with a trustee who refused an inheritance. Ernst Abbe, the physicist who had made the Zeiss optical works of Jena great and owned much of them, faced the founder’s ordinary fate — heirs, buyers, dispersal — and wrote a deed instead. In May 1889 he created the Carl-Zeiss-Stiftung; within two years the foundation stood registered as owner of the optical works and part-owner of the Schott glassworks; and the statute Abbe completed in 1896 reads, to a reader of this chapter, like a waqfiyya in German commercial law. The enterprise itself is the corpus, never to be sold; profits serve named purposes — the sciences, the University of Jena, the security of the works — and worker protections decades ahead of their time (health and pension provision, regulated hours, paid leave) stand written into the constitution. Founder, deed, corpus, purpose, trustee: every element of Umar’s instrument, reassembled around a factory the foundation owns still, funding science from the dividends. Nor was Abbe first: J. C. Jacobsen, the Copenhagen brewer, had chartered the Carlsberg Foundation by deed of gift in 1876 and willed it the brewery itself, which passed to the foundation on his death; the charter still obliges it to hold “at least 51% of the voting rights in Carlsberg A/S” — a lock on control now well into its second century. Robert Bosch’s arrangement, completed by his executors, splits the deed’s functions with juristic precision: a charitable foundation holds roughly 94 percent of the company’s capital and spends the dividends on its programs, while some 93 percent of the votes sit with a separate industrial trusteeship of senior practitioners — an ownership structure that, the company says, “guarantees the Bosch Group’s entrepreneurial independence.” Fruit to the purpose, stewardship to the competent, extraction to no one: the foundation-owned companies of northern Europe are the waqf’s quiet industrial afterlife.

What those firms did by bespoke charter, a living movement now does on principle. Steward Ownership, as the Purpose Foundation defines it, rests on two commitments made legally binding. Self-determination: “The company cannot become an object of speculation but remains self-determined and independent in the long term. The steering wheel always remains in the hands of people who are connected to the company and its mission” — control that “can never be sold or inherited like an asset.” And purpose-orientation:

“Profits are a means to an end, not an end in themselves. They serve the company’s mission and development or can be used to fund charitable activities” — and beneath both, stated as flatly as any clause in this chapter: “The value created within the company cannot be extracted by the company owners for their personal benefit.” Strip the modern idiom and the reader has been here all chapter: the steward is the mutawalli, bread licensed and enrichment barred; the two principles are Umar’s conditions, translated. The toolkit even reinvented the external guardian. In the golden-share model, a single special share — one percent or less of the votes — sits with an independent, itself asset-locked entity and carries a veto over precisely the acts that would undo the dedication: changes to share classes, sale, liquidation. The golden shareholder is a party whose only interest is the deed’s survival, holding one key that opens nothing and locks everything — Chalcedon’s *oeconomus* in corporate form. What all the structures share — Employee Ownership trusts among them, holding the company for those who work in it — is this chapter’s title concept made contractual: the asset lock binding the company’s assets to its purpose, the mission lock binding its governance.

The old locks were sacral; at least one modern lock is statutory. The United Kingdom’s community interest company, created by statute in 2004 to serve community purposes, carries what its regulator calls a compulsory asset lock that “cannot be removed”: “a legal clause that prevents the assets of a company being used for private gain, rather than the stated purposes of the organisation.” A CIC may part with assets only on terms any student of *istibdal* will recognize — the transfer “made for full market value so that the CIC retains the value of the assets transferred,” or made “to another asset-locked body” named in its articles — value may change vessels, never escape the dedication. Even the investors’ door is built like the old rules: dividends flow, but under an aggregate cap of thirty-five percent of profits, the majority of every surplus bound to the community — the moderate sum licensed, Lateran proportions in a business registry. And in 2022 the American clothing company Patagonia executed the full deed at market scale, in public: the founding family moved every voting share — two percent of the equity — into a purpose trust bound to the environmental mission, and the nonvoting ninety-eight percent into a nonprofit collective receiving the dividends, projected around a hundred million dollars a year, for the protection of nature. The announcement’s title was its theology — “Earth is now our only shareholder” — and Yvon Chouinard’s gloss belongs beside Umar’s conditions in this chapter’s museum: “Instead of extracting value from nature and transforming it into wealth, we are using the wealth Patagonia creates to protect the source.”

Honesty requires the two cautions the old rooms taught. Locks do not administer themselves: a steward-owned firm can be captured by its stewards as surely as a *waqf* by its *mutawallis*, which is why the serious structures carry the guardian, the transparency, and the removal powers this chapter catalogued — Part X’s fund charters treat drift as a design load, not a scandal; chapter 45 weighs the temptations at full length. And locks live inside legal orders: the statute behind the CIC can be amended, the trust law holding Patagonia’s deed rewritten; the expropriations of the last section fell on institutions older and better defended than any of these. The case for the modern locks is not that they are invulnerable but that they reverse the default. An ordinary company must be defended against sale by heroic effort, every quarter, forever; a steward-owned company must be sold by heroic effort, against its own constitution, over its guardian’s veto. The lock moves the burden of proof to extraction’s side of the table. Everything this Part has shown says the default, not the intention, is what survives the founders.

Now this section’s title can be cashed. Chapter 24 watched the sale relapse toward the loan,

chapter 26 watched owning paper drift back toward owed paper, and every instrument in this Part stays honest only while someone holds it honest against the pull of its market. The endowment is the traditions' structural answer to that pull: property placed where the market's master key — the exit, the sale, the buyout — does not turn, so that no future holder can convert the mission back into a margin. The *gemach's* corpus cannot be demutualized, for it has no shareholders; the *waqf's* orchard cannot be flipped, for the deed voids the flip; Patagonia cannot go to the highest bidder, for the highest bidder cannot buy what has no owner. What cannot be sold cannot be sold out. That is the endowment's contribution to this book's problem, and why chapter 41's fund architectures will reach for this chapter's instruments by name.

So Part VII closes its workshop. Six chapters ago it set out to watch contracts and institutions try to escape usury; the tour is complete: the sale that escapes by confessing its costs and owning its interval; the lease, by keeping ownership's burdens seated with its income; the certificate, by conveying a true fraction of a real thing; the pool of chapter 27, by sharing catastrophe rather than selling protection from it; the mathematics of chapter 28, proving why the sharing works; and the endowment, which escapes by retiring the capital from the contest altogether — the limit case toward which every other instrument's discipline points, the only one whose design premise is that everyone involved will eventually be dead. Each step surrenders more of capital's claim on certainty, and each surrender purchases more honesty, until at the top stands capital with no claim at all: corpus locked, fruit flowing, trustee fed and watched. The traditions built these machines while their world still agreed that extraction was a sin with a name. The next Part enters the world that stopped agreeing. Part VIII is the story of how the wall came down in law and conscience — beginning with chapter 30, "The Secularization of Interest," where the old question of what money may do to a neighbor is reclassified as a question of what money costs — and of what each tradition did with this Part's instruments when the civilization around them declared the argument over. The workshop's tools survive the workshop. Whether anyone still reaches for them is the modern half of this book.

Part VIII — Modernity, Evasion, and the Return of Form Over Substance

Chapter 30 — The Secularization of Interest

In 1545 the Parliament of England, in the same reign that had lately stripped the monasteries chapter 29 watched fall, passed a statute whose title still reads like a clerk's joke: An Act Against Usury. The act against usury made usury lawful. Under its terms a lender might take up to ten pounds in the hundred by the year; the offense now began above that line, and everything at or beneath it was, for the first time in English history, no business of any court. For as long as this book has been running, "usury" has named a kind of act — increase on a loan, any increase, the thing itself, the anatomy chapter 7 laid on the table. From 1545 it named a quantity: too much increase. A sin is a line you must not cross; a price is a dial you may argue about. One statute took the oldest category in moral finance and rebuilt it, without a syllable of theology, as a dial.

That is the whole event of this chapter, performed in miniature, and it is the door into this book's last historical Part. For twenty-nine chapters the argument has run inside a world that agreed — across three scriptures and a dozen centuries of quarrels — that the naked loan may carry no price, and that the interesting questions begin there. Part VIII enters the world that stopped asking. The wall did not fall to a better argument; no council unsaid it, no tractate retracted, no school of jurists surrendered the definition of *riba*. It fell to statutes, ledgers, and habit — the question passing from the confessor's room to the magistrate's office, and from the magistrate's office to the market, which does not take questions. Chapter 29 closed at exactly this threshold: the old question of what money may do to a neighbor was about to be refiled as a question of what money costs. This chapter tells how the refiling was done — in law, in banking practice, in each of the three traditions — and then asks the question no modern syllabus keeps a slot for: what was lost when the answer stopped mattering.

The story can even be told as the history of two words. Latin Christendom had *usura*, the charge for the use of money, the forbidden thing; and it had *interesse*, the canon lawyers' term of art for lawful compensation — literally what "stands between" an injured man and the position he ought to occupy, the vocabulary of the titles Part VI put on trial. One word named the sin, the other named the damages. English took both, and the words divided the territory between them. "Usury," already drifting toward the exorbitant in the fifteenth century, was fixed by the statutes as the name of the crime above the ceiling; "interest" — the innocent word, the damages word — entered the money vocabulary of English in the 1520s and became the polite name for exactly the thing the old law had forbidden under the other name. The modern world did not so much license usury as promote its damages vocabulary to be the name of the product. Every rate sheet printed since is, philologically, a claim that all interest is compensation — the extrinsic titles generalized into a noun.

An Act Against Usury

Run the English statutes in order, because no other country documents the secularization so cleanly in its own legal prose. The 1545 act was the work of a commercial parliament under a spendthrift crown; it did not argue, it licensed. Seven years later the wall was rebuilt: in 1552 a parliament of hotter Protestant conscience under Edward VI repealed the license outright, and the preamble shows what the repeal believed, in the spelling of the age: "Usurie is by the worde of God utterly prohibited, as a vyce moste odyous and

detestable." All increase forbidden again; the old doctrine, whole, in an English statute — for nineteen years. In 1571 Elizabeth's parliament revived the ten percent ceiling, and here the document becomes genuinely strange, because it tried to keep both worlds in one act. The statute that restored lawful interest declared in the same breath that all usury remained sin, forbidden by the law of God, and it provided that nothing in its own text should be read as allowing usury in point of religion or conscience. The queen's law would collect ten percent and the queen's God would still condemn it. The clause is a photograph of the exact instant of secularization: a civilization splitting the question between two authorities and handing the courts the half with the arithmetic in it. The theologian Ruston's summary of the decades that followed is the sentence the surveys quote: "usury passed from being an offence against public morality which a Christian government was expected to suppress to being a matter of private conscience," while "a new generation of Christian moralists redefined usury as excessive interest."

What conscience kept, it lost by attrition. When the next great usury statute came, in 1624, the historians of the subject note something absent: for the first time, no religious argument appears in it at all — the rate fell from ten to eight percent on reasoning about trade, and only trade. Thereafter the dial simply turned. Eight became six in 1660, at the Restoration; six became five under Anne in 1713; each turn debated as one debates a tariff — merchants, wars, the price of money in rival cities. The sin had become a ceiling, and the ceiling had become policy. A ceiling is what remains of a wall when only the height still matters.

Nor was the ceiling an English peculiarity; it was the whole world's compromise instrument. Charles V had beaten Henry's parliament to it by five years, licensing interest on commercial loans up to twelve in the hundred across the Low Countries in 1540 — Antwerp's sovereign clearing Antwerp's practice. The American colonies wrote their own numbers: mostly six percent in the eighteenth century, a few as high as eight, on the English model and often with the English penalty — a Massachusetts lender proved usurious forfeited principal and interest together. By the eighteenth century almost nobody with a statute book was asking the old question — whether increase on a loan can be just — and almost everybody was administering the new one: how much.

The ceiling's last century is the strangest part of the story, because its final defender was the founder of modern economics and its executioner was the founder of utilitarianism. Adam Smith, no friend of the old theology, wanted the legal maximum kept — set just above the lowest market rate, he argued in 1776, so that sober borrowers could always outbid the "prodigals and projectors" who alone will pay any price for money. It fell to Jeremy Bentham, in the *Defence of Usury* of 1787 — a set of letters whose thirteenth is addressed to Smith directly — to run the new logic to its terminus: "no man of ripe years and of sound mind, acting freely, and with his eyes open, ought to be hindered, with a view to his advantage, from making such bargain, in the way of obtaining money, as he thinks fit." Consent, capacity, information: where those stand, the state has no further question to ask about a loan — the exact inversion of the finding at the bottom of chapter 7's anatomy, that the borrower's consent is the consent of a man with no alternatives. Bentham's proposition won. The American states began easing and repealing their caps early in the new century, and in 1854 England — where the whole statutory story had begun three centuries before — repealed its usury laws outright, twenty enactments and every usury law still standing swept out in one act. The wall had become a ceiling; the ceiling now became open sky. What survives today under the name "usury law" — consumer rate caps, criminal usury statutes aimed at loan sharks — is genuinely useful and genuinely small: a guardrail at the cliff edge of a road the old law would not have built.

Paper Before Doctrine

Statutes, though, mostly ratify; they seldom invent. The world the acts of 1545 and 1571 were licensing had already been built, on paper, by men who never read a line of Aquinas, and the instrument that built it was the bill of exchange. Chapter 21 and the Salamanca files have already opened its anatomy: a merchant pays in cash in one city for a paper promising payment in another city, in another currency, at another date — a contract of exchange, not a loan, and therefore standing outside the usury wall altogether, so long as the exchange was real. Often it was real; and folded into the rate between two currencies and two dates lay whatever the parties cared to put there, including the price of the waiting. The theologians knew it perfectly well. Their long campaigns against “dry exchange” — bills drawn and redrawn between the same parties, crossing no real distance, serving no trade, exchange in appearance only — were the form-versus-substance rule of chapter 3 applied at the counting-house door, and the best of them went to the exchanges to learn the practice they were judging. The point for this chapter is not that the seam existed but which side of it grew. The paper multiplied; the audits did not.

Then the paper learned to travel, and the traveling changed everything. A bill made payable to bearer, or endorsable from hand to hand, stops being a private promise and becomes a small circulating fact — money’s cousin. The law accommodated it step by step: the Habsburg Netherlands enacted Europe’s first national legislation for bearers and endorsees of commercial paper in ordinances of 1537 and 1541; England arrived by judge-made custom and, for promissory notes, by a statute under Anne. Negotiability quietly multiplied the lenders. Whoever holds the paper is, for the length of the holding, the creditor; a bill three months from maturity might pass through six hands, and every hand had bought time-distant money at today’s price.

That purchase has a name, and the name is the hinge of this section. To discount a bill is to buy it before its day for less than its face: a hundred due at Michaelmas sells in July for ninety-seven, and the three is nothing on earth but the price of eleven weeks — the *agar natar* of chapter 7, computed to the day, carried inside a lawful instrument’s price. The economic historians date the real career of discounting from precisely the moment the charge inside it became legal to name: the 1540s in the Low Countries, the licensed decades in England. A merchant with a drawer of bills did not hold a theory about the price of time; he held the going rate at which his drawer turned into cash. The money market existed, liquid and priced, for two centuries before economics existed to explain it. By 1694 the biggest borrower had joined the market it regulated: the Bank of England was chartered to fund a king’s war, and within a generation government stock was the respectable income of widows, colleges, and charities — the creditor’s seat, which Part II’s scriptures had hedged with commandments, refurnished as the safest chair in the kingdom. Doctrine, when it finally moved, moved into a city already furnished. The moralists of the next section did not open the door to interest. They found it standing open, propped by a hundred thousand bills, and had to decide what to preach about the traffic.

Calvin’s Letter and Weber’s Shadow

In the very year of Henry’s statute, 1545, a businessman named Claude de Sachin wrote to Geneva to ask John Calvin whether a Christian might take interest, and Calvin’s answer — a private letter — became the most consequential few pages any theologian contributed to this chapter’s story. Hold on to the coincidence of dates. The reformer was not opening a gate; the gates of Westminster and Antwerp stood open already. He was answering a

correspondent who lived among open gates and wanted to know what God's law still required of him inside them.

The letter's argument can be stated in three moves, each careful, pastoral, and closer to this book's concerns than its reputation suggests. First, Calvin declined to read the scriptures as banning increase as such: the Torah's interest laws he took as given to Israel's particular polity, aimed at the protection of the poor brother, and he found no text of scripture that forbids all usury to all men in all dealings. Second, he set the sterility argument aside — a child, he observed, can see that money shut in a box breeds nothing, and any merchant can see that money in commerce is not shut in a box; the scholastic axiom seemed to him an argument about coins that ignored what coins do. Third, with anatomy dismissed, one test remained: equity. Judge each loan by justice and charity between the parties — by the golden rule — and the gain a lender takes without injuring anyone is not the forbidden thing. Around that test he built fences, and the fences are the part posterity dropped: no interest from the poor or from need; lending at interest never to become a man's trade; the borrower ought to gain as much or more from the money as the lender takes from him. Visser and McIntosh, the literature's modern surveyors, count seven such conditions and append the epitaph: they "have been generally ignored and his stance taken as a wholesale sanctioning of interest."

What Calvin actually moved was not the verdict — statutes had preceded him — but the method, and the change of method is the deep Protestant transformation this section owes the reader. The tradition this book has walked tests the contract: chapter 7's question was what, in the loan's anatomy, a charge could possibly price. Calvin's question is relational: what does this charge do to this neighbor? There is real gain in the relocation — it speaks the prophets' own register, and it put the poor borrower back at the center of the discussion. But the relocation carried a structural cost that showed within two generations. A rule about the contract can be enforced by strangers — any court, any auditor, can ask who owns, who bears the loss, what the charge prices, and chapter 3's whole method rides on that. A rule about equity between hearts can be enforced only by the hearts, and when the jurisdiction of conscience went private — Ruston's sentence again — the fences went private with it. The wall had been public architecture. The fences were good intentions, severally held; and the paper of the last section outlived both.

Three and a half centuries later, a German scholar made the relocation itself the hero of a famous argument. Max Weber's *The Protestant Ethic and the Spirit of Capitalism* — two essays of 1904 and 1905, carried into English in 1930 by Talcott Parsons — does not claim that Protestants invented capitalism; Weber knew as well as his critics that the instruments were old and many of them Italian. His claim was about temperament: that ascetic Protestantism — the Calvinist's calling, the Puritan's discipline — bred an ethos of methodical, dutiful accumulation, work as vocation and gain as its sanctified index — the spirit modern capitalism ran on once it ran at mass scale. His chosen exhibit was Benjamin Franklin, whose advice to a young tradesman Weber quotes: "Remember, that time is money"; and again, "Money can beget money, and its offspring can beget more." Set that beside chapter 1. *Tokos*, the unnatural offspring of barren metal, has become an apprentice's maxim — breeding money is no longer what money cannot do but what a diligent boy is taught money is for. Weber's own conclusion was closer to elegy than to triumph. The Puritan's care for goods, he wrote, borrowing Richard Baxter's image, should have sat on the saint's shoulders "like a light cloak, which can be thrown aside at any moment. But fate decreed that the cloak should become an iron cage." And: "The Puritan wanted to work in a calling; we are forced to do so." The religious root, on his own telling, died in the machine it

started; victorious capitalism no longer needed the support.

The thesis has taken a century of fire, and the reader should know its shape; this book does not referee. Historians pointed to fourteenth-century Italy, Catholic and flourishing at exactly the commerce the spirit was supposed to enable; Braudel judged the theory flatly false, the north having copied the south's tools; Samuelsson ran the correlations between confession and prosperity and found them wanting; even the iron cage is partly the translator's — Weber's steel-hard housing became a cage in Parsons's English, and the most famous image in the argument is half Parsons's. This book adds only the narrow finding its own subject licenses. On usury, the confessional map does not sort the way the shadow suggests: a Catholic emperor licensed commercial interest five years before Protestant Westminster did; Geneva fenced what Rome titled, and the two arrived at working settlements more alike than either liked to hear; and the silence that finally fell over the question was ecumenical. Weber's shadow matters to this chapter mostly because it taught a century of readers to file the secularization of interest as a Protestant subplot. The record reads otherwise: one settlement, many managements.

Living Under the Settlement

Begin the survey of managements in Verona, in 1744, where the marquis Scipione Maffei — antiquarian, playwright, and friend of the reigning pope — published a treatise called *Dell'impiego del danaro*, on the employment of money, defending the moderate interest his class had long collected. Rome's answer came a year later, and it was not an accommodation. Benedict XIV's *Vix pervenit*, of the first of November 1745, restated the architecture of Parts III and VI entire: the sin called usury "has its proper place and origin in a loan contract" — in the *mutuum* itself, which obliges the return of as much and no more; the gain being moderate excuses nothing, the borrower being rich excuses nothing, the money being put to fruitful work excuses nothing; and yet the titles standing genuinely outside the contract — reasons, as chapter 7 quoted it, "not at all intrinsic to the contract" — remain as honest as they ever were. It was addressed to the bishops of Italy, a register the Church's own lawyers rank below her most solemn; in 1836 the Holy Office applied it to the whole Church. On paper, nothing had moved since Aquinas.

Off paper, everything moved within a lifetime, and the mechanism was not a document but a question box. The Napoleonic Code had made interest legal across a continent; penitents who lent at the code's rate asked their confessors; the confessors asked Rome. Through the 1820s the questions accumulated, and between 1830 and the 1870s the replies of the Holy Office and the Sacred Penitentiary settled into a single quiet formula: those who lend at the rate the civil law allows are not to be disturbed — no denial of absolution, no restitution demanded — provided they stand ready to submit to any future judgment of the Holy See. The future judgment never came. What came instead, in 1917, was the Code of Canon Law, whose canon 1543 performs the entire two-century history in a single sentence: in a loan of fungibles, "no gain can be received by reason of the contract itself" — *Vix pervenit* entire, the doctrine of chapter 7 word for word — "but in the payment of a fungible thing, it is not in itself illicit to contract for the gain allowed by law, unless it is clear that this is excessive," and more than the legal gain where a just and adequate title stands. The first clause keeps the wall; the second hands every lender in every jurisdiction a standing title the size of the civil rate. Doctrine intact, practice released, and between them a silence that has now lasted a century and a half. John T. Noonan, the tradition's own historian-judge, put the weight of the retired teaching in one sentence: it had been "enunciated by popes, expressed by three ecumenical councils, proclaimed by bishops, and taught unanimously by theologians." A

teaching with that pedigree was not refuted; it was shelved in place, and the Church's own encyclopedias explained the shelving candidly: right to permit now, as it was not wrong to condemn then, the economy itself having changed. When the Church found its public economic voice again, in the social teaching of the industrial age, the word usury was no longer among its working headings; what that silence said, and what it cost, is chapter 33's file.

For the Jews of Europe, the secularization of interest arrived inside a larger event, and the larger event must govern the telling. Emancipation came first in France: on the twenty-seventh of September 1791 the National Assembly admitted the Jews to full citizenship, and over the following decades, unevenly and revocably, the rest of Europe followed. With the civil disabilities fell the occupational cage that chapter 17 documented — the centuries in which lending was licensed to Jews because nearly everything else was barred to them. Within two generations Jewish families were in general trade, in manufacturing, in medicine and law and scholarship, and in the new joint-stock banking alongside everyone else's banks: not "the moneylenders" of the old towns but merchants and financiers among others, distinguishable chiefly by the hatreds that followed them. That must be said plainly too: the polemical figure of the Jewish usurer did not retire when the licensed trade did — chapter 2 marked its afterlife, the nineteenth and twentieth centuries armed it, and this book does not forget it. In 1806 Napoleon put twelve questions to an Assembly of Jewish Notables he had summoned to Paris, and two of the twelve were about usury — the questionnaire itself a fair measure of where the old polemic still stood; an emperor's questionnaire is not a seminar. The answers, ratified the next year by the Grand Sanhedrin the emperor convened, deserve reading past the duress, because they drew on the tradition's own resources: the assembly declared the fellow citizen a brother — Deuteronomy's line read as civic rather than ethnic, the brother being every citizen of the shared state — with lawful commercial interest open toward all alike, and extortion licensed toward no one, Jew or Christian. Inside the community's own law, meanwhile, nothing secularized at all. *Ribbit* remained ribbit; the wall of chapter 7 stood untouched. What changed was scale. As observant Jews built and used modern banks, the instrument chapter 14 anatomized traveled with them, from a deed between two merchants to institutional boilerplate — the *heter iska klali*, one master partnership standing behind an entire institution's book, posted on the banking hall's wall, folded into standard terms, until in the twentieth century the banking system of a Jewish state could run on it. Whether the partnership inside the paper survives that scaling — whether anyone, at any counter, is still genuinely asked about outcomes — is chapter 31's question, and it is held there.

Islam met the settlement last and hardest, because for most of the Muslim world it was not a domestic bargain but an import with gunboats behind it. Through the nineteenth century, European banks opened in Muslim cities, commercial codes drafted on French models arrived with colonial and semi-colonial administration, and interest came in as infrastructure — the plumbing of a world system nobody had voted to join. Honesty requires the domestic half too, and chapter 29 already carried it: the Ottoman lands had run interest-adjacent practice of their own for centuries — the cash *waqf*, lending its corpus at increase, was commonplace in Anatolia and Rumeli by the early seventeenth century, attacked and defended by the empire's own jurists and retained; and the sultans capped by decree what such lending could take, at ceilings the historians report near eleven percent. The doctrine of *riba* never fell; the craft of managing the gap between doctrine and practice was old and Ottoman. What the modern era added was totality — by the twentieth century a Muslim merchant could scarcely touch banking without touching interest, and the muftis' offices swung with the pressure; Egypt's ruled bank interest one thing and

then another and then the first thing again across a century. The revival, when it came, answered with institutions rather than rulings. In 1963, at Mit Ghamr in the Nile delta, a small savings house opened that neither paid nor charged interest, placing its depositors' money in local ventures and sharing what the ventures actually earned; it was wound up by 1967, with nine such houses by then operating in Egypt. In 1975 came the Islamic Development Bank, the member states' development bank, and Dubai Islamic Bank, commonly counted the first full-service commercial bank of the kind; in February 1976 the First International Conference on Islamic Economics at Makkah gathered several hundred scholars and economists who declared bank interest to be *riba* in all its forms — the revival binding itself to the hard line, all interest and not merely the exorbitant kind. From that trellis grew the industry whose instruments Part VII anatomized, *murabaha* to *sukuk* to *takaful*. Whether the industry's practice honors the revival's line — the benchmark problem chapter 20 named, the costumes chapters 24 and 26 watched being fitted — is chapter 32's file, and the sharpest voices in it belong to the tradition's own scholars.

Three managements of one settlement, then — not three grades on a single scale but three different bets about where a wall can survive under open sky: Rome bet on conscience, holding the doctrine and releasing the practice into a century of quiet; the halakha bet on contract, keeping its law intact and industrializing its own instrument to live beside the world; Islam bet on institutions, alone among the three attempting a parallel banking civilization. The next three chapters audit the bets in turn — the contract in chapter 31, the institutions in chapter 32, the conscience and its silence in chapter 33.

Normal Finance

What Bentham's proposition won was not a policy but the definition of the question, and the discipline that formed around the new markets took the chair theology had vacated. In the economists' hands, interest ceased to be a charge requiring a title and became a price reporting a fact — and the facts on offer were, one by one, the old refused claims returning as respectable theory. Böhm-Bawerk founded the modern account on time itself, the premium of present goods over future goods; chapter 7 watched him answer William of Auxerre's stones rather than refute them, and the point holds for the whole discipline: the old questions were not won; the new science was founded on their far side. Opportunity cost finished its journey from refused exception to first axiom — chapter 20 told that story whole, the market's elsewhere priced into every contract — and by 1894 the axiom had its name in the journals. And the working formula every student now learns first, the time value of money, is the third refused title in its new career: *carrentia pecuniae*, the lender's mere lack of money for a term, the claim Molina and Lugo would not admit because admitting it repealed everything, is the silent first assumption of every discount curve — the ghost grammar, chapter 21 called it, underneath the modern phrase. What the casuists could not concede without unsaying their tradition, the textbook assumes before page one, not as an argument but as a definition.

So the birth of "normal finance" can be dated and described without a single new invention, because it was not an instrument but a retirement. The old world made the lender show a title; the new world makes the abstainer show a reason — the burden of proof crossing the table, exactly the reversal chapter 20 diagnosed inside the doctrine, now installed as common sense. The word "usury" kept working, but at a new job: in the statute books it names a rate crime at the far margins — the loan shark's percentage, the cap in a consumer code — and in the mouth it is a scold's word meaning too much. Ruston's redefinition, complete. And with the question thus divided — legality to the legislature, quantity to

the market, morality to the private heart — the two disciplines that had owned it for their whole history closed their files in the same century: theology accommodated and went quiet, and economics redefined and had nothing left to ask, since a price can be high or low but cannot be a sin. Normal finance is finance with no slot for the question this book is about.

Say honestly what the settlement bought; the traditions never argued from nostalgia, and this book will not start now. The secularization of interest is load-bearing in the modern world's prosperity: credit at continental scale, capital assembled from strangers for works no guild or *gemach* could have financed, the poor man's mortgage alongside the sovereign's bond. Nobody in Part IX proposes to repeal the twentieth century. The loss lies somewhere more precise, and this book has spent twenty-nine chapters equipping the reader to name it. What was lost was a diagnostic vocabulary — the entire instrument panel by which a civilization once told fruitfulness from extraction. Chapter 3's taxonomy: eight ways for capital to stand in a human relationship, and one test for when the label lies. Chapter 7's anatomy: whose money, whose risk, what exactly does the charge price. Part VI's discipline of titles: real burdens, shown, sized, capped, counted once — a surplus made to name its source. Part IV's counting-room: losses in their true seats, and no profit claim before real profit exists. And beneath all of it, chapter 2's person at the door, whose need was the one thing no instrument might price. None of this machinery required the world to forgo return; it required return to answer a question. When interest became morally neutral, extraction did not stop — nothing about a category's retirement changes what the category described. What stopped was the asking. Modern finance can distinguish ten thousand instruments by yield, duration, and covenant, and has no working word for the one distinction every tradition in this book thought was the point.

The loss is not antiquarian, and one sentence of arithmetic says why. Chapter 28 ran the arithmetic the burial clubs lived by, and its finding lands here: the difference between finance that smooths the strong path and finance that abandons the weak one is invisible in the rate and visible only in the ruin column — and the neutral vocabulary keeps no ruin column. A civilization that prices everything by the first number and audits nothing by the second has not become wicked; it has become blind in exactly one eye, the eye the old vocabulary existed to keep open. The modern world did not answer the usury question; it forgot the language the question was asked in. Even the words kept score. The lawyers' term for damages survived and became the product's name, printed on every statement; the Torah's word for the thing itself — *neshekh*, the bite, which chapter 4 walked — survives in no ledger anywhere.

But a language forgotten in public can be kept in houses, and the rest of this Part visits the houses. The vocabulary did not die; it retired into the traditions' own keeping, each of which had to run its instruments inside a civilization that no longer understood why anyone bothered. Chapter 31 opens the ledgers of the heter iska at national-banking scale and asks the tradition's own hard question about what the partnership still means. Chapter 32 weighs the Islamic banking industry against the revival's hard line, with the tradition's own critics holding the scale. Chapter 33 sits with Rome's long quiet and reads what the social encyclicals said around the subject they no longer named. And chapter 34 watches the market itself, uneasy in its neutrality, invent a moral vocabulary of its own — ESG, impact, sustainability — and asks what a conscience rebuilt without the old grammar can and cannot see. Then the historical work is done, and chapters 35 to 43 undertake the constructive half: the retired vocabulary rebuilt as working arithmetic — the tests as models, the titles as line items, the ruin probability restored to the reports. In 1545 a parliament

retitled the oldest question in this book as a number. The rest of this book is about the people who kept the question, and about getting it back into the arithmetic — which is where the modern world keeps everything it still intends to mean.

Chapter 31 — Heter Iska and the Jewish Art of Legal Transformation

In the last decade of the sixteenth century, the rabbi of Cracow sat down to fix a document that had made him famous and was making him trouble. His name was Isaiah Menahem ben Isaac, though hardly anyone called him that: Polish Jewry knew him as Rabbi Mendel Avigdors, after the custom of adding a father-in-law's name, and he had come up through the working rabbinate — Praga by Warsaw, a yeshiva at Szczebrzeszyn, the rabbinate of Lodomeria — before Cracow called him in 1591. Somewhere along that road he had drafted a form: a template by which a Jew with capital and a Jew who needed it could sign their way into an arrangement that paid the first out of the earnings of the second without either of them touching the Torah's ban on *ribbit*. The form worked; forms that work travel; and it soon attracted exactly the readers a clever financial instrument least wants. Mordecai Jaffe, author of the *Levush*, objected. Joshua Falk, whose commentary every subsequent student would study as the *Sema*, objected at length. What they saw in the early draft was the thing this whole Part of the book is about — paper that said partnership while the money underneath behaved like a loan — and the rabbi of Cracow, to his lasting credit, took the criticism as a draftsman should: he amended the instrument. The revised text went out under a title that named him the way court rulings name judges: *shtar heter iska ke-tikkun Maharam* — a deed of permitted venture per the enactment of Maharam, “our teacher Rabbi Mendel.” For the next two and a half centuries, when European Jews financed each other at what anyone in the street would have called interest, this was, more often than any rival, the piece of paper that made it lawful.

The document's afterlife would have astonished its author. The *heter iska* — the “permission of venture,” to give the genre its standing name — is today folded into the standard terms of Israel's banks, litigated in the commercial courts of New York, invoked over credit lines, deposits, and overdrafts. A form drafted for two merchants who knew each other's warehouses now stands behind interest-bearing balance sheets of national scale, and an observant Jew can pass an entire financial life inside its shelter without once reading it. Whether that is a triumph of legal engineering or the slow hollowing of one is a question the tradition itself has never stopped asking, and this chapter tells both the triumph and the asking. One version it does not tell is the polemicists', in which the instrument is the toll gate that proves the wall a sham. Read from inside, the record shows something more interesting: a legal civilization that refused to abandon either of two commitments — the Torah's ban and a community's need for credit — and that engineered, criticized, re-engineered, and audited an instrument for holding them together, its own scholars conducting the audit in print down to this morning. The quarrel over the *heter iska* is a Jewish quarrel, argued in the tradition's own courtrooms in its own vocabulary. This chapter reports it from those benches.

The Wall and the Fair

Begin with why the instrument had to exist, because its necessity was built, stone by stone, out of everything Part III of this book established.

The Torah's prohibition of lending at increase between Jews is not a regulation with edges

a lawyer can live in; it is, as chapter 2 read it, a wall. It binds lender and borrower alike, and the witnesses, and the scribe who writes the bond. It forbids the fixed charge stipulated at signing — *ribbit ketzutzah*, the Torah's own case, which a court will claw back — and the rabbis fenced the fence, forbidding *avak ribbit*, the "dust of interest," and interest paid in favors, lodging, market tips, words themselves; chapters 3, 7, and 8 walked the perimeter. A borrower may not even offer his lender an unaccustomed greeting. Against that standard the ordinary instruments of European credit, the bond at a yearly rate and the pawn at a weekly one, were between Jew and Jew the machinery of the sin whose strange solemnity chapter 2 paused over: the lender at interest who, in the baraita's image, brings witnesses and a scribe and signs a denial of the God of Israel.

And yet the communities that carried this law lived, by the sixteenth century, in the middle of a credit economy — not at its margins, as the polemical picture has it, but as the merchants, leaseholders, and traders of a commercial civilization spread across the Polish-Lithuanian Commonwealth. The Council of Four Lands, the standing Jewish parliament that met at the great fairs of Lublin and Jaroslaw, governed a society in which Jews financed one another daily: the wholesaler fronting a peddler's stock, the community borrowing for its synagogue roof, the merchant capitalizing a son-in-law. Deuteronomy's boundary, chapter 2 showed, had always permitted ordinary commercial terms between the covenant community and the outside world; but here was Jewish money meeting Jewish need at every stall of every fair, and between Jew and Jew the wall stood total. A community can hold a total ban and a commercial economy, but not without engineering. Credit between Jews could simply stop, which starves the community's own traders — a result the rabbis had refused since Babylonia. The ban could be quietly ignored, which manufactures sinners by the town-full and teaches contempt for the whole law. Or the financing could be moved, honestly, onto the one road the law had always left open.

That road was older than the problem. The Babylonian sages, chapter 11 told, had engineered it centuries before Poland: the *iska*, the investment arrangement in which capital is placed with a working merchant on shared terms. The Nehardean formula, preserved at Bava Metzia 104b, defines it with the bluntness of a survey stake: *palga milveh, palga pikadon* — half loan, half deposit. Half the capital becomes the manager's own, a barren loan he owes back in every state of the world; half remains the investor's property, trading in the manager's hands at the investor's risk, and the investor's return is the fruit of that half: his own money's earnings, not a charge on anyone's need. The Talmud records the enactment's purpose in institutional language — the Sages formulated it to be satisfactory to both sides, borrower and lender alike — and even the stress test: if half the merchandise is really the manager's own, may he sell his half and drink beer with the proceeds? The money was given for business, Rava answered, and business it must do. Nothing in the heter *iska* is newer than this. What the Polish rabbis added was not a doctrine but a document: the *iska* reduced to a reusable form, its evidentiary machinery — witnesses, oath, settlement — standardized in advance, so that a transaction that would otherwise take a rabbinic consultation could be executed by any two Jews with a pen.

The standardization had the fingerprints of a whole legal culture on it, not one draftsman's. Rabbi Mendel's first version drew the Levush's fire and the Sema's; the revision answered them; and Falk himself carried the campaign to the legislature. His treatise on the interest laws went before the Council of Four Lands with model ordinances attached, and in 1607 the Council approved a body of his rules on credit and lending — a parliament of scattered communities, meeting at a fair, enacting subordinate legislation under the Torah's oldest financial prohibition. Later generations kept auditing. When doubters pressed the

tikkun Maharam a half-century on, Samuel ben David ha-Levi defended it in the *Nachalat Shivah* (Amsterdam, 1667), the scrivener's handbook whose model documents taught notaries across Ashkenaz what a proper deed should say; the heter iska entered the standard repertoire alongside the marriage contracts and the powers of attorney, and later handbooks carried refined successors, the Chachmat Adam's among them. Hold the sequence in view; it is the chapter's first finding. The heter iska was not slipped past the law's guardians; it was drafted by them, attacked by them, amended under their pressure, legislated at their synod, and republished under their scrutiny for four hundred years. Whatever else this instrument is, it is not a private evasion. It is the tradition's own public works.

Two Halves and a Wage

What does the document actually do? The classical answer has three moving parts, and each is a place where the paper either tells the truth about the money or stops telling it.

The first part is the recharacterization itself: the financing the parties would otherwise have written as a loan is written as an iska — typically half loan, half deposit, on the Nehardean template, though some versions run the whole sum as a managed venture — so that the “borrower” becomes a managing partner and the “lender” an investor whose deposit half trades at his own risk. Everything downstream depends on that risk being real, and the codes are unembarrassed about it. Maimonides, codifying the iska in his laws of agents and partners, states the two regimes with a bluntness this book met in chapter 13: the manager is responsible for the loan half “even if it is destroyed by forces beyond his control,” while the deposit half stands on the investor's account — “if the half that is considered an entrusted article is stolen or lost, the administrator is not liable to pay” (Sheluchin veShutafin 6:2, in Eliyahu Touger's rendering). The manager holds that half as a keeper, not an underwriter: negligence makes him liable, as it makes every keeper liable, but theft, loss, and the ordinary treachery of markets land on the owner. An investor under a heter iska is, on the instrument's own terms, a man who can lose half his money to a bad season without recourse. That clause is no decoration; it is the entire license for his return, and the tradition polices it from both directions: chapter 11 marked the boundary where a guaranteed deposit collapses into loan, and the codes void any drafting that seats the investor beyond the reach of loss while his yield keeps flowing.

The second part is the wage, and its smallness is its significance. The managing partner works the investor's half as well as his own, and chapter 12 built the book's account of what the rabbis saw there: unpaid management of another man's money is value moving toward a lender because of a loan — interest, in the rabbis' analysis, however it is denominated. So the Mishnah forbids installing a shopkeeper for half profits unless he is paid for his trouble, and the Talmud at Bava Metziah 68b holds the seminar on how much. Abaye's measure is the idle laborer — *k'poel batel*, priced at what it costs such a man to let his own trade stand idle while he tends the venture. Rabbi Meir lets the parties set it “a lot or a little, in accordance with the agreement between them”; Rabbi Yehuda is satisfied with a token — even dipping the manager's bread in brine with him, even one dried fig, changes the labor's legal name; Rabbi Shimon ben Yohai wants a laborer's full wage. The codes kept the ladder in working order: Maimonides requires the idle laborer's wage or, where the manager has business of his own besides, “one dinar for the entire time of the partnership”; the Shulchan Arukh's laws of interest carry the rule at siman 177, the iska's home address in the code, beside the standard alternative — no wage, and a tilted split instead, the manager taking a fatter fraction of profit than he bears of loss, on the pattern of Mai-

monides' default of two-thirds against one-third that chapter 13 admired. The halakhists' name for the category is *sechar tircha*, the wage for trouble, and the dinar is doing what chapter 12 found it doing everywhere: declaratory work. A contract that pays even one coin for management has confessed, in writing, that the manager's labor is a contribution and not a courtesy — that this is a venture with a worker in it, not a loan with a debtor.

And the third part is the pair of numbers that make the instrument commercial. The document assigns the investor his share of the venture's earnings — half, classically, riding the deposit half — and then does something that looks, at first reading, like the whole game: it names a figure. The parties agree in advance what the investor's share is expected to come to — so much per hundred per year; in the modern forms, chapter 14 found, even a rate keyed to the market's prevailing benchmarks — and the manager may discharge his entire obligation by paying it. Alone, that clause would be a fixed return with a Hebrew heading — the Levush and the Sema's whole case against the first draft. It does not stand alone. What stands between the named figure and a naked rate is the instrument's evidentiary architecture — the machinery of witnesses and oaths that is the heter iska's true signature, and the next section's subject. But mark how much substance the paper has already moved. Under a loan, chapter 13's arithmetic, the borrower owes the sum in every possible world; his shop can burn and the debt survives the fire. Under the tikkun Maharam, the same man holding the same coins is a trustee of half of them: if the venture is robbed, if the stock rots, if the season fails — and he can show it — the investor's half is the investor's loss, and the "borrower" owes the loan half alone. The default has different consequences; the ruin lands in different seats; the heirs inherit a different balance sheet. At the scale of two merchants, whatever one concludes at banking scale, this is no relabeling: it redistributes catastrophe.

Witnesses, Oaths, and the Settlement Sum

The evidentiary clauses are the part of the heter iska outsiders never quote and insiders never stop arguing about, and they descend from the problem chapter 13 left standing: the venture's truth lives with the manager. An investor whose return rides real outcomes must somehow learn what the outcomes were, and the classical partnership's answer — the sworn accounting, the books opened before witnesses — is expensive, intrusive, and, between a wholesaler and forty peddlers, impossible. The tikkun Maharam's answer became the standard, and its two conditions must be stated exactly, because everything admirable and everything dangerous in the modern instrument lives inside them.

Condition one: a claim of loss is believed only on the testimony of qualified witnesses — two, the halakha's ordinary standard for establishing a fact — who know the loss of their own knowledge. The manager who says the deposit half perished must prove it the way the law proves things. Condition two: a claim of low profit — earnings below the named figure — is believed only under a solemn oath, the *shevu'ah*, sworn in the gravity the tradition reserves for invoking the Name. The calibration is exquisite, and chapter 14 admired it: losses are public events, the kind witnesses can in principle know; profits are ledger events, invisible from outside, and for those the instrument reaches past evidence to conscience. Then the third clause, the one that made the instrument's fortune: the manager may decline the whole apparatus — no books, no witnesses, no oath — and simply pay the named figure, the *sechum ha-hitpashrut*, the settlement sum, in full and final discharge. The drafting even bends courteously toward him: the payment is styled not as penalty or confession but as compromise, the price of everyone's peace and the manager's privacy.

Run the machine, as chapter 14 first ran it, and watch what comes out. The observant borrower reaches term. He can prove a loss with witnesses, if there was one and it was witnessable. He can swear, before God, to the penny of his profits — an act men of a community that takes oaths seriously will go to great lengths to avoid; the reluctance to swear even to the truth runs deep, and the drafters knew their material the way an architect knows stone. Or he can pay the figure and go home. Nearly everyone pays the figure. The halakhic literature is perfectly candid about this — candid in a way that should raise, not lower, the reader's respect: paying in lieu is no abuse the drafters failed to foresee but, as the compendia put it, a basic premise of how the instrument functions. The result, across any large book of contracts, is a stream of fixed, periodic, benchmark-shaped payments flowing from “managers” to “investors” under documents whose every clause is partnership. Chapter 20, weighing the tradition's answer to the lender who cries forgone profit, saw the named figure for what it structurally is: a conservative model of expected gain, agreed in advance, collectable only because the capital that earns it genuinely stands in the wind. The oath and the witnesses are the wind gauge: they keep the model corrigible by reality — the narrow channel, as chapter 14 called it, through which the venture's actual truth can always, in principle, reach the money.

The tradition's drafters patrolled that channel with a jeweler's suspicion, and their case law shows what they understood themselves to be building. A document naming as acceptable witnesses only men in a far city — men who cannot possibly know the venture — is invalid: verification that cannot occur is not verification, and the poskim struck the device down. An iska written over inherently unverifiable holdings — hidden investments no witness could ever attest — cannot carry a witnesses-clause at all. The oath may be made weighty — some forms stipulate it be sworn publicly, in the synagogue — but a form engineered so the oath is certain to deter, drafted never to be sworn, has crossed from using the manager's conscience to counterfeiting it. And the deepest probe in the literature is a question the poskim put to the settlement clause itself: may a manager who knows there was no profit — who could swear truthfully and cheaply — pay the settlement sum anyway? The majority answers yes, on the instrument's own logic: the fee was bargained as the price of not verifying; the investor cannot know the books; both parties signed with open eyes. A minority balks exactly there, and the balk deserves precision: where the payment stops summarizing an uncertain truth and starts replacing a known one, the settlement sum has stopped compromising about evidence and become simply a rate, and the partnership around it has gone ornamental. Chapter 14 reported the dispute; this chapter needs its geometry. The whole moral weight of the heter iska rests on the evidentiary clauses staying live — witnesses findable, oath swearable, loss provable when there is one. Everything the tradition's critics will say about the instrument at scale is a variation on one worry: that somewhere the channel to the truth has been drawn in a way that can never carry it.

A Partnership at Every Counter

Scale arrived in two steps, and the second was bolder than the first.

The first step was the ordinary spread of a good form: the tikkun Maharam and its descendants, signed transaction by transaction, a document apiece, as the *Nachalat Shivah's* notaries intended. Modern commerce strained the retail model (a wholesaler carries dozens of accounts, a community fund lends weekly, nobody signs a fresh deed for every invoice), and the drafters answered with the instrument's furthest development: the *heter iska klali*, the general permission of venture. One master document, executed once, declares that ev-

ery financing the signer extends or receives shall be governed by iska terms — every future loan pre-characterized, wholesale, before either party knows the other exists. The proposals came with the earnestness of the pious: let every man of affairs lodge one with the court, so that no Jew's credit should ever run naked of the law. The objections came with the realism of the poskim, as chapter 14 sampled them: a document can hardly convert a transaction whose other party never agreed to it, and most authorities held a one-sided klali ineffective. Some proposed hanging the text on the wall of the establishment; others answered that a sign nobody reads binds nobody. A serious minority denied the klali's efficacy altogether — a partnership, on their view, is a meeting of minds, not a filing — and the person-level klali, the compendia note, is not widely practiced today. What survived, and conquered, was the institutional form.

Because the second step was the banks, and the banks changed the question. The financial institutions of the modern Jewish center in the Land of Israel faced the wall at industrial width: a banking system is a machine for credit at interest, and here was one whose depositors, borrowers, and shareholders were substantially Jews. The answer, adopted with variations across Israeli banking and by institutions serving observant communities elsewhere, was the klali in institutional dress: the bank executes a master iska document and folds a governing clause into its standard terms, so that every account agreement, every loan contract, every mortgage incorporates the partnership characterization by reference. The poskim who upheld the arrangement reasoned, as chapter 14 put it, with impeccable contemporary realism: a customer is bound by an institution's known standard terms whether or not he has read them — that is what standard terms are — and so the institutional klali binds both sides in a way the wall sign never quite did. On the instrument's own terms the achievement is real. The depositor is, in form, an investor placing capital with the bank as managing partner; the bank's borrower is, in form, a manager trading the bank's deposit half; the interest on both legs is, in form, the settlement sum on a venture's projected earnings. A country's whole credit apparatus — overdraft, car loan, thirty-year mortgage — runs on paper descended from the Nehardeans' half-and-half and a Cracow rabbi's amended form. Chapter 26 met this paper from the certificates side and named it exactly: one master partnership document standing behind an entire bank's book, ownership paper at national banking scale.

Now let the chapter's title earn its noun: before the cautions, the genius of the mechanism deserves its full hearing, in the tradition's own terms.

The genius is that the recharacterization is real law, not costume. It would be easy to file the heter iska beside the 'ina sale of chapter 3 (two honest forms stapled into a dishonest payoff), but the comparison fails at the load-bearing joint. A court that takes the document seriously — and courts of both kinds, rabbinic and civil, have been asked to — will hold the investor to it: his claim is not for a debt but for a venture's accounting; the manager who proves the loss the document contemplates owes the loan half and no more; the estate of a ruined borrower answers a different claim than it would under a bond. The evidentiary clauses, so long as they are live, are not scenery; they are the terms on which the named figure may be collected at all, and a manager willing to open his books, produce his witnesses, and swear owes less, under the instrument's own words, whenever the venture truly failed. The theology underneath is Part IV's whole thesis, and it is no fiction either. The tradition's claim, from Bava Metziah 104b forward, is that capital's lawful yield comes from ownership at risk — the investor eats from his own field or not at all — and the heter iska does not exempt a financing from that claim; it subjects the financing to it, then manages the resulting uncertainty with evidence law. Where Latin Christendom, chapters

19 through 22 showed, came to license a lender's compensation by pricing what lending cost him, the halakha's instrument prices nothing of the kind: it writes the lender into the earning, as chapter 20 said, at his own risk, and then, because commerce cannot audit every venture, lets the parties compromise on a number while holding the audit perpetually available. That is the art in the chapter's title — not the art of hiding a loan, but of building a partnership so procedurally complete that a loan's convenience can be recovered inside it without asserting a loan's claims. Elsewhere in this Part, Christendom's casuists will stack three licit contracts toward the same payoff without achieving anything so unified; chapter 32 watches the fiqh's modern engineers work the same problem under the whole world's markets. Judged as legal engineering, the tikkun Maharam is among the most successful financial instruments ever drafted: four centuries in continuous service, one architecture serving two merchants at a fair and a national banking system, and not one clause of the Torah's prohibition formally breached.

Judged as legal engineering. The tradition's own authorities supply the other judgment, and it is time to hear them.

The Poskim's Reckoning

The worry is as old as the instrument — the Levush and the Sema saw it in the first draft — and in the modern literature it has a standing form: the parties' course of dealing can un-write the document. A heter iska says venture; the parties, in every observable behavior, mean loan. The officer extending the credit has never heard of the deposit half; the borrower learns he is a "managing partner" only if he reads the fine print; the "projected profits" come off a rate sheet that knows nothing of his affairs; and no one in the chain expects an accounting, contemplates an oath, or imagines that a proven loss would shrink the obligation. The poskim's question is the question this book learned in chapter 3 and has asked of every instrument since — whether the substance still answers to the form's description — and their answers, it must be said plainly, do not converge on comfort.

Rabbi Moshe Feinstein, by common consent the preeminent halakhic authority of twentieth-century America, put the sharpest versions on record, and two responsa frame the modern debate. In *Igrot Moshe*, Yoreh De'ah II:62, weighing the instrument's scope, he draws, in substance, the line this chapter has been drawing all along: the heter iska is not an incantation. It characterizes a financing as a venture; a venture requires something ventured; and where borrowed money simply vanishes into consumption — the wedding, the old debts, the daily bread of a man with no business and no income-bearing property — the paper's partnership language has nothing in the world to be true of. The responsum is equally clear about the repair, and the repair is instructive: the venture need not be the borrowed coins themselves. A house can serve as the investment; the iska can attach to the borrower's other profit-bearing assets, the financing freeing his own money for the venture the document describes; and the later literature wrote the solution into the standard forms. But the principle beneath the repair cuts wherever it is carried: somewhere in the borrower's affairs there must be a real asset, a real enterprise, a real exposure for the investor's money to stand in, or the heter iska, on the reasoning of the very authority most cited in its defense, is words. The practical handbooks say it without adornment: a borrower with no profit-generating holdings, borrowing to consume, cannot mend the matter with this document. The community's answer for him was never a better form but a different institution — the *gemach* of chapter 15, where the loan is free because it is mercy, and no recharacterization is needed because nothing is charged.

The second responsum, Yoreh De'ah II:63, cuts from the other side, and its fame is deserved. Asked about interest and the modern corporation, Rav Moshe reasoned that the Torah's prohibition attaches to a borrower — a person who owes personally, whose own estate stands behind the debt — and that a corporation, owing from its assets and nothing else, has no such person inside it. On this view a Jew may receive interest from a corporate borrower — the corporation may pay what no Jewish person may, though the person who borrows still owes personally, inside the prohibition's full grammar — and for a wide class of institutional credit the heter iska is unnecessary before it is even invoked. The ruling's boldness measured the tradition's honesty about its own categories (the era's leading decisor concluding that modernity's standard debtor was a shape the ancient prohibition had never contemplated), and the pushback measured it too: Rabbi Shlomo Zalman Auerbach, his Jerusalem counterpart in stature, came out sharply against the corporate leniency, and the *machloket* stands unresolved. This book takes no side, and needs none. The dispute itself establishes the point: the tradition's first-rank authorities treat the reach of ribbit into modern finance as a live legal question, argued from the sources — not as a solved problem some form disposed of.

On the instrument itself, the modern poskim's cautions gather at three pressure points, each a place where scale has thinned something the classical form kept thick. The first is knowledge: a partnership is a meeting of minds, and the handbooks warn, against their own convenience, that a heter iska signed as a talisman, by parties who would be startled to learn what it says, sits uneasily on its foundations; the klali jurisprudence (wall signs that bind nobody, one-sided filings most authorities rejected) is one long record of the tradition refusing to let the meeting of minds evaporate entirely, even as standard-terms incorporation stretched it very thin. The second is the evidentiary channel. The classical clauses assume a manager who could, if he chose, actually account: books, witnesses, an oath with something to attest. A consumer with a card balance has no venture to account for; a salaried mortgagor's "profits" are a rate sheet's arithmetic; and the minority position on the settlement sum — that a manager who knows there was no profit may not simply pay it — stands as a permanent interrogation of the retail apparatus: whether a channel no one could ever use is a channel at all. The third is the direction of drift. The classical sequence ran venture first, number second: real iska arrangements, with the settlement sum as their convenience. The institutional sequence runs number first, venture afterward, if ever: the rate is set by the market, the partnership recited to license it, and the instrument's defenders themselves (the striking thing about the responsa) keep requiring the attachable asset, voiding the unverifiable clause, repeating that the document is not magic. A tradition maintains that drumbeat about an instrument for one reason: it knows what the instrument is tempted to become.

The tradition speaks the failure state in examples rather than abstractions; this book's word for it is ribbit-washing, and the term means something precise. The heter iska is not ribbit-washing — it is a partnership instrument with a settlement option, and this chapter has spent its length on the reality of what it transforms. Ribbit-washing is what remains when the working parts are removed one by one — the wage waived into boilerplate, the witnesses named in a far city, the oath drafted to terrify, the venture nowhere — until nothing is left but the label and the rate: a fixed return, collected in all states of the world, from a counterparty whose ruin is contractually nobody's business, under a document that promises an accounting no party could render of a venture no party can locate. The line between the instrument and its failure state does not run between centuries, or between retail form and bank form, or between Poland and Tel Aviv; it runs, case by case, where this tradition always said it runs: through the substance. When the deposit half genuinely

stands in a real venture's wind — when a proven loss would truly shrink the obligation, when the accounting is available though rarely invoked, when the manager is a merchant with a warehouse rather than a wage-earner with a wedding to pay for — the heter iska preserves the Torah's substance while sparing its subjects the audit's full cost, and it is what its drafters built: the law's own bridge between the wall and the fair. When those conditions fail, the same clauses in the same order are a costume, and the tradition's own authorities say so in the tradition's own categories — words without a venture, a partnership without a meeting of minds, a settlement sum replacing a truth everyone knows. The instrument does not choose between these fates. The parties do, financing by financing, and the poskim's four-hundred-year vigil over the form is the tradition declining, under every commercial pressure, to stop asking which fate has occurred.

Which leaves the answer chapter 20 was promised. The question held open there was exact: the halakha's reply to the lender's "I could have earned more elsewhere" is the iska recharacterization — compensation sized to a real venture, collectable because the capital genuinely stands in the wind — and the open question was whether the instrument keeps that condition true at banking scale. The honest answer, assembled from the tradition's own record, is: not by itself. The condition is not self-executing. Paper, however brilliant, cannot hold itself to the world. What the instrument does — and no tradition in this book achieved more against the same problem — is keep the condition stated, justiciable, permanently collectable-against. The loss clauses can be invoked, the accounting demanded, the limiting rulings applied; and a banking system that runs on this paper carries, embedded in its own standard terms, the full grammar of the critique its practice must answer. Whether the condition is true in a given financing is a question of fact, not of form, and at national scale the facts are what a sober reader will already have concluded: mostly the settlement sum is paid, the venture is notional, and the wind never reaches the money. The tradition knows. It wrote the candor into its own compendia, and it keeps the form anyway, for the oldest of legal reasons — because the alternative to a form that states the right condition, imperfectly kept, is a market with no condition stated at all. Chapter 30 tells the story of the world that made such bargains feel inevitable — the centuries in which the surrounding civilization stopped asking the substance question altogether; chapter 32 watches the jurists of Islam meet the same pressure on their own instruments, the heter iska's dilemma replayed in another sacred legal language; and the closing Parts of this book ask what the poskim's vigil leaves standing for every tradition at once — whether the substance test can be made not merely statable but computable, so that the wind's actual reach to the money becomes a number somebody must report. Chapter 35 begins that arithmetic. The rabbi of Cracow, who rewrote his own document because two colleagues told him the substance had slipped, would recognize the project at a glance. It is his, at scale.

Chapter 32 — Islamic Finance, Shariah Arbitrage, and the Westernization Problem

In 1963, in a cotton town on the Nile Delta called Mit Ghamr, a young Egyptian economist named Ahmad El-Naggar opened a savings bank for people who had never trusted a bank. Its model came from the savings and cooperative banks of Germany, rebuilt for a pious countryside: the bank neither paid nor charged interest. A farmer's deposits earned, when they earned, as a share of what the bank's investments in local trade and workshops actually returned, and the money worked in ventures a depositor could walk past on market day. It was small, direct, and by every account beloved; within four years there were nine such banks in Egypt. Then, in 1967, the experiment was wound up — the historians mostly blame the political weather rather than the books — and its story scattered outward into what became an industry. Islamic finance now manages, by its own standard-setters' count, some 3.88 trillion dollars. Nearly every history the industry writes about itself begins in that delta town, rightly, because Mit Ghamr is the promise in miniature: finance without *riba*, tied to real activity, run for people the existing system had failed to reach.

The idea had books before it had banks. Through the 1960s and 1970s a generation of Muslim economists — Muhammad Nejatullah Siddiqi, whose *Banking Without Interest* became the movement's manual, Muhammad Umer Chapra, whose *Towards a Just Monetary System* (1985) gave it its fullest architecture, and their peers across South Asia and the Arab world — worked out on paper what a banking system inside the wall would look like. Siddiqi, looking back across the whole project in 2006, compressed the founding vision into one sentence: "What can be characterized as the 'theory' of Islamic banking was, till the end of the nineteen-seventies, largely a plea for replacing interest in bank lending by profit sharing." The moral center of the plea was the arrangement Part III watched the Qur'an condemn — the financier claiming a predetermined return while the working entrepreneur bears every risk of the business. The blueprint ran the logic of chapter 11 through an entire banking system — the depositor a *rabb al-mal* to his bank, the bank a *mudarib*, and the bank in turn a *rabb al-mal* to the town's enterprises: two tiers of *mudarabah*, capital earning only as a partner, paid from outcomes and exposed to them, with the naked loan reserved for mercy, where chapter 16 found it. It was the most complete attempt any modern community has made to build what this book's title asks for, and it deserves saying plainly, before any criticism is heard: the attempt was sincere, learned, and aimed at justice.

The banks followed the books. Dubai Islamic Bank opened on the fifteenth of September, 1975, among the first commercial banks anywhere to run under the new banner; the intergovernmental Islamic Development Bank went to work in Jeddah the same year; and through the oil decade the model spread — banks, subsidiaries, the "Islamic windows" of conventional giants, whole national systems — until the industry stretched from Kuala Lumpur to London. With the industry came a quarrel. Chapter 20 left it standing at this chapter's door in the shape of a question — whether the industry's condition is "prudence, contradiction, or an honest transition still under way" — and named the terms of the hearing: this is "the tradition's own quarrel, conducted by its own scholars in its own vocabulary." That framing is the chapter's law. The sharpest critics of Islamic finance are not its enemies; they are its founders, its judges, its board scholars, and its economists, and what they accuse the industry of is not impiety but imitation. Their name for the problem is Westernization, and in their mouths the word has nothing to do with geography — sev-

eral of the most unsparing critics are Westerners by post or passport. The word names a drift of substance: conventional finance rebuilt inside Islamic wrappers, *riba* avoided in vocabulary and reproduced in payoff, the promise of *Mit Ghamr* kept in form and surrendered in content. Whether that charge is fair — and what still stands even if it is — is the hearing this chapter conducts. Part VIII has already visited chapter 30's road, the secularization of interest, and chapter 31's, the *heter iska* and its art of legal transformation. The *fiqh* took the third road: it rebuilt the wall at industrial scale, hired guards for it, and then had to ask, in its own committee rooms, what exactly had been built. Nobody in the room is a villain. The depositors are pious savers; the practitioners are bankers asked to do something genuinely hard; the scholars are jurists doing the tradition's oldest work under the newest pressure. The quarrel is about structures, and structures are this book's whole subject.

The Landscape the Markup Conquered

The practice began to leave the theory within a few years of the first banks' opening. The instrument was chapter 24's: *murabaha*, the trust sale with its cost and markup declared, stretched into a financing triangle by the customer's promise to buy. It was honest, familiar to the jurists, and — decisively — it gave a bank margins that looked like margins: fixed, schedulable, fundable. Siddiqi again, looking back: "Within a couple of years of the introduction of *murabahah* in late nineteen-seventies, it conquered the landscape of Islamic finance, assigning *mudarabah* or profit-sharing to a corner accounting for less than ten percent of the operations." The conquest never reversed. Half a century on, the industry's own stability report draws the conquered landscape as a pie chart: of the financing on Islamic banks' books in 2024, forty-five percent was commodity *murabaha* and a further thirty-four percent *murabaha* proper — seventy-nine percent in the markup family, counting its two forms together — with *ijara* at nine, and the partnership instruments the theory was written for sharing an "others" slice of eight percent with everything else in the toolbox. The *murabaha* towers chapter 3 promised this chapter stand exactly where it said they would, visible from any skyline in the Gulf.

Two things must be said about that chart. The first is the practitioners' defense, which Siddiqi reported even while grieving the result. The bankers did try. "Practitioners of Islamic finance report they tried to push through sharing based finance but the results were not encouraging," he wrote; the surrounding legal systems offered a financier none of the protection against false profit-reporting that they offered a lender against default, leaving a bank that chose partnership to be lied to at scale without remedy. Chapter 14 watched the *halakha* price the same problem with witnesses and oaths sized to a community; a bank auditing ten thousand small ventures has no such machinery, and secular courts would not run the *fiqh*'s evidence rules for it. Some financings, Siddiqi added dryly, simply defy profit-sharing — his example was a municipality laying sewage pipes, an asset with no revenue to share. And the depositors voted, year after year, for what the theory deplored: security of capital, Siddiqi observed, mattered more to the fund owners than the size of its returns. Chapter 26 met the same appetite among certificate holders and declined to mock it; neither will this chapter. Nothing in the conquest of the markup requires a villain — only the pressures this Part keeps finding: information that will not verify, law that will not cooperate, and a crowd that wants principal protected first and enterprise admired second.

The second thing is the critics' point, and it survives every syllable of the defense. A cost-plus credit sale, honestly run, is lawful — chapter 24 spent its length on the disciplines

that keep it so: the ownership interval, the confessed cost, the price that never grows, the costume test it gave the whole Part. But lawful is not the same as different, and difference was the promise. An industry that runs four-fifths of its financing through deferred-price sales has built its balance sheet out of obligations that behave, cash flow by cash flow, like the loan book across the street: fixed claims, maturity dates, collateral, and a margin set off the price of money. The tradition's juridical summit has ruled on the family's darkest member: when the Fiqh Academy examined organized *tawarruq* in 2009 — the warehouse-metal round trip of chapter 24, sale and resale prearranged so that only cash and obligation actually move — it refused the device as “a deception... in order to get the additional quick cash.” Sixteen years later, the single largest line in the industry's own chart is commodity *murabaha* — the instrument family that resolution described, much of it still running in the organized form through the same metal warehouses. That coexistence — verdict upstairs, volume downstairs — is the quarrel of this chapter in a single exhibit, and no outsider had to point it out. The insiders printed both pages.

Deposits in a Partnership's Clothes

The promise's other half was the depositor, and here the drift is subtler and, the critics argue, deeper, because it runs through an instrument that still wears the theory's own name. Walk into an Islamic bank today and the savings product on offer is, in law, a *mudarabah*: an “investment account” whose holder owns the capital and whose bank is the working partner. Every rule of Part IV applies. The holder's return may only be a ratio of actual profit — a guaranteed sum voids the contract, chapter 11's rule; his principal is not owed back, for loss follows capital, the maxim carried there from the caliph Ali; the bank answers for losses only through the fault grammar chapter 13 anatomized — *ta'addi, taqsir*, breach of terms — and no profit exists to share until the capital stands whole, as chapter 19 found the industry's own standard-setters still printing. On paper, several hundred million people hold venture positions at their neighborhood bank. On paper, the two-tier vision was built.

Then look at the payout history, which is flat as a deposit rate, and ask how a portfolio of ventures produces it. The industry's prudential standard-setter answered in December 2010, in a guidance note whose subject line is its own summary: the practice of smoothing the profits payout to investment account holders. The Islamic Financial Services Board — Kuala Lumpur's prudential counterpart to Bahrain's AAOIFI — wrote down, with admirable candor, what the banks were doing and why. The why is competitive: an account holder whose returns lag the market may “withdraw their funds at short notice and place them with other financial institutions,” and retail holders are, in the note's own words, “‘defensive’ investors seeking low-risk investments with steady but modest returns.” The apparatus has three levers. The bank may quietly shrink its own share: entitled by contract to, say, a third of the profits as *mudarib*, it takes less in a lean year so the depositors' payout holds. It may draw on a profit equalization reserve — money set aside in fat years “before allocating the *Mudarib's* share,” in the standard's definition, “to maintain a certain level of return on investment for investment account holders and to increase owners' equity.” And beneath that sits an investment risk reserve, carved from the depositors' own profits “after allocating the *Mudarib's* share,” held “to cushion against future investment losses.” Fat years are clipped; lean years are topped up; the venture's weather never reaches the passbook. The industry even has a name for the pressure that drives the whole machine — displaced commercial risk, the risk that the bank must make its own shareholders eat the volatility of the depositors' portfolio because the market will not let it pay a partnership's

true and lumpy returns. The critics have sat with that term, because it is the industry's own vocabulary for a strange fact: the bank behaves as if it had guaranteed what its contract scrupulously refuses to guarantee. What the contract dare not promise, the treasury quietly delivers.

The critics' reading follows. The lawyers hold the line perfectly — no guarantee is ever written, and the tradition's standard-setters police stipulated guarantees of *mudarabah* capital with the severity chapter 13 documented — while the reserves, the forgone shares, and the benchmark-watching treasury produce, year in and year out, the payoff of the thing that may not be written. A partnership whose partner is never allowed to feel the venture is, in substance, a depositor; a profit share managed to track the market's deposit rates is, in substance, that rate; and the "investment" in the account's name is real on paper only — precisely the confrontation chapter 11 promised this chapter, the bond's payoff rebuilt inside a partnership's legal shell. Note what the criticism is not: not that anyone lied — the smoothing is disclosed, standardized, and supervised, the guidance note existing precisely to govern it, cap it, and drag it into the annual report, self-documentation that is the tradition's conscience working in the open — and not that the depositors are wronged in their pockets. They get the steady modest return they demonstrably want, and the widow keeping her remittances at an Islamic bank because her conscience requires it is entitled to safety, not to a lecture on variance; the sober fact under this whole section is that the system's most defensive money belongs to its most faithful savers. The criticism is about what the arrangement does to the claim. The industry's flagship deposit was supposed to be the proof that banking can run on genuine risk-sharing. What the smoothing apparatus proves instead, the critics say, is that the industry itself does not believe its retail liabilities can carry real venture weather — and so the gate of Part IV, celebrated in every prospectus, carries a sliver of the asset side and, on the liability side, a partnership managed until it behaves like the deposit it lawfully cannot be.

The Method Gets a Name

The certificate ran the same arc at wholesale scale, and this book has already walked it. Chapter 26 took the *sukuk* from its honest definition — undivided ownership shares in real assets — through the market's drift: Usmani's finding, delivered from within the industry's own standards body, that "most of the Sukuk that have been issued are identical to conventional bonds with regard to the distribution of profits," their payouts "not linked to the expected profits from the enterprise, but to the costs of financing or to the prevalent rates of interest"; the AAOIFI board's answering resolution of February 2008; the Fiqh Academy's Resolution 178, requiring that returns be "stemming from income-generating assets"; and the Dana Gas affair, in which an issuer impeached its own certificates' compliance and an English court enforced the purchase undertaking — the promise, not the ownership — as "valid and enforceable in accordance with its terms." What belongs to this chapter is the aftermath's shape. The 2008 intervention was the tradition policing itself in public, at real cost to the market that retained its jurists — and it worked as honest policing works: the worst devices retreated. And the market then did what markets do: it repapered, kept its benchmarks, and went on selling certificates whose holders wanted, and mostly got, the payoff of a bond. Usmani's five-word repair standard — "reward always follows after risk" — remains the sentence the industry's product desks must be measured against, and the measuring is a running argument inside the industry itself.

Underneath the certificates, as underneath the markups, runs the number. Chapter 24 gave the benchmark question its instrument-level hearing: the *murabaha* desks priced

their margins off LIBOR for decades, Usmani defended the practice with a jurist's precision — “merely using the interest rate as a benchmark for determining the profit of murabahah does not render the transaction as invalid, haram or prohibited, because the deal itself does not contain interest” — and then, in the same breath, told his own industry it “should get rid of this practice as soon as possible.” The habit outlived the advice. When LIBOR itself died — the last bank panel ended in June 2023 — the industry migrated to the successor rates alongside everyone else, repapering, as chapter 24 noted, without missing a payment date. Chapter 20 stated what the habit means at the level of doctrine: the benchmark problem is “*lucrum cessans* generalized — the market's elsewhere, priced into every contract, unexamined.” An industry built to refuse the price of money, steering every product by the price of money, is either navigating by a star it refuses to worship — the defenders' reading, and it is not absurd — or it has told you, in its own pricing screens, which market it actually inhabits.

The man who gave the whole pattern its name is no outsider. Mahmoud El-Gamal holds the chair in Islamic economics and finance at Rice University in Houston, has spent his career inside the industry's conferences and texts, and writes as a believer who wants the law's purposes served. His name for the method is *shariah arbitrage*, and his definition deserves quoting whole: “Shari'a arbitrage is a particular form of regulatory arbitrage, wherein a captive market of pious Muslims voluntarily chooses not to use certain financial products. Lawyers, in partnership with bankers and jurists, strive to provide them a reengineered version of those products.” The engineering method follows: “Conventional financial products are used as building blocks for the reengineered Islamic products approved by jurists” — the interest-bearing loan parked in a special purpose vehicle, the customer's own contract kept clean, the juristic gaze trained on the one document in the stack with an Arabic name. And the spread has a payer: the reengineering costs real money — “otherwise unnecessary transactions,” plus “the additional legal and jurist fees required to structure a product and certify it” — and the costs land on the captive market itself. The pious customer pays more than his conventional neighbor for the same economic position, the premium buying not a different payoff but a different description of it. A gap between form and substance is, to a moralist, a scandal; to a market, a spread. El-Gamal's charge is that an entire industry took the spread: that the wall this book has watched three traditions defend became, in the hands of the structurers, a price differential to be monetized — and his proposed cure, it should be said, is not surrender but mutuality, banks and insurers owned by the very savers and policyholders they serve, the pool of chapter 27 offered as the antidote to the arbitrage. As competition thins the spread, he observed, the engineering only coarsens — the industry reaching for *tawarruq*, its bluntest instrument.

The scholars saw the method's limit case arrive in their own docket. When the swap engineers built the *wa'd*-based total return swap — a promise structure keyed to the performance of a portfolio the Islamic investor could never lawfully hold, delivering him its returns through a compliant wrapper — it was an American-born shariah scholar, Yusuf Talal DeLorenzo, a veteran of the industry's supervisory boards, who circulated the paper against it: “The Total Returns Swap and the ‘Shariah Conversion Technology’ Stratagem.” His argument, reported here in substance, was that a jurisprudence which certifies the wrapper while ignoring what flows through it has traded the law's purposes for its instruments — that the device must be judged by its consequences, which were, with certainty, the delivery of returns from assets the law forbids. The industry's own conscience, once again, spoke first and sharpest. The pattern by now has a name, a mechanics, and a case file — and each of the three was supplied from inside the house.

The Scholars in the Middle

Every product in this chapter passed across a desk where a jurist sat, and the office deserves description before examination: it is the tradition's living machinery, and its holders are heirs of every court this book has admired. A shariah supervisory board is a committee of jurists retained by a financial institution to rule on its products — fatwa by fatwa at design, then in annual review. The office concentrates enormous trust: the industry's entire claim to difference runs through it, and its occupants carry a burden closer to a judge's than a consultant's. The structural problem is that they are paid like consultants. The scholars are selected, retained, and compensated by the very institutions whose products they judge — an arrangement World Bank researchers examined two decades ago and found, in measured language, to compromise the boards' independence. The market for their services is astonishingly narrow at the top: a network study published in 2011 counted roughly four hundred scholars holding some fifteen hundred board seats across the industry, and found the ten busiest men holding four hundred fifty of the eleven hundred forty-one positions at the institutions studied — two of every five seats — with the two most sought-after scholars sitting on eighty-five boards apiece. The study's own worry was generational as much as juridical: "Where is the room for mentoring and apprenticeship systems when the same people of similar seniority tend to occupy the same boards?" The fees are what a market that narrow pays: figures reported in the scholarly literature run to tens of thousands of dollars a year for a single seat and as much as half a million for certifying one large capital-markets transaction. These facts were assembled by the industry's own researchers and are reported here as they report them — plainly, with no imputation against any man's sincerity, for the concentration has an innocent explanation that happens to be true. When the industry exploded, the jurists fluent in both classical fiqh and a derivatives term sheet numbered a few dozen on earth, and every issuer wanted the same trusted names for the reason medieval merchants did: reputation is quality assurance, the name-economy of chapter 11 at work.

But pressure finds a narrow market, and the ugliest quotation in this chapter's file comes from the demand side, not from the scholars. In December 2009 the BBC published a report by John Foster, a former editor of the trade magazine *Islamic Business & Finance*, quoting an investment banker in Dubai who worked for a major Western financial house: "We create the same type of products that we do for the conventional markets. We then phone up a Sharia scholar for a Fatwa... If he doesn't give it to us, we phone up another scholar, offer him a sum of money for his services and ask him for a Fatwa. We do this until we get Sharia compliance." The quotation is anonymous, one banker, and a description of the buyer's conduct rather than the sellers'; weigh it accordingly — and weigh it against the documented record of the men being dialed, which cuts the other way more often than the anecdote suggests: the 2008 AAOIFI resolution condemned the sukuk market's favorite structures to their holders' faces; the Fiqh Academy refused organized tawarruq against the plain convenience of the industry's entire liquidity apparatus; DeLorenzo's paper was aimed at the most lucrative engineering of its day. Fatwas of refusal exist in quantity, and some of them moved markets. What the anecdote actually documents is the market structure the critics keep pointing at: where certification is privately purchased, dispersed across competing boards, and paid for by the certified, the temptation to shop does not need to corrupt many scholars to corrupt many products — it only needs the phone numbers.

The tradition's answer has been standardization, which is quality control practiced as jurisprudence, and the least-told story in this quarrel. AAOIFI in Bahrain writes the shariah

standards this book has been quoting for nine chapters, and governance standards for the boards themselves; the IFSB in Kuala Lumpur writes the prudential rulebook, the smoothing note of the last section among its papers; several jurisdictions have stacked a national or central shariah authority above the institutional boards, so that a product refused at the top cannot be shopped sideways; and the standards bodies publish, examine, and certify, slowly building the apprenticeship the network study found missing. None of this is decoration. Harmonized standards shrink the space in which a stratagem can hunt for its fatwa, much as the Majalla's codifiers shrank the space between Hanafi courtrooms — and the 2008 sukuk resolution demonstrated that a standard-setter can bite the market that feeds it. The quarrel chapter 27 promised this chapter completes the picture, because it is the same question one instrument over: whether a *takaful* operator may take performance incentives out of the participants' underwriting surplus — the steward paid from the pool's good years — or must be paid wages for stewardship, full stop. The industry's own jurists are divided; the diagnostic, chapter 27's indifference test, is the one this section has been applying to the scholars themselves. A certifier, like a pool operator, is honest in the measure that his compensation is indifferent to the answer he gives. The tradition knows how to write that rule. The open question, stated as the insiders state it, is whether an industry that pays its judges by the product can bring itself to adopt it.

The Word, the Payoff, and the Ideal

The whole quarrel comes down, in the end, to a rule this book met in its third chapter, and it is the tradition's own. In contracts, says the Majalla's third article, effect is given to intention and meaning, not to words and phrases — a principle every school of the fiqh states in its own way, and the rule an American court rediscovered in *Gregory v. Helvering*, disregarding paperwork assembled "as a disguise." Chapter 3 made it the method of this entire book: the names are hypotheses; the payoff is the evidence. Now apply it, as the critics do, to the industry's shelf. The murabaha tower's cash flows are a loan book's. The smoothed investment account's payout series is a deposit's. The standard sukuk's coupon and par redemption are a bond's, and when Dana Gas put the question to a court, what proved enforceable was the promise, exactly as a bondholder's counsel would have drafted it. The swap delivered the forbidden portfolio's return to the pious investor's account. At the level of what each party pays, receives, and can lose in every state of the world, the critics observe, the products are built to be indistinguishable from their conventional templates — that is what a "reengineered version" of a conventional product means. If the payoff is the evidence, the critics conclude, then *riba* avoided in word and reproduced in payoff is *riba* still — and the Qur'an's own distinction, "God has allowed trade and forbidden usury" (2:275, in Abdel Haleem's rendering, as throughout this book), was never a distinction between vocabularies.

The defense deserves its full weight, because it is not sophistry. Form is not nothing; it is where law lives. A murabaha's ownership interval is real — the crane can drop the machine on the bank's watch, as chapter 24 insisted — and realness at the margin is what separates a sale from a rate. The contracts differ in enforcement, in bankruptcy, in what happens when the paper is stress-tested — differences that sometimes bite in the customer's favor. A tradition that policed intention alone would dissolve into conscience with no law at all — the Shafi'i caution chapter 3 met in the 'ina dispute, and it is an honest jurisprudence, not a loophole. And the payoff-similarity test proves less than it seems: the lease of chapter 25 and the loan it must not become can also be tuned to resemble each other, yet the tradition keeps both, because anatomy, not resemblance, is the law's subject. So runs the

defense, and between it and the critics stands a genuine *ikhtilaf* — kin, at bottom, to the argument chapter 6 found the tradition conducting over whether the modern bank's interest is the Qur'an's *riba* at all, alive in its own registers for a century now. This book, which issues no rulings, will not issue one here. But it can report where the argument's weight has been moving, because the movers are the tradition's own senior courts: the Academy that refused organized *tawarruq* judged an assembly by what it does; the 2008 resolution judged the *sukuk* market by its payoffs; DeLorenzo's test was consequences; Usmani's was where reward sits relative to risk. When the tradition's own umpires reach repeatedly for the substance test, an industry optimized for the form test should expect to keep losing its hardest cases at home.

So state honestly what the critics get right, because the ledger is the point of the hearing. They are right that the industry has, in bulk, replicated the thing it was founded to replace — Siddiqi's sentence is the finding of this book's whole Part: "While theory aspired to prove Islamic finance was different from the conventional one, practitioners were busy searching for ways to make it similar to it. Convergence rather than divergence was the order of the day." Right, too, that the replication was no accident of villainy but the work of ordinary pressures on an under-defended design — the same pressures at work on chapter 30's road and chapter 31's — which is why the diagnosis of this chapter is the diagnosis of this book: extraction survives renaming, and no tradition's vocabulary is armor against it. Right that the costs of the arbitrage are real and regressive, paid by the pious in fees for the reengineering — in the words of the industry's own economist, "to structure a product and certify it." Right that the certification market's incentives are misbuilt, and that the repairs on offer — standardization with teeth, judges not paid by the judged, El-Gamal's mutuality, Siddiqi's return to the theory — are the tradition's own instruments taken seriously rather than imports. And right, above all, about the stakes, which Siddiqi stated with a founder's bluntness: "Meanwhile the credibility of Islamic finance as the bearer of a new promise is at stake." An industry that marketed itself as the alternative and delivered the replica has spent something no balance sheet records, and the people it was spent by — the depositors who chose it on conscience — are the ones this chapter has kept in view.

And yet the ideal stands where it stood, unembarrassed by the industry's report card, and the happiest part of the hearing remains: what it still has to teach. Strip the prohibition down to its positive content — this book has been doing so for thirty-one chapters — and it teaches three things no conventional textbook teaches. It teaches risk-sharing as justice and as mathematics: capital paid from outcomes, standing where the loss stands, which chapter 28 showed is not sentiment but survival arithmetic — the difference between the ensemble average a portfolio enjoys and the time average a borrowing household actually lives, where a fixed claim compounds ruin and a shared one absorbs it. It teaches asset-linkage: finance made to answer chapter 3's first question, what is this money doing — a thing owned, a use delivered, a venture fed — so the financial economy cannot drift from the real one without its own contracts telling on it. And it teaches the mercy layer: the *qard hasan* of chapter 16 and the *takaful* pool of chapter 27, capital deliberately exempted from growth and stationed at the community's worst moments — the layer the industry's own critics insist is infrastructure rather than a boutique, in the *ikhtilaf* chapter 16 flagged. On each count the ideal is not behind the modern world but ahead of it — ahead of the industry that carries its name and ahead of the conventional system that industry shadows — and the proof is where its arithmetic keeps resurfacing: in the post-crisis literature on equity-like finance, in the mutual and cooperative revivals, in the ESG movement's attempt to make money answer moral questions, which chapter 34 will meet as this chapter's secular sibling, complete with its own boards, its own screens, and its own form-over-substance

weather. The blueprint was never the failure; the institution-building was, so far — and that failure is solvable, which is why this book's answer to chapter 20's question is the third alternative: an honest transition, but one that must now be engineered rather than awaited. Parts IX and X take the ideal at its word and do what the founders' generation could not: chapter 35 turns the doctrines into models, chapter 40 into a return formula, chapter 41 into the design of a fund that cannot drift because its payoffs are audited against its names — the non-extractive architecture this whole book exists to specify. The tradition's quarrel with its own industry, heard in full, ends not in a verdict but in a work order.

Hold the last image of Mit Ghamr, because the industry's origin story is also its standing instruction. A small bank in a cotton town, German engineering and Delta piety, no interest paid or charged, savings working in ventures the savers could see, trusted by people no bank had ever bothered to deserve — it existed, it worked, and its memory reads today less like nostalgia than like specification. The critics of Islamic finance, who are its most faithful readers, are not asking the industry to become something foreign; they are asking it to become what it already says it is — and their patience in the asking, resolution after resolution, paper after paper, is itself the most Islamic thing about the whole affair: a tradition that has never once, in fourteen centuries, agreed to stop auditing its own instruments. Whether Christianity's modern teaching still audits its own — or has gone quiet at the very moment its oldest doctrine became newly urgent — is the next room in Part VIII, and chapter 33 opens the door.

Chapter 33 — Catholic Social Teaching and the Uneasy Silence on Usury

On the fifteenth of May, 1891, Leo XIII — eighty-one years old, elected thirteen years earlier as a caretaker and by now a decade past the brief reign anyone had planned for him — published the letter that founded modern Catholic social teaching. *Rerum Novarum*, “of new things,” took as its subject the condition of the working classes, and near its beginning it compresses the industrial world into a three-sentence diagnosis. Working men, it says, have been “surrendered, isolated and helpless, to the hardheartedness of employers and the greed of unchecked competition” — the sentence chapter 27 met when it watched Leo bless the mutual-aid societies as the guilds’ lawful heirs. Then this: “The mischief has been increased by rapacious usury, which, although more than once condemned by the Church, is nevertheless, under a different guise, but with like injustice, still practiced by covetous and grasping men.” And then the hiring of labor concentrated in few hands, laying on the poor “a yoke little better than that of slavery itself” (*Rerum Novarum* 3).

Hold the middle sentence to the light, because this chapter lives inside it. Everything this book has walked is folded in there. “More than once condemned” is a courteous undercount — chapter 30 weighed the pedigree: popes, three ecumenical councils, the unanimous schoolroom. “Under a different guise” is the form-versus-substance rule of chapter 3 in four words, asserted as a present fact about 1891. “With like injustice” refuses in advance the idea that new paperwork changes the moral object. The founding document of the Church’s modern economic teaching states, in its opening diagnosis, that usury is alive, disguised, and unrepented — and then it changes the subject. The encyclical turns to wages, property, the state, and association, and the word returns only once, in a list of the employer’s sins: the rich must not cut down workmen’s earnings “by force, by fraud, or by usurious dealing” (*Rerum Novarum* 20). Two appearances, both in passing; no definition, no anatomy, no test. The tradition’s last full-dress statement of its loan doctrine had come in 1745, in *Vix pervenit*; its next great economic document is this one; and between the two, the subject itself had been quietly refiled. Chapter 30 told how — the confessors’ questions, the not-to-be-disturbed replies, canon 1543 holding doctrine in one clause and releasing practice in the next — and it closed with Rome’s wager named: Rome bet on conscience. Part VIII has been auditing the bets. Chapter 31 opened the ledgers of the halakha’s contract; chapter 32 sat through the fiqh’s hearing on its own institutions; and that hearing ended at this chapter’s door, asking whether Christianity’s modern teaching still audits its own. So read the social encyclicals — a century and a quarter of them, from Leo’s factory gates to a Vatican office writing about credit default swaps — and keep one question running underneath: what does this imposing series say about money, and what did it stop saying about loans?

The Condition of Labor

Begin with honesty about the world Leo answered, because the encyclical’s choice of subject was not an evasion; it was triage. The nineteenth century’s industrial cities held human suffering at a scale and of a kind the old moral manuals had no page for: twelve-hour shifts, child labor, wages set by the hunger of the applicants, tenements, no insurance against the machine that took a hand or the fever that took a father. The guilds that had once orga-

nized work — and that had carried, chapter 27 showed, the burial funds and sick-boxes of their members — had been legally dissolved across Europe, and nothing had replaced them. *Rerum Novarum* says exactly this in its own voice: the old workingmen's associations destroyed, no protective organization raised in their place, and working men handed over, one by one, to a market in labor that priced them like a commodity. A church that had answered that world with a treatise on the loan contract would have deserved to be ignored. Leo answered the wound in front of him.

And the answer, on its own ground, is contract-level analysis of a high order — that is the detail this chapter most needs the reader to see. The encyclical's most consequential teaching concerns the wage bargain, and it proceeds exactly the way chapter 7 proceeded against the loan. Classical opinion held that a wage freely agreed was by definition just: willing worker, willing employer, nothing further to examine. Leo denied the premise. A man who accepts hard terms out of necessity, because the alternative is hunger, has not bargained freely, whatever he signed; beneath every labor contract there runs "a dictate of natural justice more imperious and ancient than any bargain between man and man," and a wage insufficient to support a frugal and well-behaved worker fails it (*Rerum Novarum* 45). Consent under necessity is not consent; the stronger party's advantage is not a price; there is a floor beneath the bargain that no signature can sign away. Readers of this book have met that argument before — it is the finding at the bottom of chapter 7's anatomy, that the borrower's consent is the consent of a man with no alternatives, transposed from the loan to the wage. The same document defends the worker's right to organize, praises the record of the old guilds — "History attests what excellent results were brought about by the artificers' guilds of olden times," in the sentence chapter 27 carried — and commends the "societies for mutual help" among the age's genuine consolations. Property it affirms against the socialists, but as stewardship, with the patristic lien running beneath it: the universal destination of goods, the doctrine chapter 18 traced from Basil's barns to the Catechism, holds every surplus answerable to need.

So the Church had not forgotten how to audit a contract. It audited the century's new contract — labor — with the old tools: consent examined against necessity, the weaker party's floor established, the stronger party's power named as power. What it did not do, in 1891 or after, was reopen the file on the old contract. The usury of paragraph 3 is a scene-setting villain, not a subject; the encyclical that proves the tradition's analytical machinery still worked chose not to point that machinery at lending. There were reasons. The doctrinal file had been closed within living memory — the replies chapter 30 dated had told confessors, decade after decade, that lenders at the legal rate were not to be disturbed, and an encyclical on the condition of labor was no place to disturb the settlement. The respectable rate was now the savings of the respectable poor as much as the income of the rich; the enemy of the workman in 1891 was the sweatshop and the company store more than the moneylender. But mark what the choice cost, because the rest of this chapter is the invoice. From *Rerum Novarum* forward, the Church's economic voice would speak of conditions, structures, systems, and duties — the register of social justice — and the register of contract anatomy, which the tradition had owned for six centuries, went into the archive with the doctrine it served. The word usury survived the transfer. The question usury named — what, in this instrument, is the charge actually buying? — did not. Usury survived as an accusation and retired as an analysis.

The Reign of Credit

Pius XI chose the anniversary itself for the sequel: *Quadragesimo Anno*, “in the fortieth year,” appeared on the fifteenth of May, 1931, an encyclical on the reconstruction of the social order drafted with the instrumental hand of a German Jesuit economist, Oswald von Nell-Breuning. The date matters: two years into the Depression, with parliamentary economies failing visibly and the totalitarian remedies recruiting, the Church’s second charter document faced not the factory but the system. It supplied the vocabulary the tradition still uses for getting scale right — subsidiarity, the rule that the higher body exists to help the lower and never to absorb it — and it renewed the wage and property teaching of 1891. All of that belongs to the social-teaching survey any textbook gives. What belongs to this book is the passage where Pius XI looked directly at finance.

It comes late in the document, in the diagnosis of what forty years of concentration had produced, and it remains the most unflinching thing any pope has written about the money power. Wealth, Pius wrote, was no longer the story; “an immense power and despotic economic dictatorship is consolidated in the hands of a few,” and those few are often not even owners but “trustees and managing directors of invested funds,” administering other people’s capital at their pleasure. Then the sentence that names the seat of the throne: “This dictatorship is being most forcibly exercised by those who, since they hold the money and completely control it, control credit also and rule the lending of money. Hence they regulate the flow, so to speak, of the life-blood whereby the entire economic system lives, and have so firmly in their grasp the soul, as it were, of economic life that no one can breathe against their will” (*Quadragesimo Anno* 105-106). Three paragraphs on, the autopsy of the age: “Free competition has destroyed itself; economic dictatorship has supplanted the free market,” and among the fruits stands “a no less deadly and accursed internationalism of finance or international imperialism whose country is where profit is” (*Quadragesimo Anno* 109). Nineteen thirty-one, and the description is already complete: ownership separated from control, credit the chokepoint of a whole economy, capital with no homeland but its return.

Now set the passage beside the tradition it descends from, and feel the displacement. The old doctrine asked about the loan: what does this charge price, whose risk is this, what may be taken from a man in need. *Quadragesimo Anno* asks about the lenders: who holds credit, and what does holding it do to everyone else. Both are real questions, and the second is one the scholastics barely possessed the categories to ask — there is genuine development here, the Church’s analysis growing to the size of the thing analyzed. But notice the substitution. The 1931 encyclical that anatomized the credit system does not contain the word usury at all — not once, in a document occasioned by the anniversary of the letter that had called usury rapacious and alive. The lending of money appears as an empire to be broken up, not as a contract to be audited; the analysis runs entirely at the level of power, and the level of the instrument — where the tradition had built its finest machinery — stays dark. A reader in 1931 could learn from Rome that the men who ruled lending strangled the world’s breath, and could not learn from Rome what, in any given loan on any given desk, would distinguish an honest charge from a bite. The Church now had a telescope where it once had a microscope, and the two have never since been mounted on the same instrument.

Development and the Debt of Nations

The next great enlargement came in 1967, and it moved the question to the scale of nations. Paul VI's *Populorum Progressio*, "on the development of peoples," is the encyclical of decolonization: written to a world suddenly full of new states born poor, owing money, and selling raw materials into markets they did not set. The line everyone quotes from it stands as a heading in the Vatican text — "Development, the New Name for Peace" — and its moral center is a patristic sentence this book has already carried. To ground the claim that the goods of creation answer to need before they answer to title, Paul reached back to Ambrose, in the quotation chapter 18 carried and weighed: "You are not making a gift of what is yours to the poor man, but you are giving him back what is his. You have been appropriating things that are meant to be for the common use of everyone. The earth belongs to everyone, not to the rich" (*Populorum Progressio* 23). Restitution, not largesse — the grammar of the fourth-century sermon promoted, sixteen centuries on, to the charter of development economics.

The encyclical also quotes this Part's own villain by name. Surveying the system that had built the colonial economies, Paul condemned the concepts that treat profit as the engine, competition as the supreme law, and ownership as an absolute: "This unbridled liberalism paves the way for a particular type of tyranny, rightly condemned by Our predecessor Pius XI, for it results in the 'international imperialism of money' " (*Populorum Progressio* 26) — the 1931 phrase requoted in 1967, the tradition citing its own audit and finding it still current. And then, in its program, the document does something this chapter has been waiting for: it talks about loans. Development would need capital; capital would come, in part, as credit; and the encyclical asks for terms that do not devour the borrower. A genuine dialogue between lending and receiving nations, it says, would mean that "developing countries will thus no longer risk being overwhelmed by debts whose repayment swallows up the greater part of their gains," and that "rates of interest and time for repayment of the loan could be so arranged as not to be too great a burden on either party, taking into account free gifts, interest-free or low-interest loans, and the time needed for liquidating the debts" (*Populorum Progressio* 54). Read those clauses slowly. The concern is ancient and true: repayment that swallows a debtor's gains is the millstone chapter 4 watched the Torah put off-limits as a pledge, scaled to a hemisphere. But the grammar has crossed over. Rates of interest are now a dial to be set considerably — arranged, balanced, moderated between parties — not a category to be examined. *Vix pervenit* had rejected the adjective as the question: moderation excuses nothing, because the issue is the charge's source, not its size. Two centuries later the Church's own development charter speaks entirely in the adjective's world — fair burden, reasonable rate, patient schedule — the world every decent regulator inhabits, and the world the old doctrine was built to see beneath.

What followed showed both how much that concern could do and how little the dial-grammar could see. The debt crisis the encyclical feared arrived on schedule; by the 1980s the poorest states were exporting capital to their creditors, and John Paul II's *Sollicitudo Rei Socialis* (1987) named the perversity plainly enough — the instrument created to finance development had turned into a mechanism that drained it, debtor nations shipping out the means of their own subsistence to keep current. In 1991, in *Centesimus Annus*, the centenary encyclical written with the command economies in rubble, John Paul carried the point to its edge: "The principle that debts must be paid is certainly just. However, it is not right to demand or expect payment when the effect would be the imposition of political choices leading to hunger and despair for entire peoples" (*Centesimus Annus* 35). That sentence is the release logic of chapter 4 — the sabbatical interruption of debt, the refusal

to let obligation compound into ruin — speaking at sovereign scale, and it was kept alive in the debt-cancellation campaigns of the millennium years that marched under the Jubilee's own name. The same encyclical weighed the newly victorious market with a jurist's care: asked whether capitalism had won, John Paul answered that it depends what the word means — an economy of enterprise, markets, property, and free creativity, yes; a system in which freedom in the economic sector escapes every juridical and ethical frame, no — and he preferred to speak of a business or market or free economy, warned that the market can become an idol, and observed flatly that “there are many human needs which find no place on the market” (Centesimus Annus 34). It is a superb ledger of ends: what markets are for, whom debt may not be allowed to crush. And in all of it — the debt passages are where this chapter's question presses hardest — there is no analysis of the debt contract itself. When repayment would starve a people, payment may not be demanded: the tradition's mercy, intact and operative. Why the claim had grown to where it could starve a people — what the original loans priced, whose risk they carried, which of their charges had titles and which were the compound arithmetic of chapter 21's *caementa pecuniae* generalized — no encyclical asks. The Church, having been the civilization's auditor of loans, had become the world's most credible advocate for debtors; and an advocate, however holy, argues the outcome, not the instrument.

Gratuitousness and the Return of a Word

Then the system itself failed an audit nobody ordered, and the tradition's voice changed register. Benedict XVI's *Caritas in Veritate* appeared in June 2009, nine months after the collapse of the system chapter 30 watched become “normal finance,” and it is the encyclical of the crisis: a long argument that an economy cannot run on contract and calculation alone. Its signature teaching sounds, at first hearing, like the gentlest thing in this book, and is in fact one of the most radical: “the principle of gratuitousness and the logic of gift as an expression of fraternity can and must find their place within normal economic activity” (Caritas in Veritate 36) — not beside it, not afterward as philanthropy, but inside the ordinary business of exchange. Part V of this book is the institutional record of exactly that claim — the *gemach*, the *qard hasan*, the Monti, the mercy layer of capital that chapter 18 stacked — and Benedict said as much, insisting that the crisis proved the need for finance in which giving and earning interpenetrate. He pressed the point against a temptation this Part knows intimately: it is not enough, he argued, to build “ethical” compartments — funds, labels, certified corners of the market — while the main economy runs unconverted; the whole of finance must be ethical, and by its nature rather than by an external label. Chapter 34, next door, will find the market building precisely the labeled compartments Benedict warned about, and his sentence is the standing test for them.

And in this encyclical, for the first time since 1891 in a document of this rank, the old word walks back on stage. Discussing the financial protection of the weak, Benedict commended the strengthening of micro-finance — rooting it, with a historian's precision, in the civil humanists and the birth of the pawn institutions this book met in chapter 17, the Monti di Pietà — and wrote that in hard times the vulnerable “should be protected from the risk of usury and from despair”; that “the weakest members of society should be helped to defend themselves against usury, just as poor peoples should be helped to derive real benefit from micro-credit” (Caritas in Veritate 65). Usury, named twice, as a live danger to the poor; the Franciscan credit machinery recalled by lineage; despair listed beside it as the thing usury manufactures. Whatever this chapter's title says, that is not silence.

So the record must now be stated with care, because the truth is stranger than the accusa-

tion. The word never died. The universal Catechism promulgated in 1992 contains it — one sentence, with teeth: “Those whose usurious and avaricious dealings lead to the hunger and death of their brethren in the human family indirectly commit homicide” (CCC 2269). The Church’s own 2004 synthesis of its social doctrine contains it: equitable profit may be sought, but “recourse to usury is to be morally condemned” (Compendium of the Social Doctrine of the Church 341). Benedict, commenting on a psalm at a 2005 audience, deplored in passing “the shame of usury which destroys the lives of the poor.” Francis received Italy’s anti-usury foundations in February 2018 and gave them sentences of a bluntness the subject had not heard from a pope in a very long time: “Usury humiliates and kills”; “Usury is a serious sin: it kills life, tramples on the dignity of people, is a vehicle for corruption and hampers the common good.” And Leo XIV, receiving the same council in October 2025, kept the register: “The phenomenon of usury points to the corruption of the human heart”; “It is a grave sin, at times very grave, because it cannot be reduced to a mere accounting issue.” These are not archival curiosities; they are the living pastoral voice of the Church, aimed at a real and unglamorous cruelty — the loan shark, the payday storefront, the debt that eats a family — and the foundations whose three decades of casework those addresses were honoring are doing the Monti’s work under a modern name. This register runs sober and deserves to: the families in those offices are chapter 2’s person at the door.

Now set beside it what did fall silent, and the shape of the unease comes clear. The Catechism’s one usury sentence lives under the fifth commandment — you shall not kill — as a case of indirect homicide. The word does its catechetical work in a homicide paragraph, not in the economics article: the offenses catalogued against the seventh commandment run to retention of goods, fraud, unjust wages, and “forcing up prices by taking advantage of the ignorance or hardship of another” (CCC 2409), and when the Church’s own writers want the Catechism on usury, 2269 is the paragraph they cite. The 1983 Code of Canon Law, replacing the 1917 Code, simply has no usury canon: where canon 1543 had stood — the two-clause sentence chapter 30 quoted, doctrine kept and practice released — there is now nothing, and the new Code’s one working reference to interest makes it a duty: administrators of church property must “pay at the stated time the interest due on a loan or mortgage” (canon 1284 §2). The wall’s last legal trace was replaced by a book-keeping instruction. No encyclical since *Vix pervenit* has restated, revised, or even formally revisited the loan doctrine; the settlement of chapter 30 — left undisturbed, pending a future judgment that never came — remains the tradition’s last word at the doctrinal level. The Church’s own scholars describe the arrangement candidly. The apologists’ account, given by writers like Christopher Kaczor and grounded in Noonan’s history, is development rather than reversal: the sin was always profit on the naked loan without a just title; the modern economy multiplied real titles until the naked loan became rare; the teaching stands, its matter transformed. It is a serious account, and this book has spent five chapters on the titles that make it serious. But notice what even the friendliest version concedes: if every ordinary loan now carries an implicit title, then somebody, somewhere, should be able to examine a loan and say which title, how sized, and against what shown loss — and no organ of the tradition any longer performs, teaches, or even requests that examination — the condemnation continuing, the anatomy in storage. Usury is once more a sin that everyone condemns and no one computes, which is, as chapter 30 observed of the civil law, exactly how a category retires.

Principles Without an Instrument Panel

The two most recent documents complete the arc, and they bring the story to the edge of this book's own project. Francis's *Laudato Si'* (2015) planted the economy inside creation — integral ecology, a doctrine given a chapter of its own, under which the economy, the earth, and the poor form a single case that must be judged together — and in passing delivered the tradition's tersest verdict on the sector this Part has been auditing: "finance overwhelms the real economy," the encyclical observes, among the lessons of the crisis not yet learned (*Laudato Si'* 109). Later, in its politics chapter, the knife goes deeper: "Politics must not be subject to the economy, nor should the economy be subject to the dictates of an efficiency-driven paradigm of technocracy," and rescuing banks at public cost without reforming the system "only reaffirms the absolute power of a financial system, a power which has no future and will only give rise to new crises after a slow, costly and only apparent recovery" (*Laudato Si'* 189). Pius XI would have recognized every clause; the 1931 telescope, pointed again, found the dictatorship still on its throne.

Then, in 2018, Rome did something without precedent in this whole story: it issued a document about financial instruments. *Oeconomicae et pecuniariae quaestiones* — "Considerations for an Ethical Discernment Regarding Some Aspects of the Present Economic-Financial System," issued jointly by the Congregation for the Doctrine of the Faith and the Dicastery for Promoting Integral Human Development, dated the Epiphany of 2018 — reads, in stretches, like a chapter of this book. It calls finance to its "primary vocation of service to the real economy"; it walks through securitizations and derivatives with their names attached; it finds that credit default swaps bought by parties with no stake create "a unique case in which persons start to nurture interests for the ruin of other economic entities" and condemns "gambling on the failure of others" as ethically unacceptable — Rome discovering, in its own vocabulary, the manufactured short position chapter 27 distinguished from the pool, and putting ruin itself back into the moral accounting, where chapter 28 argued it always belonged. It even reaches the benchmark: the manipulation of interbank reference rates — the LIBOR affair that shadowed chapter 20's doctrine and chapter 32's industry — is named there as a grave ethical violation with consequences reaching everywhere the rates do. Two curial offices, four signatures, and the closest thing to an instrument-level audit the magisterium has produced since *Vix pervenit*: whoever hoped the Church might again read term sheets found, in 2018, an office willing to start.

Which makes its one meeting with the old word the most precise exhibit this chapter owns. In the paragraph on credit, the document condemns lending that exploits the borrower's position, and defines the wrong like this: "applying excessively high interest rates, really beyond the range of the borrowers of funds, represents a transaction not only ethically illegitimate, but also harmful to the health of the economic system"; such practices, "along with usurious activities, have been recognized by human conscience as iniquitous" (*Oeconomicae et pecuniariae quaestiones* 16). There it is, in the tradition's own modern hand: usury as excessive interest — the rate too high, the burden beyond the borrower's range. Chapter 30 quoted the historian Ruston's summary of what the sixteenth and seventeenth centuries did to the word: a new generation of moralists redefined usury as excessive interest, and the question passed from public doctrine to private conscience. Four centuries later the redefinition is simply the working grammar of the Church's own finance office — written, it must be said, by men trying to protect borrowers, in a document of real courage. That is what makes the silence uneasy rather than scandalous. Nobody suppressed the old doctrine. It sank, so completely that the office charged with ethical discernment about credit no longer reaches for it even when writing about credit; the anatomy that once dis-

tinguished the charge's source from its size is not rejected there, merely absent, the way a tool no living hand has used goes missing from the bench. A verdict without a test is a conscience without jurisdiction.

So state the finding of the audit, in the Part VIII pattern, with the same evenhandedness the other two rooms received. What Catholic Social Teaching has built since 1891 is neither evasion nor decoration: it is as sustained a moral interrogation of economic life as any institution has kept up in the modern era — the wage floor under necessity, subsidiarity and the common good, the universal destination of goods, the debt of nations arraigned, gratuitousness ordered into the middle of commerce, the biosphere added to the ledger, and, latterly, the instruments themselves called by name. On ends, the teaching is complete. What it lacks is the middle layer its own ancestors built: the tests that connect principles to contracts — whose money, whose risk, what does the charge price, which burdens are shown and sized, when is surplus real — the instrument panel chapter 30 inventoried as the price of the settlement. And the lack is strange precisely because the panel is Catholic property in the attic. The *mutuum*'s anatomy, the just price, the titles of Part VI, Lateran V's definition of usury by the absence of work, expense, and risk — chapter 14 carried it — were substantially built in Rome's own workshops. When the American bishops campaign against payday lending, they instinctively quote the Catechism's fraud-and-hardship paragraphs and Benedict's defense of the weak against usury — ends and verdicts — while the tests that would let a diocesan credit union prove, line by line, that its own loans differ in kind sit unclaimed in the tradition's basement. The *heter iska*'s auditors and the shariah boards, whatever chapter 31 and chapter 32 found to criticize, at least exist: institutions whose daily work is holding instruments up against a doctrine. Rome retired the doctrine's institutional eye and kept the doctrine's voice — which is why its modern teaching can grieve extraction magnificently and cannot, any longer, diagnose it in a term sheet.

Renewed contract-level analysis, then, is not a demand that the Church repent of its social teaching; it is the request that the teaching's two halves be reintroduced. The principles are ready — they need no improvement, only instruments — and the old tests were built for exactly the purposes the principles now lack hands for. What would it mean? That an encyclical's heir, or a bishops' conference, or a religious lender could take *Oeconomicae*'s own instrument-courage one floor down: not only that speculation on ruin is iniquitous, but which clauses in this loan price a real title and which price the borrower's need; not only that the vulnerable deserve protection from usury, but a working definition that does not quietly concede Ruston's — one that asks, of every charge, its source before its size. The materials are all on the tradition's own shelves, and some of them — the Monti's cost-recovery rules, the microfinance Benedict blessed — are already half-deployed. The rest of this book is, among other things, an attempt to make that reintroduction in earnest: Part IX takes the principles this chapter has praised and the tests this chapter has missed and builds them into arithmetic — chapter 35 begins with the moral-title ledger, the rule that every euro of return must name its source — and Part X designs the institutions that could run it. Next door, meanwhile, stands the market's own answer to the vacancy. Into the space where religious audit once stood, modern finance has built a conscience of its own manufacture — ESG, impact, sustainability, screens and scores and boards — and chapter 34 asks the question this Part has trained the reader to ask of every such construction: whether the new vocabulary changes what the money does, or only what the money is called. Leo's subordinate clause, it turns out, was truer than its placement. Usury, more than once condemned, is still practiced under a different guise, with like injustice; what the Church of 1891 still owned, and what its heirs have yet to take back down from the attic, is the machinery for telling the guise from the thing.

Chapter 34 — ESG, Impact, and the New Language of Moral Finance

In January 2004 the secretary-general of the United Nations wrote to the chief executives of fifty-five of the world's largest financial institutions with an unusual request. Kofi Annan's letter, sent under the banner of his Global Compact, invited the banks to help work out how the concerns the world was learning to press upon business — the climate, the laborer at the far end of the supply chain, the honesty of boardrooms — could be carried into the everyday machinery of investing: research, asset management, brokerage. A working group formed, with the Swiss government helping to carry it, and the report that came back bore a title from the golf course, *Who Cares Wins*, and a roster of endorsers from the commanding heights — Goldman Sachs and Morgan Stanley, Deutsche Bank and UBS, HSBC and BNP Paribas among them, houses managing more than six trillion dollars. The drafters set aside the era's softer words — sustainability, corporate citizenship — to head off misunderstanding, and wrote instead, with committee precision, of environmental, social and governance issues. Somewhere between the letterhead and the printer the phrase compressed into an acronym, and the acronym escaped the report. ESG now names an industry. In 2006 came the Principles for Responsible Investment, a UN-convened investor network whose signatories promise first of all to carry ESG issues into their analysis and decisions; it counts today more than five thousand signatories representing, on its own tally, above a hundred and twenty trillion dollars. By the census the sustainable-investment industry keeps of itself, some thirty trillion dollars of managed money now works under the label.

Hold that scene against this Part's opening chapter, because the two are mirror images. In 1545 a parliament deleted the moral vocabulary from the price of money; chapter 30 walked the deletion to its terminus and named the result: "Normal finance is finance with no slot for the question this book is about." In 2004 a committee of bankers, convened by a diplomat, began drafting a replacement vocabulary from scratch. Between those two dates lies the whole strange arc this chapter closes. The market did not recover the old grammar — usury appears nowhere in the new lexicon, and neither do the contract-level questions of chapters 3 and 7. But the hunger for a grammar returned from inside: nobody imposed ESG on finance; finance wrote it, at the peak of its confidence, over the signatures of its largest houses. For a generation the market's official conscience had been one sentence long — Milton Friedman's 1970 pronouncement in the *New York Times Magazine* that the social responsibility of business is to increase its profits — and even that surrendered. In August 2019 the Business Roundtable, one hundred eighty-one chief executives, retired the shareholder-primacy language its own statements had carried since 1997 and committed its members to lead their companies "for the benefit of all stakeholders — customers, employees, suppliers, communities and shareholders." A civilization that had spent four centuries teaching itself that a price cannot be a sin began, within living memory, to insist that a portfolio can be a virtue.

This chapter reads the new vocabulary the way this book has read every vocabulary: from within first, in its builders' own words, and then against the payoff. The builders include some of the most serious people in modern finance, and the sharpest critics below are not the project's enemies but its own executives, founders, standard-setters, and prosecutors — the pattern of this Part, for the third time. And the chapter's claim should be stated

now, so the reader can watch it being earned, for it is precise, not cynical. The new moral vocabulary changes what gets disclosed and what gets selected: which companies capital visits, what they must report, what a fund may call itself. Left to itself it does not change who bears loss, what the charge prices, or where ruin lands — in the vocabulary chapter 28 built for this book, it does not by itself alter the payoff geometry. Whether the new conscience can learn the old questions, and what happens where it declines them, is the business of everything that follows.

The Acronym and the Anatomy

What the acronym does, in its builders' own terms, is straightforward and genuinely useful. ESG in practice is a family of instruments. Integration folds environmental, social, and governance information into ordinary analysis, on the theory the 2004 report argued: that these factors, ignored, are mispriced risk. Screens exclude what a mandate names — tobacco, thermal coal, cluster munitions. Ratings score issuers on the three letters so that portfolios can be tilted, benchmarked, and marketed. The gains are real, and this book, which has spent thirty-three chapters insisting that moral information belongs inside financial decisions, will not pretend otherwise. Companies now publish, under law in a growing set of jurisdictions, data they once refused their own boards; externalities have names, line items, and auditors; capital is steered — sometimes only rhetorically, sometimes in fact — away from harms the price system alone would never register. And the demand is the most encouraging fact this Part has met: three generations after economics declared the file closed, the owners of capital pay real money for moral information about what their money touches. The hunger whose funeral chapter 30 recorded survived it.

But watch what the instrument sees, because its field of view is the whole argument. An ESG score is a fact about an issuer — a company, rated for its emissions, its labor record, its board. The old panel this book assembled asked its questions of the contract: chapter 3's first question, what is this money doing; chapter 7's anatomy, whose money, whose risk, what exactly does the charge price; Part VI's discipline of titles, every claim made to name its burden; chapter 28's ruin column, the fraction of counterparties a payoff design destroys. Set the two panels side by side and the finding falls out at once: they do not overlap. A bank can stand in the first percentile on governance — independent directors, audited controls, exemplary disclosure — while its consumer book runs on fixed claims against fragile households at charges the old law would have named without hesitating; the score has no cell in which that fact could appear. A green bond can finance a genuinely green turbine while the claim it conveys remains exactly what it was — fixed, senior, payable in every state of the world; the label describes the turbine, not the term sheet. This is why ESG does not solve usury, and the reason deserves saying without any sneer: it was never pointed at the loan. The new conscience audits the company; the old one audited the contract. Usury is a contract-level wrong — one lender, one borrower, one schedule of payments — and an instrument built to rate issuers at portfolio scale cannot see one, any more than a census can see a bruise.

The industry's own people said this early, and the most quoted of them had held its most senior desk. Tariq Fancy ran sustainable investing at BlackRock, the world's largest asset manager, as its first chief investment officer for the subject; he left within two years, and in 2021 published a three-part essay, *The Secret Diary of a "Sustainable Investor,"* recounting how he went from evangelist to critic — "a dangerous placebo," he came to call it, comforting exactly the anxieties that might otherwise force real rules, while the incentives underneath moved not at all. His argument was aimed at the gap between what the label implies and

what the instrument does, and the reader of this Part has seen the shape before: it is an insider's substance audit, the same office Usmani and the poskim hold in the two chapters behind us, performed here without a theology.

And where the label economy was tested by authorities, the results are on file, in the case every survey now cites. In 2021 the German asset manager DWS, majority-owned by Deutsche Bank, was accused by Desiree Fixler — its own former group sustainability officer, whom the firm had dismissed — of claiming far more ESG integration than its systems actually performed. In September 2023 the U.S. Securities and Exchange Commission fined the firm's American arm nineteen million dollars for "materially misleading statements about its controls for incorporating ESG factors into research and investment recommendations": the marketing had promised an integration process the firm, in the regulator's finding, "failed to adequately implement." Police had raided the Frankfurt offices in May 2022, and the chief executive resigned in the raid's wake; in April 2025 the Frankfurt public prosecutor closed the German file with a twenty-five-million-euro fine, having concluded that advertising in which "ESG is an integral part of our DNA" did not describe the firm running the advertisements. The company settled the American action without admitting or denying the findings, said it had rebuilt its documentation and controls, and issued a closing sentence that belongs in this book's museum of concessions: "in the past our marketing was sometimes exuberant." Mark two things about the file. Every authority in it was punishing description — the fund said one thing and did another; the wrong lived in the gap between label and practice, which is the one wrong the new regime is built to see. And the label economy is capable of honesty about labels: the sustainable-investment industry's own trade alliance tightened its American definitions against exactly this inflation, and roughly nine trillion dollars of "sustainable" assets went out of the count — from seventeen trillion to under nine — the industry auditing its own signage. What no statute in the file asks, and no rating measures, is the question a fully honest label would still leave standing: whether the truthfully described product extracts. For that question the modern world built a different movement, which promised to go deeper.

Impact and the Price of the Elsewhere

In October 2007 the Rockefeller Foundation gathered financial innovators at its Bellagio Center on Lake Como — bankers, fund builders, philanthropists — to ask whether profit-seeking investment could be organized to do good on purpose. The foundation's archive remembers the meeting's working definition, "using profit-seeking investment to generate social and environmental good," and its convener remembers the breakout session at which the group chose the name. Impact investing, the movement's own histories say, was christened there; the coinage, like most coinages, has rival claimants — one industry chronicle credits a family-office summit two years earlier — and this book needs only the fact that the name found its movement. A second Bellagio meeting followed in 2008; in 2009 came the Global Impact Investing Network, the movement's standard-setter. Its definition is the field's constitution: "Impact investments are investments made with the intention to generate positive, measurable social or environmental impact alongside a financial return." The network's latest census counts about one and a half trillion dollars managed on those terms.

Read the definition with this book's eyes and the advance over ESG is immediate. Intention and measurement move the analysis from the issuer toward the investment itself: an impact investor claims to know what this money is doing — chapter 3's first question, asked at last by the modern market of its own accord — and promises to measure the an-

swer. That is real kinship, and this book honors it. The difficulty sits in the definition's last five words, because "alongside a financial return" does not say how much, and how much is where the old grammar always began. The network answers with a range: impact investments are "compatible with a range of financial returns from below market to above market rate, depending on investors' strategic goals." Its own surveys report where the money sits: most impact investors pursue "risk-adjusted, market-rate returns."

Bring in chapter 20, because this is its material returning at scale. The modern axiom that chapter traced — the market's elsewhere, priced into every contract — makes "market rate" sound like a law of nature: capital is owed everywhere what it could have earned anywhere. An impact fund that promises market-rate returns and impact has therefore promised, in the old vocabulary, that the mission will cost the capital nothing — that the clinic operator, the school chain, the smallholder cooperative will pay the market's full elsewhere and deliver the measurable good on top. Sometimes the promise is honest, because sometimes the good business is the best business; nobody needs a concession to finance a solar farm that outearns coal. The strain appears exactly where this book has always looked: at the weak counterparty, in the thin market, in the venture whose service is valuable and whose margins are not. There the arithmetic offers only three exits — the investor accepts less than the elsewhere, the enterprise squeezes itself to pay it, or the mission quietly narrows to the customers who can afford it. Concessionary capital, the below-market end of the network's own range, is the honest first exit: a price for the mission, named and paid by the party claiming the virtue. The revealing datum is that the industry's own surveys put most of its money at the other end. None of this convicts anyone of hypocrisy; it describes the return-expectation problem, which is structural. A fund must raise its capital from investors trained for a century on the axiom of the elsewhere; the expectations arrive with the capital, as much a part of the wire transfer as the dollars. What the founders underestimated was not greed but the gravity of the benchmark — the pull chapter 20 watched turn an exception into an axiom, now tugging at every term sheet with impact at the top.

The movement's own reckoning with that gravity has a place and a date, and the facts must carry this passage plainly. Microcredit began as close to this book's mercy tradition as anything modern finance has produced: tiny loans to the very poor, mostly to women, without collateral, descended in spirit from the *gemach* of chapter 15 and the *qard hasan* of chapter 16, and its founding generation ran it as mission — Muhammad Yunus's Grameen Bank lent to villagers no bank would receive, and won a Nobel Peace Prize for it. Then capital discovered the sector: loan books compounded at venture speed, private equity arrived wanting venture outcomes, and the institutions that grew fastest were those that could pay the market's elsewhere out of interest collected from the poorest borrowers on earth. In India the largest was SKS Microfinance, lending at rates contemporaries reported near thirty percent — under the moneylender's fifty, far above a mission's cost in the judgment of the movement's own founder. In August 2010 SKS went public, raising three hundred fifty-eight million dollars in an offering oversubscribed more than thirteen times. Yunus published his verdict in the New York Times within months: "Commercialization has been a terrible wrong turn for microfinance, and it indicates a worrying 'mission drift' in the motivation of those lending to the poor." And, in the sentence the movement still argues about: "Poverty should be eradicated, not seen as a money-making opportunity."

What happened meanwhile in Andhra Pradesh, the industry's Indian heartland, must be told at the pace of its facts. The state held some nine point seven million microborrowers; the average indebted household there carried, by the count in the industry's own crisis

literature, sixty-five thousand rupees of debt against a national average under eight thousand. Households held loans from several lenders at once; collection came weekly, at the doorstep, before neighbors. Through the summer and autumn of 2010, press reports and then government documents linked a series of borrower suicides to over-indebtedness and coercive recovery; official documents counted fifty-four such deaths. The counting was disputed then and since; the over-indebtedness was not. On the fifteenth of October 2010 the state promulgated an emergency ordinance against the industry, and when the legislature confirmed it the following year, the statute's recital used words this book did not expect to meet again in a modern gazette: lenders charging "usurious interest rates" and resorting to "coercive means" of recovery, resulting in impoverishment and, at times, in the suicides of borrowers. Collections across the state collapsed, and the crisis spread through the Indian industry.

Say precisely what this case is, because the vocabulary of this chapter was all over it and did not fail as polemic would prefer. The mission was real: real loans reached real women the banking system had never visited, and even researchers skeptical of the crisis narrative credit that reach. The extraction was real too: fixed weekly claims, enforced by group pressure at the doorstep, on households with no reserves, at prices set by the growth the investors' valuations required — pressure the industry's own post-mortem traced from the top of the capital stack down to the loan officer at the door. The two lived at different layers. The mission was real at the level of the institution — its purpose, its outreach, its reporting; the extraction was real at the level of the contract. The borrower met the contract. That layering is what this book means by mission-aligned extraction: alignment lodged in intention and institution, extraction lodged in payoff geometry, and no instrument anywhere in the structure obliged to reconcile them. The general temptation has a name in this book's closing audit — chapter 45 files it under impact-washing — but the arithmetic lesson belongs here: nothing in intention, measurement, or label changed who bore the loss, and who bore the loss is what the old grammar would have asked first. A statute reaching for the retired word — usurious — said in one adjective what the sector's impact reports had never found a column for.

Moving the Loss, Locking the Deed

Not everything the new movement built is a label, and this section owes the reader its two working machines — the places where the new language stops describing capital and starts re-engineering it.

The first is blended finance. The OECD, the rich world's development-policy club, defines it as "the strategic use of development finance for the mobilisation of additional finance towards sustainable development in developing countries" (the spelling is the OECD's); Convergence, the field's own network, says it plainer: "the use of catalytic capital from public or philanthropic sources to increase private sector investment in sustainable development." The machinery beneath both definitions is an ordering of loss. A development agency or a foundation takes the structure's junior position — first-loss capital, the field calls it — or sells cheap guarantees, or grant-funds the design work, so that the private tranche above enjoys a risk and return the raw project could never have offered it; in Convergence's own gloss, the concessional money stands "on below-market terms within the capital structure to lower the overall cost of capital or to provide an additional layer of protection." Convergence counts some two hundred thirty billion dollars mobilized this way to date. With that, for the first time since the acronym appeared, this book's own subject is on the table: a first-loss tranche is payoff-geometry work — the one instrument in the new

vocabulary's standard kit that exists specifically to move loss from one party to another. The old houses recognize it on sight: capital stationed in front of the blow is chapter 28's arithmetic in a term sheet — a solidarity reserve with a special-purpose vehicle around it — and the endowment absorbing the Monti's losses in chapter 17 held the same seat for the same reason. The instrument is honest about the one thing the rest of the new vocabulary keeps silent: someone must bear the worst year, and a contract can choose whom.

Which is why the audit of blended finance is not whether it changes geometry — it does — but for whom, and chapter 28 already supplied the sorting questions it asked of every pool: whose ruin does the design treat as the failure state, and who lives on the smoothed line once the pooling has done its work? Run one way, the answers come back as the traditions would have them: the junior tranche stands between a fragile enterprise and catastrophe, the mission lives on the smoothed line, and the donor's money does openly what the *waqf* corpus and the burial chest always did. Run the other way, the same clauses invert, and the field's own critics have spent a decade saying so: public and philanthropic money can be spent guaranteeing private capital its full market coupon in places where the market would otherwise have demanded a discount — the taxpayer absorbing the ruin column so that capital's elsewhere arrives intact, subsidy flowing up the waterfall, solidarity run for the strong. The field's defenders answer with a principle whose name could have been drafted in Part VI: minimal concessionality — no more subsidy than the transaction genuinely requires — which is the titles discipline applied to a donor's money: the concession shown, sized, capped, and pointed at the mission rather than the margin. Whether a given structure honors the principle is a question of fact, deal by deal, and the reader has stood at this exact point twice already in this Part. The geometry is real; that is precisely why the audit must be.

The second machine is the deed, and chapter 29 built this room before the acronym existed. In the same years ESG conquered the disclosure statements, a smaller movement set about altering not what companies report but what their owners may do. Steward Ownership, the program under which chapter 29's foundation-owned firms became a general model, rests on commitments made binding in the charter: control held by stewards bound to the mission — control that, in the movement's own words, "can never be sold or inherited like an asset" — and profit demoted to instrument, "a means to an end, not an end in themselves," with the exit bolted in plain language: "The value created within the company cannot be extracted by the company owners for their personal benefit." A golden share the size of a key, one percent of the votes or less, sits with an independent guardian — itself asset-locked — and vetoes the acts that would undo the dedication: sale, liquidation, the conversion of mission back into margin. Chapter 29 traced the lineage from Abbe's Zeiss foundation of 1889 through Bosch's separation of dividends from votes and Carlsberg's charter majority to Patagonia's 2022 announcement that "Earth is now our only shareholder," and set beside them the Employee Ownership trusts that hold companies for the people who work in them; none of it needs retelling here. What belongs to this chapter is the classification. A steward charter is not a rating and not a report; it is conveyance — a change in who may hold control, what profit may lawfully do, and which futures are contractually closed — and it therefore sits with first-loss capital on the short list of new instruments the old panel can read without translation: chapter 29's endowment logic speaking company law, the deed doing what the label cannot. The movement is honest about its limits, as that chapter taught: locks do not administer themselves, stewards can be captured, and the guardian and the audit are the standing price of the promise. The category difference stands nonetheless. A fund can wear an ESG label while its payoffs extract. A company under a genuine steward charter cannot be sold to the extractor at all,

because the deed says no and the golden share holds the pen. One changes the description; the other changes the possible.

Three Letters from the Old Houses

The new conscience is not the first to hold finance to goodness with a label; the elder experiments sit earlier in this Part, audited by their own scholars within living memory. Read as correspondence, they are three letters addressed — in effect if not in intention — to the ESG age.

The halakha's letter is chapter 31's. A community ran its financing for four centuries on a document that says partnership, the *heter iska*, and its authorities never stopped auditing whether the substance answered the description: the Levush and the Sema attacked the first draft until its author amended it; the poskim struck down witness clauses drafted so that no witness could ever qualify; the twentieth century's leading decisor held that where no venture exists there is, as chapter 31 put it, nothing in the world for the paper to be true of. The failure state earned a name in this book — ribbit-washing, the instrument with its working parts removed one by one “until nothing is left but the label and the rate” — and the tradition's candor about how often scale wins is part of the letter. Its counsel to the new movement is procedural: a moral label survives only under permanent internal audit, conducted by readers with authority to void it when the substance walks away, and even then the vigil never ends.

The fiqh's letter is chapter 32's, and it is the nearest kin, because modern Islamic finance is a certification economy — boards of jurists ruling product by product, the office the ESG raters and green-label reviewers now hold without the theology. Mahmoud El-Gamal, from inside the industry's own conferences, named the failure mode *shariah arbitrage*: “a particular form of regulatory arbitrage, wherein a captive market of pious Muslims voluntarily chooses not to use certain financial products. Lawyers, in partnership with bankers and jurists, strive to provide them a reengineered version of those products.” The pious pay a premium for the wrapper; the payoff inside is the market's. Siddiqi's sentence, the finding of chapter 32, could be filed as this chapter's thesis with two words changed: “While theory aspired to prove Islamic finance was different from the conventional one, practitioners were busy searching for ways to make it similar to it. Convergence rather than divergence was the order of the day.” And the mechanism that chapter documented — certifiers selected and paid by the certified, down to a banker who described dialing scholars until a ruling arrived — is a structural warning the secular certification economy has no grounds to think itself exempt from: wherever a label has a market price and the labeler is retained by the labeled, the label will drift toward the money unless something with teeth holds it. The tradition's own answer, chapter 32 showed, was standardization with bite and judges paid otherwise than by the judgment.

Rome's letter is the newest. In May 2018 — deep into the ESG era — the Congregation for the Doctrine of the Faith and the Dicastery for Promoting Integral Human Development issued, with papal approval, a joint document on the present financial system, *Oeconomicae et pecuniariae quaestiones*, whose fuller place in the story of Catholic Social Teaching belongs to chapter 33. This chapter needs only its two most striking sentences. On profit: “No profit is in fact legitimate when it falls short of the objective of the integral promotion of the human person, the universal destination of goods, and the preferential option for the poor.” On lending: “it is clear that applying excessively high interest rates, really beyond the range of the borrowers of funds, represents a transaction not only ethically illegitimate, but

also harmful to the health of the economic system. As always, such practices, along with usurious activities, have been recognized by human conscience as iniquitous and by the economic system as contrary to its good functioning.” There is the old word — usurious — in a curial document of the smartphone era, applied to the living system; and notice what the document asks about: not portfolios but purposes, not scores but what the profit is for and what the rate does to the borrower — contract questions, the retired grammar intact in the vault.

Three houses, one finding. Each tried the label; each learned that a label holds only where an audit of substance stands behind it with authority to say no; and each states the substance test in its own idiom as the questions this book has carried since chapter 3 — what is the money doing, who bears the loss, what does the charge price. The market’s new conscience has the labels, the raters, and now its first prosecutors. What it has not yet built is the audit the letters describe, and the letters are unanimous about what happens in its absence.

A Conscience Without a Grammar

So the distinction this chapter exists to draw can be stated cleanly. Ethical branding changes selection and disclosure. Those are real changes — selection steers capital among issuers, and disclosure makes harms visible and lying expensive, as the DWS file proved — but they leave the instrument untouched: the same loan, the same lease, the same waterfall, now described in better conscience. Altered payoff geometry changes the contract: who bears loss, as the first-loss tranche does; what the charge prices, as a cost-recovery or capped return does; who holds control, as the steward charter does; where ruin lands, as a reserve or a contingent claim does. The old instrument panel — chapter 3’s taxonomy, chapter 7’s anatomy, Part VI’s titles, Part IV’s counting-room, chapter 28’s ruin column, with chapter 2’s person at the door beneath them all — reads geometry and only geometry. Run any labeled instrument through it and the label falls away in the first minute, leaving whatever payoff was always there: a green bond reads as a bond; a market-rate impact loan to a fragile borrower reads as the fixed claim it is; a first-loss tranche reads as a reserve; a steward charter reads as an endowment lock. That is not the panel’s blindness but its calibration. The traditions and the tax court agree against the brochure — in contracts, effect is given to intention and meaning, not to words and phrases, as the Majalla’s third article has it, and as *Gregory v. Helvering* rediscovered; the names are hypotheses, chapter 3 taught, and the payoff is the evidence. Chapter 28 wrote the warning for exactly this vocabulary, and it deserves quoting on its way to being generalized: “a ‘solidarity reserve’ that does not move the ruin column is a marketing line.” So is any moral claim about capital that moves no column at all. The new conscience has begun, at its edges, to learn column-moving; at its center it remains a discipline of description, which is why its enforcement actions are about adjectives and its crises are about payoffs.

There is a name for finance built from the other end — the columns first, the description afterward — and the reader has been walking toward it for thirty-three chapters. Non-extractive capital is the program of the movement that begins where this chapter’s audit ends, and its premise can be stated in the traditions’ plain terms: capital may be compensated, fairly and openly, for real risk, real service, and real patience; what it may not hold is an unlimited claim on the enterprise’s future. In practice the program is a short stack of geometry. The investor’s return is capped by agreement, so that the claim ends the way a mortgage ends instead of running forever. Repayment is drawn from durable surplus —

what the enterprise genuinely spares after wages, maintenance, reserves, and mission have been fed — never from a surplus faked by squeezing the people or the upkeep. Control sits with stewards bound to purpose, under locks of the kind this chapter just classified, so that no future quarter can sell the mission to pay the claim. And beneath the stack stands the mercy layer the traditions never let a system omit, the *gemach*'s and the *qard hasan*'s, where need is met without any claim at all. Every clause has a home address in this book: the cap is Part VI's proportion generalized; the surplus rule is Part IV's counting-room, no profit before the capital — and here, the enterprise — stands whole; the steward is chapter 29's *mutawalli* in company law; the reserve is chapter 28's; the refusal to price need is chapter 2's oldest commandment. None of this is antiquarian; it is a build sheet.

And a build sheet needs arithmetic. This is where the historical work of this book ends and the constructive half begins. Chapter 35, "From Doctrine to Model," opens Part IX by turning the traditions' tests into working models: a ledger on which every euro of return must name its title, with extraction measured as the residual no title can claim. Chapters 36 through 39 weigh risk, waterfalls, and invisible contribution, and define the surplus that may honestly be distributed; chapter 40 assembles the return formula; chapters 41 through 44 build the fund, the contract library, the analyzer, and the accounting; and chapter 45 names the temptations that stalk every moral finance — impact-washing among them, beside the older washings this Part has cataloged — because the new architecture will deserve its own audit too. The promise chapter 30 made at this Part's door can now be kept in its own coin: "the tests as models, the titles as line items, the ruin probability restored to the reports."

Return, last of all, to the committee of 2004, because this Part should end where its evidence points rather than where its irritations do. The bankers who drafted the acronym were not wrong about the hunger; they were its proof. A market that spent four centuries retiring the moral vocabulary of finance found that it could not live without one, and reinvented labels within a generation of pronouncing them obsolete — reinvented boards, screens, certifications, even prosecutions, everything but the grammar. The grammar survived elsewhere, in the three houses whose letters this chapter read, and retrieving it has been the work of this entire book. A conscience without a grammar can select, disclose, embarrass, and occasionally move a column, and each of those is worth having. What it cannot do is the one thing every tradition in this book thought was the point: tell fruitfulness from extraction under any label whatsoever. For that, the questions must become arithmetic — and the arithmetic begins in chapter 35.

Part IX — Moral Mathematics: Making the Traditions Computable

Chapter 35 — From Doctrine to Model

Two term sheets lie side by side on a desk in Amsterdam, each offering the same young company the same half million euros. Both cap the capital's total claim at two euros and ninety cents for every euro advanced, collected over ten years. Run them through any instrument modern finance owns and they are the same animal: same multiple, same horizon, same yield to the same decimal. A rating desk would file them together. A regulator would read one and initial both.

Now ask each sheet the question this book has spent eight Parts learning to ask: what, exactly, is each of those extra euros *for*? The first sheet can answer. So much for costs it can show, invoiced and audited. A penalty clause that expects to collect nothing, because it prices only genuine delay. A charge for danger the capital actually stands inside — money that loses when the company loses. The rest, a stated share of profits that exist on the day they are shared and not before. The second sheet, pressed the same way, produces thin costs, a risk charge on a claim payable in good years and terrible ones alike, and a large allowance for profits the capital imagines it would have earned somewhere else. Pressed harder, it says: the market.

The two sheets are identical in every number and opposite in what the numbers are made of, and nothing in the modern reporting stack can tell them apart. That blindness is not anyone's malice; it is what four centuries of retiring the old questions bought, and chapter 30 priced the purchase honestly: the vocabulary that once told fruitfulness from extraction was taken out of service, and the instruments built since have had no column for it. The traditions kept the vocabulary but kept it in prose — responsa, fatwas, manuals for confessors — a case-law mode that reads one contract at a time, slowly, by hand. Between the spreadsheet that counts everything and explains nothing and the tradition that explains everything and counts nothing, the difference between those two term sheets falls through, year after year, at continental scale.

Part IX exists to close that gap, and it opens by changing the state of the book's material: from doctrine to model. Not new teaching — the teaching is done; Parts I through VIII assembled it, tested it, and watched it survive its own evasions. What changes now is the form the teaching takes. Tests become procedures; titles become line items — the first installments on the promise chapter 30 posted at this Part's door. Questions that were asked of consciences will be asked of cash flows, in a language cash flows cannot refuse to answer. This chapter builds the first and load-bearing piece of that machinery: a ledger on which every euro of return must name its title, and a remainder that measures what cannot.

The Case for Arithmetic

Begin with why the traditions need the mathematics, because the need is not obvious from inside them. For most of their history the three houses ran the most sophisticated contract-review practice in the world on prose alone. A merchant brought his paper to the rabbi, the mufti, the confessor; the paper was read, the parties questioned, the substance weighed against the form; a ruling came back fitted to the case like a key to a lock. At the scale of one desk and one contract, nothing in this book improves on that method, and nothing in

this chapter is meant to replace it.

Three things defeated it. The first is volume. A posek reviewing a partnership deed is one scholar and one document; a modern credit book is thousands of documents, revised quarterly, each one seventy pages of defined terms. The bedside method does not scale, and the traditions know it — the standardized *heter iska* hanging on the bank's wall and the standardized *murabaha* file are both, in their different ways, the case-law mode confessing that it could no longer read every case. The second is synthesis. Chapter 37 will walk the modern machinery of waterfalls and preferences, where a payoff indistinguishable from the forbidden one is assembled out of parts each of which passes inspection separately. No clause-by-clause reading catches a shape that exists only in the whole; you have to compute the whole. The third defeat is drift, and Part VIII was its documentation: the *ribbit*-washing chapter 31 audited, the shariah arbitrage chapter 32 weighed in its own scholars' words, the label economy chapter 34 walked through — every one of them survived scrutiny of language, because language was the thing being laundered. A test that lives in words can be worn as words. A test that lives in a computation has to be defeated in the computation, in the open, where the defeat leaves arithmetic behind.

There is also the plainer institutional fact: a test that cannot be computed cannot be delegated. It cannot be run by a fund administrator in Utrecht who never met a schoolman, compared across a hundred contracts, tracked across a decade, or audited by someone who distrusts the auditor before her. It stays a homily — admirable, portable, and inert. The traditions' revival on any scale that matters is therefore an engineering problem before it is a persuasion problem. The doctrine must survive translation into procedure, or it will keep losing to procedures.

Nor is the translation a foreign imposition. The traditions have been drifting toward it for centuries, whenever scale forced their hand. Ulpian's table — chapter 10 met it — valued life annuities in stated bands of years, actuarial arithmetic sitting calmly inside Roman legal doctrine. The *heter iska klali* of chapters 14 and 31 is a standardized instrument with a printed settlement figure, keyed in its modern forms to prevailing rates. And the contemporary standards bodies of Islamic finance — the resolutions and decimal-numbered articles that chapters 10, 13, 16, and 24 leaned on — are doctrine already dressed for the spreadsheet: conditions enumerated, thresholds stated, definitions written to be checked. A shariah standard with an article number is a ruling that expects to be audited by strangers. The step this Part takes is longer but points the same way the traditions were already walking: from the ruling to the rule, from the rule to the procedure, from the procedure to the number a procedure can output.

The second motion runs the opposite way, and it is the one modern readers need more: the mathematics needs the moral categories, because on its own it cannot see the difference that matters. Every number finance reports about our two term sheets — yield, duration, multiple, internal rate of return — is a measure of size. Not one is a measure of source. The sheets tie on every metric because the metrics were built to answer how much and when, never out of what. Chapter 28 showed the deeper version of the same blindness: expected value, the reporting stack's favorite summary, averages over lives no one lives, and the column that tracks who gets ruined along the way simply is not in the reports. And chapter 34 watched the market's own conscience movement inherit the blindness intact — auditing the company, screening the issuer, rating everything except the one thing the traditions ever asked about, which is what the contract does. Add categories to none of this, and computation at scale simply industrializes the not-asking.

So the two needs interlock. The traditions hold seven centuries of categories, argued out in three legal languages and stress-tested against every evasion merchants could invent, but hold them in a form that cannot keep up with the instruments. Finance holds the computational power to read every contract on earth by Tuesday, and nothing worth asking them. The obvious marriage is the work of this Part. It comes with a warning label the traditions would insist on: the model decides none of their live disputes. Where the houses disagree — and they disagree, *machloket* and *ikhtilaf* being the sound of legal systems breathing — the model's job is to hold the disagreement steady and visible, not to settle it by parameter. A conscience that cannot count is at the mercy of whoever can. The cure is not to replace the conscience with the counting, but to teach them each other's language.

Form, Label, and Payoff

Every tradition in this book converged on the same prior rule, and chapter 3 met it before any doctrine did: the label a contract wears settles nothing. The *fiqh* compressed the rule into the Majalla's third article, quoted there and worth carrying forward — “in contracts, effect is given to intention and meaning, not to words and phrases.” The rabbis enacted the same rule category by category — chasing substance, in chapter 3's phrase, past the edge of form — until even the dust of interest had a name. The scholastics asked of every clever instrument whether it was named truthfully. And the secular courts rediscovered the rule the moment revenue depended on it, striking down Mrs. Gregory's flawless three-day corporation as “a mere device” — chapter 3 told the story — because what a thing is called yields to what a thing does.

The model's first move is to make that old rule operational, and the way was shown seven centuries ago, at sea. When *Naviganti* condemned the sea loan, chapters 10 and 11 argued, the target was precise: risk priced on a claim whose payment did not actually track the venture — danger charged for, never borne. What frame a contract deserves is decided by the shape of its cash flows, not by the noun on its letterhead. That sentence, which the decretal's world could only apply by reading documents one at a time, is in ours a specification. Describe any financing not by its name but by its payoff: what does the capital's claim collect in the good year, the flat year, the catastrophic year? What the claim does across the states of the borrower's world, and nothing else, is the contract, morally speaking. Everything else is stationery.

Read by payoff, the whole taxonomy of chapter 3 reassembles itself without labels. A claim that collects the same amount in every state of the world is a loan, whatever the paperwork calls it — and if it also charges for risk, it is charging a premium for insurance it never issues. A claim paid only out of realized prosperity, that loses when the venture loses, is a partnership even if no one wrote the word — the *societas* consensus every school shared, falling out of the arithmetic before any school is consulted. Between the poles runs a spectrum, and real instruments live all along it; chapter 7's anatomy — whose money is it, who carries its danger, what precisely does the charge price — becomes, in the model, a set of coordinates rather than a set of questions. Nor did the traditions only read payoff shapes; at their boldest they engineered them. The *iska* of chapters 11 and 13 is a built object, half loan and half deposit, its two halves deliberately dividing profit and peril between the parties; the *commenda* split one voyage's uncertain return between capital and labor on stated fractions. Instruments can be designed to a moral geometry, not merely audited against one — the fact on which chapter 41 will build a fund. Out of the anatomy, two questions govern nearly everything that follows. How closely does the claim's collection track the venture's actual fortunes? And how much of the claim answers no burden anyone can

show? The first is the question of form; the second, of size. Chapter 36 and the chapters after it will put numbers on both; Appendix D names the metrics; the analyzer of chapter 43 asks them of live term sheets wholesale.

Two more things follow from reading by payoff, and both matter before the ledger is even opened. First, synthesis loses its camouflage: a payoff assembled from permitted parts is still a payoff, and if its assembled shape is the fixed-claim-plus-risk-premium shape the tradition refused, the computation says so regardless of how innocent each part looked alone — the hidden geometry of chapter 37, named early. Second, the reading is mercilessly fair in both directions. An instrument from the most pious house, wearing the tradition's own vocabulary, can compute out as a fixed claim in all weathers; an unsentimental commercial contract can compute out as honest participation. The payoff test has no ear for accents, and it cuts with the same edge whoever is holding the paper. The traditions, at their best, would recognize that deafness as their own: the desperate borrower of chapter 9 was owed substance, not diction, and the person at the door of chapter 2 could not eat a label either.

The Moral-Title Ledger

Now the construction this chapter exists to build. Take any financing. List every euro the capital will claim over the contract's life, from the first fee to the last distribution. Then open a ledger — the moral-title ledger — with one zero line and seven standing lines, and require every claimed euro to sit on a line. A euro sits on a line only if its title is genuine in fact, not merely in name: Part VI spent five chapters building that discipline fence by fence — the burden real, shown, sized, capped, and counted once — and the canonists' old suspicion of the contract *in fraudem usurarum*, usury wearing a title as costume, is the reason the fences exist. The ledger adds nothing to the doctrine; it is the doctrine, arranged in rows, with a pen waiting.

The zero line is return of capital, and it is the one claim every tradition in this book grants without argument. The Qur'an's settlement of the *riba* war, met in chapter 2 and heard in full in chapters 6 and 19, ends exactly here — "You shall have your capital," in Abdel Haleem's rendering: no increase, no loss, the principal whole and nothing owed beyond it. The Talmud's word for that protected core is *keren*, the horn; the jurists of the fiqh say *ra's al-mal*, the head of the wealth; the scholastics said *sors* — the same principal whose danger gave *periculum sortis* its name. Three legal languages, one homely intuition: what you advanced comes home to you, and its coming home is not gain. The zero line also carries Part IV's ordering rule, delivered in chapters 13 and 14: the capital is made whole before anyone's profit exists to share. Everything above the zero line is the actual subject of this book, and there are, when the traditions' files are consolidated, exactly seven honest things an extra euro can be.

Work. Somebody found the enterprise, weighed it, structured the deal, watched it, kept its books, sat with its founder in the bad month. The halakha prices this with unembarrassed precision — *sechar tircha*, the wage for trouble, chapter 12's file — and will void an iska over an unpaid dinar of it. The fiqh gives the working *mudarib* his ratio and, when the contract collapses, still pays him *ujrat al-mithl*, the wage of his like, because the labor happened whatever the paperwork did. Aquinas put the same title at the center of his defense of the honest merchant: gain sought "not as an end, but as the wage of his labor" — *stipendium laboris*. The line's fence is inherited with it: work is priced as work, shown and proportionate, not collected as a percentage of capital that swells whether or not anyone worked.

Expense. Running money costs money — clerks, premises, records, audits — and the tradition's most rigorous institutions charged for exactly this and nothing else. The Monti di Pietà of chapter 17 stood before a council on precisely that defense and won; the *qard* funds of chapter 16 operate today under a resolution permitting the service charge that recovers actual cost and forbidding the one that recovers more. The fence: costs shown, allocated to this financing, recovered once.

Loss. Where the lender has genuinely been damaged — the *damnum emergens* of chapter 19, loss arising, actual and attributable — repair is owed, and every house agrees while policing the word *actual* with its own instruments: the iska's witnesses, the fiqh's demand that the impairment furnish its reason, the schoolmen's insistence that the damage be proved, not presumed. A loss must testify before it collects.

Risk. Capital that truly stands in the venture's weather may be paid for standing there — *periculum sortis*, chapter 10's treacherous title, the danger to the principal. The fence is the sharpest in the whole tradition, because this is the line the sea loan forged: the danger must be borne, not merely priced. A risk charge attached to a claim that collects in every state of the world is the *Naviganti* pathology in modern dress. This line, more than any other, will be interrogated in chapter 36 — what the danger really is, how much of it collateral and seniority have quietly removed, and whether it has been counted once or five times.

Liquidity. Being without one's money can be a real burden — *carentia pecuniae*, chapter 21's adjacent title, the deprivation of liquidity — but the burden must be his, particular, dated, and shown. Chapter 21 posted the fence in one line: "A burden everyone can claim is not a title; it is a rate."

Opportunity. The most dangerous line in the ledger, because it prices a world that did not happen. *Lucrum cessans*, chapter 20's file: the merchant with a nameable elsewhere may, under the developed doctrine, be compensated at hope's price rather than harvest's — documented, particular, conservative — while the strongbox that has every elsewhere and therefore none may not, and the professional lender pleading lost interest on a loan he would only have replaced with another loan walks Hostiensis's refused circle. The halakha's answer was structural rather than actuarial: rewrite the financing so the lender is written inside the earning — the heter iska of chapters 20 and 31 — after which nothing is forgone and the lament "I could have earned more elsewhere" has no grammar left to stand in. The fiqh's answer is the flattest: time has no rental price at all. The ledger holds all three answers; the line exists, fenced, and some houses set it to zero.

Surplus. Last, the line the whole architecture exists to protect: a share of profit actually made, flowing to capital that genuinely ventured or work genuinely done — the *societas*, iska, and *mudarabah* shares of Part IV, measured after real profit exists and never before. Chapter 12 compressed the entire scholastic case into one demand, and the ledger adopts it as written: "every florin of return must trace to ventured property or expended work." Surplus is the only line that grows with the venture's success, because it is the only line that shrinks with the venture's failure. That symmetry is its title.

One fence runs across all seven lines rather than along any of them: a burden may be counted once. The same danger must not be charged for in the rate, removed by the collateral, shifted onto a guarantor, and penalized again at default — four collections on one risk, three of them for a burden nobody is still carrying. Part VI posted the rule line by line; chapter 36 will make it computable across them; the ledger's own contribution is bookkeeping's oldest — a line, once filled, is filled.

One famous name is deliberately absent. The agreed penalty — *poena conventionalis*, chapter 22's file — holds no standing line, and the omission is the doctrine. A genuine penalty is priced to collect nothing: it answers unwillingness, never inability, and on the day of signing its expected value is approximately zero. The moment a penalty becomes budgeted revenue it has stopped being a penalty and started being a rate wearing enforcement's clothes, and it must then find a real line — usually risk, usually falsely. So the ledger books penalties only when they happen, in delay states, against culprits; it refuses them as a standing source of return. Seven lines, then, and a zero line, and nothing else. The consolidation is this book's — each tradition kept its files by its own scheme — but the test underneath it is ancient. In 1515, blessing the Monti, the Fifth Lateran Council paused to define the thing it was against, and chapter 14 quoted the definition: usury is gain sought “without any work, any expense or any risk.” Read that definition against the ledger and it is the ledger in negative — a conviction written as three empty lines. The ledger is a listening device, not a verdict; it simply arranges a return claim so that the old questions can be asked of it euro by euro, and makes one further thing possible, which is the subject of the next section: it lets us subtract.

The Residual

Fill the ledger in for an ordinary instrument and watch what happens. The example is this book's own, kept simple on purpose. A lender advances €100,000 to a small firm for five years at 12 percent, interest only, the principal secured on the firm's premises and guaranteed, personally, by its owner. Over the term the contract claims €60,000 above the zero line. Now the lines. Servicing this loan — origination, administration, collections — can be shown to cost about €4,000 over the term: the expense line takes it. The work of making the loan is inside that figure; the work line takes nothing more. No loss has occurred; that line waits. Risk: the claim is doubly secured and payable whatever kind of year the firm is having, so the danger actually run is modest — price it honestly at a few thousand; chapter 36 will teach us to do better, but no honest pricing of a collateralized, guaranteed claim reaches far into five figures. Liquidity: no particular deprivation documented. Opportunity: this is a professional lender; the refused circle applies. Surplus: the claim is fixed, shares nothing, loses never; zero by construction. The ledger is now full, and roughly €54,000 of the €60,000 sits on no line at all.

What is that money? The tradition had names for it, all of them fighting words. The model gives it a colorless one, and the colorlessness is the innovation: it is the extraction residual — the portion of a return claim that, after every genuine title has been paid in full, answers no shown burden, no work, no borne risk, no realized share. Nothing in the definition accuses anyone. The residual is what is left when explanation runs out, and it is computed by subtraction, which is why two people who disagree about everything else can compute it together. Extraction is not an accusation; it is a remainder.

The subtraction also disciplines both sides of the desk, which is what makes it more than rhetoric with decimals. A genuine title left vague lands in the residual exactly as an invented one does: the fund that truly incurs costs but cannot show them, and the manager whose work is real but priced as a habit of percentages, watch honest euros sit in the unexplained column until the documentation exists. The ledger pays specificity a wage — Part VI's “shown” made self-interested — and it makes drift visible while it is still a number and not yet a practice, the cheapest moment anyone will ever have to correct it. It gives arguments somewhere to land, too: two auditors who disagree can now disagree about a named line by a stated amount, which is a negotiation, instead of about a word, which is

a war.

The method has a distinguished pedigree, and borrowing it openly is half the point. In 1956 the economist Moses Abramovitz consolidated what was then known about American growth and found something scandalous: between the 1870s and the 1950s, output per head had roughly quadrupled, while measured inputs per head — all the capital and labor the statisticians could count — had grown about 14 percent. Nearly the entire modern miracle, by his count, was attributable to a productivity gain of some 250 percent that the measured factors could not explain. He declined to dress the remainder up, calling it, in a phrase economists still quote on themselves, a “measure of our ignorance” — and the remainder later took a name from Robert Solow’s 1957 formalization and became the Solow residual, the most productive embarrassment in the history of the discipline, because naming the unexplained forced everyone to go explain it. The moral-title ledger runs the same operation with the moral sign reversed. Growth accounting’s residual was ignorance, and turned out to conceal the best news of the century — knowledge, technique, health. A return claim’s residual conceals no good news. Every euro in it is a euro someone pays without owing it under any title the traditions — or, chapter 34 conceded, the label economy — can produce. Their residual shrank as understanding grew. This one shrinks only when the contract changes; understanding it merely tells you whom it comes from.

Fairness requires one complication, and it is the hinge on which the model’s evenhandedness turns. A residual is always computed relative to the titles a house admits and the sizes it allows, and the houses differ. Aquinas set the opportunity line at zero; the later schoolmen opened it under conditions; the Salamancans of chapter 9 would title part of our example’s remainder to the honest *communis aestimatio* of a working credit market — that is what the settlement of chapter 30 built its house on. The ledger does not flatten this history. Run it strict and the residual is large; run it mild and it is smaller; the model can carry the strictest and the mildest readings side by side and report both, which is precisely how a measuring instrument respects a machloket it has no standing to settle. What no house on the spectrum will do is title the remainder to nothing. Somewhere between Aquinas’s zero and Salamanca’s band, every tradition in this book runs out of patience, and the euros past that point have only ever had one honest description.

Before the equation, say plainly who pays the residual, because a subtraction can launder a fact the traditions never let out of their sight. The unexplained euro is not paid by “the market”; it is paid out of the borrower’s worst year, by the household whose guarantee is called; it is paid in the founder’s unpaid seventh year, by the contributions chapter 38 will show the cap table cannot see; it is paid, chapter 34 documented, by missions sold to meet a preference. The residual lands on the people and purposes least able to refuse it, at the moments they are least able to refuse — that is what a fixed claim on a living enterprise means, and chapter 28 put the mechanism in numbers. The ledger’s politeness — measured, not accused — is a method for getting the number into the room, not a judgment that the number was ever polite to the people underneath it.

The First Equation

Everything in this chapter now compresses to a single line, the first equation of the book’s constructive half — in words, because the words are the mathematics:

Every euro of return = capital returned + work + expense + loss + risk + liquidity + opportunity + surplus + whatever is left.

The model's entire moral content lives in the last term. Whatever is left, after the zero line and the seven titles have taken everything they can honestly carry, is the extraction residual; and a financing is non-extractive exactly to the degree that the last term is zero. Every euro must have a title — the sentence is the equation, and the equation is nothing this book has not already argued; it is chapter 12's florin-rule with the loss-side doors of Part VI hung in their frames, chapter 3's form-versus-substance test made subtraction-shaped, the 1515 definition run in reverse. Doctrine, changed in state. What was a sin is, for the length of this Part, a line item — which does not make it less than a sin; it makes it findable.

Notice, finally, what the equation refuses to say. It does not say which titles your house admits, or at what size; it holds the schools' quarrels open on purpose, and the disputes stay live inside it the way they stayed live in the responsa and the manuals. It does not say that a large return is extraction or that a small one is innocence; composition, not size, was the traditions' question, and it is the equation's. The thing chapter 34 found missing — a grammar able to tell fruitfulness from extraction under any label whatsoever — begins here, and it begins, of all things, as bookkeeping. And the equation does not pretend to more precision than its lines yet have. Each line now needs its own examination, which is the rest of this Part: chapter 36 puts the risk line under oath — "Pricing Risk Without Worshipping Risk" — asking what the danger is, who really bears it, and how often it has been counted; chapter 37 follows the money through waterfalls and preferences, where repayment order does quietly what interest once did loudly; chapter 38 recovers the invisible contributions of founders and workers; chapter 39 defines the true surplus that may honestly be shared; and chapter 40 assembles the whole into the non-extractive return formula. The ruin column chapter 28 demanded gets its seat in those chapters' reports.

The two term sheets from the opening desk can wait there no longer. With the ledger between them, an afternoon and honesty suffice: the first fills its lines and zeroes its remainder; the second, asked euro by euro what its claims are for, watches fifty-four thousand answers fail to arrive. No instrument in the modern stack could tell them apart, and now a subtraction can. That is the whole promise of moral mathematics, kept small and kept honest: not a new conscience, but arithmetic the old one can finally run at the speed of the instruments. The ledger is open. The first line is capital, coming home.

Chapter 36 — Pricing Risk Without Worshiping Risk

Take chapter 35's file back out of the drawer. A hundred thousand euros, five years, twelve percent, the claim secured on the firm's premises and guaranteed by its owner; sixty thousand claimed above the principal, four thousand of it earned by shown expense, and a risk line we priced, provisionally, at "a few thousand" — with a promise that this chapter would earn the number, and now it must. But the file cannot be priced until a word is dealt with, because the word does more work in modern finance than any other in the building. Ask why the rate is twelve and the answer is risk. Ask why the fund's target is twenty-five percent and the answer is risk. Ask why the preferred stock must be paid back double before the founder sees a cent, and the answer, delivered a little more slowly, is still risk. The word is not wrong. Danger is real, lending is dangerous, and compensating the bearer of a real danger is one of the seven honest things a euro of return can be — the traditions said so, in three languages, and the ledger holds a line open for it. The trouble is that somewhere between the actuary's desk and the pitch deck, risk stopped being a fact about the world that could be measured, mitigated, and paid, and became a deity — invoked, not shown; credited with every return that has no other explanation; worshiped precisely because nobody is expected to produce it in court. This chapter does the unglamorous thing instead. It prices the danger — honestly, both of the claim and of everyone standing near it — and then asks what is left of the invocation.

The Honest Arithmetic of Danger

Begin with the craft, because the craft is honorable and old, and the traditions never feared it. Chapter 10 watched Roman jurists band life expectancy into a table without theological alarm; chapter 27 watched the first man ever to hold the title *actuary* compute a fair premium from mortality records so that a mutual club could keep its promises to its own members. It mattered, too, who owned the arithmetic: the first actuary's employer was a mutual, the priced and the pricers the same people, which kept the tables honest by construction. Pricing danger is not the worship. Pricing danger is the opposite of the worship, because a price is a claim that can be checked. And the ledger's risk line carries the tradition's own name for the object being priced — *periculum sortis*, the danger to the principal, chapter 10's treacherous title — a name that was always conditional in exactly the way a checkable claim is: a danger, to this principal, in this venture, of this size. The title never blessed the word; it blessed the fact, where the fact could be shown.

The arithmetic has two factors and refuses to run on one. Expected loss is probability times severity: how likely is the feared event, and how much does it destroy when it comes. The modern rulebooks decompose a loan's version of this into three named quantities — the probability of default, the loss given default, the exposure at default — and multiply, which is the same two-factor discipline wearing supervisory clothing: how often, how deep, on how much. Run it on our file, with assumptions announced as assumptions. Suppose small firms of this kind fail at three in a hundred firm-years. Suppose that when this one fails, the premises and the owner's guarantee recover all but a sixth or so of the principal — call the shortfall fifteen thousand euros, and hold in view what that recovery means on the other side of the table: a family's building sold, a household's savings called. Three hundredths of fifteen thousand is four hundred and fifty euros a year; over the term,

twenty-two or twenty-three hundred. That is the honest actuarial content of the file's danger to the lender — a real number, worth paying, and less than a twenty-fifth of the sixty thousand the file claims. The reader now knows, precisely, the size of the gap the word *risk* is being asked to cover.

Both factors, or neither — the discipline cuts in every direction, and each direction has its habitual sinner. The famous mortality statistics of young companies are quoted in every venture pitch as though probability were the whole of the matter — so many in ten gone within so many years — with never a word about severity's other half: what a failure actually costs this claim, at this stage, with these assets, in a portfolio of that size. Run failure rates without recoveries and every young company is a casino; run them honestly and much of the terror resolves into a number with a decimal point. The opposite sin prices severity without probability — the covenant package built as if the one-in-a-thousand catastrophe were next quarter's base case, charging every borrower always for the worst thing that ever happened to anyone. Honest arithmetic multiplies; worship selects whichever factor is largest and calls it the truth. Walk the file through the year the borrower actually faces, scenario by scenario, and the geometry shows itself. In the good year — give it fifty-five chances in a hundred — the claim collects and nobody notices. In the flat year, thirty in a hundred, the claim collects. In the bad year, twelve in a hundred, the claim still collects — the firm strains, the owner defers her own pay, and the secured lender loses nothing. Only in the catastrophic year, three in a hundred, does the claim finally lose its fifteen thousand. Notice what the two columns just revealed: the lender's expected loss lives entirely in the last row; the borrower's danger lives in the middle rows too. The claim is exposed to one scenario. The firm is exposed to all of them. Any pricing that forgets whose column is whose has already begun the worship.

And time will not stay out of the arithmetic. Capital at risk is a curve, not a number. Our interest-only file holds the full hundred thousand in the wind for all five years — maximum exposure to the last day — where its amortizing twin, principal coming home in five even steps, watches the sum at stake melt from a hundred thousand toward twenty; amortization is, among other things, a machine for making the charge track the danger, since the interest euros shrink with the balance of their own accord. The dishonest version of time is subtler, and our file exhibits it. The twelve percent was priced, presumably, to the firm as it looked at signing — young, unproven, failing at three in a hundred. Suppose the firm survives, as most do, and by the fourth year has aged into a steadier creature failing at one in a hundred. The danger has fallen by two-thirds. The rate has not moved. A charge priced to a danger and then held flat while the danger drained away is collecting on a memory — the charge outlives the danger — and the widening excess drifts, unlabeled, toward the residual. There is a deeper time problem underneath, and chapter 28 owns it: expected loss is an ensemble number, a fair summary for a pooled portfolio that lives many paths at once, while the borrower lives one path in sequence, where a single catastrophic row is not an entry in an average but the end of the story. Severity is not exchangeable the way probability tables pretend — ten losses of a tenth are survivable arithmetic; one loss of everything is an absorbing state — and an honest risk line carries that asymmetry instead of flattening it. Hold both lessons; the fourth section needs them.

The Vanishing Danger

Now watch what the file's own paperwork does to the danger it charges for, because the modern contract is a machine for removing risk, and every removal has a moral price tag the tradition wrote out seven centuries before the term sheet.

Collateral cuts severity. The pledged premises are why failure costs fifteen thousand instead of the better part of the principal; strip the security away and the same three-in-a-hundred failure rate prices the danger north of thirteen thousand over the term, and the twelve percent would at least have an opening statement. With it, the argument shrinks by the same subtraction the collateral performs. The guarantee cuts what the collateral misses — and notice immediately that this second mitigant does not reduce the loss; it relocates it, out of the firm and into a household, a move the next section refuses to lose track of. Seniority and preference are subtler, because they remove nothing from the world either: they re-seat the loss inside the same venture's capital, so that the junior tranche, the common shares, the unsecured suppliers absorb the fall before the protected claim feels it — the queue-geometry chapter 37 will draw properly. And the quieter furniture belongs on the same page: covenants that let the lender act early shrink severity by shortening the fall; information rights shrink uncertainty itself. Each device is lawful, common, often prudent. Each also subtracts from the danger the claim can honestly say it bears. The Talmud had already graded the posture: the investor who writes himself *near to profit and far from loss*, says the baraita chapter 10 verified, is the wicked one — not because profit is wicked, but because the nearness and the farness were engineered in the same document that goes on charging as if they weren't. The fiqh built the same fence structurally: the working partner may not guarantee the capital he works, chapters 13 and 26 showed, because a guaranteed *iska* or a guaranteed *mudarabah* is a loan in the exact measure of the guarantee — the tradition's way of saying that protection and risk-charge cannot occupy the same euros.

So the fence for this section, stated once: a protection purchased is a danger sold. Whatever the collateral takes off the table, whatever the guarantee rehouses, whatever seniority hands to the junior claims, may no longer be priced as if it were still sitting in front of the capital. Remove the danger or charge for it — either is honest commerce; both at once is the *Naviganti* pathology, risk priced and not borne, wearing better drafting. None of this is hostility to security as such: collateral and guarantees do honest work in the world, calming the measurable so that credit can exist at all, and a fence is not an accusation. The dishonesty begins at the second collection, never at the first protection.

Where the Danger Goes

The physics of the thing is the section's whole argument: danger removed from capital is never destroyed; it is rehoused. The contract cannot reduce the world's uncertainty by a single basis point; it can only decide who stands where when the uncertainty arrives. So follow the file's removed risk to its new addresses, and keep the register plain, because the addresses are not abstractions.

The first address is the founder's household. The personal guarantee converts a business failure into a family event: the savings, the house, the years of future income conscripted as loss-absorbing capital for a claim that priced itself as if exposed. The signature that does this is usually collected at the closing table in a stack of signatures, one page among forty; what it signs away is not one page's worth of anything. The owner who signs is, in the contract's true geometry, the most junior investor in the venture — committed without a cap, compensated without a line. The second address is the workers. Almost no financing document names them, and nearly every one obligates them: when debt service tightens, payroll is the adjustment variable — the hiring freeze arrives quarters before any missed coupon, wages stall, positions close, and the people with the least contractual protection absorb the earliest loss. In the law's queue they may rank as creditors; in the cash flow's queue they stand behind the claim that called itself risk-bearing. The third address is the

mission, wherever the borrower has one: repayment pressure in a bad season converts purposes into assets — the client list sold, the code closed, the care rationed — chapter 34's file, at one line's length. The fourth is the community around the enterprise: the users who built their days on the thing it made, the town that organized part of its life around the payroll, the public bodies that inherit what fails and the neighbors who inherit what the public bodies cannot afford. None of these bearers signed the risk section of the agreement. All of them are in it.

Trace our own file's subtractions and the conservation law stops being a metaphor. The unsecured version of the loan carried some ninety thousand euros of severity for the lender; the pledge of the premises moved roughly three-quarters of that off the claim — not into nothingness but onto the firm itself, which now faces every future crisis knowing its place of work is the first asset to go. The guarantee moved most of the remainder onto the household. What is left in front of the capital is the fifteen thousand we priced. The lender's severity fell by four-fifths; the world's severity did not fall at all. It moved twice, each time toward a seat with fewer defenses and no line in the return.

Now the construction this chapter adds to the ledger, and it is the book's own: moral expected loss. Take the same two-factor arithmetic the first section ran for the lender — probability times severity, scenario by scenario — and run it for every seat the contract touches: the claim, the firm, the household, the payroll, the purpose, the surroundings. Two rules govern the run. Severity is measured where the loss lands, not where the contract books it — fifteen thousand euros of "loss given default" is one number on a diversified balance sheet and a different object entirely when it is a family's equity in its own life. And ruin counts beyond its euro weight: chapters 13 and 28 established that a loss is lived in its seat, as a sequence, and that some seats do not get another season — so a small probability of an absorbing loss cannot be netted away against pleasant averages, for the same reason the village chest of chapter 28 refused to price its members' ruins as rounding error. Moral expected loss is not a rival to the actuarial number; it is the actuarial number finished — the same sum, run to the end of the table instead of stopping at the first seat.

Run it on the file and the inversion appears. To the lender, our contract is nearly safe: twenty-two hundred euros of expected loss against sixty thousand of claim. To the household, the same contract holds a three-in-a-hundred chance of the house — an expected loss modest in euros and unbounded in the only currency the seat actually keeps. Two contracts can carry identical actuarial expected loss for the capital and opposite moral expected loss for everyone else — one absorbing its failures, the other exporting them — and no number in the reporting stack distinguishes them. The old graders needed no such number; the baraita's wicked posture is a moral-expected-loss verdict rendered without arithmetic. The model simply writes down what the grading was tracking, and makes it computable at the speed of the instruments.

In practice the construction is two small tables where there used to be one. The first is the table every credit desk already keeps: the claim's scenarios, probabilities, severities, expected loss. The second runs the same rows for the other seats — firm, household, payroll, purpose, surroundings — the four addresses of this section becoming rows instead of rhetoric — and it is deliberately boring to produce, because its inputs are already in the file: who guaranteed, who is junior in fact, what the collateral reaches, what the covenants can force. A fund that publishes both tables has told the truth about its own instrument in a form a stranger can audit; a fund that cannot fill in the second table has learned something about its instrument too. And the pair makes drift visible the way chapter 35's residual does — a redesign that improves the first table while degrading the second now shows

up as exactly that, a number moving between seats, at the moment the movement is still a drafting choice and not yet somebody's worst year.

The Premium and the Share

The file's twelve percent can now be read the way an auditor reads it, and the first finding is arithmetic: the same danger has been collected on more than once — priced into the rate as a premium, subtracted from the world by the collateral, rehoused by the guarantee, seated behind the covenants that let the lender act early, and charged again by the default clauses that collect extra precisely when the feared thing happens. Five instruments, one danger. Some of the five are protections and some are compensations, and the audit rule is the ledger's oldest, carried over from chapter 35's fences: put them on the same page. List every device that removes or relocates the danger; list every charge that claims to compensate it; net the columns. A danger may be paid for once, by the party bearing it, in the amount actually borne. Put numbers on the mismatch, because the numbers are the finding. A pricing memo that attributes, say, five of the file's twelve points to credit risk is claiming five thousand euros a year for danger — eleven times the four hundred and fifty the file can actually show after its own protections are netted; twenty-five thousand over the term, against twenty-two hundred borne. Whatever the risk line cannot absorb after the netting is not risk compensation; it is the residual, arriving early, wearing the one title auditors were trained not to question.

The second finding is grammatical, and it is the deep one. There are two ways to be paid for standing under uncertainty, and they are not dialects of each other. The premium is a fixed addition to a fixed claim: sized before the world reveals itself, collected however the world turns out. The share is a fraction of what actually happens: sized by the outcome, collected only from prosperity that exists. The traditions sorted the two grammars with perfect consistency across their differences. The premium attached to a loan is where every alarm in this book rings — the sea loan's charge for a danger the claim did not track, the *ribbit* that collects in the borrower's worst year as cheerfully as in his best, the posture near to profit and far from loss. The share is where every tradition relaxed: the *societas* partner earning *tamquam de re sua*, as from property of his own, the *iska* and *mudarabah* fractions, the whole architecture of Part IV — capital paid out of the upside it genuinely waited for, because it stood in the downside it genuinely faced. The *fiqh* compressed the sorting into maxims chapter 10 verified and this Part can now read as engineering: *al-kharaj bi-l-daman*, return follows liability; *al-ghurm bi-l-ghunm*, the burden in proportion to the gain — profit belongs to the seat that holds the exposure, which is why the modern standard-setters' scholar, in chapter 26's fetched sentence, could state the whole tradition in five words: "reward always follows after risk."

Why must the grammar matter so much? A twentieth-century economist supplied the missing theorem. Frank Knight, in 1921, insisted that the word risk had been covering two different objects: a "measurable uncertainty, or 'risk' proper," which behaves like a quantity — tabulable, insurable, poolable — and the unmeasurable kind, for which no table can exist because the future in question has no population of precedents to count. The distinction, he wrote, is "radically distinct from the familiar notion of Risk, from which it has never been properly separated," and he put profit itself on the far side of it: "It is this 'true' uncertainty, and not risk, as has been argued, which forms the basis of a valid theory of profit." Read his line against the traditions' line and they coincide to a remarkable degree. What is measurable can be priced honestly — that is the actuary's province, the mortality table, the failure rate, the premium in its legitimate home. The *fiqh* drew the

same boundary with its own instruments: when the Academy sorted insurance, chapter 27 showed, it refused the commercial sale of the measurable and blessed its mutual pooling as *takaful* — tables yes, but held by the people who face the peril, not retailed over them. What is unmeasurable cannot be priced by anyone honest, because a price is a measurement and there is nothing to measure; a “premium” charged for it is a number invented against *gharar* itself — the fog chapter 23 refused to let anyone sell. For the unmeasurable there is only one honest instrument: stand in it together and divide what it yields. What can be measured can be pooled; what cannot be measured can only be shared. The venture’s deep uncertainty — the kind young companies actually live in — sits past the edge of every table, which is precisely where the premium’s grammar runs out and the share’s grammar begins. The traditions put partnership there by conviction; Knight’s economics puts profit there by argument; the two placements are the same placement.

Strictly, then, the map has three territories, because the pool is a grammar of its own. Between the premium — a price paid to a counterparty — and the share — a fraction of what happens — stands the chest of chapter 28: many bearers of a measurable peril holding it together, so that the ensemble average becomes a fact for the membership without ever being sold to anyone. The traditions staffed all three territories: tables where tables are honest, chests where perils can be gathered, partnerships where nothing can be known in advance — and the fund this Part is building will use the same map in the same order, pooling at the portfolio layer what no single company should be asked to promise, a design chapter 41 assembles. Note, too, who passes this section’s test without strain: the honest venture investor who buys common shares, loses everything in the failures, and collects only from the successes is squarely inside the share’s grammar — the critique of this chapter has no quarrel with her. The quarrel begins where instruments claim the share’s compensation and the premium’s certainty at once — participation on the way up, protection on the way down — and that hybrid is not a grammar but a contradiction, and it has a chapter of its own next.

The Subordination of Risk

What, then, is the worship? It is the premium’s grammar practiced past the edge of measurement, defended by invocation instead of arithmetic. Its liturgy is familiar from every direction of this Part: return is compensation for risk; this return exists; therefore a risk must exist that earned it — run not as a hypothesis to test against the file but as a syllogism that closes itself. In the worshiped form, risk is the one title that never has to show its papers. Expense arrives with invoices, work with hours, loss with testimony — chapter 19 made even damage swear before collecting — but risk, spoken with enough gravity, is expected to be believed. That is exactly backwards from the tradition’s practice and from the craft’s: the actuary prices what frightens him; the worshiper is paid for what frightens you. And the residual knows it. Of all the lines on chapter 35’s ledger, the risk line is where unexplained euros dress first, because it is the line whose audit modern manners discourage.

So the line goes under discipline — subordinated, in the outline’s exact word, to truth. Four oaths, each one already sworn somewhere in this chapter. The danger is real: named scenarios, honest probabilities, honest severities — both factors or no charge — and the price answers the file, never the borrower’s desperation, which is chapter 9’s screen standing exactly where the Salamancans posted it. The danger is measured where measurement exists: the tables consulted, the mitigants subtracted, the exposure read as a curve through time. The danger is borne by the party charging for it: not rehoused onto a household and

billed as if retained; the bearer of the residual danger is the only party with a claim on its price — and the corollary reads strangely until it reads justly: where the household bears the netted danger, the risk wage, if anyone owes one, belongs to the household. A guarantor is an insurer writing cover without premium, and chapter 38 will have things to say about policies of that kind. And the danger is counted once: protections and compensations netted on one page, the way chapter 35's cross-line fence demanded.

One practice enforces all four oaths at once: show the table to the seat that pays for it. A borrower charged five points for danger is owed the scenario arithmetic the five points rest on — Part VI's rule that burdens be shown was never meant to stop at expense receipts, and a risk story that cannot survive being read by its own subject was never a price at all, only a posture with decimals. Nothing disciplines an invocation like an audience that knows the file. Then the two form-rules that follow from the grammar. Where the uncertainty is measurable, the premium is honest — sized to the netted danger, falling as the danger falls, pooled where pooling is the mechanism, as the mutual houses of chapter 27 pooled it. Where it is not measurable, only the share is honest — the claim converts, in whole or in that part, into participation: paid from outcomes, capped like everything in this book, standing in the weather it professes to brave. A design that wants partnership's compensation must carry partnership's exposure; that much has been true since the *societas*, and no amount of drafting has ever made the middle position — creditor certainty at partner prices — anything but the oldest item in the usury file.

Truth first, in other words; premium and share afterward, each in its province; and nothing at all for invocation. Risk is a fact to be priced, not a god to be paid. The subordination costs honest lenders nothing — their twenty-two hundred euros survive every oath in this section — and it costs the worship everything, which is the point, and which is why chapter 37 matters next: when the front door of the rate closes to unearned risk-money, the money goes around the back, into the order of repayment itself — seniorities, preferences, the quiet geometry of who is paid first and who is paid maybe. "Waterfalls, Preferences, and Synthetic Usury" follows the money there. Chapter 38 will then find the bearers this chapter located — the household, the payroll, the unpaid years — and give their contributions the lines the cap table never drew. The file, meanwhile, goes back in the drawer, lighter by one illusion: its danger has a price now, and the price is not twelve percent of anything but four hundred and fifty euros a year, honestly borne — and the fifty-some thousand that remain above it will need, per the first equation, some other name.

Chapter 37 — Waterfalls, Preferences, and Synthetic Usury

The founder signs the sale documents on a Tuesday afternoon, and everyone in the room congratulates her. The company she started nine years ago is being bought for twelve million euros — a real price, a real buyer, a success by most of the measures a life gets measured with. Her share certificate says thirty percent. On the drive home she does the arithmetic she has been putting off, and the arithmetic does not say three million six hundred thousand. It says six hundred thousand. Somewhere between the certificate and the closing statement, five sixths of her thirty percent went missing, and nothing illegal happened to it. No clause was hidden from her. No rate was charged that any regulator would blink at. The money moved the way the documents always said it would move, in the order the documents always said it would move in, and the order did the work that a forbidden interest rate would once have had to do in daylight.

This chapter is about that order. Chapter 35 built the ledger and found the unexplained euros; chapter 36 put the risk line under oath and closed the front door — a danger, it found, may be paid for once, by the party bearing it, in the amount actually borne — and warned that the refused money would head for the back of the building. The back of the building is the waterfall: the clauses that fix the sequence in which a company's value, in success or in ruin, flows out to the people with claims on it. Repayment order, chapter 35 said, does quietly what interest once did loudly, and it is time to say exactly how — with the arithmetic chapter 9 promised, in the traditions' own terms, as chapter 13 required. The founder's missing three million is not a mystery to be solved; it is a mechanism to be read.

The Geometry of the Queue

Begin with what a capital stack is, stripped of its typography. A company's obligations stand in a line. Secured debt stands at the front; unsecured debt behind it; then, in an equity-financed company, the preferred stock, round stacked upon round; then the common shares of founders and employees; and around the line, rather than in it, the losses no filing ranks — the supplier stretched from thirty days to ninety, the payroll that tightens long before any coupon is missed, the town that organized part of its life around the payroll's existence. When value flows in, it fills the line from the front. When loss arrives, it eats the line from the back. That is the entire machine. Everything else — the defined terms, the seventy pages — is the machine's paperwork. And the machine is not a metaphor waiting for a court to disagree: insolvency law enforces the line as law, under a name of engaging frankness — what the American code's lawyers call absolute priority, each rank paid in full before the rank behind it collects a euro. A sale, chapter 23 saw, runs the same waterfall on a happier day. The queue is the one clause that is always, in the end, performed.

Now read the line the way this Part has learned to read everything, by payoff across the states of the borrower's world. A seat at the front of the queue collects the same euros in nearly every future; its payoff is flat; it is, whatever its certificate says, the fixed claim of chapter 35's taxonomy — a *mutuum* with a seat number. A seat at the back collects only out

of realized prosperity and eats every early loss; that is the partnership shape, the *societas*, the seat the traditions licensed to earn precisely because it could lose. Between the two poles, position in the queue is exposure, measured in meters of drafted distance from the first euro of loss. Chapter 13 drew the map this paragraph is standing on: the liquidation waterfall, it said, is a loss map with the losses at the bottom — and it promised that this chapter would run the map’s arithmetic in the traditions’ terms. The terms are ready. The halakha graded seats before it graded rates: the investor written *near to profit and far from loss*, says the baraita chapter 10 verified, is the wicked one — not for profiting, but for the drafted nearness and farness themselves. Rome voided the arrangement outright when it found a partner who could not lose, and chapter 11 told the fable that named it, the lion’s partnership; the Digest’s one honest exception, chapter 12 showed, was earned asymmetry — the partner whose labor is worth the loss he is spared, who pays for his protection in sweat rather than in drafting. And the queue, notice, is where chapter 36’s conservation law comes home to live. Seniority and preference, that chapter found, are subtractions: they remove no danger from the world; they re-seat it. The waterfall is the seating chart. Every meter of safety purchased at the front of the queue is a meter of exposure delivered to the back, and the delivery is signed for by people who mostly never read the map — which is why the geometry stays hidden in plain sight. A rate is a number on a page, comparable across every loan in Europe at a glance. An order is a shape, visible only when you compute what every seat collects in every future, which almost nobody does. The moral information did not disappear from the modern term sheet; it moved from the arithmetic everyone checks into the geometry nobody does. So this chapter’s governing questions, companions to the pair chapter 35 posted at the payoff test, are the queue’s own: who collects first? And out of whose remainder? Every clause that follows is a drafted answer, and the drafting has a direction.

One more piece of vocabulary, because this Part’s own ledger uses the word in the opposite direction and the collision is instructive. Finance calls the common shareholder the residual claimant: she is paid from whatever remains when every prior claim is satisfied, and in the law’s eyes that remainder is her whole entitlement. The ledger of chapter 35 hunts a different remainder — the extraction residual, the euros no honest title will claim. The two residuals sit at opposite ends of the same pipe. When a waterfall is honest, the residual claimant collects the venture’s true remainder, large or small, because she genuinely stood last. When a waterfall has been engineered, the unexplained euros of the front seats are collected out of the remainder the back seats were promised — the extraction residual is funded, euro for euro, by the residual claimants. The founder’s missing three million has just acquired an address.

The Machinery of Preference

Now the clauses, one at a time, the way chapter 23 promised they would be walked — each reasonable alone, each defensible in a term-sheet negotiation, and each a component in the assembled shape the last section of this chapter will price.

The liquidation preference is the foundation stone. In its simplest form it provides that when the company is sold or wound up, the preferred investor takes his invested capital back — once — before the common shares receive anything. Stated so, it has an honest face, and even-handedness requires the face be looked at: an investor who pays five times the founder’s price per share, for a minority of a company whose books he did not keep, is buying protection against arithmetic he cannot see, and a modest preference is a way of saying the founders may not price their own optimism entirely at his expense. The traditions

would recognize the instinct; they spent chapters 19 through 22 licensing genuine protections against genuine burdens. But read the clause by payoff and its second face appears. A preference is a guarantee of capital, written not by a bank but by the other shareholders; it moves first loss from the money onto the founders and employees, exactly as chapter 13 watched the guaranteed *iska* move it, and the traditions' verdict there was unanimous and structural: guarantee the capital home whole and the partnership has dissolved into a loan in the measure of the guarantee — the deposit half becomes debt, and the *mudarabah* with a guaranteed principal is void, its profit share *riba* — the rule the fiqh's standard-setters still print — and the *sukuk* market of chapter 26 rediscovered it when its par-value repurchase undertakings — capital back “regardless of their true or market value at maturity,” in Usmani's fetched phrase — were refused by the tradition's own boards. The preferred certificate says shares; the preference says not yet, not while it matters.

Participation is the second story built on the foundation. A non-participating preference at least makes the investor choose at the exit — take the guarantee back, or convert and share as an owner; protection or participation, the premium's grammar or the share's, chapter 36's two ways of being paid for uncertainty, pick one. The participating preference declines to pick. It takes the invested capital off the top and then shares in the remainder as if fully converted — collecting twice from the same exit, as chapter 23 put it. Run the founder's Tuesday through both drafts and watch the clause earn its keep — the file is staged, like chapter 35's, and kept dice-simple; the clauses are the market's own. Her company carried ten million euros of invested preferred capital, fifty-five percent of the shares, all of it participating. At the twelve-million sale: ten million off the top, then the remaining two million shared across all the shares — fifty-five percent of it, another one million one hundred thousand, to the investors, who leave with eleven million one hundred thousand of the twelve; six hundred thousand to her thirty percent; three hundred thousand to the employees' fifteen. Now redraft with the same preference, non-participating: the investors take their ten million and stop; the two million remainder belongs to the common alone, and her two-thirds of the common is one and a third million. One word — *participating* — moved more than seven hundred thousand euros from the woman who built the company to the money that was already guaranteed, in a sale where that money had already been made whole. And the certificate's thirty percent, notice, was never a share at all under the participating draft; it was a ceiling she approaches and cannot reach. At a hundred-million triumph she would collect twenty-seven percent; at twelve million, five; at ten million or below, nothing. The number on her certificate is true in one kind of future and the drafting decides which kind arrives first in the queue. Multiples compound the same move without changing its grammar — two times, three times the capital off the top before anyone else's euro — and each multiple is the same guarantee, thickened. Preferences stack, too: each financing round customarily takes its seat senior to the one before, so that a company which has raised four times carries, as chapter 23 counted, a tower of prior claims between the employees' certificates and any euro of proceeds — a tower whose height appears on no cap-table percentage anywhere. Chapter 11, watching the modern limited partnership rediscover preferred returns and guaranteed fees, called the family “the lion's clauses, notarized”; this section has merely been reading the notary's file.

The cumulative dividend is the third story, and the oldest. A preferred dividend that accrues whether or not it is declared — eight percent, say, on the ten million, eight hundred thousand a year, unpaid in lean years but never forgiven — adds itself quietly to the preference; five lean years and the queue in front of the founder has grown by four million euros while nothing was paid to anyone. Read by payoff, an accruing dividend on a protected principal is a coupon: a fixed sum, growing with the calendar, indifferent to the venture's

fortunes, payable ahead of the people whose work must generate it. The tradition met this instrument in 1586 and did not hesitate. Sixtus V's bull — chapter 13 quoted its condemnation — aimed at partnership pacts that kept “the very principal, or capital be always safe” or paid a fixed sum “yearly, or monthly” as gain, and declared them “illicit and usurious”; the honest partnership, the same bull said, casts up fruits, costs, and losses together and divides what is actually there. A cumulative dividend on participating preferred stock is the condemned instrument's modern draft, improved in exactly one respect: it no longer calls itself gain on a loan, because it no longer needs to. The queue collects it instead.

The Clauses That Change the Weather

The preference machinery fixes who is paid first. A second family of clauses does something stranger: it changes what the instrument is, or when, or at whose choosing — and every change runs in one direction.

Redemption rights let the investor put his shares back to the company after a term — equity with a maturity date. For years the instrument behaves as stock; then, at the holder's election, it presents itself for repayment like a note that had merely been traveling under another name. The Ottoman drafters of the Majalla, chapter 3 noticed, chose exactly this hybrid as the opening illustration of their code's third article — a sale subject to a right of redemption has the force of a pledge; effect is given “to intention and meaning, not to words and phrases” — and the *fiqh*'s modern boards applied the same rule when they forbade the sukuk manager's par-value repurchase promise: fix the exit price in advance, chapter 26 showed, and the certificate's holder no longer stands in the venture at all; he holds a dated claim on a counterparty. A redemption right is rarely exercised, its defenders say, and the defense misses its own point. A put need not be exercised to govern. It hangs over every board meeting in its redemption window; the company that cannot fund the redemption must refinance it, or sell itself, or bend its plans toward whoever can — the option's weather arrives years before its rain.

Anti-dilution provisions, in their full-ratchet form, are the queue's answer to bad news. An investor bought a fifth of the company at ten euros a share. A hard year later, new money will come in only at five. The ratchet reprices his old shares as if he had paid five — his four hundred thousand shares become eight hundred thousand, and the new shares are minted not from the treasury but out of the founders' and employees' percentage, which is the only place shares can come from once the price is fixed. Term-sheet practice keeps a gentler draft on the same shelf — the weighted-average adjustment, which reprices the old shares only in proportion to how much cheap new stock actually arrives — and the distance between the two drafts is a measurement: it shows how much of the clause is insurance against a real wound and how much is appetite wearing insurance's vocabulary. Walk the full ratchet through the states of the world and the shape is familiar to the point of monotony: in the good future the clause sleeps; in the bad future it transfers ownership from the people who lived the bad year, at desks, on reduced salaries, to the capital that was already senior, already preferred, already cumulating — a claim that collects most precisely when the venture does worst. Chapter 2 met this payoff in its oldest form, and the sober rule it filed there applies here without decoration: instruments that harvest the counterparty's worst year are the ones every tradition in this book built its walls against, whatever the harvest is called on the certificate.

Drag-along rights complete the family by adding a clock. A drag-along lets specified holders — in practice, the preferred majority — compel every shareholder to join a sale the

dragged shareholders did not choose. Like every clause in this chapter it has an honest use, the prevention of the ten-percent holdout who would tax a transaction everyone else needs; and like every clause in this chapter its payoff must be computed, not assumed, because bundled with a preference it becomes something else entirely: the power to choose, unilaterally, the future in which the preference pays best. A fund approaching the end of its life does not need the company's best decade; it needs the company's next transaction. The drag turns that need into everyone's obligation — the mission sold to meet a preference, as chapter 34 documented, the buyer's purposes nobody in the building would have chosen, the founder signing a Tuesday she cannot refuse. Forced liquidity is where the waterfall stops being a diagram about money and becomes a diagram about time: whose horizon governs, whose patience is structural and whose is revocable, who may convert nine years of a company's life into this quarter's realization event. The traditions' word for uncertainty deliberately sold to the party least able to refuse its terms was *gharar*, and chapter 23 read these very clauses as a consent problem — the engineered fog. This chapter's addition is only that the fog has a payoff, and the payoff can be computed, and it lands, every time, on the seats the next chapter is about.

Say plainly, before the synthesis, where the weather goes when these clauses change it, because the addresses are chapter 36's and they have not moved. The ratchet's minted shares come out of the employee pool as surely as the founder's stake. The redemption that must be funded in a tight year is funded from the payroll's growth, the maintenance deferred, the reserve never built. The drag delivers the workers to an acquirer as line items in a data room. None of these people appear in the waterfall's defined terms. All of them are in its queue, unnamed, at the back, where the losses eat first.

Synthetic Usury

In the late 1860s, the New York financier Russell Sage — Wall Street would come to call him the Money King, and his biographers kept the title — ran into the one instrument his career could not digest: a usury statute. New York capped interest by law, and Sage, the standard account runs, had been convicted under the cap. What followed is the part of the story that belongs to this book. Sage did not petition to change the law, and he did not, in any sense the statute could see, go on lending. He bought stock from men who needed cash, sold each man a call — the right to buy his shares back at a fixed higher price — and bought from him a put, the right to force the shares back on him at that same price. Watch the machine run: the customer has his cash today, exactly as if he had borrowed; whatever the stock does, one option or the other will be exercised, so the shares return to him at the fixed price, exactly as if he had repaid; and the fixed price stands above today's by whatever margin Sage chose — twelve percent, to give it chapter 35's number — a rate of interest distributed among a sale, a premium, and a strike, appearing on no page as interest at all. Financial historians credit Sage as the father of the American options business. The options business should note its paternity honestly: puts and calls entered American finance at scale as a machine for manufacturing loans the usury law could not see.

A century later, in 1969, the economist Hans Stoll proved in the *Journal of Finance* what Sage had banked on: the put-call parity theorem, the tidiest result in the theory of instruments. Own the share, own the right to sell it at a fixed price, owe away the gains above that price, and you hold — not approximately but identically, state by state — a riskless claim to the fixed sum: a bond; a loan. Payoffs add. Any fixed claim can be assembled out of pieces none of which is a fixed claim, and the assembly is not a defect of financial

engineering; it is the theorem financial engineering is built on. Every desk on earth now uses it without a second thought — which is precisely why this book must give it a second thought, because the theorem cuts in a direction its users rarely face. If a bundle of equity and options is identical, payoff for payoff, to a loan at interest, then everything the traditions concluded about the loan attaches to the bundle. Equivalence has no favorite direction. The market's own mathematics closed the gap between form and substance from the substance side, and in doing so proved, with more rigor than any canonist ever commanded, what the canonists had asserted all along: that the payoff is the contract, and the names are stationery — chapter 35's specification, arriving this time from the trading floor. The options desks still call Sage's package a conversion, and the parity is now checked by machines, continuously, across every listed market, enforced by the market's own bailiff: any gap between the loan and its synthetic is free money to whoever closes it first. A rule the traditions enforced with confessions and *responsa* is enforced today by arbitrage — the market verifies the equivalence to the eighth decimal and declines, at every decimal, to draw its conclusion.

And here is the turn this chapter has been building toward: the traditions had already found the theorem — not its algebra, its arrests. Walk their case files in order. The fiqh's *‘ina* sale, chapter 3's specimen: two impeccable sales in one afternoon that net, as chapter 3 showed, to money today against more money owed next year — a loan assembled from sales, condemned by the majority of the schools for the net and not the parts; the rabbis, for their part, had fenced the same borderland with a category built for it — *avak ribbit*, the dust of interest, chapter 3's file for arrangements that are not the forbidden loan itself but carry its afterglow. The *mohatra* of the Christian casuists, the same machine in velvet; when its defenders argued that each sale, taken alone, was valid, the Holy Office condemned the defense itself, in 1679, in the proposition chapter 24 put on display. The *contractus trinus*, the perfected instrument: a partnership, plus insurance of the capital, plus the sale of an uncertain future profit for a certain present one — three contracts, each defensible in its own literature, assembling to a guaranteed return on a guaranteed principal. Read it with Stoll's eyes and the *trinus* is a parity trade, term for term: the partnership is the share; the capital insurance is the put; the certain-for-uncertain exchange writes away the upside like a call; and the assembled position is a loan at interest, which is what Sixtus V's bull called it in 1586, four centuries before the notation existed — the fight over that bull, chapter 13 noted, ran another century, Lessius and his successors defending the assembly until the doctrine itself gave ground. And the modern fiqh completed the set in 2009, when the Academy refused organized *tawarruq* — warehouse metal bought and sold within minutes to net a cash advance against a deferred debt — because the prearranged legs amount, in the resolution's words chapter 24 carried, to “a deception... in order to get the additional quick cash.” Four traditions' files, one finding, and none of it required a formula: parts lawful, assembly forbidden, because the assembly's payoff is the forbidden shape. The scholastics and the jurists were doing put-call parity by hand — slowly, contract by contract, ruling by ruling — and mostly losing, because a hand audit cannot keep pace with a drafting industry. What they lacked was never the principle; it was the computation.

Which returns us, deliberately, to the founder's term sheet, because the modern waterfall is the triple contract at scale, and the mapping is not a metaphor but an inventory. The participating preference insures the principal: the put. The cumulative dividend pays the certain sum, yearly or monthly: the coupon the 1586 bull named. The participation collects the upside; the ratchet re-strikes the position when the price falls; the redemption right adds the maturity; the drag-along adds the enforcement. Assemble them and price the package across the states of the company's world, and the preferred position's payoff

lies closer to a secured, coupon-bearing, convertible loan than to anything the *societas* consensus would have recognized as ownership — participation's compensation riding on protection's chassis, the hybrid chapter 36 refused as a contradiction in the grammar of risk. No single clause is the usury; the stack is the usury — synthetic, in finance's own honest adjective: assembled from parts, equivalent by construction, and deniable clause by clause, which is the assembly's entire commercial appeal. The old confessors could only interrogate the parts. The parity theorem interrogates the whole, and it changes sides the moment it is asked to: the same mathematics that builds the synthetic loan detects it, because equivalence, once proved, convicts in both directions. Sage's trick, run backward, is an audit.

Reading the Waterfall

So the audit, plainly stated, because Appendix E will run it on worked examples and the analyzer of chapter 43 will run it wholesale. Take the whole stack — every instrument, every preference, participation, dividend, ratchet, redemption, drag — and compute the package's payoff in each state of the company's world, exactly as chapter 35's payoff test demands of any contract; synthesis, that chapter promised, loses its camouflage here, because a payoff assembled from permitted parts is still a payoff. Then run chapter 35's ledger on what the front seats collect, with chapter 36's netting rule enforced across the queue's whole geometry: a danger removed by preference, participation, ratchet, or drag may not also be priced as if borne. Seniority is protection, and protection is the opposite of the risk title; the seat that engineered its distance from loss holds no claim on loss's wages — the near-and-far grading, run at the speed of a spreadsheet. Whatever return the protected seats collect beyond their shown burdens is the extraction residual, arriving through the order instead of the rate, and the ledger prices what the founder's Tuesday only felt: her missing three million was the residual, and the waterfall was its delivery mechanism. The two-tables practice chapter 36 proposed extends down the queue unchanged. The first table prices the front seats' dangers; the second finds those dangers again, rehoused, in the seats that never negotiated them — and a stack that cannot fill in its second table has already told the auditor which kind of stack it is.

Honesty about the instrument cuts both ways, and the model must carry both edges. Nothing in this chapter condemns order as such. A queue is unavoidable wherever claims outnumber euros; the traditions themselves ordered claims — wages before distributions, the capital whole before profit exists, repair before gain — and a modest, disclosed, honestly priced seniority is among the oldest tools by which cautious money reaches risky ventures at all. The test is the one every clause in this chapter has now taken: protection may be bought, at protection's price, which is surrendered upside and a thinner claim on the venture's success — the front seat that accepts a lender's modest yield is telling the truth twice over. What the test refuses is the assembled instrument that books partnership's compensation from a creditor's bunker, insured against every future and paid as if exposed to all of them. And order, the traditions would add, is a design surface, not merely a hazard: the same geometry that quietly extracts can openly protect. A waterfall can be drafted that fills wages, suppliers, and a resilience reserve before any capital distribution; that caps the capital's total claim; that seats the investor's return behind the enterprise's ability to survive collecting it. Chapter 41 — “Designing a Non-Extractive Fund” — will draft those clauses, Appendix E will publish the worked waterfalls, and chapter 40 will fold this chapter's finding into the non-extractive return formula, where repayment order enters the arithmetic as a first-class term.

What the queue's geometry cannot see, because no cap table records it, is who actually filled the seats at the back. The founder's unpaid seventh year, the engineers' below-market decade, the guarantee signed at somebody's kitchen table — the queue prices these contributions at zero and then charges them the first losses, and the pricing is exactly wrong in both directions at once. Chapter 38 — “Founder Labor, Worker Claims, and Invisible Contribution” — goes to find the missing line items; chapter 39 will then ask what surplus honestly remains once every contribution is on the ledger, and there this Part's constructive arithmetic begins its climb. The file this chapter closes is thinner than the one it opened. The founder's twelve-million Tuesday holds no unexplained forces now, only an order, drafted years earlier, doing what orders do. She was told her share was thirty percent. The waterfall knew better, and now, so does the ledger: a preference is a rate that learned to stand in line.

Chapter 38 — Founder Labor, Worker Claims, and Invisible Contribution

On the day the company was born, a notary read its capital into permanent memory. The deed named the founders, the shares, and the money; the money arrived by wire, time-stamped to the second, and entered the register at one hundred cents on the euro. Every round since has entered the same way — the ten million of chapter 37's file sits in that register now, dated, priced to four decimals, senior in its preferences and exact in its percentages. Ask the register who built this company and it answers without hesitation: money did, on the following dates, in the following amounts.

The register is not lying; it is doing something subtler, which chapter 12 caught the moment this book's partnership law met a modern capitalization table: it is remembering everything that arrived by wire and nothing that arrived any other way. The founder's decade is not in it. The engineers' foregone salaries are not in it. The guarantee a spouse signed at a kitchen table — the policy chapter 36 promised this chapter would have things to say about — is not in it. Chapter 37 followed the repayment queue to its logical end and found the seats at the back priced at zero and charged the first losses; it closed by sending this chapter to find the missing line items. Chapter 35's ledger showed where they belong: a work line the halakha would void a contract over, priced to the dinar by traditions that could not have imagined a term sheet and would have read one in an afternoon. What remains is to do the finding — to walk through what the register forgot, price it the way Part IX prices everything, and then ask what the venture's arithmetic looks like once everyone who filled it appears in it.

The Overcounted Input

Begin with a concession the whole chapter depends on: the register's precision about money is a virtue, and the traditions would be the first to say so. A wire has a time-stamp because coins can be counted, and counting them protects everyone — the investor against the vanishing deposit, the founder against the phantom commitment, the auditor against the beautiful story. When chapter 12 observed that capital contributions "enter the venture's permanent memory the moment they arrive," it was not describing a conspiracy; it was describing what is easy to verify, being verified. Nobody designed the asymmetry. Labor trickles in nightly, passes through no bank, and leaves no instrument behind; money arrives once, loudly, with paperwork. An accounting system built on evidence will always see the wire first.

But seeing first has quietly become counting fullest, and the overcounting runs four ways at once. First, the money is counted gross of its protections. The register carries chapter 37's investors at ten million euros of contribution, and ten million did leave their accounts; yet the claim that came back was preferred, participating, ratcheted, and drag-equipped — a claim that, as the file's own arithmetic showed, could not lose a euro at any sale price above the round itself. Chapter 36's netting rule applies to contributions exactly as it applies to risk charges: what capital contributes is not the amount wired but the amount genuinely exposed to the venture's weather, and a euro that has arranged its own way out in every scenario but catastrophe has contributed something — but not a euro's worth of exposure.

The register books the gross; the venture receives the net; the difference is counted anyway.

Second, the same euro is counted more than once. It appears as contribution when it buys the shares, as protection when the preference attaches to it, as yield-base when the cumulative dividend accrues on it, and as control when the covenants key to it. Chapter 35 posted the fence — a burden may be counted once — and its mirror governs here: a contribution may be counted once. A term sheet that prices the money's arrival, then prices its protection, then prices its patience, then prices its permission, has sold the same ten million to the company four times.

Third, capital's hypothetical alternatives are priced while labor's documented ones are not. Every hurdle rate and discount factor in the file is, at bottom, an opportunity claim — the money could have earned elsewhere, and the venture must clear that imaginary elsewhere before its own results count. Chapter 20 spent a whole file on how carefully the traditions fenced that thought, and chapter 12 noted the irony that has only sharpened since: the founder is the rare claimant who can document her elsewhere to the euro — the salary declined, the offer letter in a drawer — and she is precisely the claimant for whom the modern instruments make no line at all. The strongbox's daydreams are priced into the arithmetic; the person's turned-down job is not. The traditions ran that asymmetry in exactly the opposite direction, paying witnessed months readily and imaginary ones almost never.

And fourth — the deep one — money is counted as the whole of what a venture is made of, when it was never more than half. Chapter 12 put the two-ingredient truth at the center of Part IV: a venture is money and a stretch of human life, and the traditions refused to let the second ingredient be priced at zero. The refusal was not sentimental; it was bookkeeping: a contribution priced at zero does not stop existing — it is simply collected by whoever the arithmetic favors, without appearing in the arithmetic at all. Zero is also a price — the only one no one is ever asked to defend. The rest of this chapter is about the people who are paid it.

The Missing Line Items

Take the file's founder, and give her decade the itemization the register never did. The venture ran ten years from deed to sale. Assume — the file is staged, like chapter 35's and 37's, and kept dice-simple — that the role she actually performed would have paid one hundred thousand euros a year in the market she left; she drew fifty, every year but the seventh, when the bridge round was closing and she drew nothing. The difference is not a sacrifice in the devotional sense, whatever it felt like on the seventh year's nights; it is a number: fifty thousand a year for nine years, a hundred in the seventh — five hundred and fifty thousand euros of wages earned by work actually performed, never paid, and left standing in the company. The folk term, chapter 12 noted, has already confessed the substance: sweat equity, equity bought with sweat. The register just never opened an account for the currency. Booked properly, her unpaid half-salary is the most honest capital contribution in the whole file — earned first, in labor the venture consumed, then lent back, month by month, without security, without interest, without a maturity date, and without a single line in the instrument that governs everyone else's money. Chapter 36 found the household holding the netted danger; this chapter finds it also holding an unbooked loan.

Below the salary line sits a stratum of contribution harder to see and just as real. She staked her professional name — the collateral chapter 11 watched the Geniza merchants trade on,

the asset that cannot be re-minted — and the venture did not just hold that name as security; it spent it daily. Customers signed because of her, hires moved cities because of her, the bank's facility carried her signature before it carried the company's numbers. A name in daily use as working capital is a contribution by any test this book's lawyers would recognize, and it degrades differently from coins: a failed venture returns the investor's reputation intact and improved ("we back bold things"), while the founder's carries the failure in its file. She spent a decade off the career ladder she would otherwise have climbed, which is why her documented elsewhere grows more expensive every year she stays. And in year two she signed the guarantee. The bank's facility was half a million euros; the signature made her household the claim's insurer of last resort. Chapter 36 priced exactly this shape of promise: take the early years' honest failure rate — call it five in a hundred, per year, for the four years the facility ran — against what the bank would actually have taken from the household after the firm's assets went first, say three hundred thousand euros, and the cover the household wrote was worth some fifteen thousand euros a year in premium. A commercial guarantor writing the same cover would have charged about that — three euros per hundred per year, sixty thousand over the facility's life — and no commercial guarantor on earth would have written it for zero. Her household did. The premium was never paid, never accrued, never booked; chapter 36 located the seat, and the corollary it flagged for this chapter can now be cashed: where the household bears the netted danger, the risk wage belongs to the household. Sixty thousand euros of it, on this file, collected by nobody.

Now the arithmetic that makes the itemization matter. At chapter 37's exit — twelve million, participating preferences operating as drafted — she collected six hundred thousand euros, and the room applauded. Set her decade against the schedule instead. Wages drawn, four hundred and fifty thousand; exit, six hundred; total, one million and fifty thousand euros. The salaried decade she declined would have paid one million, with her evenings free and her house unpledged. The whole premium for ten years of founder work, founder hours, a year at zero, a staked name, and a household underwriting a bank facility comes to fifty thousand euros — five thousand a year — and even that evaporates the moment you price the guarantee alone at its sixty-thousand-euro market rate. Run it as chapter 35 runs everything, contributions in, collections out: five hundred and fifty thousand in unpaid wages plus sixty thousand in unpaid premium is six hundred and ten thousand euros contributed; six hundred thousand came back. In the success case, the sale that made the newspapers, the founder's own capital came home ten thousand euros short. The register recorded none of this, which is why nobody in the room could see that the person being congratulated was the only investor in the company who had just realized a loss on principal.

The Quiet Underwriters

The founder at least appears in the register, however junior her stock. Chapter 12 promised this chapter would give an accounting to the stratum below her — the contributors the documents see even less — and chapter 36 named them as the addresses where the file's removed danger actually lives. Keep the telling plain here; these are the seats where the losses land first and the applause never arrives.

Start with the engineers. Ten of them, across the decade, each earning on staged average ten thousand euros a year under their market rate — the discount they accepted against options, against the mission, against the founder's late-night conviction. That is one million euros of salary left in the company: earned, worked for, and converted nightly into the

venture's runway. The options were the promised consideration, and the option is worth pausing over, because it is the register's one sincere attempt to see labor — and a lesson in how the seeing fails. An option is a share of the residue, standing behind every preference in chapter 37's queue, vesting on a schedule the employer sets, expiring when the employment does, and worth the most on paper exactly when the venture needs sacrifice least. Set it beside the instrument the *fiqh* built for the same purpose and the difference is the whole argument: the *mudarib*'s ratio is a first-class claim on real profit, defined at the front of the contract; the option is a junior claim on whatever the front seats leave, defined at the back. The traditions gave the working partner the share's grammar; the modern file gives the working seats the share's vocabulary and the creditor's position to everyone else — chapter 36's hybrid, inverted, with labor holding the honest half. At the twelve-million sale the fifteen percent of common the engineers collectively held paid three hundred thousand euros. Thirty cents on the euro of their documented contribution — in the success case. No covenant protected their million. No ratchet adjusted it when the price fell. No preference stood in front of it. In the law's queue, chapter 37 noted, their earned wages rank as creditor claims, often with priority — but the discount was never booked as wages owed, so there was nothing for the queue to rank. The loss arrived years before any filing could have seen it, in the gap between what their work was worth and what the register was told it cost.

The mission discount deserves its own line, because it is the contribution most cheerfully collected and least often repaid. People accept less to work on something that matters — the fair-trade sourcing, the open code, the care that is not rationed — and the discount is a real transfer from the worker to the purpose. The transfer is tolerable, even honorable, exactly as long as the purpose holds. But the purpose is not theirs to keep. When repayment pressure converts the mission into an asset — chapter 34 documented the sales, and chapter 36 traced the pressure — the discount is retroactively re-priced: what the workers gave to a cause turns out to have been given to a cap table. The mission discount is a loan whose collateral is the mission itself, and nobody holding it has any say in the foreclosure.

Around the payroll stand the contributors with no contract at all. The contractor who carried an invoice ninety days through the tight quarter made a bridge loan and was paid in apologies — an unpriced benefit traveling beside a debt, which chapter 12 would file as the contribution without a charge: chapter 8's crime, chapter 12 said, with the arrow reversed. The spouse whose salary floated the household while the venture consumed the founder's — chapter 12 counted the household among the co-investors, and the guarantee section above shows how right it was. The suppliers who stretched, the users who built their days on the thing and will inherit its failure, the town around the payroll: chapter 36 walked these addresses as bearers of rehoused risk. Walk them again as contributors and the same names come up, because bearing a venture's danger without a premium and supplying its inputs without full payment are the same act in two grammars. Contribution, like danger, is conserved: unbooked is not unmade, merely collected elsewhere — and the elsewhere, as the residual's incidence paragraph in chapter 35 said plainly, is always the seats least able to refuse.

None of this is an argument that every sacrifice is a claim. The sole owner who pays herself nothing is investing in her own property, and owes herself only honesty about it; the volunteer at the *gemach* of chapter 15 is giving, and the gift — chapter 3 kept the forms distinct for exactly this reason — seeks no return and books no debt. The line the traditions drew, and the line this chapter draws, runs elsewhere: where one party's unpriced input stands behind another party's priced claim, the zero is not modesty but a subsidy, flowing

uphill, from the seats without instruments to the seats with them.

Three Terms, Revisited

The traditions saw all of this. That is the burden of chapter 12, and the reason its three terms come back now, by name, exactly as its closing paragraph promised — not as history this time, but as the specification for a line the model must be able to fill.

Sechar tircha, first: the wage for trouble. The Mishnah would not let an investor set a man trading on half profits without paying him for the toil spent on the investor's half; the Talmud sized the wage — “as an idle laborer,” the price of setting aside one's own trade to watch over another's — and the Shulchan Arukh, at siman 177, let the sum shrink to a single dinar while refusing, to the end, to let it shrink to nothing. Chapter 12 read the dinar correctly: it is declaratory. It forces the contract to say, in its own text, that the working partner's labor is a compensable contribution and not a courtesy, and once the contract has said it, the substance can ride honestly in a tilted split. And before anyone objects that a rabbinic partnership rule has nothing to say to a preferred-and-common capital stack, look at the shape the rule was written for. The *iska* the halakhists were policing is half loan, half deposit: the financier's money half fenced as a loan the worker must repay regardless, half ventured as a deposit traded at the financier's risk, with the working partner minding all of it. A stack in which the preference cannot lose at any price above the round, while the common seats absorb whatever comes, has re-derived that geometry, a fenced half and a ventured half under one roof, without re-deriving the rule that made it lawful, which was the wage. The codifiers' gloss even marks the boundary from the other side: where both parties truly venture capital, the wage requirement relaxes, because a genuine two-sided partnership is not a creditor-debtor pair in disguise. The halakha, in other words, keyed the wage rule to the money's protection — where the capital is fenced, its keeper's labor must be priced — which is chapter 36's netting rule, discovered from the labor side, several centuries early. Read against the modern file, the dinar rule is a drafting instruction: the term sheet must carry a line, with a number on it, for every working seat's contribution — the founder's foregone salary, the engineers' discount — even where the parties then agree to venture that value rather than draw it. Priced and ventured is partnership. Unpriced and absorbed is the arrangement the rabbis called by its name. The halakha's wage rules came with an enforcement theology the register should envy: wages are a debt, not a distribution — “You shall not keep for yourself the wages of a laborer until morning,” chapter 22 quoted from Leviticus, and Deuteronomy's sunset deadline arrives with the reason attached, “because they are poor and their livelihood depends on them.” A tradition that would not let a day-laborer's wage sleep overnight in the employer's house was never going to smile on a decade of it sleeping in a cap table.

The *mudarib*'s portion, second. The *fiqh* built the working partner's compensation into the instrument itself — the ratio is the wage, negotiable, first-class, immune to being fixed away — and then, in the doctrine chapter 12 called the part modern summaries most often miss, made the claim survive the contract's own death. Capital promised and never delivered, a venture void for defective terms: the remedy, echoed in the modern standards, is *ujrat al-mithl*, the wage of his like — the venture that consumed a man's months owes him a fair wage though every other clause should fail. Now look at the file. The founder's five hundred and fifty thousand is precisely the wreck case the jurists drafted for: labor actually performed inside an arrangement whose paper never priced it. The *fiqh*'s answer is not that she should have negotiated better. The *fiqh*'s answer is that the claim exists anyway, because the work happened — a wage floor beneath the waterfall, senior to

the argument. And the tradition's severity ran in both directions, which is what keeps it honest as a specification: the same jurists who guaranteed the worker's floor forbade him a fixed draw on top of his ratio, because a wage that collects regardless of outcome is a creditor's claim in a partner's seat. Priced, yes; guaranteed against the venture's truth, no. And around the contract doctrine stands the tradition's wage theology, which does not whisper. The Prophet's instruction, carried in chapter 22 with its locus, sets the clock: give the worker his wages before his sweat dries. Sahih al-Bukhari (2270) preserves the stakes as the tradition heard them, in God's own first-person speech: three people will find Me their opponent on the Day of Resurrection — and the third, after the covenant-breaker and the man who sells a free person and eats the price, is the employer who takes a laborer's full work and does not pay his wage. The company the third keeps is the commentary. Withholding earned work, the tradition is saying, belongs to the family of selling people.

Stipendium laboris, third. Aquinas rehabilitated the merchant by reading his margin as a paycheck — gain sought “not as an end, but as the wage of his labor,” chapter 12's verified line — and the Gospel had already put the principle in the instruction Jesus gave his own traveling workers: “the laborer deserves to be paid” (Luke 10:7, NRSV). The Monti of chapter 17 ran on the doctrine: clerks paid, buildings kept, cost and labor recovered, profit never taken. What the scholastic term adds to the file is the fence on both sides. One side faces capital: a structure whose returns cannot all be traced to ventured property or expended work is collecting for a phantom — and a cap table that prices the founder's decade at zero while paying its own imagined alternatives has failed the trace in the labor column, whatever its yield. The other side faces labor's own institutional forms, and Part IX needs the discipline as much as any fund it audits: the management fee that swells with assets whether or not anyone worked, the monitoring charge that monitors nothing — the wage title covers work performed, shown, and proportionate, not a percentage that collects like capital while calling itself labor. The stipendium is a title, and titles, this book has now said in every register it owns, are tested by substance — the wage line included. The same honesty governs the schedule's other direction, because a construction built to recover invisible work can be abused to inflate visible legend: the founder's line enters at the market rate for the role she actually performed, not at the valuation her myth would prefer; the schedule prices work, not aura, and a founder who wants more than the role's wage must earn it where everyone else does, in the ventured half. Three traditions, one specification, and it reads almost tautologically once said aloud: work is a contribution; contributions are priced; a price of zero is not a price but a confession.

The Contribution Schedule

So the model gains its next construction, and like the ledger it extends, it is doctrine arranged in rows. Call it what chapter 12 called Ulpian's three-word list — *pecuniae, operae, cuiuscumque alterius rei*, money, labor, or any other thing — a contribution schedule: every input the venture actually consumed, entered by name, dated, and priced at entry. Wages foregone, at the market rate declined. Guarantees written, at the premium a stranger would charge. Reputations staked and careers parked, estimated honestly where estimation is all there is — the schedule prefers a rough number to an exact zero, because the rough number can be argued with, and the zero, as the whole chapter has shown, simply pays whoever the paperwork favors. Chapter 12 compressed the principle into the sentence this section exists to operationalize: “the schedule of contributions is longer than the wire transfers.” The register kept only the wires. The schedule keeps the venture.

Like chapter 36's second table, the schedule is deliberately boring to produce, because its

inputs are already lying around: the payroll ledger beside the market surveys, the guarantee's face amount in the loan file, the aged invoices, the offer letter the founder declined. Nothing about it requires new law — only a few lines of drafting the traditions would recognize on sight. A shareholders' agreement can carry the dinar's modern descendant: a stated schedule, revisited annually, naming each working seat's contribution at its market price and recording whether it was drawn, deferred, or ventured. A fund that publishes the contribution schedules of its portfolio has told the truth about who builds its returns; a fund that cannot fill the schedule in has learned the same thing chapter 36 said of the second table. And the schedule forces nothing. It does not make employees shareholders by decree, or forbid a founder from working cheap, or outlaw the option; it makes the price of every input visible, so that whatever the parties then agree to operates on truth instead of on the strongest party's silence. The investor with nothing to hide loses nothing here — the share-grammar investor of chapter 36, who funds honest ventures and collects only from real prosperity, gains a document proving exactly that.

Run it on the file, and run the zero line first — capital home before anyone's profit, the ordering every tradition agreed on — but run it for every seat, because sweat capital is capital. The investors contributed ten million; they collected eleven million one hundred thousand; their capital is home, one million one hundred thousand to the good. The founder contributed six hundred and ten thousand in unpaid wages and unwritten premium; she collected six hundred; her capital is short ten thousand. The engineers contributed a million; they collected three hundred thousand; their capital is short seven hundred thousand. Add the shortfalls: seven hundred and ten thousand euros of working-seat principal simply never came home — at a sale everyone toasted. And now subtract honestly, the way chapter 35 taught. Total contributions, wired and sweat together: eleven million six hundred and ten thousand. Sale: twelve million. The venture's entire true surplus — the only money in the story that is gain rather than somebody's input returning — is three hundred and ninety thousand euros, a little over three percent, total, for the decade. The protected seat's excess of one million one hundred thousand decomposes exactly: three hundred and ninety thousand of genuine surplus, and seven hundred and ten thousand that is not surplus at all but transfer — the working seats' unreturned principal, changing hands, renamed on arrival. What the exit calls profit, the schedule may call repayment. The register's version of this story had one number and a hero; the schedule's version has three numbers and a bill.

What would honesty have looked like instead? Not a formula this chapter is entitled to impose — the ratios were always the parties' business, the *mudarabah's* split negotiable, Ulpian's shares set by honest judgment of what each brought, and the *machloket* and *ikhtilaf* over the edges stay as live in the model as they were in the responsa. The order, though, was never negotiable, and the schedule makes it computable: all capital home first, the wired and the sweat alike; then the true surplus — all three hundred and ninety thousand of it — shared by stated fractions over stated contributions. On a plain pro-rata reckoning the money's share would run to something over three hundred thousand, its total finishing a little above ten million three hundred thousand — and the working seats would keep, for the first time, everything else: the founder's principal home with a modest share atop it, the engineers' million home with theirs. Smaller numbers for the front seat, and truer ones; the difference between the two term sheets of chapter 35 was never size but composition, and the difference between these two exits is the same. Nor does the schedule only redistribute; it re-describes. It tells the investors what their one point one million was made of before they spend the story of having earned it. It tells the founder, before the next deed is signed, what the last decade actually paid. And it hands chapter 39 its opening question

in computable form, because once every contribution is on the ledger and every principal has its seat in the queue, the number that remains — the money that is not somebody's input on its way back — is the only thing in the file that was ever honestly available to share. Chapter 39, "True Surplus," takes that number apart; chapter 40 will seat contribution beside composition, risk, and order as a first-class term of the non-extractive return formula. The register, meanwhile, can keep its virtues: let it remember the wires to the cent. The schedule remembers the company — and somewhere in its rows, ten years late and sixty thousand euros light, the register finally learns the name of its most senior creditor. She is the same person chapter 36 found seated as the file's most junior investor, and both descriptions are true, which is the finding: most junior by the paper, most senior by the doctrine — the seat every tradition in this book moved to the front of the line, holding wages the arithmetic was never told about.

Chapter 39 — True Surplus

The meeting, for once, is a happy one. The company — say it roasts coffee on a canal in the east of the city and trains its crew out of long-term unemployment, because this file, like chapter 35's and chapter 38's, is staged and kept dice-simple — has finished the best year in its short life. The accountant shares her screen, and at the bottom of a tidy page, above the words *available for distribution*, stands the number: one hundred and twenty thousand euros. The word does its usual work around the table. Available sounds like a fact about money, the way a full shelf is a fact about bread; it suggests that the only remaining question is how to divide the loaf. This chapter is about that word, and its claim is blunt: available is almost never true, and the distance between what a till holds and what a venture may honestly share is where the oldest sin in this book now does most of its quiet living.

Part IX has been climbing toward this table since it opened. Chapter 35 built the ledger on which every euro of a capital claim must name its title, and assigned this chapter its constructive twin: defining the true surplus that may honestly be shared. Chapter 36 put the risk line under oath; chapter 37 followed the money down the waterfall; chapter 38 put the contribution schedule beside the cap table and reached this chapter's question from the exit side, decomposing a celebrated twelve-million-euro sale into seven hundred and ten thousand euros of transfer booked as profit and three hundred and ninety thousand that was really new — then handed over, in its words, “the money that is not somebody's input on its way back” as the only thing ever honestly available. And the question itself is older than the Part. Chapter 14, closing the partnership arc, watched every tradition refuse the profit written in advance and promised that this chapter would take its question for a title: what, truly counted, is surplus? At a settlement table in a Fustat courtyard the answer was counted out in coins and witnessed oaths. The modern version is asked in January, of a bank balance, by people holding term sheets — and the answer still runs through chapter 14's procession — capital, costs, wages, provisions, reserves — only now the procession stops at the till, and every marcher presents a bill.

Available Is Not Surplus

Begin, as chapter 38 began, with a concession: the number on the screen is honest about what it measures. One hundred and twenty thousand euros of cash, cleared and spendable, sits in the account; no one is cooking anything. The register's cash column is the most truthful column it owns, for the reason chapter 12 named: what is easy to verify gets verified. The trouble starts when a custody fact is promoted into a verdict, because surplus was never a fact about the account. Surplus is a conclusion about the year, and chapter 14 already wrote its grammar: profit sits at the end of an audited story, and every attempt to move it forward in time — toward the signing table, toward the January meeting — is the same attempt the traditions spent seven centuries refusing.

Modern accounting, to its credit, half knows this. Its accrual half exists precisely because cash misleads: revenue is not earnings, receipts are not income, and the profession's whole apparatus of provisions, accruals, and depreciation is a machine for correcting the till's first impressions. The *fiqh's* standards bodies, chapter 14 showed, built the same corrections into partnership law with a jurist's precision — interim profits are retained for the protec-

tion of the capital, the working partner's entitlement "is not absolute" until valuation, early distributions ride "on account," provisional, callable back when the real story arrives, and receivables enter the count at honest collectibility with nothing added for the waiting. Every one of those rules is a way of saying that the pile in the till is not yet anyone's increase. Some of it is the venture's own capital, resting there on its way through. Some of it belongs to bills the year has already incurred and not yet seen — the maintenance the machines have earned, the wages the schedule shows below standard, the reserve still hollow from last year's flood. Some of it is next year's story arriving early, or last year's losses not yet admitted. And some of it may not be the venture's story at all: cash borrowed into the account in December looks, on the screen, exactly like cash earned there, and the fiqh's standards closed that door by name, chapter 14 found — the manager smoothing a lean year may draw on a disclosed reserve, never on fresh debt, because a distribution financed by borrowing declares, in that chapter's words, "a surplus no operation ever produced." Cash answers where the money is; surplus answers what happened, and the two questions part company exactly as often as ventures are complicated, which is always. Available says the money can leave; surplus says it may.

The halakha carries the distinction in the most domestic formula it owns, and the formula is worth importing whole. The Mishnah's tractate on the corners of the field opens with a list of deeds "whose fruits a person eats in this world while the principal remains for him in the world to come" — honoring parents, deeds of kindness, making peace — *ha-keren kayemet*, the principal standing (Mishnah Peah 1:1). The Gemara expanded the list, and the tradition set its version into the daily morning prayers, where it is recited to this day: a community that begins every morning, before breakfast, by distinguishing the fruit, which may be eaten, from the keren, which may not. Chapter 35 met the keren as the zero line of the moral-title ledger — the principal that comes home before anything is gain. The till's version is the same rule run forward: what a venture may consume, distribute, and celebrate is fruit; the keren is not food. Even the modern register remembers a version of the rule when it is being watched — its capital-maintenance doctrines exist to stop distributions from consuming the capital a company's creditors rely on — and then forgets it in the metrics it actually steers by, several of which are engineered to look away from exactly the lines this chapter is about to walk. The most popular of them announces the maneuver in its own name: earnings before, among other things, the cost of keeping the body whole.

So the number on the screen is not a lie; it is a question wearing a statement's clothes. The rest of this chapter is the interrogation — the procession of claims that stand senior to any distribution, taken one seat at a time, each with its tradition and its arithmetic. At the end, whatever no claimant can name will still be standing in the till, and that remainder, at last, will deserve the word.

Wages Before Distributions

The first claimants are on the payroll. The roastery employs ten people, and the contribution schedule chapter 38 taught us to draft shows what the register never volunteered: measured against the stated rates for the work actually done, the crew runs thirty thousand euros a year under standard. Chapter 38 named what such a gap is — wages earned, unpaid, and left standing in the company, the most honest capital in the file, lent nightly on no security. What the gap adds up to at an exit, that chapter computed to the cent. What it does at the till is simpler and sharper: it stands first in line, because in every tradition this book reads, a wage is not a distribution. It is a debt, and a debt with a clock on it.

The clocks are scripture. Leviticus, in the clause chapter 22 carried, forbids an employer to keep a laborer's wages "until morning"; Deuteronomy moves the deadline to sunset, "because they are poor and their livelihood depends on them." The Prophet's instruction, carried in the same chapter, sets an earlier clock still: the worker's wages before his sweat dries. Sahih al-Bukhari's warning about the employer who takes full work and withholds the wage, met in chapter 38, files that employer among God's own opponents on the Day of Resurrection. The Epistle of James, addressed to the comfortable in the voice the prophets used, gives withheld wages a physics all their own: "The wages of the laborers who mowed your fields, which you kept back by fraud, cry out, and the cries of the harvesters have reached the ears of the Lord of hosts" (James 5:4, NRSV). The verse's economy is exact — the wage kept back does not rest in the account where the register shows it; it cries out from there, and something hears it. And Leo XIII, whom chapter 33 heard argue the wage bargain like a canonist, set beneath every labor contract "a dictate of natural justice more imperious and ancient than any bargain between man and man" (Rerum Novarum 45) — a floor older than the signatures above it. A tradition that talks this way about a day's mowing was never going to bless a distribution declared across an unpaid payroll.

Run the doctrine into the January meeting and it cuts one way only. A venture that declares one hundred and twenty thousand euros available while its schedule shows the crew thirty thousand under standard is not distributing surplus; it is distributing the wage line — chapter 14's inversion, now with the schedule to prove it. The money moves from the seats that earned it first to the seats that rank ahead of them on paper, and arrives under the one label the surplus line was built to protect. Precision matters here, and chapter 38 supplied it: not every gap is theft. A crew may knowingly price its discount and venture it — priced and ventured is partnership, as that chapter put it — and a cooperative's members may vote to leave their own wages standing in the walls. But the choice must be priced, recorded, and theirs; and even then the gap is their capital, holding a seat in the queue, not a courtesy that evaporates at the first distribution. Unpriced, unrecorded, and outranked, it is — chapter 38 again — the arrangement the traditions already had a name for.

There is one more wage claim, and it is the one this Part has learned to check last because the person holding it checks it never. The founder's line on the roastery's schedule shows ten thousand euros of deferred draw this year — modest, deliberate, and hers. Chapter 38 found her twin at the exit: the founder the paper ranked last and the doctrine ranked first, whose own capital, by that chapter's arithmetic, came home ten thousand euros short at the celebrated exit. The traditions priced her seat long before term sheets did — the *mudarib's* keep billed to the venture itself, the wage for trouble over which, chapter 12 showed, an iska could be voided, the Monti's clerks paid before the lending was called clean: chapter 12's file, chapter 17's institutions, one rule. Founder sustainability is that rule stated as a going concern rather than a settlement: the venture must carry the cost of the person carrying it. A company whose plan only balances if its founder is consumed is not lean; it is running on an unbooked subsidy with an expiry date, and the year at zero — chapter 38 priced one — is not a business model but a countdown. Sustainability even has a plain test the traditions would recognize: price the role so that a stranger could hold it. If the numbers work only while the seat is filled by someone who will not charge for it, the venture does not have a wage line; it has a benefactor, and the schedule is carrying a fiction. The discipline cuts both directions, as it did in chapter 38: the line pays the role's wage, not the legend's, and a founder who wants more than the work's price must earn it where everyone else does, in the ventured half. But a till that distributes past her deferred line has done to its own center what it did to the crew, one seat quieter.

The Three Reserves

The next claimants do not attend meetings. The first is the venture's own body. Machines wear at a rate a maintenance log can state to the month: the roastery's two drum roasters and its delivery van shed about twenty thousand euros of working life this year, invoiced by no one, and the roof over the training floor keeps its own slow ledger. The register, in fairness, sees this one — depreciation is accrual accounting's honest admission that the body is being spent — and the till ignores it anyway, because depreciation is not a payment and January votes on payments. Distribute the cash the accrual already spoke for and nothing happens, for a while. The machine keeps the books now; its ledger is patient, compounds quietly, and presents its bill all at once, in the language of breakdowns, at the moment chosen by the machine. Maintenance deferred is not expense avoided but a loan taken out against the equipment, at terms nobody read. So the till's discipline runs one step past the accountant's: not merely to admit the accrual but to fund it — cash moved on the log's schedule to a line that exists for nothing else, so that the roasters' next overhaul is a withdrawal rather than a crisis. An entry confesses the wear; only a funded line pays for it. The scriptures' oldest treasury story is on the reserve's side here: Joseph reading seven fat years as storage instructions for seven lean ones (Genesis 41), the harvest of the good years standing, unconsumed, precisely because good years are when the storing is possible.

The second reserve guards against the years no log can schedule. Chapter 28 built the arithmetic in a model village and it does not need rebuilding: paths that pool their blows survive; paths that face them alone are absorbed; and the difference between a young pool that fails and one that holds is the solidarity reserve — chapter 10 named it, this book's anti-ruin instrument — standing in the chest before the doubled year arrives, sized in the rule chapter 28 carved: "to the pile-up, not the average year." The roastery's charter states the size plainly: fifty thousand euros, one genuinely bad quarter. Last year's flooded cellar drew the chest down by half, and the apparatus did exactly what it exists to do — the flood cost the members a chest instead of a company. This year's till question is therefore already answered by the charter's own grammar: the chest's twenty-five thousand euros of replenishment stand senior to any distribution, because a reserve refilled only in years nobody needs money is a reserve that will never be refilled. The traditions' living instruments run the same seniority in modern paragraph numbering. The *iska's* codifiers blessed the disclosed reserve, chapter 13 found, in the same breath that they forbade the guarantee; the *fiqh's* standards retain interim profits for the protection of the capital before any entitlement hardens; and its banks operate standing reserves — chapter 32 read the prudential guidance — that hold back strong years to cushion weak ones as ordinary machinery. Chapter 28's warning travels with the mathematics: a chest harvested in the calm years has been converted, quietly, into somebody's yield, and the conversion sells the one thing the chest existed to hold. Ruin probability is the honest name for what rises when the replenishment line loses its seniority, and it rises for every seat at once — including, the investors around the table might notice, theirs.

The third reserve holds something no invoice can name. An institution's purpose, chapter 28 observed, walks one path through time exactly as a household does, and it can be ruined the same way — not by one great betrayal but by a run of bad quarters, each survived by shaving the thing itself: the training places cut back, the free seats withdrawn, the patient sourcing traded for whatever keeps protected capital whole. Chapter 34 documented the endgame, missions sold to meet a preference; chapter 14 named the bookkeeping that enables it, purpose counted at zero so that its liquidation books as gain. A mission reserve is

the same chest built for the purpose's worst year — capital stationed between the mission and the quarter that would otherwise buy it — and the roastery's charter funds it on a written schedule, ten thousand euros this year, toward a declared ceiling. Grown permanent, locked past the reach of any board's bad quarter, it becomes the endowment, and chapter 29 has already told that story whole: Umar tying up the orchard itself — “keep the corpus intact,” as the instruction is carried there — so that only the produce could ever travel. The fruit-and-keren rule, institutionalized against everyone, including its own trustees.

Three reserves, and one discipline that keeps them honest, because this section hands management no blank check. A reserve is a claimant only while it can answer the ledger's questions: a stated size, a stated purpose, a stated schedule, drawn by rules everyone has read and visible to every seat at the table. The roastery's fifty thousand is one bad quarter, written in the charter, auditable by a stranger — not whatever the treasurer fears this year. Money parked past its declared size is not prudence; it is a distribution being denied rather than shared, extraction's mirror image, run by the inside against everyone else's lines — and institutions slide there easily, because a swelling cushion feels like virtue to the people whose comfort it cushions. The instrument for the distinction has existed since chapter 13: disclosure. A disclosed reserve at a stated size is the pool's honest machinery. An undisclosed cushion of no particular size is a number with no title — and this Part has a column for those.

The Claims Outside the Walls

The till's next claimants never signed anything, which is precisely why the traditions wrote their claims into law. Begin with the ones the enterprise can see from its own windows. Chapter 18 established the grammar with some care: in all three houses, the community's share is owed, not offered — *zakat* a levy with chartered recipients, *tzedakah* a debt collectible by the court, the patristic lien on surplus a matter of justice before charity ever enters. The roastery's charter takes the tradition at its word and writes the obligation as a line: ten thousand euros, named, scheduled, and computed before the distribution question is reached, because a debt by the tradition's own grammar cannot rank behind the creditors' celebration. The fence from chapter 18 travels with the line: the transfer must be real and its recipients stated — the *zakat*'s eight chartered classes were that chapter's exhibit in how a tradition writes a distribution list — not a sponsorship budget filed under the tithe's name; and chapter 28's two-questions test reaches charters too, since a community line an enterprise quietly runs as marketing has already answered whose flourishing the pooling serves. Around the stated line stand the quieter neighbors chapter 36 walked as addressees and chapter 38 walked as contributors — the town that carries the payroll's weather, the suppliers who stretched, the users who built their days on the thing. A till that has never once counted them is not wrong about a number; it is wrong about who was in the venture.

Then there are the claimants no window shows. The economists' term for them is external — costs a venture causes and does not carry, landed on seats outside the contract — and the word's own geometry confesses the problem, because the seats are external only to the paperwork. The costs are as real as diesel and as datable as invoices: the wastewater the town treats, the carbon the century will, the neighborhood's air run through the roaster's stack. The conservation law chapter 10 first stated applies without amendment — chapter 36 traced where removed danger goes, chapter 38 where unpriced contribution goes, and unpaid costs travel identically: nothing vanishes; it is collected elsewhere, from the seats chapter 35's residual already taught us to expect, the ones least able to refuse. The

traditions' modern documents say it with unusual flatness. "Businesses profit by calculating and paying only a fraction of the costs involved," Francis observes in *Laudato Si'*, in a paragraph noting that nobody prices what the cleared forest actually cost (*Laudato Si'* 195). The Qur'an's diagnosis is older and wider: "Corruption has flourished on land and sea as a result of people's actions and He will make them taste the consequences of some of their own actions so that they may turn back" (30:41, in Abdel Haleem's rendering). And the covenant's labor law reached the land itself long before either document — the *shmita* of chapter 4, the seventh year in which the field's own rest is written into the same code that will not hold a laborer's wage past sunset. A tradition that would not price the mower at zero did not price the meadow at zero either. Stewardship in the *fiqh* runs on the same appointment — chapter 18 carried the Prophet's word that the world is sweet and green and its people are installed in it as God's vicegerents, to see how they act — and the church's universal destination of goods, chapter 18 again, holds every surplus answerable to a wider table than the one that votes on it.

What can a till do with a claim whose collector never comes? Exactly what chapter 38 taught it to do with the founder's unpriced decade: prefer the rough number to the exact zero. The roastery's charter states an internal price on its own footprint — ten thousand euros this year, retired against the wastewater contract and the offsets the members chose after an afternoon's argument — and the number is imperfect, disputable, and infinitely more honest than its absence, because zero is not a measurement but a shrug, and it is the one estimate guaranteed to be wrong. The rough line does what pricing always does in this book: it makes the cost visible while it is still negotiable, it disciplines the operation that must now budget against it, and it converts a moral embarrassment into a negotiation the parties can actually have. A "surplus" computed by leaving the town and the river off the schedule is chapter 14's inversion at the widest radius the book has yet drawn — extraction from claimants who cannot even attend the meeting, booked as prosperity by the people who can.

What May Be Shared

Now the procession can be counted, and the counting is short, because everything hard happened on the way here. Capital's own line stands where the traditions put it and nothing in this chapter has moved it: the zero line, every principal, wired and sweat alike, home or provided for before anything on earth is surplus — settled doctrine since chapter 13 seated the losses and chapter 14 the remainder, with chapter 38 widening only the roll of who counts as having principal. But between the zero line and the distribution stands everything this chapter has walked: wages to standard, the founder's named line, the body's maintenance, the chest refilled to its charter size, the mission's reserve on its written schedule, the community's owed share, the rough-numbered line for the seats outside the walls. Call the sum what the traditions would have called it: the life-preserving claims — the venture's food and tools. The *halakha's* collection law, chapter 22 showed, hands exactly those back to a debtor before even an unquestioned creditor collects: the court may take everything and must return the thirty days of food, the bed, the craftsman's two awls and two planes, because — as that chapter put it — "Collection may empty the estate, but it may not unmake the person." The Qur'an commands the creditor holding a genuine claim on a debtor in difficulty to wait for ease, and better, to remit. Neither tradition doubted the debt; both subordinated its collection to the debtor's remaining alive. The venture's till deserves the same two-queue clarity. On the ledger of claims, capital is senior, and the doctrine keeps it so. In the queue of actual cash leaving a living enterprise, the claims that

keep the enterprise alive collect first — and far from robbing capital, the ordering is what keeps capital's own zero line worth holding, since a distribution taken out of the payroll, the roof, and the chest is a distribution collateralized by the venture's next collapse. The honest investor of chapter 36 loses nothing at this table and knows it; a seat that objects to funding the reserve that protects its own principal has told the room which of chapter 28's two pools it thought it was running. The design consequence belongs to Part X, but it fits in a clause: capital that wants distributions has an interest in wanting the senior lines funded, which is why the fund chapter 41 will draw caps its return and recycles its surplus — an instrument shaped so that its appetite and the venture's survival point the same way.

So run the roastery's January, line by stated line. Wages to standard, thirty thousand. The founder's deferred draw, ten. Maintenance as logged, twenty. The chest's replenishment, twenty-five. The mission reserve's schedule, ten. The community's owed line, ten. The priced footprint, ten. One hundred and fifteen thousand euros of the one hundred and twenty on the screen were never anyone's to distribute — ninety-six cents of every available euro already had a name on it, which the label on the page simply declined to read. What remains is five thousand euros, and the remainder is the chapter's whole finding: small, true, and clean. It answers every question the ledger knows how to ask. It is increase a living venture can part with and remain whole, sharable by the stated fractions over the stated contributions, chapter 38's schedule doing at the till what it did at the exit. The board that distributes it has extracted nothing from anyone — which no board distributing the larger number could say, because the till's version of chapter 38's decomposition is that the other one hundred and fifteen thousand would reach the shareholders as a transfer, collected from the crew, the machines, the chest, the purpose, the town, and the river, and handed over under profit's name. What the account calls available, the procession has already spoken for.

And with the remainder named, the Part's affirmative theology can finally be said out loud, because this book has never been against gain. Aristotle's old shudder at coin breeding coin, chapter 1's inheritance, was answered across three traditions not by forbidding increase but by insisting it be real — chapter 14 compressed the answer into three words, surplus is fruitfulness, and chapter 35 gave the words a ledger's grammar: "Surplus is the only line that grows with the venture's success, because it is the only line that shrinks with the venture's failure." What this chapter adds is the test that makes the word operational, and the book has been holding the test since its lease chapter: the usufruct of chapter 25 lets its holder take the fruit while the substance stays whole — *salva rerum substantia* — and a distribution is nothing but the owners' usufruct on their own venture, bound by the same ablative clause. Fruit is what the tree can spare; everything else is the tree. The rabbis' morning formula knew it — eat the fruits, the keren stands. Umar's orchard knew it — the produce travels, the corpus never. Hosea's farmer, whom chapter 15 heard through the Talmud, knew which harvests a person actually eats. Nor is the honest till's remainder doomed to stay small: the senior claims weigh heaviest exactly where ventures are young — wages catching up to standard, reserves filling toward their charter sizes, maintenance steadying — so true surplus widens as a venture matures, which is the shape fruitfulness has always had; orchards bear little in their planting years. Extraction runs the arc backward, largest at the start and brief, and the roastery that toasts the full one hundred and twenty thousand spends its next years paying the distribution back — with interest, in breakdowns, departures, and an empty chest when the bad quarter comes — the false distribution being what chapter 14 said such numbers always are: a claim paid out of the venture's substance rather than its increase.

One line of Part IX's machine remains to be built. Chapter 35 gave return a ledger; chapter 36 priced its danger honestly; chapter 37 exposed the order of its collection; chapter 38 completed the roll of its contributors; and this chapter has defined the only pool it may honestly be paid from. Chapter 40 assembles the pieces into the non-extractive return formula — return of capital, real costs, real losses, real work, and a bounded share of exactly this chapter's remainder, with the reserves standing and the residual at zero. The formula's last input is the till's last number: whatever survives the procession. In the meeting on the canal, the accountant changes one word on the page, and the board votes the five thousand with something unfamiliar in the room — the knowledge, line by stated line, of what the money it is not distributing is doing instead. It is the smallest distribution in the company's history, and the first one nobody else paid for.

Chapter 40 — The Non-Extractive Return Formula

Go back to the desk in Amsterdam. Two term sheets, one young company, the same half million euros, the same cap of two euros and ninety cents for every euro advanced — chapter 35 opened this Part with them and found that no instrument in the modern reporting stack could tell them apart. Five chapters later, everything can. The moral-title ledger of chapter 35 asks each of their euros for its title. The oath of chapter 36 makes their risk charges show the danger, netted of every protection, borne where it is billed. The geometry of chapter 37 reads their repayment order as the rate it quietly is. The schedule of chapter 38 counts their founders' and engineers' unpaid years as the capital those years always were. The procession of chapter 39 stands between their claims and the till and lets through only what a living venture can actually spare. Each chapter built its instrument, used it, and set it down. What none of them did — what this chapter exists to do — is pick all five up at once and turn them around, from reading contracts to writing one.

That reversal is the whole announcement. Part IX has so far been an audit; from here to the end of the book it is a workshop. The pieces assemble into a single design rule — the non-extractive return formula — and the rule can be displayed the way chapter 35 displayed the first equation, in words, because the words are the mathematics. This is that equation's constructive twin, the same line solved for the only unknown the traditions ever cared about:

Permitted return = capital returned, wired and sweat alike + costs shown + losses proven + work and service priced + liquidity deprivation that passes its tests + opportunity bounded to a particular elsewhere + a capped share of true surplus — with the reserves standing senior, and the residual held at zero.

Every term arrives from a chapter that already built its fences, which is why this chapter can be short on derivation and honest about what it is doing: assembly. The formula adds no new doctrine and settles no live dispute; where the houses disagree about a line's size, the disagreement rides inside the line, exactly as it rode inside the responsa and the manuals. What the formula changes is the direction of travel. A test that could only convict contracts after signing can now draft them before it, and the drafting, walked line by line, turns out to be the most compact summary the traditions of this book have ever been given.

Principal and Repair

The first term is the zero line, and it needs the least argument, because chapter 35 built it from all three legal languages at once — the Talmud's horn, the scholastics' *sors*, the fiqh's head of the wealth, one homely agreement that what you advanced comes back to you, the same and whole, and that its coming back is not gain. The Qur'an's settlement of the *riba* war, carried verbatim in chapters 6 and 19, ends on exactly that line. The formula takes the zero line with both of the disciplines this Part attached to it. From chapters 13 and 14, the ordering: the capital is made whole before anyone's profit exists to share, the first rule both partnership traditions wrote into their settlement tables. From chapter 38, the widening: every principal comes home before anything on earth is surplus — the wired principal the register remembers to the cent, and the sweat principal it never opened an account for, the founder's unpaid decade and the engineers' discount standing in the schedule at

the market rates declined. A formula that returned only the money's capital would repeat the modern instrument's first error in the traditions' own vocabulary. Return of capital, in this book's arithmetic, is a roll call, not a wire confirmation. And the line polices its own imitations, because chapter 26 met capital "returned" by promise rather than by fact: the par-value repurchase that guarantees an exit price to a claim dressed as ownership is not the zero line honored but the zero line counterfeited — return of capital names what the venture actually gives back, not what a counterparty undertook to pay regardless.

Above the zero line sit the two repair lines, and chapter 19 wrote their specification in advance — its promise, kept here, was that this formula would carry "actual cost recovery and actual loss recovery as named, separate, bounded lines." Cost is the planned burden of running money: origination, administration, accounts, audit, the clerk and the premises. The tradition's most rigorous institutions stood on this line and nowhere else — the Monti before their council, the *qard* funds under the resolution chapter 16 carried, permitting the service charge that recovers actual cost and forbidding the one that recovers more — and the line's grammar has not moved since: shown, allocated to this financing, recovered once, at zero margin. Each of those words is a fence with a history. *Shown* is the invoice and the timesheet, the modern descendants of the witnesses the *iska* posted around its claims. *Allocated* divides the burdens this financing caused from the institution's own keep — a fund's fundraising, its investors' reporting, the overhead of its existence belong to the fund's side of the desk unless a stated policy honestly assigns them, because a borrower billed for whatever happened to be nearby is carrying the institution, not the loan. *Once* is chapter 35's oldest fence run through the accounts. And *zero margin* is what keeps the line repair instead of revenue: honest counting is not free — chapter 14 priced the sworn accountings and valuations that make partnership possible — but the moment a cost line clears more than it spends, whatever it clears belongs to some other line's discipline or to no line at all.

Loss is the unplanned burden — the wound an event actually caused — and it enters the formula with chapter 19's whole method attached: name the event, trace the cause, produce the papers, measure the repair, count it once, and watch the direction of the lender's hope. The two lines stay separate because their temptations differ. Cost inflates by habit, percentages quietly outgrowing the work; loss inflates by imagination, injuries asserted where none can testify. Both are bounded at the same edge — the burden proven, the repair ending where the wound ends — and neither may carry a target. A fund that needs its damages to hit a budget has already told the auditor what the damages are for; repair priced to a return is not repair, and the formula refuses it the way Rava's netting rule and the standards' provisions refused it, centuries apart, in the same spirit. Nothing in these two lines makes capital richer, which is their point, and the reason the traditions trusted them first.

The Wage and the Wait

The fourth term prices work and service, and Part IX has already argued it twice — once as doctrine, once as arithmetic. The doctrine is the oldest three-way agreement in this book: the halakha voiding an *iska* over an unpaid dinar of the trouble-wage, the fiqh giving the working *mudarib* his ratio and, in the wreckage cases, *ujrat al-mithl*, the wage of his like, the schoolmen reading the honest merchant's gain as *stipendium laboris*, the wage of his labor. The arithmetic is chapter 38's, and the formula inherits it whole — contribution seated, as that chapter promised, "beside composition, risk, and order as a first-class term." The contribution schedule is how the work line gets its numbers: every working seat's input

named, dated, and priced at entry, the founder's role at the role's market wage, the stewardship of a fund priced as the work it actually performs — sourcing, diligence, accounts, governance, the sitting-with-the-founder in the bad month. Service earns the same way work does, by happening: a stewardship fee answers for what was done, for whom, and to what use, and support the enterprise neither asked for nor felt is not service but a subscription the stronger party wrote for the weaker one to pay. So the line comes with chapter 38's two fences built in. Facing capital: a return that cannot trace to expended work or ventured property has no business here. Facing labor's own institutional forms: a fee that swells with assets whether or not anyone worked is a percentage collecting like capital while calling itself labor, and the line prices work, not aura — the founder's legend and the manager's letterhead both pay for themselves. Between the fences, the line is generous on principle. Moral finance runs on real hours, and a structure that will not pay for them is not purer than one that will; it is planning to consume its own people, an extraction with the direction reversed, and chapter 39 already named where that plan ends — in a benefactor where a wage line should be, and a countdown where an institution should be.

The fifth term is the one the formula could most easily have left out, and the reason it stays in is the reason it rarely pays: honesty about the tradition's own long argument. Being without one's money — *carentia pecuniae*, chapter 21's file — can be a real burden, and the seventeenth century's best minds disagreed honestly about whether that burden could ever be sold. The halakha had answered early and structurally: waiting itself takes no wage — the *agar natar* of chapter 7's file, payment for waiting, is the thing forbidden, not a title needing fences. The formula carries the entry chapter 21 nonetheless promised it would carry, "an entry called liquidity deprivation," and carries it under that chapter's full discipline: the line must be capable of reading zero, and it reads zero wherever no test puts a number in it. The tests are the file's — the deprivation particular, dated, and shown, a burden this lender actually carried for this financing, not the season's inconvenience every lender can recite. Patience, where it is real, is better paid in weight than in rate — the longer-committed capital taking the deeper share of realized surplus, chapter 21's grammar — and a fund that fears what waiting could do to it builds the chest of this chapter's eighth term rather than billing its nerves to the borrower. On the mercy layer of the stack, the *gemach* and *qard hasan* floors of chapters 15 and 16, the line reads zero by design, because there, in chapter 21's words, "the deprivation is the gift"; price it and the instrument is unmade. A line that usually reads zero may look ornamental; it is the opposite. Writing it down is what keeps its residue honest — whatever cannot pass the line's tests may not ride the contract under some vaguer word for time.

The Elsewhere and the Share

The sixth term is the most dangerous line the formula carries, because it prices a world that did not happen, and the formula's answer is to keep chapter 20's whole cage around it. Opportunity enters bounded to a particular elsewhere: a named alternative, in the claimant's real practice, already in train, causally severed by this financing — mourned afterward, on proof, at hope's price and never harvest's, in chapter 20's rule; discounted for every contingency the real venture would have faced; capped at the loss it claims to repair; counted once against the files of chapters 19 and 21, which name different wounds or name nothing. The elsewhere must also be lawful, and chapter 20's cage said what that refuses in five words — "a forgone extraction is not a compensable injury" — which reaches further than it looks: a mission fund measuring itself against yields its own charter forbids is mourning a world it had already renounced. A clause that appears in every contract fails the

whole cage at once, having stopped compensating anyone for anything in particular. The professional lender still walks Hostiensis's refused circle — money whose only elsewhere was another loan has forgone nothing any of the three traditions would price — and the modern fund inherits the circle's logic upstream: its own promises to its own investors, the hurdle it must clear, the cost of its capital are decisions about whether to invest, not debts the borrower contracted. A fund that has promised extraction to its funders has not acquired a title; it has only chosen, in advance, who will be asked to perform the extraction. The formula's answer to that arrangement is the same at every scale: the promise upstream must bend to the lines downstream, or the mission language is a letterhead. And the line's size stays where the houses left it, which is to say disputed. Aquinas set it at zero; the later schoolmen opened it under conditions; the halakha, rather than pricing the forgone world, redrafted the financing so the lender stood inside the earning — the *heter iska* of chapters 20 and 31, after which nothing is forgone; the fiqh keeps the line flat at nothing, having refused time a rental price altogether. The formula does not vote. A house that zeroes the line runs the formula with the line at zero, and the formula is still the formula — which is precisely how a design rule respects a *machloket* it has no standing to settle.

The seventh term is the one the whole architecture exists to protect, and the only one that can grow: real surplus participation. Its pedigree is Part IV entire — the *societas*, the *iska*, the *mudarabah*, capital earning as from property of its own because it stood to lose as an owner loses. Its base is not revenue, not cash, not the January number that says *available*; it is chapter 39's remainder — “the till's last number: whatever survives the procession” — after wages to standard, the founder's line, the maintenance funded, the chests refilled, the community's owed share, the priced footprint. Its form is chapter 14's distinction, on which the whole partnership arc turned — “A ratio asks the world what happened; a rate tells the calendar what is owed” — and the formula has no line where a calendar can collect. Its timing is late by construction: profit is a remainder, found by counting and never promised, so the share is discovered at the settling, exactly as the Fustat quays and the standards' valuations always insisted. Its license to grow is bought with its exposure to shrink — the symmetry chapter 35 wrote into the ledger's seventh line — and the fraction itself stays where every tradition left it, with the parties: the *mudarabah*'s split was always negotiable, Ulpian's shares were set by honest judgment of what each partner brought, and the formula constrains the base, the timing, and the cap while leaving the ratio to the people whose venture it is. And its relation to danger is the settlement chapter 36 negotiated. Where a danger is measurable, netted of every protection, and genuinely borne, its honest small premium survives — the twenty-two hundred and fifty euros of that chapter's file, not the twenty-five thousand the pricing memo wanted. Past the edge of measurement, where young ventures actually live, only participation pays honestly, because a share is the one instrument that cannot claim the venture's weather without standing in it. The share is capped. On this the traditions' modern boards and this book's model agree without strain: a bounded multiple, stated at signing, after which capital's claim is complete and further surplus belongs to the venture, its people, and its purposes — the design chapter 41 will draft and Appendix E will work in numbers. A cap does not make a share honest by itself, as chapter 37's stacked certificates proved; but a share that refuses any cap has told the room it wants partnership's upside with perpetuity's appetite, and the formula, which asks every euro its errand, has no line for appetite.

The Reserves That Stand

The eighth term is not a way capital gets paid; it is the condition under which anyone gets paid, and writing it into the return formula — not beside it, in it — is half of what makes the formula non-extractive rather than merely tidy. Before the surplus line opens, the reserves stand: the maintenance the machines have already earned, funded rather than merely admitted; the solidarity reserve refilled to its charter size — the anti-ruin reserve chapter 10 named for the whole book and chapter 28 sized, “to the pile-up, not the average year” — so that ruin probability falls for every seat at once and the bad quarter costs the venture a chest instead of a life; the mission reserve on its written schedule, standing between the purpose and the quarter that would otherwise buy it. Chapter 39 established their seniority at the till, and the formula makes the seniority a term of the instrument itself: a distribution that reaches capital across an unfunded reserve is not a return under this formula at all; it is the residual, collected early, out of the venture’s next collapse.

The same term runs at the portfolio layer, where it does the work chapter 36’s map assigned to the pool. A fund holds many paths; each of its companies holds one. Dangers that are measurable across the many — the failure rates, the bad quarters, the ordinary mortality of ventures — belong in a chest at the fund, funded from the fund’s own arithmetic, so that no single company is asked to promise its survival to smooth the ensemble’s year. The traditions’ living instruments already run the seniority in modern paragraph numbering — the disclosed reserve the *isla*’s codifiers blessed in the same breath that they forbade the guarantee, chapter 13’s finding; the interim profits the *fiqh*’s standards retain for the protection of the capital before any entitlement hardens, chapter 14’s; the standing reserves its banks operate as ordinary machinery, chapter 32’s — and the formula asks of a fund only what those instruments ask of a venture. Chapter 28’s two-questions test travels up the layers with it: a chest is judged by whose ruin it prevents, and a fund reserve that exists to steady the fund’s reported year while its companies each face their own weather alone has answered the test in the wrong direction — the pooling must reach the paths that can actually be ruined, or it is the ensemble comforting itself. The term carries its own fences: a stated size, a stated purpose, a stated schedule, drawn by rules everyone has read, per chapter 39; money parked past its declared size is a distribution being denied rather than shared. And chapter 28’s warning travels with it — a chest milked in the calm years has been converted into somebody’s yield, and a solidarity reserve that never moves the ruin column was, in that chapter’s verdict, a marketing line. Capital, notice, is not the term’s victim but its beneficiary of record. The reserves are what keep the zero line worth holding — a fact the honest investor of chapter 36 grasps without help, and the extractive one discovers in the only year it matters.

From Theology to Term Sheet

Now the ninth term, which is less a line than a verdict standing at the end of every line. Run chapter 35’s first equation on any claim: whatever is left, after the zero line and the honest titles have taken everything they can carry, is the extraction residual. The formula is that equation with the last term held at zero — by construction at the drafting table, by audit ever after. Held at zero at signing, it is a design rule: no term in the instrument may collect what no line can explain. Measured at zero in operation, it is a verification: the promise kept, year by year, in arithmetic a stranger can re-run. And behind both uses stands the fact chapter 35 refused to let the subtraction launder — a residual is never abstract: it is paid by the borrower’s worst year, the household whose guarantee is called, the

crew whose gap was never booked, the purposes chapter 34 watched sold to meet a preference — the seats, in chapter 35's words, least able to refuse it. Holding the term at zero is not bookkeeping fastidiousness; it is the whole of the book's case, stated as a constraint. Between the two uses, the formula stays honest about the one thing it cannot do, which is end the old arguments; a strict house and a mild house will size the liquidity and opportunity lines differently, and the formula, like chapter 35's ledger, reports both runs side by side rather than crowning either. What no house will bless is the unexplained remainder, and the worked file shows how much of one an ordinary instrument carries. As written — chapter 35's hundred thousand for five years at twelve percent, doubly secured — the file claimed sixty thousand euros above principal and could explain about six: four thousand of shown expense, twenty-two hundred and fifty of honestly netted danger, nothing on any other line. The formula would have signed the six and struck the rest. The same contract, redrafted in the formula's plainest mode — cost-recovery credit, the Monti's mode of chapter 17, the serviced qard of chapter 16 — carries a claim about a tenth the size and a residual of exactly nothing. Redrafted as participation, it converts the refused fifty-four thousand into a capped share of remainders that may pay more than that or nothing, depending on what the venture actually does — which is the entire moral difference between a return and a rent. One name is deliberately absent from the walk. The agreed penalty of chapter 22 holds no line here, exactly as it held no standing line on chapter 35's ledger: a genuine penalty is priced to collect nothing, booked only in the delay states, against culprits, and a penalty budgeted as revenue has already failed both chapters' tests at once. The formula leaves it where the doctrine left it — at the door, useful, and watched.

The tenth term is the translation itself — theology into LP distributions, with no softening of either word. The vocabulary waiting on the other side is settled market furniture: limited partners who fund, a general partner who manages, the management fee run at a habitual two percent of committed capital, the carried interest at a habitual twenty percent of the gains, the preferred return near a habitual eight that the fund must clear before its carry begins, and the distribution waterfall that sequences it all — whole-fund in the drafting the market calls European, deal by deal in the American. None of this furniture is wicked, and none of it is a title; they are customs waiting to be asked the formula's questions, and the asking re-derives every clause. The management fee is a work line, so it answers work's tests: priced to the stewardship actually performed and shown, capped against the drift chapter 38 named, in which a percentage collects like capital under labor's name — two percent of committed capital is an answer to what the work costs only by coincidence. Fund expenses are a cost line: shown, allocated by stated policy, recovered once, at zero margin, and never collected from both sides of the fund at once — investors funding an operating capacity the portfolio is then billed for again. The preferred return is this Part's oldest acquaintance under its newest name: a number written in advance, accruing with the calendar, indifferent to any venture's actual year — the certain sum the 1586 bull condemned in partnership dress, chapter 13's file, chapter 37's cumulative coupon, here promoted to the fund's own constitution. The formula's redraft does not abolish the investors' protection; it relocates it into truth: the fund's investors hold the zero line — their capital home, wired and sweat alike, before any carry exists — and above it a stated share of true surplus, in place of a clock that ticks against companies that had different years. The carry itself, so redrafted, becomes the most defensible euro in the building: a remainder share, collected after every principal is home and every reserve stands, capped like everything else — the general partner as working partner, paid in the share's grammar for work and judgment, which is what the twenty percent always half-remembered it was. And the waterfall is where chapter 37's finding folds in, as that chapter promised: "repay-

ment order enters the arithmetic as a first-class term.” Ask the two market draftings the queue’s own question — may carry collect while any seat still holds unreturned principal? The deal-by-deal order says yes, paying the manager on the winners while the losers’ capital is still in the ground; the whole-fund order says not yet; and the formula’s order says it in full, writing itself down the page in one motion — operating claims and wages, the reserves to charter, every principal, wired and sweat, then shared remainders to the cap, then nothing: surplus past the cap stays with the ventures or recycles to the next fund’s floor, per the design of chapter 41. One coherence rule governs the whole translation, and it runs upstream. A fund can be no more patient with its companies than its own capital is with it, unless it absorbs the difference in its own design — which is what the reserves, the caps, and the recycling are for. The gift and mercy layers chapter 41 will draw exist precisely so that the stack’s patience is funded somewhere real rather than promised everywhere and held nowhere; a fund that signs the formula downstream while owing the old arithmetic upstream has not translated the theology, only relocated the extraction to its own investor meetings. And the translation is a reporting practice before it is a boast. A fund that runs this formula publishes the runs: the fee against the work performed, the expenses against the policy, every reserve against its charter size, every distribution against the remainder it came from — the investor’s letter carrying the formula’s lines the way chapter 36’s honest fund carried its two tables and chapter 38’s carried its schedules, so that the claim of non-extraction is, every year, a set of subtractions a stranger can check rather than an adjective the marketing renews. A creed that cannot reach the distribution clause is a preamble. This one reaches it, term by term, and the terms hold in both directions — the same lines that draft a new fund’s waterfall will read an old fund’s, which is the analyzer’s whole business when chapter 43 arrives.

So the desk in Amsterdam can be cleared. The first term sheet — the one that could answer — was never a mystery; it was this formula, met before the book had taught the reader to recognize it: costs invoiced, a penalty that expects to collect nothing, danger charged only where the money actually loses, the rest a stated, capped share of profits that exist on the day they are shared. What chapter 35 could only tell apart by interrogation, this chapter can write on demand, and Part X will spend five chapters building the institutions that write it — the fund of chapter 41, drafted layer by layer; the contract library of chapter 42; the analyzer of chapter 43, which runs this formula wholesale on term sheets that never heard of it; the accounting of chapter 44, which keeps the answers honest after signing; and the temptations of chapter 45. The formula’s finding fits in a sentence, and it is the book’s oldest conviction made computable: a non-extractive return is not a smaller number; it is a number made entirely of answers. Seven titles above a zero line, the reserves standing, the remainder at zero — and every euro beyond the principal arriving, for the first time in the instrument’s long history, with its errand written on it.

Part X — Institutions for the Future

Chapter 41 — Designing a Non-Extractive Fund

An architect who wants to know whether a building will stand does not linger over the facade. She draws the section: the cutaway that shows the foundations, the bearing walls, what rests on what, and what happens to the whole if any one floor fails. Finance has such a drawing too. The market calls it the capital stack — every source of a venture’s money, drawn in order of who gets paid first and who absorbs loss first, from the senior debt at the top of the queue down to the common equity holding the bottom. Chapter 37 spent a chapter reading such drawings and delivered its verdict in one line: “a preference is a rate that learned to stand in line.” The order is never decoration; a stack can extract through its seating what no clause would dare charge as interest. But that chapter ended on the instrument’s other face. The same geometry that quietly extracts can openly protect, and order is a design surface. This chapter takes up the pencil.

The commission is the one Part IX prepared. Chapter 35 established that instruments can be “designed to a moral geometry, not merely audited against one” — its own words, and the fact on which this chapter now builds a fund. Chapter 40 assembled the non-extractive return formula, the building code every drawing in Part X must satisfy: capital home before gain exists, costs and losses recovered at their proven size, work priced as work, patience weighed only where it carried a real burden, surplus shared by ratio under a cap, the reserves standing senior, the residual held at zero. What remains is the architecture — the actual floors, in the actual order, with the actual money. So take a fund invented for the purpose, ten million euros of committed capital serving a community of young ventures, and draw its section. The proportions that follow are the designer’s to argue about; the order is not, because the order is where the theology lives.

The stack, bottom to top — bottom meaning first to absorb loss and last to claim return, the ground the rest stands on:

1. The gift layer (€500,000) — granted outright; expects nothing back; absorbs first losses; funds the floors above.
2. The *gemach* / *qard hasan* layer (€700,000) — revolving zero-return loans; principal back, nothing more, ever.
3. The cost-recovery credit layer (€2,300,000) — bridge and working-capital loans repaid with documented costs and not a euro beyond.
4. The *takaful* / mutual-aid / solidarity-pool layer (€800,000) — the community’s shared chest for defined perils, owned by its members.
5. The risk-sharing layer (€4,200,000) — *mudarabah*, *iska*, *societas*: capped participations that earn only from realized surplus and lose when the venture loses.
6. The mission-locked equity layer (€1,000,000) — patient ownership stakes in steward-owned companies, exit only by mission-preserving buyback.
7. The community reserve layer (€500,000) — the fund’s own standing chest: never deployed, never distributed, sized to the bad stretch.

Seven layers, and not one of them invented for this book. Every floor is an instrument from an earlier chapter, built centuries ago, tested in use, and waiting, which is why the drawing can be walked layer by layer with credits instead of blueprints. Chapter 36 mapped the territories where the traditions put honest money to work — “tables where tables are honest, chests where perils can be gathered, partnerships where nothing can be known in

advance” — and promised that the fund this Part was building would use the same map in the same order. Here it is, drawn to scale. Notice, before the walk begins, how the drawing quarrels with the market’s own section. In the conventional stack, the bottom absorbs loss in exchange for unlimited upside: equity eats the first wound because equity owns the last euro of any triumph, and every seat above it buys protection by surrendering appetite. In this stack the logic of the bottom is inverted. The floors that absorb the most expect the least, and what they are compensated in is not upside but purpose — the gift layer and the mercy floor exist so the community survives its worst months, and their return is that the survival happens. What the stack adds to the old instruments is the architecture they were always implying: each layer renounces something exactly so that the layer above it can exist without pretending. Every layer earns its seat by what it refuses to collect.

And the layers are not a menu; they are a path. Follow one venture through the drawing. A two-woman workshop takes a bridge loan from the credit floor in its first winter, repays it with the documented forty euros it actually cost to make, and joins the pool that spring against the perils every workshop shares. In its third year it wants a second lathe and a hire, and the risk-sharing floor buys in: a participation, capped, collected only from settled remainders. In its seventh year, grown into the town’s anchor employer, it converts to steward hands with the mission-locked floor financing the buyback, and somewhere in that decade there was a January when the founder’s child was in the hospital and the mercy floor carried the payroll for six weeks at zero, because that is what the floor is for. Five instruments, one company, no seam at which anyone’s emergency became anyone’s yield. A conventional fund meets that workshop once, prices it once, and needs its exit; the stack meets it at every altitude of its actual life, which is what the traditions’ whole inventory was assembled to do.

Where the Stack Begins

The bottom of the stack is a gift, and the placement is the whole doctrine. All three traditions rank giving as the economy’s true ground floor — the *tzedakah* that is owed rather than felt, the *zakat* that is calculated rather than moved, the almsgiving chapter 18 weighed as commanded infrastructure rather than optional warmth — and all three, chapter 15 showed, rank the loan above the gift precisely because rescue that expects repayment leaves the rescued standing. A fund built on their ledger begins the same way. The gift layer is capital that has already said goodbye: donated in, spent as granted, or held as the stack’s first-loss ground. Because it demands nothing, it can do the two jobs no priced layer can — catch the falls no structure can price, and stand under the mercy floor so that the mercy floor never has to earn. The market has been rediscovering this seat under its own vocabulary: catalytic first-loss capital, the blended-finance drafting in which a foundation or a public funder takes the junior position so that warier money can stand above it. The label is honest as far as it goes, and the traditions would add two fences to it. The first is chapter 34’s warning: a gift that exists to sweeten someone else’s extraction has been enlisted, not given — a first-loss layer that lets the layers above collect the old arithmetic in safety is subsidizing the very thing this stack was drawn to refuse, so the gift stands under mercy floors and capped claims or it stands under nothing. The second is that the gift layer is governed like every other floor, not scattered from it: grants move on published criteria, are recorded like disbursements, and answer to the same reporting as every priced layer, because generosity without bookkeeping decays into patronage, and patronage is a claim in disguise. Chapter 40 wrote the deeper rule as upstream coherence — patience must be funded somewhere real rather than promised everywhere and held nowhere. This is the

somewhere.

One floor up sits the mercy loan, and here the fund is not innovating; it is franchising. The *gemach* of chapter 15 and the *qard hasan* of chapter 16 are the same machine in two liturgies: a pool of given and deposited capital, lent at absolute zero, guarded by guarantors and community knowledge, repaid at rates that embarrass the professionals, and revolving — out, back, out again, a founding gift still rescuing strangers a century later. The machinery travels into the stack intact. Depositors, where the layer takes deposits, may reclaim their principal and not a cent more; the layer's administration is waged and priced at documented cost, chapter 16's receipts-never-percentages specification; underwriting, guarantors, and collection stay as rigorous as any bank's, because the home chapters proved that rigor and mercy are not trade-offs — what mercy forbids is only that the rigor become a revenue line. Its job in the stack is the one chapter 16 specified when it told the builder to seat the layer deliberately, sized to the shocks that actually arrive, “upstream of the growth layers and unembarrassed beside them.” The shocks in a venture community are not exotic: the founder's two bad months, the bridge between an invoice and its payment, the repair that cannot wait for a board cycle. Each is exactly the emergency the beautiful loan was built for — principal for a season, nothing riding back — and each, left to the layers above, would be priced into a permanent claim. Dignity is engineered in rather than hoped for, standard terms and confidential rolls, because clientage is a design failure, not a price of charity. On this floor the return formula's liquidity entry reads zero by design, chapter 21's rule, since here being without the money is the very thing being given; a fund that cannot bear that zero has no business drawing this floor, and a fund that can has bought something the priced floors cannot make — a community whose worst months do not compound. And it was this floor, before any other, that taught funds to revolve. What the market calls a revolving loan fund, and what Part X will keep meeting as the evergreen structure, is — as chapter 15 promised this chapter would say explicitly — “the *gemach*'s shape scaled up”: capital designed to circulate serially instead of extracting and exiting, the wheel as a fund's whole body plan.

Credit at Cost, Perils in Common

The third layer is where the fund starts running priced instruments, and the price is the strictest one the traditions ever blessed: everything shown, nothing beyond. Cost-recovery credit is the Monti's mode from chapter 17 and the serviced *qard* of chapter 16's resolution — the loan whose service charge recovers actual, documented expense and forbids itself anything more — and chapter 40 has already fenced its grammar into the formula: shown, allocated by stated policy, recovered once, at zero margin. In the stack this layer is the workhorse: the working-capital line, the inventory bridge, the equipment loan — credit for ventures that need a lender, not a partner. That distinction is a dignity as much as a mechanic. A bakery that needs six months of flour financing should not have to sell a share of itself to buy a season of liquidity, and a stack with an honest credit floor never forces participation on a venture that only needed a bridge. What the layer refuses is as load-bearing as what it provides. It refuses margin, so its receipts can be audited against invoices rather than appetites; it refuses percentages that scale with size and term while the work does not, chapter 16's oldest specification; and it refuses to let the fund's own overhead migrate downward onto borrowers as a “cost of capital,” because the fund's keep belongs to the fund's side of the desk. The layer also carries the Part's insolvency posture, because a credit business is defined at its worst tables: distress is met first as discernment rather than enforcement; the penalty clause keeps chapter 22's grammar — expecting to

collect nothing, armed only by true delay, aimed only at culprits; and a late borrower meets a redrawn schedule before a compounding claim. Priced and policed this way, the floor does something quietly radical to the whole drawing: it proves the stack can run a credit business — underwriting, collections, defaults and all — with the extraction residual at a standing zero, which is the demonstration the Monti made before their council and the reason this book keeps returning to them.

Beside it stands the pool, because some of what a venture community fears is not a financing need at all but a peril: the fire, the lawsuit, the founder's illness, the client that vanishes owing a quarter's revenue. Chapter 36's settlement sorts the stack here, and it is worth quoting as the sorting rule: "What can be measured can be pooled; what cannot be measured can only be shared." Perils that are nameable, pre-existing, and measurable across many members belong in this layer, gathered, in chapter 27's grammar, as the members' own — the books the chevra, the confraternity, the friendly society, and the takaful fund all kept, that chapter's census showed; the unmeasurable — whether the venture itself will find its market — belongs two floors up, where only participation pays honestly. The design rules arrive from chapter 27 intact, a specification rather than a sentiment: contributions enter as the members' own money, *tabarru'* or dues, never as a seller's revenue; the fund belongs to those it shelters; its administrator is paid a disclosed *wakala* wage for work, indifferent to whether claims run light or heavy, and fenced off the surplus; the light year's remainder thickens the reserve or goes home to the members; exclusions police conduct, never misfortune; pools pool upward, retakaful and mutual reinsurance, so that no single chest's weather is final; and outside capital may serve the pool — a backstop, an operator's interest-free advance in the industry's standard rescue — but never own it. One honesty the design owes its own doctrine, plainly: a pool defends members, so the gate is where its mercy stops, and chapter 28 warned that a community which pools brilliantly inside and prices or excludes at the door has moved the extraction to the gate. A fund's pool is judged there — at the smallest member, at the venture whose dues are hardest to pay — before it is judged on its actuarial tables. Carried honestly, the layer takes the sharpest modern perils off every other floor's price. The risk-sharing layer above need not smuggle insurance into its participation ratios; the credit layer need not pad its costs against catastrophe; each instrument can charge for exactly what it is, because the dangers that can be counted are standing in the one room built for counting them.

The Partnership Floors

Now the floors the whole Part IV arc was for. The risk-sharing layer is the stack's largest commitment and its plainest doctrine: capital enters ventures as the *societas*, the *iska*, the *mudarabah* always insisted — as a partner, earning from realized surplus by ratio and standing in losses as an owner stands — under the full discipline the book has already built. Profit is a remainder, found by counting, never promised, chapter 14's law, with the interim retentions and sober valuations its modern standards require. No guarantee dressed as partnership: the stack may hold disclosed reserves against bad seasons, chapter 13's licit pool, but the moment any instrument promises the capital's safety plus a return, it has reassembled the stacked contract the 1586 bull in chapter 13's file met with the word detestable, and it belongs to a different book. Patience is paid in weight, not rate, chapter 21's grammar: the longer-committed participation takes the deeper share of what actually happened, never a percentage of the calendar. The fund's own stewardship of these positions — sourcing, diligence, the accounting that makes ratios collectible, the sitting-with-the-founder in the bad month — is priced on the work line at the work's wage, chap-

ter 38's fence, not skimmed from the participations as a percentage collecting like capital. And every participation is capped: a bounded multiple, stated at signing, after which the claim is complete. Run the arithmetic on the file this book already owns — chapter 35's half-million-euro participation, capped at two euros and ninety cents per euro advanced — and the ceiling is one million four hundred fifty thousand euros, ever. If the venture flourishes, the fund collects its capped share from true surplus, chapter 39's remainder, and is done; everything after that belongs to the venture, its people, and its purposes, or turns the wheel below. If the venture fails, the fund's capital fails with it, netted only by whatever protections were honestly priced in the one layer built for pricing them. That symmetry is not a marketing stance; it is the license — the thing that lets the return exist at all in three legal languages at once.

The mission-locked equity layer is smaller, slower, and structurally the most interesting floor in the drawing, because it holds the instruments chapter 29 built the case for and this chapter, as that chapter promised, now reaches for by name. The layer takes ownership stakes only where the charter has retired the exit: the asset lock, binding the company's value to its purpose so that mission-built worth cannot be converted into a seller's prize; Steward Ownership's two commitments, control held by people bound to the mission — never marketable, never heritable — and profits serving purpose rather than owners; the golden share, the one-percent guardian holding, in chapter 29's image, "one key that opens nothing and locks everything"; the Employee Ownership trust, holding the company for those who work in it. Where a company arrives at the fund's door not yet locked, the lock is the deal: conversion to steward hands is written into the investment itself, priced and scheduled, so that the fund's money finances the dedication rather than waiting politely for it. Equity in such a company cannot be venture capital's equity, because the payoff that model needs — the appreciating exit — has been deliberately welded shut. What remains is exactly what the traditions said ownership honestly was: a share of fruit, for as long as the tree is held, with departure by redemption — the mission-preserving buyback, at a formula written the day the money went in, returning the shares to steward hands on a schedule the company can survive. This is the endowment's logic from chapter 29 — *waqf*, *hekdes*, church patrimony, the whole family of deeds that retired capital from the market — running at business speed: what cannot be sold cannot be sold out, in that chapter's words, and a fund that wants its portfolio's missions to outlive its own involvement buys into locks, not in spite of them. The layer will never post the stack's largest number. It exists because some of what a community needs held — its clinic, its grid cooperative, the workshop that anchors a town — should be held by capital that has agreed, in writing, to be the last money that could ever hollow it.

The Chest the Fund Keeps

The seventh layer never leaves the building, and the drawing is incomplete — worse, dishonest — without it. The community reserve is the fund's own standing chest — the anti-ruin reserve chapter 10 named and chapter 28 sized, stationed now at fund level: capital held against the pile-up, drawn by written rules, and counted a working layer even though it earns nothing, because what it manufactures is every other layer's ability to keep its promises. Chapter 28 proved the mechanism at village scale and named the design consequence this chapter was built to receive: portfolio resilience is nothing but "the village model wearing a term sheet," and the industry is now held to it. A fund holds many paths; each venture holds one. So the fund layer, which genuinely lives across the ensemble, is where the smoothing honestly belongs — the steady cadence, the reserve, the shock ab-

sorption — precisely so that no single venture's worst year is mortgaged to a calendar. Schedule the chest, never the household: promises of regularity are made at the pool, and patience is granted at the path, which is the exact inversion of the instrument chapter 37 audited, where the schedule sat on the company and the flexibility belonged to capital.

The chest's rules travel up from the home chapters unchanged. It carries a stated size, set against the worst plausible stretch rather than the typical year, chapter 28's sizing discipline; a stated purpose; and a stated schedule for filling and drawing, because an undocumented reserve is a discretion, and discretions drift. It fills the way chapter 39 ordered the till — senior to every distribution, replenished before any surplus exists to share — and it can begin the way the old pools began, chapter 28's founding-cohort device: an entry contribution at the door, so the chest opens with a bad year already banked instead of promising to grow brave later. It obeys the two-pools test: a reserve is judged by whose ruin it prevents, and a chest that exists to steady the fund's reported year while its ventures each face their weather alone has answered in the wrong direction. It must actually move the ruin column — chapter 28's verdict on the reserve that never does is that it was a marketing line — and, the same chapter's warning, it may never be milked in the calm years, since a chest converted to yield has quietly sold the one thing it existed to hold. Held to those rules, the layer pays for itself in a currency the drawing has already spent twice. It is what makes the fund's own patience survivable, chapter 21's condition, so that waiting never has to be billed to anyone; and it is what lets the mercy floors run at zero without fragility, because the institution above them is never one bad quarter from becoming a creditor with regrets. Capital, here as in chapter 40, is the chest's beneficiary of record — the honest investor's principal is safest in exactly the year the chest was sized for — and the crowning irony of the whole section drawing is that its least productive floor is the one holding the roof up.

The Cap, the Wheel, and the Guardian

Now the top of the stack, where the investors stand, and the two clauses that make the whole drawing a single instrument. The first is the cap. The fund's funders hold the zero line first — every principal home, wired and sweat alike, before any carry or distribution exists — and above it a stated share of true surplus to a bounded multiple, stated at signing, after which the fund's obligation to capital is, in the oldest sense, finished. Nothing about the seat is grudging: the investor's terms are written in the same grammar as every floor below — capital whole, work priced, a capped ratio of what actually happened — which is what makes the stack one instrument rather than a mission wrapped around an ordinary fund. The honest investor of chapter 36 recognizes the trade at once, because it is the one the traditions always offered: less than the market's promise in exchange for claims that are all answers — and the investors such terms select are the ones whose own charters can hold them, the foundation, the endowment, the family office with a purpose clause, the public funder whose mandate is the region's ventures surviving. The waterfall beneath the promise is the one chapter 37 commissioned: operating claims and wages filled, the reserves funded to charter, every principal returned, then shared remainders to the cap — the investor's return seated, in that chapter's words, "behind the enterprise's ability to survive collecting it." Chapter 39 named the alignment this ordering buys: an instrument shaped so that "its appetite and the venture's survival point the same way." An investor in this stack has no seat from which to profit by the portfolio's squeezing; the only route to the capped share runs through funded reserves and living ventures, so the appetite itself has been conscripted into the architecture. Appendix E will run these waterfalls in worked

numbers, and Appendix D holds the metrics the runs are graded by.

The second clause answers the question every cap creates: where does the rest go? Surplus past the cap either stays with the ventures that made it or flows to the stack's own floor — replenishing the reserve, refilling the mercy layers, funding the next wheel's turn — and this is the point at which the fund stops being a portfolio with better manners and becomes a different body plan. The market's own label for funds without a terminal date is evergreen: no fixed end, proceeds turned back into new deployments rather than paid out toward a closing. The traditions have been running the reference implementation for centuries — the *gemach* that never closes, the *waqf* whose corpus outlives its founder by half a millennium, the Monte still lending five hundred years on. A terminal fund must, by its charter, convert everything to cash and leave; its final act is liquidation by design, however gentle the decade before it. An evergreen stack built on this chapter's floors points the other way: capital that finishes its capped errand is either retired or turned back into the ground floors, the wheel of chapters 15 and 16 running at institutional scale, so that success compounds into thicker chests, fuller mercy floors, and further ventures rather than into a farewell distribution. Nothing about the shape is soft. The wheel still underwrites, still prices its credit at shown cost, still caps and collects; it has simply been pointed at continuation instead of exit, and a fund pointed at continuation can make promises — to founders, to workers, to a town — that no self-liquidating vehicle can honestly sign.

Which leaves the last obligation on the drawing, and the oldest lesson in this Part: none of it administers itself. Chapter 29 taught that drift is a design load — the *waqf* captured by its *mutawallis*, the steward-firm captured by its stewards — and chapter 45 will spend a chapter on the temptations peculiar to religious finance. The stack's answer is architectural, not exhortatory. Give the deed a guardian: an independent, asset-locked holder of standing, the golden share's logic raised to fund level, whose consent is required for precisely the acts that would undo the drawing — unlocking a layer, raising the cap, milking the chest, redrafting the waterfall. Give the floors a voice: the borrowers, stewards, and members the layers exist for sit where they can be heard, in the pool's assembly and the fund's council, because the people at the thin end of a structure feel its drift before any auditor books it. Publish the runs, chapter 40's reporting practice: every fee against its work, every reserve against its charter size, every distribution against the remainder it came from, so that non-extraction stays a set of subtractions a stranger can check. Watch the side doors: the related-party fee, the affiliated management company, the service contract that collects like capital while calling itself administration — chapter 38's fence, patrolled on a schedule. And re-ask the layers the formula's questions annually, because a claim that was honest at signing can rot in place: the fee that outgrew the work, the reserve parked past its stated size, the participation that collected its cap and kept collecting. The interfaith contract library of chapter 42 supplies the clauses for all of it; the analyzer of chapter 43 will read this same stack back, term sheet by term sheet, and grade the drawing against the code it was built from.

Step back from the section, then, and read the building whole. At the bottom, money that expects nothing, so that the floors above never have to pretend. Rising through it, money that expects less than the market at every station — principal only; costs only; the pooled peril only; the capped share of what actually happened — each layer refusing exactly the claim the floor above would otherwise have been forced to invent. At the top, investors whose road to a return runs through funded reserves, living ventures, and a written cap. And standing beside the whole drawing, a guardian whose single share exists to say no. What the traditions ran from closets, courtyards, and chests, a fund can now run from a

waterfall — provided the waterfall was drawn by someone who knew what the closets knew. Nothing in the section is new money; it is old mercy, given a modern org chart and told to hold the line. The gemach's counter, the Monte's chest, the takaful pool, the partnership's honest ratio, the waqf's unsellable deed: one building, seven floors, every euro in it arriving — as chapter 40 taught this book to demand — with its errand written on it. A capital stack, it turns out, is a creed read from the ground up. The next chapter opens the drawer of contracts that make each floor signable.

Chapter 42 — The Interfaith Contract Library

Every financing in the world, whatever its theology, ends at the same piece of furniture: a table with pages on it. The pages carry small plastic flags where the signatures go. Nothing the parties believe is in the room unless it is in the pages — not the sermon, not the pitch deck, not the handshake warmth, not one of the forty-one chapters behind this one. A conviction that has not become a clause stops at the door of the closing room, and the door is soundproof. The last chapter drew a fund as a section drawing, seven floors of layered mercy and discipline, and ended by promising to open “the drawer of contracts that make each floor signable.” This chapter is the drawer. It holds ten documents, and the claim made for them is deliberately narrow: nothing in them is new. Every clause descends from an instrument some tradition built, tested, policed, and wrote down centuries ago. What is new is only the assembly — the old fences, in modern enforceable language, on pages with flags where the signatures go.

The traditions themselves would recognize the project before they recognized the paper, because all three of them, sooner or later, stopped trusting doctrine to travel on its own and wrote it into forms. Doctrine was written twice: once to be believed, once to be signed. The rabbis of Poland, chapter 31 told, standardized the *heter iska* into a printed instrument precisely so that a merchant who could not follow the Talmudic argument could still transact inside it — the wall democratized, compliance at the price of a signature. The notaries of Genoa, chapter 11 showed, kept formularies whose compressed boilerplate carried the *commenda*’s whole risk grammar from one shipowner to the next without either of them reopening the theology. The Monti of chapter 17 lent under chartered *capitoli* that fixed what the counter could charge before the first pledge crossed it. And the modern fiqh, chapters 13 and 14 reported, maintains a standards body whose model provisions tell an Islamic bank’s drafting department, article by article, what a *mudarabah* may and may not say. A standard form, in every one of these houses, is how the tradition made its discipline affordable to people who would never afford a scholar. The library exists so that the fund of chapter 41, and any smaller venture with none of its apparatus, can do the right thing at the marginal cost of doing anything at all.

Chapter 40 made the promise this chapter keeps. Its formula, it said, could now be written on demand, and Part X would build the institutions that do the writing: the fund first, then this library. So here are the ten instruments, the drawer’s full inventory, each named for the work it does:

1. The interest-free loan agreement — the mercy floor’s paper: principal out, principal home, nothing else, ever.
2. The cost-recovery loan agreement — credit at documented cost: everything shown, nothing beyond.
3. The solidarity-pool participation agreement — membership in the shared chest, not a policy purchased from it.
4. The *mudarabah*-inspired profit-sharing agreement — capital and effort, joined by ratio.
5. The *iska*-inspired investment agreement — the two-halves instrument, with its wage and its witnesses.
6. The *societas*-inspired partnership agreement — consent, contribution, and a common fate.

7. The mission-preserving buyback agreement — the exit that returns a company to its purpose.
8. The asset-lock covenant — the deed that takes value off the market.
9. The stewardship governance charter — the offices, vetoes, and voices that keep the drawing drawn.
10. The anti-extraction disclosure schedule — the exhibit on which every other document confesses.

Ten documents, three parent traditions, one skeleton. (Appendix C holds the wider field, a classified map of the traditions' whole contractual inventory; the drawer is the signable subset, drawn to one fund's needs.) Before walking the shelves, it is worth stating what every instrument in the library answers in its own terms, because the recurring anatomy is the point. Each names its parties and the path the money takes. Each states what may be collected, and, the harder discipline, out of what: invoiced cost, proven loss, waged work, a ratio of realized surplus, never the bare calendar. Each says where losses land, in writing, before there are any. Each carries a bad-month protocol, because a contract that has never imagined its counterparty's worst quarter has been drafted for a different world. And each ends with a short list of clauses the parties alone cannot amend — the locked terms, answerable to a guardian — because the library learned from chapter 29 that drift is a design load. The word *interfaith* in the chapter's title is meant exactly, not decoratively. These are not Jewish documents, Christian documents, or Islamic documents; they are secular, enforceable instruments drafted so that a signatory from any of the three houses finds the fences of home already inside. On the deepest rules the houses simply agree, and the drafting is easy: all three insist that capital's gain ride a realized outcome, so the ratio clauses below would read as native law in any of them. Where the houses genuinely differ, the library does not referee. It drafts to the strictest common floor and leaves each tradition's further permissions where they belong, with each tradition; that is a drafting posture, the designer's own, offered for argument rather than announced as doctrine. A merchant of the Cairo Geniza, chapter 11 found, "could tell you, precisely, which tradition's rules his money was traveling under." A signatory in this library enjoys a stranger comfort: the honest answer is all three at once.

The Mercy Papers

The first drawer holds the two lending instruments, and the shorter document is the stricter one. The interest-free loan agreement is the *gemach* and *qard hasan* machinery of chapters 15 and 16 reduced to operative clauses. Parties: the fund's mercy floor, or any free-loan pool, and a borrower with a bridge to cross. Principal: stated, dated, and scheduled. Mercy names its dates — the halakha's default term for an unspecified loan, thirty days, is the tradition's oldest reminder that even kindness runs on a calendar. Consideration flowing back: the principal — all of it, and only it; no fee, no charge, no gift-in-expectation, no benefit smuggled through softer channels. The perimeter clauses do what chapter 15's design grammar demanded: the borrower's file stays confidential; terms are standard across borrowers, so that no one's rescue is bespoke charity; and the document is deliberately mute about gratitude, because the tradition that coined *ribbit devarim* — increase by words — knew that a thank-you exacted is interest paid in a devalued currency. Where the floor takes deposits, the deposit clause is one line: depositors take back exactly what they placed and nothing riding on top, the deposit earning nothing because it risks nothing. And the pool writes its own continuation into the paper: repayments return to the fund and go out again, the revolving clause, so that the instrument's afterlife is the next borrower — the

wheel of chapter 15, here a term rather than a metaphor.

What gives the paper its spine is the security package, and here the library copies the free-loan societies at their least sentimental: guarantors, named and signed; documentation a bank would recognize; the pool's rolls kept like a lender's, not a donor's. Chapter 15's finding travels intact — rigor is not extraction's property, and the discipline that collects a mercy loan back is what lets it be lent again. What the instrument refuses is the meter. There is no late-fee clause, because *agar natar*, reward for waiting, is the exact thing this paper exists to refuse; the remedies for the slow borrower are a redrawn schedule, then the guarantor, then the community's long memory — in that order. A mercy loan that starts charging for delay is no longer a mercy loan, whatever its heading still says.

The cost-recovery loan agreement is the same paper with one additional permission, fenced on every side. The lender here may run a service charge, and the charge lives inside chapter 40's four fence-words, received through the last chapter as the credit floor's grammar: shown, allocated, once, and at zero margin. The schedule of recoverable costs is an exhibit, not a discretion: origination, administration, collection, audit, each at documented actual expense, allocated to the loan by a stated policy — and the policy itself is attached, so that a borrower can see not only what was billed but why it was billed here, the exhibit standing guard against costs that wander toward whichever account complains least. Two further refusals complete the fence. The charge may never be a percentage: a charge set as a share of principal has already confessed, chapter 16 showed, since "real service costs do not double when the loan does," and a percentage charge on a mercy instrument is interest that has changed its name and kept its appetite. And the lender's own general keep may not migrate downward: the fund's housekeeping is the fund's expense, chapter 41's rule, not a solemn-sounding cost of capital on the borrower's statement. This is the Monti's counter from chapter 17 rendered in modern clauses — its verdict, quoted, that "a charge pinned to costs inherits a ceiling" — and it is chapter 16's serviced *qard*, priced by receipts and never by rates.

Then the two clauses this instrument is proudest of, because they govern the bad month. First, hardship. The paper distinguishes the borrower who cannot from the borrower who will not, and for the first it writes chapter 22's rule into operative text: the insolvent is "entitled to time, not merely eligible for it." The clause descends from the verse that chapter carried, in Abdel Haleem's rendering: "If the debtor is in difficulty, then delay things until matters become easier for him; still, if you were to write it off as an act of charity, that would be better for you, if only you knew." Deferral is the debtor's right under the document, triggered by demonstrated inability; write-off is named in the drafting as the act the parties agree is better. Beneath both stands the collection floor, the schedule of what may never be seized — the tools of the borrower's trade, the bed, the month's food, the mercies chapter 22 carried out of the old codes — written as an unwaivable clause. Second, the penalty. Against the strategic delayer, chapter 22's wealthy procrastinator whose delay is a choice, the document carries a default clause built on that chapter's grammar: it fires on fault, never on time; it names its culprit conditions; the drafting expects to collect nothing, because its work is deterrence; and whatever it does collect is routed, by the clause's own text, to a named charity and never to the lender's account. A penalty routed past the lender's pocket disciplines the debtor without ever becoming a product, and a lender who cannot profit from lateness has no reason to hope for it.

The Pool's Charter

The third instrument is not a loan paper at all, and its first clause exists to say so. The solidarity-pool participation agreement admits a member to a shared chest; it does not sell a customer a promise. Everything chapter 27 specified arrives here as drafting. The contribution clause: what the member pays enters as the member's own gift to the common fund — the *takaful* fund's *tabarru'*, the chevra's box money, the friendly society's dues, whichever liturgy the pool descends from — and never as a seller's revenue. That single clause decides whether the instrument holds or becomes a sale of insurance under a softer name. The ownership clause: the chest is owned by the sheltered, collectively, and the books are theirs to inspect. The administration clause: the operator, where the pool retains one, serves for a disclosed *wakala* wage, fixed to the office and its work, paid the same in the heavy claims year as in the light one, and fenced off the surplus entirely; an administrator paid more when members collect less has been handed a quiet stake in refusal, and refusal is exactly where the drafting must keep appetite out. So the claims clause completes it: a member's claim is assessed on stated criteria, by persons whose compensation cannot rise by denying it, and paid as the pool doing its work rather than as the operator's loss. The surplus clause: the light year's remainder is named in advance — to the reserve until the reserve meets its stated size, then back to the members, or rolled into the coming year's floor — so that a good year never becomes anyone's margin. The exclusion clause polices conduct, never misfortune: the arsonist leaves the pool; the burned do not. The reinsurance clause sends the pool upward into pools of pools, mutual all the way up, so that one chest's bad winter is never the members' last word. The outside-capital clause lets a backstop serve, including the operator's own interest-free advance against a deficit year, recovered from future surpluses — the industry's standard rescue, chapter 27 reported — and forbids it to own; rescue capital enters as a donor or a lender at zero, never as a shareholder of the members' weather.

And two clauses come straight from chapter 28's mathematics, with the sobriety that chapter earned. The founding clause: the chest does not open empty; members buy in with an entry contribution, chapter 28's device, so that the pool is holding its first bad year before it has had one. The gate clause: entry terms are set, and annually re-answered, under chapter 28's two-pools test, which the review must answer in writing — name the members the chest carried this year, and name the applicants the gate turned away — because that chapter's finding was that a pool's honesty is measured at both lists.

The Partnership Papers

The middle of the drawer holds three instruments that are one family, and the library drafts their shared grammar once, as common boilerplate, before it lets them differ. Capital enters each of them as a partner, never as a creditor. Return is a ratio of realized profit — a share of what actually happened, established by the count once the venture has answered, never a sum certain, never a percentage of the capital, never a number the calendar could compute on its own. No clause may guarantee the capital's safety and its return together; the moment a draft does, it has rebuilt the guaranteed partnership that Sixtus V's bull of 1586, sitting in chapter 13's file, condemned as "illicit and usurious," and the library's boilerplate voids it in advance: a seat that cannot lose voids the table. Interim distributions are provisional, held in the modern standards' phrase chapter 14 carried — "subject to the retention of interim profits for the protection of the capital" — so that no partner eats profit while the principal is still in the weather. The venture keeps books to a stated standard and

opens them on the terms the instrument names, because a ratio can only bite on an answer, and the whole participation apparatus, chapter 11 warned, runs on verification; the boilerplate treats the cost of honest counting as part of the venture's keep, not a surcharge on trust. The managing party answers for negligence, fraud, and breach of mandate, and for nothing the weather does — the *amin* standard the fiqh gave chapter 11; the fund may keep a disclosed reserve against the bad season, the pool chapter 13 ruled licit, but no drafting may quietly convert the manager into a guarantor of outcomes. And every participation carries the cap, the bounded multiple stated at signing, after which capital's claim is complete — at the worked ceiling the last chapter set on chapter 35's half-million file — 2.90 per euro advanced, a lifetime ceiling of one million four hundred fifty thousand euros — after which the wheel.

On that common chassis, the three instruments keep faith with three parents. The *mudarabah*-inspired profit-sharing agreement is the purest division of labor: the fund brings capital and nothing else; the venture brings the skill, the effort, and the time; and the profit ratio, stated on the signature page as prominently as the amount, bites on actual profit alone. A monetary loss falls on the capital alone; the working party loses its months and its forgone wage, a loss the jurists of chapter 11 counted real. The venture's reasonable expenses run on the venture's account, but the manager draws no salary alongside the share — paid twice, once for labor and once from profit, the share stops measuring anything — and the mandate clause scopes the money to a purpose, a market, or a season, or grants the unrestricted commission: the instrument chapter 11 watched Mecca's caravans run on trust and reputation, now fitted with modern boilerplate.

The *iska*-inspired investment agreement carries the Nehardean geometry into modern drafting: *palga milveh*, *palga pikadon*, half loan and half deposit, the split chapter 11 called "a small masterpiece of moral engineering." Half the advance is a loan, barren by definition, owed back whole and earning nothing; half remains the fund's own property, trading in the venture's hands at the fund's risk, and only that half earns. The venture's minding of the fund's half is paid, a wage the document names; the authorities let it be nominal but never absent — the Mishnah's old rule that the investor pays for the management of his own half, lest free labor become the loan's hidden yield — and the instrument's evidentiary machinery is its signature: losses proven by named standards, profits attested on the venture's books, and, if the parties elect it, a disclosed settlement figure the venture can pay in lieu of opening its books entirely, the *shtar iska*'s classical buyout. There the library writes in its own margin, because chapter 31 already wrote it in the tradition's voice: the settlement figure must stay a conservative model of the venture's actual expected gain, and the deposit half must genuinely stand in the venture's weather, or the paper has begun lying about the money. The *poskim*'s own warning belongs in the drafting file verbatim — "Paper, however brilliant, cannot hold itself to the world." — and the library's answer to it is structural rather than pious: the disclosure schedule at the drawer's end, and the analyzer of the next chapter, exist to keep this instrument's elegant machinery honest about the world it is modeling.

The *societas*-inspired partnership agreement is the Roman chassis for ventures in which each side brings capital, or capital meets capital and labor in any mixture. Formed by consent; scoped surgically — one venture, one facility, one voyage of whatever modern kind, dissolved at the accounting — or standing, at the parties' election. Ownership is never transferred: each partner's contribution remains that partner's, at that partner's risk, the clause on which Aquinas rested the whole acquittal, and the shares are whatever the partners agree, equality the default where the paper is silent, each share answerable to a

real contribution — money, labor, anything that is actually real — chapter 14’s doctrine of proportion. The care standard is the Digest’s *diligentia quam in suis*, chapter 13’s finding rendered as a covenant: each partner owes the venture the diligence he shows his own affairs, a standard the drafting states plainly so that the party who chose a careless partner remembers, in the bad quarter, whose choice that was. Loss allocation obeys the ownership arithmetic, and no clause may redraw it; the exit provisions run through chapter 14’s valuation machinery, sober appraisals and disclosed methods, so that no partner’s departure is priced on hope; and the lion clause from the common boilerplate stands over the whole: nobody signs into a seat that cannot lose.

The Deeds That Hold

The seventh, eighth, and ninth instruments are not financing papers; they are the papers that keep the financed thing whole, and the library shelves them together because chapter 29 taught them as one family: the deed, the lock, and the offices.

The mission-preserving buyback agreement is the exit instrument for the stack’s equity floor, and it delivers in clauses what the last chapter promised in one line: redemption “at a formula written the day the money went in, returning the shares to steward hands on a schedule the company can survive.” The formula clause fixes the redemption price’s method, not its hopes, at signing: a stated multiple or a stated appraisal method with disclosed inputs, chapter 14’s sobriety applied to the one valuation everyone is tempted to inflate. The schedule clause subordinates every redemption payment to the company’s actual breath: installments run from free cash after wages, suppliers, and the reserves at their charter sizes — chapter 39’s procession made a condition precedent — and stretch, without penalty, when the till says wait. An acceleration clause is exactly what this instrument refuses to carry: the company’s bad year postpones the buyback; it never triggers it. The initiation clause names who may set the wheel turning — the company, its stewards, the fund at a stated horizon — and pointedly omits anyone hostile; and where a company comes to the fund still unlocked, the last chapter’s design travels into this paper: conversion rides the investment from its first page, priced and calendared at signing, so that the money, in that chapter’s words, “finances the dedication rather than waiting politely for it.” The destination clause does the mission-preserving: shares redeem to steward hands, the trust, the foundation-held class, the steward-buyer named or nameable, never to the open market, so that the exit is a homecoming rather than an auction. Capital leaves such a company the way a good tenant leaves a house: paid in full, on notice, with the home intact.

The asset-lock covenant is one page, and the page is a wall. Its operative clause binds the company’s assets and their value to the stated purpose: no distribution, transfer, or winding-up may convey mission-built worth to private hands, and on dissolution whatever remains passes to another asset-locked body, named or defined. This is the *waqf*’s deed and the *hekdes*’s dedication and the church’s patrimony running in company law — the family chapter 29 called “the drawer where property goes to stop being property” — and its living statutory cousin is the community interest company’s compulsory lock, the clause that chapter quoted from the regulator’s own guidance. The covenant’s honesty is that it knows its own jurisdiction. A lock binds everyone inside the legal order that enforces it and cannot bind the sovereign who rewrites the order, chapter 29’s second enemy, which is why the library drafts the lock redundantly — in the articles, in the covenant, in the guardian’s consent rights — so that undoing it requires not a signature but a campaign.

The stewardship governance charter is the longest document in the drawer, because offices are wordier than prohibitions. Its first part writes Steward Ownership's two commitments into binding text: control sits with people bound to the mission, steward shares that are never marketable and never heritable, passing by appointment and never by sale or blood; and profits serve the purpose — reinvestment, repayment of capped capital, reserves, mission spending — with no residual pipe to private pockets. The selection clauses make stewardship an office with a doorway: eligibility stated, appointment by a defined process answerable to the mission rather than to the outgoing steward's affections, and removal for cause by an authority that does not report to the removed. Its second part builds the guardian: the golden share of chapter 29, a holding of almost nothing that may forbid almost anything, held by an independent, asset-locked body of standing — itself steward-owned, so the guard is built of what it guards — whose consent is required for the short list chapter 41 drew of "the acts that would undo the drawing." The charter names them: amending the lock or the purpose; creating or converting share classes; sale, merger, or liquidation; and, at fund level, the last chapter's list delivered as promised — unlocking a layer of the stack, raising the cap, drawing the reserve outside its charter, redrafting the waterfall. Its third part seats the voices: the workers, borrowers, members, and stewards the structure exists for hold named seats, in the pool's assembly, in the fund's council, in the Employee Ownership trust where the workforce holds the company, with information rights drawn wide — because chapter 41's rule holds here in its own words: "the people at the thin end of a structure feel its drift before any auditor books it." And its fourth part is maintenance, chapter 29's exportable cargo made schedule: dual control at the points where money moves; a trustee who is not the beneficiary, answering to an authority who is neither, and removable by it; compensation for every office fixed to the office and never to the flows it oversees; and the annual re-asking, the charter's requirement that the institution put the formula's questions to its own instruments once a year and publish the answers.

The Schedule That Confesses

The tenth instrument is the shortest and governs all the others: the anti-extraction disclosure schedule, an exhibit attached to every document in the drawer, renewed annually, drafted to be read by a stranger. Chapter 37 supplied its reason in one sentence — "No single clause is the usury; the stack is the usury." — and clause-level virtue is exactly what the schedule refuses to settle for. Nine honest documents can still assemble one extractive arrangement if their claims stack wrong, so the schedule discloses the stack. Its lines are the formula's lines, chapter 40's published-runs practice turned into a form. Every claim above the zero line appears with its title and its proof: costs against invoices and the allocation policy; losses against the netting that proved them; work against the wage, the hours, and the office; patience against the shown burden, or against the honest zero the mercy floors bear by design; the surplus share against its ratio, its cap, the cumulative amount collected under the cap, and the remainder it was collected from. The repayment order is disclosed in plain sequence — who stands where, and whose remainder pays them — so that the queue — the instrument chapter 37 caught doing a rate's work in the dark — does whatever it does in daylight. The contribution schedule from chapter 38 rides along: what money arrived, on which dates, in which amounts, and what arrived unwired — the founder's decade, the workforce's forgone wages — priced on the work line or confessed as unpriced. The reserves are listed against their charter sizes. The related-party register names every affiliated fee, service contract, and management company, chapter 38's side doors, because the side door is where extraction goes when the front hall is watched. The

amendment log lists every change made to any instrument in the file since the last issue, with the guardian's consent noted beside each, so that drift cannot happen quietly even where it happens legally. And the last line is the verdict line: whatever any party collects that no titled line explains is published, by name, as the extraction residual. The schedule exists so that the line reads zero, and so that a stranger with an afternoon and the exhibit can check that it does. Appendix E runs sample waterfalls in worked numbers, the schedule's arithmetic demonstrated end to end; the schedule itself is the blank the fund fills in every year, in public. A term the parties cannot show on the schedule is a term the library will not let them sign.

Close the drawer, then, and notice what has happened to the book's argument. For forty chapters the traditions' rules lived in treatises, responsa, decretals, and fatwas: binding on the conscientious, invisible to the closing room. The library moves them into the room. A founder can now sign a mercy loan whose kindness is enforceable, a funder can take a participation whose cap is a clause rather than a hope, a town can hold its anchor company through a charter no buyer can charm open, and every party to all of it can read, on one exhibit, exactly what everyone is collecting and out of what. None of this requires the signatories to share a theology; the instruments carry the agreements of three houses that could not read each other's books and built the same fences anyway. What paper cannot do, the drawer's own contents keep saying — the iska file's warning, the lock's candor about sovereigns — is hold itself to the world. Documents are promises about conduct, and conduct will need examining as long as people sign things. So the library's last service is to its own examiner: these ten instruments, drafted to the formula, are also the reference set for the analyzer the next chapter builds, the machine that reads any term sheet, including these, against the code the drawer was built from and says where the theology leaked out. Chapter 44 will keep the ledgers telling the truth long after the closing table is cleared, and chapter 45 will study the hands that reach into the drawer with other intentions. A contract library is a tradition's conscience, filed for reuse. The instruments are ready. What remains is to find out what happens when paper this honest meets money that isn't.

Chapter 43 — A Moral Geometry Contract Analyzer

The two term sheets of chapter 35 have had a long career for two documents offering a single young company an identical half million euros. They entered this book unreadable — “same multiple, same horizon, same yield to the same decimal,” so alike that “A rating desk would file them together” — and were told apart only by a chapter of patient hand labor: each claimed euro interrogated for its title, the dangers netted against the protections, the queue followed to its end. Chapter 40 identified the first sheet as its formula already walking around in the world, and cleared the desk in Amsterdam. This chapter puts both sheets back on it, deliberately, because they are about to be read a third time, by something that is not tired and not impressed and has the whole book for a memory. The hand interrogation worked; it also took a chapter. A working fund sees hundreds of such documents a year, most of them longer than these, none of them drafted by anyone who has read this far, and the gap between one careful reading per chapter and the paper output of a single quarter is the gap this chapter exists to close. The last chapter finished its drawer of ten instruments and ended by commissioning, in so many words, a machine “that reads any term sheet, including these, against the code the drawer was built from” and says — the phrase is worth keeping — “where the theology leaked out.” This chapter builds that machine: a moral geometry contract analyzer, the examiner the tradition was always trying to staff.

The commission is older than the software. For most of their history, the three houses ran contract review as a personal service: a merchant with a doubtful paper walked it to the rabbi, to the mufti, or to the confessor, one trained reader worked through it against a tradition carried in his head, and the answer came back as a ruling, a *responsum*, a penance, a redrafted clause. Chapter 23, watching that practice meet the modern closing room, drew up the job description this chapter inherits: an instrument “that takes the stack as input and returns the moral geometry as output: who is paid, in what order, at every exit value; which clauses moved the losses, and onto whom; what the thirty percent is actually worth across the futures the parties can foresee; where the drafting departs from the plain design and by how much.” And it set the bar such an instrument must clear, in a distinction this Part has honored ever since: “Transparency describes the documents; justice describes the parties.” A data room is transparent. The founder of chapter 37 had access to every clause that took five sixths of her certificate away from her — hers is the thirty percent in that job description — and the access rescued nothing, because disclosure was never sight. What the traditions demanded — the second reading the Mishnah’s window bought the buyer, the named number that armed the waiver, the description that stood in for sight — was that the signer see the bargain whole before it binds — and at modern documentary scale that old demand has exactly one honest implementation. It must be computed.

Say plainly why, because “theological forensics” sounds like an accusation and is actually a kindness. A contract does not hide its morals; it stores them. Every allocation of gain and loss, every seat in the queue, every trigger and floor and fee sits in the drafting in retrievable form, which means the moral questions of this book are not matters of motive or mood but of geometry — readable off the paper by anyone with the patience to compute what the paper does in every future it touches. No one has that patience. That is not a slur on lawyers or founders; it is chapter 37’s arithmetic — a stack’s behavior is the joint product of every clause acting on every other, and the interactions multiply past what any afternoon holds.

The confessor's method did not fail; it was outbuilt. Forensics is what the old pastoral care looks like when it returns properly equipped: not suspicion of the signer but respect for the drafting, taken seriously enough to be read all the way down. The machine's parentage should be stated just as plainly. Chapter 40 promised "the analyzer of chapter 43, which runs this formula wholesale on term sheets that never heard of it"; chapter 41 promised the fund's own stack would be read back "term sheet by term sheet" and graded against its own building code; the last chapter handed over its ten instruments as the reference set. The analyzer is those promises kept — the formula of chapter 40 turned from a drafting standard into a reading standard, pointed at paper from anywhere.

One honest disclaimer before the modules, and it is the book's, not the vendor's. An analyzer is an instrument, and this chapter describes one in the spirit of chapter 41's drawing: a design the traditions' rules specify, not a vendor's product. Nothing below requires artificial cleverness. Every module is a procedure a diligent clerk could run on paper in a long week; the machine contributes speed, memory, and the refusal to get tired, and it inherits the method whole from the chapters behind it. The design is the doctrine. What follows is the walk through it, module by module, in the order the ten questions arrive.

The Intake

The first working module chooses what to read, and it opens with the question chapter 26 assigned to it in advance: ask of every certificate in the file "what its holder can actually reach." That chapter's four-way audit is the intake's sorting bench. Paper may hold the thing itself, title running to the holders; or a usufruct, the fruit of a term; or a named stream, tolls and rents assigned while they flow; or — the fourth thing, the layer chapter 26 warned devours the other three — "a promise about all of the above," the guarantee that pays at par "whatever becomes of thing, fruit, or stream." The analyzer's first pass tags every instrument in the stack with which of the four it is, because everything downstream — which dangers are real, which titles are available, whose weather the money stands in — turns on that tag. A "participation" whose reach is a guarantee is reclassified at intake, whatever its cover page says, before a single euro is modeled.

Intake also draws the boundary of the file, and the boundary is a finding in itself. The unit of analysis is the stack, never the document — that was chapter 37's hardest lesson — so the module demands the whole household: the side letters, the fee agreements, the management contract that lives two folders away from the data room's tidy index, the guarantee signed the same afternoon under a different heading. A stack missing its side letter is a different file wearing the first one's name, and the intake report says which documents were requested, which arrived, and what the analysis therefore cannot vouch for. And what the parser cannot classify it must never invent: an unrecognizable clause is flagged for human eyes and held, not guessed at — a reading machine that quietly papers over what it failed to read has joined the problem it was built to examine.

Then comes clause selection, which is the art of knowing where moral freight travels in a legal document. Most of a contract is plumbing: notices, definitions, governing law, the furniture of enforceability. The freight rides in a short list of clause families, and the analyzer's parser is essentially this book's table of contents turned into a search pattern. Claim clauses — what is collected, and out of what, which chapter 42 called the harder discipline. Security clauses — collateral, guarantees, seniority: the protections that will have to be netted before any danger may be priced. Queue clauses — preference, participation, ratchet, redemption, drag: chapter 37's machinery, the geometry that does a rate's

work without one. Penalty and default clauses — chapter 22's territory, where the test is fault. Control clauses — vetoes, boards, consent lists, amendment paths: chapter 29's territory. Exit clauses — who may force a sale, at what price, toward which hands. Fee and affiliate clauses — chapter 38's side doors, the related-party channels where claims travel without ever walking past the queue. And disclosure clauses — what the paper promises to keep showing after signing. Selection is not neutral bookkeeping; it is the reading rule of chapter 3 running at parse time. Names are read past, per the Majalla's third article and *Gregory v. Helvering*, both carried since that chapter: effect is given to intention and meaning, not to words and phrases. The clause called "advisory fee" is selected into the claim family if it collects like a claim; the "profit share" whose floor and collar flatten its payoff is selected into the queue family with the other fixed seats. The parser, in other words, is the first place the analyzer refuses to be lied to by a heading.

The third module classifies the dangers, and it is chapter 36 made procedural. For every claim clause the analyzer asks the risk grammar's questions in that chapter's order. What is the feared event — named, dated, sized? Is it measurable — does it sit inside a table of frequencies, where the two-factor arithmetic of likelihood and severity can run — or is it the deep uncertainty past the table's edge, where nothing can be priced and everything can only be shared? Then the netting, before any pricing is honored: subtract the collateral, the guarantors, the seniority, the covenants — every protection the same drafting installed — and price only the danger that survives its own contract. The module also refuses to read exposure as a snapshot — "Capital at risk is a curve, not a number," chapter 36 ruled — so an interest-only loan holds its full principal in the wind to the last day, while an amortizing loan sheds exposure with every installment as principal walks home early, and a risk charge that ignores the shedding is billing for danger already gone home. And then the column check, the one chapter 36 taught the book to run last and remember longest: compute the exposure twice, once for the claim and once for the counterparty, because a loan that is nearly safe to the lender can hold, that chapter found, "a three-in-a-hundred chance of the house" on the other side of the table. On the book's own hundred-thousand-euro case file, this module reproduces in milliseconds what the chapter found by hand: a pricing memo attributing five points to credit risk claims five thousand euros a year for danger, and the file, netted, can show about four hundred and fifty — the claim running eleven times the shown burden, the excess already on its way to the residual line. "What can be measured can be pooled; what cannot be measured can only be shared," chapter 36 concluded; the module's whole function is to notice, clause by clause, which side of that sentence each claimed euro is standing on, and whether it is dressed for the wrong side.

The Ledger and the Waterfall

The fourth module is the oldest machinery in the building. It is chapter 35's moral-title ledger run as code — the zero line, the seven titled lines, and the requirement that every euro the paper will collect across its life find a seat whose title is, in that chapter's phrase, "genuine in fact, not merely in name." The module walks the claim schedule euro by euro. Costs go to the expense line against invoices and an allocation policy, at zero margin, or they do not go at all. Work goes to the work line against wage, hours, and office. Losses go to the loss line when they have happened and been netted honestly. Danger goes to the risk line at the size the previous module could show, and no larger. Patience goes to the liquidity line against a shown burden — or against the zero the mercy floors carry on purpose, patience there being priced at nothing by design — since, as chapter 21 ruled, "Duration may deepen a stake; it may not draw a wage." Foregone opportunity goes to its

line under the discipline of chapter 20: named, evidenced, and never presumed from the mere existence of a market. Realized surplus goes to the surplus line by its ratio, under its cap — and the cap is arithmetic the module tracks, not sentiment it admires: cumulative collections run against the bounded multiple stated at signing, so that on the book's staged half-million file the counter stands visibly somewhere beneath the €1,450,000 where — at €2.90 on each euro advanced — capital's claim completes, and the memo says how close, because a cap nobody is measuring is a cap in name only. The surplus line polices its source as well as its size: a distribution is surplus only if it left through chapter 39's procession — wages, suppliers, provisions, the reserves at their charter sizes, every senior claimant paid before the word applies — since "Available says the money can leave; surplus says it may," and the module grades distribution clauses on which of those two sentences they actually wrote down. A single fence runs across the whole ledger, and the analyzer patrols it with special attention because hand-readers miss it most: "a burden may be counted once." No danger may be collected in the rate, extinguished by the collateral, exported to a guarantor, and then collected a second time at default; the module de-duplicates burdens across clauses the way an auditor de-duplicates invoices, and it books the overcounts where they belong. Then the subtraction that gives the machine its verdict line: the euros left standing when every titled line has taken all it can prove are the extraction residual — computed, chapter 35 insisted, not accused, which is what lets two readers who agree on nothing else arrive at the same number. On the hundred-thousand-euro file the module returns the committed arithmetic exactly: of the €60,000 claimed above the zero line, about €4,000 belongs honestly to expense and €2,250 to risk — €6,250 of titled claim, roughly a tenth — and the rest, some €54,000, in that chapter's words, "sits on no line at all," named at last by the colorless word chapter 35 coined for the purpose.

The fifth module answers for the fact that almost nothing arrives as one contract. It is the waterfall model, and it delivers the promise chapter 37 lodged by name — "the analyzer of chapter 43 will run it wholesale" — along with that chapter's work order, quoted here because it is the module's requirements document: "Take the whole stack — every instrument, every preference, participation, dividend, ratchet, redemption, drag — and compute the package's payoff in each state of the company's world, exactly as chapter 35's payoff test demands of any contract." This is the module where the machine's advantages stop being conveniences and become the point, because payoff-by-states is precisely the computation human readers cannot hold. The module builds the state grid — the company's futures laid out from catastrophe to triumph, year by year, the same walk the fund's own planning models take through a portfolio company's life — and runs the closing statement in every cell: who collects what, in what order, out of what remainder. Flat payoffs betray their fixed-claim nature whatever the certificates say — a "*mutuum* with a seat number," chapter 37 called the front of the queue. Payoffs that bend where the venture bends are participation, wherever the paperwork filed them. And equivalences are detected rather than debated: the module assembles each seat's payoff from the clauses that build it, and a package that sums, state by state, to a riskless fixed claim is one, whatever assembled it — detection being construction run in reverse, the put-call parity of chapter 37 reporting for its second shift, this time on the audit's side. On the founder's file from chapter 37, the module does in a second what cost that chapter a drive home: certificate says thirty percent; payoff at the twelve-million sale says six hundred thousand euros, not three million six hundred thousand; and the memo names the clauses that moved each tranche of the missing three million — so much to the participation, so much to the ratchet, so much to the order of the line itself, since a preference, as that chapter put it, "is a rate that learned to stand in line." The waterfall module is also where the fund's own paper reports for duty.

Chapter 41 asked that its stack be read back annually and graded “against the code it was built from”; the module runs the seven layers as one file, checks each floor’s collections against what that layer agreed to refuse and each reserve against its charter size, and posts the whole-stack verdict the disclosure schedule of chapter 42 promises to publish.

The Steering and the Weather

The sixth module reads the clauses that decide who can do things to whom, because extraction, chapters 29 and 41 taught, is as often a governance event as a pricing one. The control model maps the file’s steering: which acts require whose consent; where the amendment paths run and whether any of them bypasses the people the amendment would injure; whether the mission’s locks exist and whether they can be picked — an asset lock that dissolves on a simple resolution is a wall with a gate in it, and the module says so. It checks for the guardian: is there a golden-share holder or its equivalent, independent and itself asset-locked, whose consent stands in front of “the acts that would undo the drawing,” in chapter 41’s list — unlocking a layer, raising the cap, drawing the reserve outside its charter, redrafting the waterfall? It checks the offices against chapter 29’s maintenance cargo: dual control where money moves, trustee and beneficiary never the same person, compensation fixed to the office rather than to the flows passing through it. It reads the drag and the ratchet a second time, as steering rather than payoff — a drag-along is the power to sell other people’s company, and the module prices who holds it and what stands between that power and the workforce whose futures ride in the same signature. And it walks chapter 38’s side doors: the related-party register, the management contracts, the affiliate fees — every channel through which control converts into claim without asking. The output is a map with pressure points marked, and the sobering, useful discovery on real files is how often the map is the finding: money flawlessly titled, and steering concentrated so that every title can be revised next year by the party collecting under it. Drift, chapter 29 said, is a design load. The governance module is the wind gauge.

The seventh module models the weather itself. Liquidity pressure is the file’s behavior in the bad month, and the module’s method is to simulate the bad month rather than admire the covenant page. Walk the state grid’s low road: revenue misses, a payment is late, a covenant trips. Now read what the drafting does. Does the paper distinguish the borrower who cannot pay from the one who will not — chapter 22’s boundary — or does one missed week accelerate the whole balance regardless of cause? Do penalties fire on fault, with culprit conditions the drafting names, routed away from the collector’s own account so that no one profits from a stumble — or do they fire on time, metering delay like a parking garage, in which case the module books them as what chapter 22 taught: a rate in enforcement’s clothing, destined for the residual? Is there a collection floor — the trade’s tools, the bed, a month of food, the exemptions all three houses drafted centuries before modern statutes rediscovered them — or does the security package reach the person? Does deferral exist as a right where inability is shown, as the traditions of chapter 22 made it, or only as the creditor’s mood — and if the paper does grant the deferral, is anything standing behind the grant? A hardship clause with no reserve behind it is a promise the first hard winter will break, which is why chapter 21 set a fund-level anti-ruin reserve behind the mercy layers: patience, that chapter found, is survivable to the one extending it only when someone has provisioned for it, and the module checks the provisioning, not just the promising. It also reads the file’s weather as one system rather than page by page: cross-default clauses wire every document’s bad month to every other’s, so that a covenant tripped in one agreement can detonate remedies across the stack — the liquidity

model runs the cascade and reports what the whole file does to the borrower's worst quarter, not what each page does to its own. And then the dials that Comparative Interlude I bequeathed, the ones the Dharmaśāstra stated more explicitly than any tradition in the braid: does any chain of rollovers, accruals, and penalties carry the total claim, in the interlude's words, "past a fixed multiple of what was actually advanced," or is there a hard ceiling, *damdupat*-fashion, beyond which the meters stop? The module also checks the accrual stops the interlude specified — they must engage automatically at demonstrated hardship and at tender, the moment the borrower cannot and the moment he offers what he owes — and it scans the pricing for identity: anything keyed to who the borrower is rather than what the venture is, a person's station doing the work that documented hazard should do. "Price the road, never the man," the interlude distilled, and the module runs that sentence as a check. The liquidity model's summary line for any file is a single honest paragraph: here is what this paper does to a person having the worst quarter of his life, stated as machinery, before anyone signs.

The Missing Lines

The eighth module hunts for what the register forgot, and its textbook is chapter 38. Every file arrives with its wired contributions perfectly remembered — the register keeps the money to the cent and the second. The founder-labor model rebuilds the other schedule, the one that chapter specified: "every input the venture actually consumed, entered by name, dated, and priced at entry." Wages foregone against the stated rate for the work actually done — the roastery's crew, chapter 39 found, ran "thirty thousand euros a year under standard," wages earned and unpaid, standing in the company overnight on no security at all. The founder's declined salary, priced at the market rate she turned down, not at the zero the cap table printed — "Zero is also a price," chapter 38 ruled, "the only one no one is ever asked to defend," and the module refuses to let the file defend it by silence. Guarantees written on the house; equipment contributed; the decade of evenings — whatever was real, entered and priced, because chapter 14's doctrine of proportion makes reality the only test a contribution must pass. Then the confrontation: run the waterfall module's exit arithmetic against the completed schedule and see whose inputs came home and whose funded the celebration unthanked. "What the exit calls profit, the schedule may call repayment," chapter 38 found, and on files like its own staged one, the module finds precisely that: a founder standing last in the queue for money that was, by the schedule, hers already, her missing line items having financed everyone senior to her. The module's output is the contribution schedule as exhibit, chapter 42's disclosure form pre-filled, with every unwired input either priced on the work line or confessed, in writing, as unpriced.

The ninth module asks the oldest question in the book's mathematics: can everyone in this file survive it? It is chapter 28 made a procedure, and it begins by refusing the average. Ensemble arithmetic — the expected value across many parallel outcomes — is the grammar in which every pitch deck already speaks, and it is the grammar, that chapter proved, in which a system can flourish while most of the people inside it are destroyed. So the module computes on the time axis: each participant's path, sequence by sequence, compounding on itself, with ruin modeled as what it is — an absorbing floor, a locking door, a state no later good year reopens. It prices ruin beyond offset, because a bankruptcy is not a debit to be netted against someone else's credit; and it reads every claim at the worst year, not the mean one. Where the file contains a pool — a solidarity layer, a *takaful* wrapper, a mutual chest — the module runs chapter 27's clause tests first: contributions entering as the members' own and never as a seller's revenue; the administrator's wage fixed and indif-

ferent to refusals; exclusions policing conduct, never misfortune; outside capital serving, never owning. And then it runs the two books side by side, solo and pooled — the same ventures with and without the chest, the entry reserve, the contingent claim — and reads the three numbers of chapter 28's village: collections nearly unmoved, median paths alive, and the ruin column, which is the only place solidarity can prove it is architecture rather than adjective. The village itself rides along as the module's calibration case, its committed numbers unchanged: households provisioning a hundred and twenty units a year against blows averaging a hundred, prosperous in every ensemble ledger, and still ruined one by one because a jar holding ninety-six or less loses to a hundred-unit blow every time it comes — until the chest, and then not. That chapter's specification travels here in one sentence, quoted because it is the module's requirements document: "judge every contract on the time axis, at the worst year, with ruin priced beyond offset, and let pooling prove itself in the ruin column or not at all." And its verdict on decoration travels too — "a 'solidarity reserve' that does not move the ruin column is a marketing line, and the simulator will say so" — a sentence this module exists to make true in print.

The Memos

Which arrives at the tenth module, the one the other nine exist to feed, and its form matters as much as its findings. The analyzer does not return a score. Scores are for rating desks, and the rating desk is the instrument that filed the two Amsterdam sheets together in the first place; compressing a moral geometry to one digit is how the geometry was lost. The analyzer returns memos — one per document, one for the file — written for the person about to sign, in the order her risk arrives, in language that never requires this book. The document memo walks the instrument: what kind of paper this is and what its holder actually reaches; what it claims and on which titles, with each title's proof attached or its gap named; what it does in the bad month; what it costs at the worst year. The file memo assembles the stack: the waterfall at every exit, the whole-stack ledger, the residual line with its addressees — because the extraction residual, chapter 37 established, "is funded, euro for euro, by the residual claimants," and a memo that names the payers has told the founder what her certificate is worth in a way no percentage ever did. And every finding closes with its remedy, because the last chapter built the shelf the remedies live on: the penalty that meters time points to the mercy papers' fault-and-charity clause; the naked downside points to the participation drafted to lose; the unpriced decade points to the contribution schedule; the pickable lock points to the covenant and the guardian. A finding without a cure is a complaint; the library is what lets the memo end, every time, with a page number.

The memo is the oldest genre in this subject wearing modern formatting: the *responsum* answered a merchant's file; the fatwa answered a fund's structure; the confessor's counsel ended, at its best, not with a verdict but with what to change. Three houses that agreed on almost nothing agreed that paper deserved a learned second reading before it bound the unwary, and none of them ever pretended the reading was the judgment. Neither does the machine. The analyzer prepares the case; the deciding stays with people — the signer who may sign anyway, the fund council reading its own grades, the guardian with the consent pen, the communities whose rules these were first. What the machine removes is the one excuse every era of this history has heard at the closing table: that nobody could have seen it coming, in a document nobody could have read in time. The memo is delivered before signature, in the tradition of the Mishnah's window — and so the promise chapter 23 made for this instrument comes due exactly as written: that it would stay beside the

person at the pen until “the moment the signature becomes true, because this time, before she signed, she saw.”

So run the machine, finally, on its own commissioning paper. Fed the last chapter’s ten instruments, the analyzer returns ten short memos and one file memo reading, in effect: titles proven, queue published, residual zero — the reference set behaving as a reference set, which is calibration, not celebration; an examiner must know what innocence reads like before it can recognize anything else. Fed the two Amsterdam sheets, it files them apart in the time it takes the pages to load, and prints for the first sheet the memo this book took eight Parts to earn. And fed the world’s ordinary paper, it does the only thing forensics ever promised: it makes the geometry visible while the pen is still capped. The metrics themselves live in Appendix D; Appendix E works the waterfalls in full numbers. What the analyzer cannot do is keep anyone honest after the signing — that is a ledger’s work, and chapter 44 builds the accounting that does it — and it cannot stop a determined hand from gaming the code it grades against, which is why chapter 45 studies the temptations last. A machine can read the theology out of the paper. Getting it into the paper was the last chapter’s work; keeping it there, it turns out, is the rest of the book’s.

Chapter 44 — Accounting for Moral Economy

Every machine this Part has built stops working at the signature. The formula of chapter 40 is a drafting standard; the fund of chapter 41 is a drawing; the library of chapter 42 is a drawer of unsigned paper; and the analyzer of chapter 43 reads the file before the pen comes out. Each of them, finished, said so. Chapter 40 promised an accounting that “keeps the answers honest after signing.” Chapter 42, closing its drawer, promised that this chapter would “keep the ledgers telling the truth long after the closing table is cleared.” And the analyzer’s last memo conceded the limit of all forensics: keeping anyone honest once the ink is dry — “that is a ledger’s work.” This chapter builds the ledger’s work. It is about the working life of the money: the years in which the beautiful geometry of the signing day meets rent, payroll, a bad quarter, a good exit, a tired bookkeeper, and a board that has half forgotten what the clauses were for.

Begin with why the subject deserves a chapter rather than a clerk’s appendix. An institution does not remember the way a person does. The founder who could recite every promise retires; the guardian who negotiated the cap is replaced; the clauses stay in a drawer that no one opens because nothing has gone visibly wrong. Chapter 29 named the load-bearing fact — drift is a design load — and the load arrives annually, whether or not anyone is wicked. And drift’s first footprints are always in the books. Before a mission is betrayed it is reclassified: a penalty starts life in a delay state and ends it in a revenue budget; a reserve is drawn “temporarily”; a fee migrates from invoiced cost to standing percentage; a founder’s unpaid decade quietly finishes becoming nobody’s liability. None of these requires a villain. They require only an accounting system with no place to notice. That is what this chapter means by calling accounting moral memory, and the phrase is meant literally. The books are where an institution keeps what it promised, in a form that someone who was not in the room can check after everyone who was in the room is gone. Intentions cannot be audited; entries can.

The traditions understood this long before there were funds. The Qur’an’s longest verse concerns neither prayer nor pilgrimage nor war; it is about writing down debts. “You who believe, when you contract a debt for a stated term, put it down in writing,” it opens, and then legislates the machinery for a page — “have a scribe write it down justly between you,” in Abdel Haleem’s rendering, which chapter 6 read whole: scribes who may not refuse the work, the debtor dictating his own obligation, witnesses who must answer the summons and may not conceal what they saw. The Talmud’s charity law, chapter 18 found, choreographed collection day like a liturgy of internal controls: two officers, never one, each walking where the other could see him; a found coin dropped straight into the purse before witnesses; the coins told out singly, never in pairs, so that no bystander could imagine a trick; and distribution by a panel of three, because to allocate the poor-fund is to judge a monetary case, and such cases take a court. And the Council of Chalcedon, in 451, in the same sitting that gave the greater part of Christendom its Christology, ordered that “every church having a bishop shall have also a steward from among its own clergy, who shall manage the church business under the sanction of his own bishop” — canon 26, as chapter 29 carried it: the *oekonomus*, installed beside the creed. None of the three houses trusted virtue where money moved, including its own. They trusted entries, witnesses, and coins counted in daylight — not because they thought bookkeeping holy, but because they knew memory is mortal and money is patient.

So the chapter ahead walks the books such a memory requires, with chapter 41's fund as the standing example — and, like the fund and the analyzer before it, what follows is a design specified by the traditions' rules, nobody's product. Eight kinds of account, taken in the order the money meets them: the fund's own books; the books it requires of its companies; the second set that maps every euro to its title; the chest's ledger and the mission's reserve; the accounts of the people the wire register never saw; the line for the world outside the file; and, last, the counting in public — audit, disclosure, and what happens when the numbers confess.

The Books of the Fund

Chapter 41 drew the fund as seven layers, ten million euros committed from gift floor to community reserve, and the drawing's rule was that every layer earns its place by what it declines to collect. The fund-level ledger writes those refusals down as accounting policy — for each floor, an account that exists and several that deliberately do not. The gift layer's grants are booked as spent for purpose, full stop: no receivable is created, nothing accrues, and the layer's return is reported in the mission's terms rather than in euros, because a gift that books a claim has stopped being one. The mercy floor's deposits sit as liabilities repayable at par — the depositors' money, never the fund's income — and its repayments are booked as the wheel turning: redeployment, not yield. Its income line reads zero on purpose, the honest zero chapter 21 taught the liquidity title to bear on the mercy layer. The cost-recovery floor may book income equal to invoiced, allocated cost and not a cent of margin — chapter 19's discipline, loss and repair in their own lines at zero markup, kept as a posting rule rather than a sentiment. The solidarity pool's money does not appear as the fund's at all, for reasons the next section spends properly. The partnership floor recognizes profit only when profit is realized: interim distributions ride as provisional advances in the fiqh's manner that chapter 14 verified — retained against the capital's protection, callable back until a real valuation speaks — and nothing accrues for hoped-for gains, hope having no account here at all. The mission-locked equity floor's distributions run under their cap. And the community reserve's whole budget is its charter size: never deployed, never distributed, its only honest transactions being replenishment and the rare, logged draw.

Two running accounts belong to this fund in particular, and both exist so that a clause elsewhere in the file can operate. First, the cap counter. The bounded multiple of chapter 40 is a number stated at signing — on the book's staged half million, capital's claim completes at one million four hundred fifty thousand euros, two euros ninety on each euro advanced — and a stated number is only a fence if something measures the herd against it. So the counter is an account: every collection above the zero line posts to it, cumulatively, instrument by instrument, and the year-end statement publishes each certificate's counter beside its ceiling. When the counter arrives at the cap, the recycle-or-stay clause of chapters 40 and 41 fires — and it can only fire because the books were keeping score. The second is the residual account. The extraction residual — chapter 35's remainder, the euros a claim collects for which no titled line can answer — is not left to auditors to discover; it is a named account in the fund's own ledger, computed by subtraction every time the titles are posted, and its expected balance is zero. The account exists to be empty. A posting to it is not a bookkeeping event but an institutional one, with addressees, and what must happen next is the business of the chapter's final section.

One more fund-level rule turns the whole apparatus from description into governance: the gate. No distribution leaves for the members while the ledger is dirty — while any titled line rests on proof not yet produced, any reserve stands below its charter size, any counter

is unposted, or any balance sits in the residual account unaddressed. The clean ledger is a defined accounting state, checkable by a clerk, and the fund's own planning discipline runs the same way: surpluses recycle into new deployments with the reserve kept whole at its protected size, so that the evergreen wheel of chapter 41 turns on posted numbers rather than optimism. A distribution, in this fund, is not a decision the board is in the mood for; it is a state the books are either in or not in. Print the fund's year-end statement, in fact, and it is chapter 40's formula restated as balances: the zero line's principal home or provided for; each titled line at its proven size; the reserves standing; the counters beneath their caps; the residual row at zero. The formula told the fund what it might collect. The statement is the fund answering, once a year, under its own letterhead, what it did collect — and out of what.

Beneath the fund sit its companies, and the fund lends its chart of accounts downward. The portfolio-company ledger is chapter 39 made a bookkeeping standard. Its ordering rule is the procession: the company's books are built so that the claims senior to any distribution stand as accounts with balances, not as sentiments the January meeting must reconstruct. Wages to standard are booked monthly as what that chapter found them to be — "It is a debt, and a debt with a clock on it" — so the crew's under-standard gap accrues as a liability, visible, dated, senior. Maintenance runs on a funded line, since "An entry confesses the wear; only a funded line pays for it." The chest's replenishment, the mission reserve's schedule, the community's owed line, and the priced footprint each carry their own account, posted through the year. Do this, and the roastery's famous January changes genre. On the screen, one hundred twenty thousand; spoken for by the procession, one hundred fifteen; distributable, five thousand — but where chapter 39 had to argue that reading into existence, a procession-ordered ledger simply closes the year and prints it, because "ninety-six cents of every available euro already had a name on it" and now the names are account names. What that chapter performed as an interrogation, this ledger performs as a closing entry. And the fund needs the company's books this honest for a reason that is arithmetic before it is virtue: the fund's own surplus line is a ratio of exactly this remainder. A funder entitled to a share of true surplus can collect honestly only from a company whose ledger can compute one — which is why the procession-ordered chart of accounts arrives in the deal documents beside the term sheet, not as a courtesy afterward. Distributable surplus stops being a claim anyone makes and becomes a balance the books compute.

The Second Set of Books

"Keeping two sets of books" is the oldest shorthand for fraud in the language of money: one ledger for the taxman, one for the truth. This fund keeps two sets of books and publishes both, and the inversion is the whole design. The first set is statutory — whatever the law of the seat requires, kept exactly, because a fund that quarrels with the tax code has chosen the wrong fights. The second set is the moral-title ledger, chapter 35's construction promoted from a chapter's exercise to a living record: one zero line, seven titled lines, and a rule that every euro the fund collects, on any instrument, from any counterparty, finds its seat or falls to the residual. The second set is not a shadow ledger, hidden and rival; it is a mapping, open and total. Every account in the statutory books answers to a title — this fee income is expense recovery, proven so; this gain is surplus under such a ratio and such a cap — and whatever answers to none is not smoothed into a category but posted where the fund must see it.

The posting rules are the doctrine, compressed. A euro of collected cost reaches the ex-

pense line only with its invoice and its allocation policy attached. Work enters against the wage actually stated, the hours actually kept, the office actually held. A loss enters after the event and after honest netting, every recovery already subtracted. Danger enters at its demonstrated size — chapter 36's netting done first — never at its advertised one. Patience enters against a documented deprivation, or bears the deliberate zero of the mercy floors. Foregone opportunity enters under chapter 20's discipline: a named alternative with evidence behind it, never an inference from the bare fact that markets exist. And realized surplus enters by the stated ratio, beneath the stated cap, out of chapter 39's remainder and nowhere else. Across all of it runs the fence the ledger was born with — chapter 35 wrote it as bookkeeping's oldest rule, "a line, once filled, is filled" — so a burden charged in the rate cannot be charged again in the collateral, the guarantee, or the default clause; the de-duplication the analyzer ran across clauses at reading time, the second set runs across documents at posting time, every month, as a matter of routine.

Penalties get a paragraph of their own because their accounting is their theology. The second set carries no standing penalty-income line, and the absence is deliberate: "A genuine penalty is priced to collect nothing," chapter 35 ruled when it refused the *poena* a seat — so a fund whose plan shows penalty revenue has stopped planning to be obeyed and started planning to be paid for disobedience. When a delay state actually arrives, the books sort the event the way that chapter's two-tier rule sorts it: documented repair — the cost of the chasing, the proven loss of the waiting — may be kept, and posts to its lines against its evidence; everything above repair leaves the house entirely, bound for charity with the oversight real, and a routing entry proves the leaving, event by event. The rule, as chapter 22 compressed it: "repair may be kept; punishment must be given away." The second set makes that sentence a posting path. The clause did the promising; the account does the keeping.

What the second set changes, in the end, is where a fund's creed lives. Classifications are commitments. An institution that books a penalty as yield, a repair as return, or a fee as margin has amended its theology — without a vote, in the middle pages of a report nobody reads — long before any charter is touched. Run the mapping and the amendment has nowhere to happen: every reclassification is visible as a transfer between titled lines, and the year ends, every year, at the same bottom row — the residual, computed by subtraction, expected at zero, published either way. Appendix D houses the metrics that grade these balances; the worked waterfalls of Appendix E run their arithmetic end to end. The second set of books is the fund's conscience issued a closing procedure — and unlike a conscience, it cannot be talked out of what it remembers.

The Chest and the Reserve

The solidarity pool keeps the strangest books in the file, because the first thing its ledger must record is whose money it is not. Contributions enter the chest as what they are, the members' own — dues in the burial-club manner, *tabarru'* in the *takaful* manner, chapter 27's genus-move performed in the opening entry — and never as anyone's revenue. The fund administers the pool; it does not own it; and the consolidation refusal is itself an accounting rule: the chest's assets do not appear on the fund's balance sheet — the bishop of chapter 29 kept the books of the *patrimonium pauperum* and owned none of it, and the fund holds the chest the same way. The operator's compensation posts as a fixed expense of administration, agreed in advance, indifferent to the year's weather — the old rule of the pools, as chapter 27 gathered it: "pay the steward wages, never a share of the misfortunes his members were spared" — and the indifference is checkable in the books, because a

wage line that rises when claims are refused has confessed in numbers what no charter would confess in words. Claims paid are booked as the purpose performing, not as damage suffered. The commercial vocabulary calls a heavy year a bad “loss ratio”; the chest’s ledger has no such line to write, because — chapter 27’s verdict — “A member’s claim is the pool at work; a customer’s claim is the seller’s loss.” A chest that paid nothing all year did not have a good year. It had no year. Two more entries complete the pool’s chart. A light year’s surplus has its destination written in advance — “home to the members or forward to the mission, never out as the light year’s yield,” as chapter 27 specified — so the books post it where the founding papers said, not where the good year tempts. And the founding entry itself, the reserve contributed to open the chest per chapter 28’s architecture, sits as what it is: the members’ common cushion, or a served loan from outside capital that may stand behind the pool and may never own it.

The chest’s annual statement also reads on the axis the whole institution lives on: time. Its balance stands against its charter size; its year is tested at the worst plausible stretch, not the average one; and the number the report exists to publish is the ruin column — how many member paths the design kept short of ruin. Chapter 28’s village calibrates the report as it calibrated the model: households provisioning one hundred twenty units against blows averaging one hundred looked rich in ensemble average and were ruined household by household, jars of ninety-six losing to hundred-unit blows, until the common chest turned fifty-seven ruins in a hundred into about nine. Those are the numbers a solidarity ledger is for. A pool’s books that report contributions, expenses, and investment income but never the ruin column have measured everything except the reason the chest exists.

The mission reserve keeps the quietest books and needs the strictest ones, because nothing tests an institution like money that is supposed to sit still. Its accounting is three rules. It is funded on its written schedule, ahead of distributions — the procession seat chapter 39 gave it, kept as a standing order rather than a January mood. It is drawn only for its charter’s purposes, and every draw carries the guardian’s logged consent, since an off-charter draw on the reserve stands on chapter 41’s short list of acts no officer may take alone. And it is held at its charter size, or the fund-level gate stays shut. Behind the three rules stands the oldest earmarked accounting in the book’s story. The church’s fourfold division, chapter 29 found, split revenues into standing portions — bishop, clergy, poor, fabric — with the poor written into the division as co-beneficiaries, not remembered at the administrator’s discretion; the *waqf* ran on its deed, the trustee entitled to his reasonable bread and to nothing that the purpose owned; and the same tradition supplied the caution, watching family endowments decay into dynastic shelters run by trustees who inherited the office and mined it. Perpetuity, that chapter concluded, is “cheap to declare and expensive to administer, and the administration is the institution” — and the institution, day to day, is bookkeeping before it is anything grander. A mission reserve with a schedule, a charter, a logged consent trail, and a published balance is the *hekdes* chest and the poor’s quarter given a modern chart of accounts; a mission reserve without them is a promise compounding, unwatched, into a discretion. Chapter 41 said it of every reserve, and the books exist to make it false: “an undocumented reserve is a discretion, and discretions drift.”

The Unwired Accounts

The wire register remembers money perfectly and remembers nothing else, and chapter 38 measured what the forgetting costs: a twelve-million-euro exit, celebrated in every register but the honest one, that turned out — once every input stood on the schedule — to be seven hundred ten thousand euros of the working seats’ own principal coming home under

profit's name, beside a true surplus of three hundred ninety thousand. That chapter built the corrective as an exhibit; this one keeps it as accounts, opened at founding and posted for as long as the venture consumes anything it does not pay for. "The register kept only the wires. The schedule keeps the venture."

The founder contribution account is the schedule's first standing page. It is posted at entry prices, in the discipline chapter 38 set: the salary she declined, at the rate she declined it, month by month, not reconstructed a decade later from grievance; the guarantee written on the house, at what an unrelated guarantor would have charged; the equipment at its door price; the decade of evenings priced at an honest estimate, because the account, in that chapter's words, "prefers a rough number to an exact zero" — a rough figure invites argument, and a zero ends it, in favor of whoever wrote the paperwork. The balance is not a sentiment; it is principal. It stands in the zero line's roll with the wired money, senior to surplus by the ordering every tradition in this book agreed on, and it collects the way principal collects: at the till, on the procession's schedule; at an exit, ahead of any share of gain. What the founder's account ends is the oldest silence in the venture file — the one in which her decade finishes becoming, by pure bookkeeping, nobody's liability.

The workers' accounts run on the same page and answer an older chapter still. Chapter 12, watching the shopkeeper mind the investor's half of the stock unpaid, gave this book its first accounting verdict: "What the ledger cannot see, the ledger will not pay." The workers' contribution accounts exist to make the ledger see. Each month the crew runs under standard — the roastery's thirty thousand a year, chapter 39 measured — posts to named accounts as wages earned and unpaid: a debt senior at the till, and, where the venture's paper says so, a participation at the exit by the stated fractions. The account does not make the workers investors by courtesy; it records that they already were, nightly, on no security, whether or not anyone wrote it down. Bookkeeping is the difference between a workforce whose sale-day toast was poured from its own unpaid wages and one that stood in the schedule all along.

The last of the unwired accounts belongs to no one at the table, and its name has been waiting since chapter 2 counted the figures a loan must answer to — the poor, the brother, the neighbor, the stranger. Call it the stranger's line: the account for what the venture takes from, and owes to, the world that signs nothing. The roastery's ledger already carried two of its postings — the community's owed line and the priced footprint — and the externality account generalizes them: the river the process warms, the road the trucks wear, the town whose air carries the roast — each named, each dated, each priced at an honest rough figure under the same rule that governs the founder's decade. Two fences keep the line honest. It is measured, never assumed — a footprint priced by wish is decoration wearing a number — and it is a price, never a license: the account records what is taken and funds what is owed, and paying the entry does not consecrate the taking. Chapter 34 drew the boundary this account sits on: "The new conscience audits the company; the old one audited the contract." The stranger's line is where the two consciences finally share a page — the company's footprint, the sort of fact the new conscience gathers, posted into the contract-level ledger as a claim with a seat in the procession, paid to the town's line and the river's remediation the way any senior claim is paid: before the distribution, or the gate stays shut. And because a rough number is only as honest as the people allowed to argue with it, the line is read aloud where the strangers have voices: the community seats chapters 41 and 42 built into the fund's council and the pool's assembly exist, among their other work, to tell the bookkeeper what the road and the river actually cost this year.

Counted in Public

Chapter 19, closing the file on repair, promised that this chapter would “give the ledger its audit,” and the audit is where every account above earns its keep. Its tests are the book’s, gathered. Proof per line: every titled balance rests on its evidence — invoice, wage record, netting, shown danger, documented deprivation — or is reclassified while the auditor watches. Counted once: the de-duplication the analyzer ran at reading time, re-run across the year’s actual postings, so no burden collected twice under two headings. Appetite: chapter 22’s test, quoted here because no audit test in this book bites harder — “if next year’s budget balances only because default charges are projected as revenue, the model has inverted” — run not on the penalties alone but on every mercy in the file: a hardship process budgeted as a profit center, a collection floor priced for breach, a deferral clause the plan assumes will be waived. Funded, not confessed: the reserves at their charter sizes, the maintenance line wet, the mercy floors provisioned — the anti-ruin reserve chapter 21 set behind the mercy layers standing at strength, patience being survivable only where it is provisioned. And the gate: no distribution left on a dirty ledger, checked against the postings, because a gate the books cannot close is a covenant in decoration. Last, the bottom row: the residual account at zero, or the year’s real work begins.

The office of auditor is older than the office of accountant in this story, and the ancestry sets the register. The Talmud’s collectors walked in pairs and counted the coins singly, in the open — internal controls as halakha, suspicion institutionalized precisely so that no person need be suspected. Chalcedon seated a steward beside every bishop so that church goods never moved on one man’s say-so. The Monti kept salaried offices — the appraiser, the notary, and the *depositarius* at the till — audited their registers, folded any profit back into capital, and ran even forfeiture in the open: the pledged pan went to public auction and the surplus over the debt went home to the borrower, because, as chapter 17 put it, “The Monte was forbidden to profit even from her default.” The pattern across the houses is uniform and a little humbling: the traditions audited their mercy harder than their commerce, on the theory that the places built for the poor are exactly where quiet money would go to hide.

Disclosure is the audit’s public half, and Part X has already drafted its instrument: the anti-extraction schedule of chapter 42, renewed annually, “drafted to be read by a stranger.” The books of this chapter are what fill it — the titled lines with their proofs, the queue in plain sequence, the contribution schedules, the reserves against their charters, the related-party register, the amendment log with the guardian’s consents, the counters against their caps, and the residual line, published by name. The fund’s own paper takes the same medicine: the whole-stack verdict the analyzer posts each year rides the annual re-asking the charter requires, so the institution interrogates itself with the instrument it points at others. Publication runs at the granularity a stranger needs to re-run the arithmetic — the whole test of disclosure being not whether the institution said something, but whether an outsider armed only with the exhibit could catch it lying. One boundary holds, and it is mercy’s: the mercy floor’s borrowers are published as aggregates and never as names, chapter 15’s confidentiality carried into the reporting, because a hardship whose relief requires a public confession has been priced in dignity. The analyzer reads the paper before the pen; the books testify after; between them, a promise has nowhere quiet to age.

Which leaves accountability — the word this chapter has saved for last, and the only one of the three that costs anything. An audit that finds a residual has not found a talking point; it has found a debt. The residual is funded, chapter 37 established, by the residual claimants — it came from identifiable seats at the table, euro by euro — so the finding

arrives with addressees attached, and the tradition's answer to that arrival was never a disclosure footnote. It was restitution. The scholastics' court of conscience, chapter 23 found, held gainers to a stricter ledger than any judge would enforce, person to person; and chapter 9's rule for the honest seller who discovers his goods were defective is the rule here for the honest fund that discovers its numbers were: "the moral fault vanishes but the debt does not." So the books gate the correction the way they gate the distributions. The found euros go home first — repriced, refunded, or credited forward against the counter, at whatever granularity the accounts allow, which is why the accounts were kept at granularity. The clause that produced the residual is amended, with the guardian's consent in the log. And no new distribution passes the gate while found money is still in the house. An institution's accountability is not its willingness to apologize; it is the standing rule that what the subtraction finds, the wire returns.

Close the books, then, on the year and on the Part's machinery, and notice what the chapter has actually built. The formula told a fund what to promise; the library made the promises signable; the analyzer caught the paper that broke them in advance. All three assume someone is looking. The books are for the years when no one is — the long, uneventful stretches in which institutions actually live, when the founder is gone, the guardian is new, and the only voice still telling the fund what it vowed is a chart of accounts that refuses to hold a euro without a reason. Moral memory is not nostalgia; it is machinery with a place for everything and an alarm on the place that must stay empty. In the end, a fund believes what it books. And that is precisely as far as bookkeeping can carry a conscience — every account in this chapter assumes hands that want the books true. Some hands do not. Chapter 45 is about them: the washings and costumes by which religious finance is tempted to keep the vocabulary and lose the geometry — the last examination this book owes its own project.

Chapter 45 — The Temptations of Religious Finance

Every machine in Part X was built for honest hands, and every chapter that built one said so on its way out the door. The analyzer of chapter 43 conceded that it could not stop “a determined hand from gaming the code it grades against.” The library of chapter 42 closed its drawer wondering aloud “what happens when paper this honest meets money that isn’t,” and promised that this chapter would study “the hands that reach into the drawer with other intentions.” And chapter 44, having built the books that keep a fund’s promises countable, ended on the same doorstep: some hands do not want the books true, and the last chapter of this book would be about them — about “the washings and costumes by which religious finance is tempted to keep the vocabulary and lose the geometry — the last examination this book owes its own project.” This is that chapter, and it should be said plainly at the threshold: nothing in it is about villains, who are rare, expensive, and usually caught; it is about temptations, which are cheap, respectable, and annual.

Begin, then, with the shape of the thing. Nobody sets out to found a counterfeit. The people who build religious finance — the *poskim* drafting partnership deeds, the shariah boards reviewing structures, the friars founding pawn banks, the founders of mercy funds and mutual pools in every century this book has visited — are, on the record, among the most sincere people in the history of money. That is precisely the problem. Comparative Interlude II, reading the Buddhist monasteries’ drift from renunciation to treasury, left Part X a warning it asked the reader to hold as a null hypothesis — “assume drift” — and a sentence that deserves requoting at the door of this chapter: “The sincerest founders in this history built the biggest treasuries.” Chapter 29 said the same thing as an engineering rule: drift is a design load, due annually, villain or no villain. What drift wears while it arrives is the subject here. A washing, in this book’s usage, is drift’s costume: the tradition’s vocabulary retained — the Hebrew heading, the Arabic contract name, the encyclical’s cadence, the impact report’s photography — while the geometry underneath goes back to the market’s default. The label stays; who bears loss, who receives surplus, who controls time, and who survives the path revert to the answers the label exists to refuse. What a washing cleans is never the money; it is the question.

One more word about the word, because the traditions themselves would insist on it. Temptation, in every moral vocabulary this book has read, is not sin; it is the ordinary weather sin arrives in, and the tempted are not the fallen. The chapter keeps that distinction deliberately. A fund that discovers its fee has drifted toward the market’s default and reprices it has not been exposed; it has been maintained — that is what the machinery of chapters 41 through 44 is for, and institutions that run such machinery will trip its alarms regularly, for the good reason that the alarms work. A washing, fully grown, is something else: the alarm unwired, the question retired, the vocabulary kept on as staff after its authority has been dismissed. Between maintenance and washing runs the same line the *poskim* drew between the *iska* that needs re-drafting and the *iska* that was never meant: not the presence of failure, but the presence of the audit.

The washings deserve study one language at a time, because each tradition has already caught its own — that is the remarkable fact, and the chapter’s method. No criticism below is an outsider’s. The *poskim*, the *fiqh* academies, Rome’s own dicasteries, the impact movement’s own founders and prosecutors: every washing in this chapter was named first by the house it tempted, which is both the most even-handed thing that can be said and

the strongest evidence that the temptations are real. Traditions that police themselves this hard are not hiding something; they are guarding something.

Washing in Three Languages

Catholic-washing is the easiest of the three to fall into, because the Catholic vocabulary is the one modern finance already half speaks. Dignity, stewardship, the common good, care for creation, the preferential option — the words have escaped the encyclicals and entered the brochures, and a fund can assemble them into a mission statement in an afternoon without changing one clause of its waterfall. The tradition's own history shows both the real thing and the test for it. The Monte di Pietà of chapter 17 was Catholic finance with its geometry converted all the way down: salaried appraisers instead of margin, audited registers, forfeiture run in the open, the auction surplus walked home to the borrower — "The Monte was designed to make no one rich, including itself." Strip those operating rules and keep the name, the sermon, and the papal blessing, and what remains is an ordinary pawnshop with excellent signage. The distance between those two institutions is the exact width of Catholic-washing. And the tradition knows it, in its own words, at its own expense: chapter 33 audited the social encyclicals and found a body of teaching that kept the conviction and retired the instrument — "Usury survived as an accusation and retired as an analysis." Rome's modern documents still condemn usury by name; what they no longer field is the contract-level test that made the condemnation operational, which leaves usury, in the tradition's present condition, "a sin that everyone condemns and no one computes." Catholic-washing lives in that gap. Where there is no computation, a citation is as good as a conversion — better, in fact, since it costs nothing and photographs well. None of this makes the vocabulary fraudulent. The words built the Monti, wrote the wage teaching, moved real capital toward real mercy for centuries; Rome's own offices have condemned, in this decade and by name, in documents chapter 33 read closely, the financial practices that price and sell need. The washing is not the creed; it is the creed worn as decor by instruments the creed would fail. And the cure is not humility but inventory, because the tradition's own attic — chapter 33's closing argument — already holds everything a working audit needs: the mutuum analysis, the just-price tests, the four titles with their proofs and proportions, five centuries of contract-level jurisprudence that Rome built, shelved, and never repudiated. A Catholic fund that wants to be more than its adjectives does not have to invent its examination; it has to fetch it. Until it does, the gap between the accusation and the analysis stays open for business, and the brochures will keep being written by whoever finds it first.

Shariah-washing has the largest paper trail of the three, because the industry it tempts is measured in trillions and its critics publish. Chapter 32 walked the file at full length, and its findings need only their summary here. The method has a name from inside the house: *shariah arbitrage*, Mahmoud El-Gamal's coinage, defined by him as a form of regulatory arbitrage in which "a captive market of pious Muslims voluntarily chooses not to use certain financial products" — and is therefore sold the same products reengineered, at a premium, with the jurists' attention drawn to the single document in the pile that bears an Arabic name. The demand side of the temptation was stated, anonymously and unforgettably, by a Dubai investment banker in 2009: conventional products are built first, then a scholar is telephoned for a fatwa, then another, then another — "We do this until we get Sharia compliance." Chapter 32 weighed the anecdote for what it is — one banker, anonymous, confessing the buyers' conduct rather than the scholars' — and set against it what the scholars on the other end of those calls actually did, which, more often than not,

points the other way: the AAOIFI resolution of 2008 told the sukuk market, to its face, that its favorite structures failed the standard; the Fiqh Academy said no to organized *tawarruq* while the industry's whole liquidity apparatus leaned on it. Refusals are plentiful on the record. But a certification economy in which the certified choose and pay their certifier, across any number of competing boards, does not need many weak scholars to yield many weak products; as that chapter put it, "it only needs the phone numbers." The verdict that stings most is the founder generation's own. Muhammad Nejatullah Siddiqi, the theorist whose plea helped launch the industry, looked back at what practice had done with the theory: "While theory aspired to prove Islamic finance was different from the conventional one, practitioners were busy searching for ways to make it similar to it. Convergence rather than divergence was the order of the day." Shariah-washing is that convergence sold at a markup — the wall, as chapter 32 concluded, monetized into a spread.

The halakhic version is the oldest and the most precisely anatomized, because the tradition that runs on it has audited it for four hundred years. Chapter 31 must not be misread here, and this chapter will not misread it: the *heter iska* is not the washing. Drafted with its working parts — a genuine venture, a deposit half genuinely in the venture's wind, a manager's wage, an accounting that could actually be rendered — it is what that chapter concluded it is: "a partnership instrument with a settlement option," built by its drafters as "the law's own bridge between the wall and the fair." The washing is what that chapter named ribbit-washing: the working parts stripped in sequence — the manager's wage waived away in boilerplate, the witnesses a formality of geography, the venture nowhere to be found — until what remains under the Hebrew heading is "a fixed return, collected in all states of the world, from a counterparty whose ruin is contractually nobody's business." Form without substance, in the categories that chapter gathered from the tradition itself: "words without a venture, a partnership without a meeting of minds, a settlement sum replacing a truth everyone knows." What keeps the instrument honest is not the instrument; it is the vigil — the poskim's four centuries of refusing, in chapter 31's words, "under every commercial pressure, to stop asking which fate has occurred" — case by case, financing by financing. The Levush and the Sema attacked the first draft until its author amended it; the case law has never stopped sorting real iskas from costumes. A tradition that could have declared victory when the clauses were printed chose instead to keep reading the deals. That choice, chapter 31 showed, is the whole difference between a legal instrument and a legal fiction.

Line the three up and the grammar is identical, which is why this book has treated washing as one phenomenon with three accents. In each house the label is retained because the label is valuable — it commands trust, loyalty, and in two of the three cases a captive clientele — and the geometry is surrendered because the geometry is expensive: real risk-sharing costs the financier bad years, real cost-pricing forfeits the market spread, real mercy forgoes the penalty stream. And in each house the alarm was raised from inside, in the tradition's own technical vocabulary, by people who wanted the thing itself and not its poster. The washings differ only in age. The halakhic one has case law, the Islamic one has a literature, the Catholic one has a gap where its audit used to be — and the newest washings, to which this chapter now turns, have prospectuses.

The New Costumes

Impact-washing arrived where chapter 34 said this book would file it. That chapter's microfinance case is the type specimen, and its facts must stay plain here as there: the mission was real — the loans reached women no bank had ever visited — and so was the extrac-

tion: weekly claims, fixed in advance, collected at the doorstep, from households holding no reserves, priced for the growth that the investors' valuations demanded. The mission lived in the institution: in what it aimed at, whom it reached, what it reported. The extraction lived in the contract. Nothing in the structure was assigned to reconcile them, and the name chapter 34 gave that unreconciled layering — mission-aligned extraction — is the anatomy of impact-washing generally: "alignment lodged in intention and institution, extraction lodged in payoff geometry," and the label structurally unable to see the thing it labels. The modern conscience, that chapter found, examines companies; the elder one examined contracts — and a conscience aimed at the company can be fully satisfied there while every question this book asks is being answered wrongly one signature down. Selection and disclosure are not nothing — steering capital and making harms visible are real work, and the sector's first prosecutors have made lying about even the labels expensive, as the enforcement file chapter 34 read attests. But the boundary that chapter drew holds and is the test: ethical branding changes the description of an instrument; altered payoff geometry changes what can happen to the people inside it. "One changes the description; the other changes the possible." Impact-washing is the industry of the first pretending to the second — the report photographed where the waterfall should be.

Stewardship-washing borrows the newest vocabulary in this book, the one chapter 29 took seriously as the modern heir of *waqf* and *hekdes*: Steward Ownership, mission locks, golden shares, the deed that cannot be sold out because it cannot be sold. The real thing, that chapter showed, is a conveyance — control actually parted with, an asset lock that actually binds, a guardian able to say no and survive saying it. The washing is the same wardrobe worn over unconverted power: a mission statement vague enough to bless anything, a steward board appointed by the founder and removable by him, a golden share parked with a foundation that has never once voted against the hand that endowed it, a return cap generous enough in timing to out-collect the uncapped. Chapter 29 supplied the failure mode in advance — the steward-firm captured by its stewards as surely as the *waqf* by its *mutawallis*, the office inherited and mined — and the movement's own literature, to its credit, says so as bluntly as the poskim ever did: locks need governance, or they are scenery. The economics of the temptation deserve one more sentence, because stewardship-washing is the only washing that pays in control rather than cash. A founder who steward-washes keeps what exits would have forced him to share — the company, the story, the chair — while acquiring the one asset ordinary ownership cannot buy: a reputation for having renounced it. That is why the structures that meant it put the renunciation in the deed, at the notary, where it cannot be repossessed by appointment, and why the locks that have actually held did their holding in bad years, when holding cost something. The test is the same one this Part has used everywhere. Ask what the lock refuses, concretely, this year; ask who can say no to the money, and whether the people they say no to can remove them. A stewardship structure that has never cost anyone anything is not yet evidence of stewardship; it is evidence of weather that has not yet turned.

Solidarity-washing is the one this book is honor-bound to name most carefully, because solidarity is the book's own favorite word. The genuine article was built in chapters 27 and 28: a pool owned by its members, holding their contributions as their own, existing to pay the claim, its operator on a fixed wage indifferent to refusals, its year measured on the time axis — how many member paths ended short of ruin. The washed version keeps the warm vocabulary — mutual, solidarity, community, *takaful* — and turns the geometry around, and the inversion states itself in one line: pooling risk to protect capital rather than people. It looks like this. A first-loss layer gets marketed as solidarity while working as the senior money's shock absorber — what chapter 34 called "a solidarity reserve with a

special-purpose vehicle around it,” concessional money positioned so that the concession flows up the stack instead of down to the borrowers. An operator’s fee is structured to grow when claims are refused, so that the fee schedule admits in arithmetic what no charter would ever put in writing. A “mutual” publishes its dues collected, its costs, and its investment earnings in immaculate detail, and nowhere the one number the chest exists for — the ruin column — because its actual design brief was smoothing the capital’s returns, not shortening the members’ worst years. The members, examined closely, turn out to be customers: their claims are booked as the seller’s losses, their surplus in light years leaves as someone’s yield, their assembly meets annually to receive a presentation. Chapter 28’s village can serve as the audit standard here as it served as the model’s calibration: the question was never whether the word mutual appears in the papers; it is whose ruin the pool is engineered to prevent. A chest that protects people pays out in the bad year and holds the line of jars above the flood. A chest that protects capital somehow always has a reason, in the bad year, why this claim fell outside the rules. Between those two chests runs the whole difference between solidarity and its costume.

Compliant and Unconverted

There is a washing that needs no theology at all, and every religious financier meets it first because it is the cheapest: the statute. Since the 1540s the civil law has regulated lending by ceiling — a maximum rate, a disclosure schedule, a license — and chapter 30 measured what that settlement keeps and what it surrenders: “A ceiling is what remains of a wall when only the height still matters.” Compliance with such a law is genuinely worth having; the predatory storefront that violates even the ceiling is not this chapter’s subject. The temptation is subtler: to let compliance stand in for conversion — to meet every cap, file every disclosure, pass every examination, and convert no power at all. A fund can be flawlessly legal while everything this book audits stays exactly where the market put it: the same party bears every loss, the same party takes every surplus above a decorative cap, the same party holds the clock — the acceleration right, the redemption window, the forced exit that chapter 37 showed can harvest a decade of someone else’s patience in an afternoon — and the same party walks away whole in the state of the world where the other party is ruined. Legal compliance without conversion of power is the washing bureaucracies are most comfortable with, because it can be certified. Time is where it hides best, and the reason is chapter 21’s: of everything a contract distributes, the clock is the hardest to see on a disclosure form. A rate is a disclosure; a deadline is a power. The party that can accelerate, refuse to roll over, shorten the window, or sit out the storm holds the other party’s liquidity in its hand — the very deprivation the *carentia pecuniae* title priced when it was honest — and no ceiling anywhere regulates it. A financing can carry a compliant rate and an unconscionable calendar. But the traditions’ tests were never about ceilings, and the book’s whole Part VI stands behind the point: a charge is justified by what it prices, not by what it is beneath; a small extraction is an extraction with good manners. Conversion of power is checkable, though — that is what Part X built. Power has converted when the consent that matters has moved: when the guardian can veto what would unmake the deed and cannot be removed for the veto; when the distribution is what chapter 44 made it, a state of the books rather than a mood of the board; when the borrower’s bad year triggers a procedure she was shown at signing instead of a discretion she must beg for; when the clock has more than one owner. Drift never asks for a vote; it asks for a waiver. An institution converts power by deciding, in advance and in writing, who may not grant the waiver alone.

And the reason conversion keeps failing to happen is not usually anyone's design; it is pressure — the market's standing pressure to replicate itself, which is the last temptation and the strongest because it never stops and never apologizes. Chapter 32 documented its purest form: desks pricing the permissible instrument off the impermissible one's benchmark until the two payoffs converged, a whole industry, in its own founder's words, busy making the different similar. The mechanism is not corruption; it is gravity. Every intermediary in a financing is paid in the market's coin — deal flow, assets under management, fee percentages, league position — so every intermediary, honestly and without malice, sands the strange edges off the strange instrument until it runs through existing systems: the risk-sharing note acquires a floor, the floor acquires a schedule, the schedule acquires an enforcement clause, and the instrument that began as a partnership ends as a bond with a glossary. The certification economy compounds the pull, since the certifiers are paid by the certified, in every tradition and in the impact market alike. Benchmarks compound it again: whatever the mission says, the fund is compared quarterly against funds that refuse nothing, and the comparison arrives with a fiduciary vocabulary attached. "A gap between form and substance is, to a moralist, a scandal; to a market, a spread" — and the market closes spreads; that is its job. It does not argue with the wall; it prices a way around it and calls the price innovation. Success makes the pressure worse, not better: the model that works attracts imitators who want its returns without its refusals, and the second generation of any moral finance is always its first crisis. Normal finance, chapter 30 found, has no slot for this book's question — so the pressure to become normal is the pressure to lose the slot. Nothing in that pressure can be repealed. It can only be designed against, which is why the traditions' own answer, everywhere this book has looked, was never exhortation. It was architecture.

The Final Test

Why does every religious finance need anti-drift governance? Because everything above is annual, and nothing above is anyone's fault. That is the finding of the whole chapter, and each house reached it independently, in its own idiom, about itself. The poskim concluded it four centuries ago and instituted the vigil — the standing refusal to let the printed clauses replace the audit of the deal. The fiqh academies and standards bodies concluded it in this generation and started issuing the refusals chapter 32 recorded, to their own industry's face. Rome's own auditors concluded it in the negative, by demonstration: chapter 33 found a tradition with a living voice and a retired instrument, and the finding is a sentence every fund in this Part should have framed on a wall — "A verdict without a test is a conscience without jurisdiction." The Buddhist interlude concluded it about the sincerest institutions in this history and set Part X its null hypothesis — the hypothesis this chapter has run from its first page. And the design response is already on the table, because Part X built nothing else. A guardian with real standing and real tenure; seats with real voices for the people the layers exist for; runs published to a stranger's granularity; the side doors watched on a schedule; every layer re-examined against the formula each year — chapter 41 drew that architecture, chapter 42 made it signable, chapter 43 made it readable, chapter 44 made it countable. This chapter adds only the reason: because the temptations are not exotic. They are the default. An institution that has not decided, structurally, who can say no to its money has decided that no one can.

There is also a reason the watch must be sharpest exactly where the institution is kindest. Chapter 44, assembling the auditors' ancestry, noticed that "the traditions audited their mercy harder than their commerce"; this chapter can add the reason. Mercy institutions

hold money on behalf of people who cannot audit it, under a vocabulary that discourages everyone else from trying — almost no one examines a charity's spread, a mutual's fee, a mercy fund's waterfall. Every washing in this chapter exploits that deference, which is why the old houses withdrew the deference and kept the mercy, making the institutions built for the defenseless the easiest in the city to examine rather than the hardest. Religious finance needs anti-drift governance, in the end, because religious finance asks for trust — and structures that ask for trust must be built as if they will not deserve it, or one year they won't.

Which leaves the test itself — the final test, this book's whole audit folded to four questions: who bears loss, who receives surplus, who controls time, and who survives the path? The questions are not new; they are the book, compressed. Costume questions have been asked here before — chapter 24 put four of them to every sale-shaped financing, and this test is that habit grown to the scale of institutions. Chapter 3 taught that "the moral analysis of money begins with names rather than amounts," and every tradition's own audit has run on some form of the questions since — chapter 34 heard all three old houses ask, in their own idioms, "what is the money doing, who bears the loss, what does the charge price." The fourth question is the one the ergodic thread of this book added arithmetic to: survival is measured along the path, in time, where a person actually lives — "An average survives its losers; a person does not." Run the four against any instrument, any fund, any institution wearing any vocabulary whatever, and every washing in this chapter fails by name. The ribbit-washed iska and the shariah-washed note fail the first question: loss, whatever the heading says, is contractually the borrower's in all states of the world. Impact-washing fails the second: the surplus, measured honestly against chapter 39's remainder, flows to seats the mission's own reports never audit. Stewardship-washing fails the third: time — the acceleration, the exit, the waiver — turns out to be controlled by exactly the hands the lock was said to bind. Solidarity-washing fails the fourth: the pool's members, run forward along their own paths, ruin at the market's ordinary rate while the ensemble's average looks splendid in the annual presentation. And an institution that passes all four has, by that fact, answered this book's question, whatever it calls itself: its return is "a number made entirely of answers," its extraction what chapter 35 declared it: "not an accusation; it is a remainder" — computed, published, and expected at zero, in the account chapter 44 opened, the account that "exists to be empty."

What makes the final test final is not its severity but its portability. It requires no seminary and no spreadsheet. A parish council can run it on the diocesan fund's annual exhibit; a community board can run it on the mutual's; a member of a shariah committee, a rabbinical court, a workers' assembly, a town holding its anchor company — each can take the one disclosure chapter 42 drafted to be readable by exactly them and ask the four questions of the actual numbers: whose names are on the losses in the bad-year column, whose on the surplus above the cap, who signed the waivers in the amendment log, and what the ruin column says about the members with the least behind them. Every instrument in Part X was built so that the answers would be checkable from outside the room — by the stranger of chapter 42, the auditor of chapter 44, the reader holding this book. The washings, without exception, depend on nobody asking. The test is four questions that anyone may ask, and the deepest anti-drift device the traditions ever built was the habit of asking them out loud, in public, of their own institutions, every year, forever.

The test's last application is the one this book promised before it was ever drafted, when chapter 34 filed its warning that "the new architecture will deserve its own audit too." The non-extractive fund of Part X is religious finance. It carries a creed, claims a lineage

in three traditions, asks for trust on the strength of both — and is therefore heir to every temptation in this chapter, plus the special one reserved for reformers: believing that having diagnosed drift is the same as being immune to it. It is not exempt because its formula is published; the formula can be gamed by a determined hand — its own analyst said so. It is not exempt because its books are honest; the books assume hands that want them true. It is not exempt because its founders mean it; so did the founders of every treasury in the interlude's warning. So the four questions apply to this book's own design first, annually, in public, on the instruments Part X itself supplied: the guardian's consent log, the published runs, the counters against their caps, the ruin column, the residual row. If the answers hold, the vocabulary has earned its geometry for one more year. If they do not, then whatever the letterhead says, the thing on the letterhead has joined the washings — and any reader of these chapters will be equipped to say so, which was the point of writing them. The traditions never asked money to be eloquent. They asked it to answer. Chapter 3 handed this book its assignment, and the assignment closes here: "The rest of this book is an audit of names." The names have been audited. What remains — for the Conclusion, and for anyone who builds — is the shorter, harder work of keeping the answers true, year after year, under pressure, in public, on the books: capital examined into service, and held there by design.

Conclusion — Capital That Serves Life

This book opened with a question that sounded, at the time, almost naive: what is money allowed to do? Forty-five chapters, two interludes, and some four thousand years of legal argument later, the question should sound different — not naive but load-bearing, the kind of question civilizations answer whether or not they mean to, in their contracts if not in their creeds. Chapter 45 finished the audit and handed this conclusion its assignment: “the shorter, harder work of keeping the answers true, year after year, under pressure, in public, on the books,” compressed into the book’s last working line, “capital examined into service, and held there by design.” What a conclusion owes the reader who has crossed all of that is not a tour of it again. It owes a settlement: what the traditions actually teach, now that they have been read together, mechanism by mechanism, with none of the three as the foil.

The method deserves one sentence of farewell, because it did most of the work. This book braided rather than surveyed. It took the loan, the partnership, the sale, the lease, the pool, one instrument at a time, and asked what Torah and Talmud, Gospel and canon, Qur’an and fiqh each made of it, with the Sanskrit lawbooks and the Buddhist Vinaya as guests at two turnings. Read separately, the traditions look like three walled gardens with private vocabularies. Read on the mechanism, they turn out to have been working the same bench, asking of every instrument that came to hand the four questions the last chapter folded the whole audit into: “who bears loss, who receives surplus, who controls time, and who survives the path?” The settlements below are what four millennia of that bench-work established. They are teachings in the plainest sense: things learned once, at cost, that do not need relearning — only keeping.

Three Answers to One Question

Judaism teaches that lending is kinship in action — that the loan, before it is a financial instrument, is the legal form of the command to hold onto a falling neighbor. The support law of Leviticus stands at the center of everything the tradition built: “you shall support them; they shall live with you as though resident aliens,” in the NRSV’s rendering, and chapter 18 went underneath the English to the Hebrew verb, *vehechezakta bo*, whose root is strength: “you shall strengthen him, take hold of him, brace him.” Strengthening the hand is the tradition’s own name for the whole department, and its timing is the teaching, in the image chapter 18 carried from the halakhic midrash: one hand can steady a stumbling man before he falls; five cannot raise him after. That is why the ban on *ribbit* is written in kinship language rather than market language — “A rate is a number; a brother is a claim,” as chapter 2 put it — and why the four defenses chapter 4 toured all ring the borrower, not the loan: the banned increase, the protected floor of cloak and millstone, the scheduled release, the future no one may sell. “Israel’s ledger was commanded to forget, on a schedule, what every other ledger in the ancient world was built to remember forever” — chapter 4’s summary of the sabbatical architecture, and still the most radical sentence in the book’s Jewish strand. None of this made the tradition dreamy about commerce; its genius was to be merciful and exact at once. The same halakha that sealed the loan against increase, and chased the ban in chapter 8 all the way down to the borrower’s greeting, built the *iska* so that capital could share a venture’s fate honestly, priced the working partner’s

wage so the arrangement would not lie, and policed even lawful sales with *ona'ah* so that no market price could overreach the one-sixth the law would believe. When the sabbatical release began strangling the credit it was meant to sanctify, the tradition amended its own machinery with the *prozbul* rather than let the poor go unlent; and it has kept four centuries of vigil over the *heter iska*, case by case, lest the partnership decay into plain interest beneath its Hebrew paperwork. It staffed the charity box like a courtroom — chapter 18's opening puzzle — because *tzedakah* is not warmth but law: "Generosity measures what the giver can spare; justice measures what the poor man lacks," as chapter 18 found. And where the market could not reach at all, it built the *gemach*, the free-loan fund still turning in Jerusalem and on Henry Street, whose teaching chapter 15 kept: "Interest teaches money to breed; mercy teaches it to circulate."

Christianity teaches mercy pushed past every boundary a lender would draw for himself, and then, because mercy must eventually keep books, a jurisprudence of price to keep commerce honest wherever mercy's writ ran out. The Gospel's instruction is the shortest lending law in this book: "lend, expecting nothing in return." Luke's sentence does not regulate the loan; it converts it, commerce into mercy, and posts the converted instrument even across enemy lines. The tradition prayed its finance daily — "forgive us our debts, as we also have forgiven our debtors" — and chapter 5 drew the working conclusion: "In this economy the one way to keep forgiveness is to spend it." From the collision of that prayer with ordinary commerce came the long casuistry that fills the book's middle. The *mutuum* was anatomized: the naked loan on which nothing may ride — not interest, not gifts, not hope with a memory. Then, over centuries and against every appetite, the Church licensed the exceptions that were not exceptions: the extrinsic titles, narrow doors through which a lender could be made whole for actual loss, for gain provably forgone, for a peril genuinely retained; each a compensation with a proof attached, none of them a price for the neighbor's need. Chapter 19 drew the discipline's plainest boundary — a priced expectation is one thing; a loss with an address is another — and chapter 20 kept the harder line behind it, that the market return is "what anyone could have earned, which is why it is proof that no one in particular lost it": the titles repair what happened to this lender, and refuse to price what money in general might have done. Beside the loan stood the sale, and the just-price doctrine whose rule chapter 9 traced from the Roman chancery to Aquinas's desperate buyer: "no one may keep, as profit, what the other side could not see." What Christianity refuses, root and branch, is the sale of need itself — the charge that grows precisely as the borrower's alternatives shrink. Its proof that the refusal can be operated, not merely preached, stands in the hill towns of fifteenth-century Italy: the Monti di Pietà, mercy engineered at cost, salaried, audited, and open, designed, as chapter 17 showed, to enrich no one, including themselves. Even the tradition's modern silence teaches. Chapter 33 found Rome still condemning usury by name while the contract-level test sits in the attic, a voice that kept its conviction and retired its instrument — the one loose end this book's last Part was written to tie.

Islam teaches that money earns its increase only inside a real venture: no lawful profit without real trade, a real asset, or genuinely shared risk. The Qur'an's line is bright and double: "God has allowed trade and forbidden usury," in Abdel Haleem's rendering, and the entire architecture of Islamic finance is the working-out of that distinction. Chapter 6 ended the scriptural story just where the law begins: "The war against *riba* ended the way wars end — in treaties. The treaties are the contracts, and the rest of this book is largely spent reading their clauses." The clauses, read, all enforce one grammar. The *qard* is sealed; a loan may carry no benefit home. Capital that wants a return must enter a *mudarabah* or *musharakah* and wait for the venture's truth, because, as Part IV established across all three

traditions, “Profit is a remainder, and a remainder cannot be promised.” The permitted instruments each confess what their charge prices: the *murabaha* markup “earned in the hours the goods are the seller’s problem,” as chapter 24 put it; the *ijara* rent licensed by the repairs and hazards the lessor keeps; the *sukuk* certificate honest only while the paper stays married to the asset it names. The tests of *gharar* and *maysir* guard the border no rate can: a bargain must be visible to both parties whole, and a peril must exist before anyone may be paid for carrying it. Risk pooled in *takaful* belongs to the members who pooled it, not to a seller who prices it. Zakat runs a purifying current through the whole estate, a share of wealth owed outward every year as worship. And the tradition polices its own frontier in public: the sharpest catalog of Islamic finance’s modern shortcuts — the benchmark that converged, the fatwa gone shopping — was written, chapter 32 showed, by the industry’s own founders and jurists, which is what a living law looks like when it means to stay one. Beside the market, the mercy instrument again: the *qard hasan*, the beautiful loan, made to the neighbor and booked, in the tradition’s own astonishing arithmetic, to God’s account — of which chapter 16 recorded the economics: “Mercy capital collects its wages in things that do not happen.”

Set beside one another, the three teachings are not three rules but three emphases of one refusal. Judaism guards the borrower’s personhood, Christianity the mercy of the giver, Islam the honesty of the money — and each, pressed, contains the other two.

The Guests and the Mathematics

The book’s first guest answered the same question with a different instrument. Hindu Dharmaśāstra teaches lawful measure: where the Abrahamic traditions raised a wall, the Sanskrit lawbooks published a tariff: rates banded to the venture’s real exposure, doubled for the forest road, doubled again for the sea, capped in total by *damdupat* so that no debt could compound past the size of its own principal, pledges regulated down to the rule that a pledge in use bears no interest, and the whole schedule enforceable before the king. Interlude I treated it as the one structurally serious rival to the wall, the measure grammar beside the wall grammar, and kept its distillation: “A ceiling is a prohibition that has learned to count.” The lawbooks’ teaching is that increase can be civilized by proportion: priced to danger, stopped at a ceiling, answerable to context. Their warning is the measure grammar’s characteristic corruption, which the interlude handled plainly and this book has carried soberly since: the same texts that banded rates to the road’s peril also banded them to caste, pricing the borrower’s rank instead of the venture’s risk, and a rate card for persons is exactly what the interlude’s design lesson forbids a modern reader to rebuild. Where wall and measure overlap, though, they are one tradition: neither will tolerate what Interlude I called “increase answerable to nothing — the rate with no ceiling, the claim with no stop, the price that reflects only what the desperate will sign.”

The second guest teaches from the far side of the argument, and teaches by biography rather than by statute. Buddhism begins where no other tradition in this book does, with renunciation, money refused at the root as an attachment the path cannot carry, and then demonstrates, across two millennia of monastic institutional history, what happens to renunciation when it acquires an endowment: “What the monk renounces, the monastery must administer.” The interlude traced the drift from begging bowl to temple treasury to monastic lending at interest, watched the Vinaya’s drafters answer with an administrative craft so exact that a robe fund reads like a contract, and drew from the tradition’s own psychology the diagnostic this book has used ever since: “Institutions crave too; their craving compounds.” What Buddhism adds is holy suspicion: the standing assumption that

accumulation works on the accumulator, that sincerity is no vaccine against drift (the interlude's grandest treasuries were raised by founders who meant every vow), and that — in the interlude's own warning — “an ideal that refuses to specify its own administration will be administered anyway, by drift.” Its question has hung over every ledger in the book's second half: “what the money, so carefully minded, is doing to its minders.” Part X's governance machinery — the guardian, the published runs, the annual re-examination — is, among other things, that suspicion drafted into bylaws.

The last addition is the newest and the strangest, because it arrived from mathematics, owing nothing to any scripture, and landed on the traditions' side. Ergodicity economics teaches that the average and the person part company. A return that looks splendid averaged across an ensemble of ventures can still ruin, one by one, most of the people who actually walk its path, because a person lives sequentially — this year's loss becomes next year's smaller capacity, multiplying forward — and ruin is an absorbing state: the household that hits the floor does not rejoin the average. Wealth over time is non-ergodic; the ensemble forgets its losers, the path cannot. Chapter 10 planted the distinction by name; chapter 28 did the arithmetic, running a village's bad year twice, separate jars and then a common chest, and found the traditions' instruments waiting at the end of the equations. The pool does what chapter 2 promised no fairly priced loan could do — it defeats ruin; the solidarity reserve is anti-ruin engineering; the fixed claim that must be paid in the bad year is the very instrument that walks its holder toward the floor, which is what the traditions had been saying about the naked loan all along, in Hebrew, Greek, Latin, and Arabic. A pool that means it, chapters 27 and 28 agreed, is owned by its members, pays its operator a wage indifferent to the claims he grants, funds its reserve before declaring any surplus, and measures its year on the path: how many members ended it standing. The burial club at Lanuvium and the friendly societies' fourteen million members were living the correct theorem centuries before anyone could write it down. What ergodicity adds is the demonstration that solidarity is survival mathematics rather than sentiment — in chapter 28's words, “Mutual aid is the art of building something that lives in the ensemble so that no one has to” — and it supplies the reports' most honest missing line, the ruin column, which chapter 28 put into the model and chapter 44 refused to let out of the books. The traditions, it turns out, were never bad at finance. They were early.

The Shared Grammar

Beneath the three Abrahamic answers lies a grammar so common that one label has usually served this book for all three traditions. Its first rule is a claim about ownership — the earth belongs to God, “The earth is the LORD's and all that is in it,” as the Psalm has it — and human title is therefore real but junior: tenancy in Israel's idiom, trusteeship in Islam's, stewardship in Christianity's. Its second rule follows from the first, and chapter 4 gave it its lasting form: a neighbor's need obligates my surplus and never owes me a yield. Where those two rules hold, the loan drawer seals itself: *mutuum*, *milveh*, *qard*, three names in three legal languages for the same sealed instrument, discovered independently and matched almost clause for clause. Around the sealed drawer stands the same taxonomy: sale, lease, partnership, gift, pool, each lawful just insofar as its charge can say what it prices. And over the whole cabinet runs the same method, the book's oldest rule: substance over form, the audit run past the label to the geometry. Ask of any instrument, whatever its name and paperwork: who bears loss; who receives surplus; who controls time; who survives the path. Increase is legitimate exactly as far as it can answer for itself; past that point it is extraction, under any letterhead whatever. The grammar traveled because of what it is

made of — it crossed legal systems and centuries, Mishnah to code, canon to statute, fiqh manual to standards board, without losing its questions, because questions survive translation better than rates do. And it stayed honest by disagreeing in public: halakha and fiqh are live legal systems that recorded their disputes, *machloket* and *ikhtilaf*, beside their rulings; the schoolmen published the objections before the answers; the traditions kept their quarrels in the file where later generations could re-run them. A grammar of questions, asked aloud, at home, on a schedule: that is the shared Abrahamic inheritance, and it is why chapter 2's summary of its purpose has never needed updating — “the community could not keep ruin from coming; it could refuse to let ruin find anyone alone.”

From Prohibition to Design

The grammar was nearly lost. Part VIII told the story without nostalgia: the sixteenth century relocated the usury question from the confessional to the statute book, the wall became a ceiling, the sin became a price, and, in chapter 30's verdict, “The modern world did not answer the usury question; it forgot the language the question was asked in.” What followed the forgetting was not freedom from the old problem but unaccountability to it. Form triumphed over substance on every front this book audited: the fixed return in Hebrew dress, the benchmark wearing Arabic, the encyclical quoted by funds it would fail, the impact report standing in for the waterfall it never shows. The washings of chapter 45 differ by tradition and rhyme in structure, and every one of them depends on nobody asking the four questions.

This book's answer to the forgetting has been retrieval, not invention. From prohibition to design: the conviction that the traditions' rules were never mere avoidances but specifications — a design brief four millennia in the drafting, waiting to be treated as buildable. That required making the doctrines computable, because, as chapter 35 put it at the opening of the book's last movement, “A conscience that cannot count is at the mercy of whoever can.” So the tests became arithmetic. Every euro above principal was made to name its title, the old titles becoming line items with proofs attached; extraction was measured as a remainder rather than hurled as an accusation; danger was priced only where it is actually borne; labor and deferred wages entered the ledger as the contributions they always were; the moral-title ledger of chapter 35 gave the audit a place to happen, and the ruin column gave it a conscience. And from usury avoidance to non-extractive capital: the arithmetic assembled into instruments. The return formula of chapter 40 wrote the whole tradition onto one term sheet — “Seven titles above a zero line, the reserves standing, the remainder at zero” — and delivered, now computable, the conviction the whole book had been carrying: “a non-extractive return is not a smaller number; it is a number made entirely of answers.” The fund of chapter 41 stacked mercy, patience, and market capital in the order their claims deserve, each layer earning, in chapter 41's rule, “its seat by what it refuses to collect,” the stack “a creed read from the ground up” — with the mercy tier seated as designed infrastructure, the *gemach* and the *qard hasan* and the Monte scaled to real shocks, load-bearing and unembarrassed, never an annex. The library of chapter 42 drafted the doctrine to be signed; the analyzer of chapter 43 reads any term sheet, in any vocabulary, against the whole tradition at once; the accounting of chapter 44 keeps the promises as balances, with the extraction account expected at zero and watched because it is expected; and the standing audit of chapter 45 turns the four questions on the new architecture itself, annually, because it inherits every temptation of the old. Non-extractive capital is not a softer deal or a discount for the deserving. It is capital that has agreed to answer — the traditions' whole demand, translated into instruments a notary can witness

and a stranger can check. Chapter 29 closed the workshop of the old instruments with a sentence that was half elegy, half dare: “The workshop’s tools survive the workshop.” The book’s last Part was the dare accepted — the tools taken down, cleaned, and put back to work on modern money.

Stewardship, Not Dominion

The last teaching is the first premise, and every tradition in this book arrives at it: capital is a stewardship, not a dominion. Dominion is capital answerable to nothing — increase with no ceiling, a claim with no stop, title mistaken for sovereignty, the owner’s hand on the clock and his name absent from the losses. Stewardship is ownership that answers: to the owner of the earth, in the traditions’ full theology; and even in the barest secular translation, to the people inside the instrument — the borrower in her bad year, the worker whose contribution the ledger cannot see until it is made to, the member whose one path through time is the only average she will ever live. Chapter 18 found the three vocabularies — tenancy, trusteeship, stewardship — and the one structure beneath them: “title is real, and title is junior.” Chapter 29 found the structure’s limit case in the endowment, capital dedicated past the reach of any future seller — “What cannot be sold cannot be sold out” — and found its modern heirs in the asset locks and Steward Ownership structures now being drafted for companies; the old drawer is open again, *waqf* and *hekdes* and the patrimony of the poor in notarial clothes, and the deed still does what deeds do best — it remembers, on behalf of people who will forget. Part X built stewardship the tools of its trade: a formula so the steward knows what to promise, contracts so the promise binds, an analyzer so anyone can check, books so the institution remembers — “In the end, a fund believes what it books” — and an annual audit of temptations, because stewards drift and the deed must outlast the mood. The audit was built portable on purpose, and chapter 45 called the roll: a parish council, a rabbinical court, a shariah committee, a workers’ assembly, a town holding its anchor company — any of them can put the four questions to the disclosure chapter 42 built for their hands. None of that machinery is the point. The point is what the machinery keeps: money in the office the traditions assigned it — servant of exchange, fuel of ventures, carrier of mercy — and out of the throne it keeps climbing toward.

So the question the book opened with can be answered in the traditions’ one voice. What is money allowed to do? It is allowed to serve: to move surplus to need and come back no larger; to enter a venture and share its fate; to buy, sell, lease, and pool at prices that can say what they price; to strengthen the falling hand, expect nothing in return, and collect its wages in things that do not happen. It is forbidden to reign: to harvest need, to compound past its principal’s claim, to be owed the same sum in the neighbor’s best year and in his worst, to walk away whole from a ruin it priced and helped to cause. Between service and reign runs the line that Torah and Gospel and Qur’an, lawbook and Vinaya, ledger and theorem have each drawn in their own hand, and the drawing of it is the oldest continuous project in the history of finance. This book has been that project’s biography, and its ending is not an ending, because the line has to be redrawn every year, on real books, against real pressure, by people who mean it. The traditions supply the grammar; the mathematics supplies the proof; Part X supplies the tools. What remains is the choosing — the annual, unglamorous, entirely possible choice a fund, a family, a congregation, a community makes when it examines its capital into service and holds it there by design. Money is the most obedient thing a civilization owns; it becomes whatever its instruments permit. Write the instruments so that the only things capital can do are the things the traditions taught — bear loss where it lodges, take surplus only where it is true, share the

clock, keep every path alive — and the oldest question answers itself, year after year, on the books: capital that serves life.

Appendix A — Comparative Glossary

This glossary is the catalog the preface promised: the working vocabulary of four legal languages (Hebrew and Aramaic, Latin, Arabic, Sanskrit and Pali), together with the English terms this book has used to read them side by side. The entries follow the order of the table of contents, which groups the words by the work they do: the names of the forbidden thing, then the loan, the partnerships, the mercy institutions, the pools, the endowments, the doctrines of consent, the wager, the Latin titles, the mathematics of survival, and the guest traditions' terms. Definitions are deliberately short and carry no argument. Each word carries the sense this book gives it, in the spelling this book gives it, and each entry points to the chapters that hold the full treatment — sources, disputes, cases and all. Where traditions' terms translate one another almost exactly, the entry says so; where a resemblance is only apparent, it says that instead. A reader who wants the doctrine should follow the pointers; a reader who only needs to recall what a word meant three hundred pages ago should find it here in under a minute.

Usury, *riba*, *ribbit*. The three names of the thing this book is about. English *usury* descends from the Latin *usura*: “use,” as chapter 1 has it, “with a suffix: a payment taken for the use of money.” In the doctrine the word originally named, the offense begins at the first coin — to lend and take back anything past the principal, however little, is usury (chapter 1). Only in the modern era did English narrow the word to *excessive* interest, leaving lawful rates morally unexamined, a shift chapter 1 dates and chapter 30 diagnoses. *Riba* is the Qur'an's word, from a root meaning to swell or increase: forbidden gain on wealth lent, the growth that fed on deferral and doubling — trade permitted, *riba* forbidden is the scriptural dividing line, and the scriptures' fiercest warning on any financial subject falls on the lender who persists (chapters 2 and 6). *Ribbit* is the rabbis' legal name for forbidden increase on a loan between Jews, built from the Torah's own pair of words, *neshekh* (the bite) and *tarbit* (the increase), which the Talmud reads as one sin under two names, binding lender, borrower, scribe, and witnesses alike (chapters 2, 4, and 7). The rabbinic shorthand for what all three traditions forbid is *agar natar*, payment for waiting (chapter 7); the fence around it runs as far as *ribbit devarim*, interest paid in words (chapter 8). The three words are not identical in scope (each tradition draws its own boundaries of person and community, walked in chapters 2 and 4), but they name the same offense: increase demanded of a borrower simply because money was lent and time ran.

***Mutuum*, *qard*, *milveh*.** One contract in three languages: the loan for consumption, the anatomy on which the whole prohibition stands. Money, or any good consumed in the very act of use, passes to the borrower, ownership and all; what comes back is the *tantundem*, the same in quantity and kind, never the selfsame coins. The Roman schoolbooks derived *mutuum* from *ex meo tuum*, what was mine becomes yours; the Talmud's rule is that “a loan is given to be spent” (*milveh le-hotza'ah nitnah*; chapter 3 gives the folio); the Arabic *qard* is said to come from a root meaning to cut — a slice of the lender's wealth cut off and handed over (chapter 3). Because what was lent no longer exists (spent, eaten, burned as fuel) and only its like returns, all three traditions concluded that the lender retains nothing on which an increase could honestly be charged. That is why the loan, and only the loan, carries an absolute bar on gain — and why so much of financial history is the search for a contract that is honestly not one (chapter 7).

***Societas*, *mudarabah*, *iska*.** The three grammars of partnership: the traditions' shared

escape from usury, lawful because the investor's money keeps sharing the venture's fate. The Roman and Christian *societas* is the consensual partnership of gain and loss shared, with the lion's partnership (profit to one, loss to the other) void from the classical jurists onward (chapter 11). The *mudarabah* of the fiqh (the Hijaz school said *qirad*) joins the capital of the *rabb al-mal* to the labor of the *mudarib*: profit divides by a ratio agreed at the start and paid only from profit actually earned (a stipulated fixed payout voids the contract), while money losses land on the capital and nowhere else, the agent standing as an *amin*, a trustee who answers for negligence, never for the sea (chapter 11). The rabbis' *iska* runs half loan, half deposit (the Nehardean formula of Bava Metzia 104b), so that half the money trades at the investor's risk and earns his return, while the other half is the manager's own debt whatever the venture does (chapters 11 and 13). The *heter iska*, the "permission of venture," is the later document that turned the *iska* into a standard form, its proof machinery (witnesses, oath, settlement) drafted in advance; its uses and its perils have a chapter of their own (chapter 31). The Mediterranean *commenda* — the *stans* who stays entrusting capital to the *tractator* who sails — is the same architecture in a merchant's hand (chapter 11).

Musharakah, partnership, joint venture. Islamic law's sharing partnership, riding the root of *shirka*: capital comes from both sides, profit may be set by agreement, and loss tracks the capital shares exactly — the rule the lawbooks trace to the caliph Ali (chapter 11). "Partnership" and "joint venture" are the plain modern English for arrangements of this family, and the book uses them wherever the deep structure chapter 11 finds beneath all three grammars holds: "ownership retained, risk retained, gain therefore honest." The family's tests do the sorting work of Parts IV and X: who bears loss (chapter 13), whether profit comes only from profit (chapter 14), and whether a modern risk-sharing layer deserves the name (chapter 41).

Gemach, qard hasan, Monti di Pietà. Three institutions of mercy credit: lending stripped of gain and built to revolve. The *gemach* (the word compresses *gemilut chasadim*, deeds of loving-kindness) is the Jewish free-loan fund: donated capital lent at zero, secured by guarantors and sober paperwork, repaid and lent again — money, wedding dresses, mid-night wheelchairs (chapter 15). The *qard hasan* is Islam's beautiful loan, the loan the Qur'an describes as made to God: a real debt, repayable in full, on which the lender may accept nothing beyond the principal; the borrower's freely added gratitude is virtue, the lender's expectation of it is *riba*; the institutional forms run from household lending circles to statutory bank deposits (chapter 16; the outright gift is *sadaqah*, whose law is its own, chapters 6 and 18). The Catholic entry is the Monti di Pietà: charitable pawn banks preached into existence by the Franciscans, lending modest sums on pledge at a charge sized to bare operating cost, a practice contested for a generation until the Fifth Lateran Council settled it in 1515 — expenses may be recovered, profit may not (chapter 17). Three institutions, one design conviction: mercy can keep books, and a fee that keeps the fund alive is not a yield (chapters 15 through 18).

Takaful, mutual aid, confraternity, friendly society, solidarity pool. Here is the family of arrangements by which members carry one another's catastrophes. The solidarity pool is this book's own generic term, introduced with the taxonomy of chapter 3, which defines it as "many small, certain payments against one large, uncertain blow; shared protection against individual ruin." *Takaful* is the fiqh's engineered version — the word is built on *kafala*, guarantee: mutual standing-for — in which contributions enter as *tabarru'*, donation, the fund is the participants' common property, the operator serves as hired agent (*wakil*) or profit-sharing manager (*mudarib*), and pools reinsure upward through *retakaful* (chapter

27). Confraternities are the medieval Christian version: parish and trade brotherhoods holding common funds against burial, sickness, and dowry. Friendly societies are their industrial-age heirs, the member-run safety net of the working classes, praised by name in the founding document of Catholic social teaching (chapter 27). “Mutual aid” names the whole genus, and one fork sorts it from its commercial cousin — chapter 3’s line, cashed in chapter 27: “protection pooled among members versus protection sold at a margin by someone who profits when claims stay low.” Why a pool can stop ruin that fair pricing cannot is the business of the mathematics entry below (chapters 27 and 28).

Waqf, hekdesb, Church endowment, asset lock. Deeds written against the future: property dedicated beyond private ownership. The *waqf* detains a productive thing forever and spends its fruit on a named purpose — the jurists’ shorthand, chapter 29 reports, is “detain the origin, let the fruit go free” — with founder (*waqif*), deed (*waqfiyya*), trustee (*mutawalli*), the three bars that are its signature (no sale, no gift, no inheritance), and *istibdal*, audited substitution, as the lock’s one controlled exception (chapter 29). *Hekdesb* is the Hebrew word for property dedicated away from ordinary use: the Temple treasury in its first life, the communal endowment thereafter, and — late and grimly — the poorhouse of the East European towns; a people under conditional tenure kept its permanence moving, revolving funds under covenant rather than land under title (chapter 29). The church’s endowment is its inalienable patrimony, locked and stewarded by conciliar law since Chalcedon in 451: the monastery that must remain a monastery, the *oeconomus* who must watch the books (chapter 29). “Asset lock” is the modern genus — the statutory locks of community interest companies, the golden shares and purpose trusts of Steward Ownership, the mission lock on governance: the old dedication rewritten in corporate instruments (chapters 29 and 41).

Gharar, ona’ah, fraud, defective consent. Doctrines for policing whether a bargain’s consent was real. *Gharar* is excessive uncertainty in exchange, the sale of what neither party can see or count, and the fiqh’s response is structural: it condemns the blind contract itself, no wrongdoer required, the way a building code condemns a staircase (chapters 6, 9, and 23). *Ona’ah* is the rabbinic law of overreaching, price-wronging measured against the going rate, with a one-sixth threshold: beneath it the market is left alone, at it the surplus returns, past it the injured party may undo the sale — strict liability, no inquiry into the seller’s heart (chapters 9 and 23). Fraud is the Latin tradition’s *dolus malus*, defined by the jurists at its widest as every cunning employed to circumvent another — a net wide enough to catch the staged and the arranged even where no false word is spoken (chapter 23). “Defective consent” is the umbrella over all of it: chapter 9’s sorting of the corrupted bargain — the lie, the fog, and the squeeze — runs the doctrines in parallel, and chapter 23 hands each its finest machinery; together they are the ancestry of every disclosure rule a term sheet has ever met (chapters 9, 23, 37, and 43).

Maysir, speculation, gambling. *Maysir* is the Qur’an’s word for gambling: originally the arrow-game of pre-Islamic Mecca, portions of a slaughtered camel allotted by a blind draw of marked arrows, and forbidden on grounds not of probability but of fellowship — the winner eats exactly what the loser lost, nothing produced, nothing exchanged, enmity the game’s reliable output (chapter 6). The insurance chapter draws the line the traditions were groping toward: a wager invents the peril it prices, while a genuine pool answers a peril the world has already delivered — which is why mutual protection and gambling, superficial cousins, are opposite answers to chance (chapter 27). “Speculation” is the modern word the book uses sparingly and tests rather than assumes: exposure manufactured for the sake of the payoff sits under maysir’s shadow; the bearing of a real venture’s real

risk does not (chapters 6, 27, and 37).

Damnum emergens, lucrum cessans, periculum sortis, carentia pecuniae, stipendium laboris, poena conventionalis. The Latin file: the extrinsic titles and their neighbors — claims standing beside the loan, never inside it, weighed by the scholastic tradition as grounds for recovering something past the principal (chapter 10 opens the file). *Damnum emergens*, the loss arising: compensation for damage the loan actually inflicted on the lender; it faces the past alone, prices harm already done and measurable, and repairs without enriching — the title the tradition trusted first (chapter 19). *Lucrum cessans*, the gain that ceases: the profit the loan prevented; it faces the future, was fought over accordingly, and won admission only where the missed gain was specific and showable — never on the bare ground that money always has somewhere better to be (chapter 20). *Periculum sortis*, the peril of the principal — *sors* is the capital sum, named for the fall of the lot: the question of whose loss the money's loss would be, and the most treacherous of the titles, since an uncapped price on risk reopens everything the prohibition closed (chapter 10). *Carentia pecuniae*, the deprivation of liquidity: the lender's own moneylessness for the loan's term, pressed late in the tradition as a title in its own right; the argument's real force, and the short step from it to ordinary interest, have chapter 21 to themselves. *Stipendium laboris*, payment for real work: Aquinas's name for the merchant's margin read morally as a wage; it stands beside the loan rather than upon it, with exact siblings in the rabbis' *sechar tircha*, the wage for trouble, and the *mudarib*'s profit share (chapter 12). *Poena conventionalis*, the agreed penalty: teeth for the due date, licit while it punishes a failure and corrupt once the lender profits by hoping for one — the clause every modern default-interest schedule descends from (chapter 22). Behind them all stands *interesse, id quod interest*, "that which is between": Rome's compensation word for the gap an injury opens, which spent a millennium naming harm before it became, as chapter 19 puts it, "the modern world's blandest word for the price of money" (chapters 19 and 22).

Ergodicity, ensemble average, time average, ruin probability, anti-ruin reserve. Five terms carry the book's survival mathematics, laid down in chapter 10 and worked to a proof in chapter 28. An ensemble average is taken across many parallel cases at one moment, the cohort averaged at a stroke; a time average is the accumulation of one lone sequence, step by step, each year compounding on the last. *Ergodic* is the mathematicians' word for processes in which the two coincide; the processes of livelihood (seed, stock, herd, fund) multiply rather than add, and for them the averages part company, the lived path settling below the cohort's mean. Real paths also have floors: an absorbing state is a place a path enters and never leaves, and ruin probability is the chance of reaching one within some horizon. On those definitions stands the book's central mathematical finding: a system can pass every test run on its average and still deliver most of the lives inside it to the floor, because the average belongs to whatever is big enough to hold a thousand lives at once, and a household holds exactly one. The anti-ruin reserve is chapter 10's name for the instrument aimed at that gap — the common chest "standing between each member's path and the floor" — which, as chapter 28 demonstrates, leaves every ensemble number where it stood and transforms what a single life can survive (chapters 10 and 28; the metric registry is Appendix D's).

Dharma, artha, pledge, lawful increase, renunciation, aparigraha. The guest traditions' terms, from the two comparative interludes. *Dharma* is right order: the measure by which the Sanskrit lawbooks, the *Dharmaśāstra*, judged increase, licensing the *dharmya* rate and damning the lender who exceeds it; *artha* (wealth, power, statecraft) names the domain of the *Arthaśāstra*, the manual that speaks the same debt law in a statesman's voice (In-

terlude I). The pledge of those codes, *adhi*, splits into custody and use forms, and where the pledge itself yields fruit the loan may bear no interest: one waiting may not be charged twice (Interlude I). “Lawful increase” is the architecture the interlude sets against the Abrahamic wall — interest itself lawful (*kusida* is the old word for lending at increase; *vridddhi*, growth, the later term for interest), but graded by circumstance, capped in published tariffs, watched by the crown, with over-rate stipulations refused by the courts (Interlude I). Renunciation is the Buddhist starting point: the monk forbidden money outright, down to the touch; the order’s necessities handled through the steward protocols of the Vinaya, its book of discipline; and the endowed monastery, needing to eat, discovering finance inside that same book (Interlude II). *Aparigraha*, non-grasping, non-possessiveness, is the old Indic vow against holding-as-hoarding, the disposition beneath the interludes’ shared warning: accumulation tempts every institution, including those built on renouncing it (Interlude II; the drift lesson lands in chapter 45).

Appendix B — Historical Timeline

This timeline gathers the book's dated record into one place. Its seventeen sections follow the table of contents, which sorts the history by theme rather than by strict sequence; the themes overlap in time, so each entry lives with its subject and the dates carry the traffic between sections. Within a section the entries run chronologically, every one compressed from the chapter named beside it, where the sources, the disputes, and the full circumstances live. Every date stands here at exactly the precision its home chapter verified — a day where the book pinned a day, a century where it pinned a century — and a few entries lead with a name instead of a year, because the book deliberately leaves certain foundations undated (the rabbinic redactions, the hadith collections, the classical lawbooks) and this appendix asserts nothing the chapters have not already established. A reader who wants the argument should read the chapters. A reader who wants to know when the Monti reached Mantua, or whether Nicaea's canon precedes the Qur'an's ultimatum — it does, by three centuries — can settle it here in under a minute.

Biblical prohibitions. Covenant law, written against a very old background of debt.

Around 1750 BC. In Old Babylonia interest is furniture: Hammurabi's laws set ceilings on rates already customary (silver at a fifth, barley at a third), and kings periodically proclaim the clean slate that wipes the board (chapters 1, 2, and 4).

The Torah's three statements. Exodus 22 forbids playing the creditor toward the poor; Leviticus 25 ties the loan to the brother's survival and the jubilee's great reversal; Deuteronomy restates the law with a release fixed to the calendar every seventh year and the disputed brother-and-foreigner boundary. The book, like the laws themselves, gives these no dates (chapters 2 and 4).

Eighth century BC. Amos indicts the whole cascade at once: the shaved measure, the rigged scale, the poor sold off for debt — the market conduct the release laws exist to reverse (chapter 4).

594 BC. The same crisis runs Mediterranean-wide: Solon's *seisachtheia*, the shaking off of burdens, cancels Athens's debts, frees the debt-enslaved, and bans the enslaving; Rome ends its own debt bondage, the *nexum*, by the traditional date 326 BC (chapters 1 and 2).

Middle of the fifth century BC. Nehemiah confronts creditors taking fields, vineyards, and children from their own kin, and forces a public act of restitution that returns even the interest (chapter 2).

Rabbinic development. Fences built around the Torah's bite; the book dates none of the redactions and keeps the tradition's own order.

Last decades before the Common Era. Hillel the Elder, watching credit dry up as each seventh year neared, enacts the *prozbul*, which lodges the creditor's claim with the court and so carries it across the release — the ancestor of every later device in this book (chapter 4).

The Mishnah's interest chapter. The fifth chapter of Bava Metzia sets the fences: a present sent to win a loan and a present sent after repayment are both *ribbit*; an investor must pay the shopkeeper who trades on his money; and the closing census of liability reaches everyone the paper touches, down to the scribe (chapters 8, 11, and 12).

The Babylonian elaboration. The Talmud prices the working partner's wage at what an idle laborer would take, compresses the offense into *agar natar*, payment for waiting, stretches the fence as far as *ribbit devarim*, interest paid in words; and, in the Nehardean academies, hands the *iska* its lasting skeleton, half loan and half deposit, each half with its own loss rule (chapters 7, 8, 11, and 13).

Patristic condemnation. The early Church legislates where it can, preaches where it cannot.

325. The council of Nicaea, first of the ecumenical councils, forbids the clergy to lend at interest (canon 17), and the councils after it widen the ban to the whole Church (chapters 1 and 7).

Late 360s. Famine in Cappadocia; Basil, soon bishop of Caesarea, takes the parable of the rich fool into the pulpit and tells the hoarders in the pews that their stored grain already belongs to the hungry. Ambrose preaches the twin argument in the *De Nabuthae*, under whose name the canon lawyers will later file the sentence (chapters 1 and 18).

Early Islamic jurisprudence. Revelation ends the old *riba* by name and leaves the jurists rules concrete enough to build a law on.

Among the final passages revealed. So the tradition itself places Qur'an 2:275–281: trade permitted and usury forbidden, the ultimatum of war against the lender who persists, ease for the strained debtor, capital kept without wrong done or suffered — the case that went out against Ta'if's old deferral-and-doubling (chapters 2 and 6).

Spring 632. At the farewell pilgrimage, on Arafat's plain, the Prophet declares the age of *riba* closed and starts with his own house, the *riba* owed to his uncle Abbas; the tradition likewise remembers taker, payer, scribe, and witnesses cursed alike (chapter 6).

The rulings the jurists build on. The collections carry the working rules without dates: the six commodities traded like for like, hand to hand; the ban on selling *gharar*, radical uncertainty; the advance sale confined to known measure, known weight, known term; *al-kharaj bi-l-daman*, the yield goes with the liability; and the plain pledge: the Prophet dies with his armor pawned to a lender for barley (chapters 6, 9, 10, 23, and 24).

The *qirad*. When the Italian notaries first record the commenda in the tenth and eleventh centuries, the *fiqh*'s capital-and-labor partnership has by then run for some three hundred years on the same chassis: profit by agreed ratio, money losses on the capital alone, the manager liable for negligence and never for the sea (chapter 11).

Hindu Dharmaśāstra and Arthaśāstra debt rules. The guest tradition of Interlude I: increase regulated and graded rather than banned.

Last centuries before our era. The Dharmasutras of Gautama, Baudhayana, and Vasishtha already hold the full apparatus: lawful rate tables, the usurer listed among sinners, doubling rules that stop a debt's growth at a ceiling (Interlude I).

Around 320 BC, by ascription. The Arthaśāstra, credited to Kautilya, its present form a composite of the first centuries of our era, opens its debt law with a four-rate tariff graded by danger: dearer through the forest, dearest over the sea (Interlude I).

Second or third century of our era. The Manusmṛiti fixes the verse-code form (rates carded by *varna*, the creditor's lawful and unlawful means enumerated), and the Yajñavalkya Smṛiti follows it, later and tighter (Interlude I).

Around 1100. Vijnaneshvara's *Mitakshara* commentary explains the danger premium candidly — where life and goods stand in doubt, so does the principal — and remains a working authority for Indian courts into the twentieth century (Interlude I).

Buddhist monastic money prohibitions and temple lending. The guest tradition of Interlude II: renunciants forbidden the coin, and treasuries that lend it anyway.

The Vinaya's money rules. The discipline bars a monk from accepting gold and silver, from letting anyone hold it for him, from letting it so much as lie beside him; and the tradition's gloss widens the bar to any medium of exchange (Interlude II).

A century after the founder's death. By the texts' own count, the monks of Vesali set out a collecting bowl for coins on observance days, and the quarrel over that bowl becomes the young order's first great money crisis (Interlude II).

Around the second century before our era. A railing medallion at the Bharhut stupa carves the laity's side of the bargain, Anathapindika's purchase of the Jeta Grove, the ground drawn as a tiling of coins; around the year 120 of our era, the donor Ushavadata records at Nasik the developed instrument: imperishable principal lodged with the town guilds, its interest clothing and feeding the monks (Interlude II).

End of the fifth century. A man named Chen Pin pawns a bolt of hemp cloth at a Chinese monastery treasury — the earliest pawnshop transaction the historians can point to — and until the Tang the pledge-loan trade is effectively a monastic monopoly (Interlude II).

Sixth century. Xinxing's Sanjie movement builds the Inexhaustible Treasury at the Huadu monastery in Chang'an, boundless giving channeled through one institution, until the emperor Xuanzong liquidates it in 713 (Interlude II).

1255. Japan's records first name the *mujin*; the temple rings run for centuries as mutual aid without interest, then commercialize by degrees; a business statute reaches them in 1915, and in 1951 a mutual-bank law converts the survivors (Interlude II).

Medieval Jewish commercial law. The halakha of credit equipped for a commercial civilization.

Maimonides' codification. A millennium after the Mishnah's sages, the Mishneh Torah catalogs the interest laws with a codifier's completeness and states the iska's two halves at their starkest: the loan half on the manager come what may, the deposit half on the investor (chapters 7, 8, and 13).

1437. Florence charts its Jewish lenders under the condotta (the word Italian cities also used when they engaged a mercenary captain), with the licensed rate capped at twenty percent; within a generation almost every central and northern Italian city but Venice and Genoa holds such a charter (chapter 17).

The Shulchan Arukh at siman 177. The code files the iska machinery at the address where every later drafter finds it, the working partner's wage shrinkable to a single dinar but never to zero (chapters 12, 13, and 31).

By the sixteenth century. The Jewish communities of the Polish-Lithuanian Commonwealth sit at the center of a working credit economy (leaseholders, merchants, traders), not at its margins (chapter 31).

Scholastic extrinsic titles. The *mutuum* stays barren by definition while compensation grows around it, title by argued title.

Twelfth century. Gratian's compilation carries the blunt arithmetic, that whatever exceeds the principal is usury; and in the mid-1180s Urban III's *Consuluit*, answering a confessor's query, rules that the lender's hope alone can be the usurious price (chapters 1 and 8).

Thirteenth century. William of Auxerre gives the debate its most beautiful sentence: the usurer sells time, the common possession of all creatures. On 5 September 1234 Gregory IX promulgates the *Liber Extra*, compiled by Raymond of Peñafort, and its decretal *Naviganti* rules that even the lender who takes the voyage's peril on himself may not charge past the principal. Around 1260 Moerbeke's Latin *Politics* reaches the schools, and within a generation Aquinas joins Aristotle's sterile coin to the theologians' case (chapters 1, 7, 10, and 11).

Mid-1290s. Peter John Olivi (1247–1298), writing the *Tractatus de contractibus* at Narbonne, defines the merchants' word *capitale*: money aimed by its owner's settled purpose at probable profit already does capital's work, and deserves capital's name (chapter 20).

Fifteenth century into the sixteenth. The pastoral theologians of the banking cities — Antonino in Florence, Bernardino of Siena (1380–1444) — concede compensation for a trade genuinely torn away while writing chiefly against the abuses; Conrad Summenhart's seven-part anatomy of contracts is printed at Hagenau in 1500 (chapter 20).

25 October 1586. *Detestabilis avaritia* condemns the *contractus trinus* — partnership, capital insurance, and an uncertain profit swapped for a certain one, stacked into a guaranteed return — and rules the assembly usurious (chapters 13 and 37).

1605–1642. Lessius, who cross-examined the merchants on the Antwerp exchange, formulates and accepts the deprivation title that Molina (1535–1600) had refused; Juan de Lugo's synthesis of 1642 reads foregone gain and endangered principal as one family of compensable detriment (chapters 19, 20, and 21).

1679. The Holy Office under Innocent XI condemns the casuists' defense of the *mohatra*, the buy-dear-on-credit, sell-back-cheap-for-cash pairing that nets a loan at interest (chapters 24 and 37).

1 November 1745. *Vix pervenit*: usury arises in the loan contract as such (moderate gain, rich borrowers, and fruitful use excuse nothing), while genuine titles outside the loan stand. The Holy Office applies the letter to the whole Church in 1836; from 1830 to the 1870s the replies converge on the not-to-be-disturbed formula; and canon 1543 of the 1917 Code restates the doctrine while releasing the practice, both in one sentence (chapters 7, 14, and 30).

Islamic contract elaboration. The fiqh builds out its lawful instruments and patrols their edges.

The classical grammar. The lawbooks develop what the rulings founded: *mudarabah* profit divided by ratio and only from profit, the *musharakah* loss rule the jurists trace to the caliph Ali, sale and lease and advance sale each hedged with its conditions of possession and risk (chapters 11, 23, 24, and 25).

Around 1100. Al-Ghazali's *Ihya* supplies the money theory: coin exists to measure and move other goods, and *riba* is the perversion that turns the measure into the merchandise (chapter 1).

Sixteenth century. Ottoman founders begin endowing money itself: the cash *waqf*, its corpus lent out to fund the teacher and the lamps. Civizade attacks the practice on clas-

sical grounds before his dismissal in 1541; Birgivi answers in 1572 with a treatise titled as a drawn sword; practice settles in the funds' favor, and the courts' registers of the early seventeenth century record whole towns among their debtors (chapter 29).

The Majalla. The Ottoman codification writes the classical sale law into modern articles (*ma'lum*, known, at every joint) and shows how codification narrows the room a stratagem needs, a lesson the modern standard-setters relearn (chapters 23 and 32).

Confraternities, guilds, and mutual-aid institutions. The pool keeps the books of every era in between.

9 June 136. At Lanuvium the worshipers of Diana and Antinous cut their by-laws into a temple porch: entry at a hundred sesterces and an amphora of good wine, dues of five asses a month, three hundred sesterces from the chest at a member's death — the burial club dating itself by the consuls (chapters 3, 27, and 28).

The *aqila*. Islamic law keeps the older custom under which a man's kinsmen share the blood-money he owes: liability pooled long before anyone priced a premium (chapter 27).

Confraternity and guild chests. Medieval parish brotherhoods and trade fraternities collect dues against sickness, against a daughter's dowry, against burial, and escort their dead to the grave as the Lanuvium club once did — the same equipment, rededicated (chapter 27).

1564. Eliezer Ashkenazi organizes the first of the modern *chevra kadisha* societies, patterned deliberately on the guilds around it; Judah Loew, the great rabbi of Prague, codifies its rules (chapter 27).

1891. *Rerum novarum* reads the guilds' abolition as the workman's abandonment and commends the mutual-help societies by name as their lawful successors (chapters 27 and 33).

Monti di Pietà. The charitable pawn bank, and the era's mercy and cruelty bound into one story.

1361. A London fund endowed to make pledge loans without charge pays its costs out of its own capital and closes as the arithmetic dictates, the failure mode the Monte's small charge is designed against (chapter 17).

13 April 1462. Perugia's council adopts its Lenten preacher's proposal and votes three thousand florins for the first Monte; Orvieto follows within a year, Monterubbiano in 1465 — a Franciscan and a Dominican founding it together — Viterbo in 1469, Siena in 1472, Genoa in 1480, Milan in 1483 (chapter 17).

1484–1494. Bernardino of Feltre (1439–1494) plants Monti from Mantua onward and becomes the institution's engine. The record stays whole here, as chapter 17 keeps it: his Lent of 1475 in Trent is followed by the death of a child named Simon, and the town's Jews are tried on the ritual-murder libel and executed; expulsions follow at his urging: Perugia 1485, Gubbio 1486, Ravenna 1491, Brescia 1494, the year he dies at Pavia, venerated at once. The cult built on the libel is not suppressed until 28 October 1965 (chapter 17).

1494–1498. The opposition goes to print: Bariani's *De Monte Impietatis* at Faenza, then Cajetan's treatise, the most thorough of the attacks; the objectors' strongest card is equity, the rescue billed to the rescued (chapter 17).

4 May 1515. The Fifth Lateran Council's *Inter multiplices* closes the question: the Monte may charge its borrowers its running costs, and nothing for gain (chapters 14 and 17).

1534 onward. Ypres from 1534, Rome in 1539, Avignon for France in 1577, Madrid in 1702, Mexico City in 1775 on a silver fortune, Paris re-founded in 1777 and still lending at the same address; Italy counts five hundred fifty-six working Monti in 1896; and Siena's Monte of 1472, re-founded in 1624 on the pledged Maremma pastures, trades today as Banca Monte dei Paschi di Siena (chapter 17).

Heter iska expansion. Hillel's kind of move, made standard form.

1591. Cracow calls Rabbi Isaiah Menahem ben Isaac, Mendel Avigdors to Polish Jewry, who sets about repairing the venture deed his own fame had spread (chapter 31).

1607. The Council of Four Lands enacts his model rules on credit: communal legislation drafted in the shadow of the Torah's oldest prohibition (chapter 31).

1667. The *Nachalat Shivah*, the notaries' handbook printed at Amsterdam, seats the *heter iska* in the standard repertoire beside the marriage deeds (chapter 31).

The *heter iska klali*. Drafted for modern volume, the general permission of venture rides standard terms into Israel's banking system and the New York courts; Rabbi Moshe Feinstein's responsa set the twentieth-century debate's frame (chapter 31).

Friendly societies and modern mutual aid. The pool professionalizes without giving up its members.

1793. Parliament passes the first friendly-societies act; the registry that grows from it counts nearly fourteen million members by 1905, and in 1945 the societies still cover some eight and a half million Britons (chapters 27 and 28).

1810. The Oddfellows organize and put a surgeon in every lodge; the Rechabites pool the temperate; Tredegar's miners build the society that doctors the whole town (chapter 27).

1892. Eleven immigrants pool ninety-five dollars in a Henry Street synagogue; their free-loan society has since revolved hundreds of millions of dollars, interest-free, with repayment running near ninety-nine percent (chapters 15 and 28).

1983 and 1986. Iran writes the *qard hasan* into its banking statute, and the Fiqh Academy at Amman confines loan fees to actual service expenses, nothing beyond (chapter 16).

December 1985. The Islamic Fiqh Academy at Jeddah refuses the commercial insurance contract for its gharar and names cooperative insurance the lawful form; *takaful* and *re-takaful* answer, and the Academy's criteria follow at Oran in 2012 and Riyadh in 2013: a non-profit fund, a manager who is only an agent (chapter 27).

2001. In Lahore, a widow's request that no interest-bearing loan could honestly meet sets Dr. Muhammad Amjad Saqib and his friends lending at zero from their own pockets; Akhuwat, named for brotherhood, grows from that seed to millions of interest-free loans, repaid at better than ninety-nine percent by 2020 (chapter 16).

Modern Islamic banking. The wall rebuilt at industrial scale — and the tradition auditing its own instruments in public.

1963. Ahmad El-Naggar opens the Mit Ghamr savings house in the Nile Delta, paying and charging no interest and placing deposits in local ventures; the experiment is closed down by 1967, by which time Egypt has nine such houses (chapters 30 and 32).

1975. The Islamic Development Bank starts work in Jeddah, and Dubai Islamic Bank opens on 15 September, by most counts the first full commercial bank of the type; in February 1976 the Makkah conference of scholars and economists declares bank interest *riba* in every form (chapters 30 and 32).

1988. Meeting in Kuwait, the Fiqh Academy licenses the *murabaha* triangle on the classical conditions — the institution must genuinely possess the goods and carry their risk until delivery (chapter 24).

February 2008. AAOIFI's *sukuk* resolution turns Usmani's audit (most issued *sukuk* pay like conventional bonds) into standards: no capital guarantees, no repurchase at par; in 2009 the Academy at Sharjah refuses organized *tawarruq*, and in December 2010 the prudential standard-setter takes on payout smoothing by name (chapters 13, 26, and 32).

2017–2018. The Dana Gas affair: an issuer disowns its own certificates' compliance, an English court enforces the par repurchase as an ordinary contract, and the holders learn which layer of their paper was real (chapter 26).

June 2023–September 2024. LIBOR is switched off and the markup industry repapers to the successor benchmarks on schedule; the 2024 stability report counts seventy-nine percent of the industry's financing in the markup family, with the partnership instruments sharing the eight-percent remainder with everything else in the toolbox (chapters 24 and 32).

Catholic Social Teaching. The tradition resumes its interrogation of economic life at modern scale.

15 May 1891. Leo XIII issues *Rerum novarum*, the founding document of modern Catholic social teaching, which names usury in its opening argument: condemned repeatedly, practiced still in altered dress (chapter 33).

15 May 1931. *Quadragesimo anno*, drafted with the economist Nell-Breuning's hand, dissects the credit dictatorship, the few who govern the lifeblood of the economy, without once using the word usury (chapter 33).

1967–1987. *Populorum progressio* carries the question up to the scale of nations; *Sollicitudo rei socialis* then names the inversion: development finance become a drain on the developing (chapter 33).

1991–1992. *Centesimus annus* presses the debt principle to its limit in the centenary year; the new Catechism keeps one sentence with teeth, counting the usurious starving of a brother as indirect homicide, and the 2004 Compendium repeats the condemnation (chapter 33).

June 2009. *Caritas in veritate*, published nine months after the crash, argues that markets need trust and gift and not calculation only, and returns the old word to a document of the highest rank for the first time since 1891 (chapter 33).

2015–2018. *Laudato si'* sets the economy inside creation and passes its terse verdict on finance outgrowing the real; in 2018 two curial offices jointly issue *Oeconomicae et pecuniariae quaestiones*, the magisterium's first document since *Vix pervenit* to read financial instruments one by one, and Francis tells Italy's anti-usury foundations in February that usury humiliates and kills (chapter 33).

October 2025. Leo XIV holds the register: usury a grave sin, and never a mere accounting issue (chapter 33).

Ergodicity economics and modern risk-pooling theory. The mathematics that vindicates the pool.

1745 and 1774. Parliament voids the wager policies, marine and then life, requiring an insurable interest in the thing insured — the law walling the pool off from the bet (chapter 27).

1762. The Society for Equitable Assurances is founded in London on an idea still novel: that the right price of a life policy can be worked out in advance (chapter 27).

1921. Frank Knight splits the word risk in two: the countable kind that tables and pools can carry, and the kind with no population of cases behind it (chapter 36).

2019. Ole Peters's *Nature Physics* paper puts the ergodicity problem before economics: what a gamble averages across many players is not what it does to one player over time: the arithmetic of ruin the traditions had guarded by rule (chapters 10 and 28).

2023. Graham Boyd and Jack Reardon's *The Ergodic Investor and Entrepreneur* carries the time-average case to practitioners: wealth compounds in sequence, so the order of gains and losses matters, and strategy should treat unpredictability as a partner rather than an enemy (chapter 28).

Contemporary non-extractive finance. Old instruments under new charters.

1876 and 1889. Jacobsen deeds the Carlsberg brewery to a foundation; Abbe vests the Zeiss works in the Carl-Zeiss-Stiftung, whose 1896 statute a reader of chapter 29 will recognize as a waqf deed in German legal dress (chapter 29).

2004. The United Kingdom legislates the community interest company, its asset lock compulsory and irremovable (chapter 29).

2007–2009. The Bellagio meetings convene finance to ask whether investment can be organized for good on purpose, and the Global Impact Investing Network follows (chapter 34).

2010. A microfinance flotation oversubscribed more than thirteen times in August and, in October, an Andhra Pradesh emergency ordinance against usurious rates and coercive recovery, drafted after officials counted fifty-four borrower deaths, put the extraction question into a modern statute book (chapter 34).

2022. Patagonia's owners place all voting shares in a purpose trust holding them to the mission and the economic shares in a nonprofit, performing the endowment move at market scale in the open (chapters 29 and 34).

Steward Ownership and the mission-locked forms. Employee Ownership trusts, golden shares, perpetual purpose trusts, the braided capital stack: Parts IX and X assemble these into working instruments, and there the timeline hands the reader back to the book (chapters 29 and 41 through 44).

Appendix C — Contract Typology

Appendix A holds the book's words; this appendix holds its forms. Fifteen entries classify the contractual inventory the chapters have walked: every arrangement by which the three traditions let capital move, and the assemblies by which people have tried to move it dishonestly. Each entry states the form's anatomy (what changes hands, what comes back, where loss sits), names its instruments across the traditions, gives the test that separates the honest specimen from its counterfeit, and routes the reader to the chapters holding the sources and the quarrels. The order is the table of contents' own: the exchange forms first, then the venture forms and their paper, the protection form, the grace forms, the counterfeit, and finally the five instruments of the book's own constructive turn, the financing floors of chapter 41's stack, classified here rather than deployed. The signable subset, ten documents fitted to one fund, is chapter 42's library; this appendix is the field those documents were chosen from. One rule governs every row, laid down in chapter 3 and applied ever since: "The names are hypotheses; the payoff is the evidence."

Loan. The transfer of money, or of anything consumed in its use, against the return of its equivalent — chapter 3's one object held in the vocabulary of three legal traditions: *mutuum*, *milveh*, *qard*. Ownership crosses with the handover; what returns is the *tantundem*, the same in quantity and kind; the claim is personal, following the borrower whatever his venture does, and its payoff stands written before the venture has an outcome to report. That fixity is the form's signature among its siblings, and the source of its law: because the lender has parted with the thing entirely and retains nothing that could earn, all three traditions bar any stipulated increase, from the first coin, and post fences well past the coin, as far as benefits in kind and interest paid in words (chapter 8). The traditions' honest deployments of the bare loan are the mercy instruments classified below; nearly everything else in financial history that calls itself lending at profit is examined in this book under some other row of this table. Anatomy and boundaries: chapters 3 and 7; persons and communities: chapters 2 and 4.

Sale. A thing for a price, both transferred at once, no claim surviving the exchange. The traditions bless the merchant's margin ("buying cheap and selling dear is how merchants eat," as chapter 3 put it) and police the exchange for equivalence instead: the rabbis' *ona'ah* band with its one-sixth trigger, the Latin doctrine of the just price, the fiqh's *gharar* rule that the thing sold must exist, be known, and be deliverable, so that a price is never set against a mystery (chapters 9 and 23). The sale is the benchmark against which every financing in this table is measured, because its gain is the cleanest kind: realized in one completed exchange of real things. Its counterfeits arrive when time or prearrangement creep in — the deferred premium (the next row) and the boomerang sale (the hybrid row). Full doctrine: chapters 3, 9, 23, and 24.

Deferred sale. A sale whose price waits (goods now, payment at term), and the seam where the sale form meets the loan's temptation, since a padded term price collects for the waiting under a sale's name. All three traditions permit the form; all three guard the same gate, each with its own instrument. The canonists allowed a term price above the cash price only while the goods' worth at the payment date remained genuinely uncertain, so that the premium prices shared ignorance, never the calendar (chapter 24). The rabbis ruled that rent may rise for time but a sale price may not, and filed the padded deferred price under *avak ribbit*, the dust of interest (chapters 3 and 24). The fiqh's *murabaha*, the

disclosed cost-plus sale, demands that the financier really own the goods for a real interval (his title, his risk), with cost and markup confessed, and a price fixed once, never to grow with delay; in the form called *murabaha* to the purchase orderer it now carries most of the industry that bears Islam's name (chapters 24 and 32). Test: whether anything but the calendar explains the premium.

Lease. The sale of a thing's use, in slices of time, possible only for things that survive their use — a field, a house, an ox, never bread or coin. The lessor keeps ownership and ownership's burdens: maintenance, structural failure, the thing's perishing are his — the rent is honest because it is income from an asset the lessor still owns and still risks. The rabbis accordingly rent courtyards without *ribbit* anxiety and refuse the "rental" of money, whose use cannot be separated from its spending; the *fiqh's ijara* is defined as a sale of usufruct, valid only if the lessor first acquires the asset and answers for its defects; Christendom's rent and usufruct doctrine runs on the same floor (chapters 3 and 25). The failure mode is the lease that walks every burden of ownership across to the user while retaining nothing but the payment schedule — at which point the traditions, like the accountants, reclassify the instrument as the secured loan it is (chapter 25).

Partnership. Capital committed to a shared venture rather than transferred against a fixed claim: ownership retained, exposure retained, and a share of surplus earned by that exposure. Loss follows an iron map — money-loss to the money's owner, labor-loss to the laborer — and no clause may walk an investor's loss onto his partner while his gain stays behind: the lion's partnership is void in Roman law, the guaranteed *iska* half collapses into loan, and the *mudarabah* whose capital is indemnified is void, its stipulated share condemned as *riba* (chapters 11 and 13). The instruments are the traditions' proudest: the *societas* of consent and common fate; the *musharakah*, capital from both sides, profit by agreement, loss strictly by capital shares; the *iska*, half loan and half deposit by the Nehardean formula; the *mudarabah* and the *commenda* (the next row). Partnership is, as chapter 3 introduced it, "the door all three traditions deliberately left open in their wall against usury" — and the most imitated silhouette in this table: the costumes of the hybrid row are cut to its pattern. Chapters 11 through 14.

Profit-sharing investment. The partnership family run as finance: one party brings money, the other work, and the financier is paid a ratio of realized profit fixed at signing — never a stipulated sum, which voids the contract. In the *mudarabah*, the investor's capital alone bears money-loss while the working *mudarib* stands as an *amin*, a trustee liable for transgression, negligence, or breach of mandate, never for the venture's ordinary fate; the *commenda* gave the same architecture to the Mediterranean, the traveling *tractator* trading on the seated investor's capital; the *iska* engineered the mix with rabbinic caution, its working partner owed a real wage — the *sechar tircha* that stands beside Aquinas's *stipendium laboris* as pay for genuine work (chapters 11 and 12). Profit here is discovered at the reckoning, never promised before it, and chapter 14's law governs the count: profit must come from profit, with interim payouts held back or sobered until the reckoning is real. The modern deployment, with its retentions, valuations, and caps, is chapter 41's risk-sharing floor.

Asset-backed certificate. Fractional ownership written on transferable paper. Chapter 26 states the row's whole test: "A certificate is honest when it conveys a genuine fraction — of a thing, of a venture, of a named stream of fruits — and its holder eats what the fraction earns and loses what the fraction loses." The form is old in every tradition: the *shtar iska* is already a certificate, its holder awaiting the venture's actual accounting; Genoa's Casa di San Giorgio cut a city's tax revenues into the *luoghi*, participation paper with a market price; the *fiqh's sukuk* are the standard-setters' modern version: certificates of ownership in

assets or ventures, built to pay from what the assets yield. The counterfeit is the certificate engineered back into the bond it was meant to replace (par value protected, coupon-like distributions smoothed, recourse running to the issuer rather than the asset), the drift the industry itself names in its distinction between asset-backed paper and merely asset-based paper (chapters 26 and 32). Test: whether the holder owns something or is merely owed something.

Insurance / mutual guarantee. Protection against a defined peril, funded by many premiums and paid to the stricken: a form with two possible owners, and the ownership decides its moral character (chapter 27). The *fiqh*'s jurists condemned the commercial contract sold at a fixed premium as an exchange built on excessive *gharar* — a known payment against an unknowable return — and engineered the objection away rather than the protection: in *takaful*, the members' payments enter as donations to a fund the members own together, an operator runs it for a fee or a profit share, and the pools reinsure one another through *retakaful* (the terms of art are in Appendix A). The mutual genus is older and broader: guild boxes and confraternities, the friendly societies of the industrial age, the communities' own burial and sickness funds — members carrying one another rather than buying a stranger's promise. The common law adds its own fence, insurable interest: protection may answer a peril the insured really bears, never manufacture a wager on a stranger's ship (chapter 27). Where the peril is shared and the chest is owned in common, the form crosses into the pool proper, the mutual-aid row below. Chapters 27 and 28.

Gift / endowment. The forms that refuse to create a claim. The gift transfers without counter-obligation: *tzedakah*, built on the root for justice — owed to the poor, not volunteered; *zakat*, the obligatory levy whose name carries purification, with voluntary *sadaqah* above it; the Gospel's instruction to lend expecting nothing, which keeps the loan's shape and removes its claim (chapters 4, 5, 6, and 18). Maimonides' ladder belongs to this row's law: the highest giving (gift, loan, partnership, or employment) is whichever restores a person to standing (chapters 3 and 18). The border needs patrol in both directions, since gifts that arrive just before or just after a loan can be interest in kindness's clothing (chapter 8). Stretched past its giver's lifetime the gift becomes the endowment: the *waqf* with its three bars (no sale, no gift, no inheritance) and its audited substitution; the *hekdesh* of Temple, community, and poorhouse; the church's locked patrimony under conciliar law since Chalcedon. The endowment exists precisely so that its capital is never again anyone's to extract (chapter 29).

Hybrid and synthetic forms. Not a form but a method: assembling individually licit contracts until, together, they pay what a forbidden one would have paid. The specimens run the width of the tradition. The '*ina** double sale — sold for one hundred twenty deferred, bought back for one hundred cash — condemned by most schools as *riba* in costume, defended in one line of analysis that weighs only the visible contracts, an honest and permanent *ikhtilaf* (chapter 3). The rabbis' *avak ribbit** category exists because the dust of interest settles where the biblical definition cannot reach (chapters 3 and 24). Latin Christendom's perfected specimen was the *contractus trinus* (partnership, plus insurance of the principal, plus the sale of an uncertain profit for a certain one, three defensible agreements stacking into a guaranteed return), condemned by Sixtus V in 1586 and quarreled over for a century more (chapter 13). The *heter iska*, by contrast, earns a wary respect: a genuine repartnering of credit, whose uses and whose perils the tradition's own critics weigh in chapter 31. The modern stack (participating preferences, cumulative dividends, redemption rights) rebuilds the triple contract at scale (chapter 37, whose verdict this table applies row by row: "No single clause is the usury; the stack is the usury."). Diagnosis: chapters 30 through 32 and 37; the method of

reading: chapter 3.

Cost-recovery credit. Lending whose charge prices the work of lending, never the loan itself: the borrower repays principal plus the documented cost of running the fund, and no one earns a margin on his need. The charter case is Christian: the Monti di Pietà, the pawn banks the Franciscans preached into being, lending the poor small amounts against pledged objects at a charge sized to bare operating cost. The Fifth Lateran Council ended a generation of dispute in 1515, ruling that expenses may be recovered and profit may not (chapter 17). The fiqh reached the same line in modern form: a service fee on benevolent lending is permissible within actual expenses, and anything beyond is *riba* (chapter 16). Chapter 40 wrote the row's modern specification as the cost line of its formula — shown, allocated, once, and at zero margin — and chapters 41 and 42 deploy it as the fund's credit floor and its loan paper. The row's characteristic corruption is the percentage that outgrows the work it claims to price; the remedy is the invoice. Chapters 16, 17, and 40 through 42.

Interest-free solidarity loan. The bare loan run as mercy and engineered to revolve: principal out, principal home, nothing else, ever. The Jewish form is the *gemach*, the free-loan fund of *gemilut chasadim* — donated capital revolving at zero, its security guarantors and good records rather than any charge, from the medieval communities to the free-loan societies of the immigrant cities (chapter 15). The Islamic form is the *qard hasan*, the beautiful loan the Qur'an casts as lent to God: a true debt, due in full, its lender entitled to nothing past the principal (gratitude the borrower adds freely is virtue; the same increase, expected by the lender, is *riba*), institutionalized from household circles to Akhuwat's national scale (chapter 16). The row's discipline surprises first-time visitors: rigor is what keeps mercy solvent, and the funds keep books like lenders precisely so they can keep lending like givers. Both systems treat repayment as part of the mercy: it restores the borrower's standing and refills the fund for the next borrower. Chapters 15 and 16; the signable version is chapter 42's first instrument.

Mutual-aid / solidarity-pool agreement. Membership in a shared chest against defined perils: the contract-form the traditions built specifically against ruin, which chapter 3 refused to file as a footnote to the others. Members pay small scheduled contributions into a fund that is their own common property; when the named blow lands on one of them, the chest answers by rule. The pool carries its members' downside and no venture's upside: nobody collects a dividend from a light year, which is what separates membership in a chest from the purchase of a policy (the insurance row above). The ancient instances are complete specimens — the Roman burial club at Lanuvium with its dues, its funeral benefit, and its effigy rite for a withheld body; the Talmud's donkey-driver and sailor compacts, replacing a lost animal or boat while excluding the negligent; the confraternities and friendly societies that carried the pattern into the industrial age (chapters 3 and 27). Its mathematics is the book's deepest argument: pooling moves none of the group's averages while sharply lowering each member's chance of ruin (chapter 28). Chapters 3, 27, and 28.

Mission-locked equity. Ownership stakes in companies whose charters have retired the exit: an asset lock ties the company's worth to its purpose, so that no sale can turn mission-built value into a private windfall; control sits in steward hands, unmarketable and unhereditary; and the investor's stake leaves only by the mission-preserving buyback, its price fixed by formula when the money enters, its pace set so the company outlives its own redemption (chapters 29, 41, and 42). The form is the endowment's logic translated into corporate instruments — the *waqf*'s bars, the *hekdes*'s covenant, the patrimony's inalien-

ability, reappearing as the statutory lock of the community interest company, the golden share's veto over the acts that would undo the dedication, the purpose trust, and Steward Ownership's binding commitments (chapter 29). Capital in this row is patient by construction: it earns from real results while it stays, and its leaving is priced to protect the purpose rather than to capture it. Design and deployment: chapters 41 and 42; ancestry: chapter 29.

Non-extractive revenue participation. The table's last row is the book's own synthesis: capital paid a stated share of a named, realized result — participation by ratio, capped at a bounded multiple fixed at signing, collected only out of the result actually counted, and therefore honestly zero in the year the venture has nothing to give. The share's honest base is chapter 39's true surplus — the remainder after wages to standard, the founder's line, maintenance, and the reserves — rather than gross revenue: chapter 40 bases the share there, not on takings, so that no participation collects ahead of wages, upkeep, or the chests. Participation in a merely named stream is as old as Genoa's tax paper (the certificate row); the discipline lies in naming the right stream. The full specification is chapter 40's formula, in which every unit of return above principal carries a title (cost shown, loss proven, work waged, surplus shared by ratio under the cap), and the whole stack is published so a stranger can check it (chapter 42's disclosure schedule). Worked numbers: Appendix E; the metrics: Appendix D; the fund that runs it: chapter 41.

Appendix D — Moral Geometry Metrics

Appendix A holds the book's words, Appendix B its dates, Appendix C its forms; this appendix holds its measures. Seventeen entries catalog the gauges the chapters built one at a time and chapter 43's analyzer computes wholesale: for each, what it measures, how it is computed, how to read it, and where its doctrine lives. The suite has a spine, laid down when chapter 35 first turned the doctrine into arithmetic, in two questions that govern nearly everything a contract can be asked. "How closely does the claim's collection track the venture's actual fortunes?" — that chapter's question of form — and "how much of the claim answers no burden anyone can show?" — its question of size. Every gauge below is one of those two questions pointed at a particular part of the machinery: the ledger, the queue, the reserves, the locks, the pools, the paths. Two rules govern the registry. First, these are computations, not accusations — each metric is built so that any two readers, however opposed, land on one number, and what the number means remains an argument the home chapters equip the reader to have. Second, the seventeen are never added up: the book's machine returns memos, not grades, because a moral geometry compressed to one digit stops being geometry, and chapter 43 rests its whole tenth module on that refusal. A reader wanting the constructions in motion should run the worked scenarios of Appendix E beside these pages; the metrics grade what those waterfalls stage.

Moral-title coverage ratio. The ledger's own gauge: of everything a capital claim will collect above return of principal, the fraction that sits on a proven titled line. The construction is chapter 35's moral-title ledger run to completion: every euro the instrument will claim across its life, seated on the zero line or on one of the seven standing lines (expense, work, loss, risk, liquidity, opportunity, surplus), a seat granted only where the title is "genuine in fact, not merely in name": the invoice for the expense line, the wage record for work, the netted shortfall for loss, the shown danger for risk, the documented deprivation for liquidity, the named and evidenced elsewhere for opportunity, the realized result for surplus. One fence crosses the whole computation before the ratio is struck — "a burden may be counted once," so a danger priced into the rate, retired by the collateral, handed to a guarantor, and billed once more at default enters the numerator one time, not four. Coverage of one means every claimed euro has named its work. Anything less is measured by the next entry, because the two gauges are one fact read from opposite ends. Construction and lines: chapter 35; the fences line by line: Part VI; the audit that keeps coverage honest after signing: chapter 44.

Extraction residual. The book's verdict quantity, and the only metric the whole suite exists to drive toward zero: the share of a claim that, in chapter 35's definition, "answers no shown burden, no work, no borne risk, no realized share." It is found by subtraction: run the ledger, let the zero line and every honest title absorb what they can evidence, and weigh the remainder. That is what makes it a measurement rather than a charge — the construction accuses no one, names no motive, and lands on the same number from either side of the table. Two committed disciplines give the reading its teeth. The residual is funded, "euro for euro, by the residual claimants," chapter 37 showed, so a nonzero residual always has addressees, not just a size; and the non-extractive formula of chapter 40 holds the term at zero "by construction at the drafting table, by audit ever after," so for any instrument claiming that formula's name, the residual is not a score to optimize but a promise to keep. Naming and construction: chapter 35; who pays it: chapter 37; the zero:

chapters 39 and 40; the yearly bottom row: chapter 44.

Capital-at-risk index. How much capital is actually out in the venture's weather, netted, over time. Chapter 36 supplied both corrections that make the gauge honest. "Capital at risk is a curve, not a number": exposure is read period by period, the whole principal exposed to the last day on an interest-only claim, a melting stake on an amortizing one, so that a risk charge priced on day-one exposure overcharges every year after the stake begins to melt. And exposure is read after netting: take off the collateral, the guarantors, the seniority, the covenants (every shelter the same paper erected) and let the risk line price only what is still exposed once its own protections have done their work. The index is that netted curve, and it is computed twice, once per side, because chapter 36's column check found a contract the lender's column showed as nearly safe holding "a three-in-a-hundred chance of the house" across the table. Read it against the fence that has policed the risk line since the sea loan: danger earns payment where it is borne, not where it is merely priced. Construction: chapter 36; the title it disciplines: chapters 10 and 35; the module that runs it: chapter 43.

Founder-labor visibility score. How completely the venture's consumed inputs made it onto the record at entry prices. The register never forgets the wired money; chapter 38's contribution schedule exists for everything else — "every input the venture actually consumed, entered by name, dated, and priced at entry" — and this score is the schedule's completeness. Wages foregone are checked against the going standard rate for the work in fact performed; the founder's declined salary is priced at the rate she turned down, because "Zero is also a price," and it is the only one the cap table never has to defend; the house guarantee, the equipment, the years of evenings enter at entry value, reality being the entrance requirement chapter 14's doctrine of proportion sets for any contribution. Low visibility predicts chapter 38's exit arithmetic: distributions recorded as profit that the completed schedule shows to be repayment, senior seats collecting applause the unrecorded lines paid for. Full visibility means nothing was priced at zero by silence. The schedule: chapter 38; the two-ingredient doctrine behind it: chapter 12; wages under standard: chapter 39; the pre-filled disclosure form: chapter 42.

Mission-reserve sufficiency score. Whether the purpose has a funded chest standing between it and the quarter that could buy it. Chapter 39 built the reserve, the purpose's own chest, funded on a written schedule toward a declared ceiling — because a purpose, like a household, gets exactly one path, and enough lean quarters can shave it to nothing without one dramatic decision. Chapter 41 wrote the rules the score checks: a stated size set against the worst plausible stretch rather than the typical year, a stated purpose, a stated schedule for filling and drawing, since "an undocumented reserve is a discretion, and discretions drift." Sufficiency is the balance and the schedule measured against that charter, with seniority verified (the reserve fills before any surplus exists to share, per chapter 39's procession) and with chapter 41's ownership test applied: "a reserve is judged by whose ruin it prevents," so a chest that smooths the fund's numbers while the mission weathers alone points the wrong way, whatever its balance. A reserve tapped in good years to sweeten a distribution scores as what it has become: yield wearing a rescue's name. The reserve: chapter 39; the rules: chapter 41; the sizing discipline: chapter 28; the endgame it exists to prevent: chapter 34.

Surplus authenticity score. Whether what the till calls surplus has earned the word — whether it is chapter 39's true surplus or merely a January balance. Chapter 39's procession is the construction: before any distribution, the claims that stand senior — wages to standard, the founder's line, maintenance as logged, the chests replenished to charter, the

community's owed line, every principal wired and sweat — each paid or provisioned, in order, out of a January number that only ever says *available*. The score is the fraction of actual distributions that left through that procession rather than around it, graded by the sentence that carries the test: "Available says the money can leave; surplus says it may." Authentic surplus traces to a realized result — profit from profit, chapter 14's rule, never from revenue, never from the borrower's principal returning in a costume, never from a drained reserve. Distributions that fail the trace are reclassified by the audit of chapter 44 and land, like every unexplained euro, on the residual. The procession: chapter 39; the base of the surplus term: chapter 40; the accrual honesty behind the January number: chapters 14 and 39; the yearly re-run: chapter 44.

Fixed-claim severity score. How fixed the claim really is, and how hard the fixity bites. The construction is chapter 35's payoff reading run across chapter 43's state grid: compute what the claim collects in each state of the venture's world, and score the flatness — a claim collecting alike in distress and in prosperity carries the loan's signature whatever the certificates say, while a claim paid only from prosperity that actually arrived has partnership's shape though nobody wrote the word. Severity weighs the flatness where it matters — chapter 28 proved that a constant subtraction swells against a shrinking base — so the claim is read at its weight in the bad states, not the average ones. Two committed verdicts calibrate the reading. A flat payoff at the front of an equity queue is chapter 37's "*mutuum* with a seat number"; and a flat claim carrying a risk charge is pricing insurance it never issues, chapter 35's rendering of the book's oldest test. The payoff reading: chapters 35 and 37; why fixity kills: chapters 10 and 28; the loan's anatomy underneath: chapter 7.

Risk-shift index. Where the danger actually went. Every protection is a transfer: collateral moves loss to the pledger, a guarantee to the household that signed it, seniority to whoever stands behind, a penalty to the stumbling. This index re-runs the losses through chapter 36's netting map to say which clauses moved them, and onto whom. It is the counted-once fence read as a flow: the four collections chapter 35 forbids on one danger are four arrows on this map, and the index totals the arrows that point away from the party charging for risk. The calibrating doctrine is Part IV's — loss belongs where capital stands — so a stack in which capital collects the risk line while the losses land on guarantors, workers, and the founder's unpaid years reads as shifted precisely to that degree. Chapter 13 supplies the test's oldest form — the loss clause is where a contract tells the truth — and chapter 37 its modern one: follow the waterfall to the seats that absorb the bad states, and compare them with the seats that were paid to. Loss allocation: chapter 13; netting and the two-sided count: chapter 36; the queue's geometry: chapter 37.

Governance-asymmetry index. The gap between who bears the venture's fate and who can steer it. The construction is chapter 43's steering map made a measure. Inventory the consequential acts (amending the waterfall, unlocking a layer, raising the cap, drawing the reserve outside its charter, forcing the sale) and record, for each, whose consent is required and whose is not; the index is the distance between the consent map and the exposure map, weighted toward every path that lets a party revise next year the very title it collects under. The committed checkpoints do the grading. Chapter 29's maintenance cargo: two keys wherever money moves, trustee and beneficiary never the same person, pay fixed to the office and blind to the flows through it. Chapter 41's guardian: an independent, itself asset-locked consent standing in front of "the acts that would undo the drawing". Chapter 38's side doors: the affiliate and management channels that turn control into claim without ever standing in the queue. And the drag read as chapter 43 reads it, "the power to sell other people's company," with the index pricing how far that power can travel before anything

with standing can stop it. Low asymmetry means the steering sits with those in the weather. The map: chapter 43; the cargo: chapter 29; the guardian and the acts: chapter 41; the side doors: chapter 38.

Form-substance divergence score. The distance between what the paper says it is and what it does. The reading rule has been fixed since chapter 3 — effect is given to intention and meaning, not to words and phrases, the Majalla's third article and *Gregory v. Helvering* converging on one sentence — and the score arithmetizes it: classify every instrument by its payoff, classify it again by its label, and measure the disagreement, because "The names are hypotheses; the payoff is the evidence." Assembled packages are tested whole under chapter 37's parity discipline — clauses whose combined payoff comes out flat in every state add up to a fixed claim, no matter how many instruments share the work — and single instruments against their own conditions: the *heter iska* run without its witnesses and oaths, chapter 31's own authorities warn, is a loan in partnership dress; the *murabaha* that never truly touches the goods is, in chapter 32's internal critique, the conventional product wearing the tradition's clothes. Zero divergence is the honest specimen: the label survives its payoff. Divergence is the counterfeit's one universal signature: every fraud in this book, ancient or synthetic, is a gap between name and geometry. The rule: chapter 3; the legal transformations and their tests: chapters 31 and 32; synthesis detection: chapters 37 and 43.

Cost-recovery discipline score. Whether a service charge is actually recovering costs. The standard is the strictest one under which a lending institution can still exist — the Monti di Pietà's, vindicated at the Fifth Lateran Council in 1515 — and the fence-words the credit floor carries from chapter 40 through chapter 42 are the score's four checks: *shown* (each cost line evidenced), *allocated* (attributed to the lending it serves), *once* (no cost recovered twice across fees), and *at zero margin* (the charge budgeted to break even, falling when costs fall). Chapter 17 supplies the diagnostic the score operationalizes: the honest charge pays for the apparatus surrounding the loan, appraisal desk to auction room, and would vanish if a benefactor covered the desk some other way; the loan itself remains gratuitous. A fee that grows with the appetite of the book rather than the cost of the desk has become a rate and is scored as one, whatever line it was booked to; the excess lands on the residual. The institution and its trial: chapter 17; the fence-words: chapters 40 and 42; the expense title underneath: chapter 19.

Mercy-capital purity score. Whether the mercy instruments are running clean. The floors this gauge patrols are the ones the traditions built to run at zero — the *gemach*, the *qard hasan*, the alms-fed pools of chapter 18 — and purity is scored against their own rulebooks, which are stricter than any regulator's. Nothing collected past principal, in money, in kind, or in words; the fences of chapter 8 run to benefits and flattery, not just coins. Queue and eligibility by need, never by donor advantage: chapter 15's separation of the giving hand from the deciding hand. The loan a real debt, mercy lending in expectation of repayment (the instrument whose repayment is optional is alms under a different name, chapter 16's boundary), but with deferral standing as a right wherever inability is shown, per chapter 22, and with the patience funded — a hardship promise that no reserve stands behind scores as the fragility it is, chapter 21's condition, answered at fund level by the anti-ruin reserve chapter 41 stations behind the mercy floors. Purity of one means the beautiful loan is still beautiful on audit: capital that circulates, collects its principal, and takes nothing for the service of being merciful. The instruments: chapters 15 and 16; the fences: chapter 8; the gift floor beneath: chapter 18; funded patience: chapters 21 and 41.

Asset-lock integrity score. Whether the lock actually locks. The doctrine is chapter 29's —

the corpus detained while the fruit alone travels, held by the waqf's three bars against sale, gift, and inheritance, by the *hekdes*'s trustees under outside authority, by Chalcedon's ban with an *oeconomus* to answer for it — and the score tests a modern lock's drafting against that standard. Walk every amendment path — the ordinary resolution, the merger, the redomiciliation, the supermajority the collecting party controls — and record which of them opens it. A lock any such path can open is what chapter 43 called “a wall with a gate in it,” and scores as a gate. Substitution is checked the way the jurists checked *istibdal*: permitted for the corpus's own preservation, under an authority distinct from the trustee, never as disguised sale. And the guardian is counted: a consent with no stake in the undoing — independent, and locked itself — seated in front of chapter 41's undoing acts, with the standing to refuse and keep its seat. Integrity of one means the corpus outlives its own trustees' worst quarter — locked, in chapter 39's words on the endowment, “past the reach of any board's bad quarter.” The three bars and their keepers: chapter 29; the mission-locked layer and the guardian: chapter 41; the steering audit: chapter 43.

Ruin-probability reduction score. What the arrangement does to the chance of reaching the floor. Ruin, throughout this book's mathematics, is an absorbing state, a place a path enters and never leaves; this score is the book's central claim about solidarity made falsifiable. Run the same ventures through the same weather twice, once solo and once under the arrangement being graded (the chest, the entry reserve, the contingent claim), and read the movement in the ruin column. Nothing else counts as the answer. Collections may stand nearly unmoved, the ensemble may not notice, and the arrangement may still have changed what one life can survive — that is chapter 28's demonstration, calibrated on its village, where provision comfortably above the average blow still lost, household by household, to the worst blow, and stopped losing only when the common chest stood. The score prices ruin beyond offset, chapter 28's specification: no one's collapse is netted away against a neighbor's good year; and its threshold is that chapter's blunt one: the reserve that leaves the ruin column where it stood was never architecture, whatever the brochure says. The floor and the reserve against it: chapter 10; the proof and the sizing: chapter 28; the pools it grades: chapters 27 and 41; the module that runs it: chapter 43.

Time-average resilience score. What the claim does to the path a venture actually lives. The book's survival mathematics runs on the distinction Appendix A carries and chapters 10 and 28 built: the ensemble average, taken across many parallel cases at a stroke, and the time average, accumulated along one compounding sequence, the only average anybody lives. This gauge works on the time axis by construction: take the venture's path year by year under the claim being graded, let each year compound on the last, and read the growth the single path keeps — graded at the worst year, not the mean one, because the mean is the property of whoever holds the whole cohort, and a venture holds one path. The finding this score exists to catch is chapter 10's: a fixed claim can look flawless in the ensemble's books and still be consuming the trajectory it collects from. A resilient claim caps its own drag: it takes from good years, waits through bad ones, and leaves the path's compounding intact. Read this score beside the previous one — ruin is the floor, resilience the slope, and an honest instrument answers for both. The two averages: chapter 10 and Appendix A; the proof: chapter 28; the pricing consequences: chapter 36; the grading module: chapter 43.

Solidarity-pool sufficiency score. Whether the pool can carry what it exists to carry. Two audits, and the pool must pass both. First the membership grammar of chapter 27, because a pool owned wrong is not a pool: contributions enter as the members' own (as dues, as donations, as *tabarru'*), never as a seller's revenue; the administrator's wage is fixed and

indifferent to refusals; exclusions police conduct, never misfortune; outside capital serves, never owns. Then the arithmetic of chapter 28, because a pool sized wrong is a promise scheduled to break: the solidarity reserve sized against the worst stretch it can plausibly be asked to meet; opened funded, by way of the entry contribution the old pools took at the door, rather than hoping to grow brave on schedule; and never drawn down in good years to flatter a distribution, a chest turned yield source having stopped being a chest. Sufficiency is provision read against the blows the pool must absorb, calibrated by the village's discipline: a jar even slightly short of the worst blow loses every time it arrives. The grammar: chapter 27; the sizing and the proof: chapter 28; the fifth form's anatomy: chapter 3; the fund-level chest: chapter 41.

Institutional-drift risk score. The standing pressure on an institution to become what it grades. Drift, chapter 29 ruled, is a design load, to be planned for the way an engineer plans for weather, and chapter 45 catalogs the temptations that are its names. This score counts the channels the drafting leaves open and weights each by who profits through it: compensation keyed to flows rather than fixed to office; reserves drawable in calm years; fees that grow with appetite; locks openable by those collecting behind them; classifications amendable without a vote, because "Classifications are commitments," and chapter 44 showed that an institution rebooking penalties as yield has amended its theology "in the middle pages of a report nobody reads". The committed antidotes are the checklist the score runs: counting done in public where the money comes in; removal power lodged with an authority that is neither trustee nor beneficiary; the disclosure schedule renewed annually and drafted for a stranger; the analyzer turned on the institution's own paper on a calendar, not on a suspicion. Low drift risk is not innocence; it is an architecture that assumes the third generation will be ordinary and locks accordingly — the oldest lesson of the endowments, from Interlude II's cave-wall benefactions to the monastic estates of chapter 29. The design load and its instruments: chapter 29; the bookkeeping where drift shows first: chapter 44; the temptations by name: chapter 45; the published self-examination: chapters 42 and 43.

Appendix E — Sample Non-Extractive Waterfalls

Appendix A holds the book's words, Appendix B its dates, Appendix C its forms, Appendix D its measures; this appendix holds its runs. Seventeen scenarios follow, each a waterfall worked in numbers: which seats collect, in what sequence, out of which euros, and what the sequence does to everyone in the line. They are the worked examples chapter 37's audit commissioned, the sketches chapter 26 ordered of what non-extractive paper "would pay and to whom," the waterfalls chapter 41 drafted and chapter 42 papered. Every file obeys the discipline chapter 28 set for its village: every rate chosen for transparency, every result checkable with a pencil, none of it a claim about any historical venture. Files the chapters already computed return here under their own committed numbers, credited as they arrive — the founder's Tuesday of chapters 37 and 38, the roastery's January of chapter 39, the village of chapter 28, the half-million-euro participation of chapters 35, 40, and 41. One file is new: call it the workshop, the young company of chapter 35's two term sheets, taken at the honest sheet's word. Its paper: €500,000 advanced as a capped participation; a total ceiling of two euros and ninety cents per euro advanced, which is €1,450,000, ever, by chapter 41's arithmetic; a stated thirty percent share of true surplus (a staged fraction); a reserve charter of €60,000, one genuinely bad quarter, sized by the roastery's rule; a crew of eight. Scenarios one through six run the workshop across six futures. Seven and eight run the cautionary stacks the honest paper exists to refuse. Nine through seventeen walk the instruments of chapter 41's ten-million-euro stack, layer by layer, each in its own liturgy. Every run obeys the commission chapter 37 wrote and chapter 41 drafted: "operating claims and wages filled, the reserves funded to charter, every principal returned" — wired and sweat alike — "then shared remainders to the cap," the investor's return seated "behind the enterprise's ability to survive collecting it." The metrics of Appendix D grade what these waterfalls stage; the analyzer of chapter 43 runs the same audit wholesale; and no run is ever compressed to a single digit, for chapter 43's standing reason.

Failure scenario. The workshop dies in year three; the market it served moved, and no clause could have moved it. Wind-down realizes €150,000. The waterfall runs the committed order: wages owed to the crew, €35,000, paid first; suppliers, €45,000; the costs of an orderly closing, €10,000. The remaining €60,000 goes to the participation, and the participation's loss, €440,000 of the €500,000 advanced, lands where the paper priced it. The shortness is the doctrine: the seat that contracts for a share of surplus is the seat that eats the loss, up to the capital and no further — the boundary chapter 13 found the figh stating as cleanly as anyone. The founders lose their years; the contribution schedule of chapter 38 prices what they lost (€60,000 of logged under-standard wages) rather than pretending the years were free. Appendix D's gauges, run on the wreck, return the one comfort the geometry allows: the extraction residual is zero. Money died here, and none of it moved to a seat that had drafted itself away from dying.

Partial recovery scenario. Year five: after a hard stretch that drew the chest down to €40,000, the workshop is sold to a competitor for €400,000. Proceeds plus remaining chest: €440,000. Wages and suppliers are already current — keeping them current is what the chest was drawn down for — and the closing costs €20,000. The participation takes home €420,000 of its €500,000. No surplus ever existed, so the thirty percent share collects thirty percent of nothing; chapter 37's ordering, "the capital whole before profit exists," makes the rest of the arithmetic unnecessary. The loss of €80,000 is capital's own. For contrast,

at this file's scale: a cumulative dividend of the kind chapter 37 priced, eight percent accruing, would have stood €200,000 of coupon (five lean years at €40,000) in front of these same proceeds, collecting hardest in exactly the future least able to pay it. Appendix D's fixed-claim severity score exists to see that coming.

Sustainable survival scenario. The workshop lives, and is never spectacular. Year six, the pattern year: €90,000 sits in the account in January. The procession of chapter 39 marches first — wages brought to standard, €25,000; maintenance as logged, €15,000; the chest refilled to charter, €20,000; the mission reserve's scheduled €10,000 — leaving €20,000 of distributable cash, split by the stated policy: €10,000 reinvested, €10,000 redeemed against the participation's principal. Seven years at that tempo return €70,000 by year ten, a seventh of the way home, and the ceiling of €1,450,000 is a figure on a document nobody rereads. The paper anticipated this future. Chapter 42's clause is the mechanism: installments run “from free cash after wages, suppliers, and the reserves at their charter sizes” and stretch, “without penalty, when the till says wait”; the remedies for a slow stretch are “a redrawn schedule, then the guarantor, then the community's long memory,” in that order, with no late-fee clause to compound the waiting. Zero in a lean year is a price, not a default. The participation may never reach its cap, and that possibility was inside the price of calling it a participation.

Mission-preserving buyback scenario. Year seven, the future chapter 41 narrated: “grown into the town's anchor employer,” the workshop converts to steward hands. The participation has collected €450,000 by then. The buyback formula, written the day the money went in per chapter 41, completes the claim at the ceiling's arithmetic: €1,450,000 less €450,000 leaves €1,000,000, paid in five annual installments of €200,000, each subordinated to the procession as a condition precedent, each stretching without penalty when a year says wait. On the last payment the shares return to steward hands, the economic rights end, and, in chapter 40's sentence, “further surplus belongs to the venture, its people, and its purposes.” Total collections: €1,450,000 on €500,000 over roughly twelve years, complete. The mission never met a buyer; the exit was what chapter 42's destination clause drafted it to be, “a homecoming rather than an auction,” priced at signing and survivable by construction.

Moderate surplus scenario. The middle future. From year four the procession leaves modest distributable cash: enough that principal comes home by year nine, while the thirty percent share of true surplus adds €180,000 across years four through twelve. Total collected, €680,000 on €500,000 over twelve years; the €1,450,000 ceiling never binds. Two readings. First, every distribution in the run passed the same gate — no arrears standing, the chest at charter, the reserve schedule current — so the payments could keep a rhythm the members could plan around without any single year choking the company to keep a date. Second, in the moderate future the cap cost the investor nothing: an uncapped honest partnership would have paid nearly the same euros. The cap is not a price paid in every future; it is a promise about the good one, which the next scenario reaches.

Large surplus scenario. The workshop compounds. True surplus crosses €300,000 a year by year five, still climbing; principal is home in year six; the thirty percent share collects €80,000, €120,000, €150,000, €180,000, €200,000, €220,000 across years five through ten — €950,000 above principal, which is the ceiling, reached. The claim is complete: two euros and ninety cents for every euro advanced, in ten years. From year eleven the venture's entire surplus, the share included, stays with the venture, its people, and its purposes, and an uncapped thirty percent would have collected €200,000 and more, every year, indefinitely. This is the future in which the design pays most of its price, and it pays on purpose:

each layer, chapter 41 said, “renounces something exactly so that the layer above it can exist without pretending.” The traditions had long since stated the shape. “A debt under damdupat can be heavy; it cannot be bottomless,” the first interlude reported from the Indian codes. The capped participation reads the same sentence to capital: a return can be handsome; it cannot be endless.

Forced-sale cautionary scenario. Now the stacks the honest paper refuses, run so the refusal has numbers. A funded company carries €2,000,000 of participating preferred at one times, fifty-five percent as converted, with a drag-along; the fund holding it is at the end of its life. An offer stands at €4,000,000 today; the plan the room believes in reaches €9,000,000 in two years. Run both futures. Today: the preferred takes €2,000,000 off the top plus fifty-five percent of the €2,000,000 remainder, €3,100,000 in all; the common splits €900,000. In two years: €2,000,000 plus fifty-five percent of €7,000,000 is €5,850,000 preferred, €3,150,000 common. Both seats collect more by waiting. The sale happens today anyway because, as chapter 37 put it, a fund at the end of its life “does not need the company’s best decade; it needs the company’s next transaction,” and “the drag turns that need into everyone’s obligation” — the mission “sold to meet a preference,” toward “the buyer’s purposes nobody in the building would have chosen,” the workers riding into the data room as line items. The numbers cannot even call the choice greedy; they show the clock, not the arithmetic, choosing the future. Whose horizon governs is a term of the contract, and it should be read as one.

Preferred-equity extraction scenario. The founder’s Tuesday, rerun under chapter 37’s committed numbers. Sale: €12,000,000. The preferred: €10,000,000 invested, participating, holding fifty-five percent as converted; the founder thirty percent; the employees fifteen. As drafted, the preferred takes its €10,000,000 off the top, then fifty-five percent of the €2,000,000 remainder: €11,100,000 of the twelve. The founder’s thirty percent pays €600,000; the employees’ fifteen, €300,000. Redraft one word — the same preference, non-participating — and the preferred takes €10,000,000 and stops; the €2,000,000 remainder belongs to the common alone, one and a third million of it to her. One word moved more than seven hundred thousand euros. Chapter 38’s schedule then decomposes the protected seat’s €1,100,000 excess exactly: €390,000 of genuine surplus, and €710,000 “that is not surplus at all but transfer — the working seats’ unreturned principal, changing hands, renamed on arrival.” Chapter 37 gave the residual its address: it is funded, “euro for euro, by the residual claimants.” A preference, that chapter concluded, “is a rate that learned to stand in line” — and this is the line, with the amounts filled in.

Non-extractive partnership scenario. The same Tuesday, redrafted as the paper chapter 26 sketched. The €10,000,000 enters as partnership capital: no preference, no participation stacked on top, no accruing coupon. At the €12,000,000 sale, chapter 38’s order runs — all capital home first, wired and sweat alike: the €10,000,000, and the €710,000 of sweat principal the contribution schedule priced at entry (the under-standard years, the deferred draws, the founder’s own ten thousand); then the true surplus, all €390,000 of it, shared by stated fractions over stated contributions — at a staged sixty–forty between capital and work, €234,000 and €156,000. The investors leave with €10,234,000, which is €866,000 less than the participating draft paid them, and every euro of the difference sits on a titled line in somebody else’s column: the crew’s principal home, the founder’s decade repaid, a surplus that had earned the word. Both drafts were legal. Only this one can fill in chapter 36’s second table; and the test Sixtus V set for the honest partnership, as chapter 37 rendered the bull — it “casts up fruits, costs, and losses together and divides what is actually there” — closes over it without remainder.

Gemach-style emergency support scenario. Chapter 15's file, under its own numbers. A *gemach* of \$10,000 lends \$500 bridges, repaid over ten months: roughly twenty households a year carried across their bad week, two hundred rescues in a decade, and the ten thousand dollars all still there for the third decade and the fourth. The instrument's grammar does the work: no interest, ever; a guarantor and community knowledge in place of collateral; a repayment plan in place of a late fee. The waterfall is a circle, out, back, out again, and the arithmetic of revolving mercy is the entry's whole point: capital given once rescues indefinitely — the founders' ninety-five dollars of 1892 still out working thirteen decades later. In chapter 41's stack the same machine holds the €700,000 second layer: at the file's scale, fourteen hundred five-hundred-euro bridges per revolution, the floor that catches what no participation should be asked to price.

Qard-hasan-style bridge support scenario. Chapter 16's anatomy at bridge scale. A member's business is stranded by a late receivable; the fund advances €3,000. Owed back: €3,000, carrying "no stipulated benefit of any kind," in cash, kind, or favor. Ten months, redrawn to twelve when repayment needs the room. If the borrower, made whole, freely adds something at settlement, the classical rule permits it — the tradition's exemplar repaid a borrowed camel with a better one, as chapter 16 told it — provided nothing was stipulated, promised, or prearranged; the excess is gratitude, not price. The same instrument works at industrial scale as the shock absorber chapter 27 found inside regulated *takaful*: when a pool runs dry, the operator advances the deficit as *qard hasan*, repaid from future surpluses. The waterfall line is one line long, principal out, principal home, and its emptiness is the design; a charge for the crossing would be the first step toward pricing the flood.

Takaful-style mutual-aid shock scenario. A staged pool. Forty members contribute €250 each; the year's chest is €10,000; every contribution enters under *tabarru'*, "a donation rather than a price" in chapter 27's phrase, so no one has sold anyone certainty. The operator stands outside the fund as hired administration at a disclosed fee, €1,200, priced as work. The shock: one member's shop floods; the assessors certify €6,000; the chest pays, that being what the chest is for. The year-end ledger keeps chapter 27's three books — "the cost of running the thing, the reserve that keeps it standing, and the surplus of the light years." This year: costs €1,200, claims €6,000, the reserve line €1,000, surplus €1,800 — which, in the committed sentence of the industry's own criteria, "can be retained in the fund or distributed, totally or partially, among policyholders." It stays home or goes home; the operator is excluded from it by seat, not by virtue. Forty people bought each other's worst week, and the price of the purchase is published.

Mudarabah-style profit-sharing scenario. €100,000 from the *rabb al-mal*; the *mudarib* brings the venture, the skill, and the months. The stated split of realized profit: sixty to capital, forty to work — a ratio, chapter 11's rule insists, that "must bite on actual profit," never a guaranteed sum. Year one, profit €30,000: €18,000 and €12,000. Year two, loss €20,000: it falls on the capital, up to the capital, and the *mudarib* is owed nothing and owes nothing. "The *mudarib* loses his months, his labor, and his hoped-for share, which the jurists regarded as loss enough," as chapter 11 put it. Year three, the venture is wound up at €95,000: less than the capital that entered, so across the whole life no profit ever existed to split; the investor's loss is €5,000, the manager's wage for three years is zero, and both outcomes were in the price of the words they signed. Now the test that guards the instrument: add one clause guaranteeing the €100,000 home whole, and the verdict chapters 13 and 37 carried is structural, not stylistic — the partnership "has dissolved into a loan in the measure of the guarantee," and the profit share it promises is *riba*. The

scenario's arithmetic is easy; the discipline is refusing the clause that would make it easier.

Heter-iska substance-test scenario. €200,000 placed under a standard *heter iska*: half loan, at the manager's risk; half deposit, at the investor's; and a named settlement sum, the *sechum ha-hitpashrut*, of €12,000 a year, payable in lieu of opening the books. Run the instrument twice. Live run: a bad year comes, and the manager does what the form contemplates — brings the witnesses, proves the loss; the deposit half's share of the loss falls on the investor, as the codified rule chapter 13 carried has it (the entrusted half "stolen or lost, the administrator is not liable to pay"), and she collects less than the settlement sum, in a hard enough year less than principal. The evidentiary machinery — witnesses findable, the *shevu'ah* swearable, loss provable — is, chapter 31 showed, the instrument's true signature; kept live, the named figure is a settlement between partners, not a rate. Dead-letter run: the acceptable witnesses are named only in a far city; the oath is drafted never to be sworn; and the €12,000 arrives on schedule every year, profit or loss, for a decade. The poskim struck the far-city device down themselves, on the ground chapter 31 quoted: "verification that cannot occur is not verification." Appendix D's form-substance divergence score is the same ruling as arithmetic, and the substance test fits in one sentence: a settlement figure that has never once been declined, varied, or verified against a real year is chapter 11's guaranteed sum in new paperwork — a loan's payoff "wearing a partnership's clothes."

Cost-recovery Monti-style credit scenario. A revolving credit fund at cost. Capital €200,000, lending small bridges against pledges and invoices; the year's audited operating costs — the two clerks' wages, the rent, the audit itself — come to €9,000, recovered pro rata across the year's lending as a disclosed charge and recalculated annually from the books. No profit line exists for costs to wander toward: the charge answers the Fifth Lateran definition quoted in chapter 17, "a moderate sum for their expenses and by way of compensation," intended "exclusively to defray the expenses" of staff and upkeep, "provided that no profit is made therefrom"; and the schedule of recoverable costs is "an exhibit, not a discretion," per chapter 42. The arithmetic that makes cost recovery a mercy rather than a fee is the counterexample chapter 17 preserved: the London fund of 1361 paid its expenses out of the capital itself, and plain arithmetic ate it, every year's costs carved from the corpus until there was no corpus. A fund that recovers its documented costs, and nothing above them, is not charging for money; it is arranging to still exist next year, at scale, for strangers.

Mission-reserve-first distribution scenario. The roastery's January, under chapter 39's committed numbers, run here as paper rather than as virtue. The screen says €120,000 available. The procession marches as a condition precedent written into the distribution clause itself, the drafting move chapter 42's buyback schedule demonstrated: "Wages to standard, thirty thousand. The founder's deferred draw, ten. Maintenance as logged, twenty. The chest's replenishment, twenty-five. The mission reserve's schedule, ten. The community's owed line, ten. The priced footprint, ten." One hundred and fifteen thousand of the one hundred and twenty were never anyone's to distribute; the true surplus is €5,000. "The board that distributes it has extracted nothing from anyone." "Available says the money can leave; surplus says it may." The scenario's addition to chapter 39 is only the enforcement: because the reserve is filled before any distributable amount legally exists, a lean year cannot be voted against the mission, and a reserve tapped in a good year to sweeten a distribution is reclassified by chapter 44's audit as what it became — in Appendix D's phrase, "yield wearing a rescue's name."

Ergodic solidarity-pooling scenario. The village of chapter 28, closing the appendix under its own numbers, because every waterfall above assumes what this one proves. Twenty households; six units saved per year against a hundred-unit blow. Jars: a household needs seventeen years of uninterrupted saving before one blow is survivable, and across twenty years more than eleven of the twenty are ruined on average, about fifty-seven in a hundred, in a village that as a whole was never poor. One chest, the same six units each: the first year's blow of a hundred lands on a chest of a hundred and twenty and is carried, that year and every year; ruin falls to about nine in a hundred. Bank one blow in advance and it falls below six; bank two, toward three; give a town of a hundred its one-blow reserve and ruin runs about one in a hundred. The sizing rule travels with the arithmetic: a solidarity reserve, chapter 28 ruled, is "sized to the pile-up, not the average year." And that chapter already drew the conclusion these pages have been staging: "portfolios are villages." The community reserve layer of chapter 41's stack, the €500,000 held "never deployed, never distributed, sized to the bad stretch," is the village chest at institutional scale, standing between every path in the portfolio and the floor. It appears in no waterfall as a payee, which is the point. It is the reason the other sixteen scenarios could afford their gentleness: the seat held empty so that everyone else's seat holds.

Appendix F — Reading Guide

Appendix A holds the book's words, Appendix B its dates, Appendix C its forms, Appendix D its measures, Appendix E its runs; this appendix holds its books. One rule governs the shelf: every source named here did verified work in a chapter of this book — its wording checked, its dates pinned, its claims read at the granularity the chapter reports — and nothing is listed on hearsay, because a guide that recommends what it has not examined would fail the audit this book runs on everyone else. The thirteen clusters follow the table of contents. Each names the texts first, then the way in for a reader starting cold, then the chapters where the book put the source to work, since the chapters carry the disputes, the case files, and the full circumstances that a shelf list cannot. Where a tradition's texts are quoted in these pages, they are quoted from named renderings — the 1989 New Revised Standard Version for the Bible, M. A. S. Abdel Haleem for the Qur'an, Eliyahu Touger for the Mishneh Torah, the book's own close renderings with exact loci for the Talmud, the Digest, and Aquinas — and a reader who wants to meet the sources in the same form should start with those.

Primary Jewish sources. Begin where the law does: the three Torah passages of chapter 2 ([Exodus 22:25](#), [Leviticus 25:35–37](#), [Deuteronomy 23:19–20](#)), and then the Mishnah's interest chapter in [Bava Metzia](#) with the Gemara that grew around it, the folios this book kept returning to: [61b](#) on the two kinds of forbidden increase, [63b](#) where any reward for waiting is named, [68b](#)'s wage seminar, [70a–71a](#) on orphans' money and precedence, [75b](#) where the lenders' own joke is filed, [104b](#) where the *iska* is born, [42a](#) on burying money properly. The Davidson edition of the Koren Steinsaltz English, freely readable on [Sefaria](#), is the way in; this book's renderings are its own, made against that text with exact loci. From the Talmud, go to the codes: Maimonides' [Mishneh Torah in Touger's translation](#) (the partnership laws of [Sheluchin veShutafin 6:1–3](#) that chapter 13 reads, the laws of sale that chapters 23 and 24 read, and [Gifts to the Poor 10:7](#), the ladder's top rung, met in chapters 3 and 18), then the [Shulchan Arukh's laws of interest](#), met in chapter 12 at [siman 177](#), with the [Tur](#) behind it. Doctrine lives in chapters 2, 4, 7, and 8; instruments in chapters 12, 13, 15, and 18; the modern practice in chapter 31.

Primary Christian sources. The scriptural spine is chapters 4 and 5; read them beside the [1989 NRSV](#), the version this book quotes. The doctrine then runs through documents a reader can hold: [canon 17 of Nicaea](#) (chapter 1); Aquinas at [Summa Theologiae II-II, questions 77 and 78](#), the anatomy chapters 7, 9, and 12 walk clause by clause; the [Liber Extra](#) that Raymond of Penyafort assembled for Gregory IX in 1234, where *Naviganti* sits at X 5.19.19 (chapter 10); *Consuluit*, Urban III's answer, where the lender's hope alone makes the usurer (chapter 8). The councils are best met in the standard conciliar English of the [Tanner edition](#), carried in full at [papalencyclicals.net](#): above all Session 10 of the Fifth Lateran, the bull [Inter multiplices](#) of May 1515, the council's formal definition of usury and its verdict on the Monti (chapters 14 and 17). Benedict XIV's [Vix pervenit](#) of 1745 closes the doctrinal file (chapters 7 and 30); Sixtus V's [Detestabilis avaritia](#) reaches English in Jeremiah O'Callaghan's [Usury, Funds and Banks](#) (chapter 13); the older collections travel under [Denzinger's numbering](#) (chapter 14). The standard modern road into all of it remains John T. Noonan's [Scholastic Analysis of Usury](#) (1957), the study behind chapters 10, 21, and 30.

Primary Islamic sources. The Qur'an in [M. A. S. Abdel Haleem's translation](#), the rendering this book quotes throughout: start with the [riba verses, 2:275 through 2:280](#), and the

charity verses around them (chapters 2 and 6). The hadith collections (Bukhari, Muslim, Abu Dawud, Ibn Majah) are readable with collection and number at sunnah.com, the carrier against which every hadith quoted in this book was checked; chapters 6, 10, 16, and 24 hold the load-bearing reports, from the curse on the whole cast of a riba loan to the Prophet's own credit purchases. For the law worked out, three modern bodies of text: the Majalla, the Ottoman civil code, in [Tyser's English](#), whose articles on sale, deceit, and intention this book cites by number (chapters 3, 10, and 23); the [AAOIFI Shari'ah Standards](#), on *mudarabah* and partnership, *sukuk*, *gharar*, insurance, and *waqf*, with the February 2008 sukuk resolution that disciplined a market (chapters 13, 24 through 27, 29, and 32); and the resolutions of the [International Islamic Fiqh Academy](#), from [the 1985 insurance ruling](#) to [the 2013 Riyadh criteria](#) for the cooperative pool (chapters 16, 24, 26, 27, and 32).

Hindu Dharmaśāstra and Arthaśāstra sources. Interlude I's texts are all readable in the old translations online. The Manusmriti in Bühler's translation (Sacred Books of the East volume 25, from 1886, at sacred-texts.com), whose vulgate numbering the Interlude follows, since later critical editions renumber, carries the eighteen titles of law, the caste-graded tariff of 8.142, and the doubling ceiling of 8.151; [Jha's edition with Medhatithi's commentary](#), at wisdomlib.org, lets a reader watch the great early commentator at work. [Yajñavalkya's code with the Mitakshara](#), the commentary scholars place around 1100, long the working law of Anglo-Indian courtrooms, sits in the same library, beside the dharma-sutras of Gautama, Vasishtha, and Baudhayana. Kautilya's Arthaśāstra, in Shamasastri's translation, prices the whole system in one chapter, [3.11 on the recovery of debts](#). Behind every secondary account stands Kane's [History of Dharmaśāstra](#), the standard authority historians of the literature lean on. The Interlude walks all of it, including *damdupat*, the doubling ceiling still enforced in the courts of the Bombay tradition.

Buddhist Vinaya and temple-finance history. Start with the rules themselves: *Nissaggiya Pacittiya* 18, 19, and 20 (the monk's renunciation of money, buying and selling, and barter) with rule 10's steward mechanics, and the Jetavana purchase at Cullavagga VI.4; the [Book of the Discipline](#) carries the rules in English. Thanissaro Bhikkhu's [Buddhist Monastic Code](#) is the modern commentary a beginner wants, and his sutta translations at [Access to Insight](#) and dhammatalks.org, with Narada Thera's rendering of the layman's discourse at Digha Nikaya 31, carry the lay-facing texts Interlude II quotes. The scholarship then tells the astonishing institutional story: Gregory Schopen's "Doing Business for the Lord" (Journal of the American Oriental Society, 1994; reprinted in [Buddhist Monks and Business Matters](#), 2004), where a Vinaya orders the endowment lent at interest; Yang Lien-sheng's "[Buddhist Monasteries and Four Money-raising Institutions in Chinese History](#)" (Harvard Journal of Asiatic Studies, 1950); and Jacques Gernet's [Buddhism in Chinese Society: An Economic History from the Fifth to the Tenth Centuries](#), written in French in 1956 and Englished by Verellen in 1995, still the standard economic history — with the Dunhuang loan documents behind it. Interlude II holds the file, from the silent steward to the Inexhaustible Treasury.

Medieval commercial history. The sea is the way in. Goitein's [A Mediterranean Society](#) reconstructs the Geniza merchants whose paperwork chapter 11 reads, down to the India traders' archive of his final years; Udovitch's "[At the Origins of the Western Commenda: Islam, Israel, Byzantium?](#)" (Speculum, 1962) puts the parentage question this book leaves unsettled because the scholarship does; the [cartulary of the Genoese notary Giovanni Scriba](#) shows the *commenda* splitting profits before a notary of the 1150s. Florence Edler de Roover's "[Early Examples of Marine Insurance](#)" (Journal of Economic History, 1945) documents the earliest insurers writing protection as loans they had never received, the costume story chapter 27 tells; Koyama's "[Evading the "Taint of Usury"](#)" reads the

same evasions as an economist; [Felloni's work on the Casa di San Giorgio](#) anchors chapter 26's oldest financial machine. For the late-scholastic exchange floor, Wim Decock's "[Les-sius and the Breakdown of the Scholastic Paradigm](#)" (Journal of the History of Economic Thought, 2009), Toon Van Houdt's study "[Lack of money](#)," and [Monsalve's 2014 companion article](#) in the same journal carry chapter 21's story at Antwerp. And because this history has a wound in it: Hsia's *Trent 1475* (Yale, 1992) is the trial-record study behind chapter 17's plain account of what the preaching campaign against the Jews of Trent cost them.

Modern Islamic finance criticism. The quarrel is the tradition's own, and the reading list is its critics in their own words. M. N. Siddiqi's "[Islamic Banking and Finance in Theory and Practice: A Survey of State of the Art](#)" (Islamic Economic Studies, 2006) is the theorist's audit; the *murabaha* that "conquered the landscape" is his phrase and his complaint. M. Umer Chapra's *Towards a Just Monetary System* (1985) states the profit-sharing vision the industry was supposed to build. Mahmoud El-Gamal's "[Limits and Dangers of Shari'a Arbitrage](#)," in the Harvard Islamic Finance Project's *Islamic Finance: Current Legal and Regulatory Issues* (2005), names the reengineering method cold, and his [companion essay on mutuality as the antidote](#) points where chapter 27's pools already stand. Taqi Usmani speaks for the jurists twice: *An Introduction to Islamic Finance*, with its benchmark defense and its plea to be rid of the practice, and the AAOIFI paper "[Sukuk and their Contemporary Applications](#)," whose finding that most issued sukuk mirror conventional bonds set off the 2008 correction. John Foster's BBC report "How Sharia-compliant is Islamic banking?" (December 2009) preserves the fatwa-shopping banker chapter 32 weighs against the scholars' refusal record. Chapters 20, 24, 26, and 32 hold the full proceeding.

Modern halakhic finance. The documents come first: the *shtar heter iska* of [Mendel Avig-dors of Cracow](#), the late-sixteenth-century *tikkun Maharam* that standardized the instrument; [Joshua Falk's Kuntres al Diney Ribbit](#) with the Council of Four Lands' credit rules of 1607; the [Nachalat Shivah](#), whose author came to the formula's defense; and the [Chachmat Adam's](#) later version, the line that runs to the wall signs in Israeli banks (chapters 14 and 31). The modern responsa then test the machine: Moshe Feinstein's [Igrot Moshe at Yoreh De'ah II:62](#), the scope ruling with its attachable-asset repair, and II:63 on the corporate borrower, a ruling R. Shlomo Zalman Auerbach opposed, a *machloket* chapter 31 reports without adjudicating. Bleich's survey "[Heter Iska in American and Israeli Courts](#)" follows the document into secular litigation, where judges must decide whether the partnership is real. Read chapter 12 first for *sechar tircha* and siman 177, then chapter 31 for the whole architecture (witnesses, oaths, settlement sum) and for what the *poskim* themselves say *ribbit*-washing looks like.

Catholic Social Teaching and modern finance. The encyclicals are all at [vatican.va](#) in official English, and the sequence chapter 33 audits reads quickly: *Rerum Novarum* (Leo XIII, 1891), where the tradition's economic voice returns with usury in its third paragraph; *Quadragesimo Anno* (Pius XI, 1931), drafted by Nell-Breuning, on credit's reign; *Populorum Progressio* (Paul VI, 1967) on development and debt; *Sollicitudo Rei Socialis* and *Centesimus Annus* (John Paul II) on the debt of nations and the market's two answers; *Caritas in Veritate* (Benedict XVI, 2009), where gratuitousness enters the balance sheet; *Laudato Si'* (Francis, 2015) for the ecology of the whole; and *Oeconomicae et pecuniariae quaestiones* (2018), the joint instrument-level document that reads term sheets, credit default swaps included, in Rome's own hand. Around them: the [Catechism](#) at 2269 and 2409, the [Compendium](#) at 341, and the two codes' quiet swap of canon 1543 (1917) for [canon 1284 \(1983\)](#), the exhibit chapters 30 and 33 explain. Chapter 34 carries the teaching into ESG's world; the addresses of [Francis in 2018](#) and [Leo XIV in 2025](#) show the old word alive in Rome's mouth.

Mutual aid, friendly societies, guilds, confraternities, and takaful. The oldest document on this shelf is a by-law: the burial society of Lanuvium, CIL XIV 2112, dated by its consuls to June of AD 136, readable in English in [Harland's collection of Greco-Roman associations](#) — dues, benefits, arrears rules, all of it (chapter 3). The 1911 Encyclopaedia Britannica's article "[Friendly Societies](#)," on Wikisource, surveys the movement Rose's Act of 1793 first encouraged, with the counts (nearly fourteen million members) that chapter 28 reads as anti-ruin infrastructure; the Marine Insurance Act 1745 and the [Life Assurance Act 1774](#) (the latter still on the statute book at [legislation.gov.uk](#)) mark the state drawing the line between pooling and wagering (chapter 27); the mathematician [Dodson's work for the Equitable](#) begins actuarial fairness in earnest. For the living instruments: chapter 15's free-loan societies, from the [Hebrew Free Loan Society](#) founded in 1892 on ninety-five dollars; [Akhawat's](#) lending through mosques and churches, with Munir's study "[Akhawat: Making Microfinance Work](#)" (2012) as the outside look (chapter 16); and the *takaful* texts already shelved in the third cluster — the Academy's [1985](#) and [2013](#) resolutions, AAOIFI's insurance standard — which chapter 27 reads beside the *chevra kadisha* and the lodge.

Mathematical finance, actuarial fairness, and ergodicity economics. The prehistory is short and startling: Ulpian's annuity table at [Digest 35.2.68](#), Rome pricing lifetimes in three bands, which chapter 10 reads with the modern actuarial scholarship, and [Dodson's mathematics for the Equitable](#), where fairness first became a calculation (chapter 27). The modern shelf is two items and a method. Ole Peters's "[The ergodicity problem in economics](#)" (Nature Physics, 2019) states the divergence this book's Part IX runs on (the ensemble average nobody lives versus the time average everyone does) in the coin-toss gamble chapter 28 stages at a kitchen table. Graham Boyd and Jack Reardon's [The Ergodic Investor and Entrepreneur](#) (2023) carries the finding into venture finance and the design of companies. The method is this book's own from there: chapter 28's village, run twice, with every number checkable by pencil; chapter 10's two averages planted in Talmudic soil; the ruin arithmetic that Appendix D gauges and Appendix E stages. A reader who wants the mathematics without the theology can start at Peters; a reader who suspects the theology got there first can start at Bava Metzia 70a.

Steward ownership and non-extractive capital. The movement's own texts are the entry: the Purpose Foundation's [steward-ownership principles](#) (self-determination and purpose-orientation, control held by stewards rather than sold or bequeathed) and its practical companion, "[Steward-ownership: a short guide-book to legal frameworks](#)," which chapter 29 reads beside the fiqh of the *waqf*. The living exhibits publish their own constitutions: the [Carl-Zeiss-Stiftung](#), founded by Ernst Abbe in 1889 and governed by his 1896 statute; Robert Bosch GmbH, whose [ownership pages](#) state the trust arithmetic plainly; [Carlsberg](#); and [Patagonia's announcement of September 2022](#), under the title "Earth is now our only shareholder." [Employee Ownership trusts](#), met in chapter 34, put the same locks in workers' hands. Chapter 29 maps these onto the *mutawalli* and the asset lock; chapters 41 and 42 then draft the instruments a Steward Ownership conversion signs: the stack, the waterfall, the ten documents. This book's Parts IX and X are, in effect, this cluster's missing manual: what the locked company owes its capital, and what its capital may honestly collect.

Contract analysis and moral-economy accounting. The shelf for the book's last discipline is mostly the book. The ledger that seats every euro on a named title is chapter 35's; the analyzer that runs the audit wholesale, in ten modules and no composite score, is chapter 43's; the audit year, with its reclassifications and its found euros, is chapter 44's; the ten signable documents are chapter 42's library; and the three appendices before this one hold the forms, the measures, and the runs. Two legal texts stand behind the whole method

and reward reading in the original: article 3 of the Majalla, in [Tyser's English](#), and the United States Supreme Court's [Gregory v. Helvering](#), 293 U.S. 465 (1935) — the Ottoman jurists and the American justices converging on the rule chapter 3 laid down for every instrument in this book, that intention and meaning outrank words and phrases. AAOIFI's Standard 31, on *gharar*, shows a living tradition grading contractual uncertainty with the same seriousness (chapter 23). The reader who has reached this page needs no further list; the method is portable, the paper is everywhere, and the last instruction is chapter 3's first one — read the payoff, not the name.